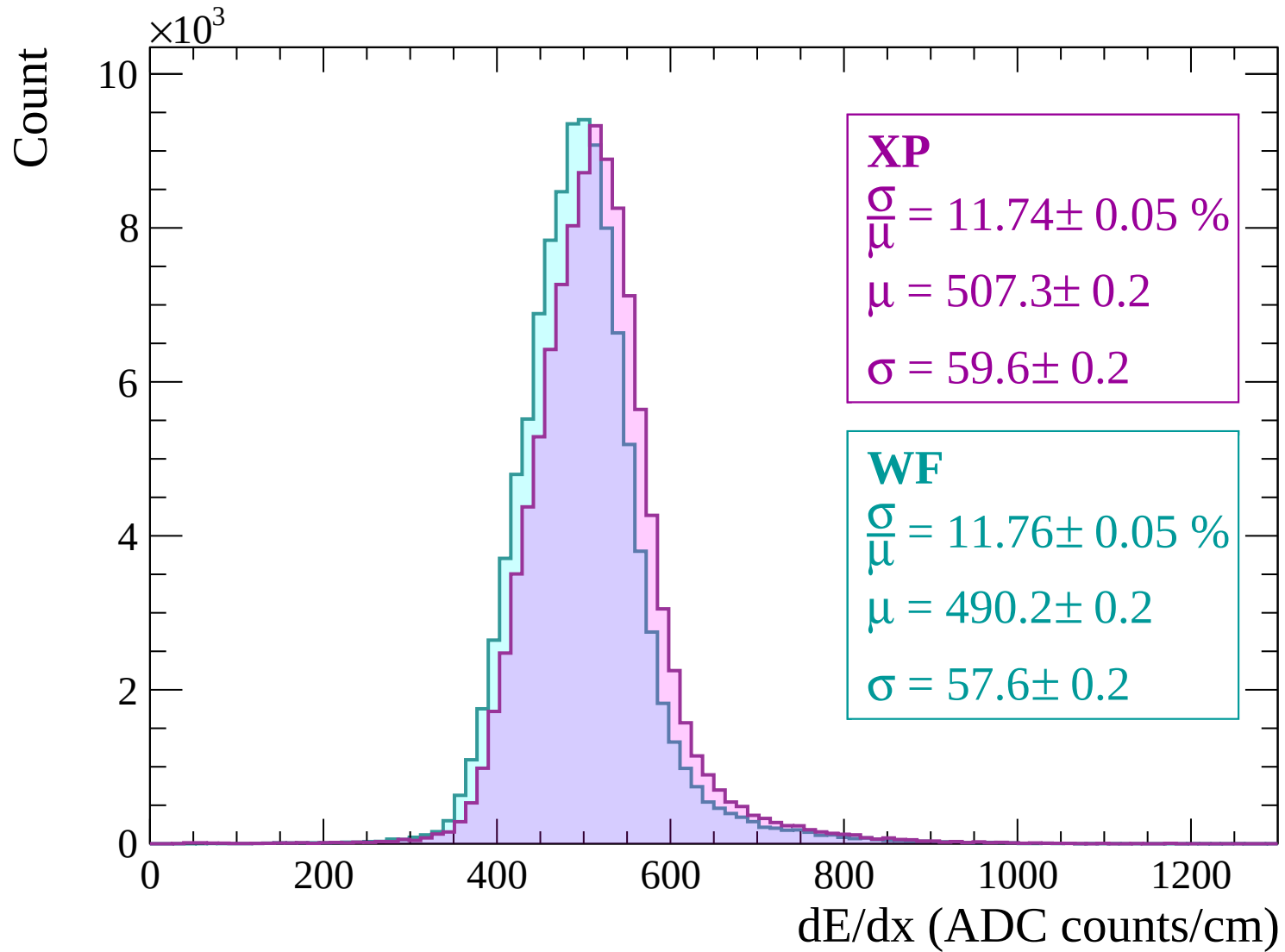
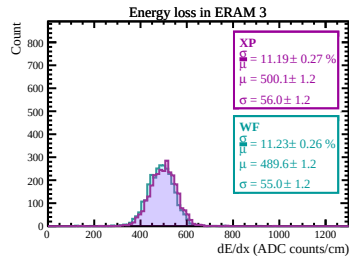
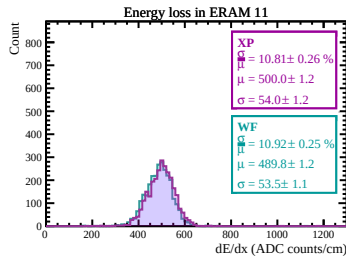
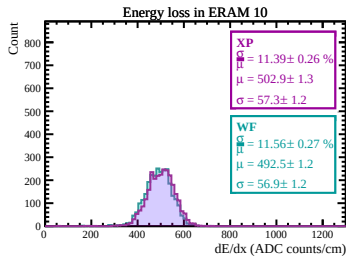
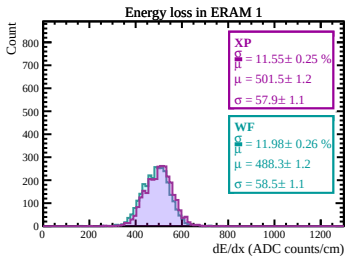
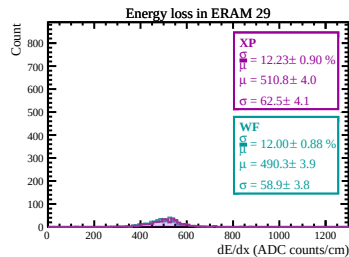
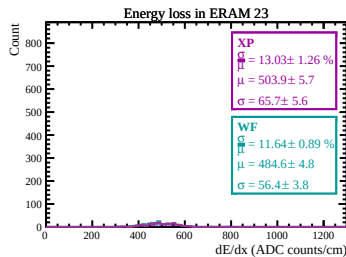
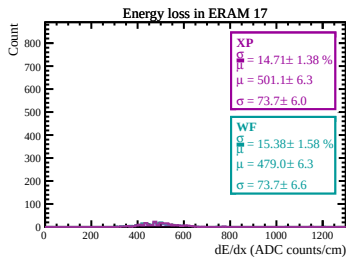
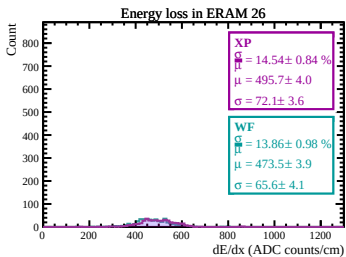
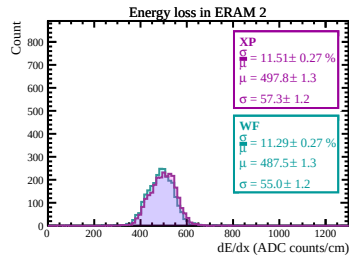
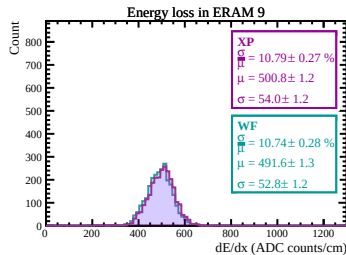
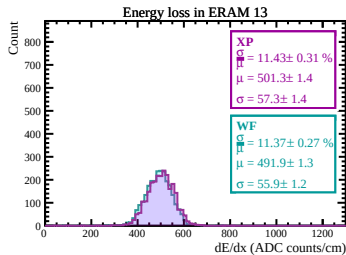
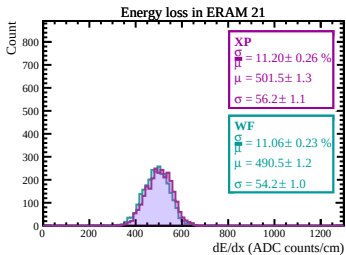
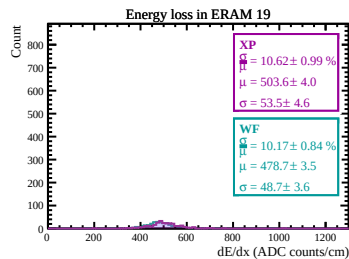
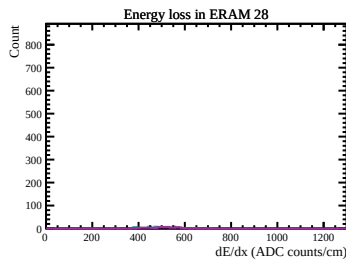
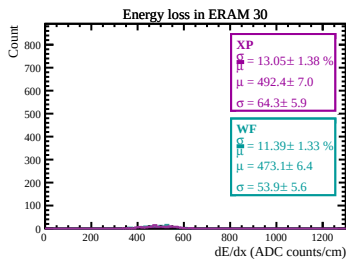
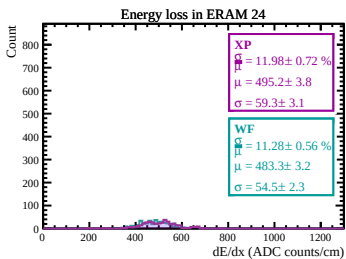
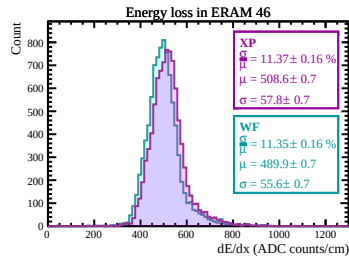
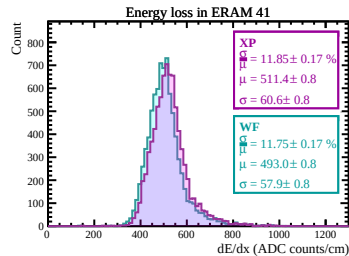
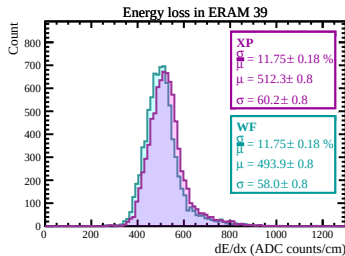
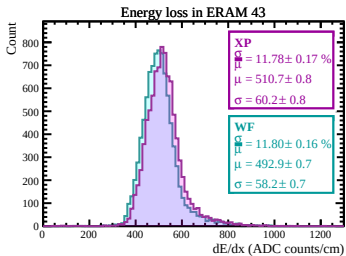
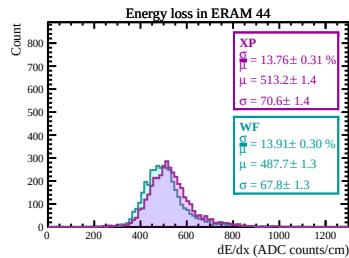
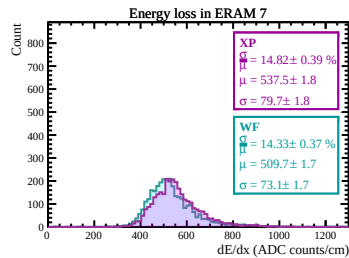
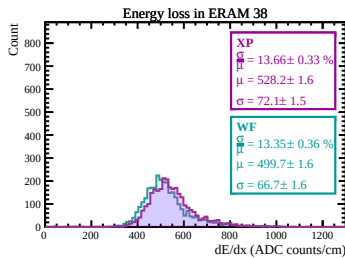
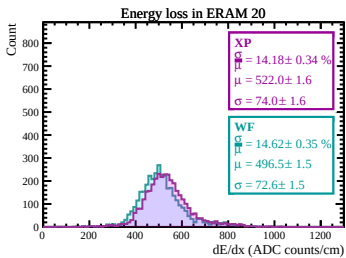
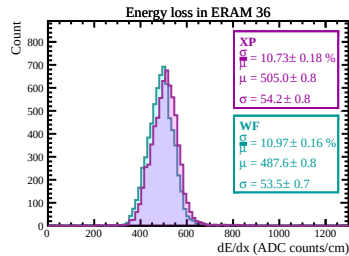
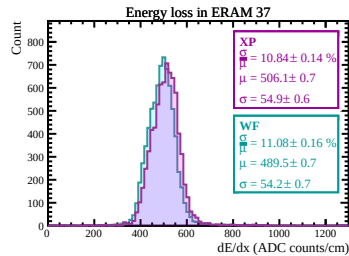
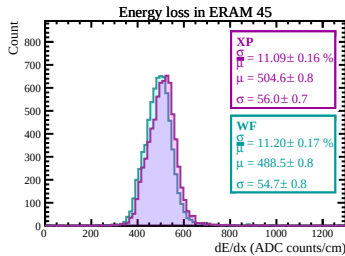
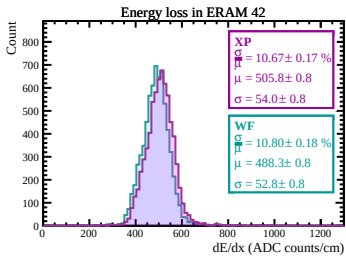
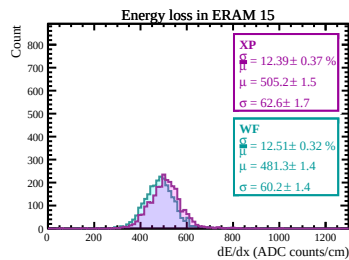
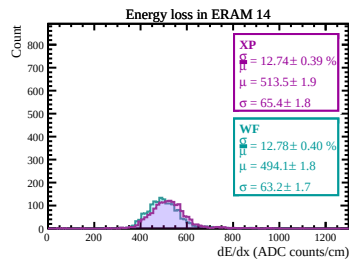
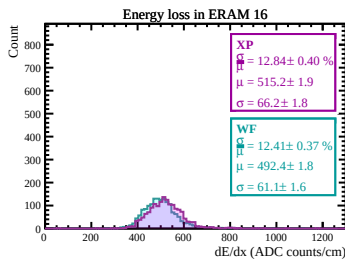
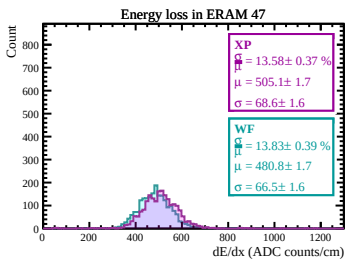
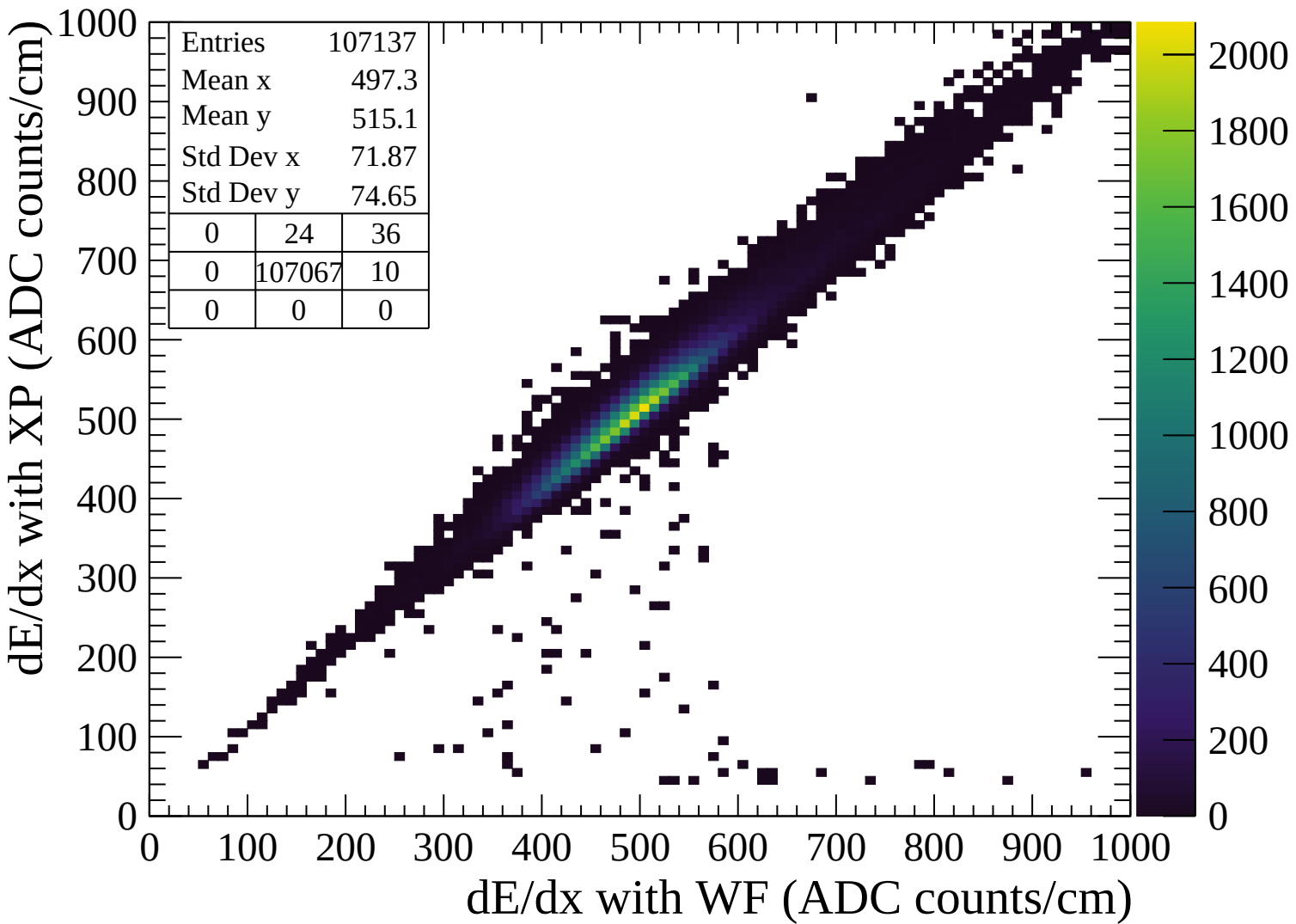
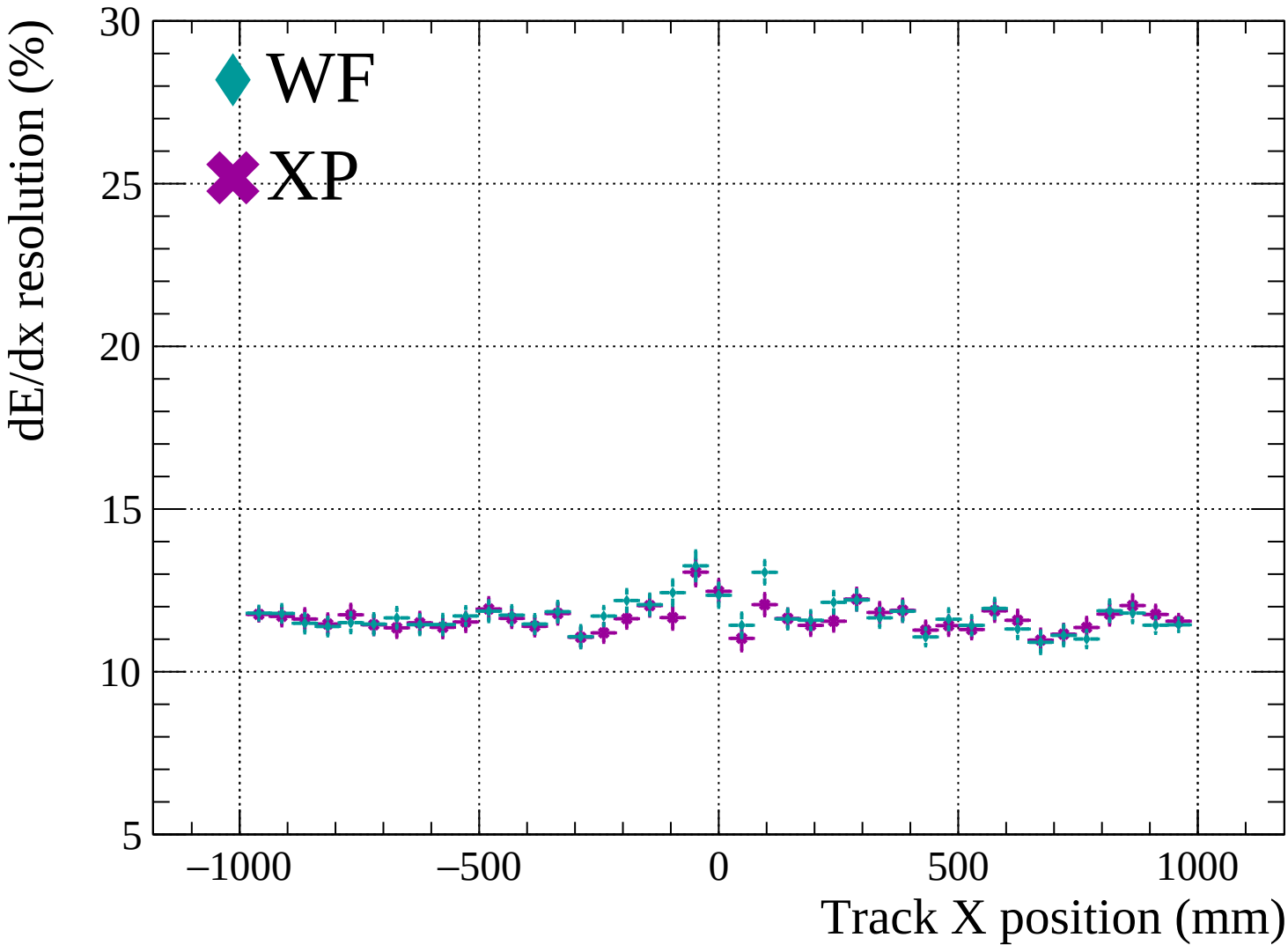


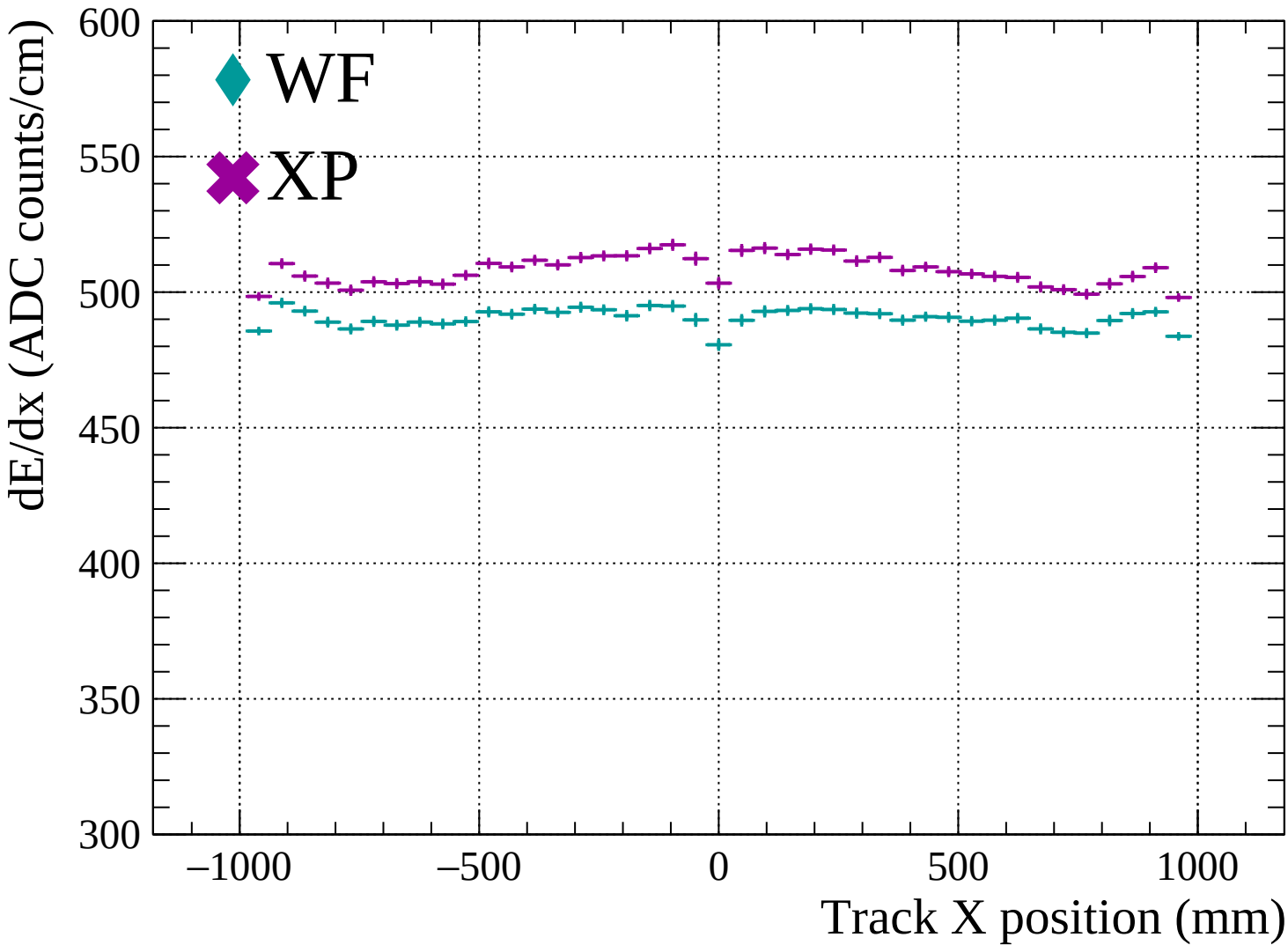


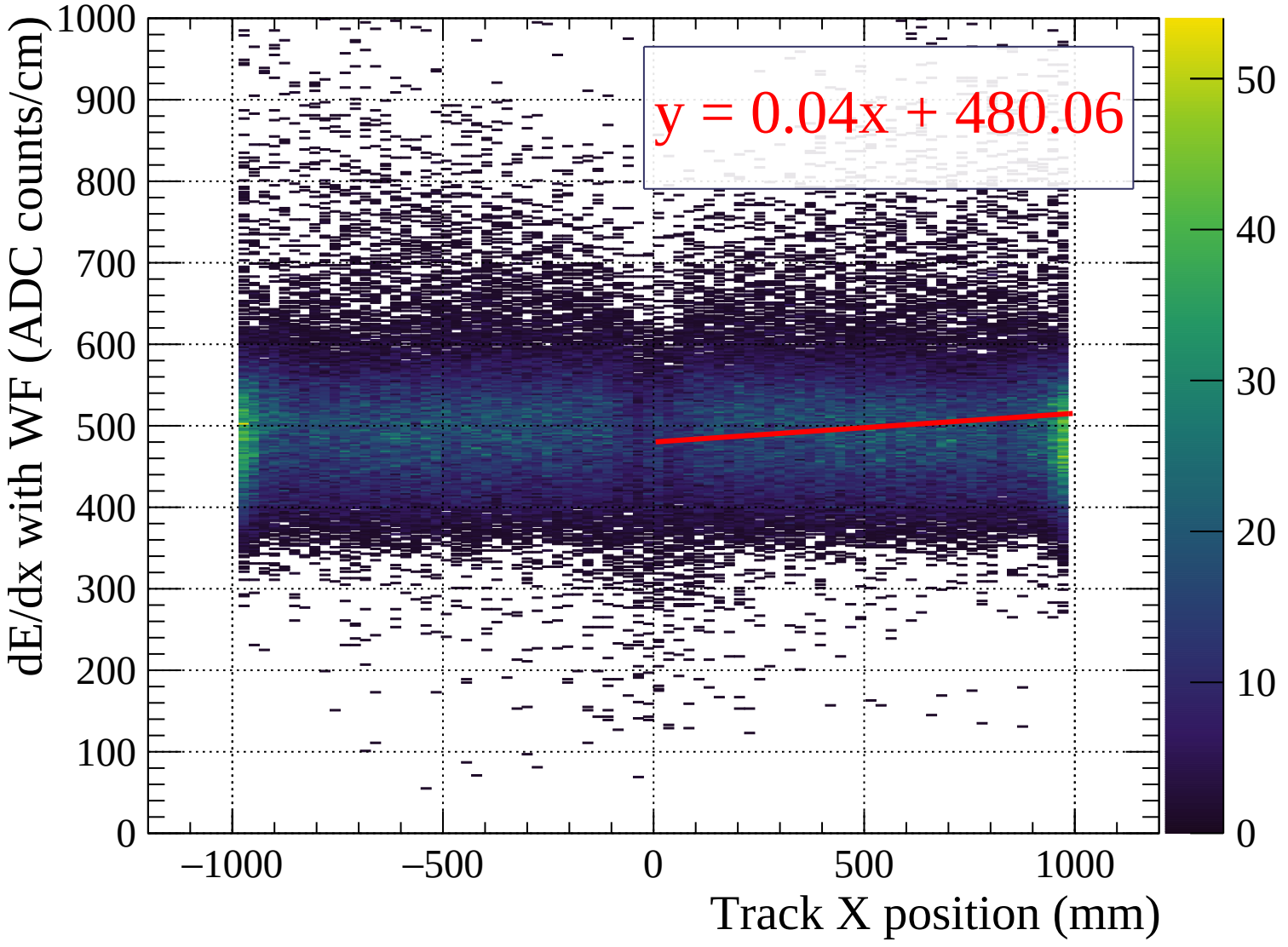


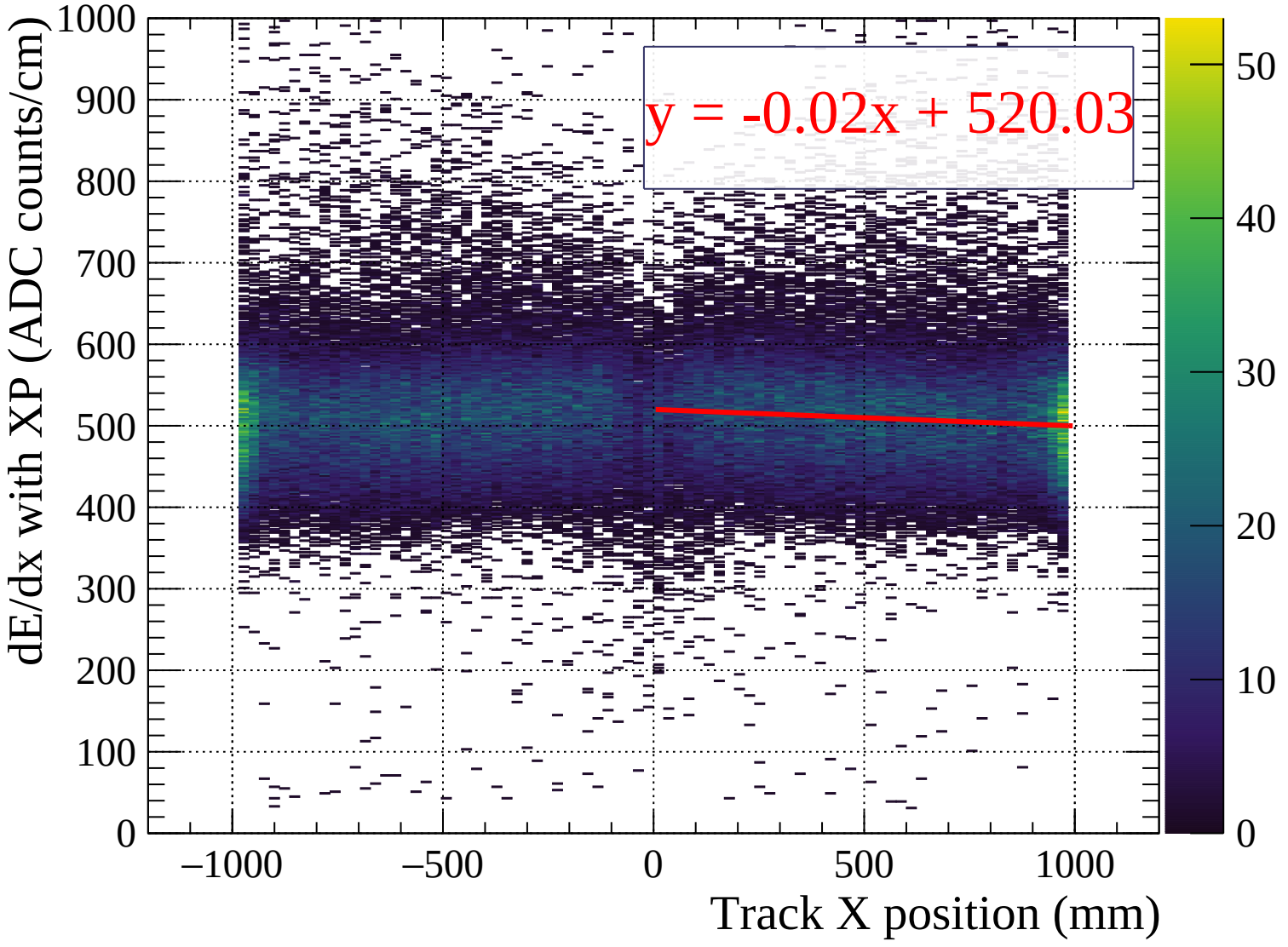


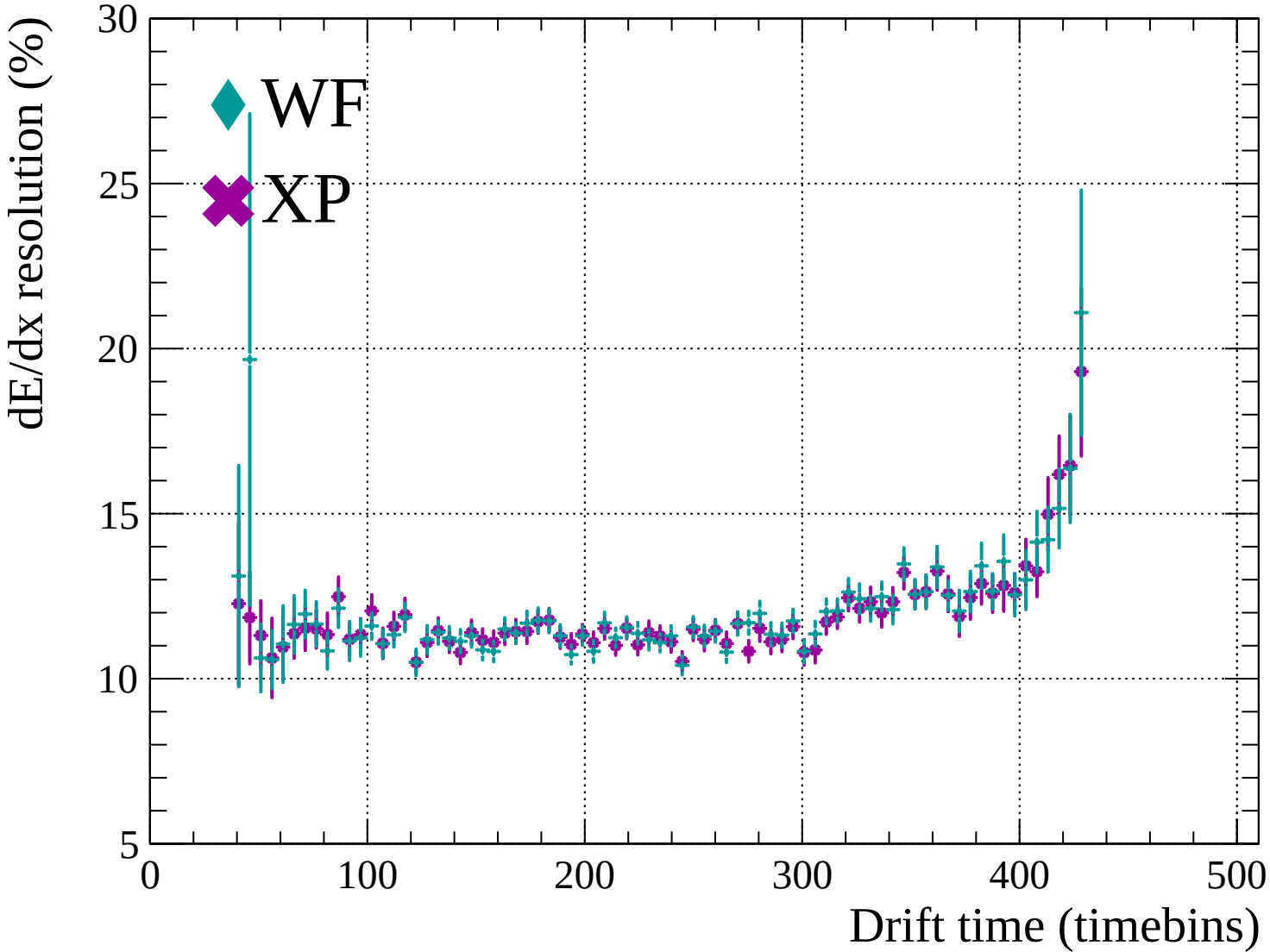


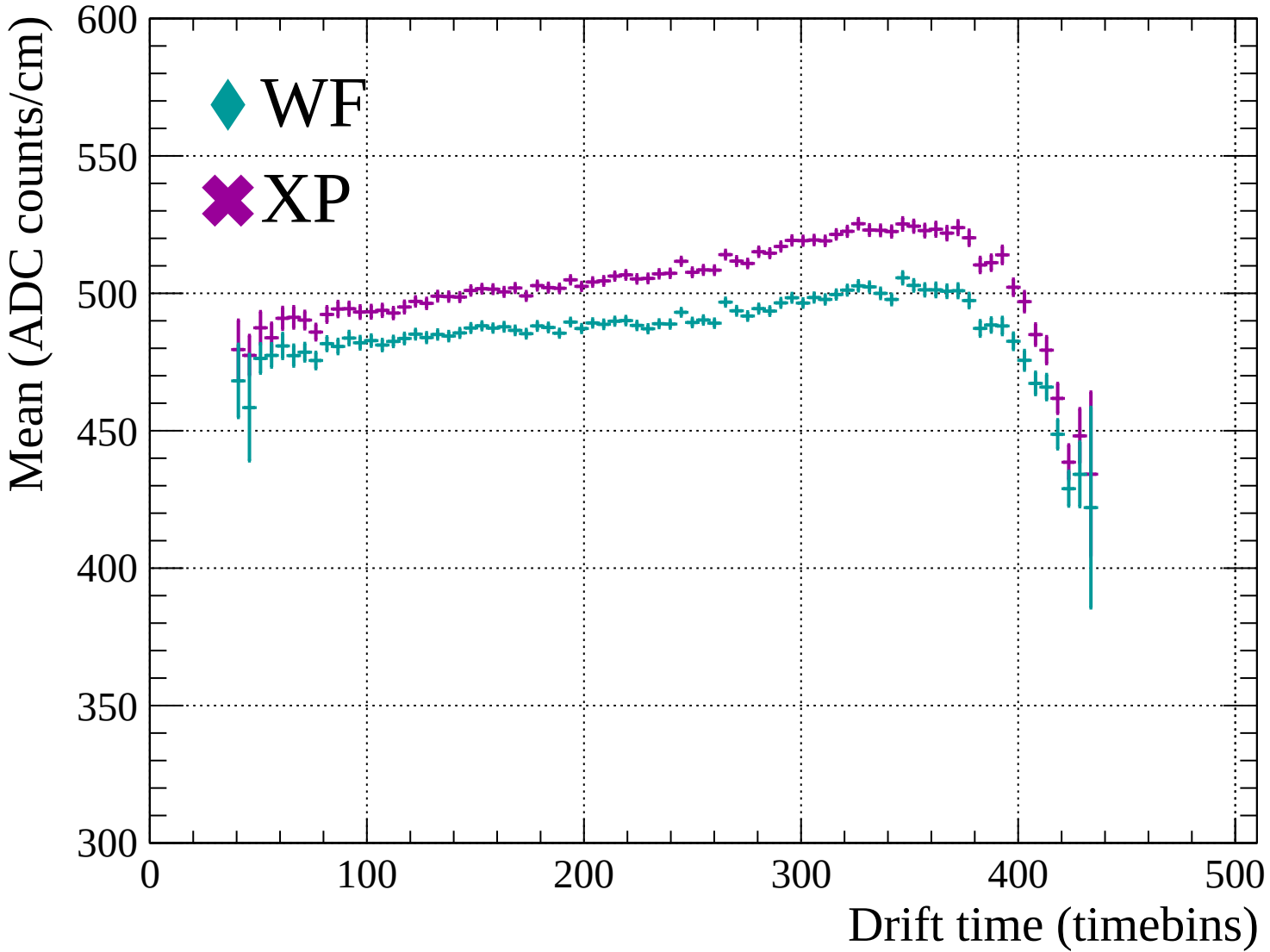


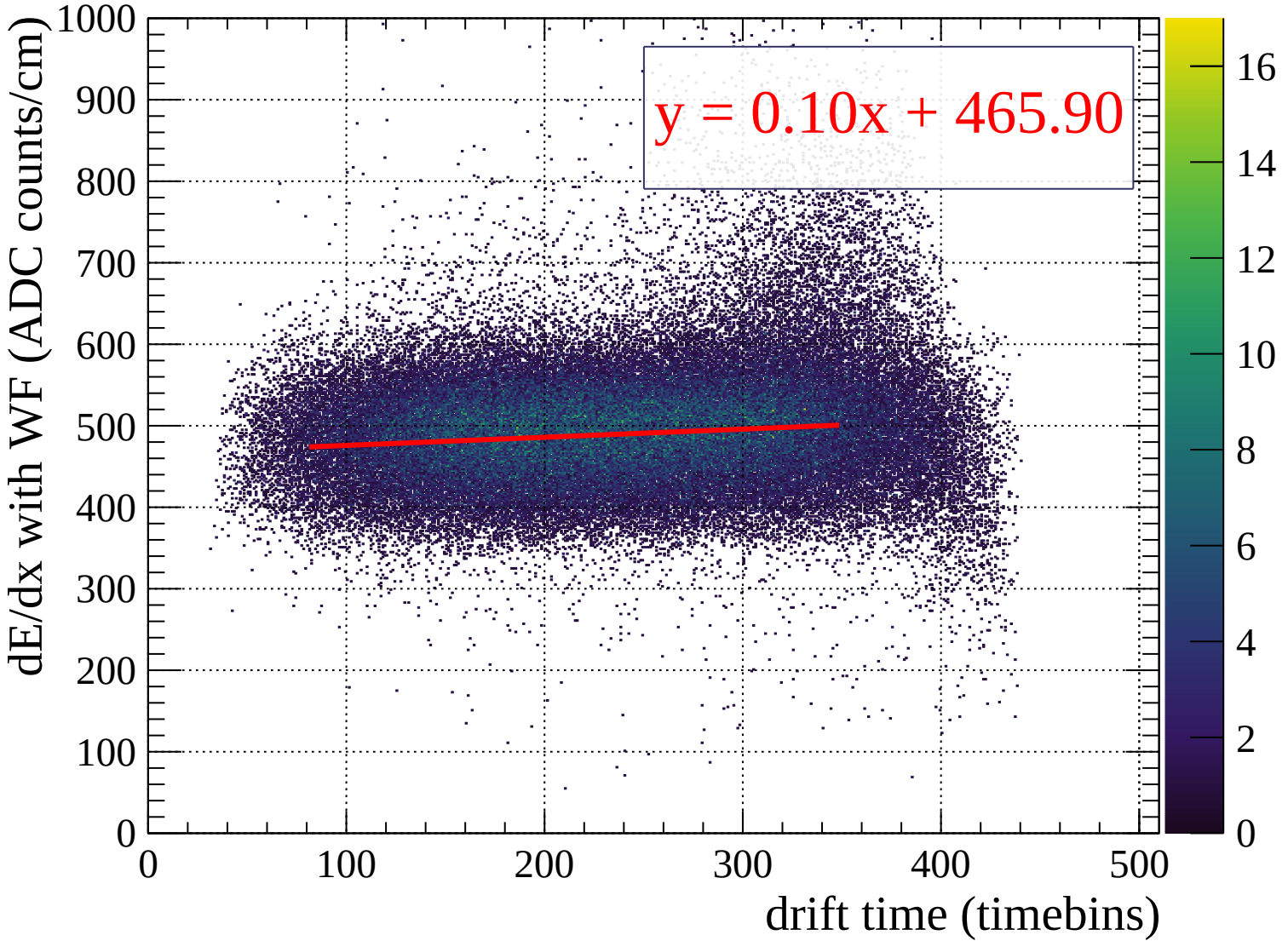


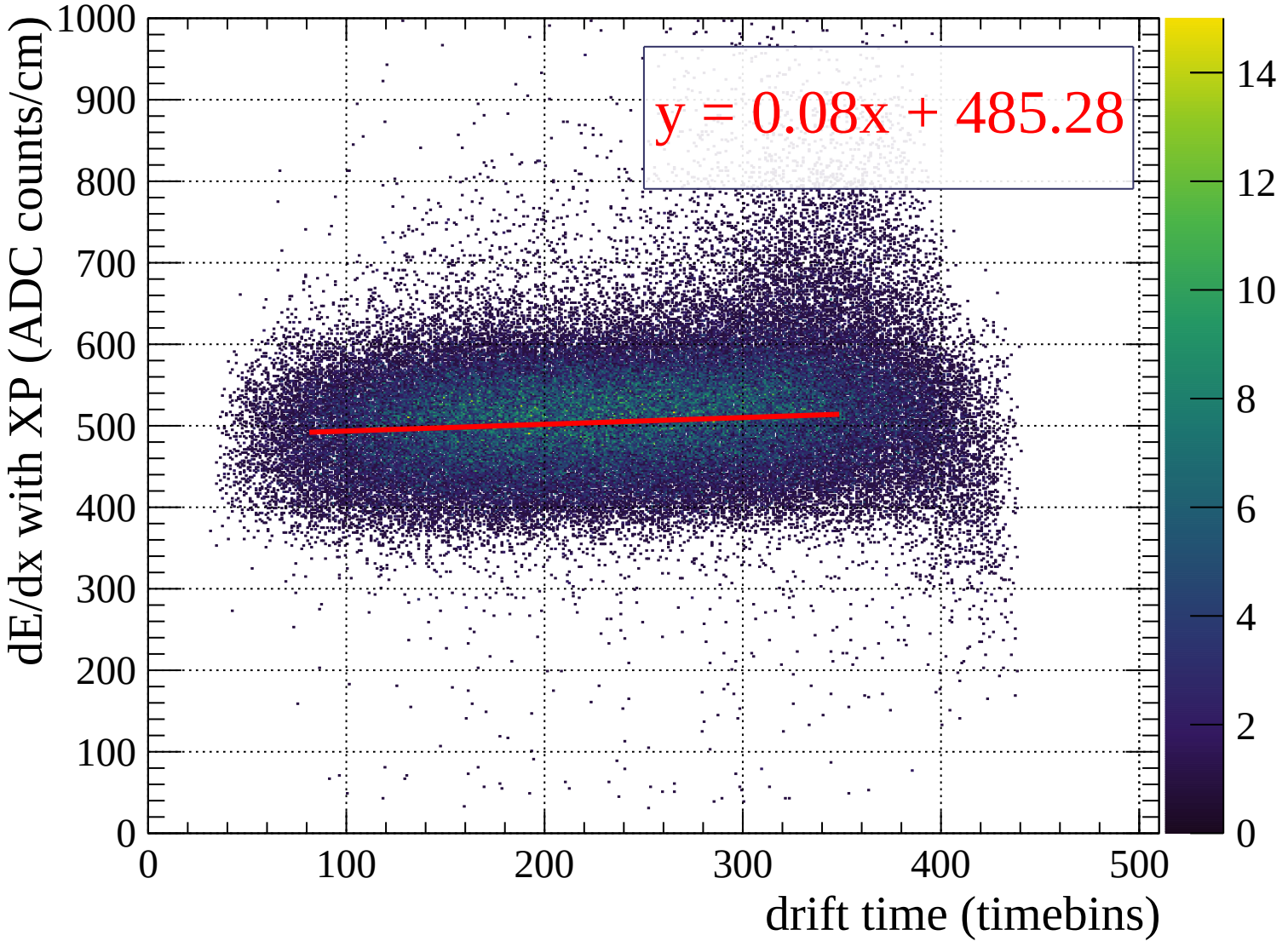


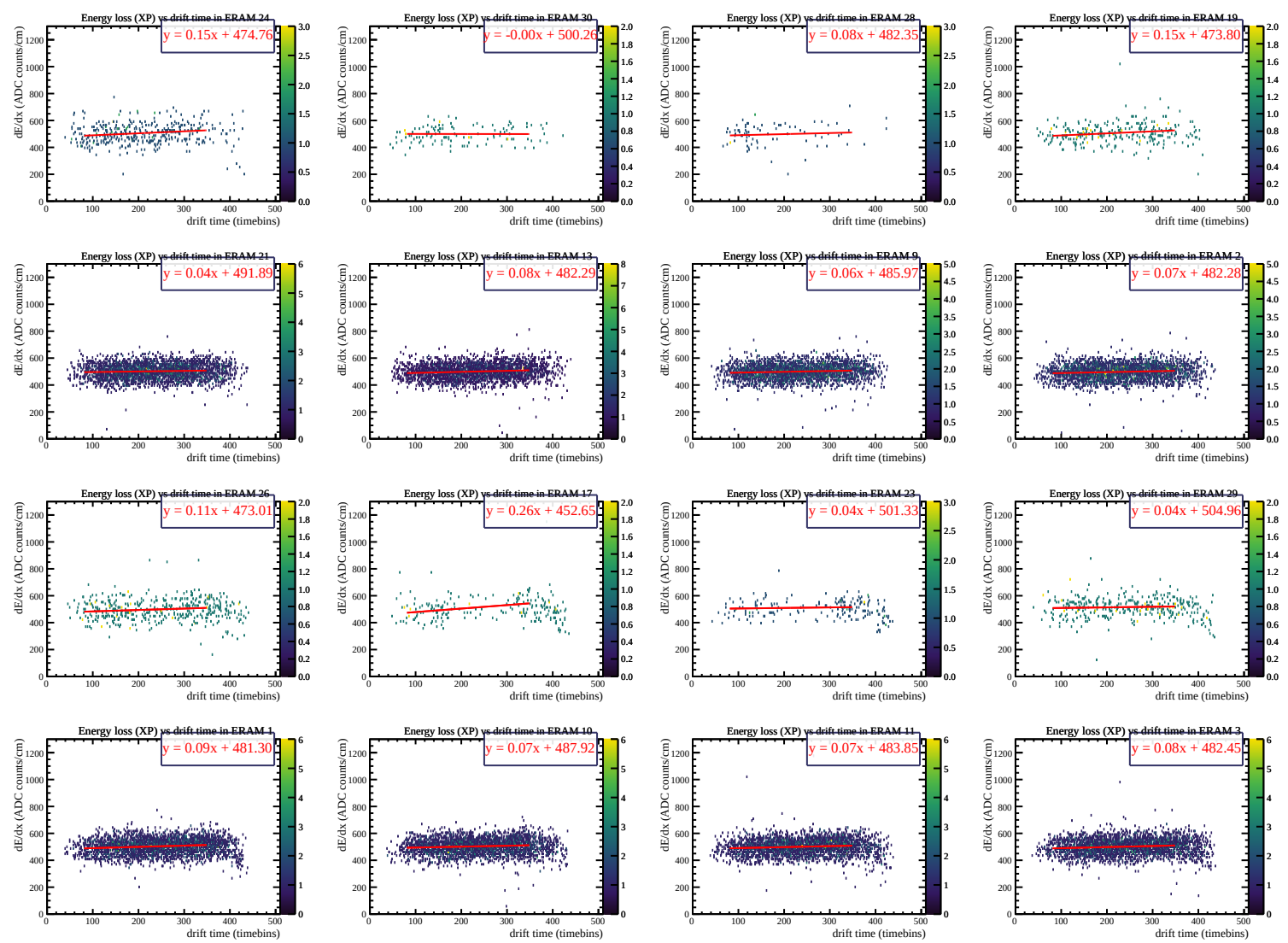


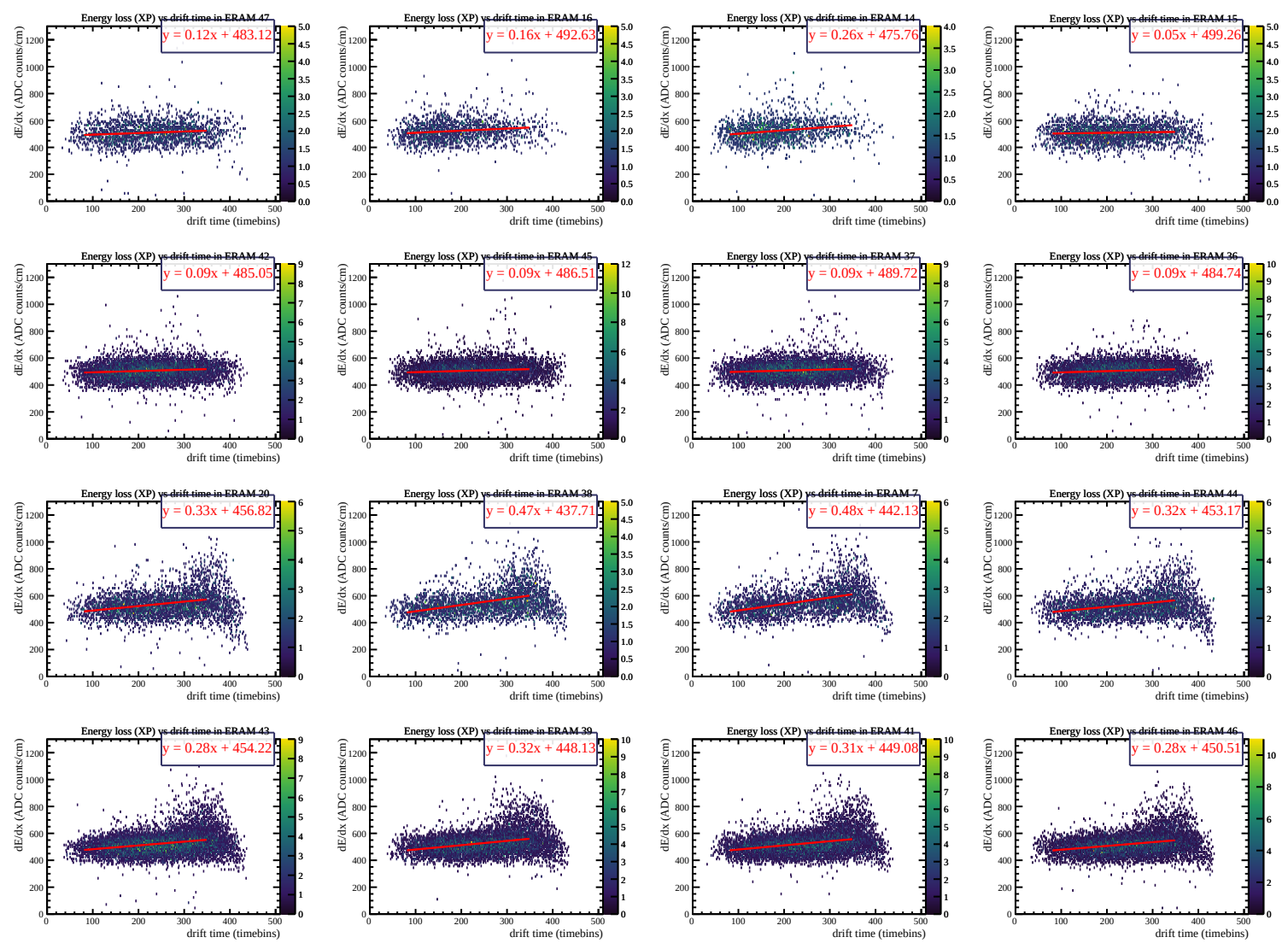


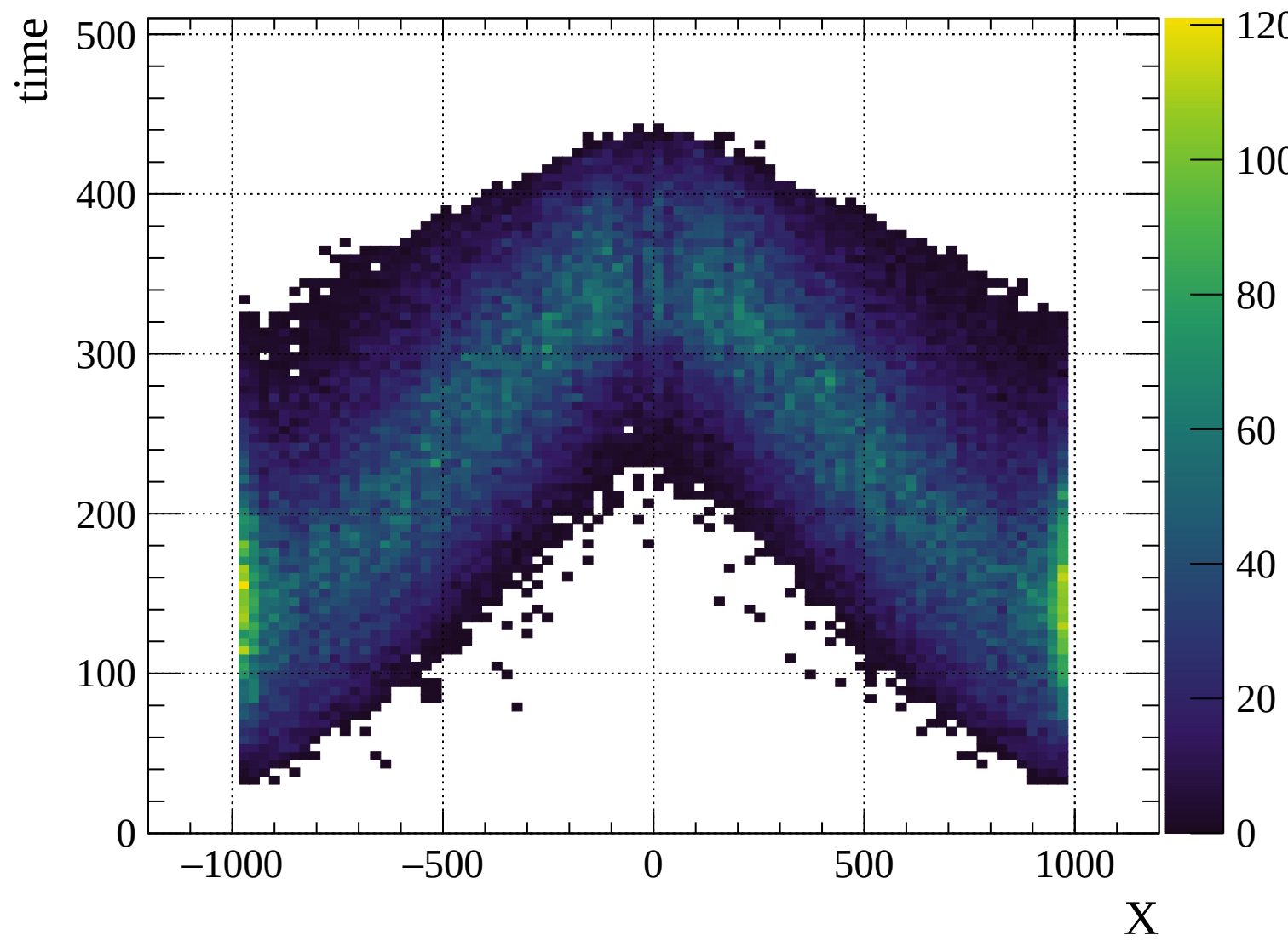


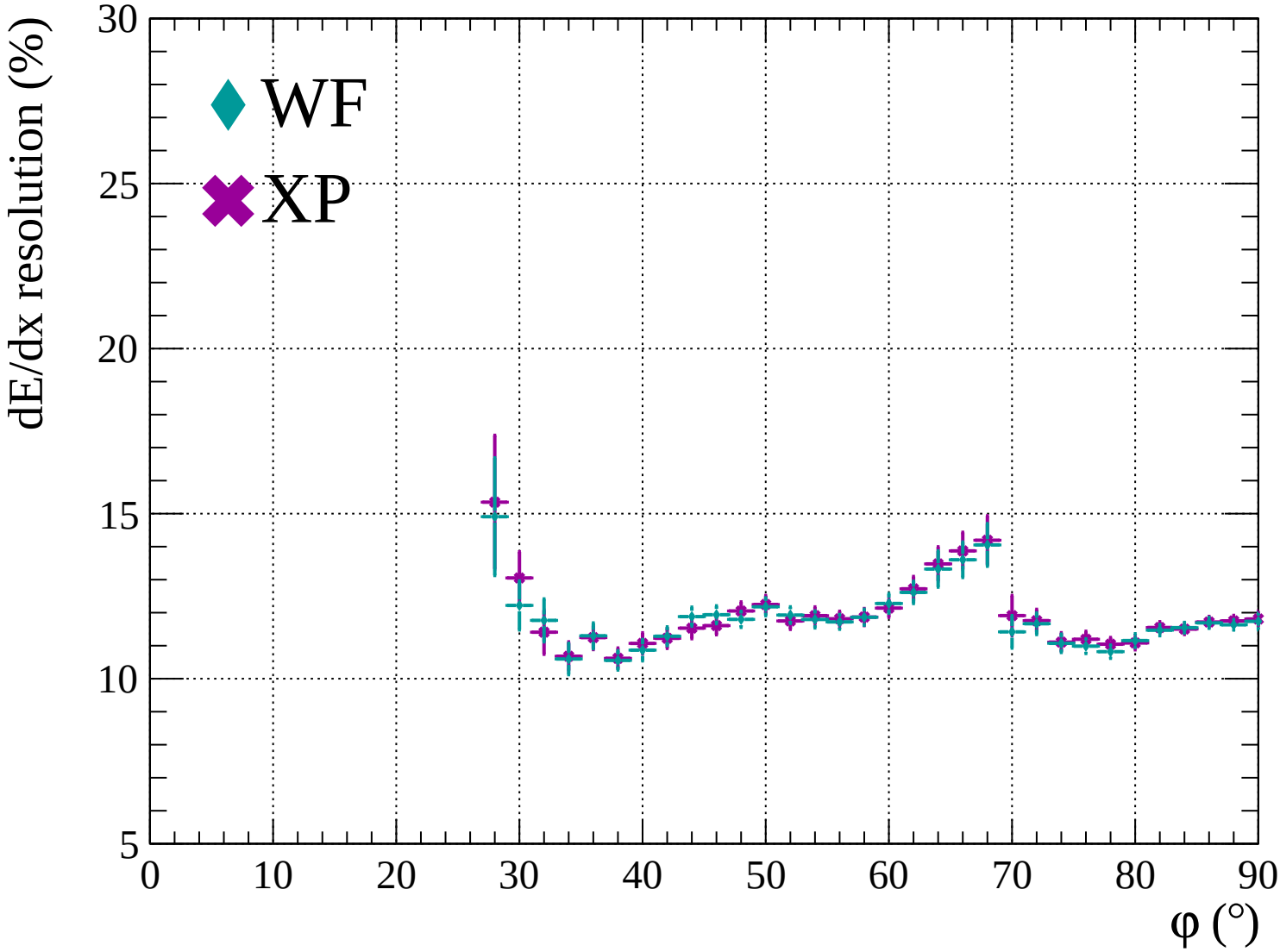


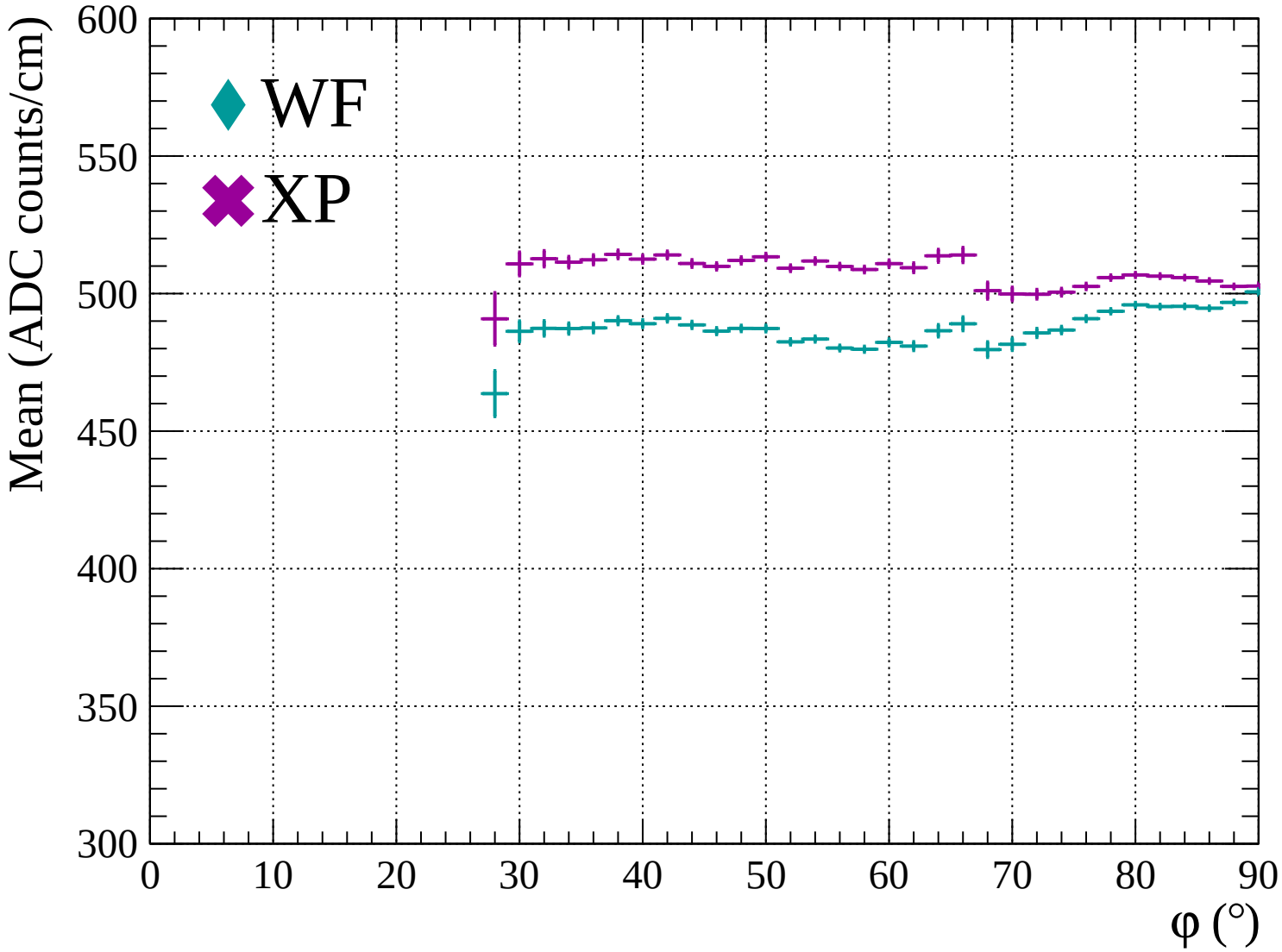


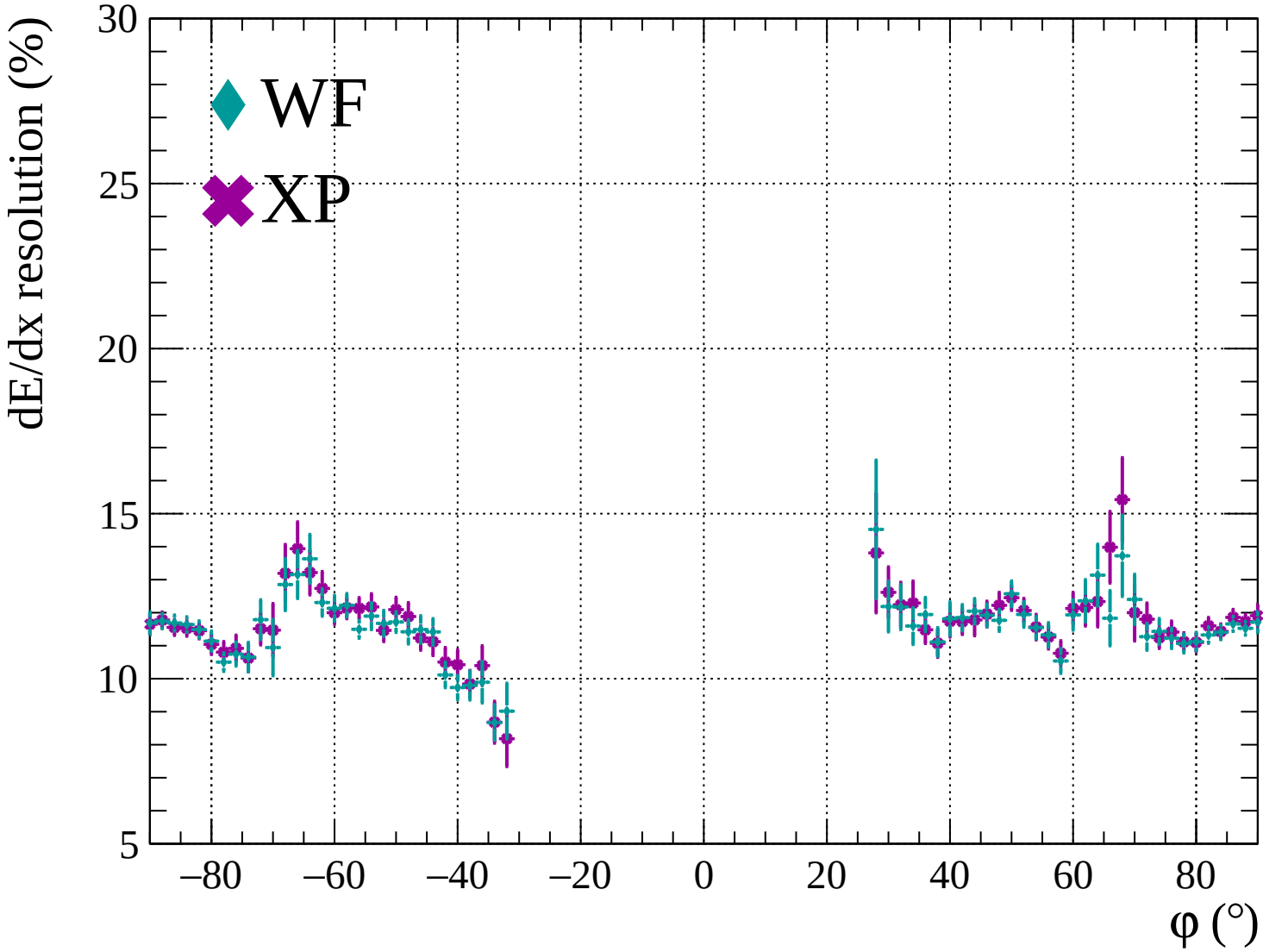


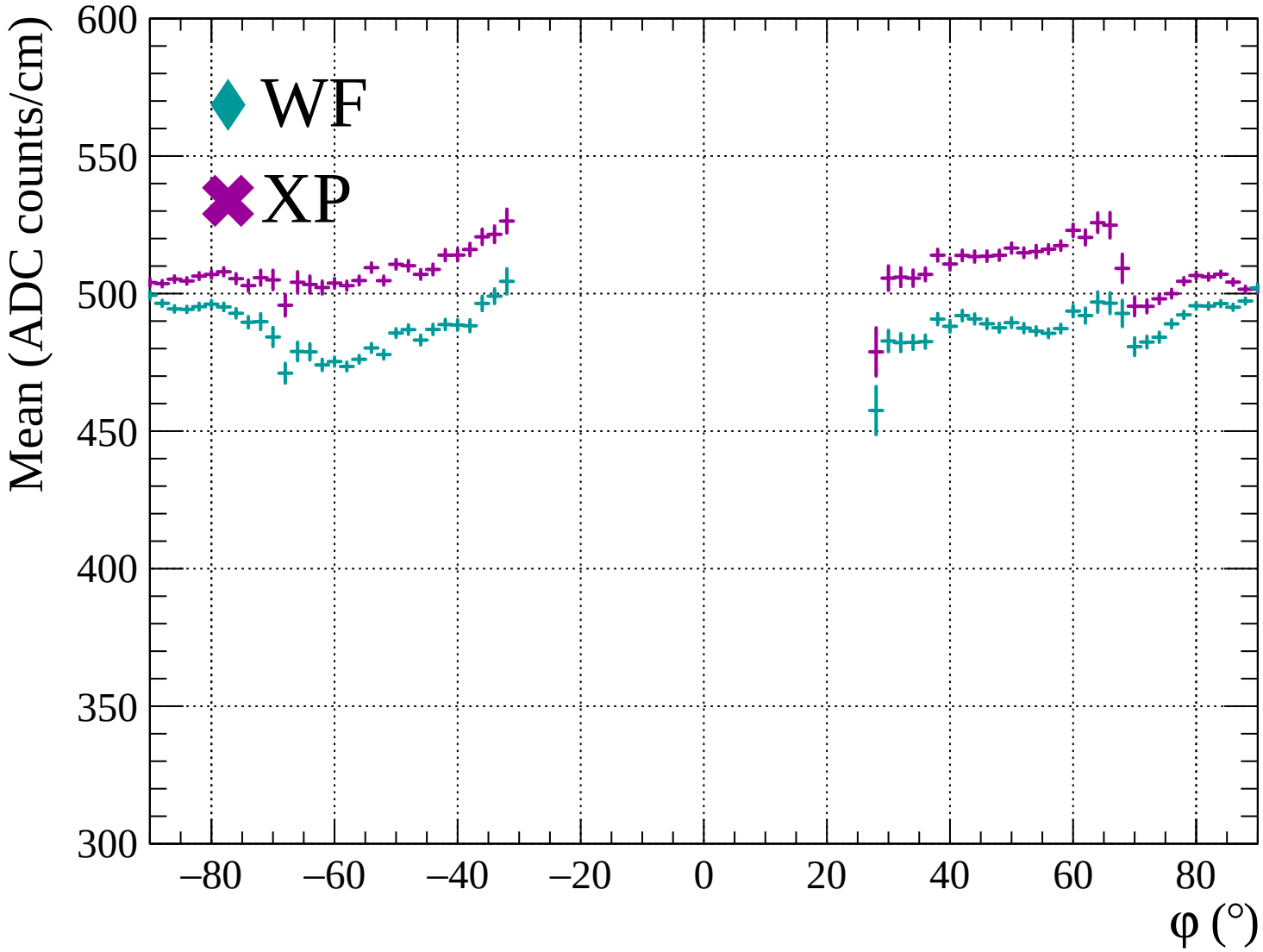


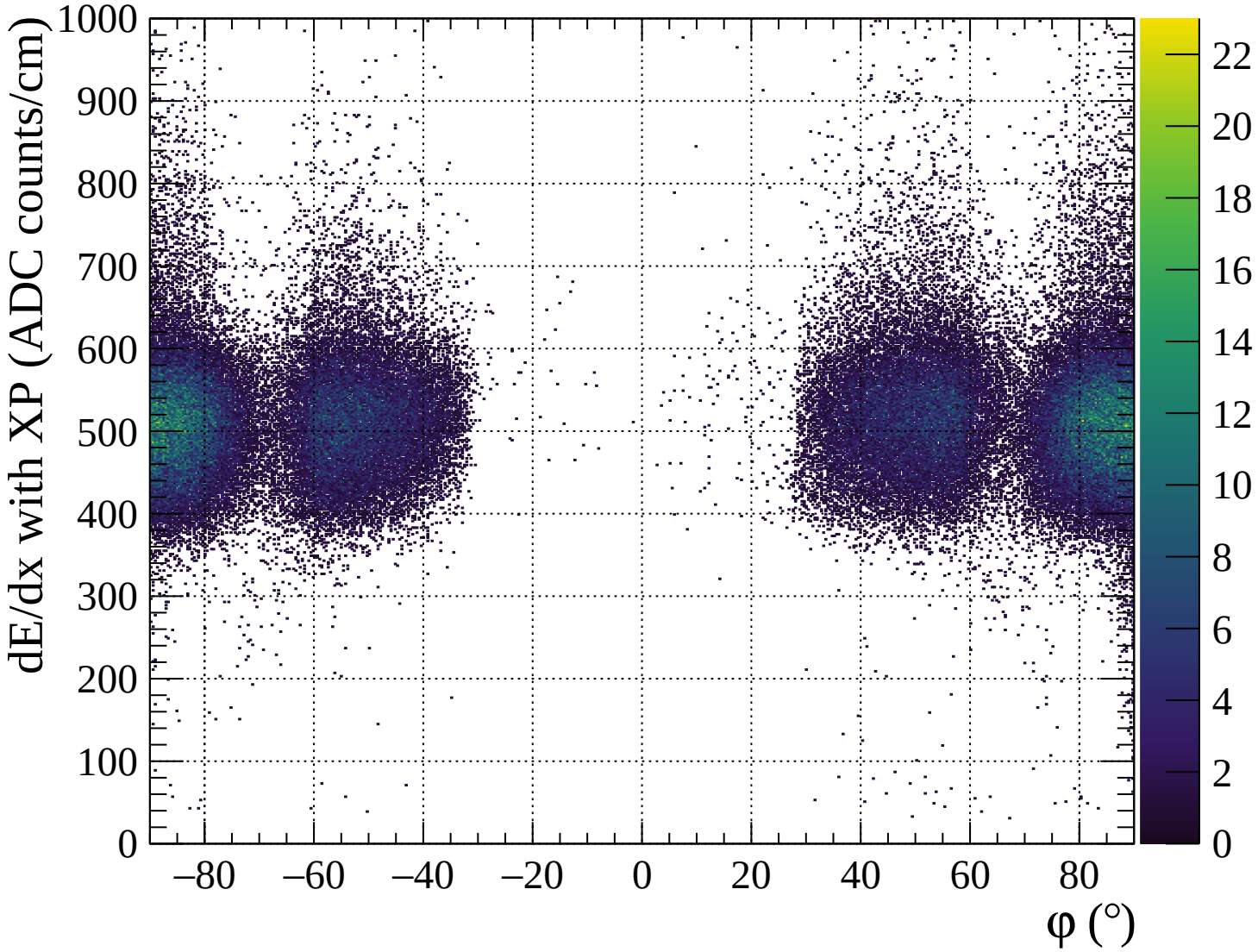


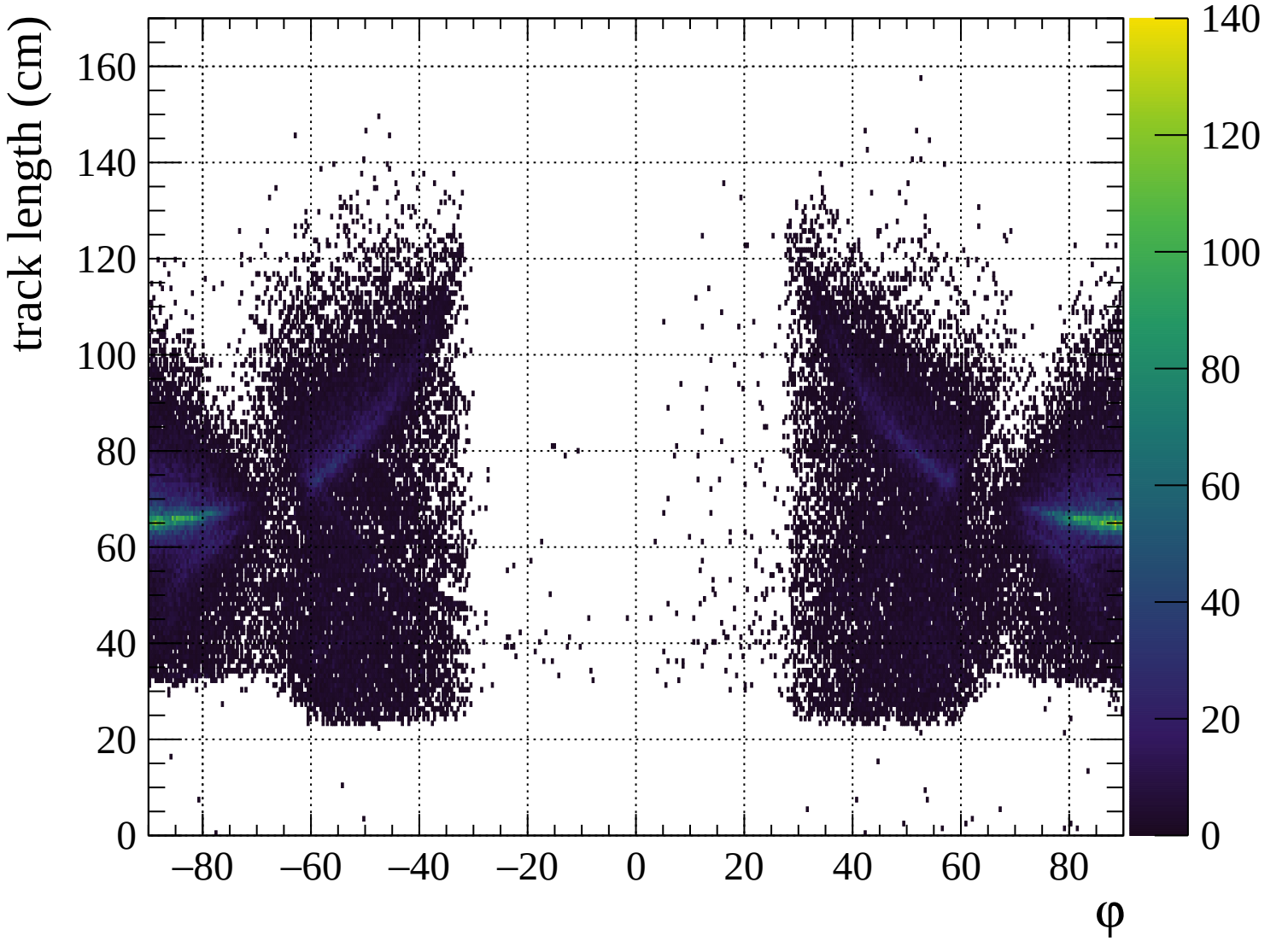


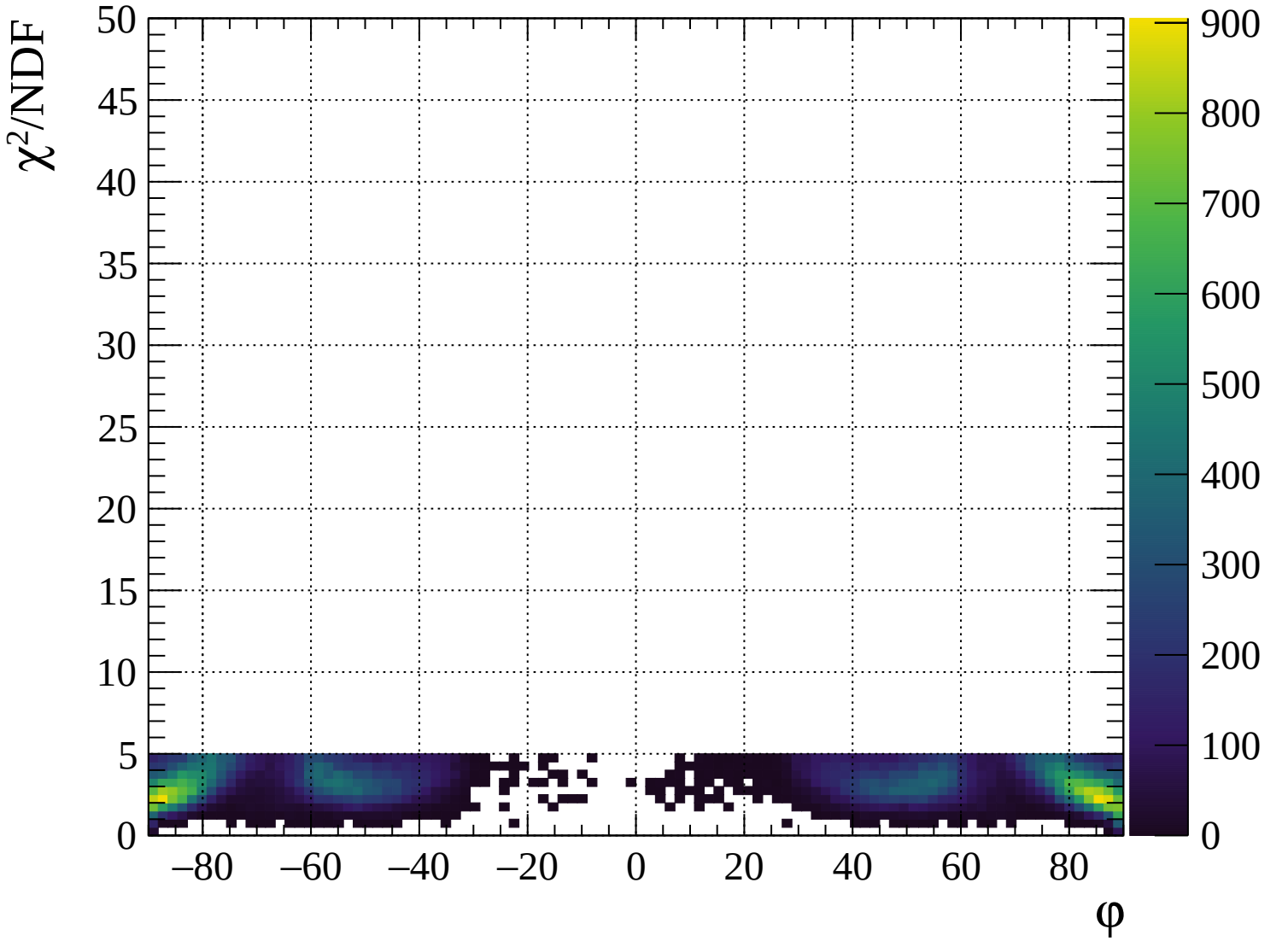


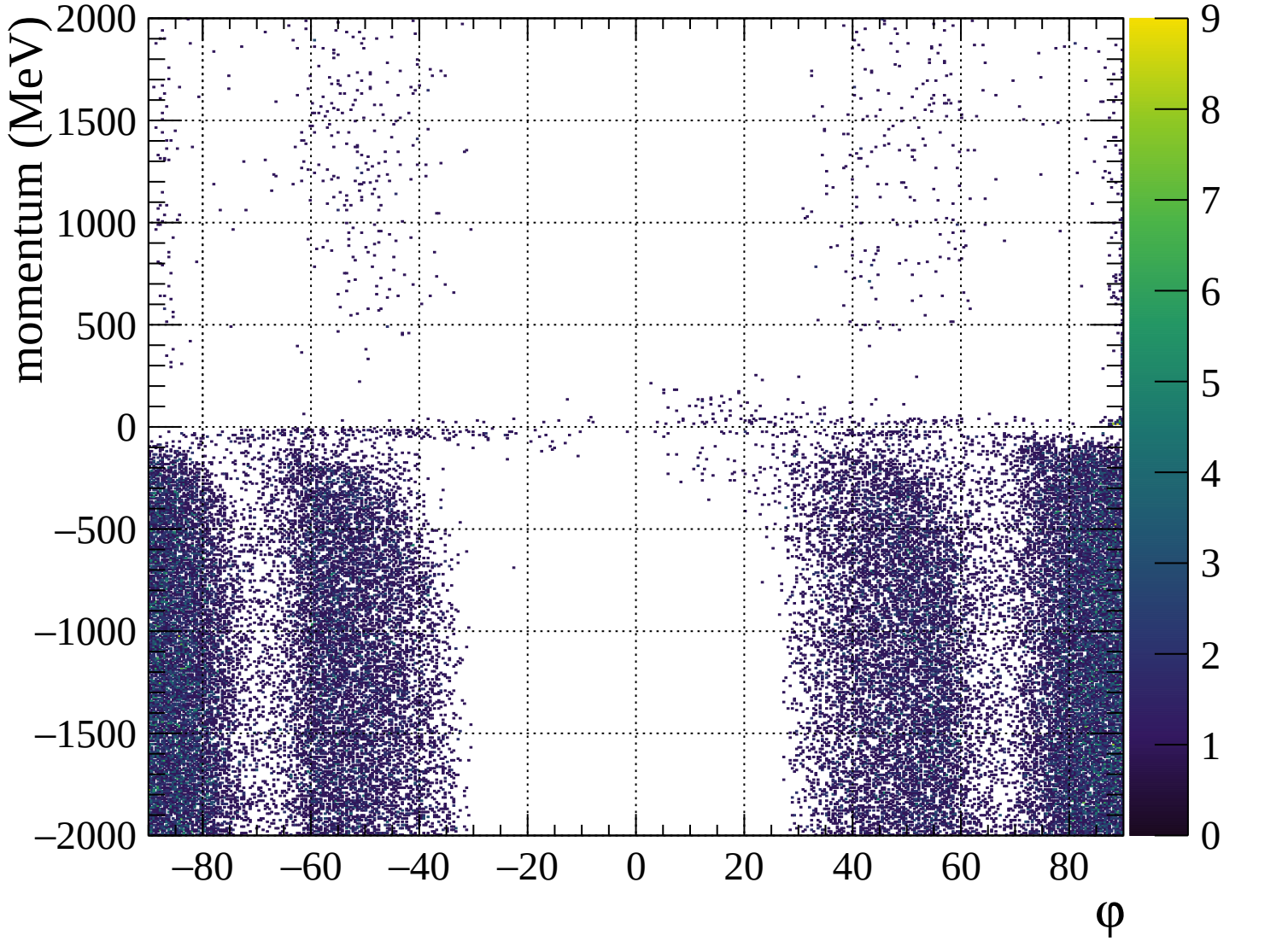


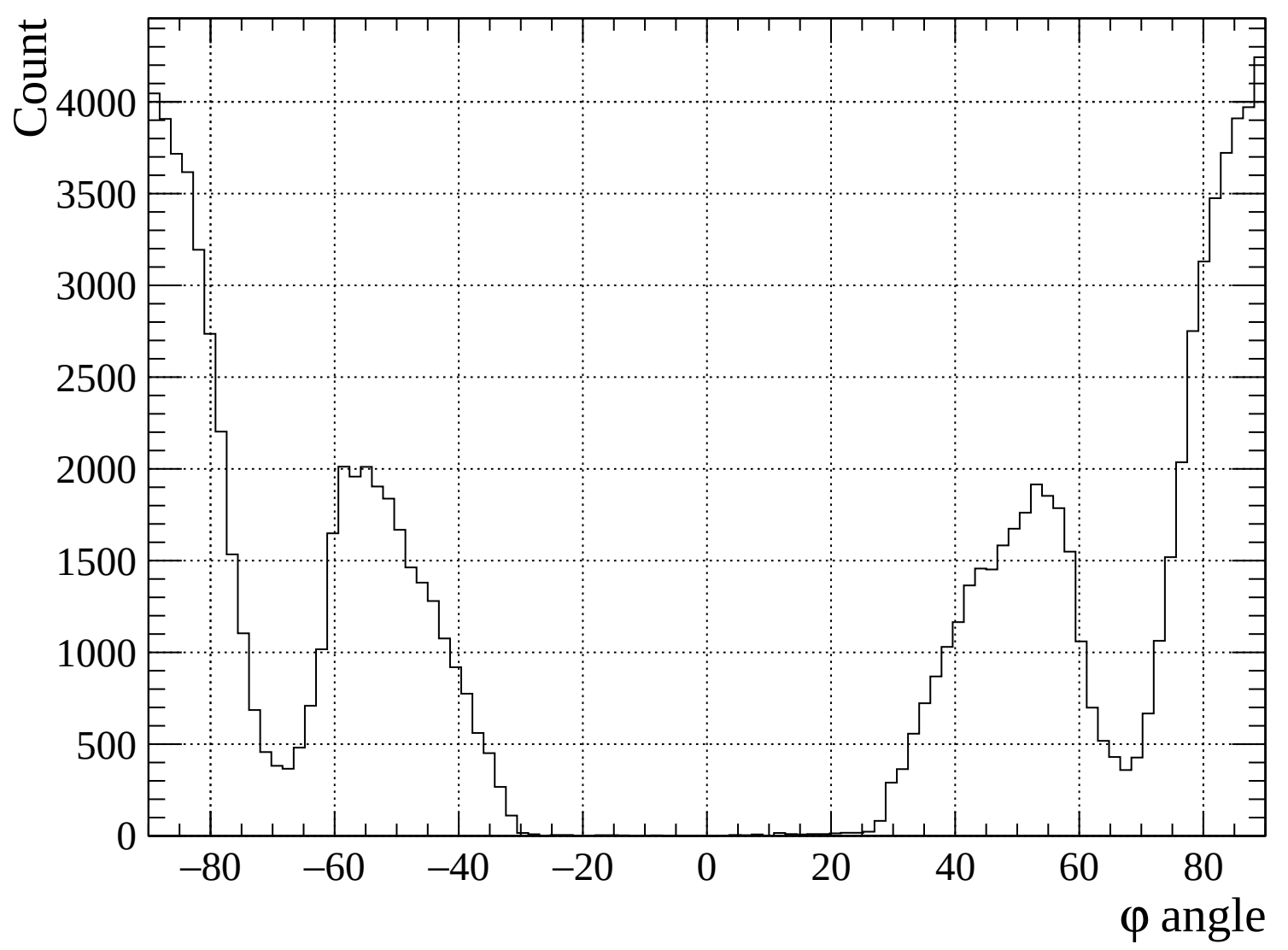


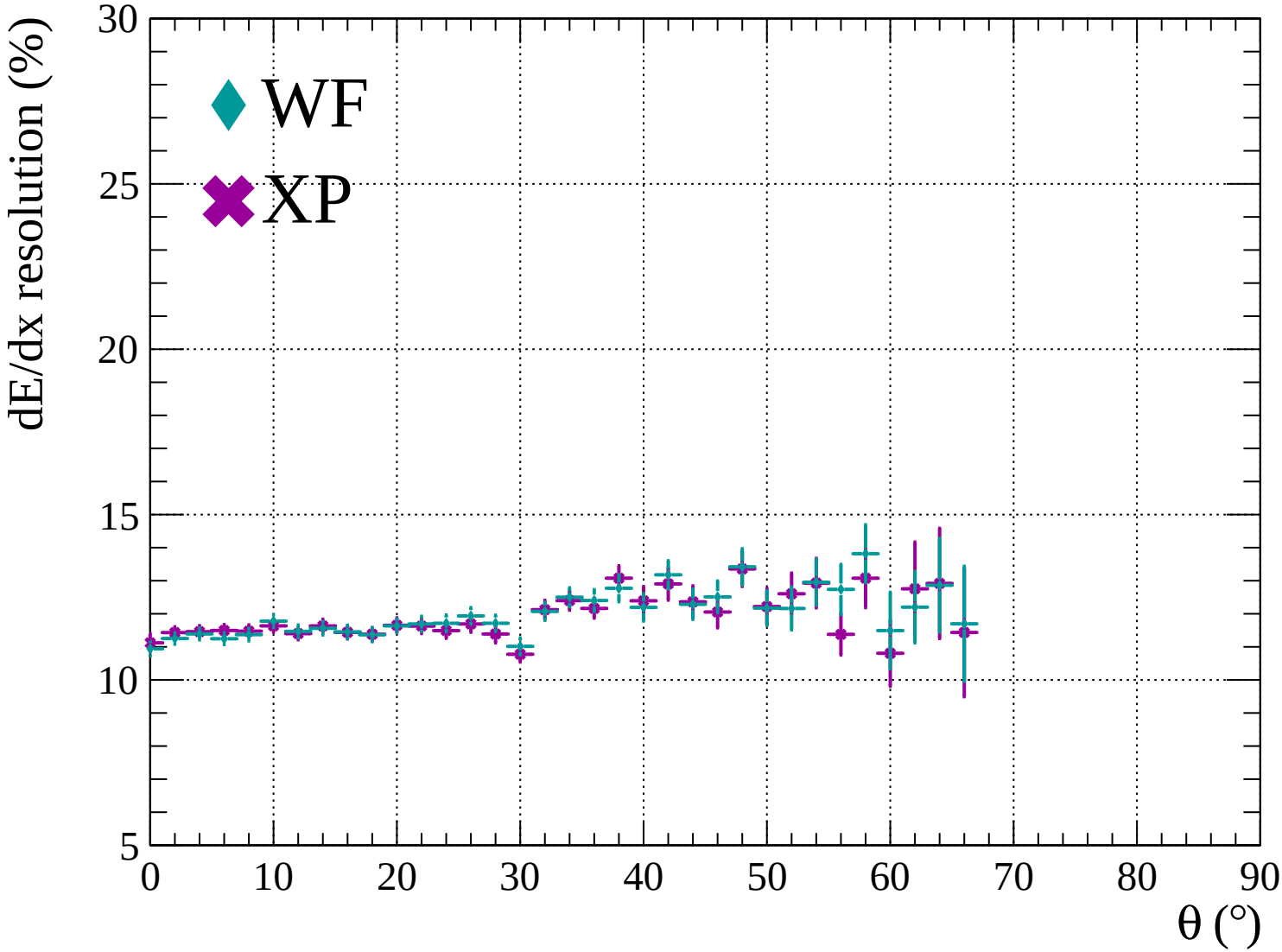


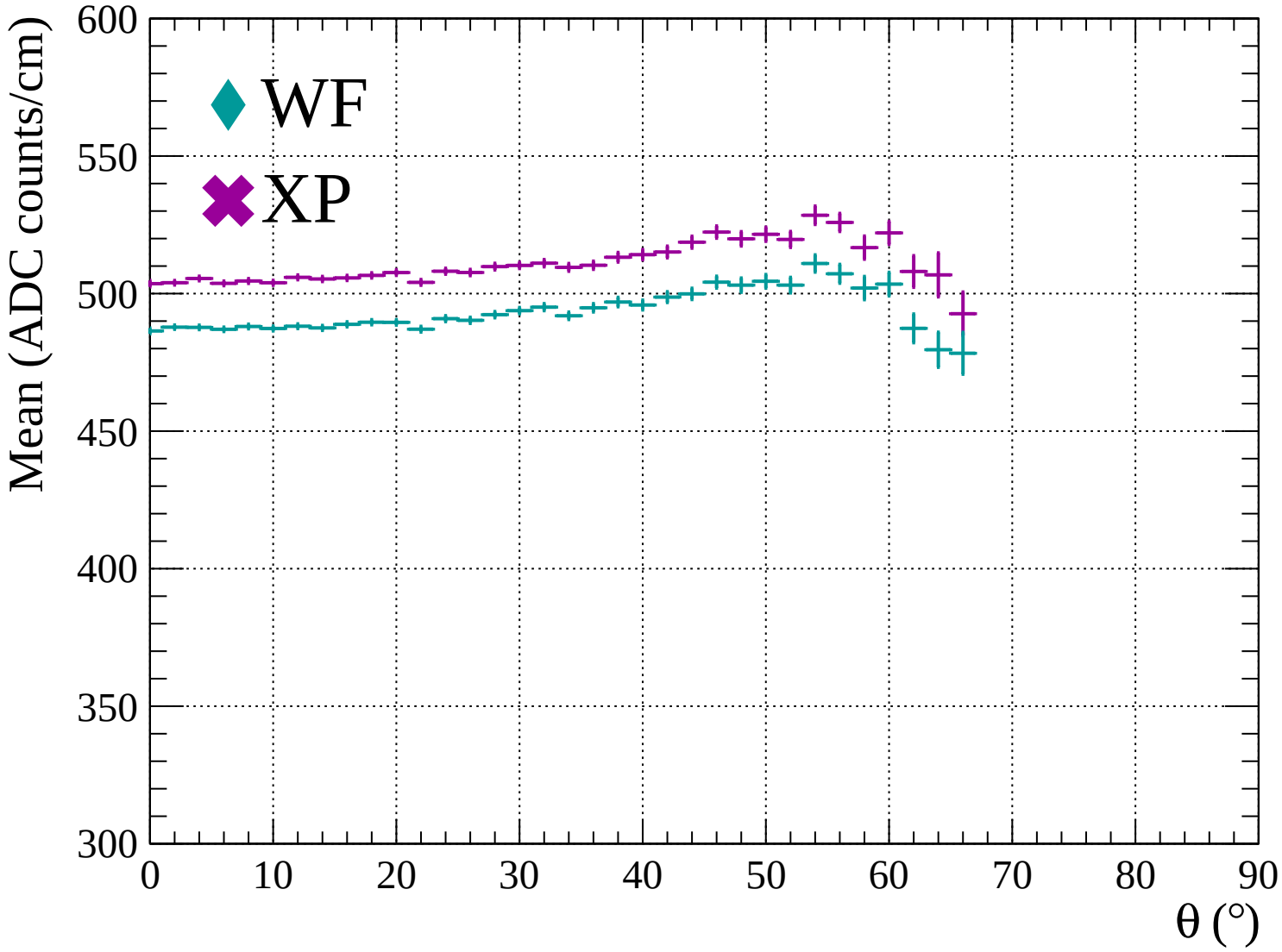


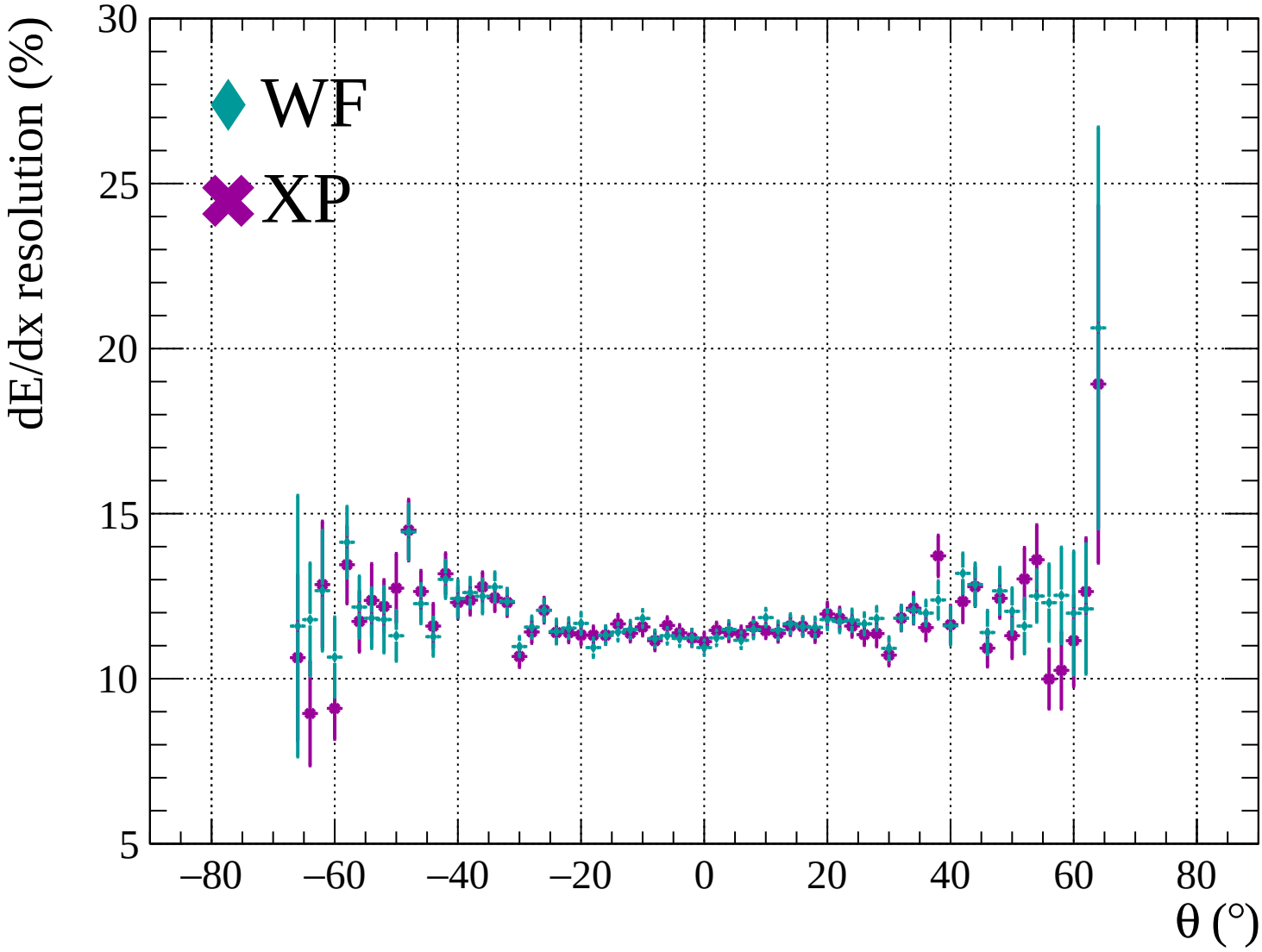


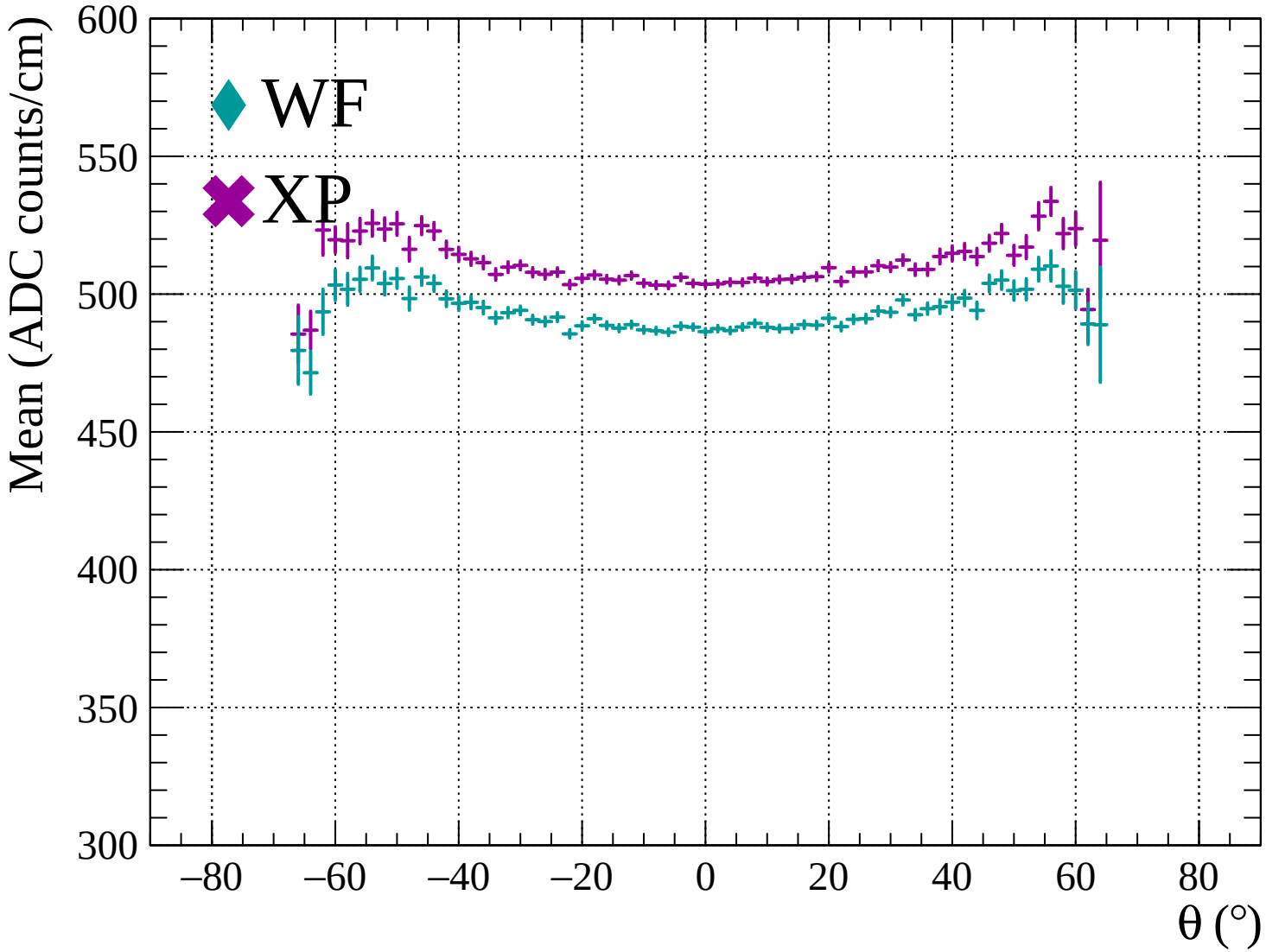


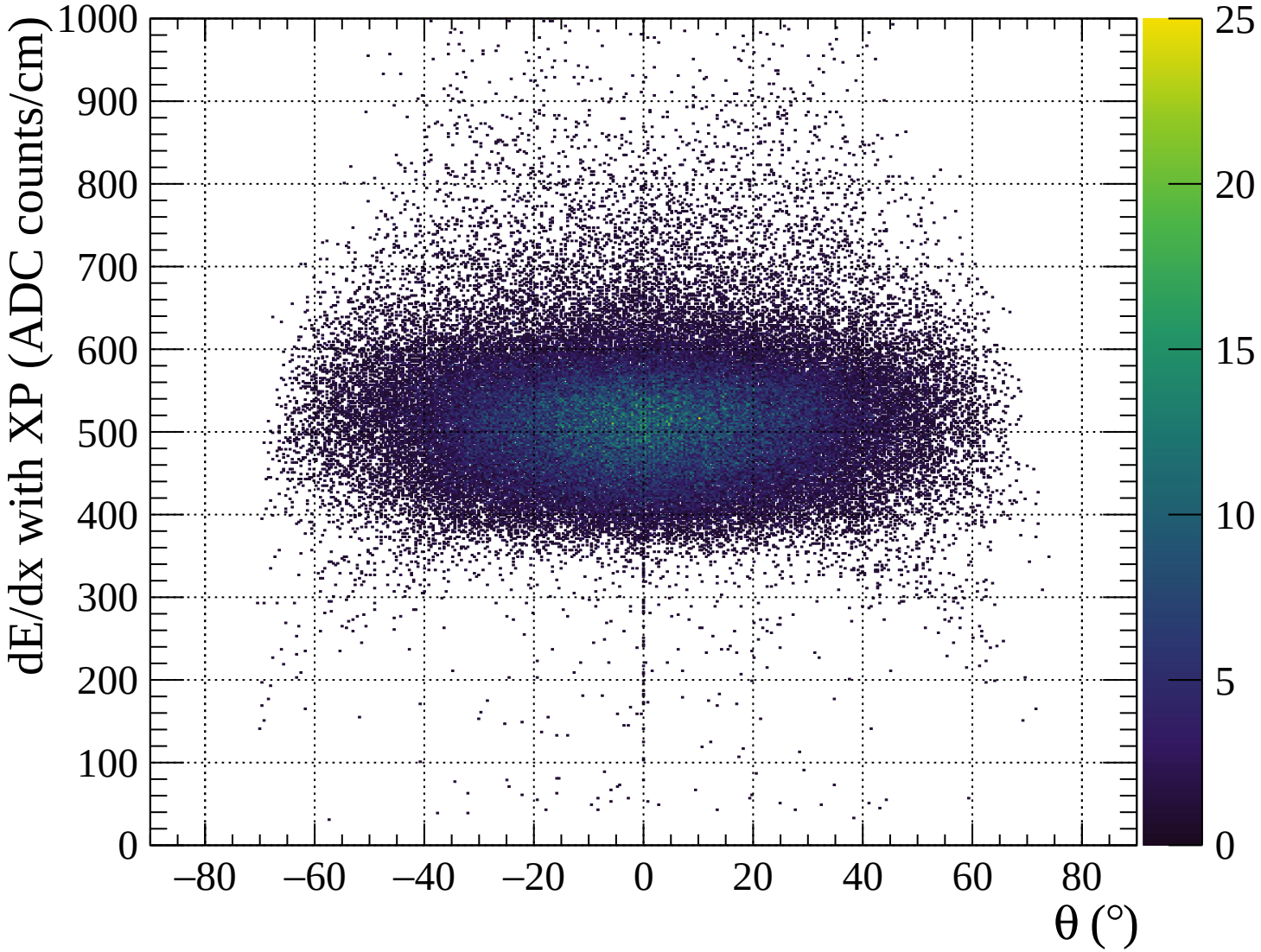


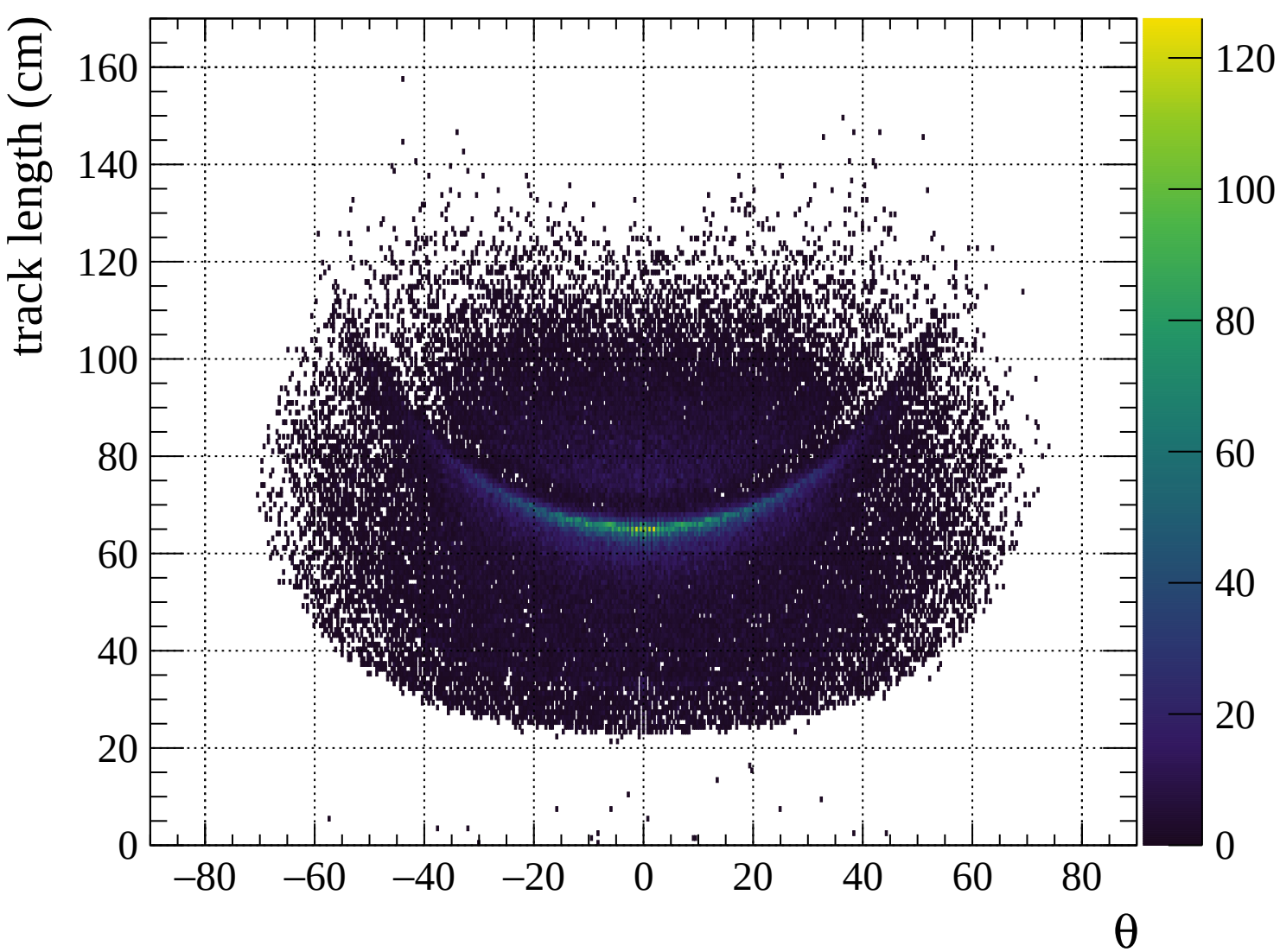


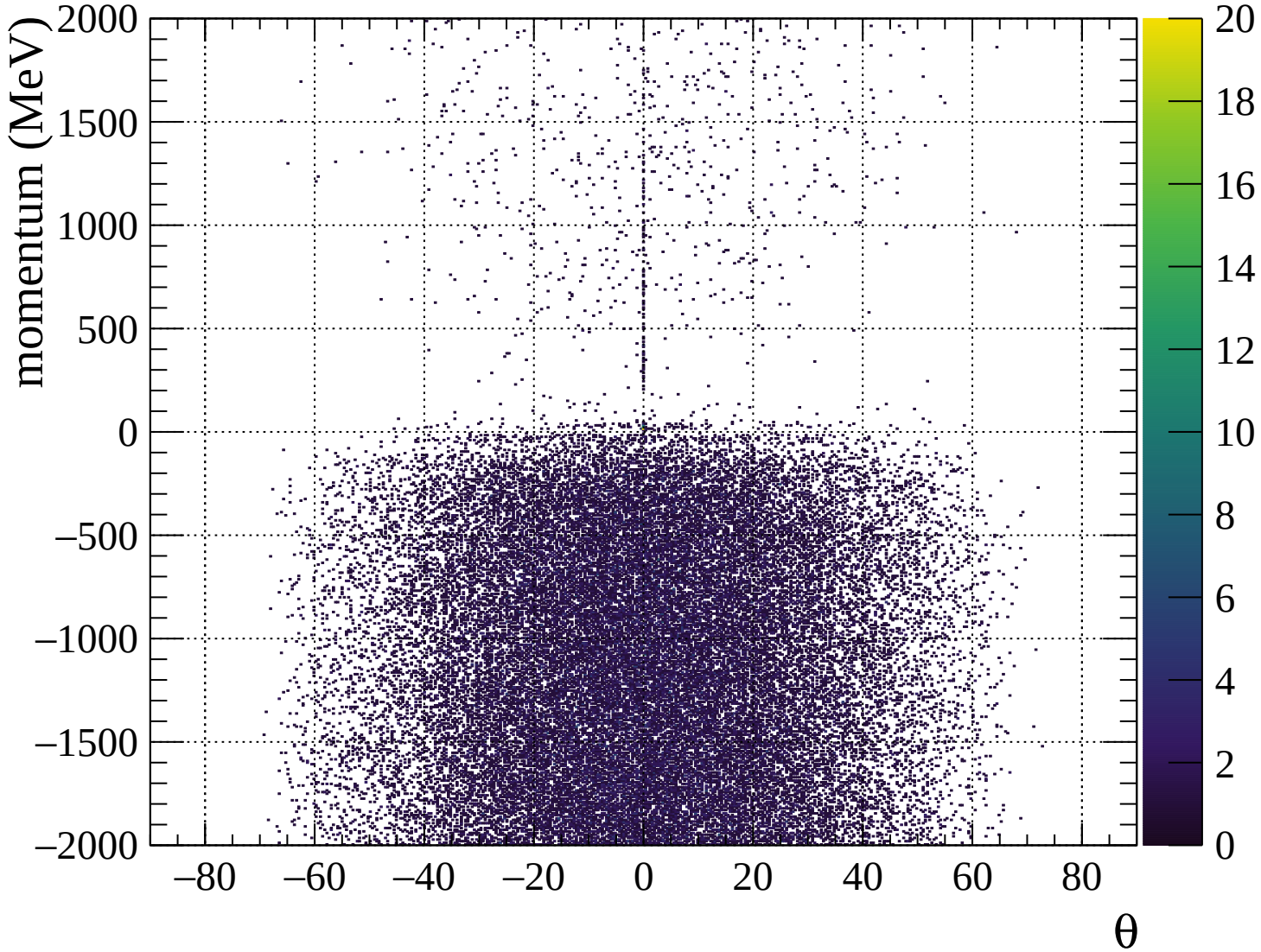


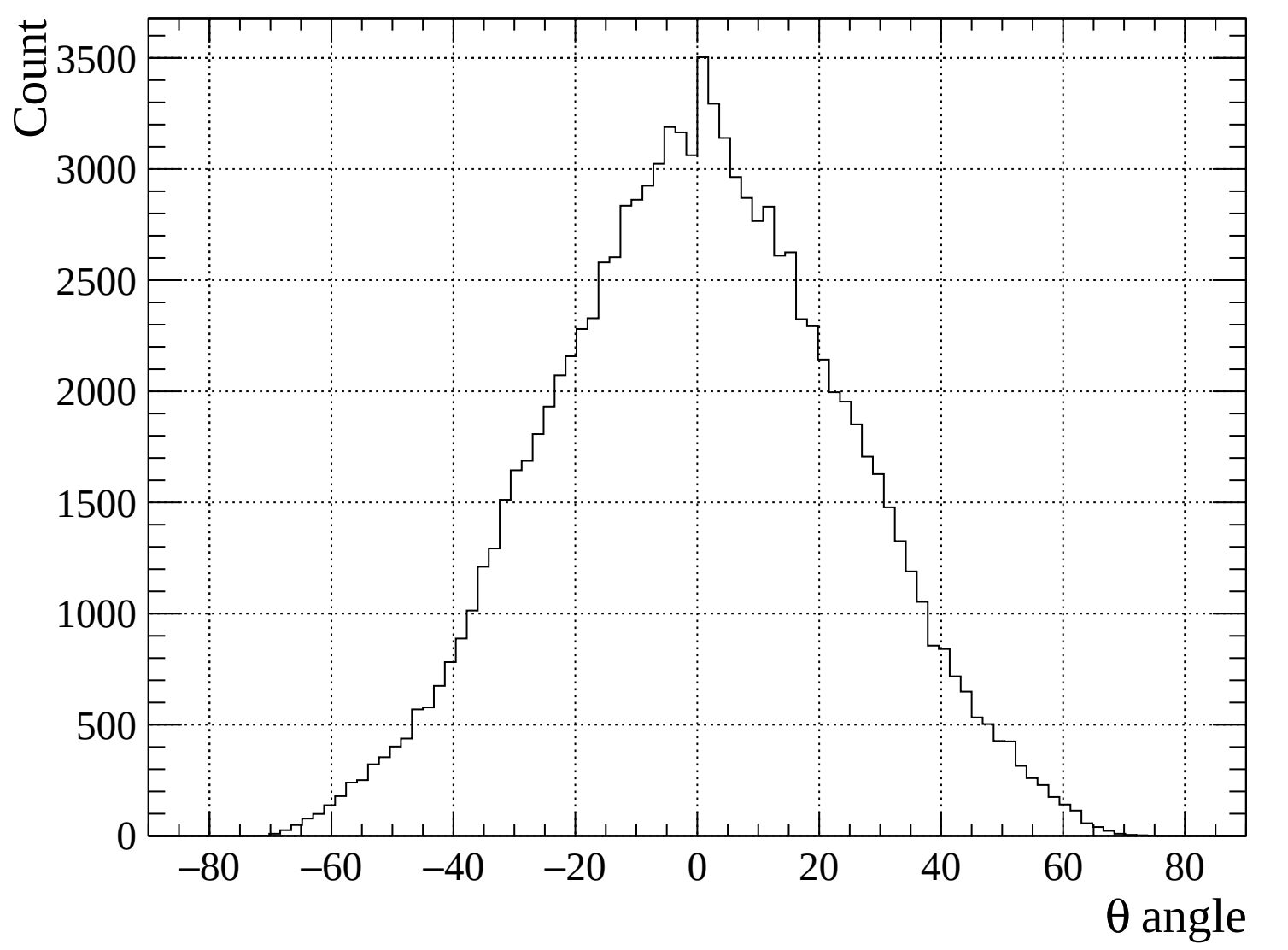


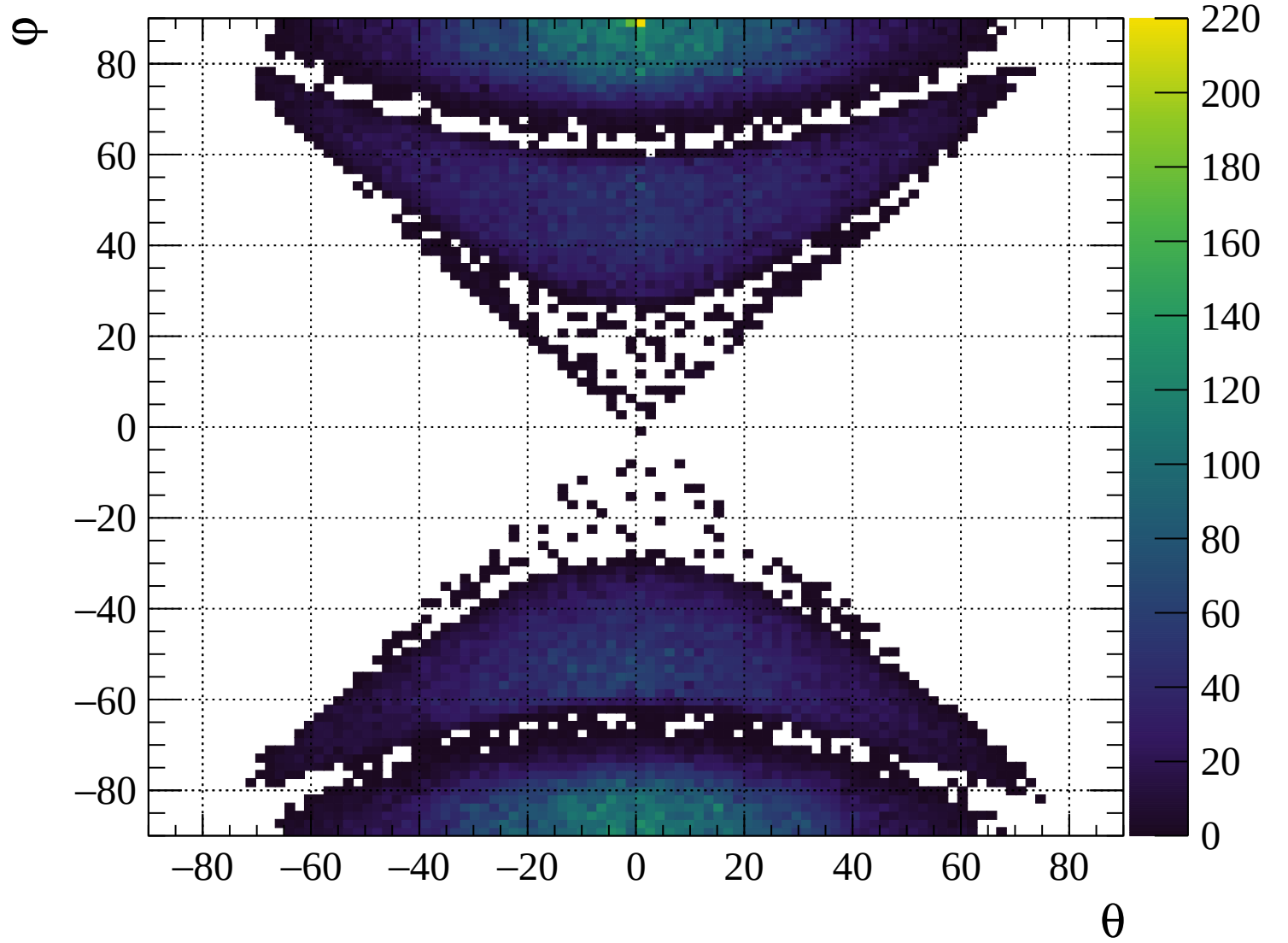


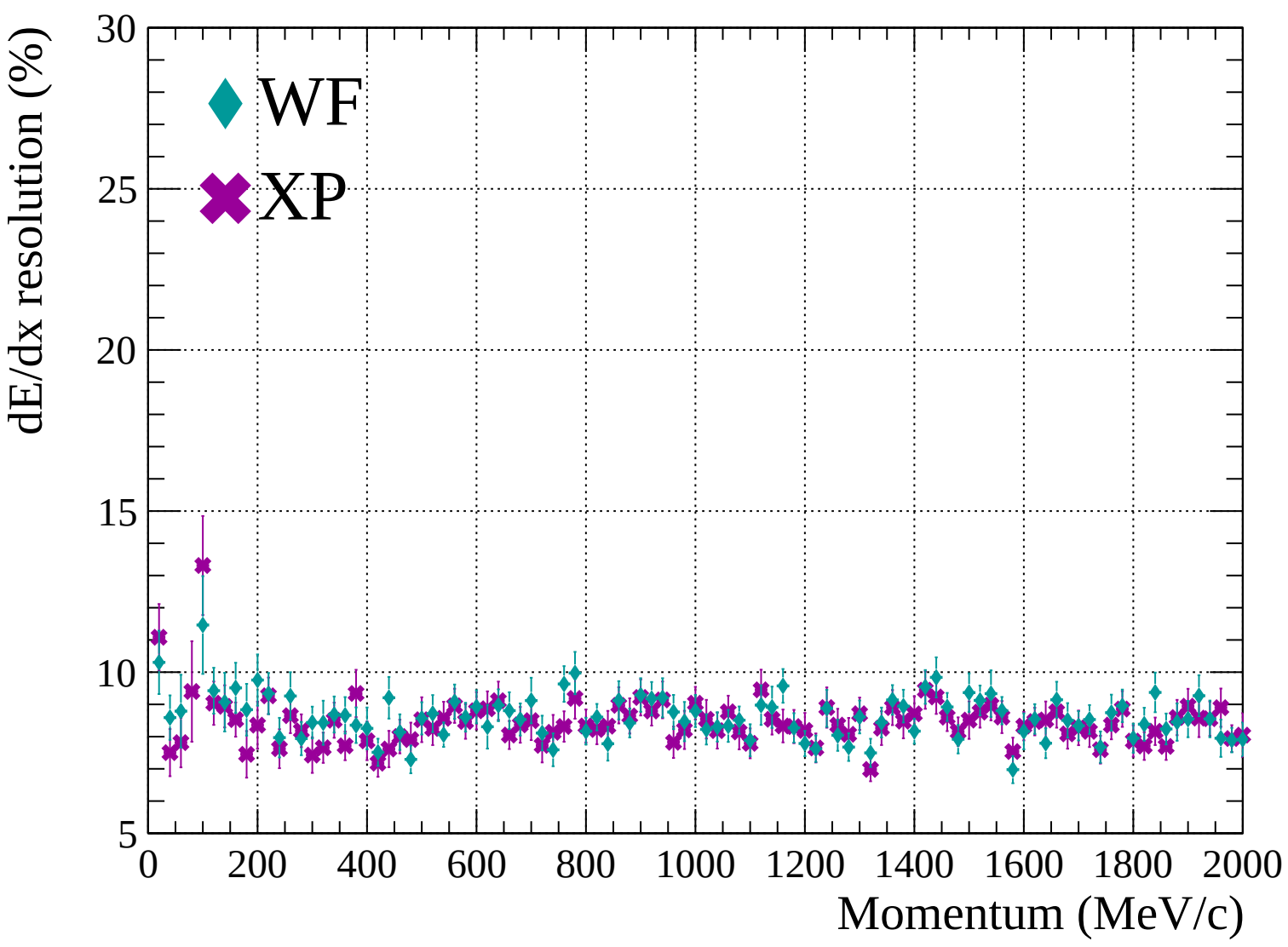


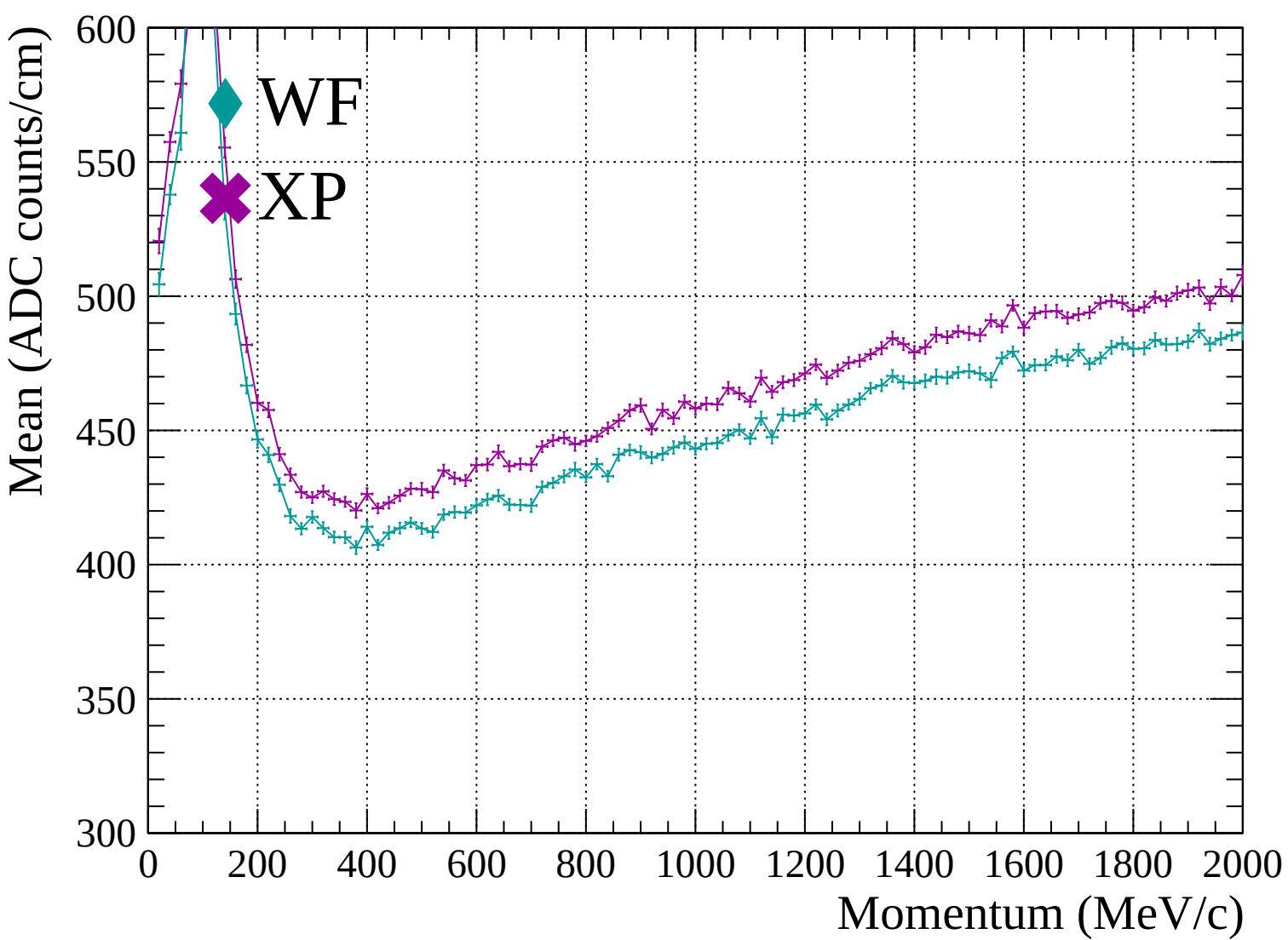


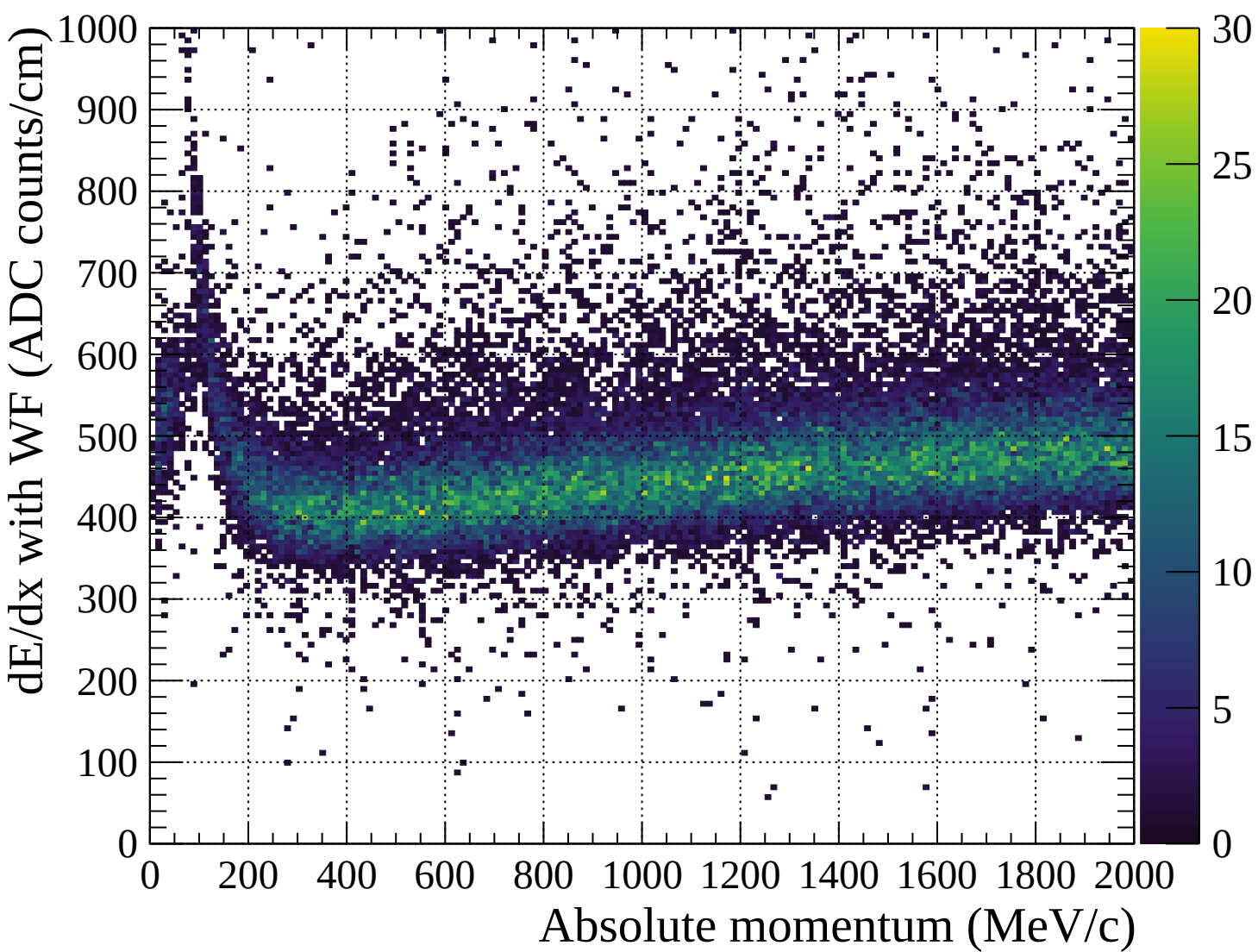


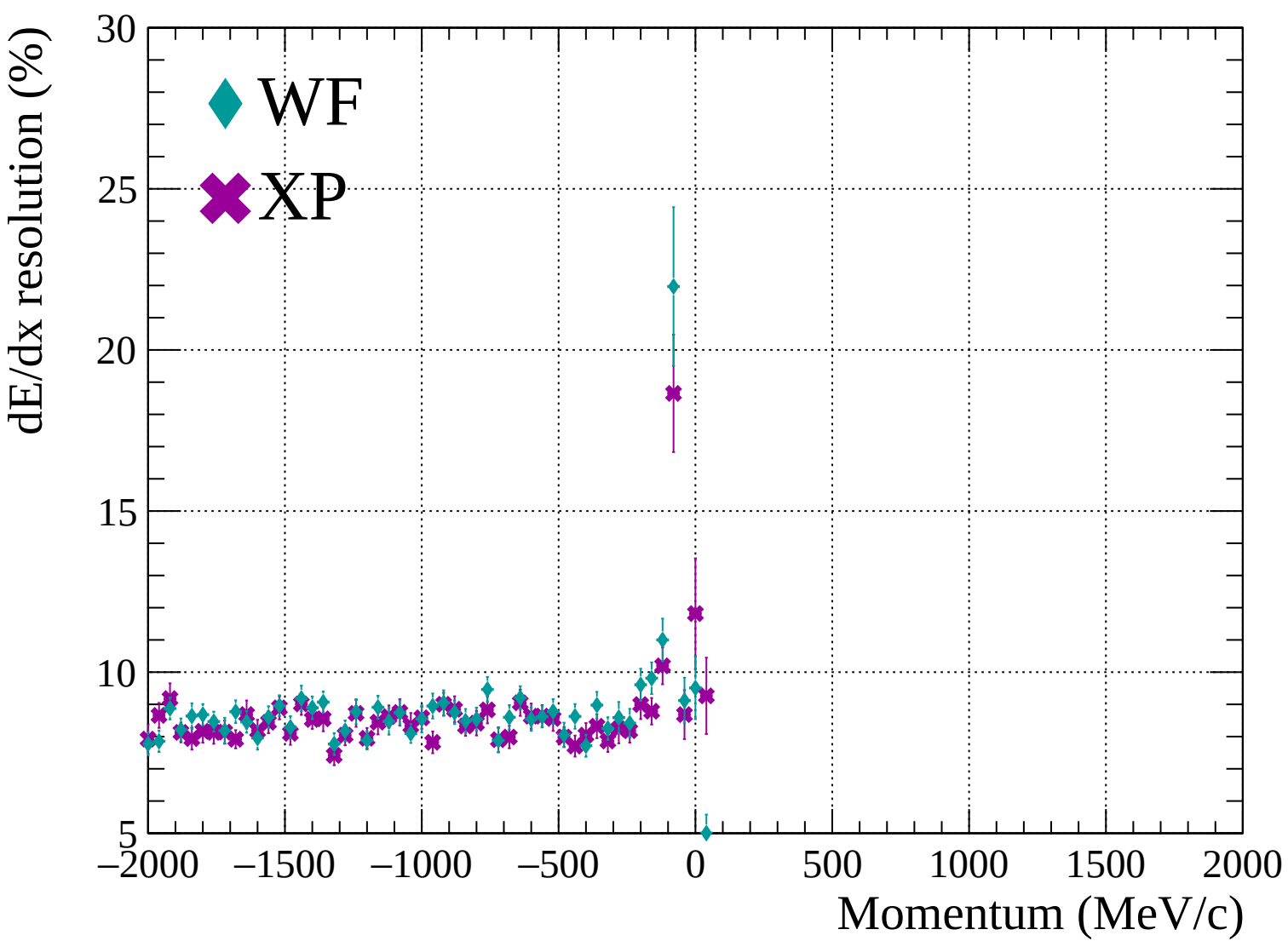


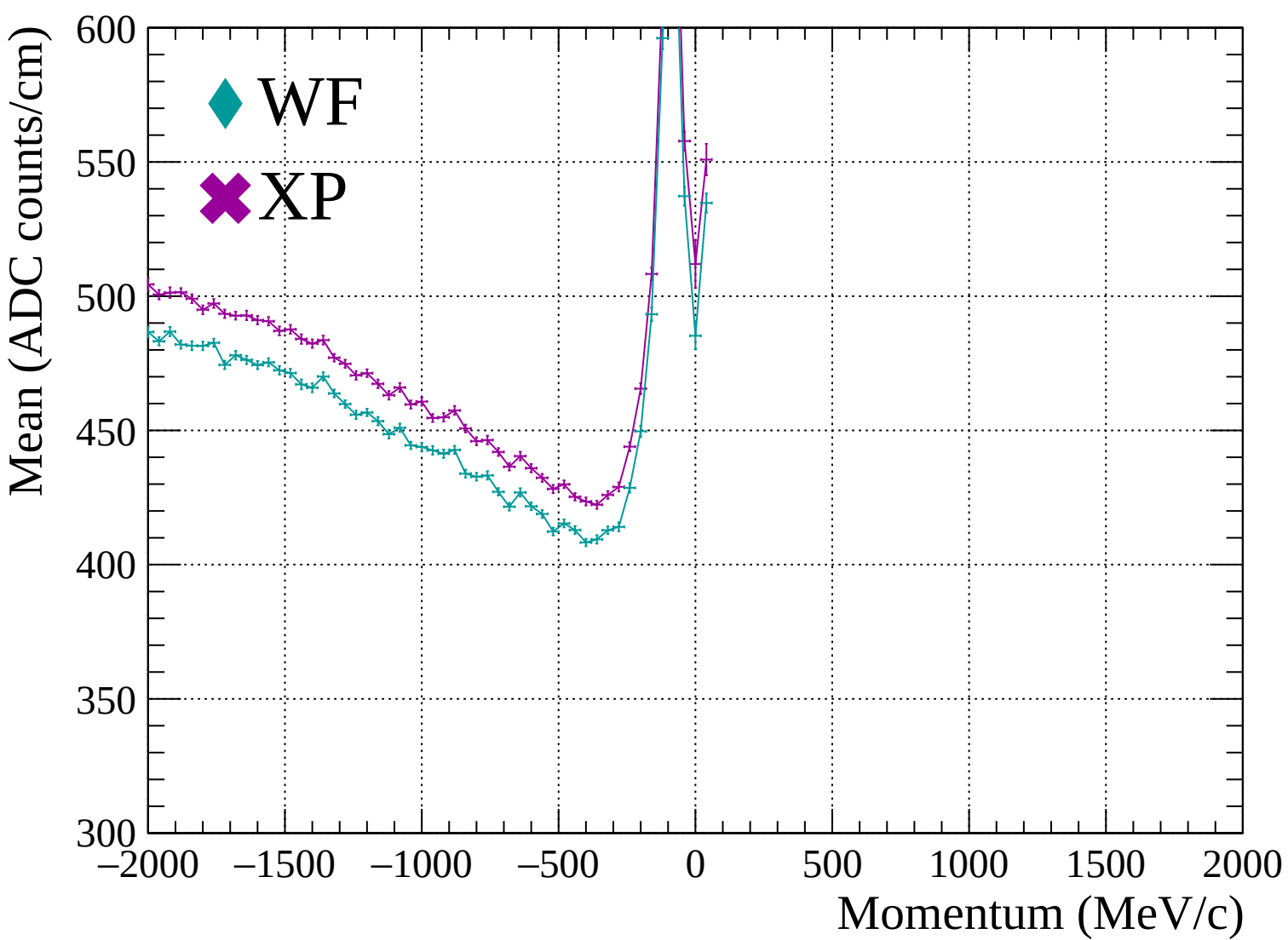


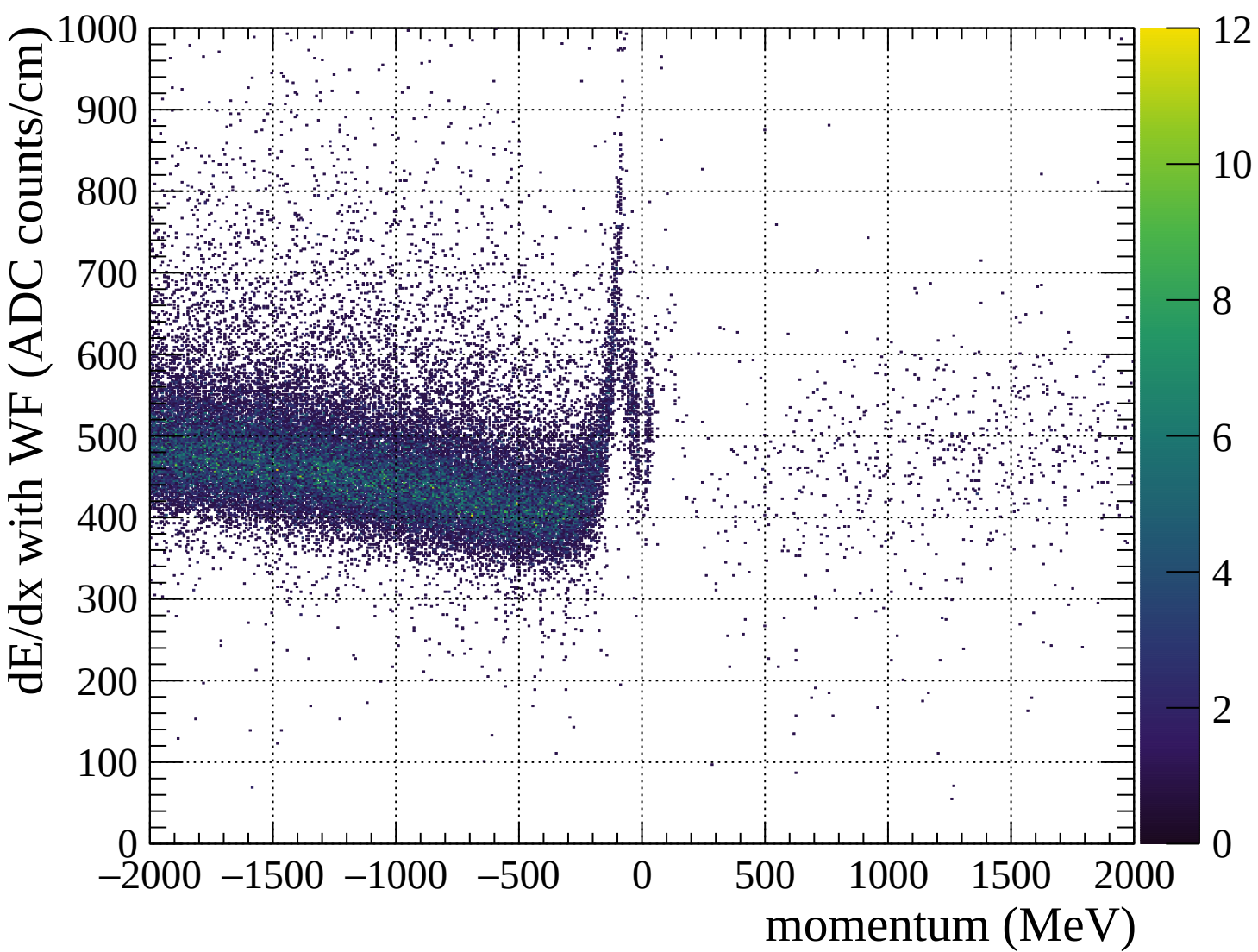


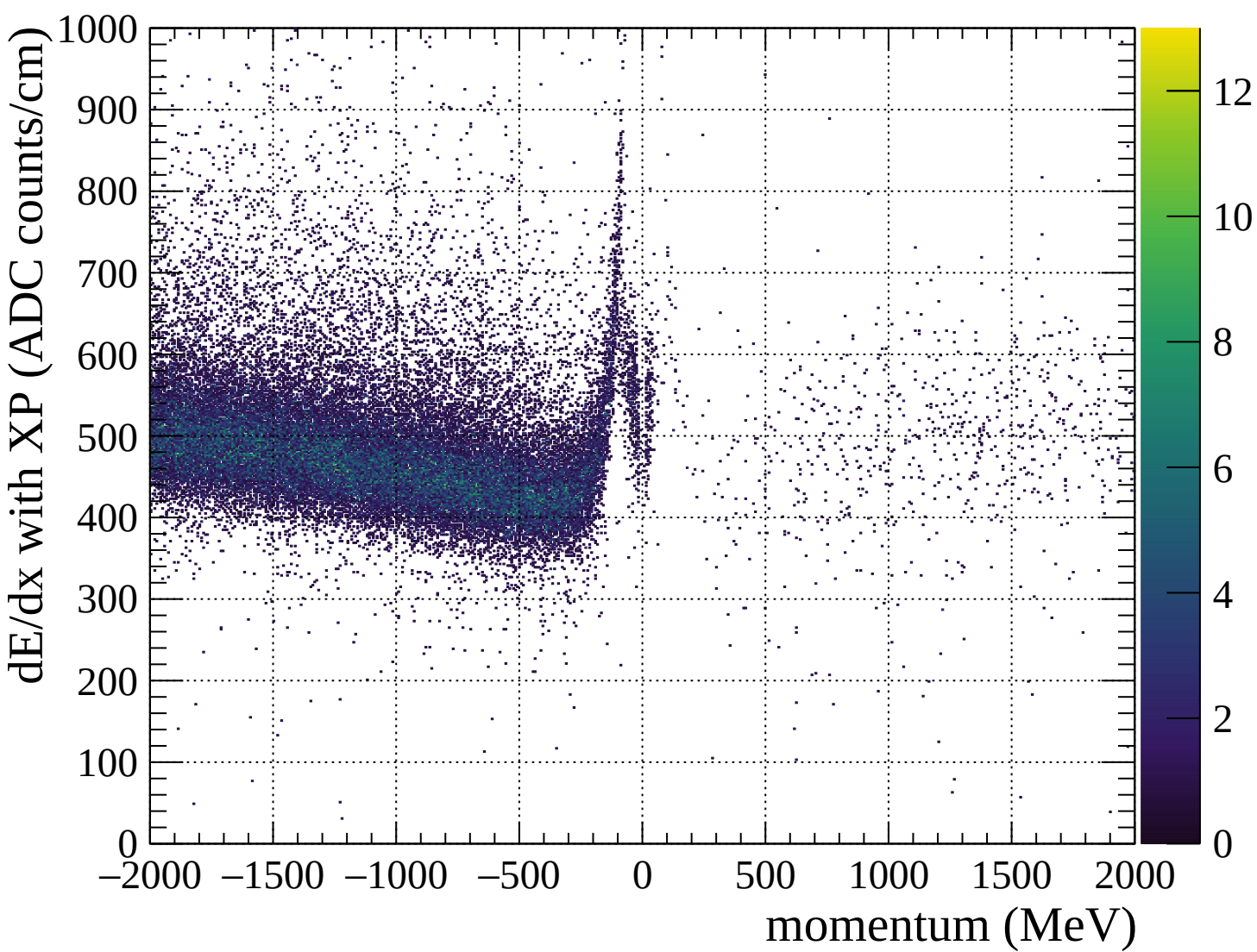


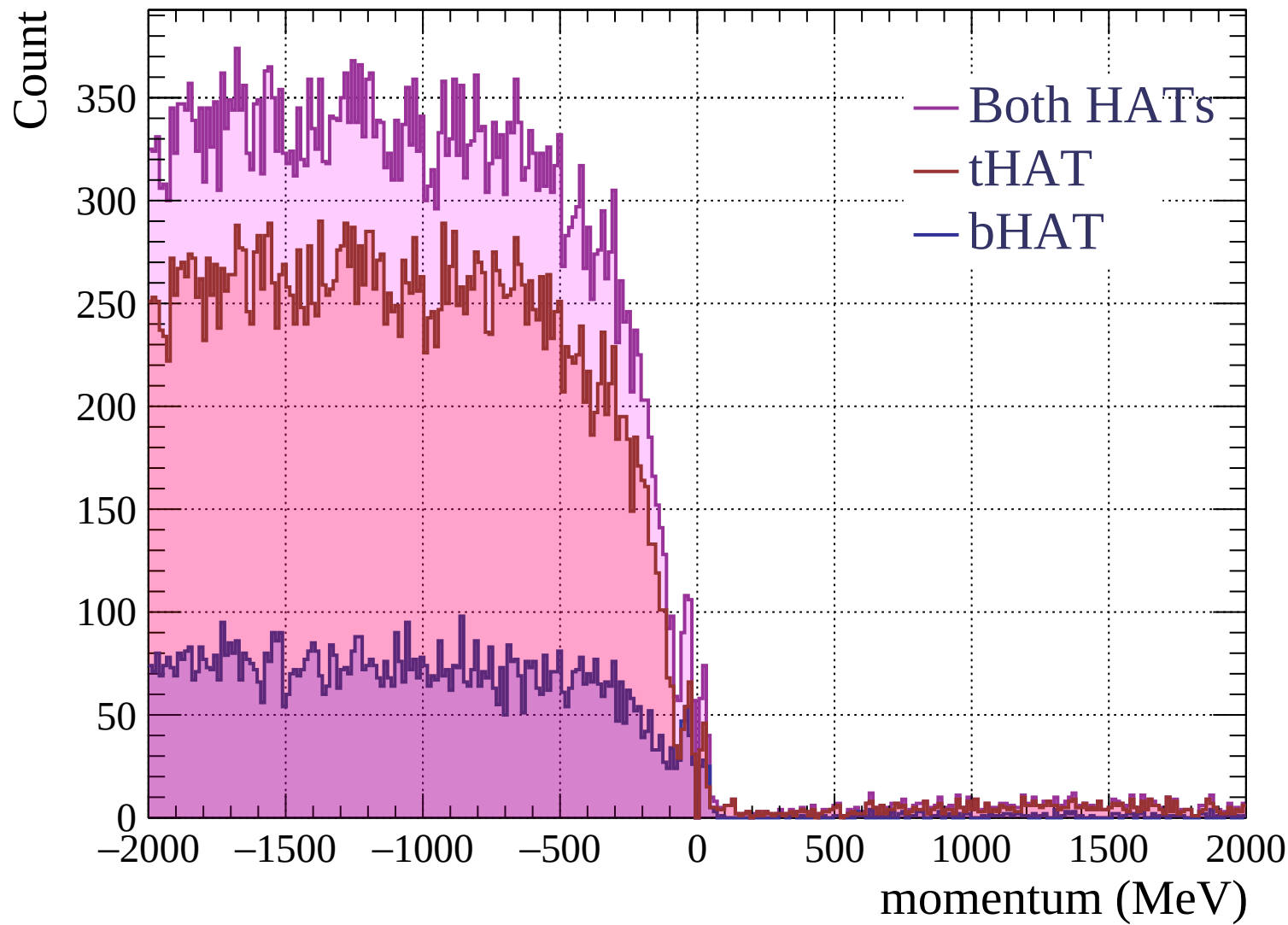


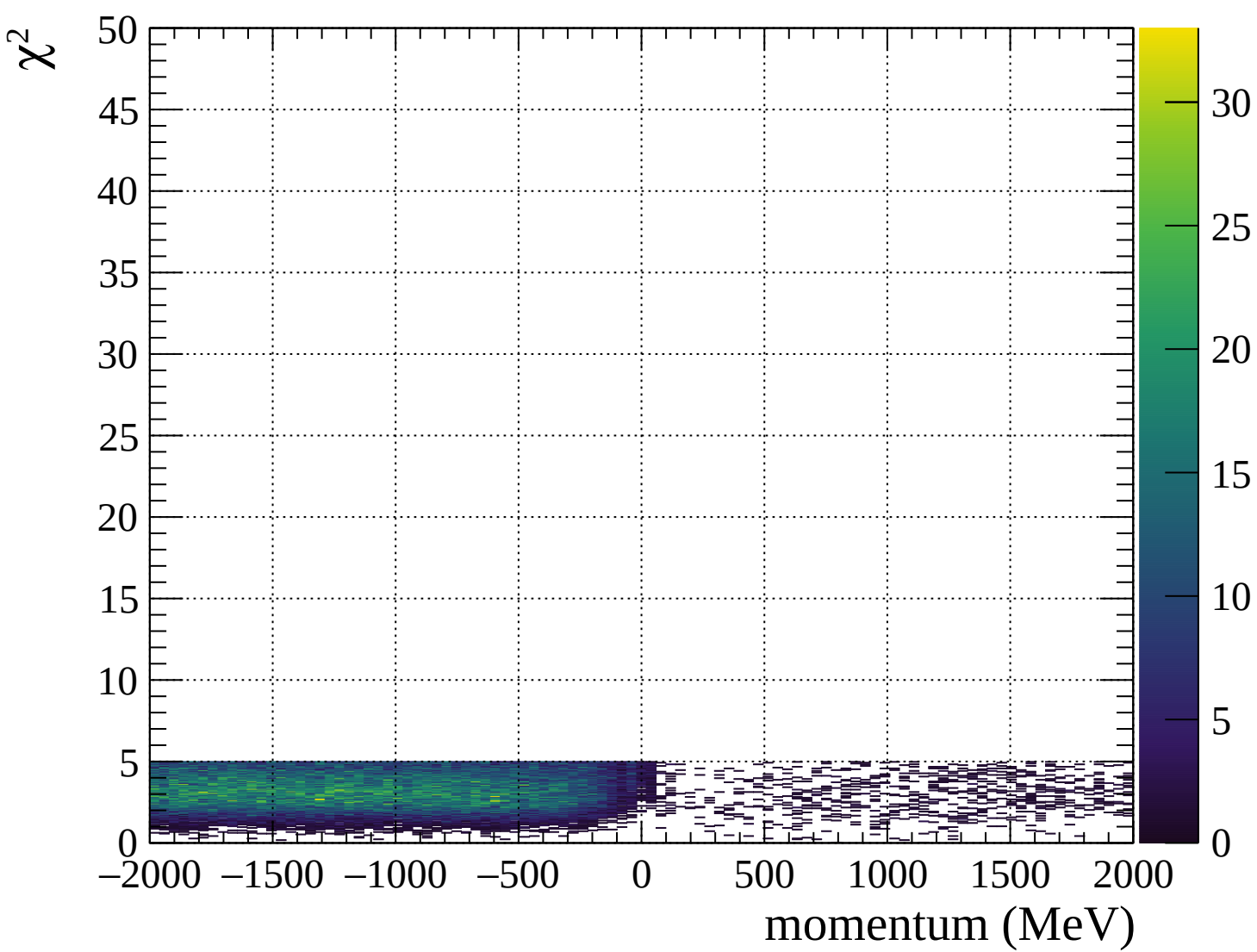




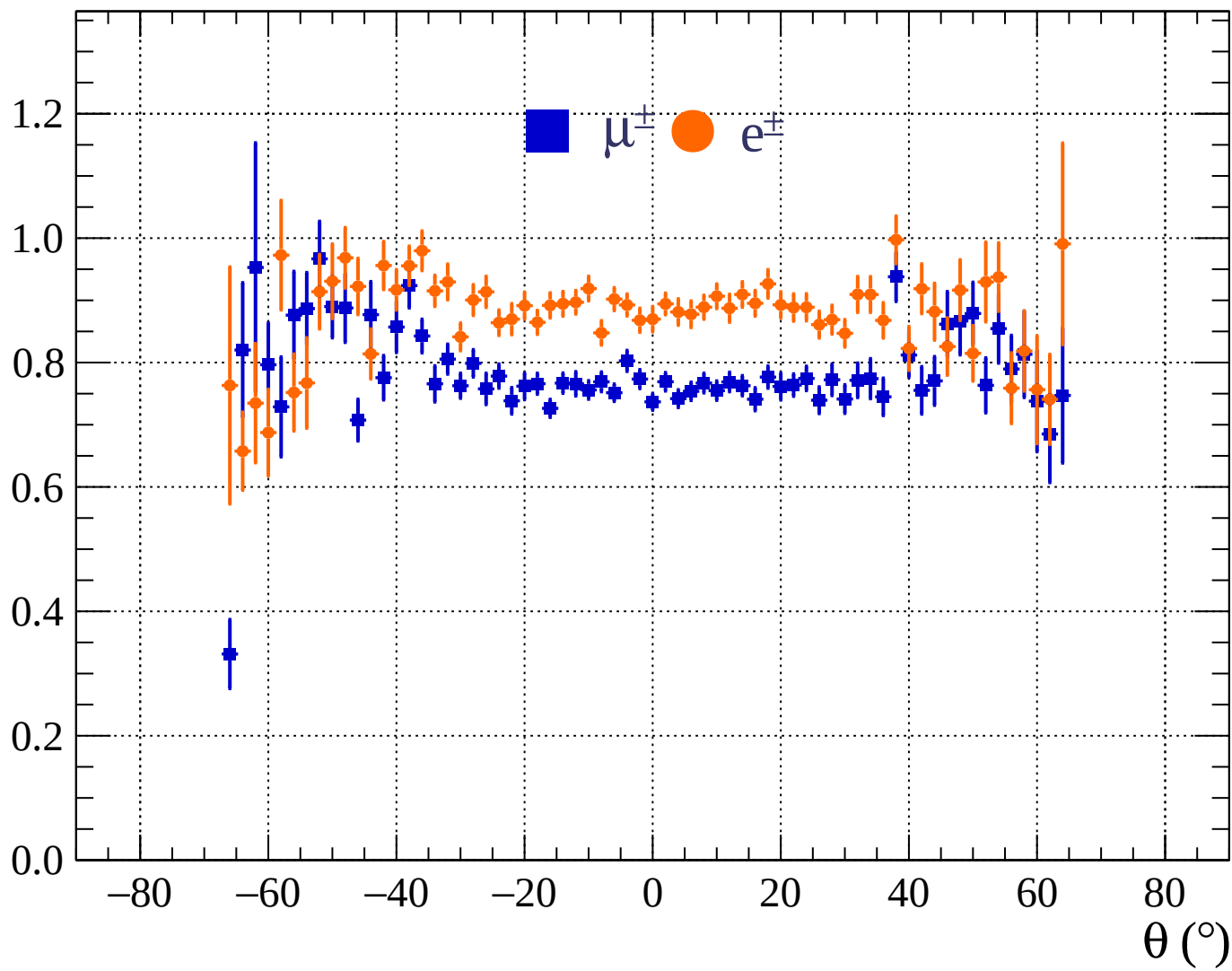


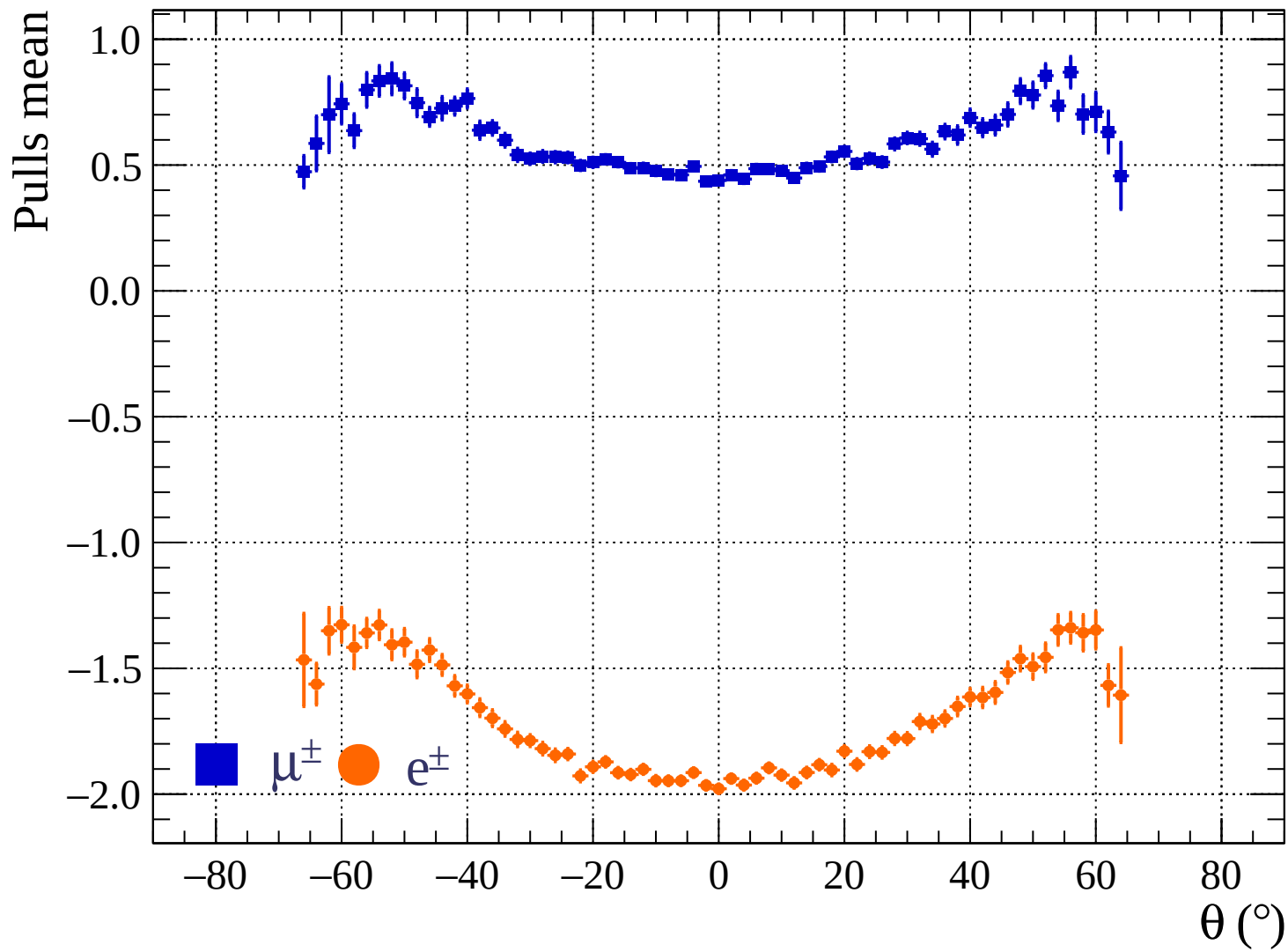




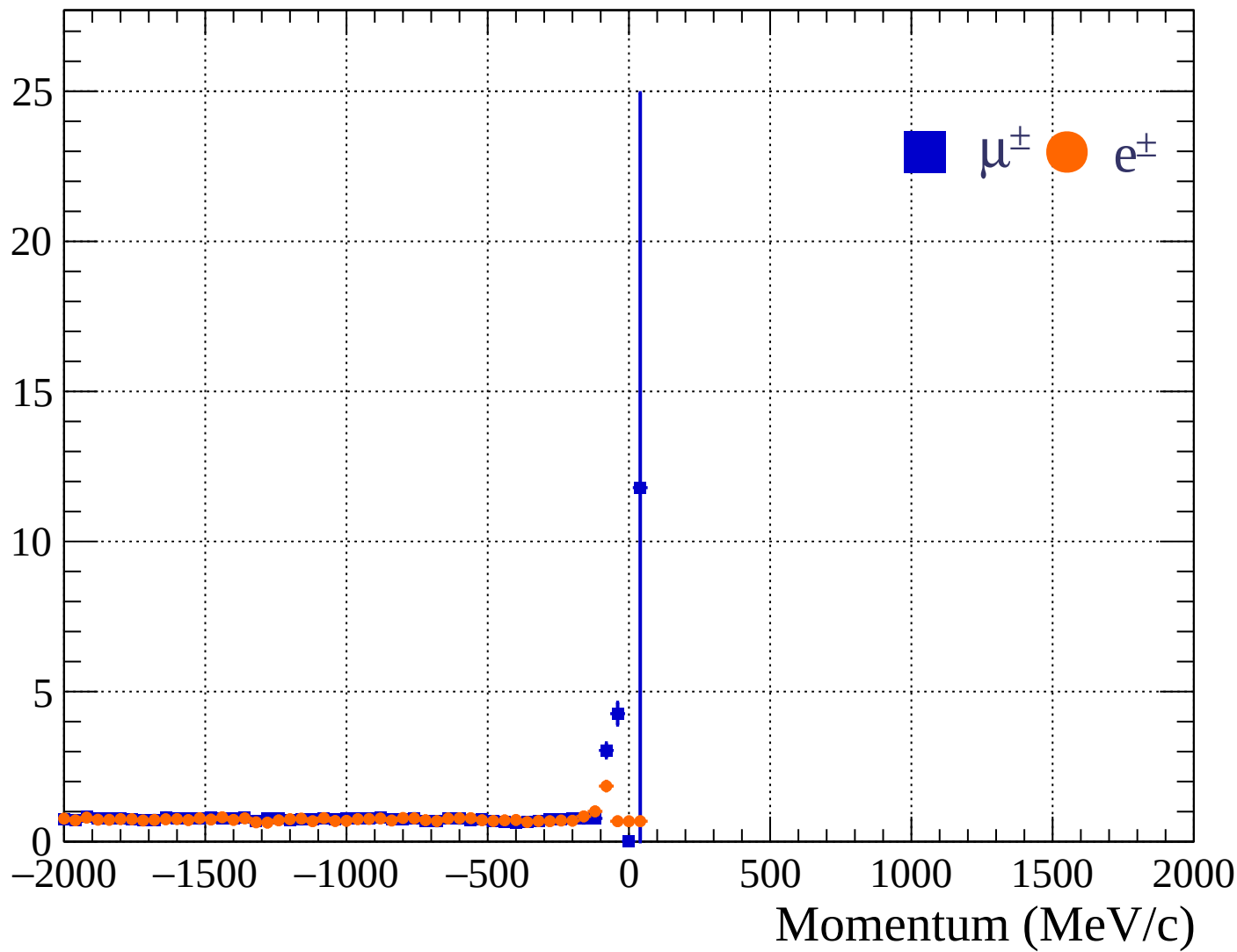


Pulls standard deviation

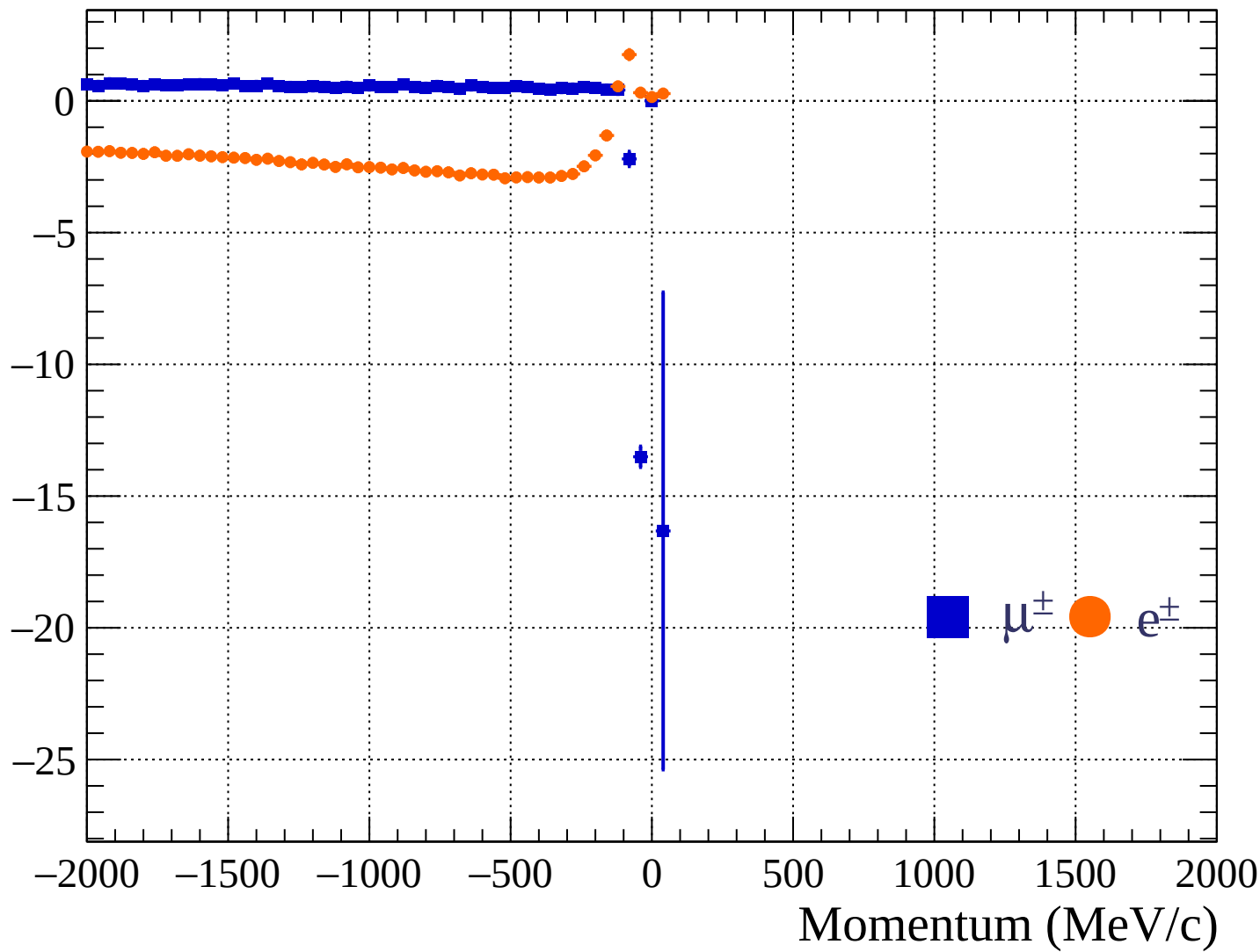


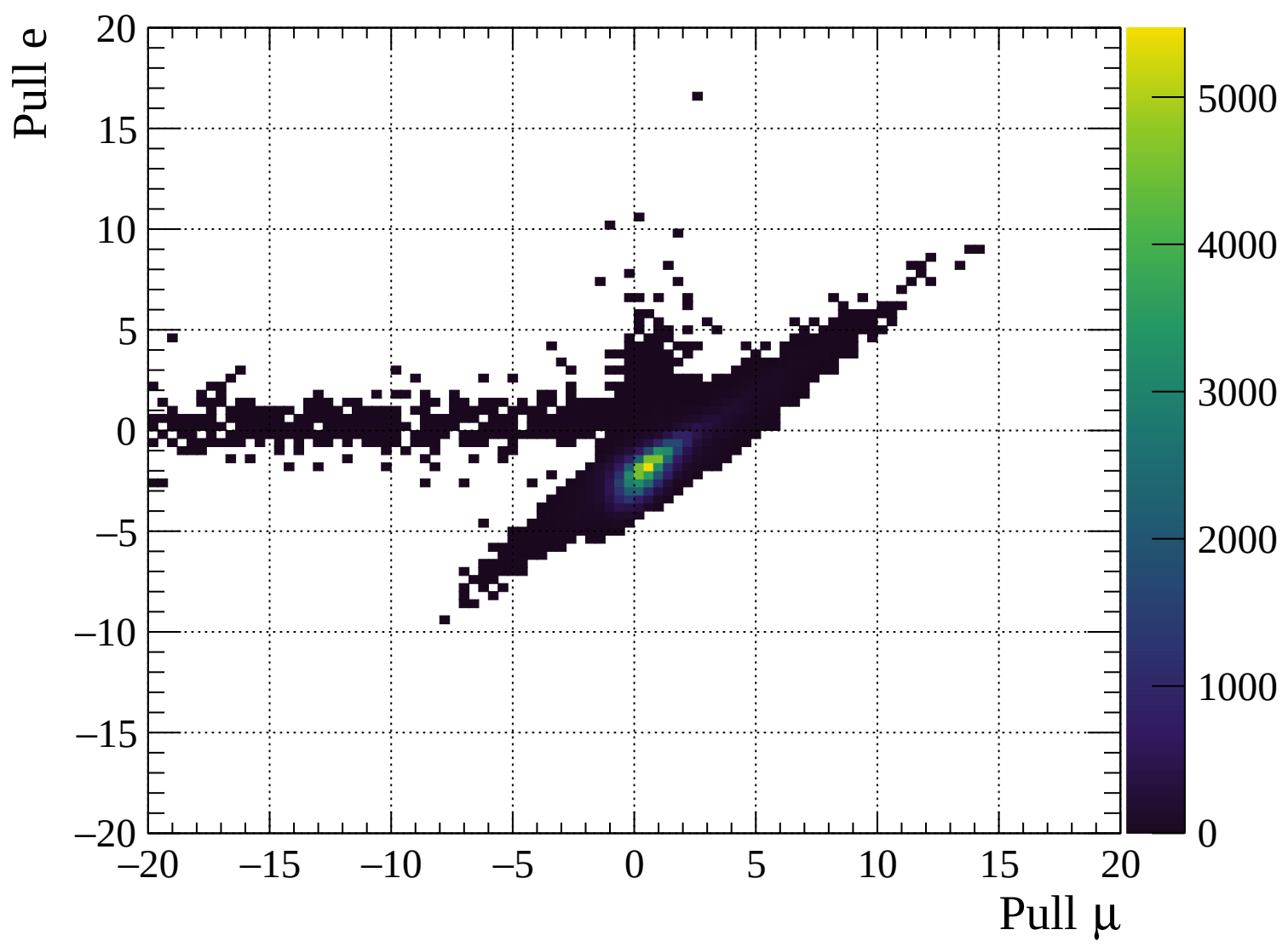


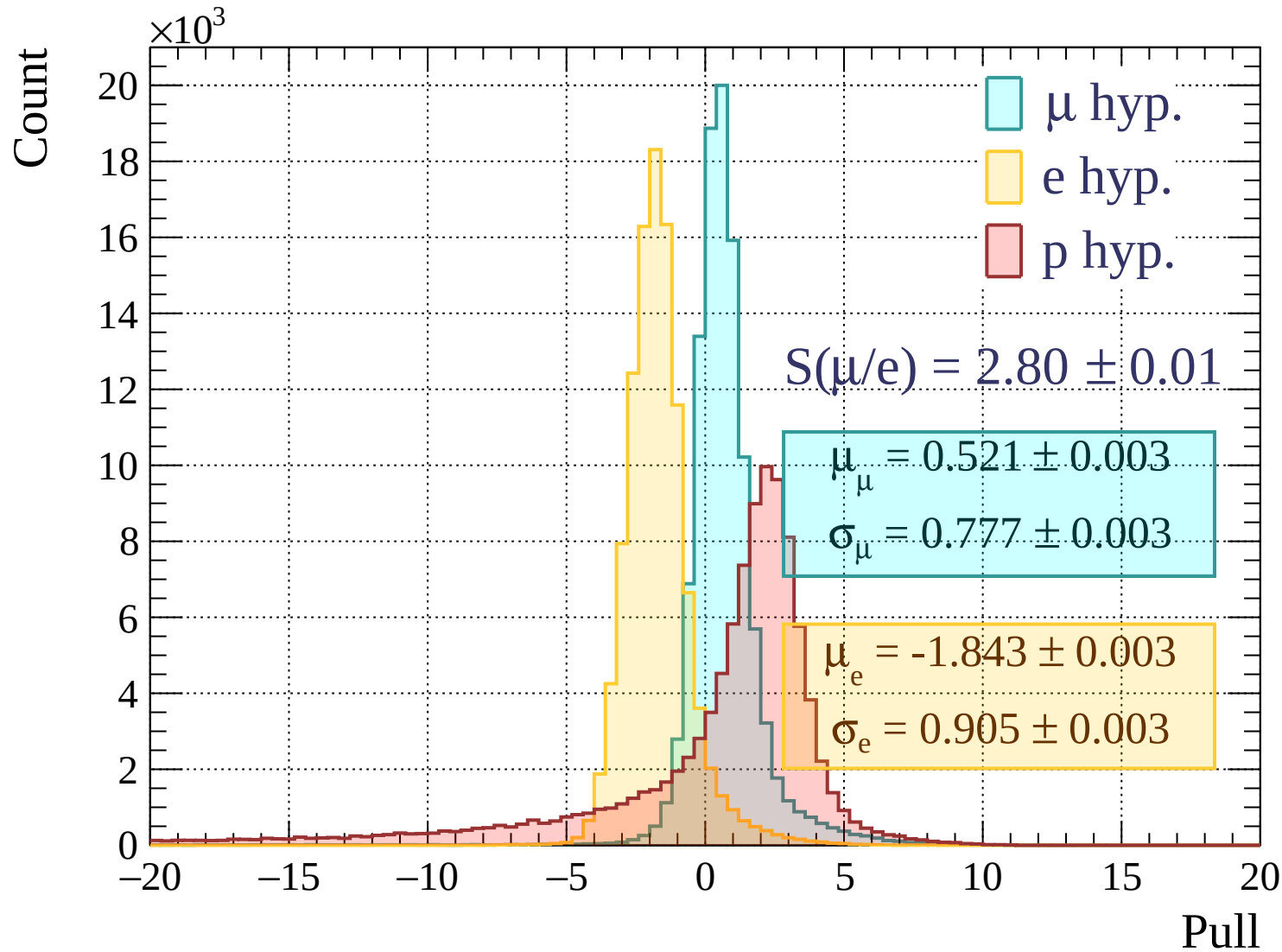
Pulls standard deviation

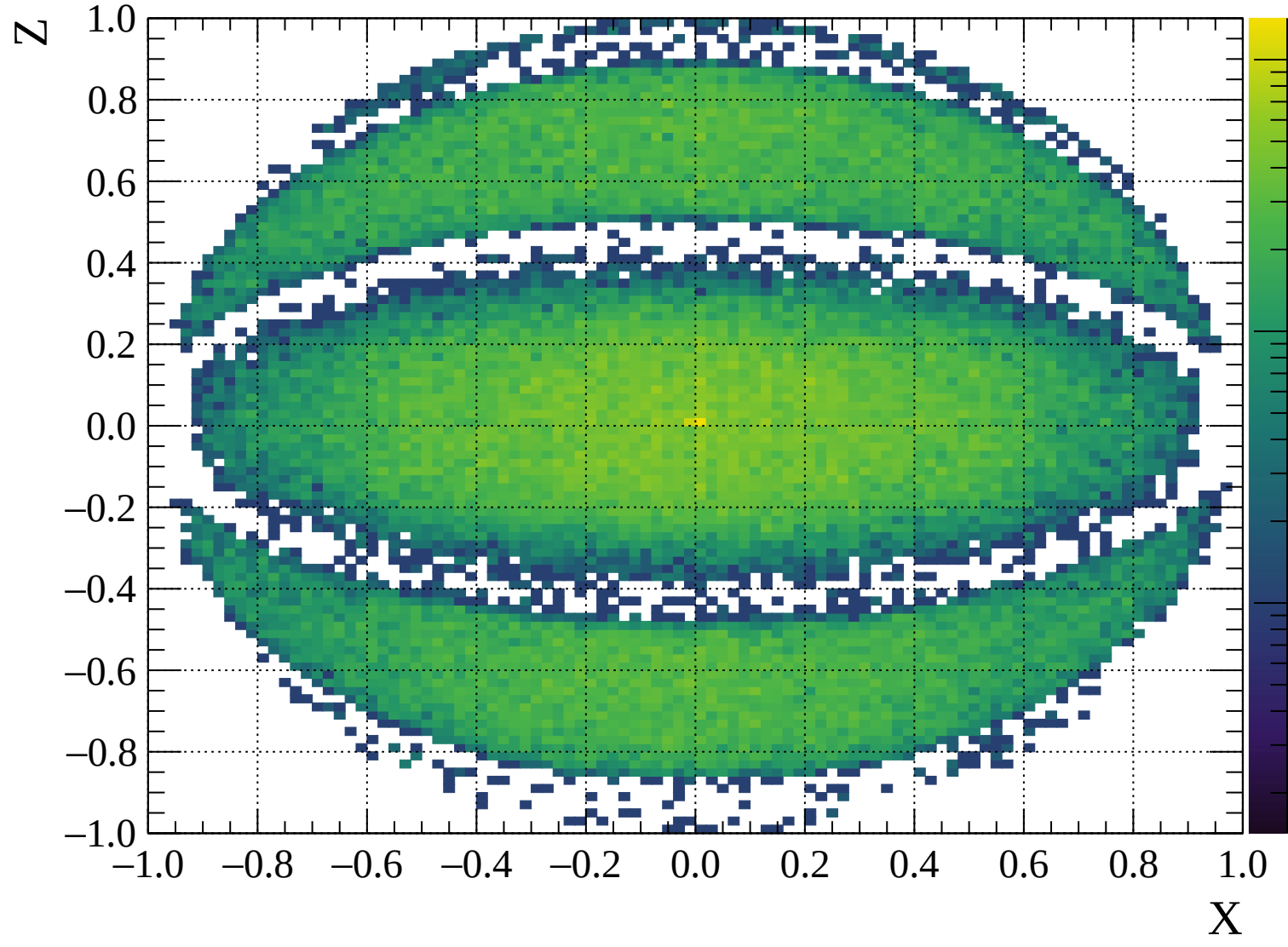


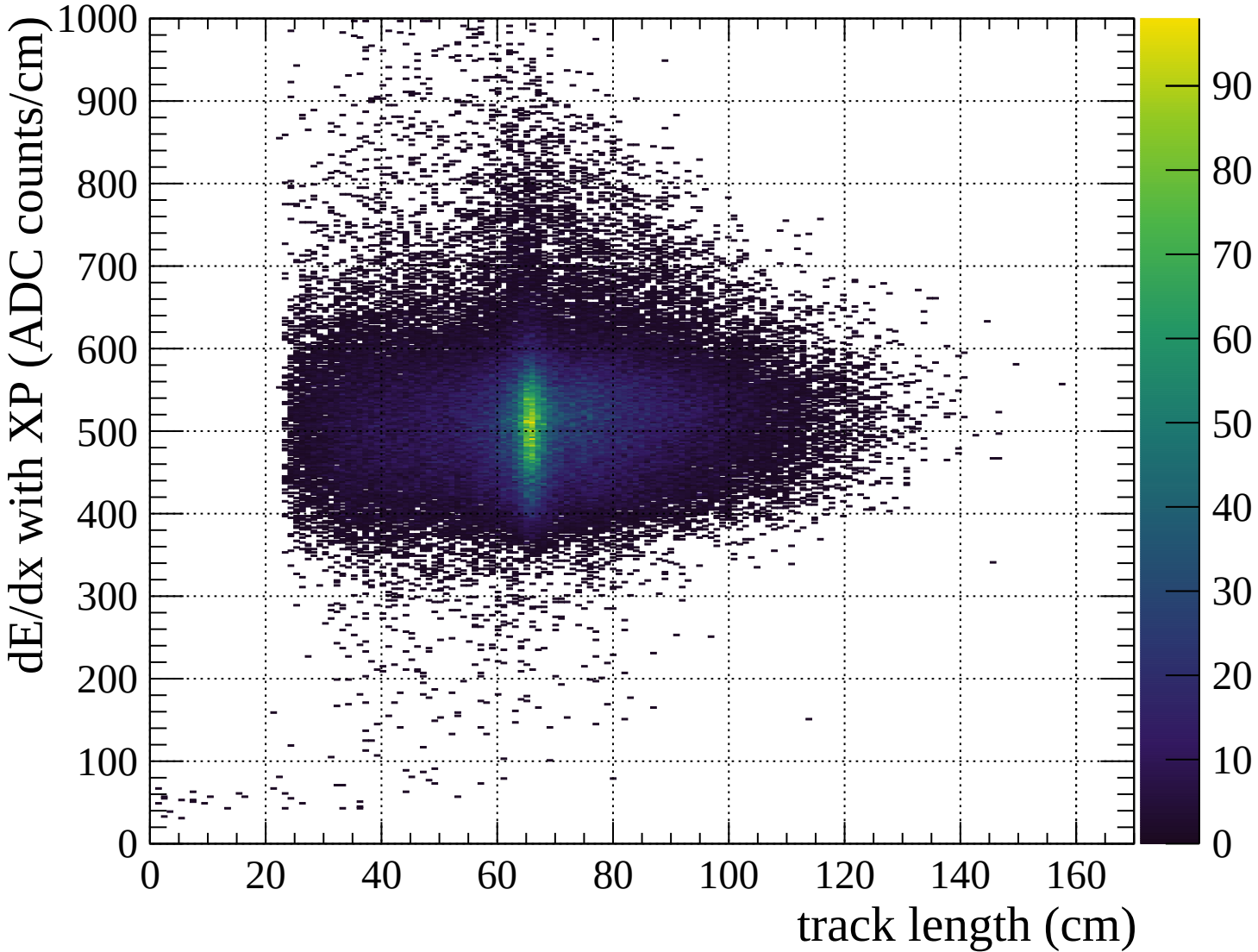
Pulls mean

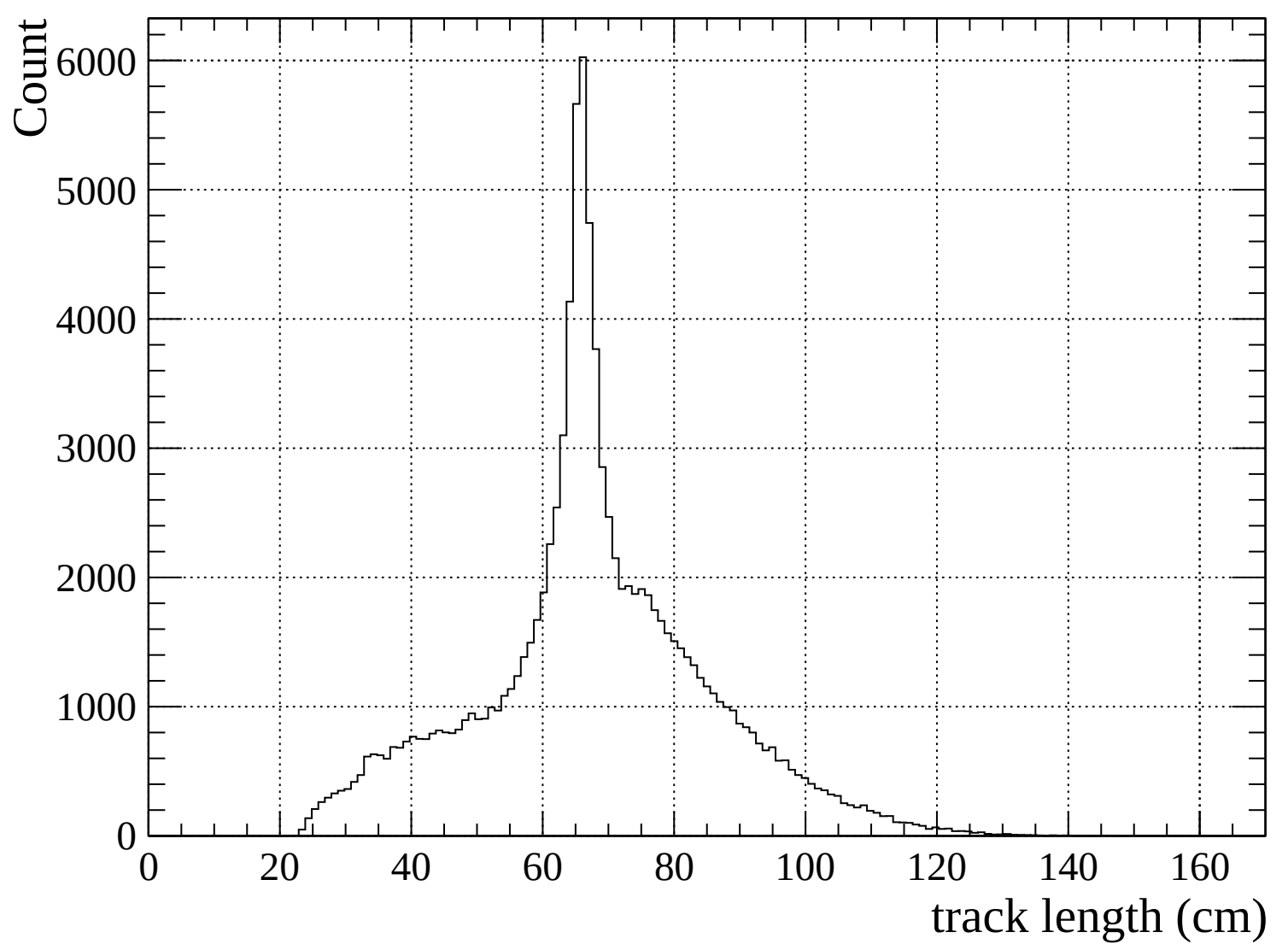


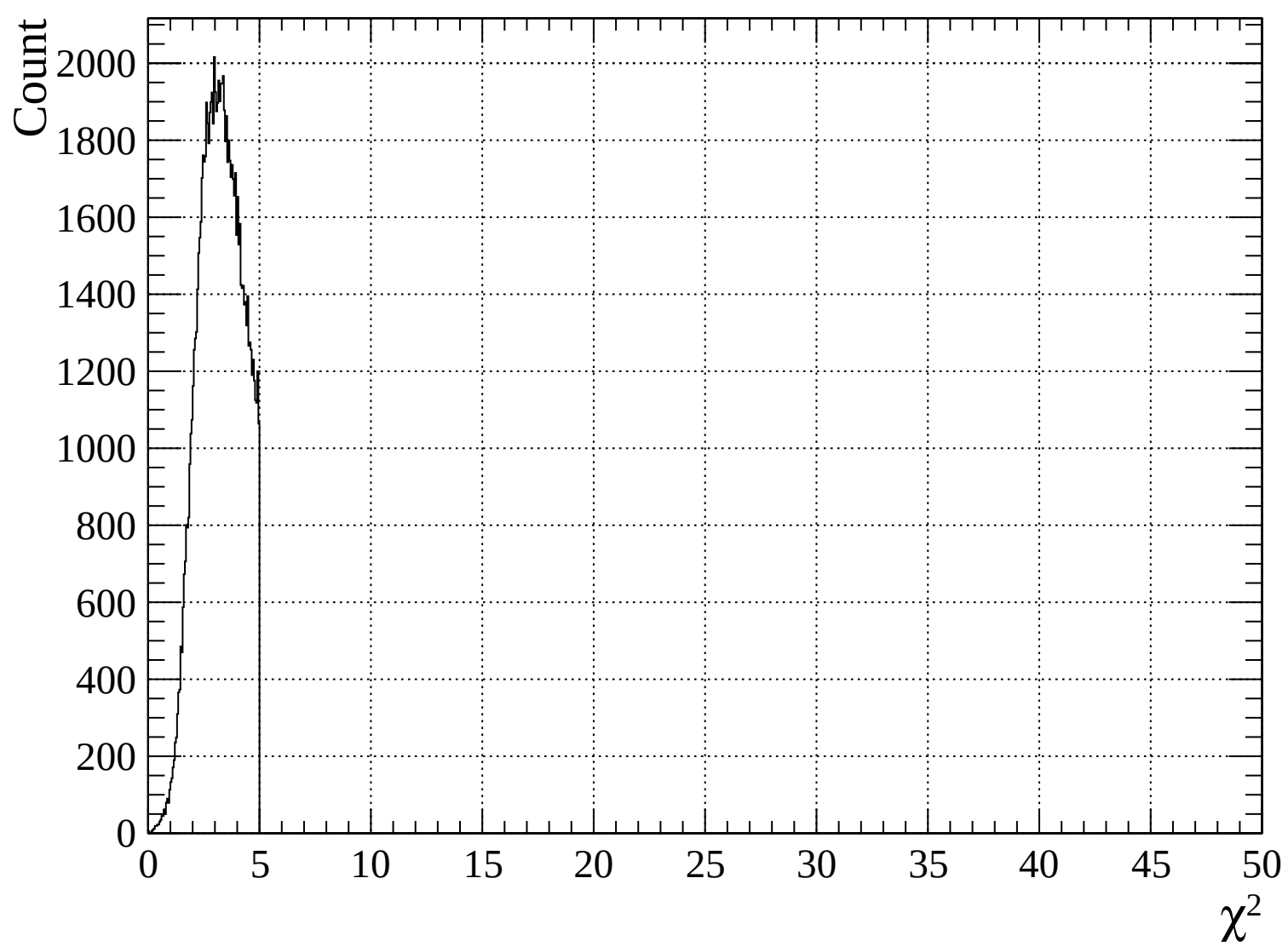


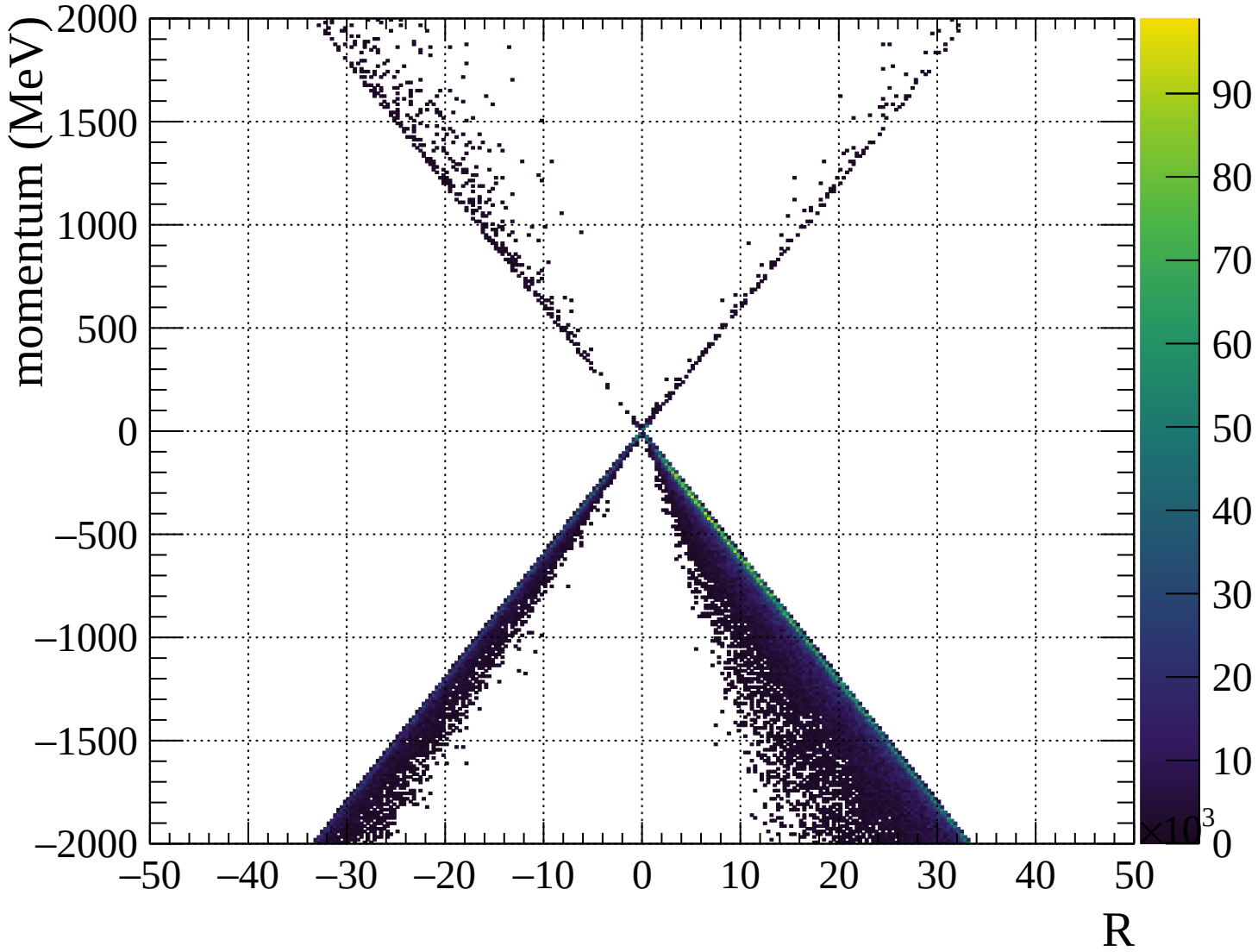


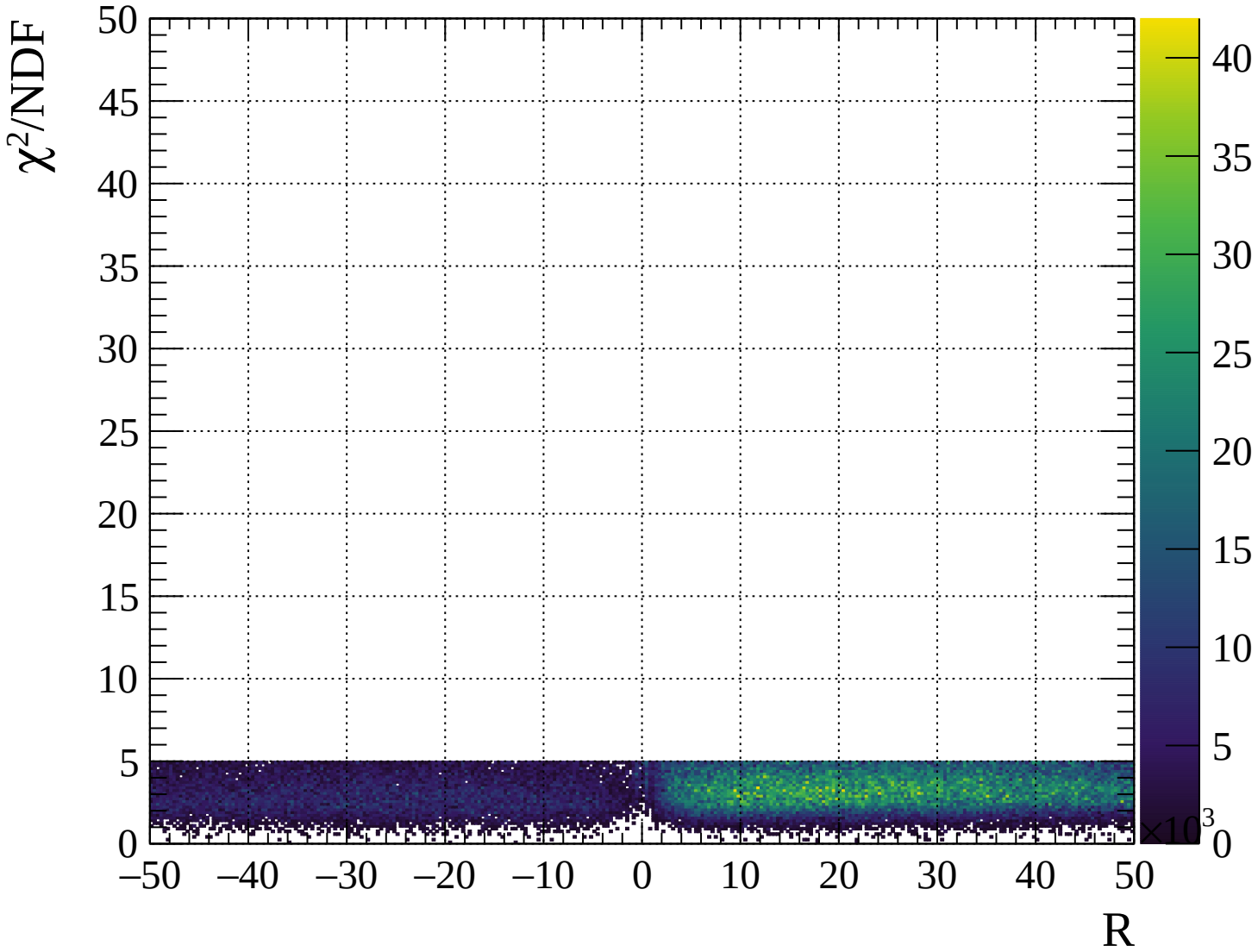






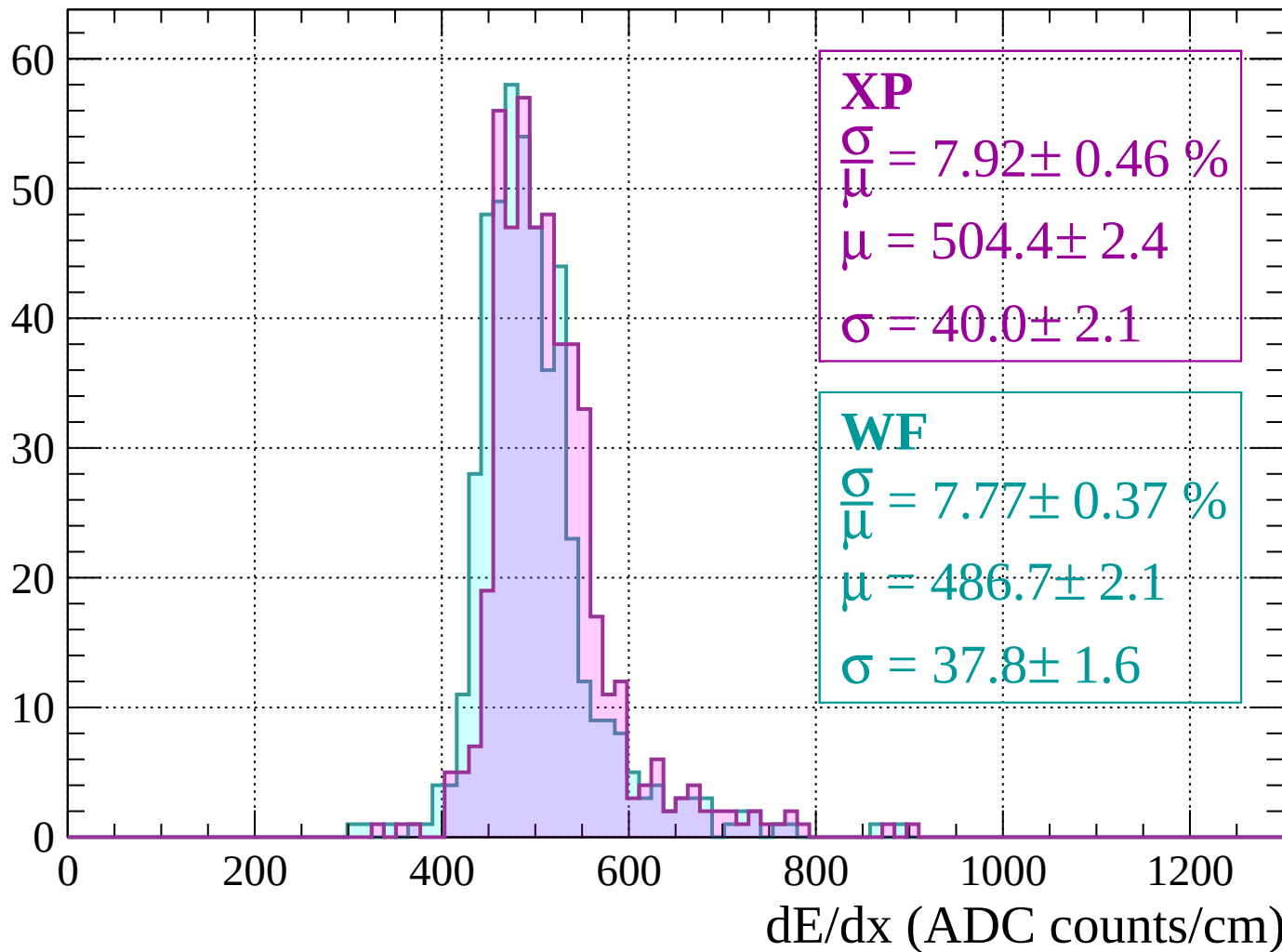






Energy loss | -2000 < p < -1960

Count



Energy loss | -1960 < p < -1920

Count

120

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 8.66 \pm 0.39 \%$$

$$\mu = 500.6 \pm 1.8$$

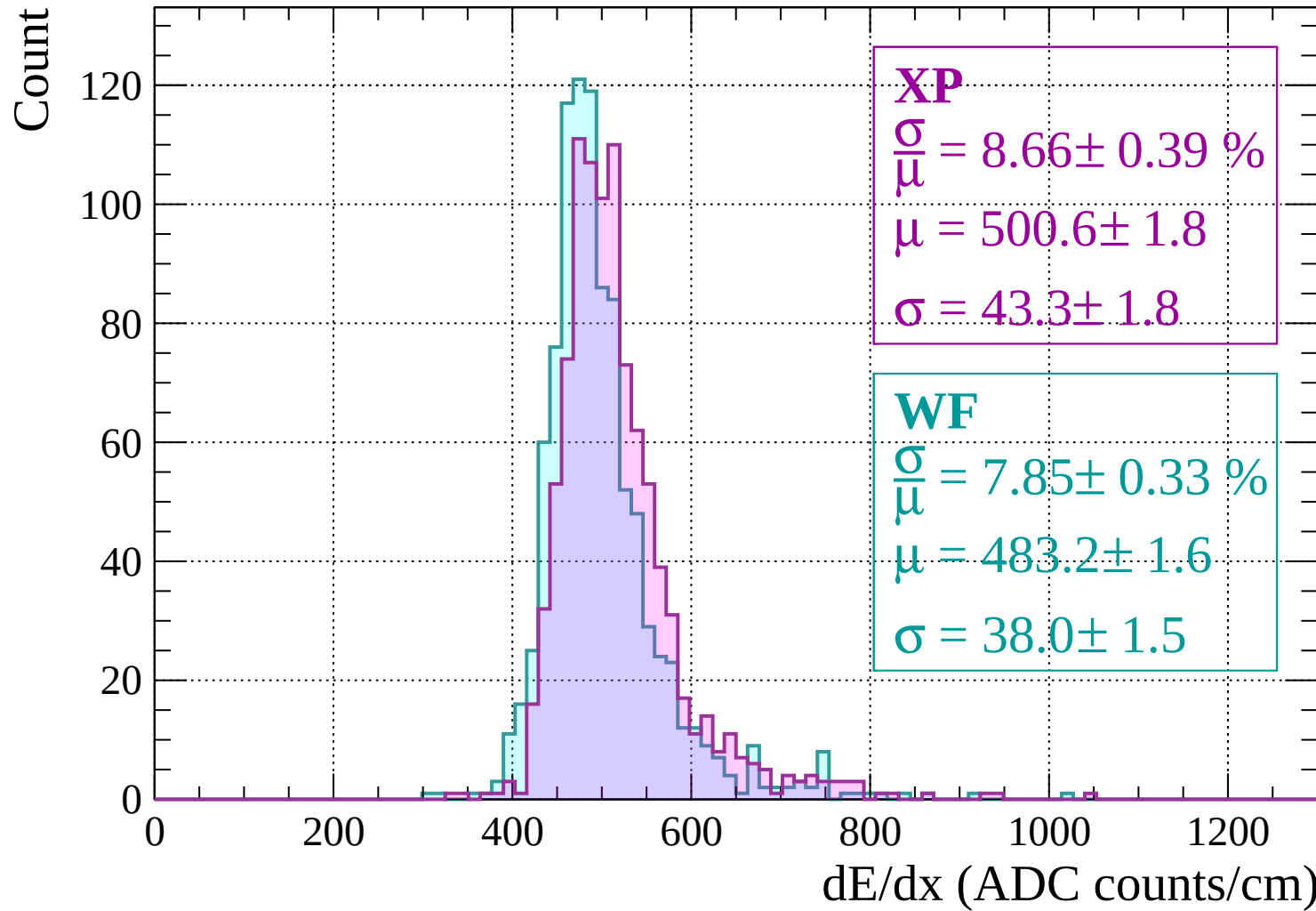
$$\sigma = 43.3 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 7.85 \pm 0.33 \%$$

$$\mu = 483.2 \pm 1.6$$

$$\sigma = 38.0 \pm 1.5$$



Energy loss | -1920 < p < -1880

Count

120

100

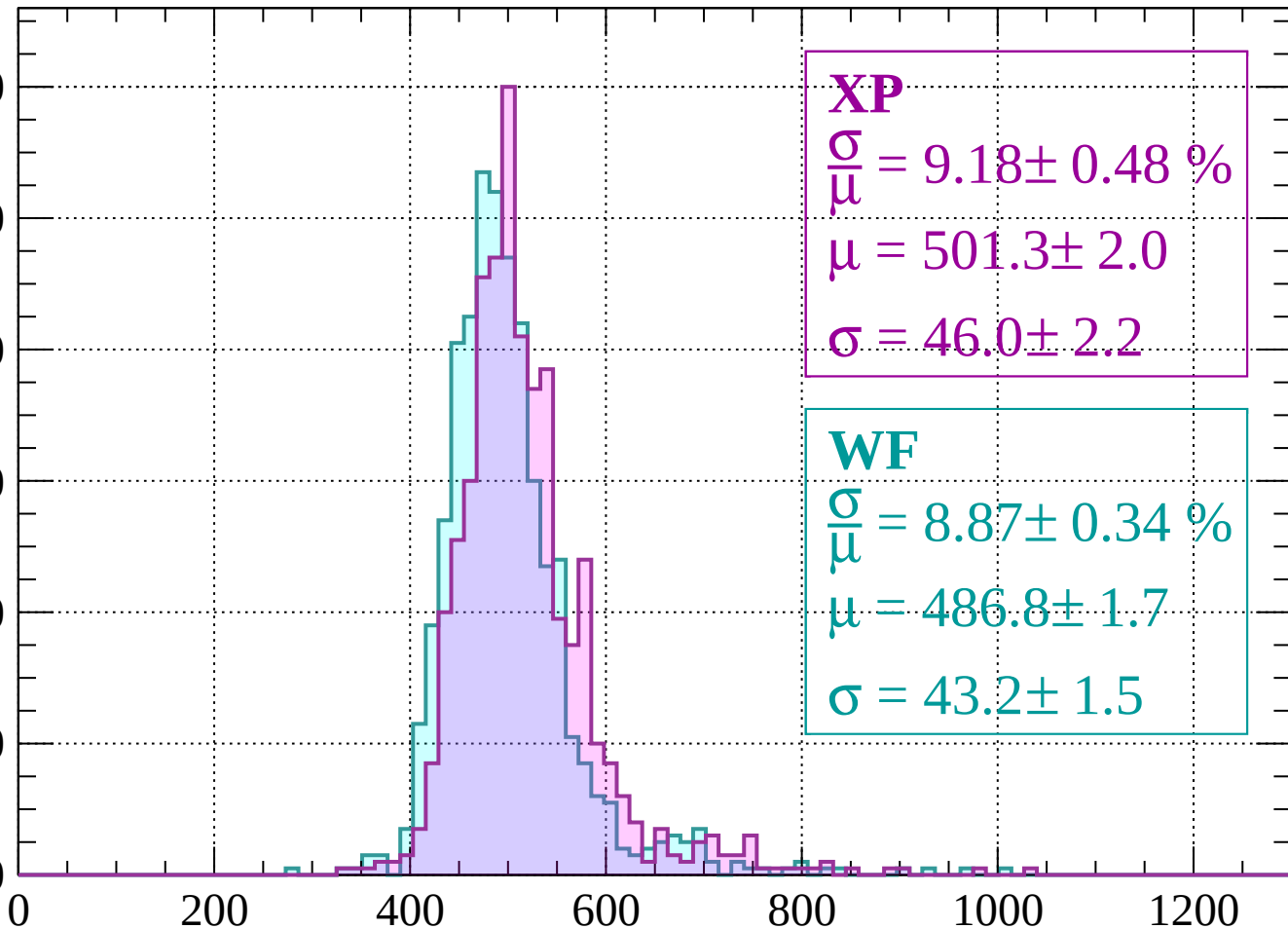
80

60

40

20

0



XP

$$\frac{\sigma}{\mu} = 9.18 \pm 0.48 \%$$

$$\mu = 501.3 \pm 2.0$$

$$\sigma = 46.0 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 8.87 \pm 0.34 \%$$

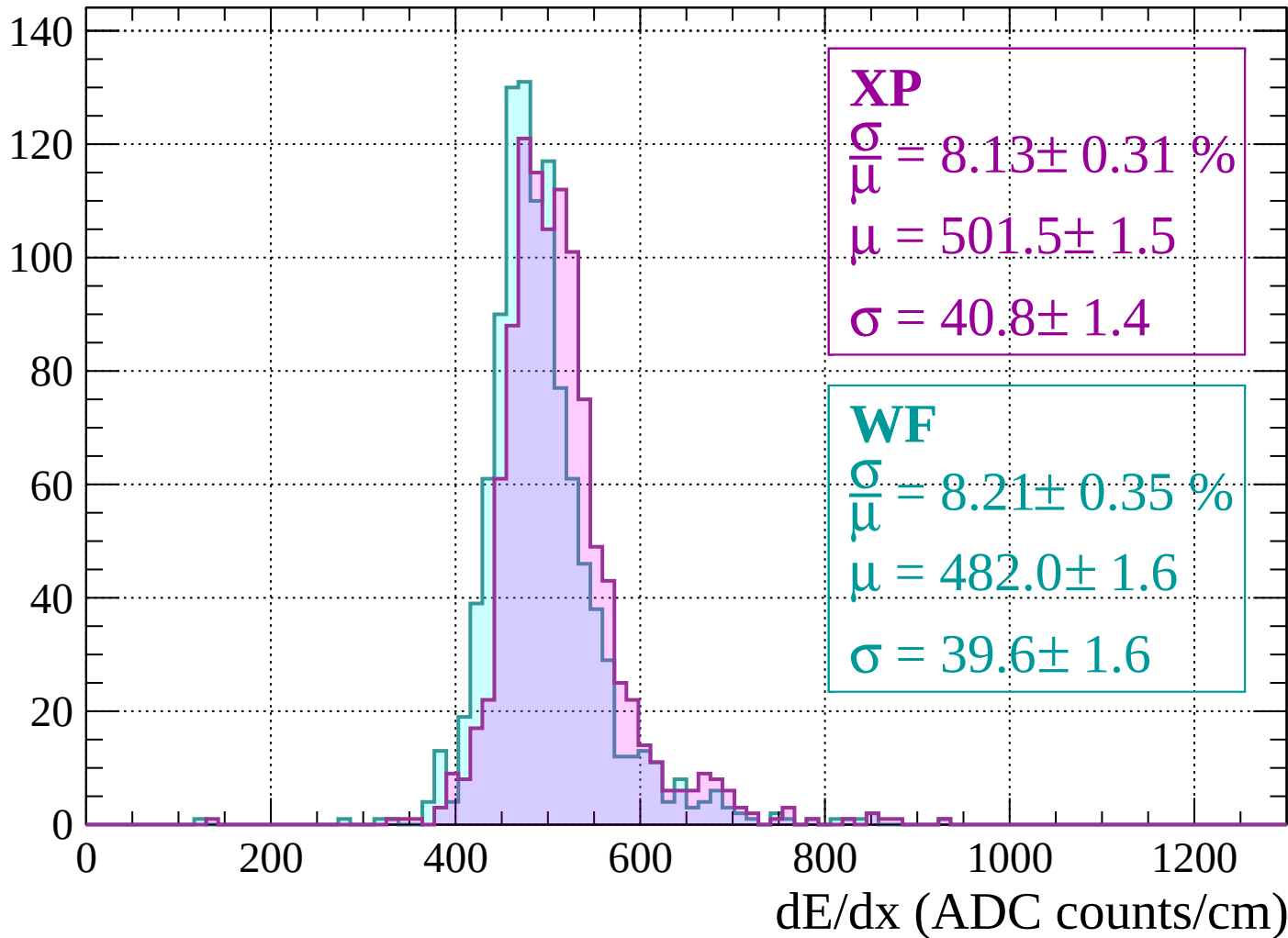
$$\mu = 486.8 \pm 1.7$$

$$\sigma = 43.2 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1880 < p < -1840$

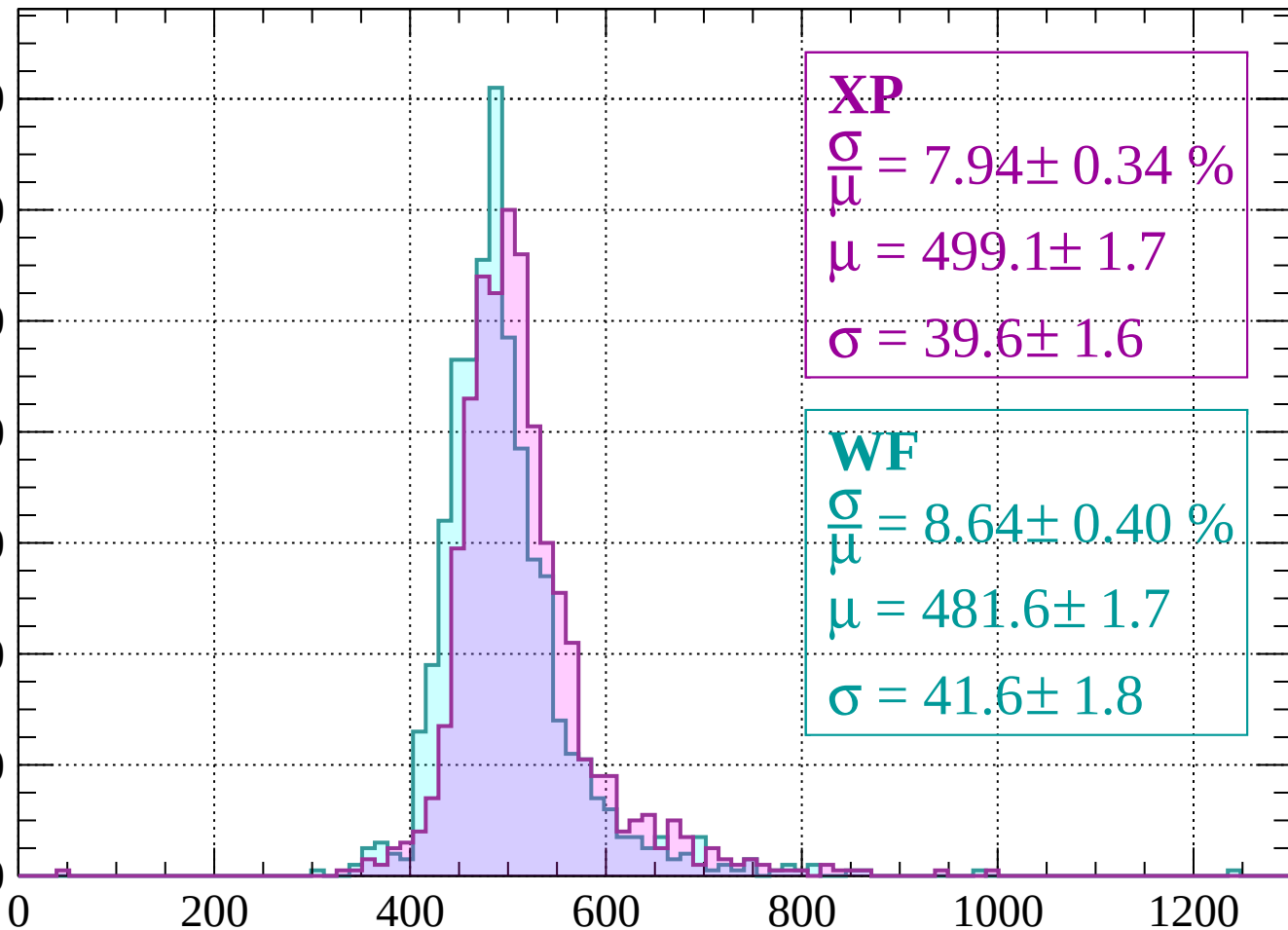
Count



Energy loss | $-1840 < p < -1800$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 7.94 \pm 0.34 \%$$

$$\mu = 499.1 \pm 1.7$$

$$\sigma = 39.6 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.64 \pm 0.40 \%$$

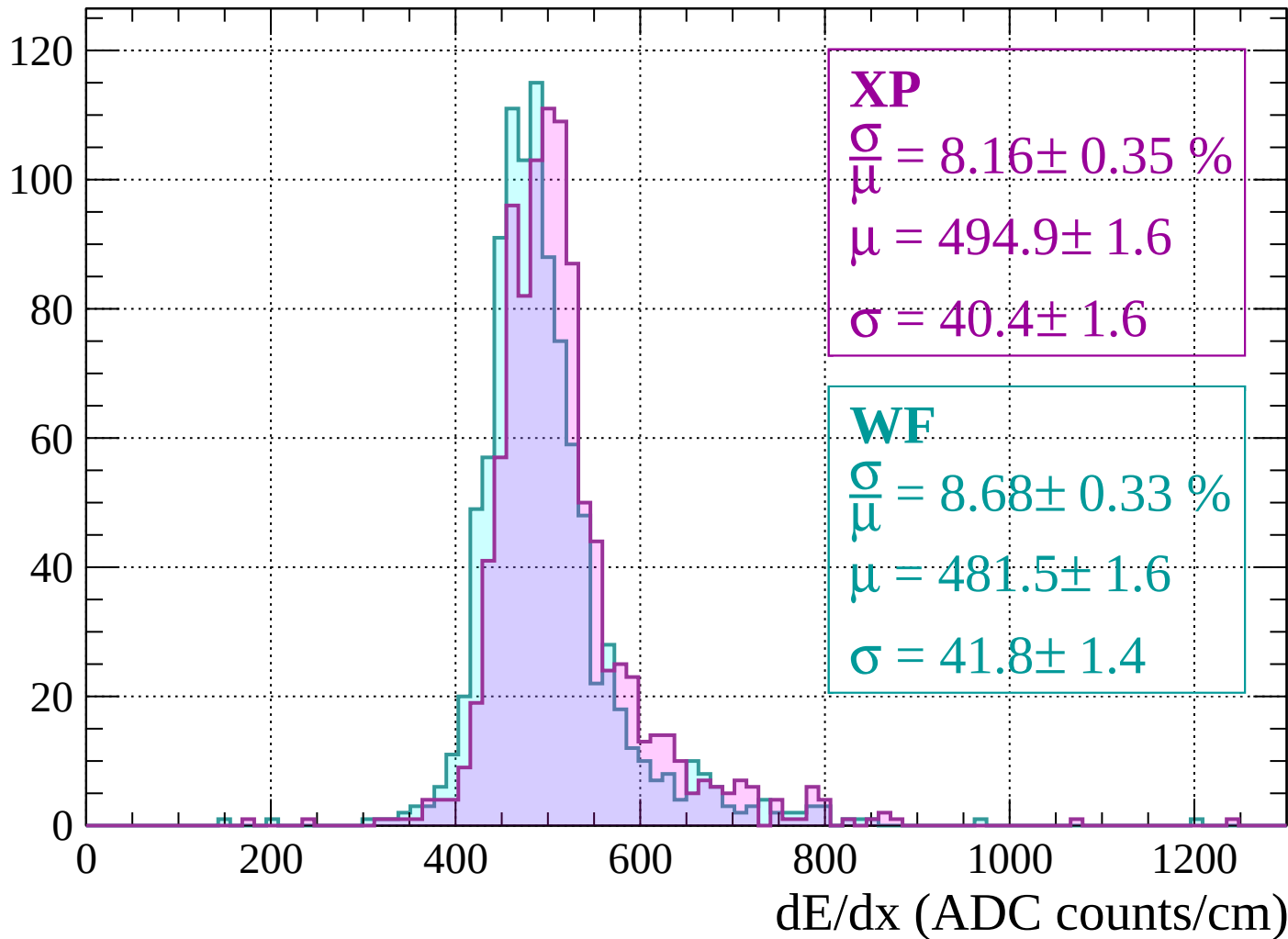
$$\mu = 481.6 \pm 1.7$$

$$\sigma = 41.6 \pm 1.8$$

dE/dx (ADC counts/cm)

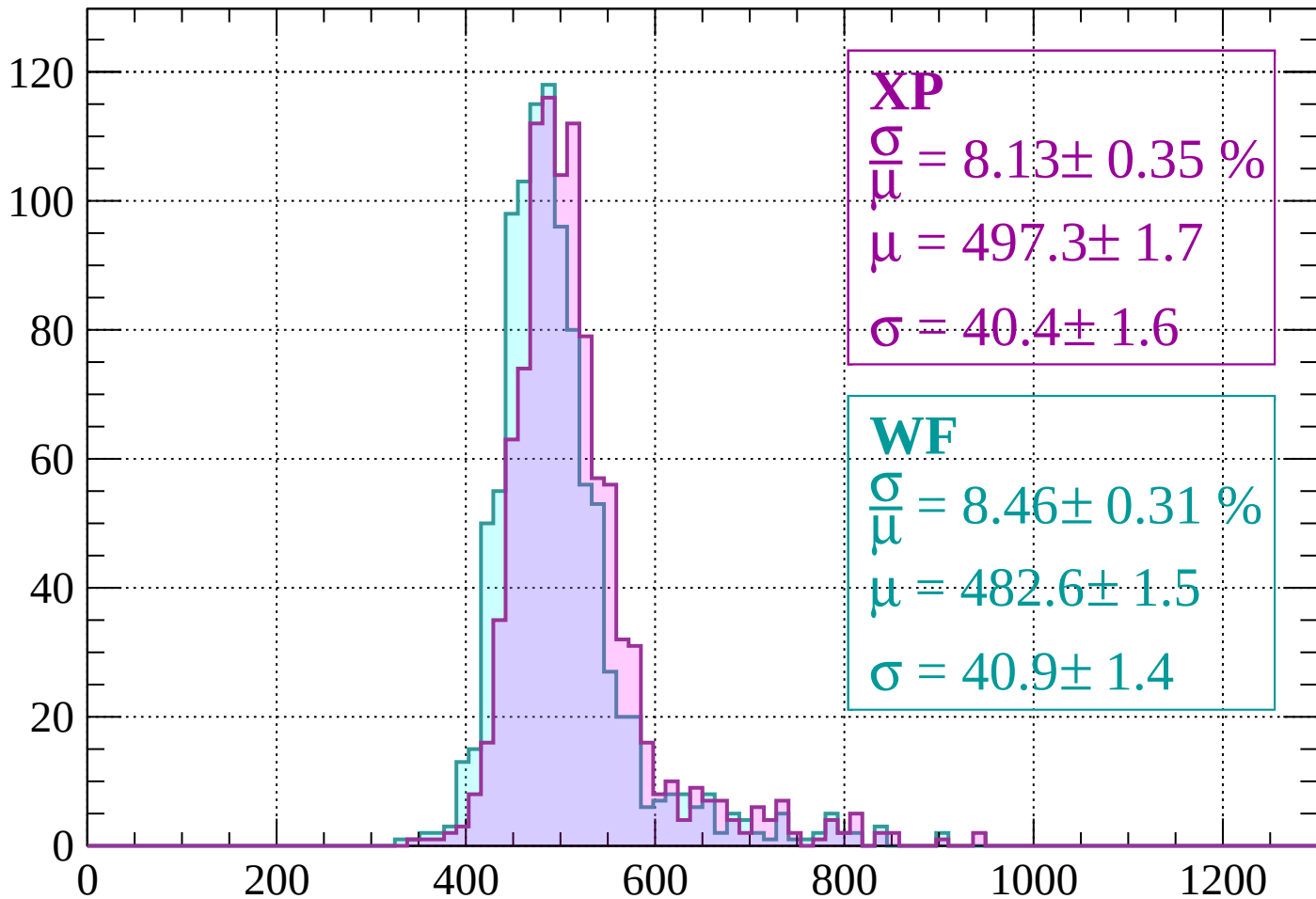
Energy loss | $-1800 < p < -1760$

Count



Energy loss | $-1760 < p < -1720$

Count



XP

$$\frac{\sigma}{\mu} = 8.13 \pm 0.35 \%$$

$$\mu = 497.3 \pm 1.7$$

$$\sigma = 40.4 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.46 \pm 0.31 \%$$

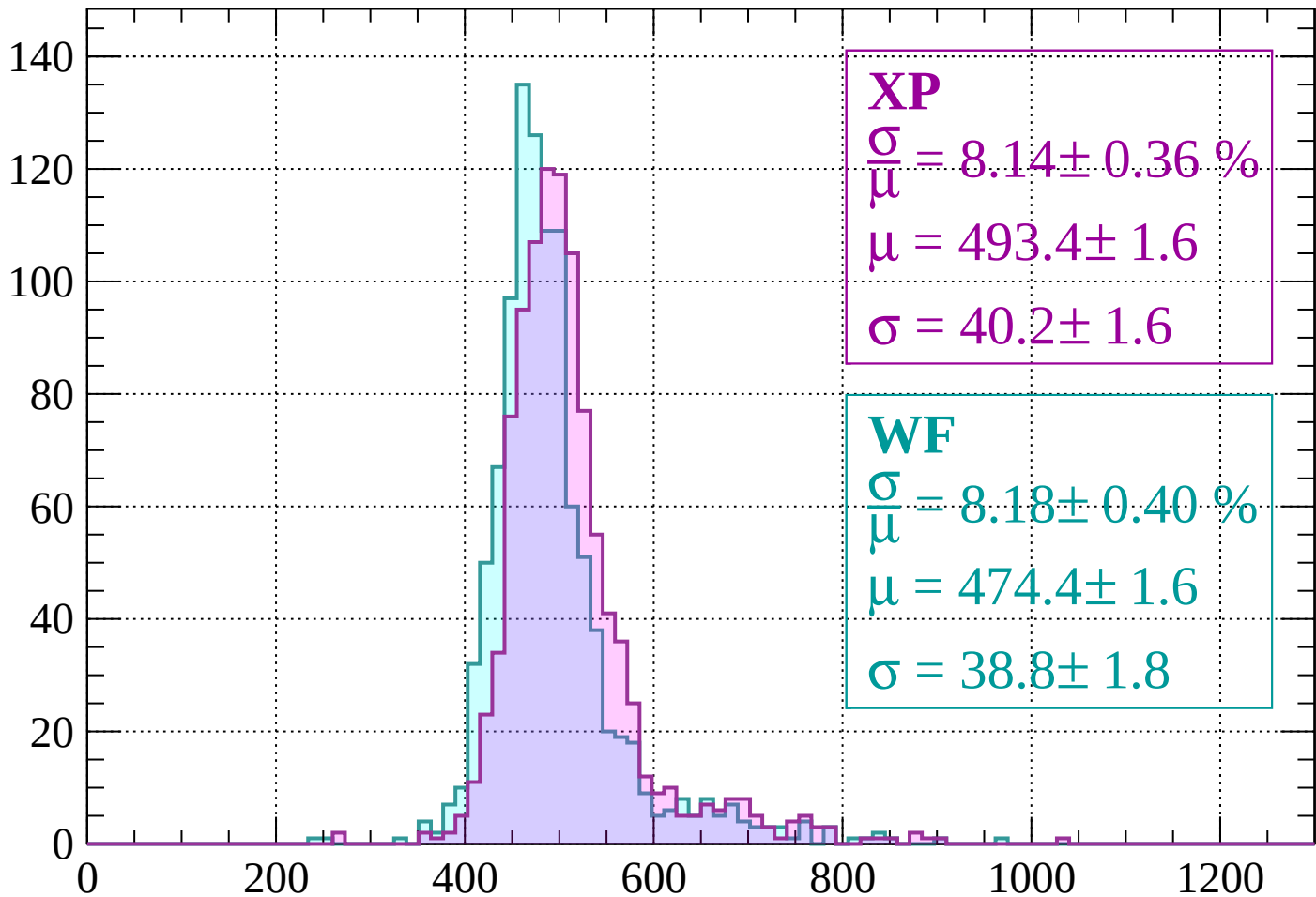
$$\mu = 482.6 \pm 1.5$$

$$\sigma = 40.9 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-1720 < p < -1680$

Count



XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.36 \%$$

$$\mu = 493.4 \pm 1.6$$

$$\sigma = 40.2 \pm 1.6$$

WF

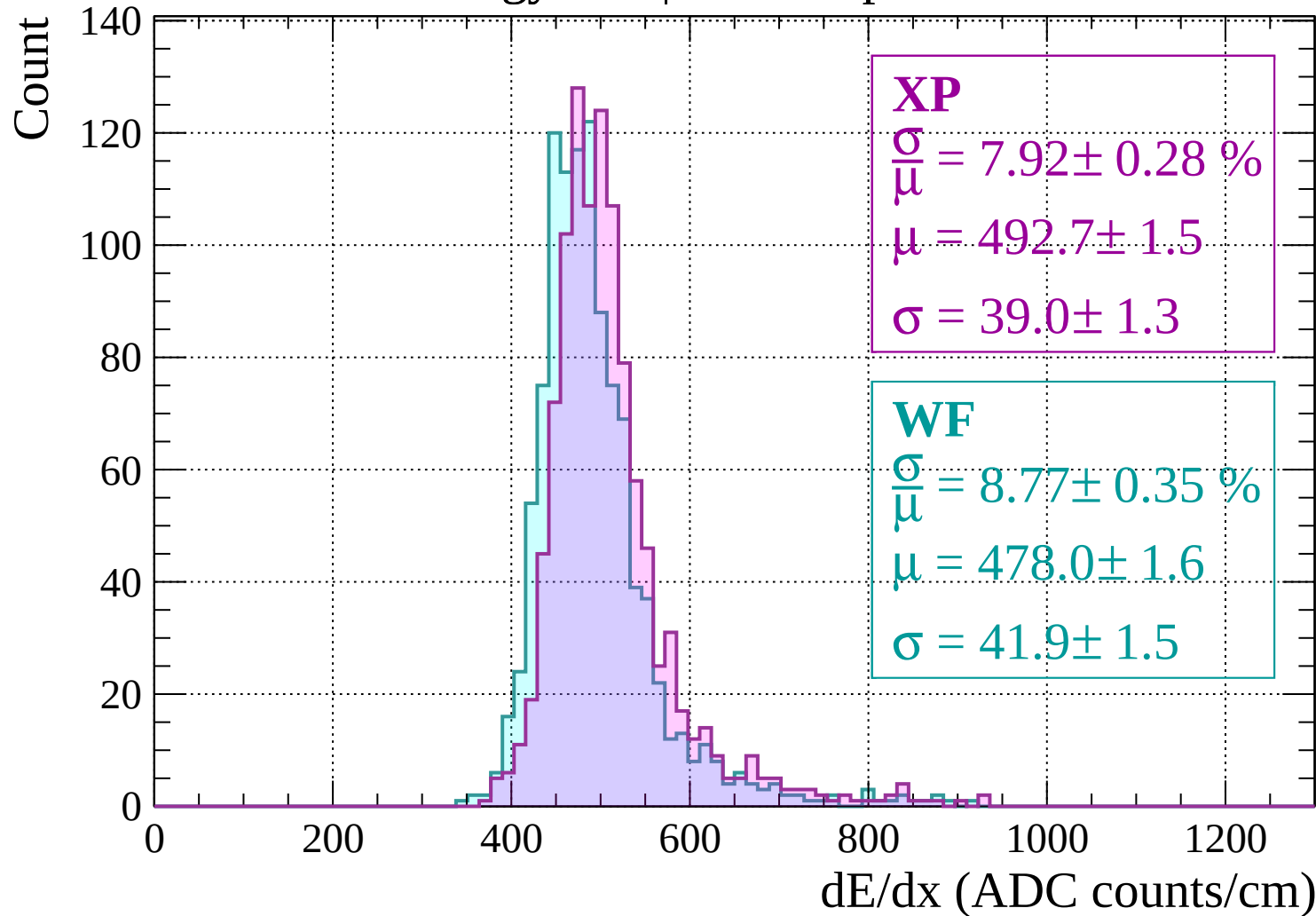
$$\frac{\sigma}{\mu} = 8.18 \pm 0.40 \%$$

$$\mu = 474.4 \pm 1.6$$

$$\sigma = 38.8 \pm 1.8$$

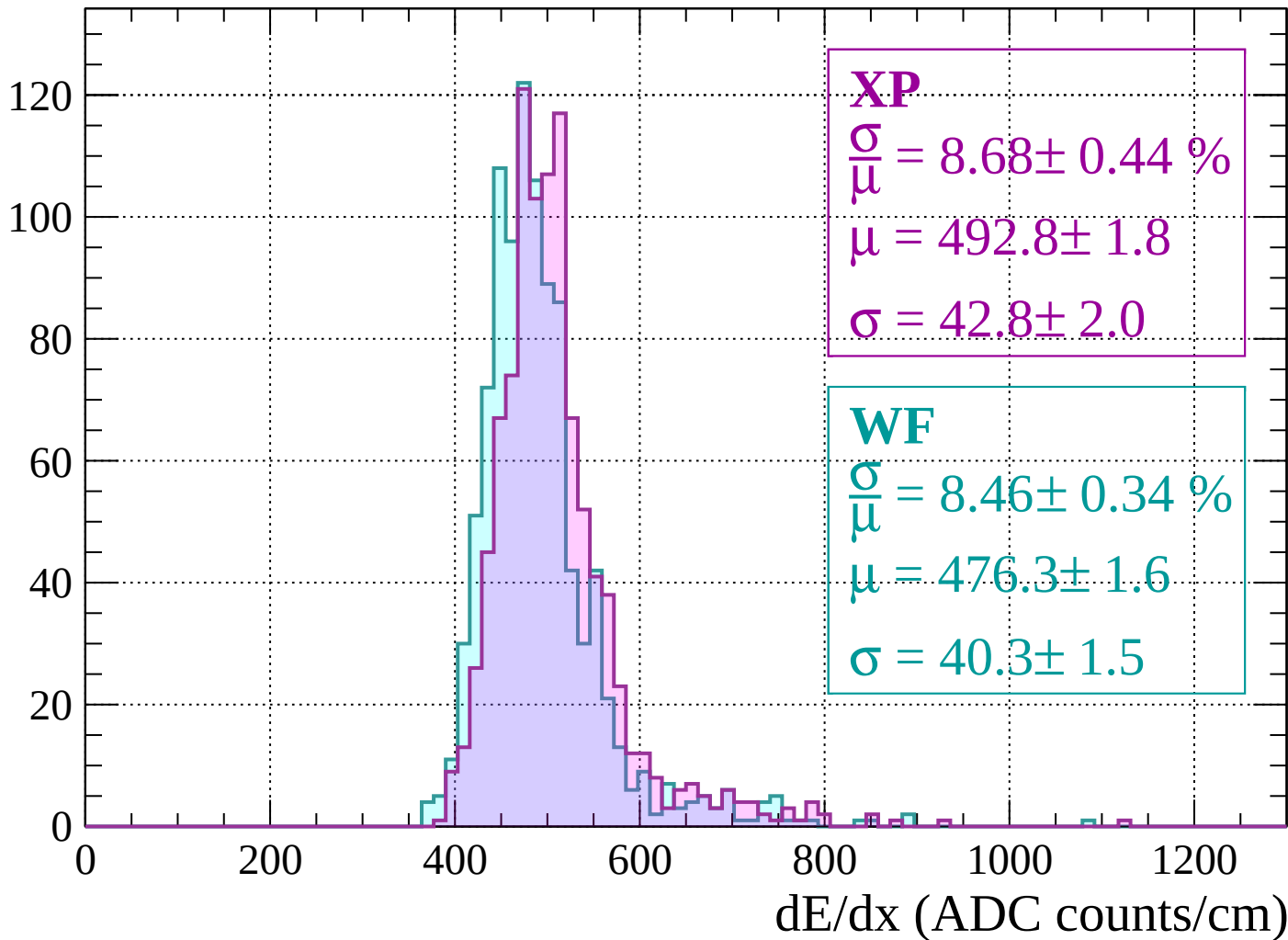
dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1640$



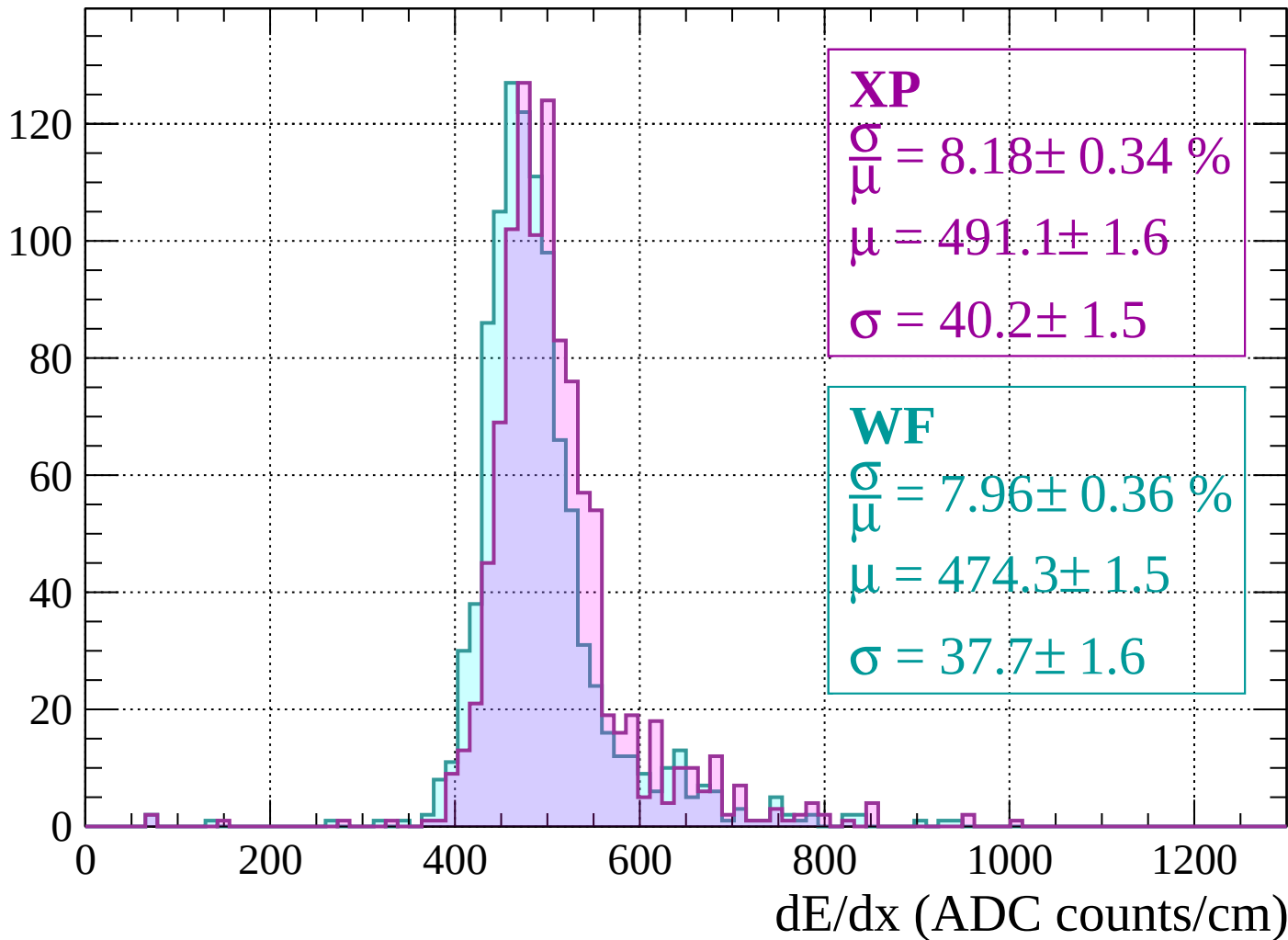
Energy loss | $-1640 < p < -1600$

Count



Energy loss | $-1600 < p < -1560$

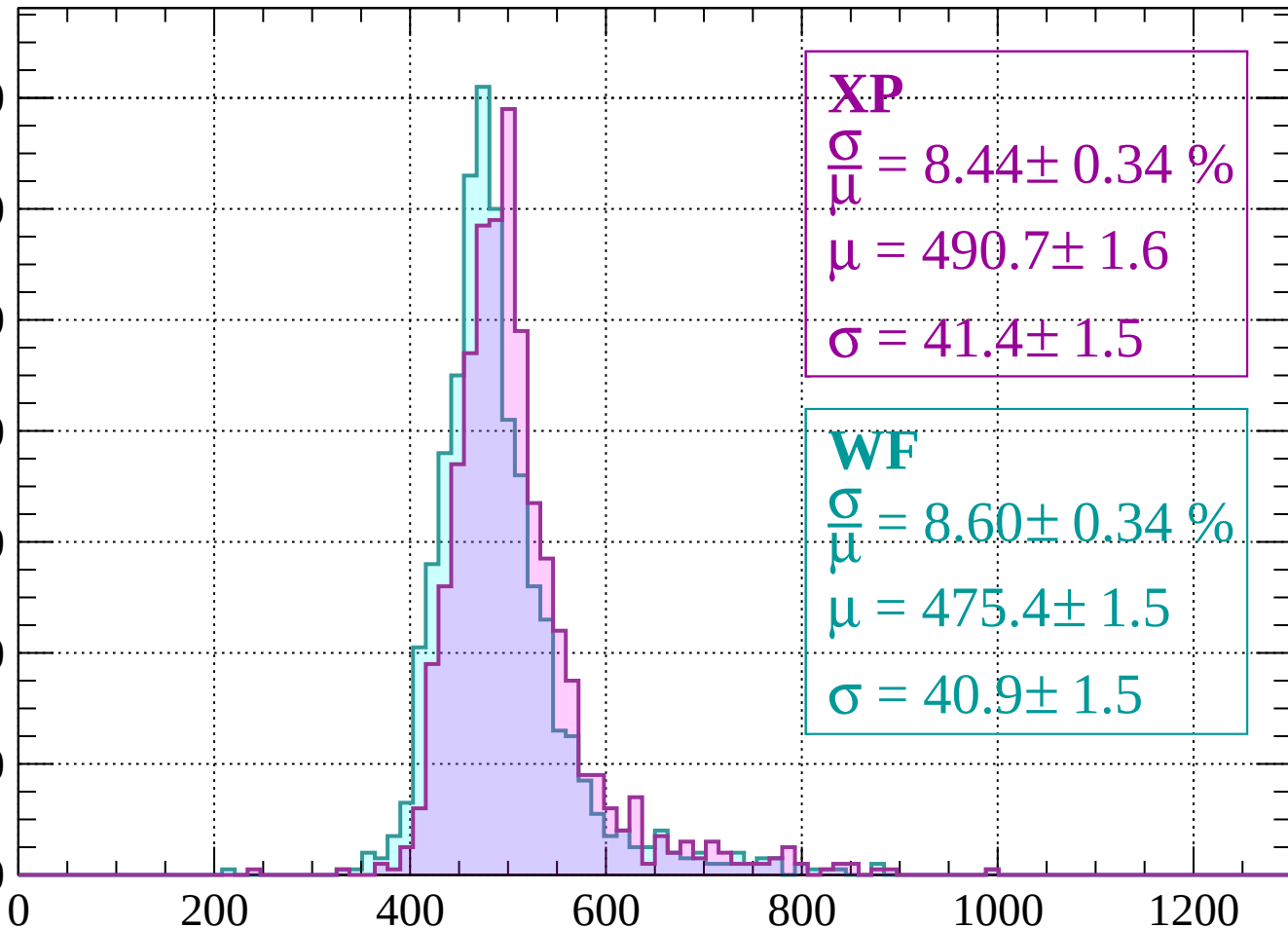
Count



Energy loss | $-1560 < p < -1520$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.44 \pm 0.34 \%$$

$$\mu = 490.7 \pm 1.6$$

$$\sigma = 41.4 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.60 \pm 0.34 \%$$

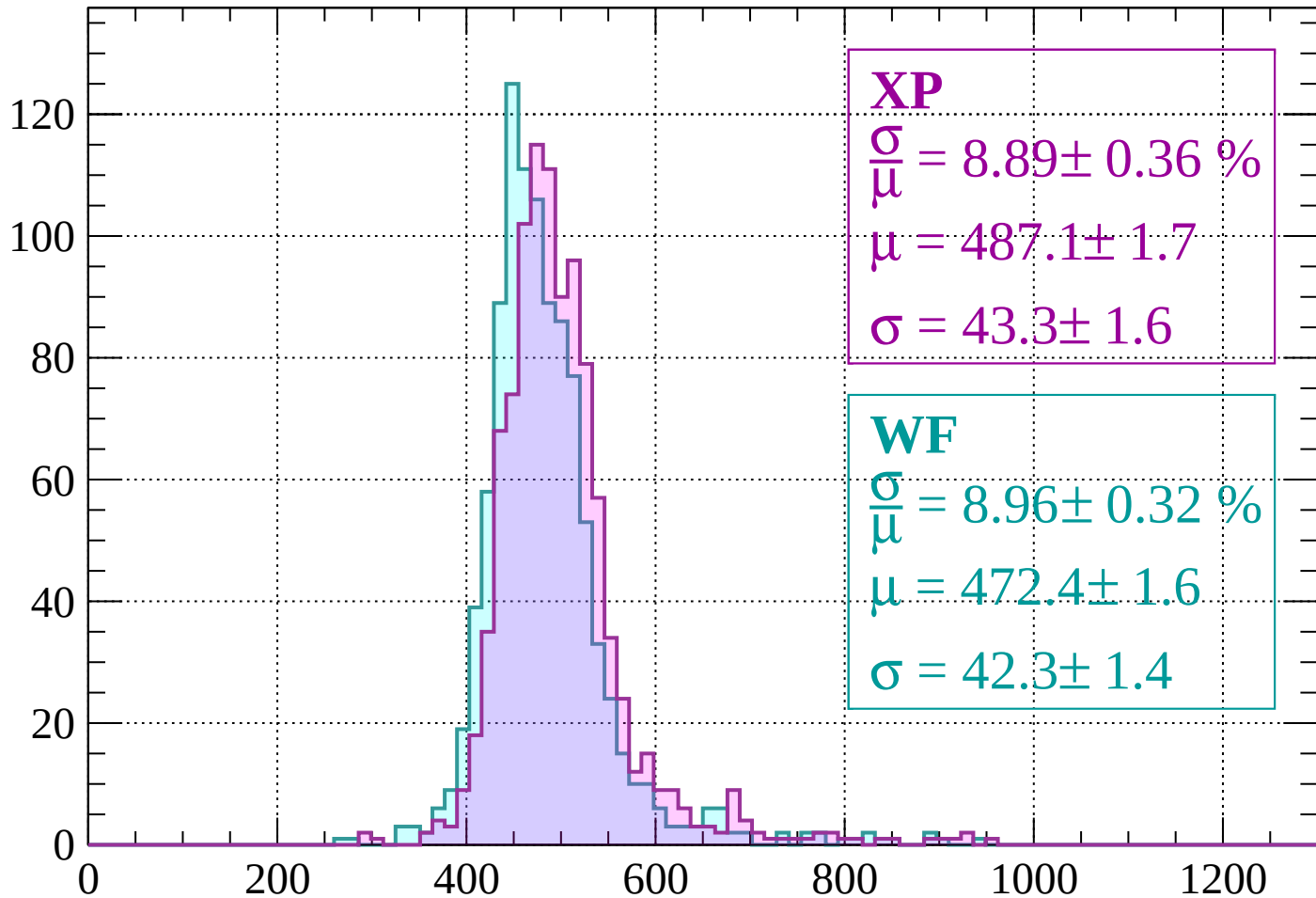
$$\mu = 475.4 \pm 1.5$$

$$\sigma = 40.9 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1520 < p < -1480$

Count



XP

$$\frac{\sigma}{\mu} = 8.89 \pm 0.36 \%$$

$$\mu = 487.1 \pm 1.7$$

$$\sigma = 43.3 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.96 \pm 0.32 \%$$

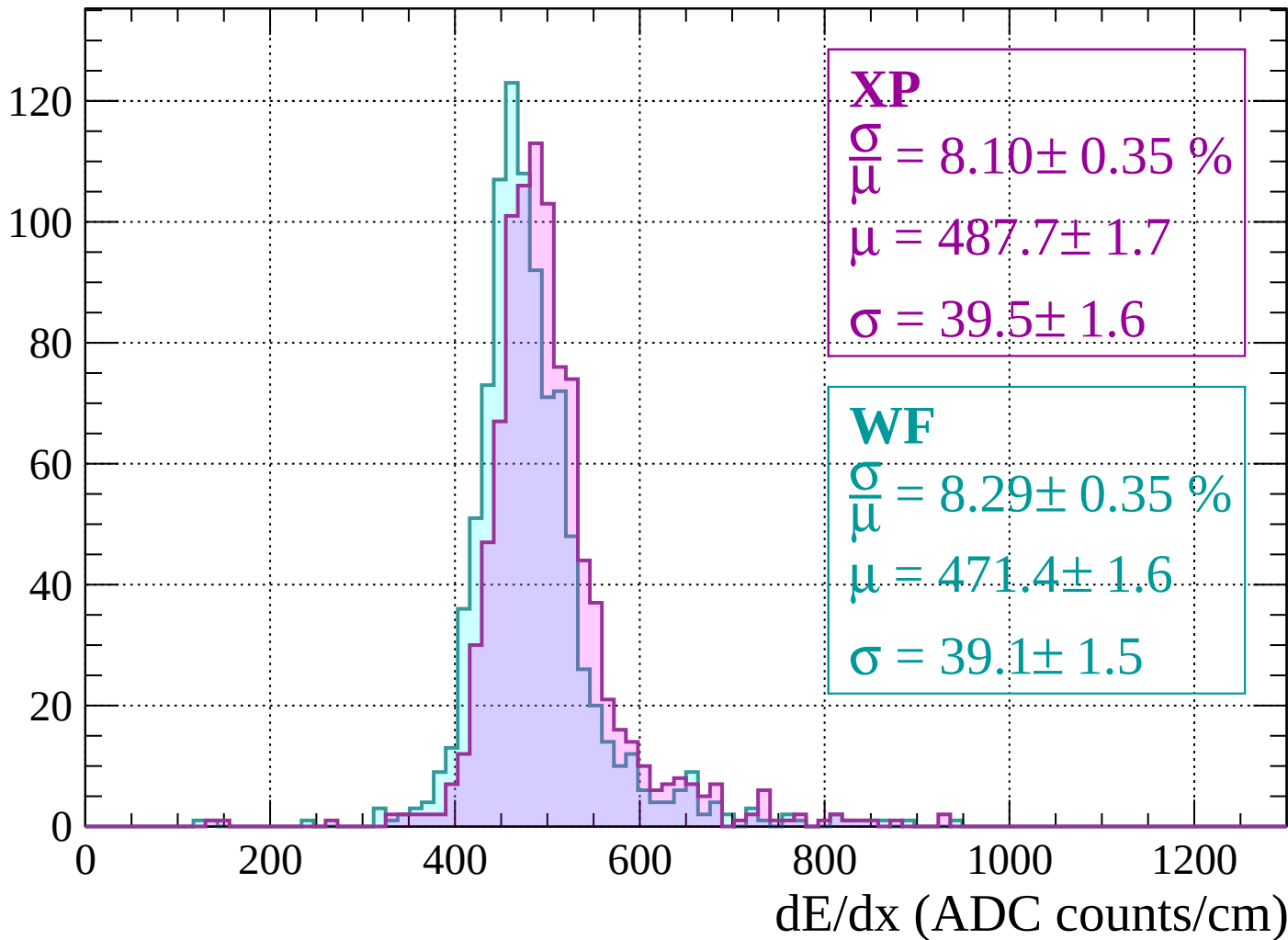
$$\mu = 472.4 \pm 1.6$$

$$\sigma = 42.3 \pm 1.4$$

dE/dx (ADC counts/cm)

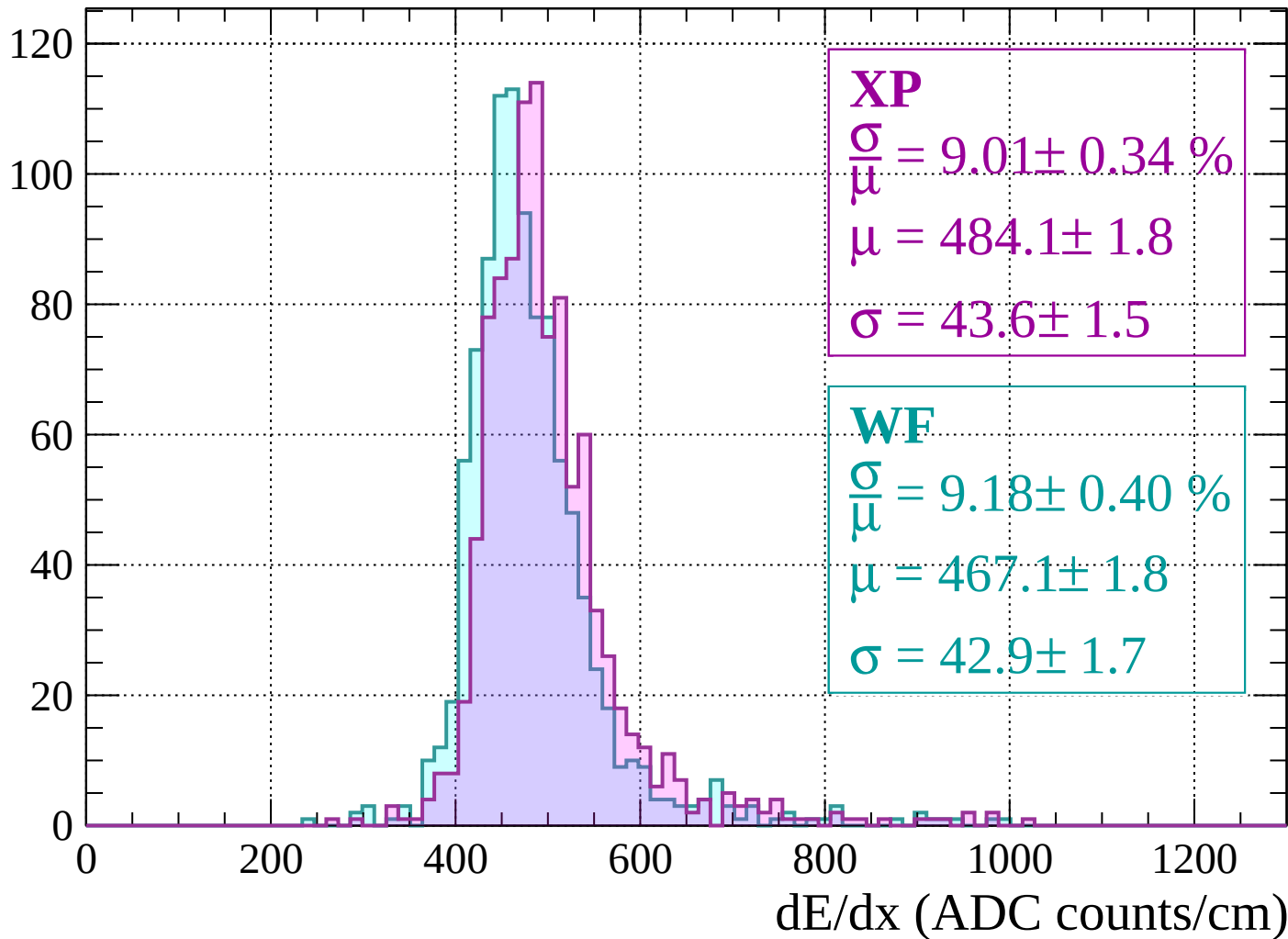
Energy loss | $-1480 < p < -1440$

Count



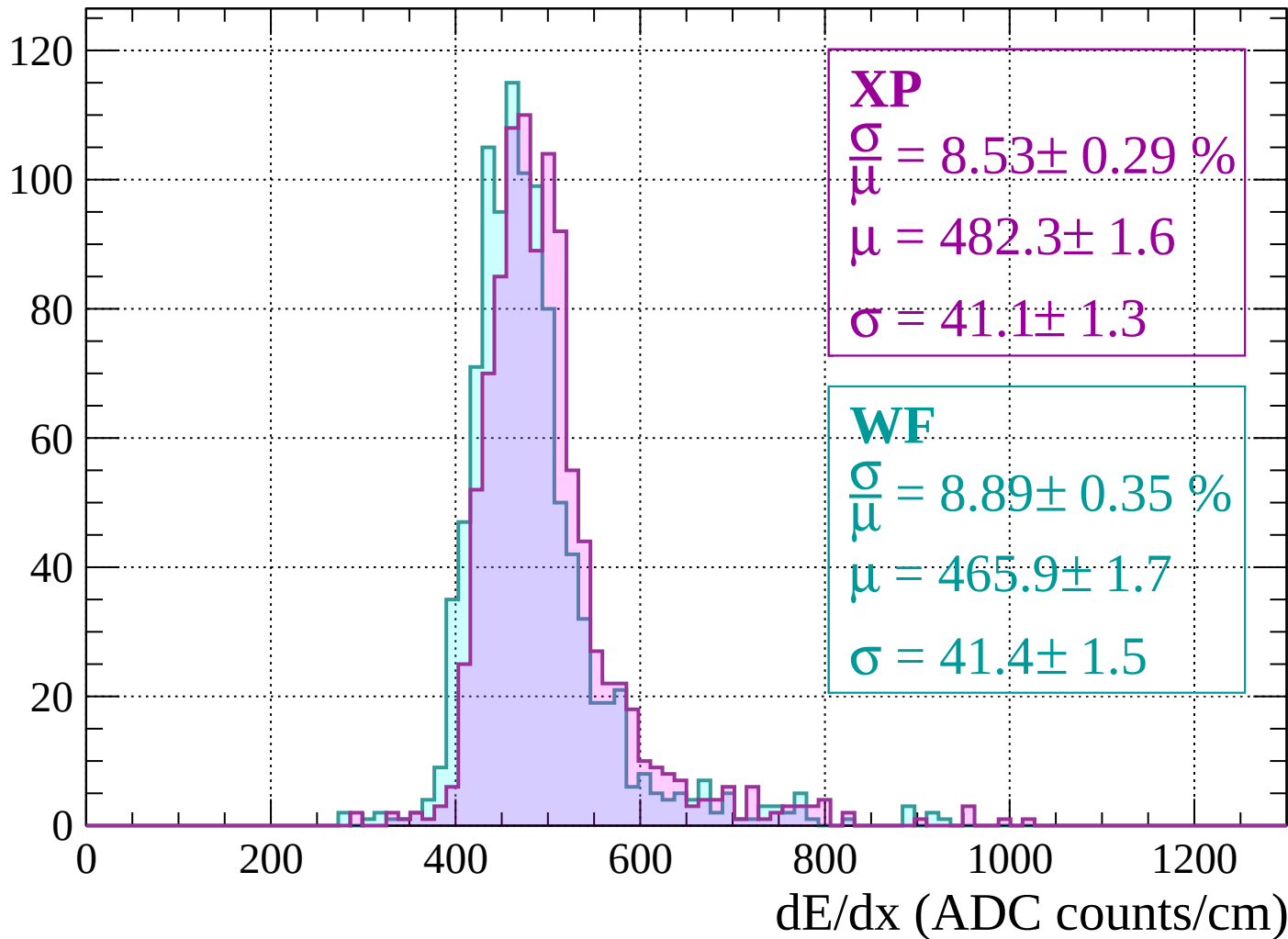
Energy loss | $-1440 < p < -1400$

Count



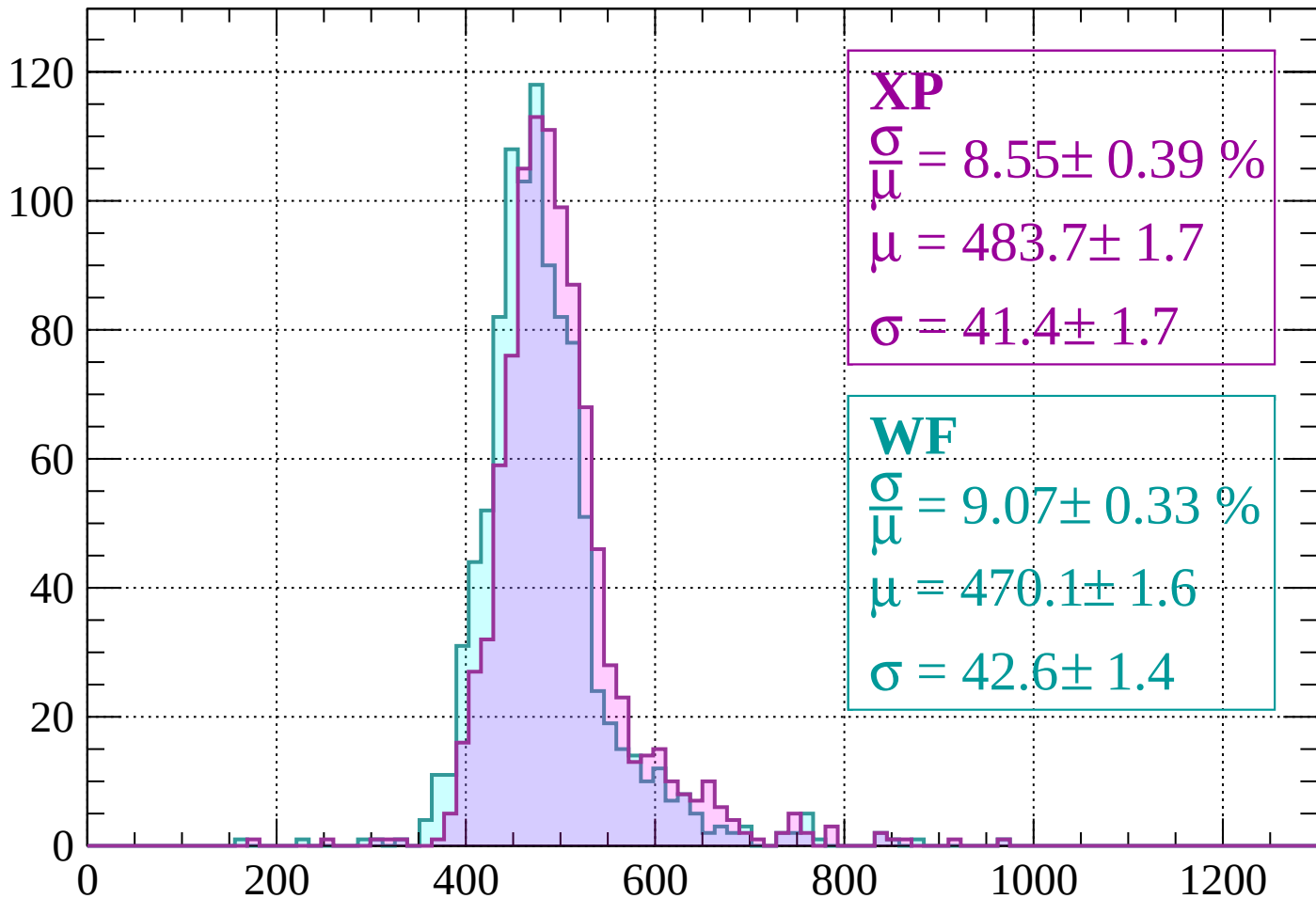
Energy loss | $-1400 < p < -1360$

Count



Energy loss | $-1360 < p < -1320$

Count



XP

$$\frac{\sigma}{\mu} = 8.55 \pm 0.39 \%$$

$$\mu = 483.7 \pm 1.7$$

$$\sigma = 41.4 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 9.07 \pm 0.33 \%$$

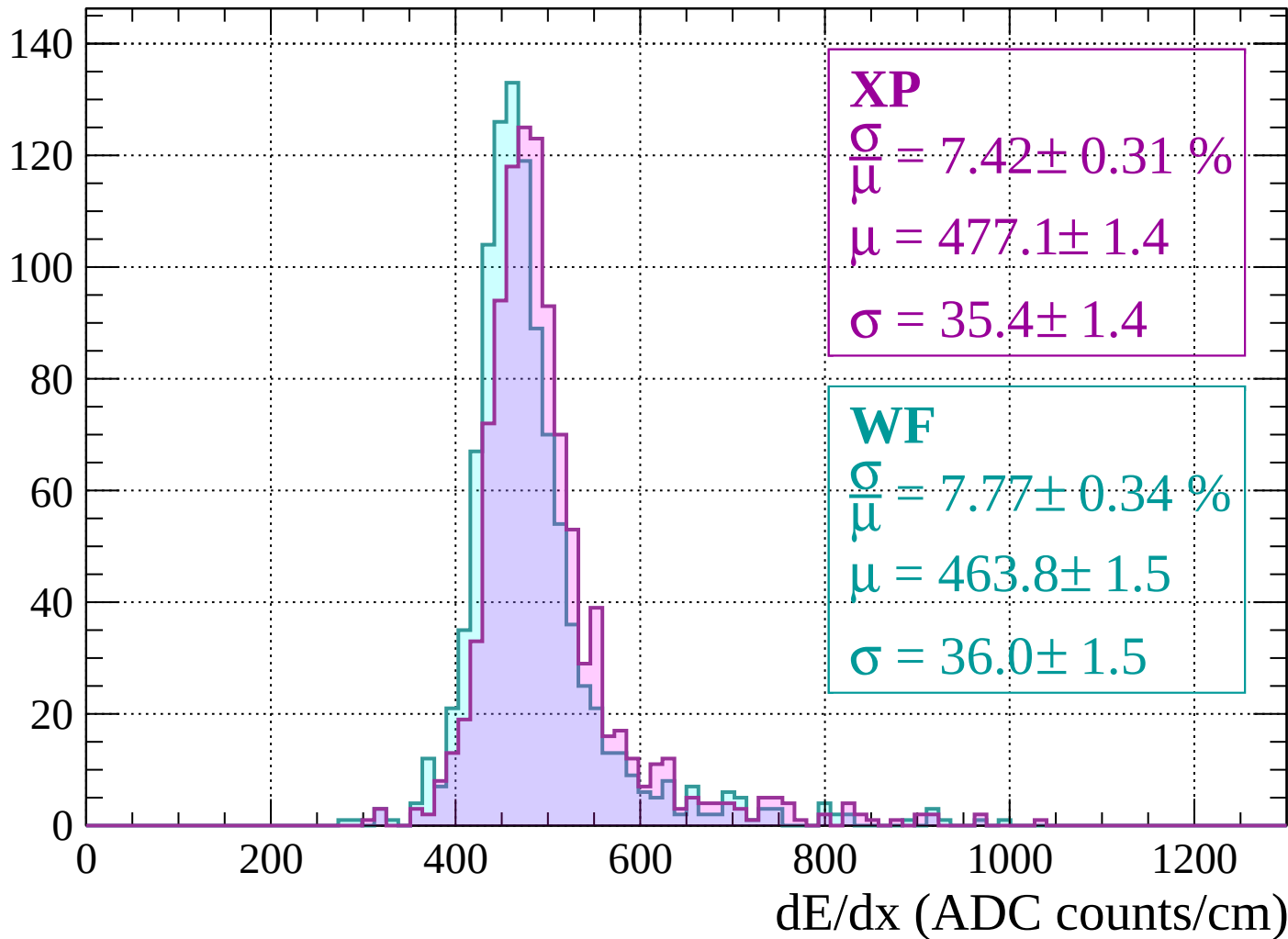
$$\mu = 470.1 \pm 1.6$$

$$\sigma = 42.6 \pm 1.4$$

dE/dx (ADC counts/cm)

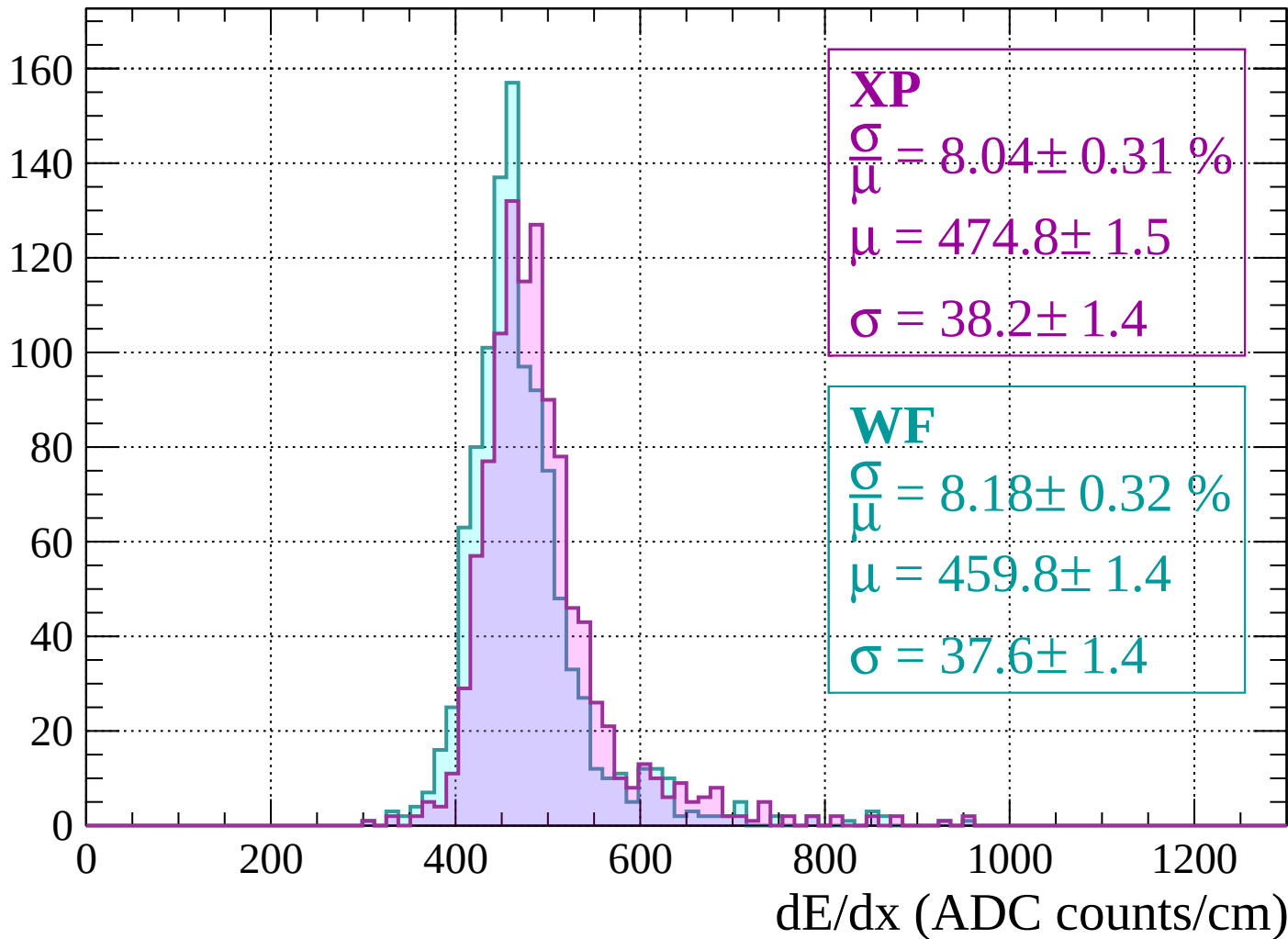
Energy loss | $-1320 < p < -1280$

Count



Energy loss | $-1280 < p < -1240$

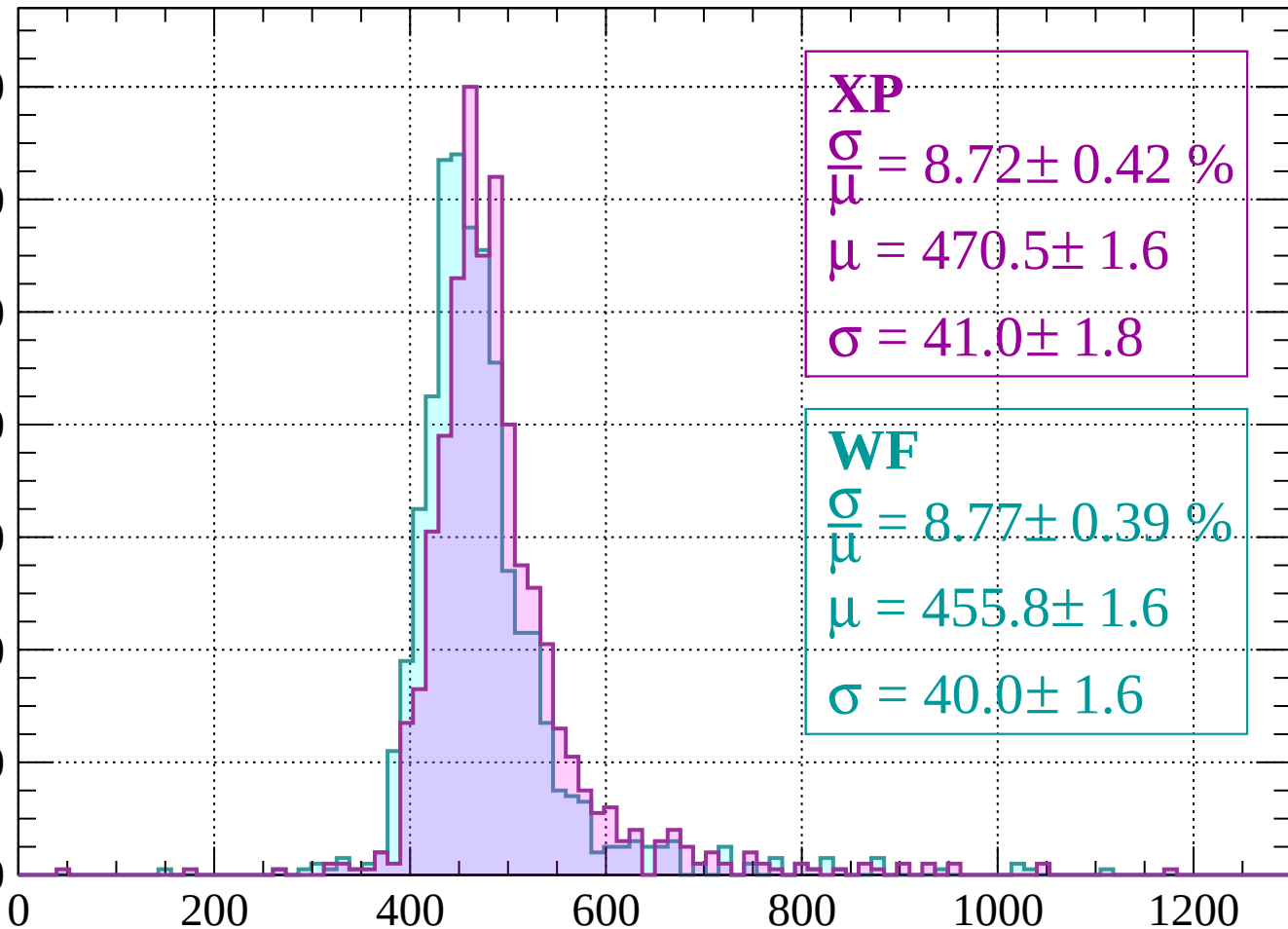
Count



Energy loss | -1240 < p < -1200

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.72 \pm 0.42 \%$$

$$\mu = 470.5 \pm 1.6$$

$$\sigma = 41.0 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 8.77 \pm 0.39 \%$$

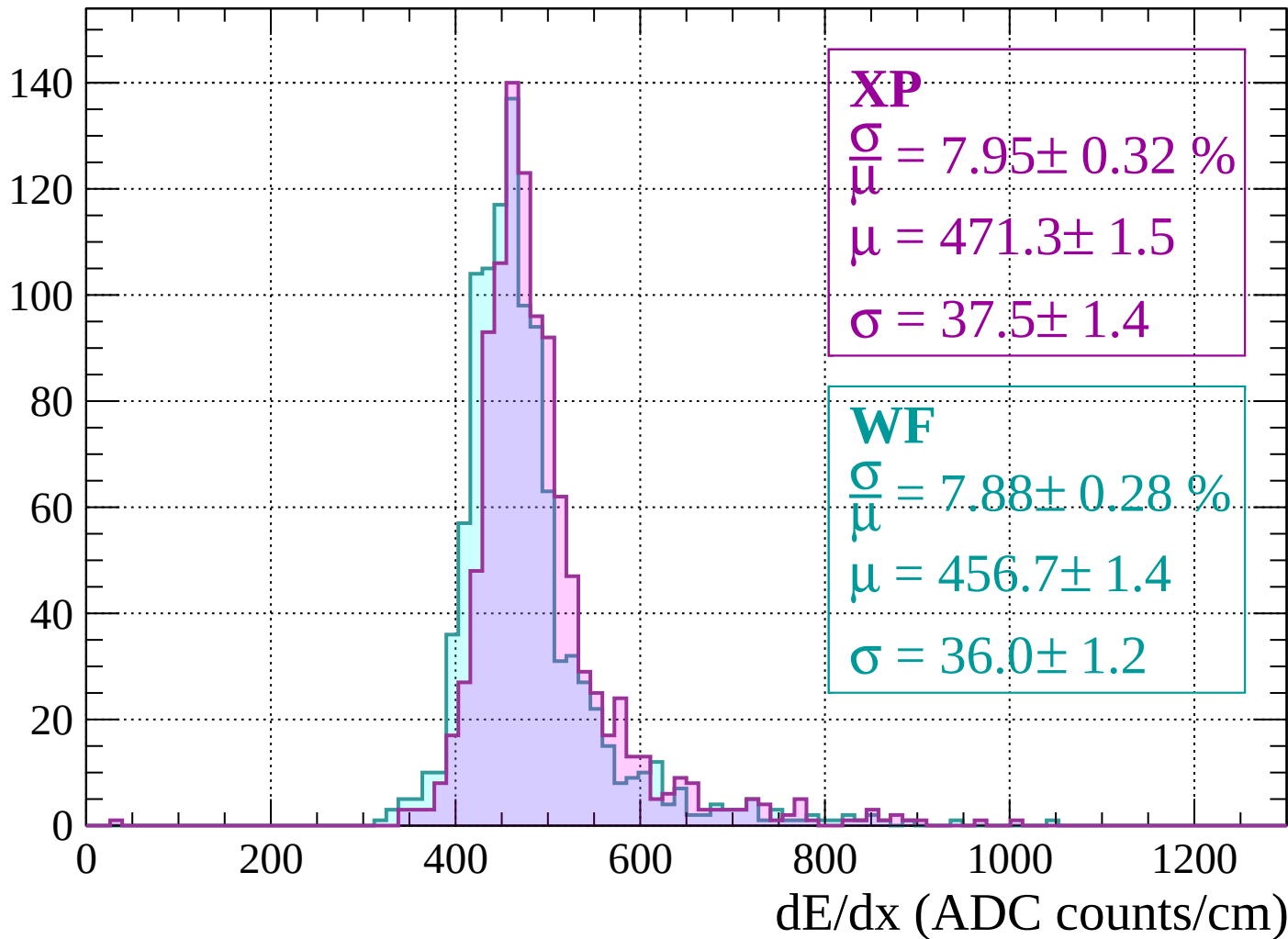
$$\mu = 455.8 \pm 1.6$$

$$\sigma = 40.0 \pm 1.6$$

dE/dx (ADC counts/cm)

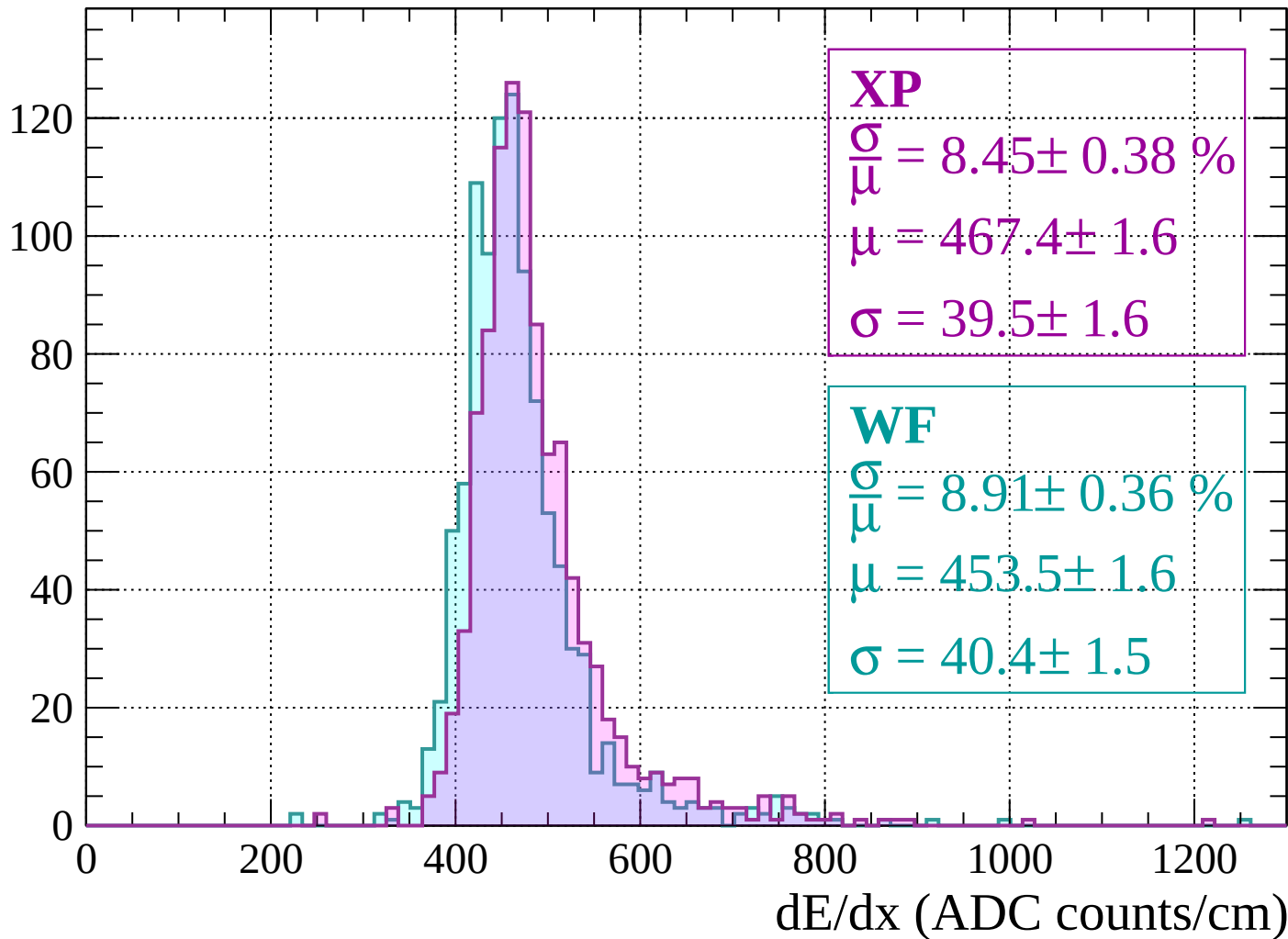
Energy loss | -1200 < p < -1160

Count



Energy loss | $-1160 < p < -1120$

Count



Energy loss | -1120 < p < -1080

Count

120

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 8.62 \pm 0.35 \%$$

$$\mu = 463.1 \pm 1.6$$

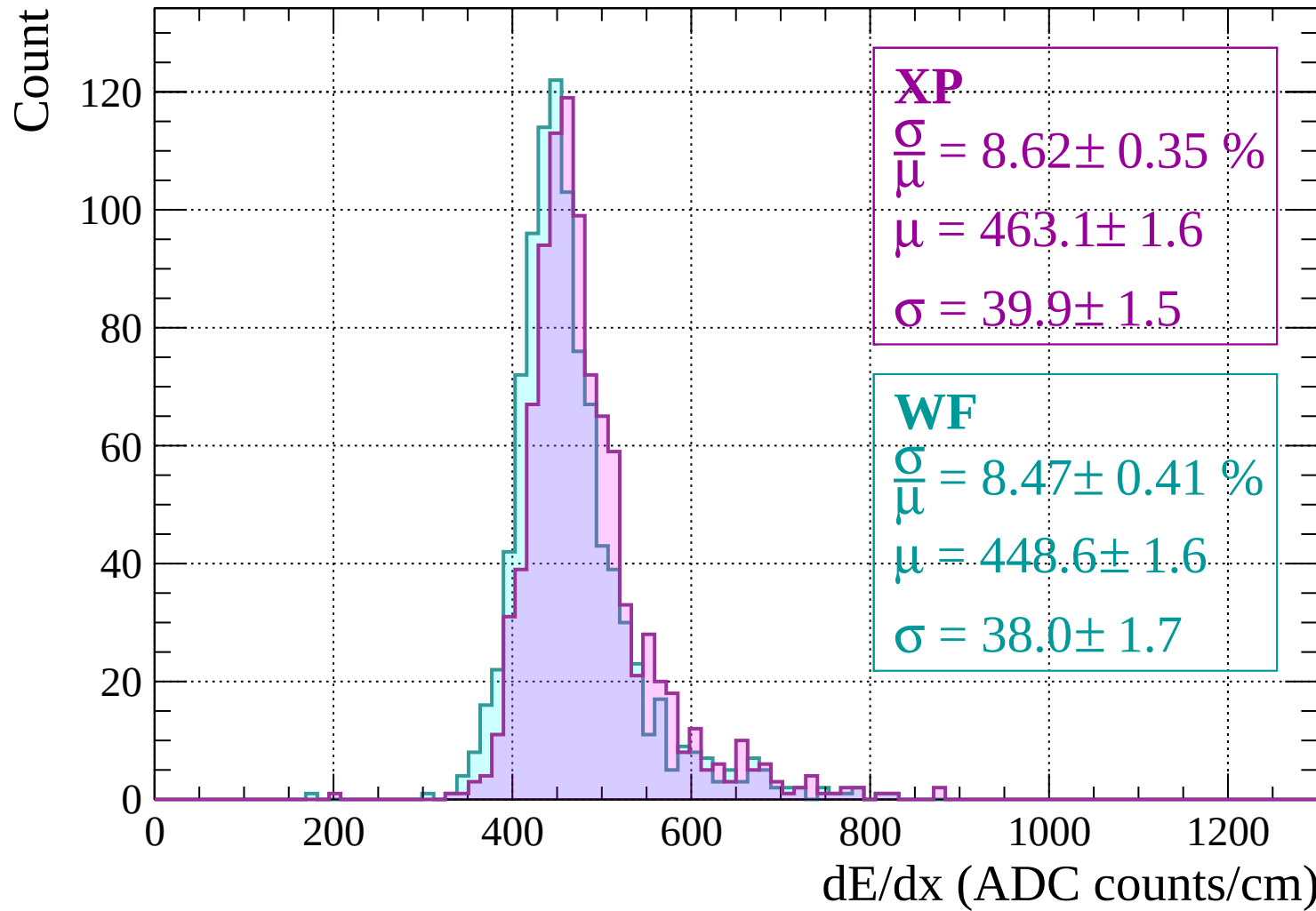
$$\sigma = 39.9 \pm 1.5$$

WF

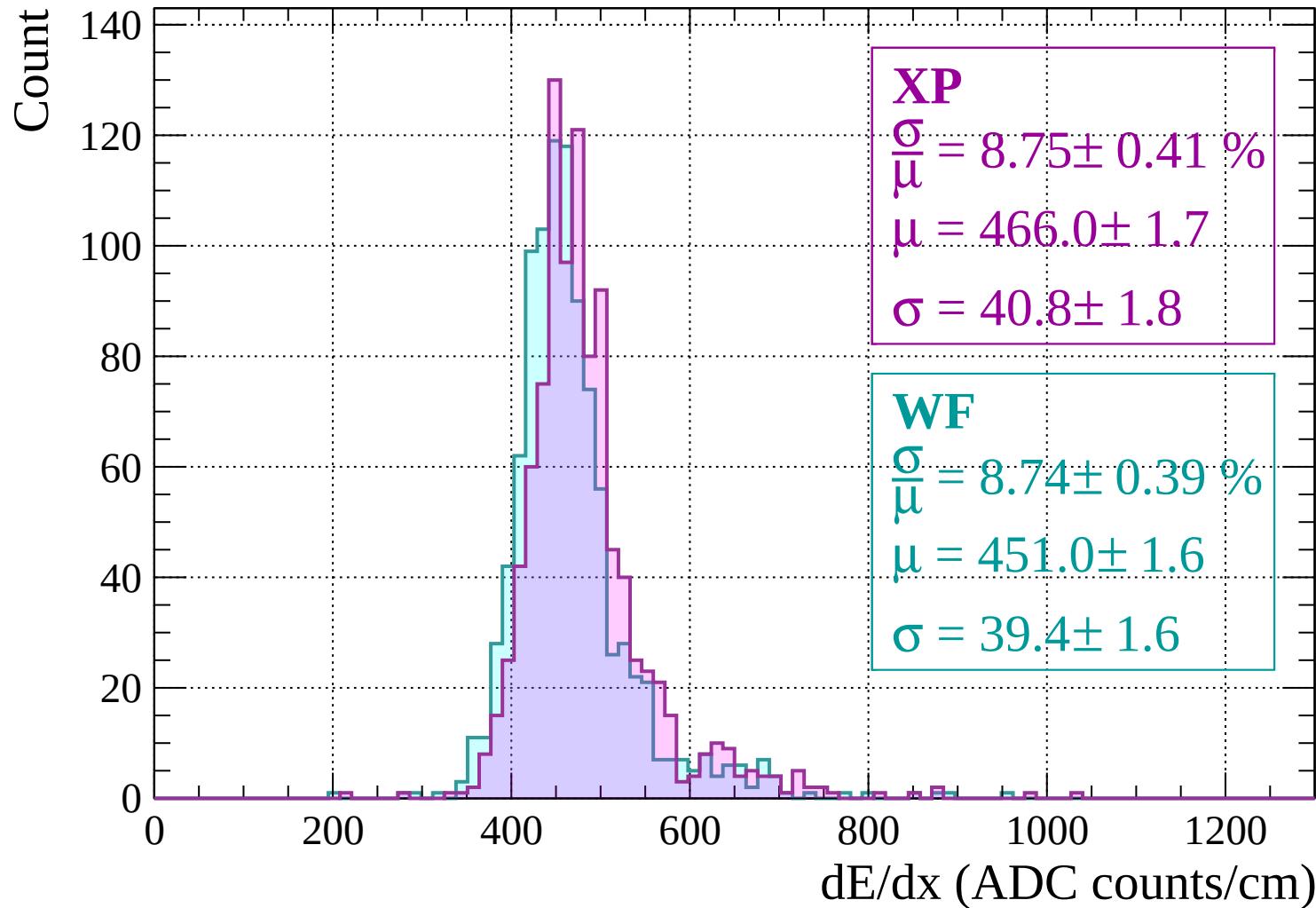
$$\frac{\sigma}{\mu} = 8.47 \pm 0.41 \%$$

$$\mu = 448.6 \pm 1.6$$

$$\sigma = 38.0 \pm 1.7$$



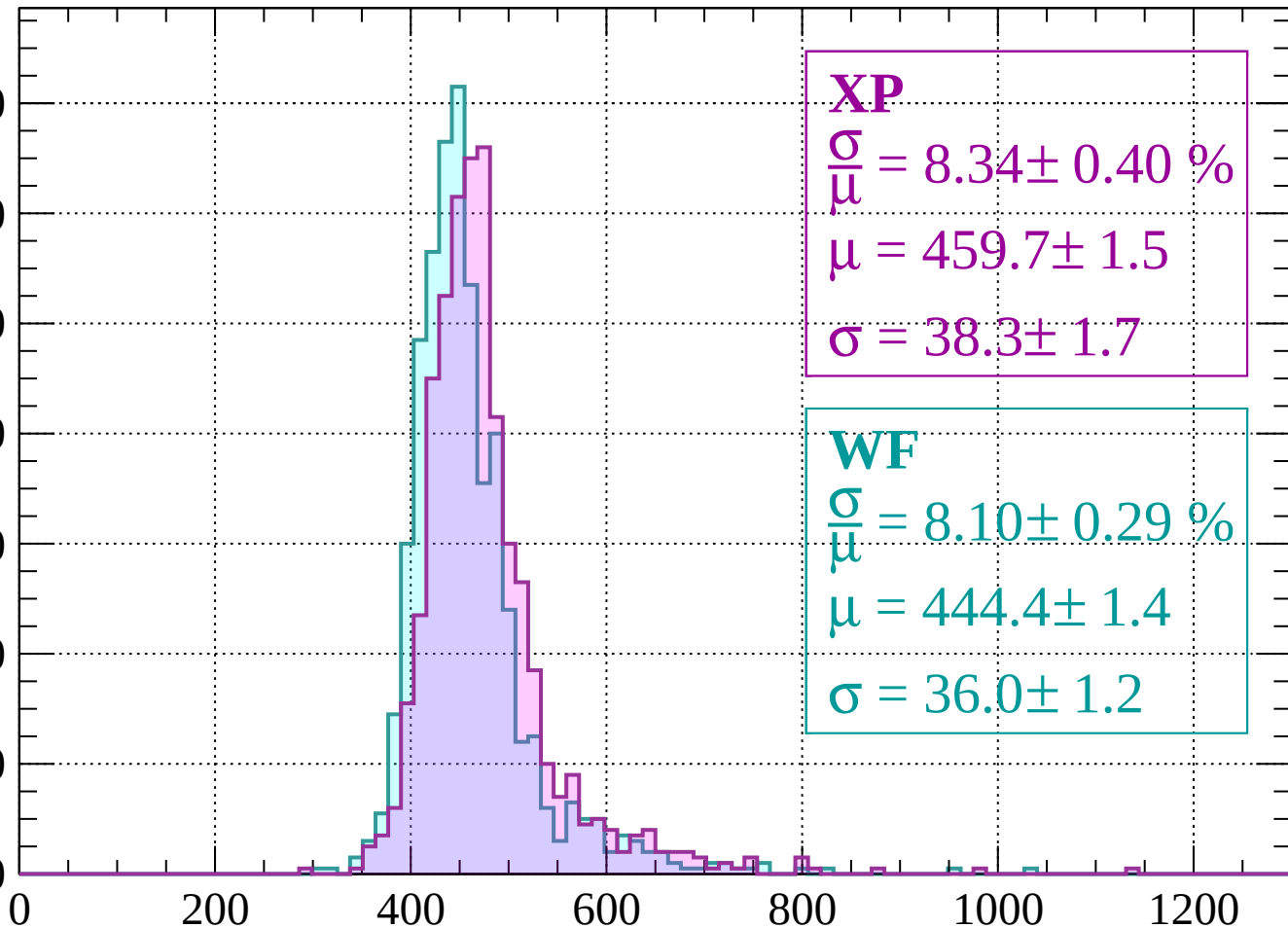
Energy loss | -1080 < p < -1040



Energy loss | $-1040 < p < -1000$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.34 \pm 0.40 \%$$

$$\mu = 459.7 \pm 1.5$$

$$\sigma = 38.3 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.10 \pm 0.29 \%$$

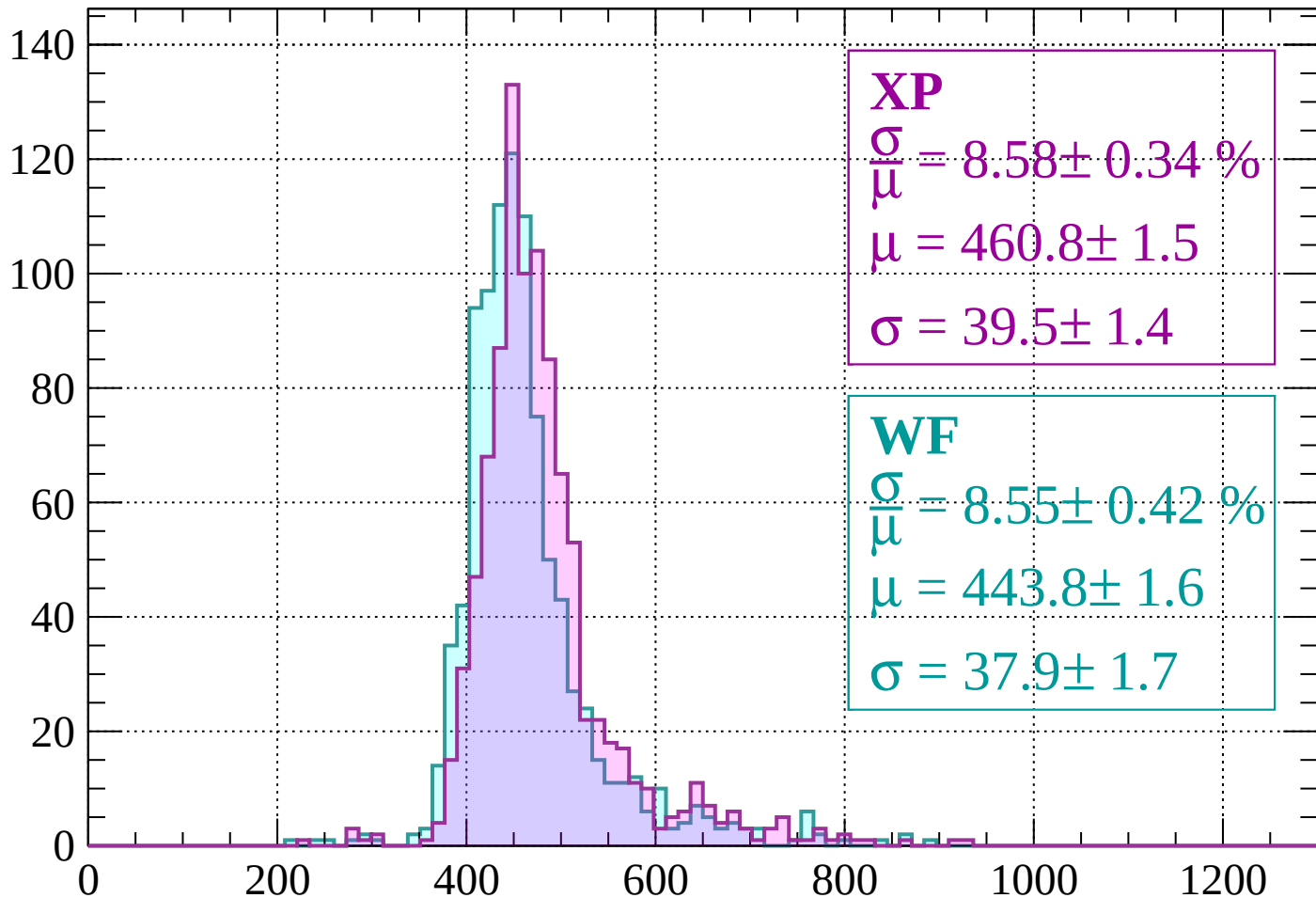
$$\mu = 444.4 \pm 1.4$$

$$\sigma = 36.0 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 8.58 \pm 0.34 \%$$

$$\mu = 460.8 \pm 1.5$$

$$\sigma = 39.5 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.55 \pm 0.42 \%$$

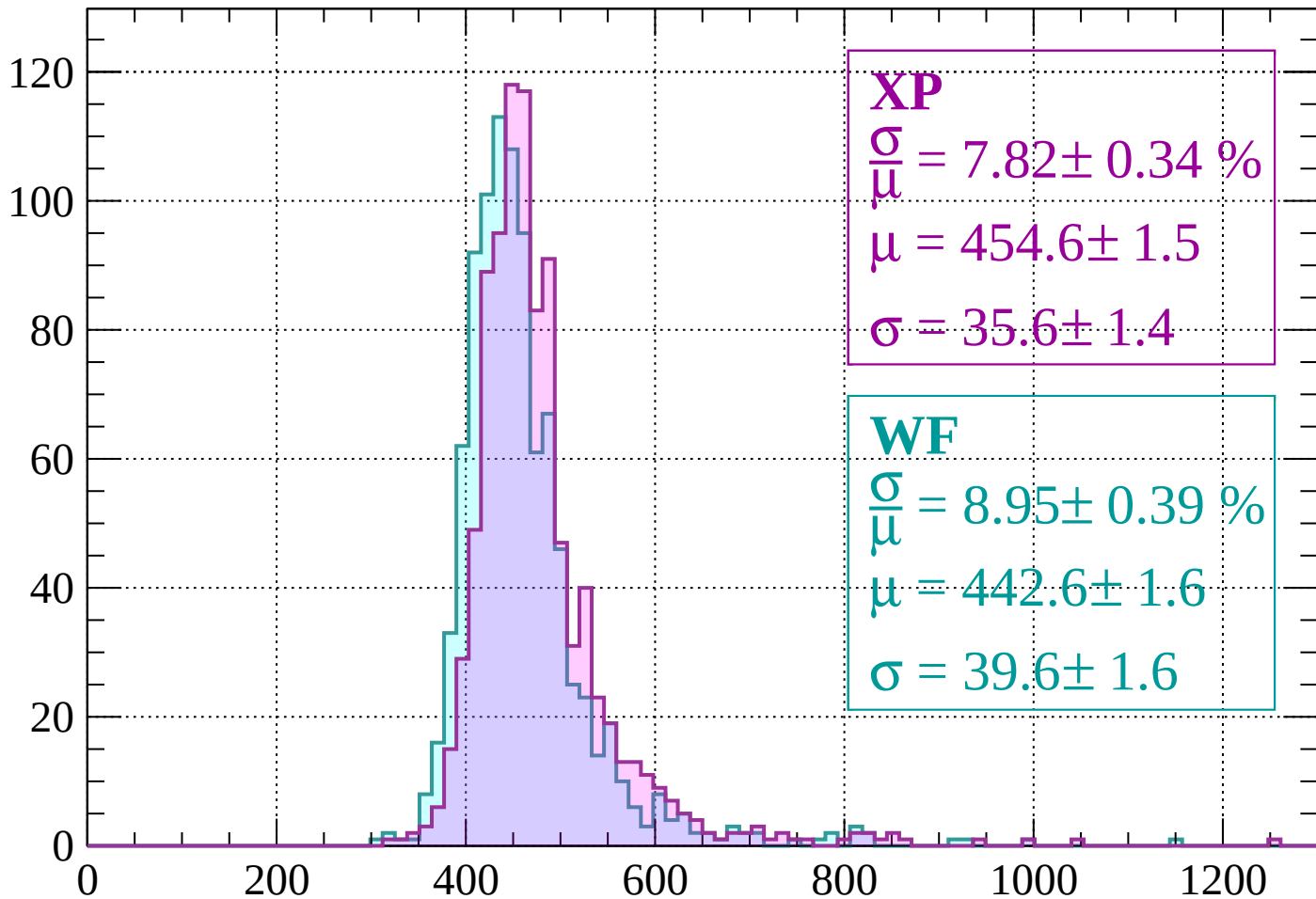
$$\mu = 443.8 \pm 1.6$$

$$\sigma = 37.9 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 7.82 \pm 0.34 \%$$

$$\mu = 454.6 \pm 1.5$$

$$\sigma = 35.6 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.95 \pm 0.39 \%$$

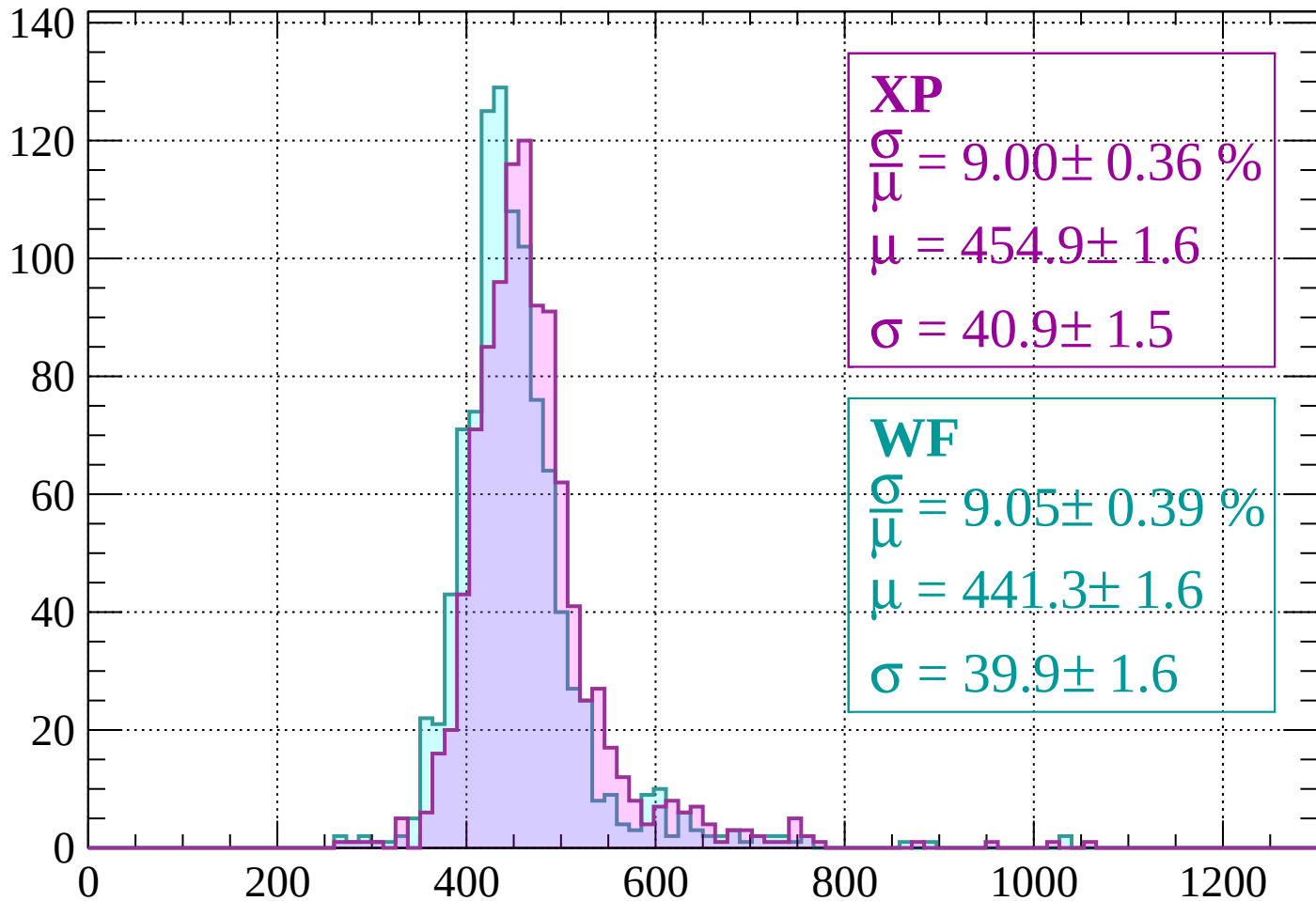
$$\mu = 442.6 \pm 1.6$$

$$\sigma = 39.6 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 9.00 \pm 0.36 \%$$

$$\mu = 454.9 \pm 1.6$$

$$\sigma = 40.9 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.05 \pm 0.39 \%$$

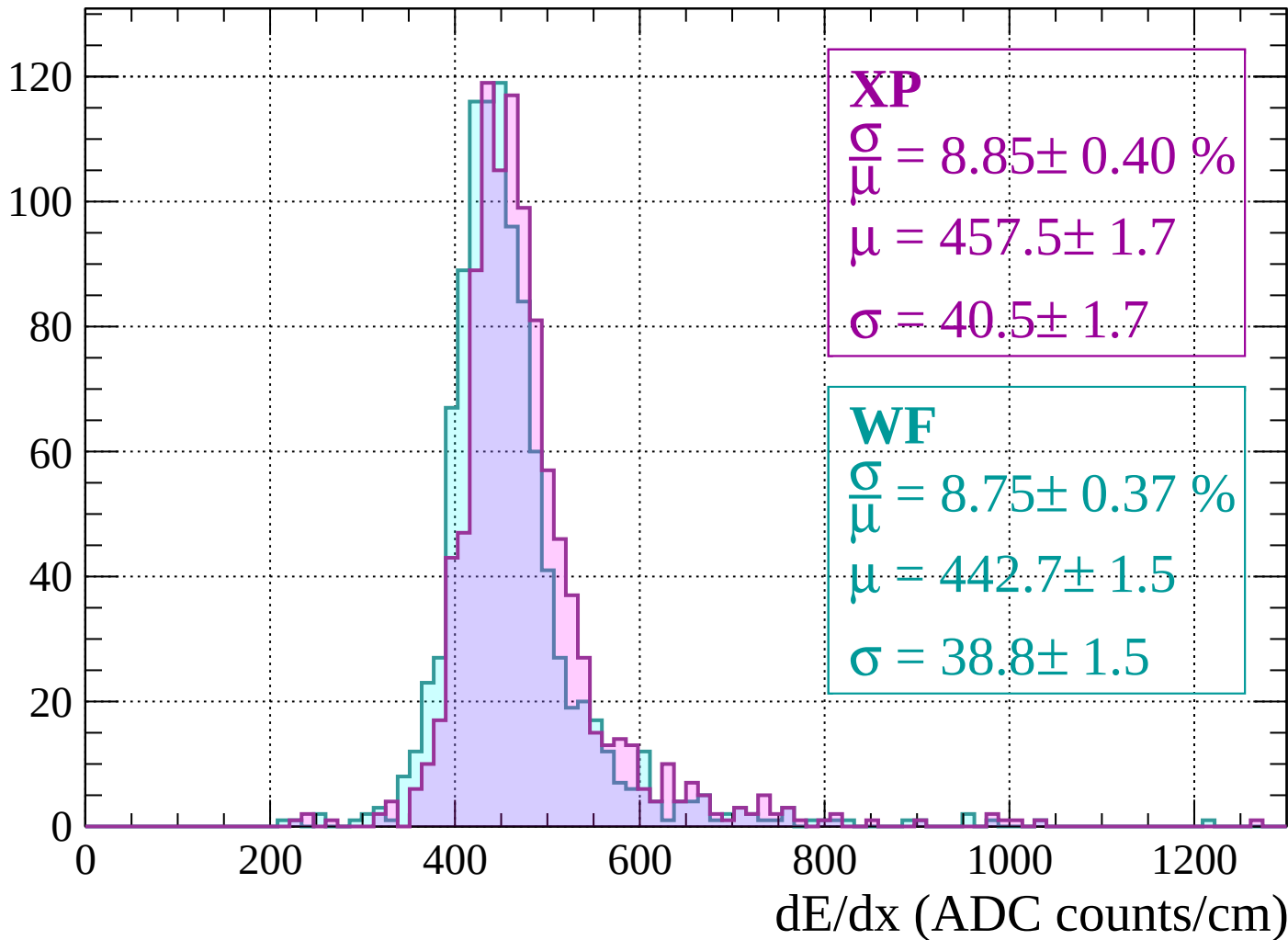
$$\mu = 441.3 \pm 1.6$$

$$\sigma = 39.9 \pm 1.6$$

dE/dx (ADC counts/cm)

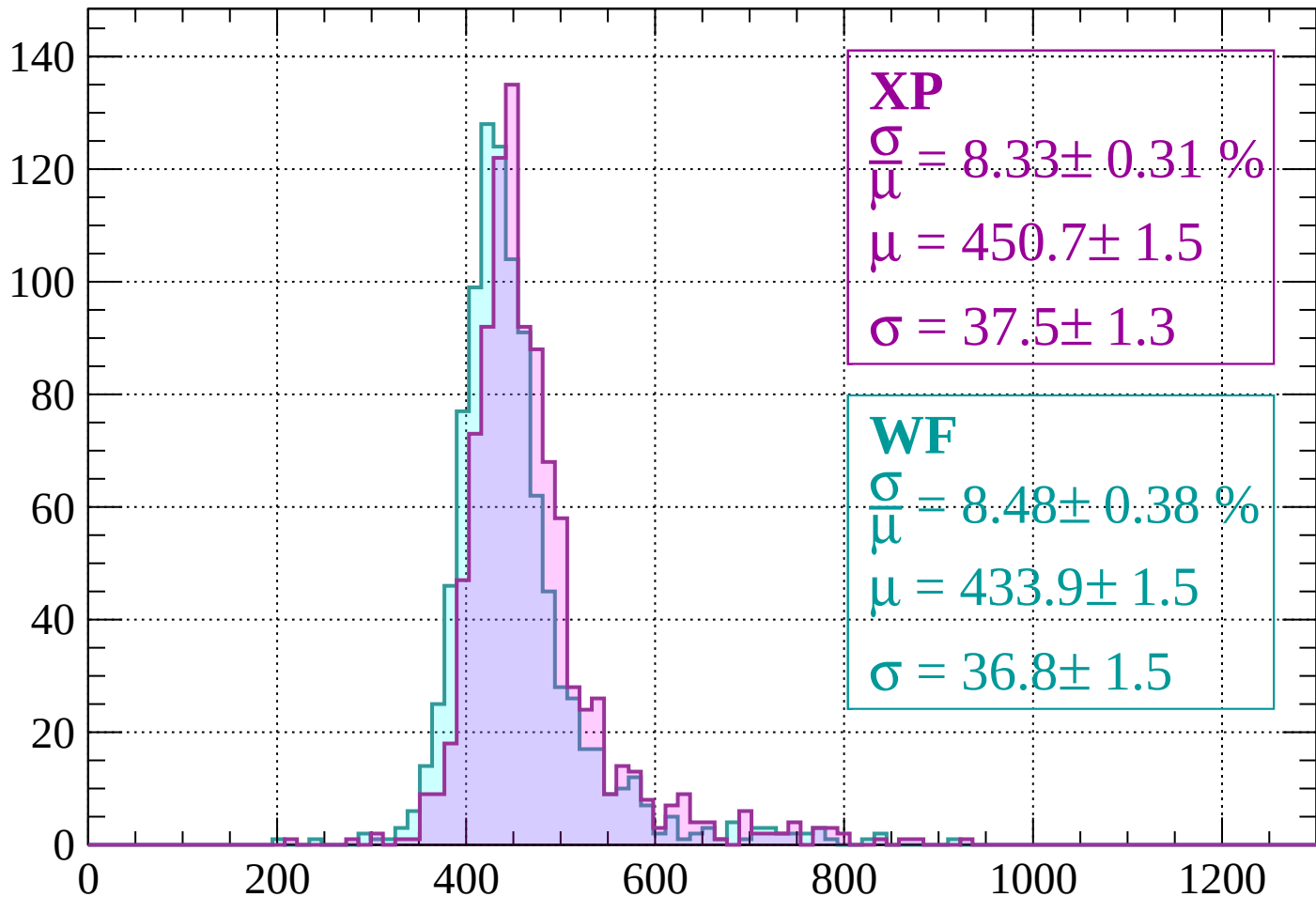
Energy loss | $-880 < p < -840$

Count



Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 8.33 \pm 0.31 \%$$

$$\mu = 450.7 \pm 1.5$$

$$\sigma = 37.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.48 \pm 0.38 \%$$

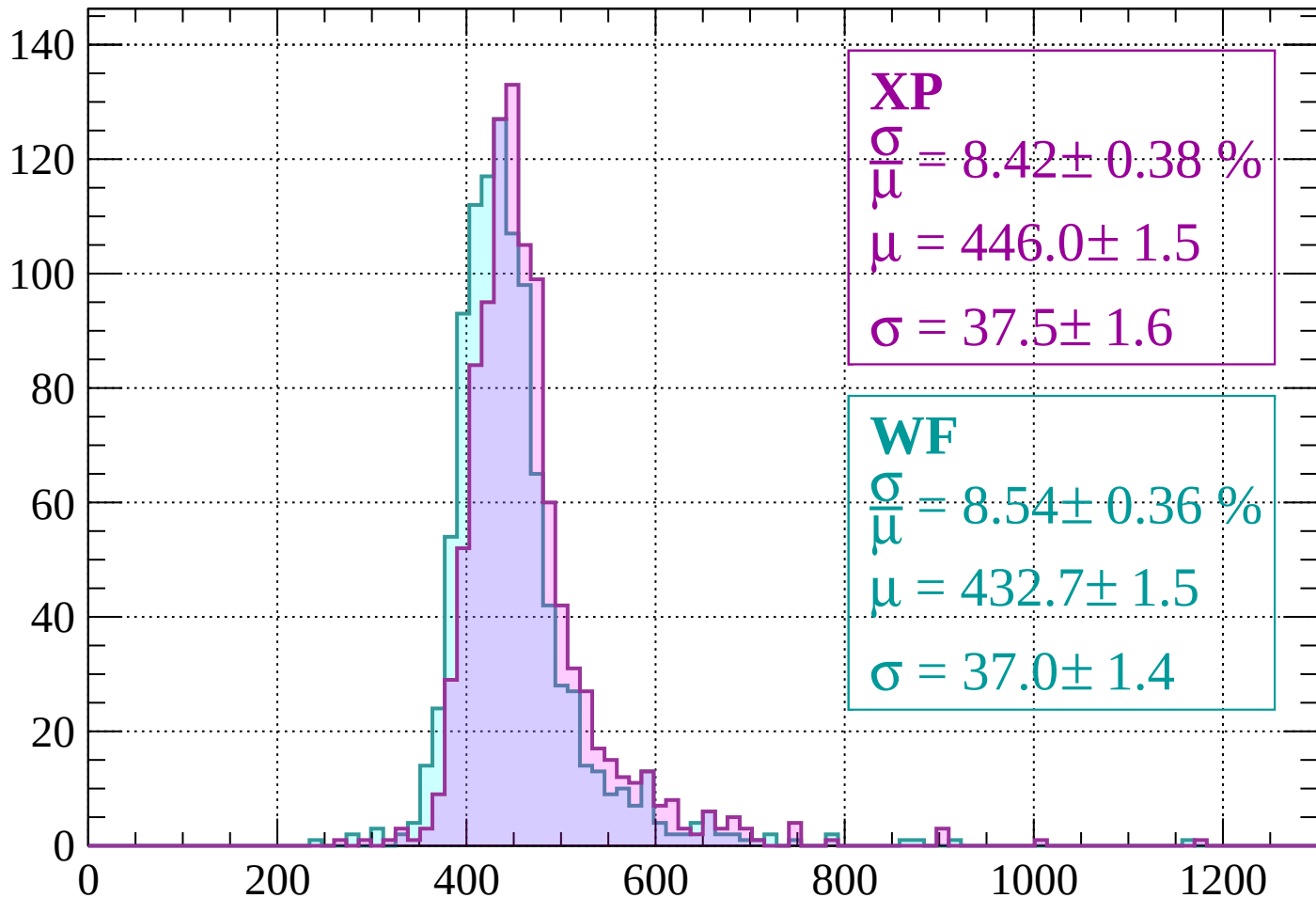
$$\mu = 433.9 \pm 1.5$$

$$\sigma = 36.8 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 8.42 \pm 0.38 \%$$

$$\mu = 446.0 \pm 1.5$$

$$\sigma = 37.5 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.54 \pm 0.36 \%$$

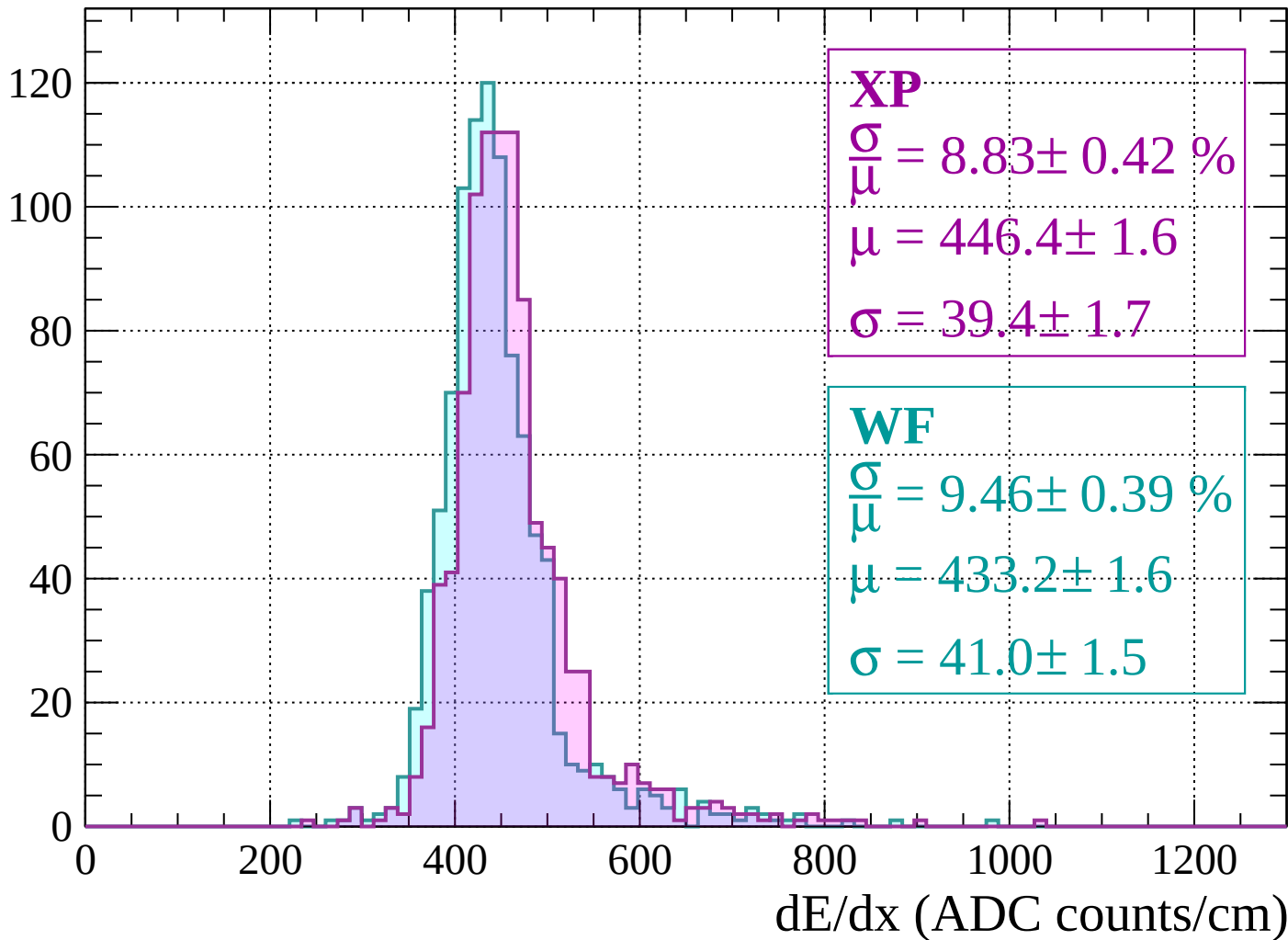
$$\mu = 432.7 \pm 1.5$$

$$\sigma = 37.0 \pm 1.4$$

dE/dx (ADC counts/cm)

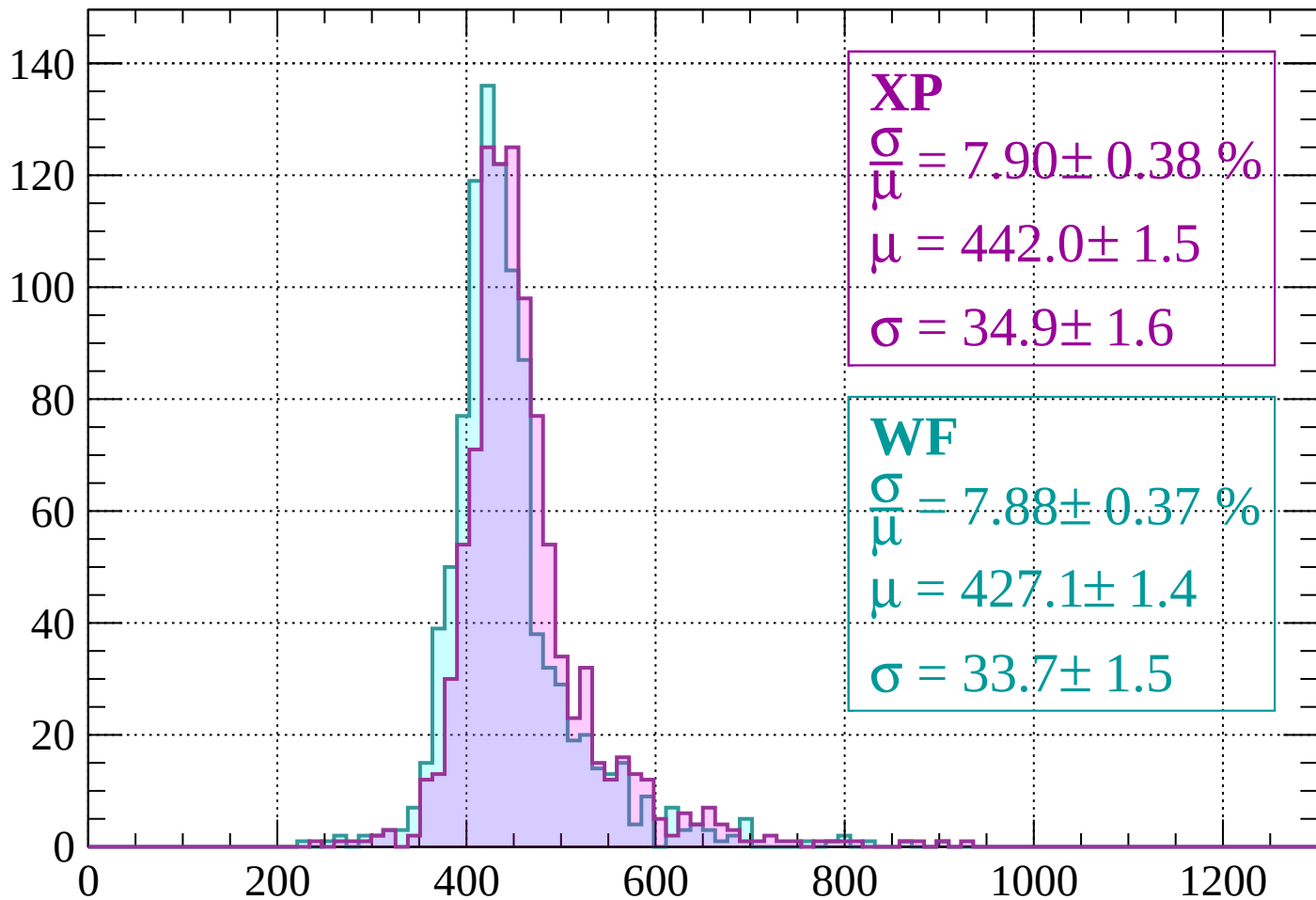
Energy loss | $-760 < p < -720$

Count



Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 7.90 \pm 0.38 \%$$

$$\mu = 442.0 \pm 1.5$$

$$\sigma = 34.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.88 \pm 0.37 \%$$

$$\mu = 427.1 \pm 1.4$$

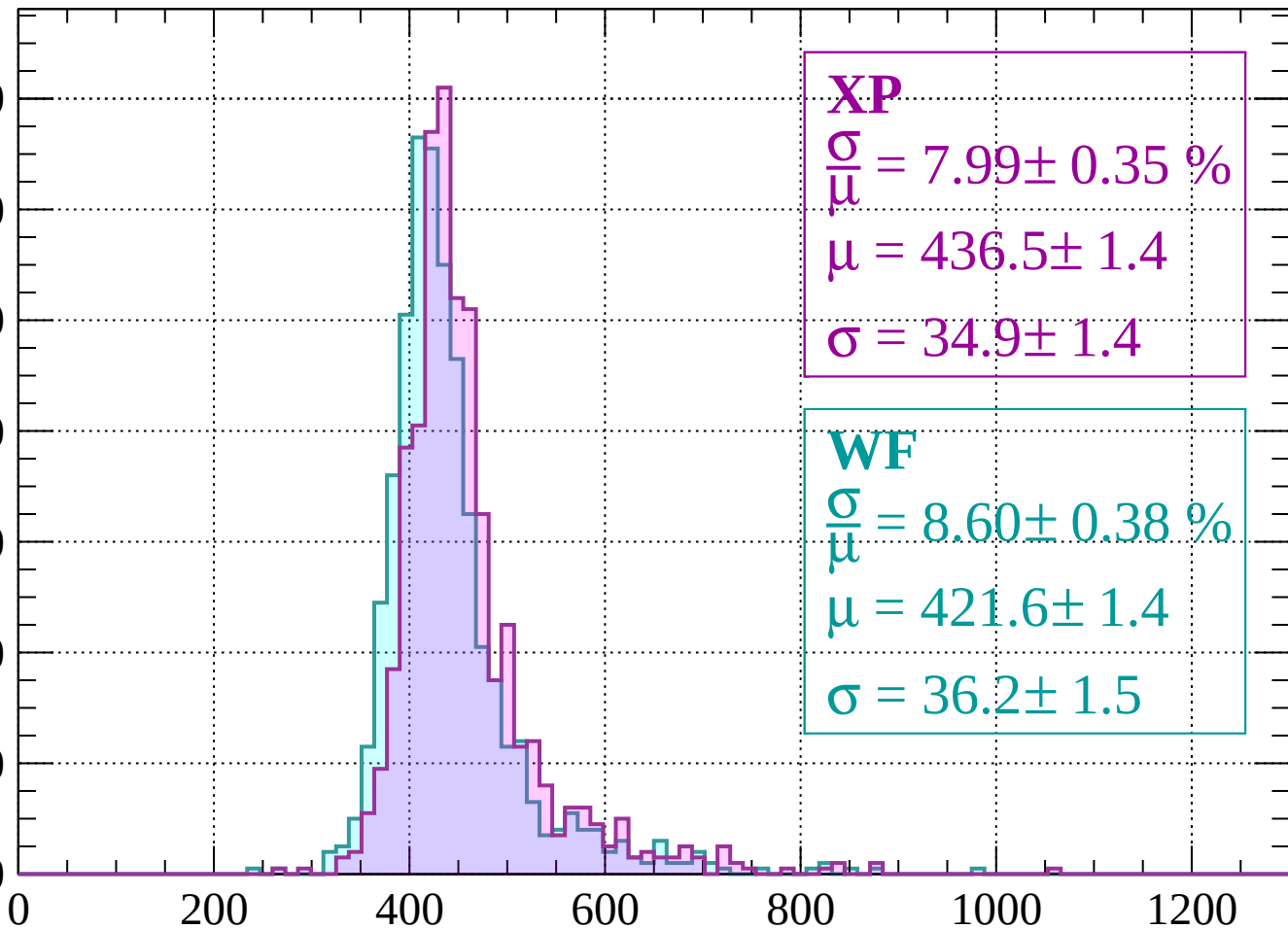
$$\sigma = 33.7 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 7.99 \pm 0.35 \%$$

$$\mu = 436.5 \pm 1.4$$

$$\sigma = 34.9 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.60 \pm 0.38 \%$$

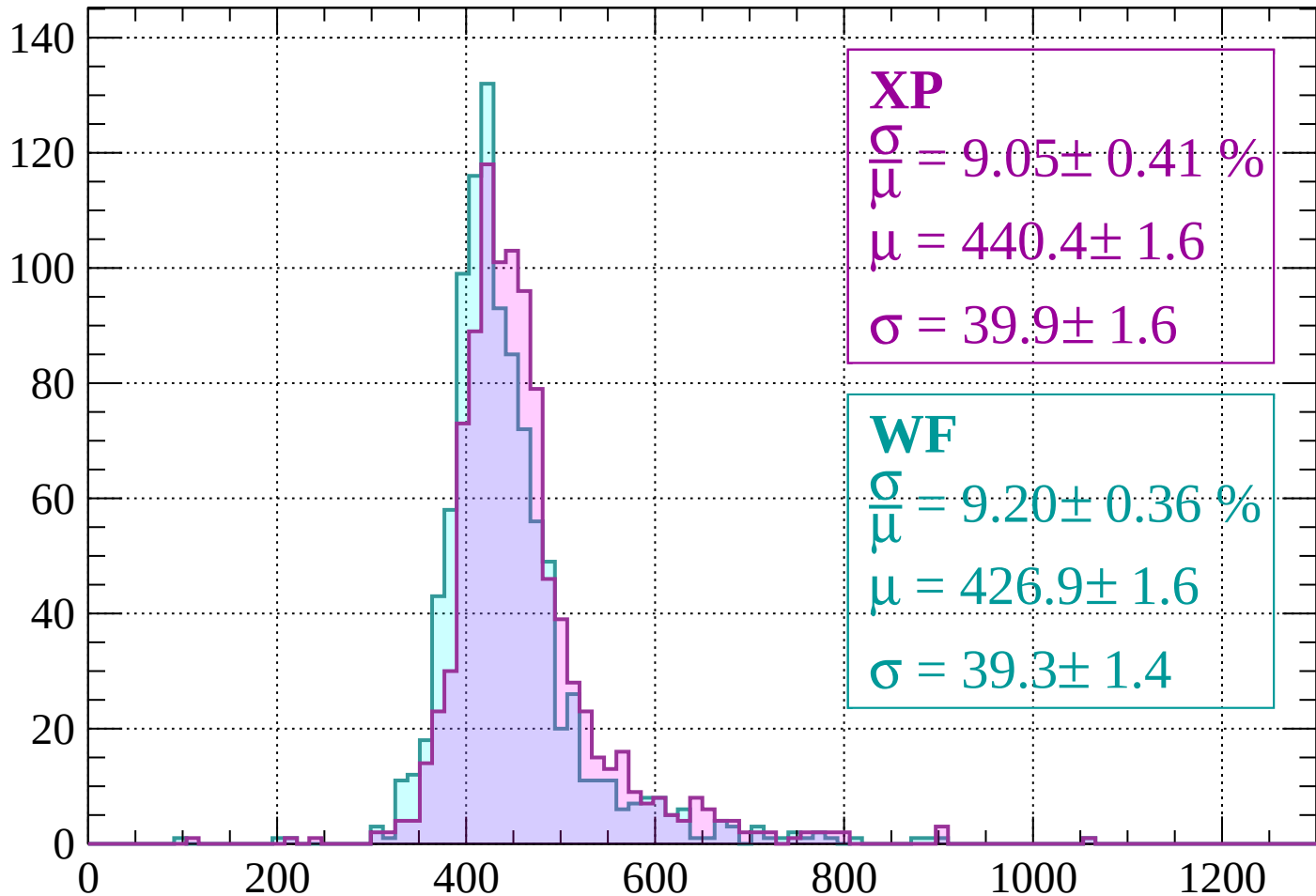
$$\mu = 421.6 \pm 1.4$$

$$\sigma = 36.2 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 9.05 \pm 0.41 \%$$

$$\mu = 440.4 \pm 1.6$$

$$\sigma = 39.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 9.20 \pm 0.36 \%$$

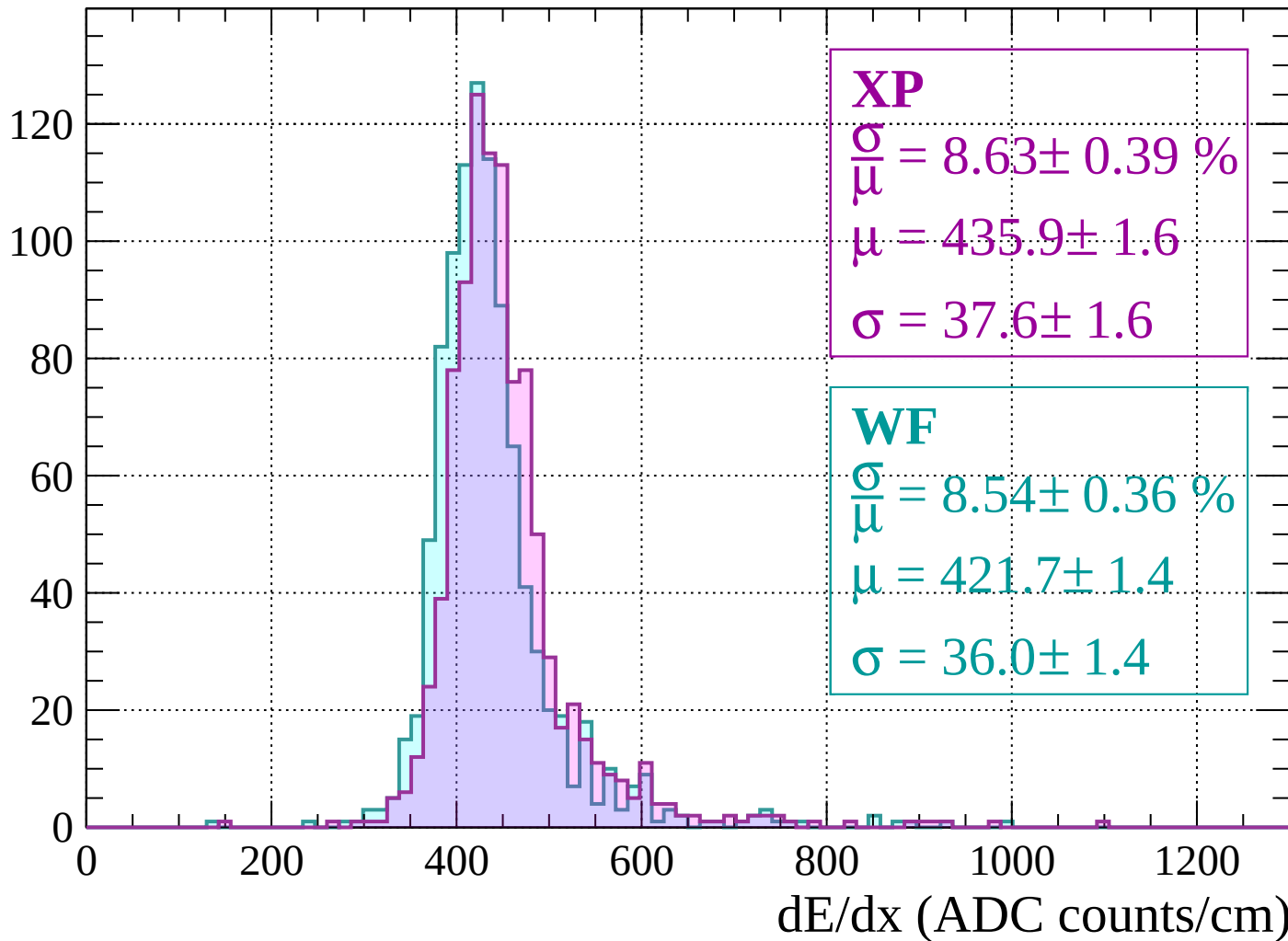
$$\mu = 426.9 \pm 1.6$$

$$\sigma = 39.3 \pm 1.4$$

dE/dx (ADC counts/cm)

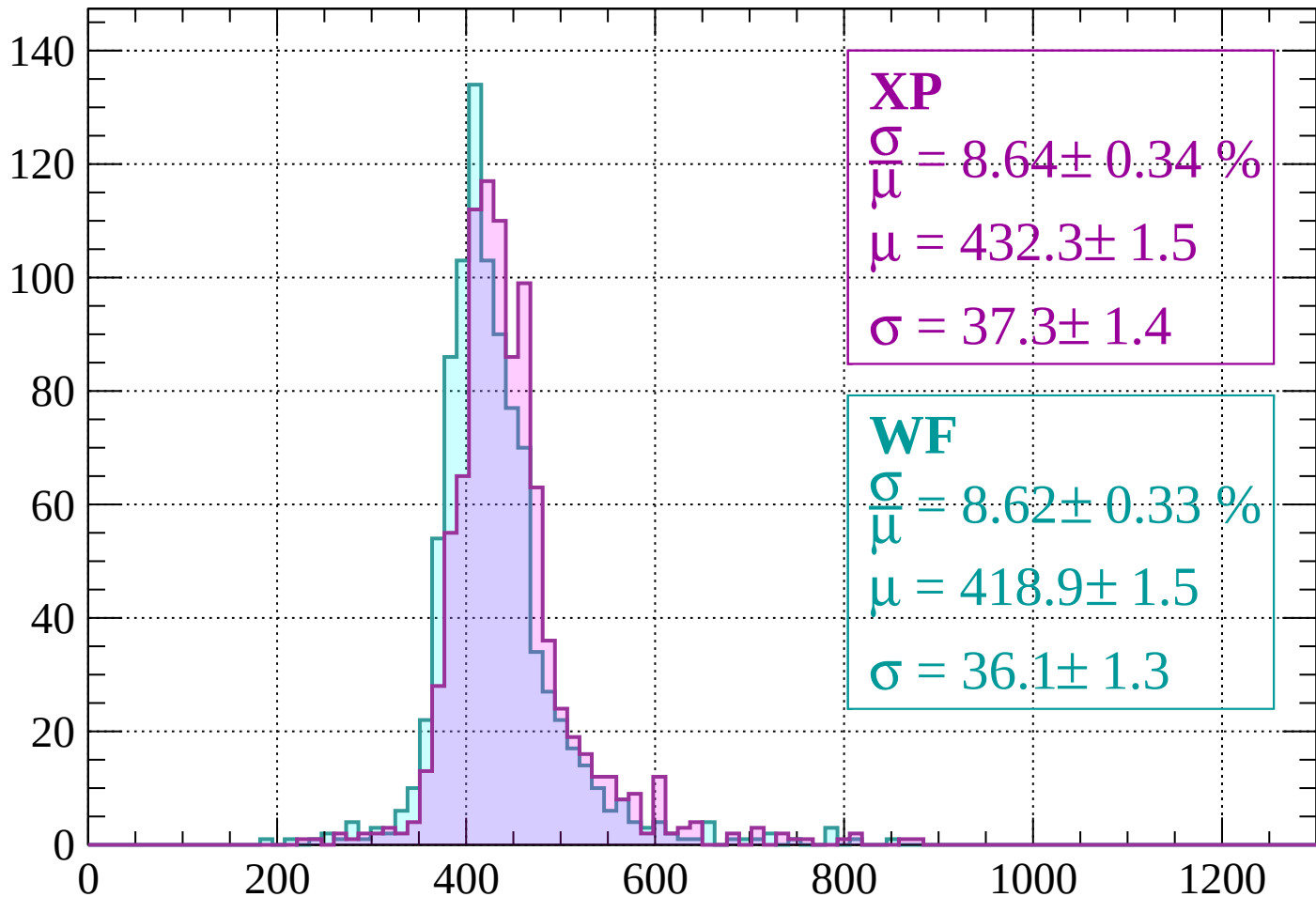
Energy loss | $-600 < p < -560$

Count



Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 8.64 \pm 0.34 \%$$

$$\mu = 432.3 \pm 1.5$$

$$\sigma = 37.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.62 \pm 0.33 \%$$

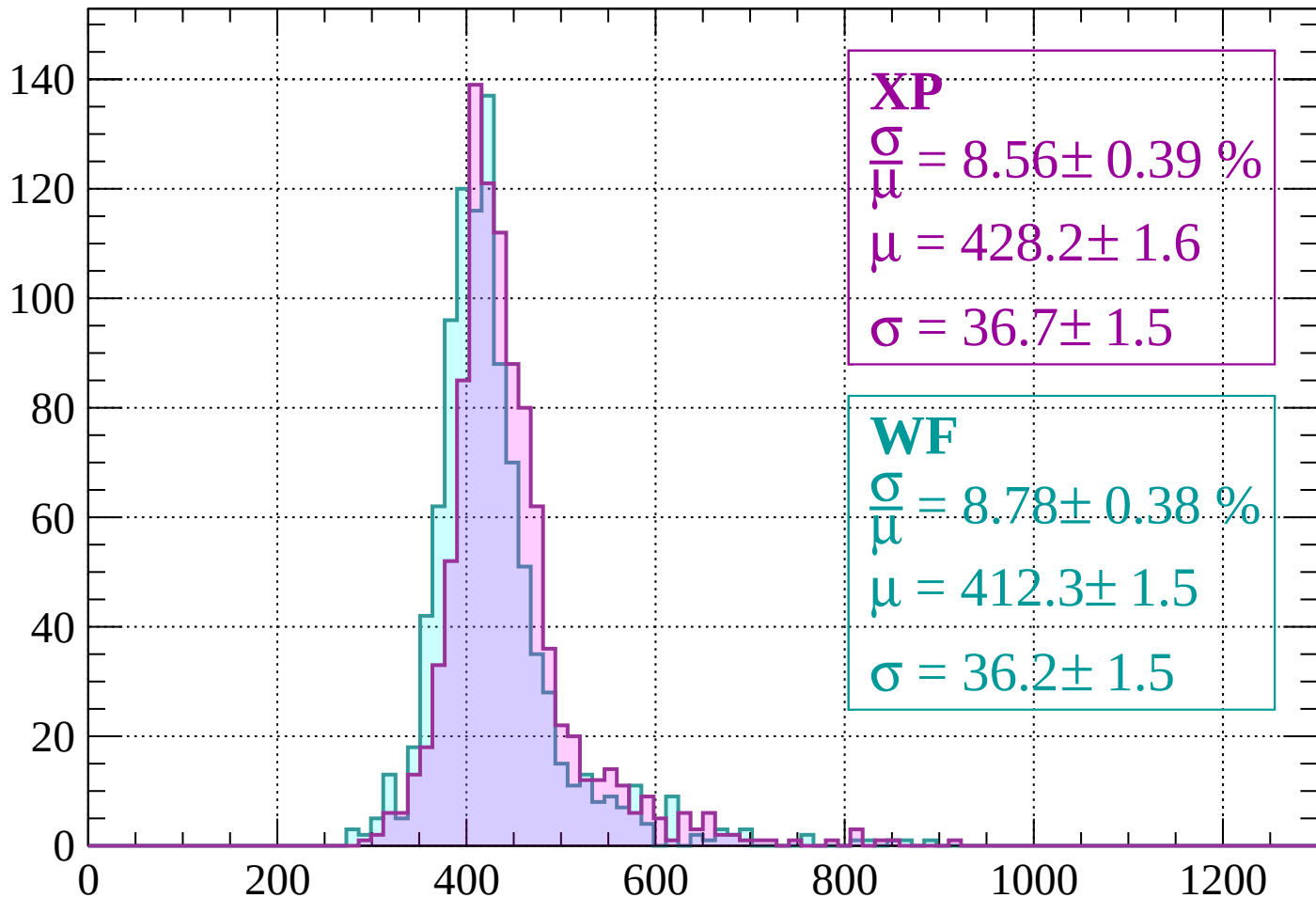
$$\mu = 418.9 \pm 1.5$$

$$\sigma = 36.1 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 8.56 \pm 0.39 \%$$

$$\mu = 428.2 \pm 1.6$$

$$\sigma = 36.7 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.78 \pm 0.38 \%$$

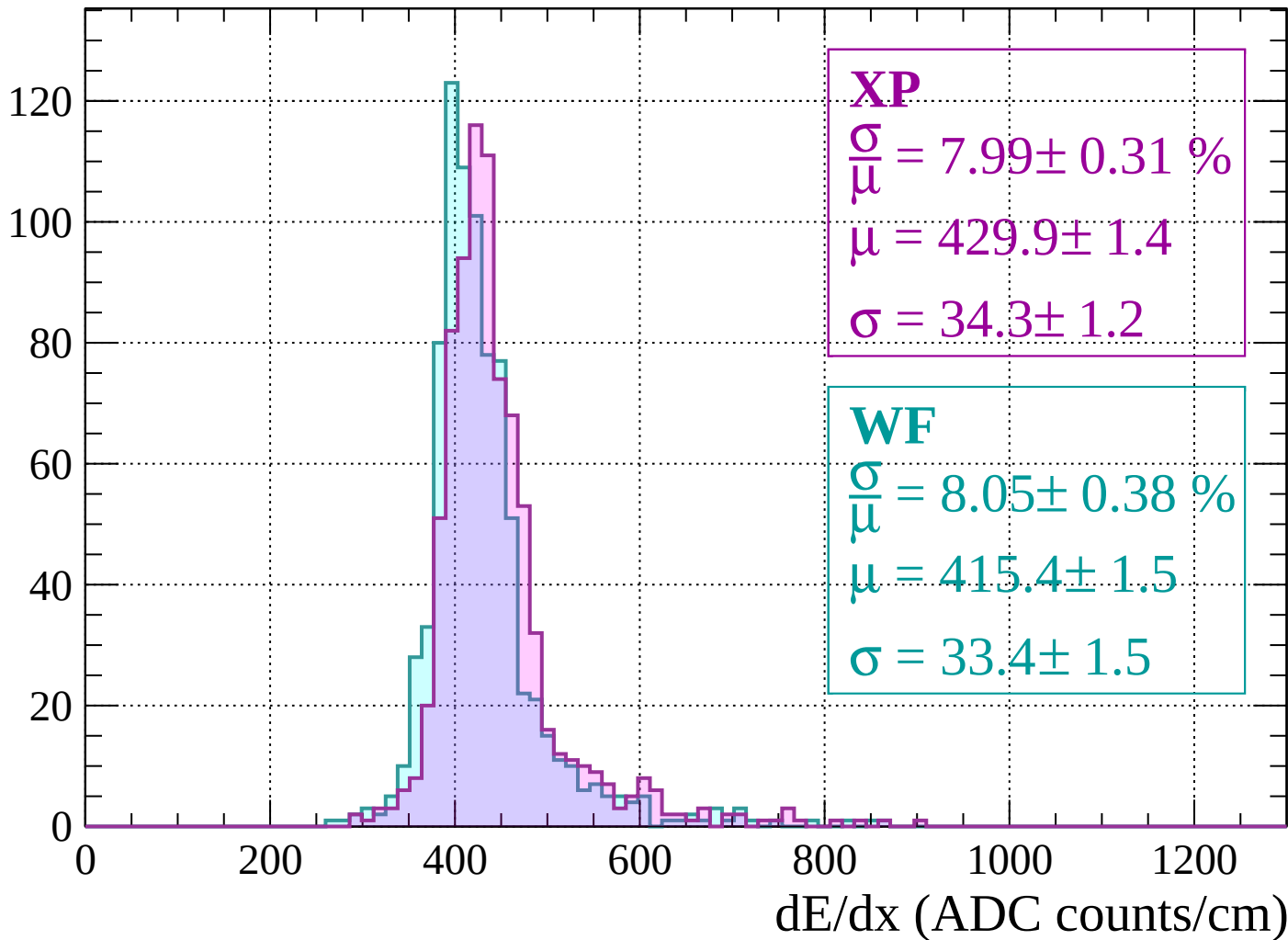
$$\mu = 412.3 \pm 1.5$$

$$\sigma = 36.2 \pm 1.5$$

dE/dx (ADC counts/cm)

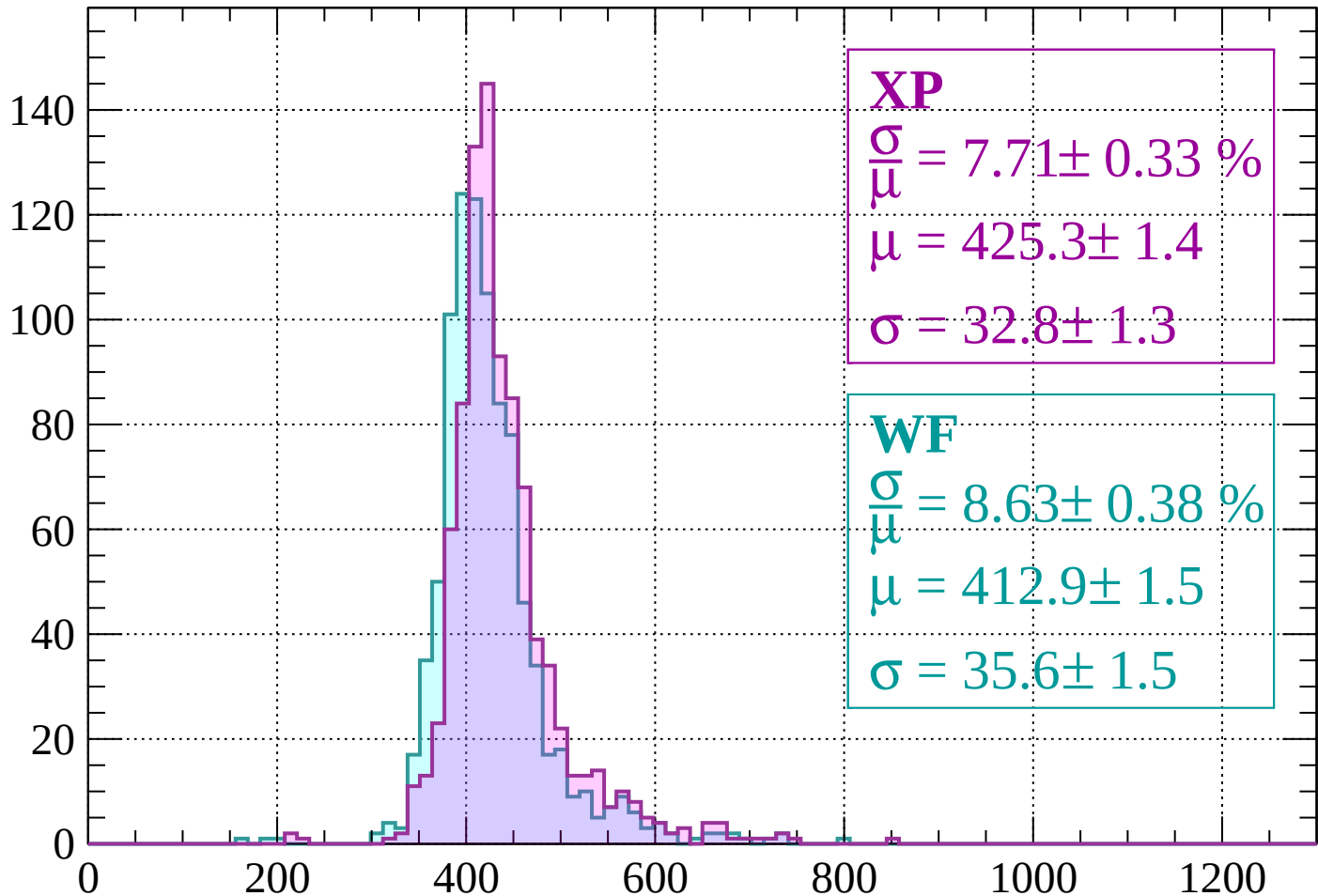
Energy loss | $-480 < p < -440$

Count



Energy loss | $-440 < p < -400$

Count



XP

$$\frac{\sigma}{\mu} = 7.71 \pm 0.33 \%$$

$$\mu = 425.3 \pm 1.4$$

$$\sigma = 32.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.63 \pm 0.38 \%$$

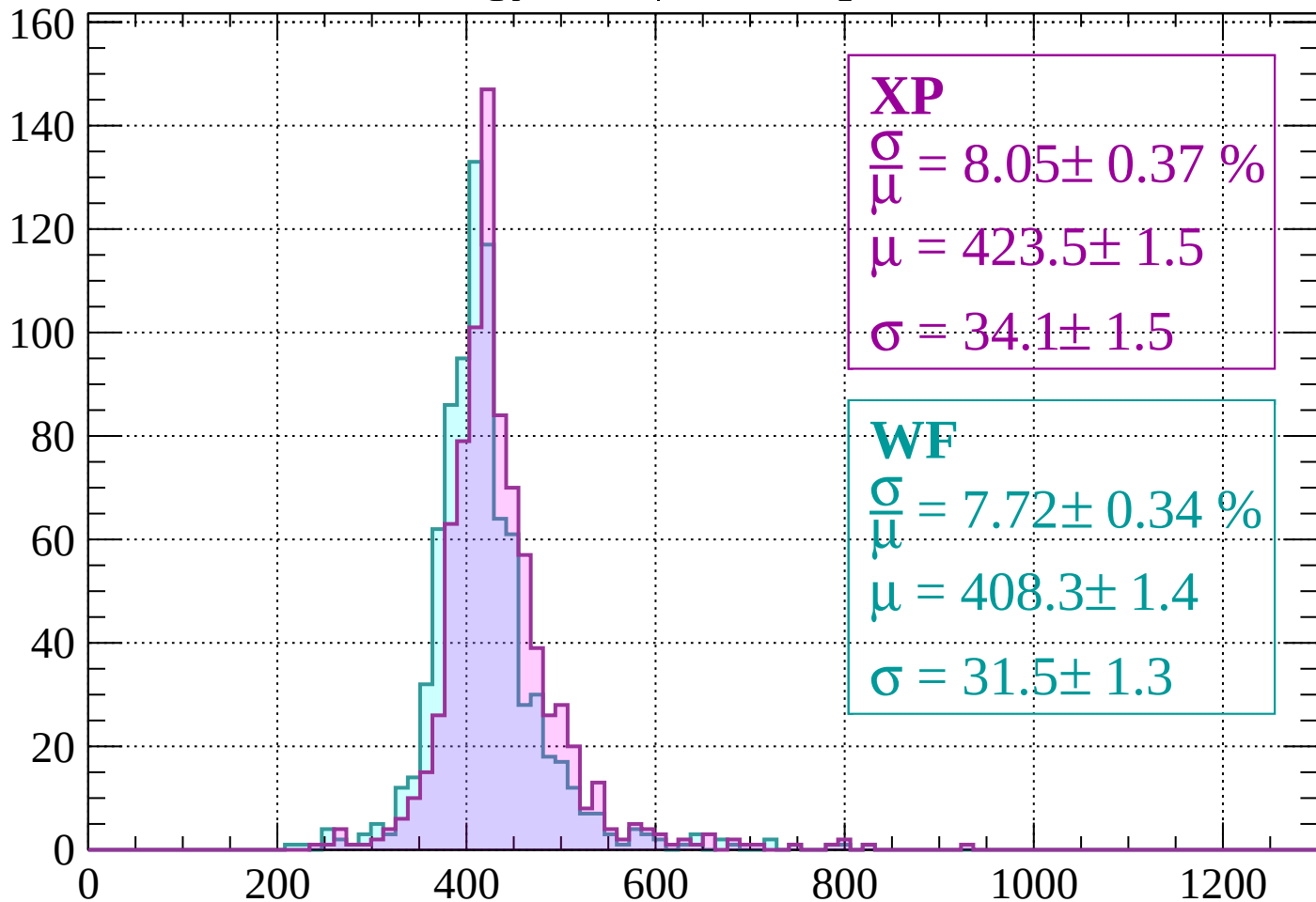
$$\mu = 412.9 \pm 1.5$$

$$\sigma = 35.6 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.37 \%$$

$$\mu = 423.5 \pm 1.5$$

$$\sigma = 34.1 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.72 \pm 0.34 \%$$

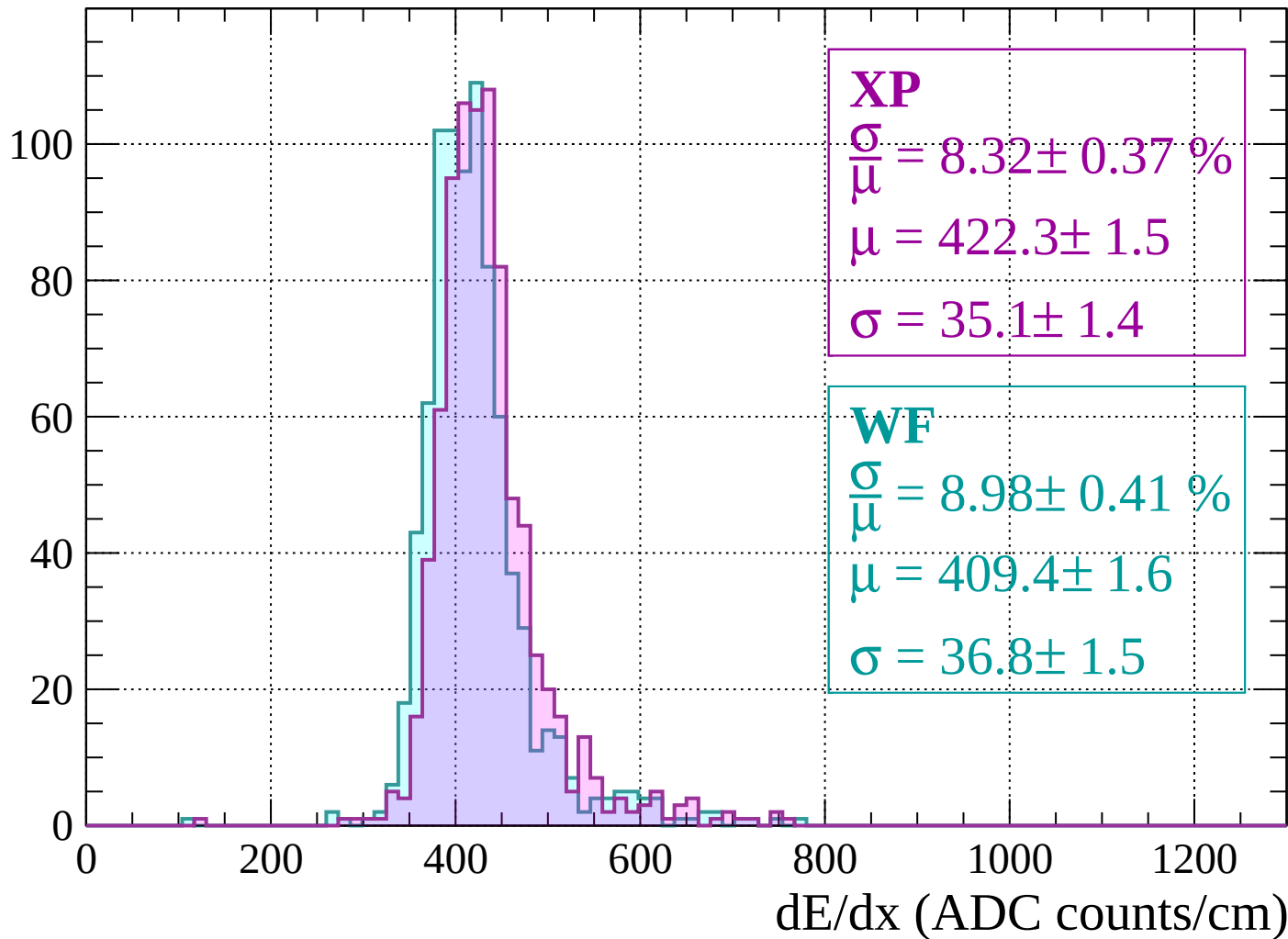
$$\mu = 408.3 \pm 1.4$$

$$\sigma = 31.5 \pm 1.3$$

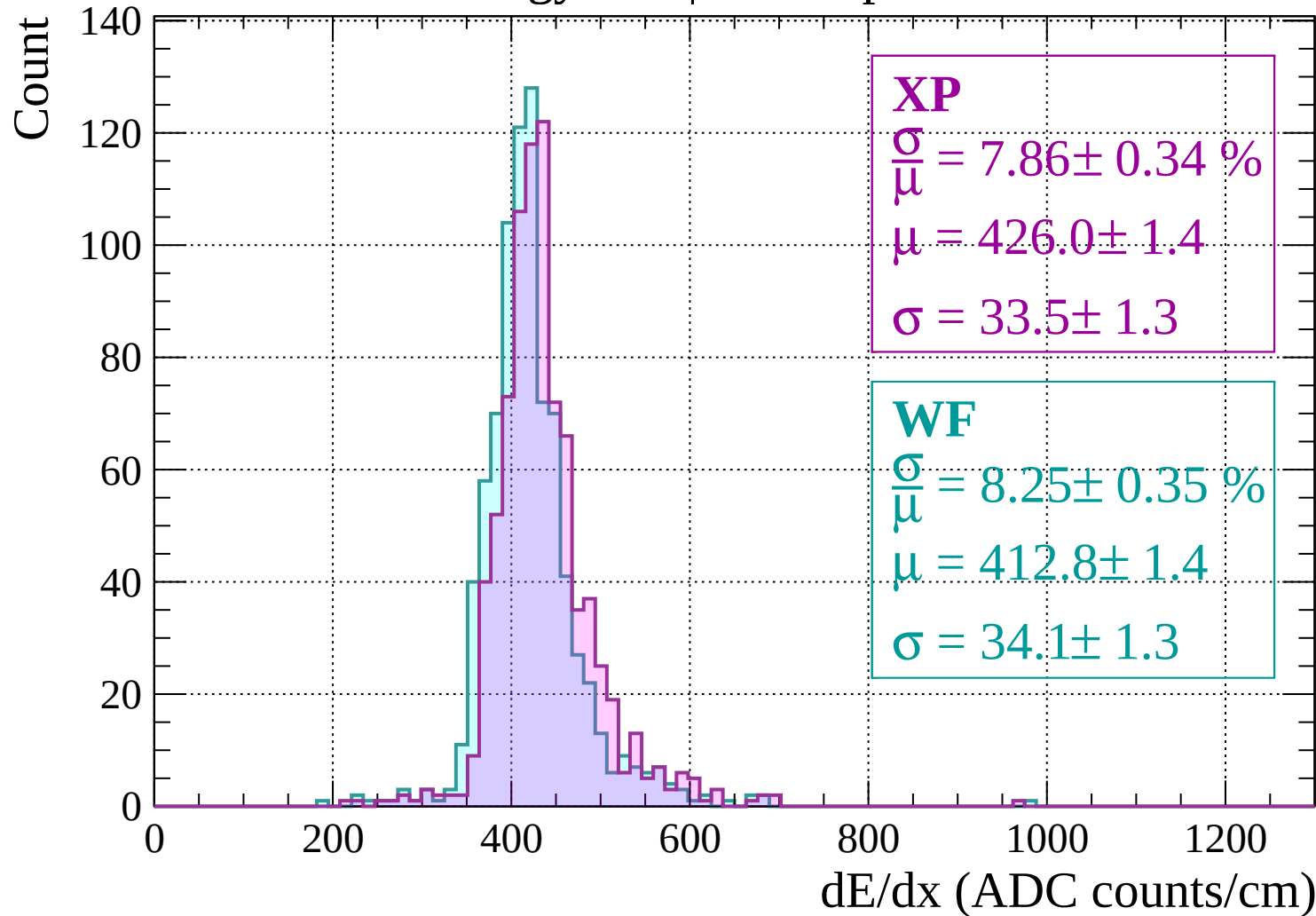
dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count

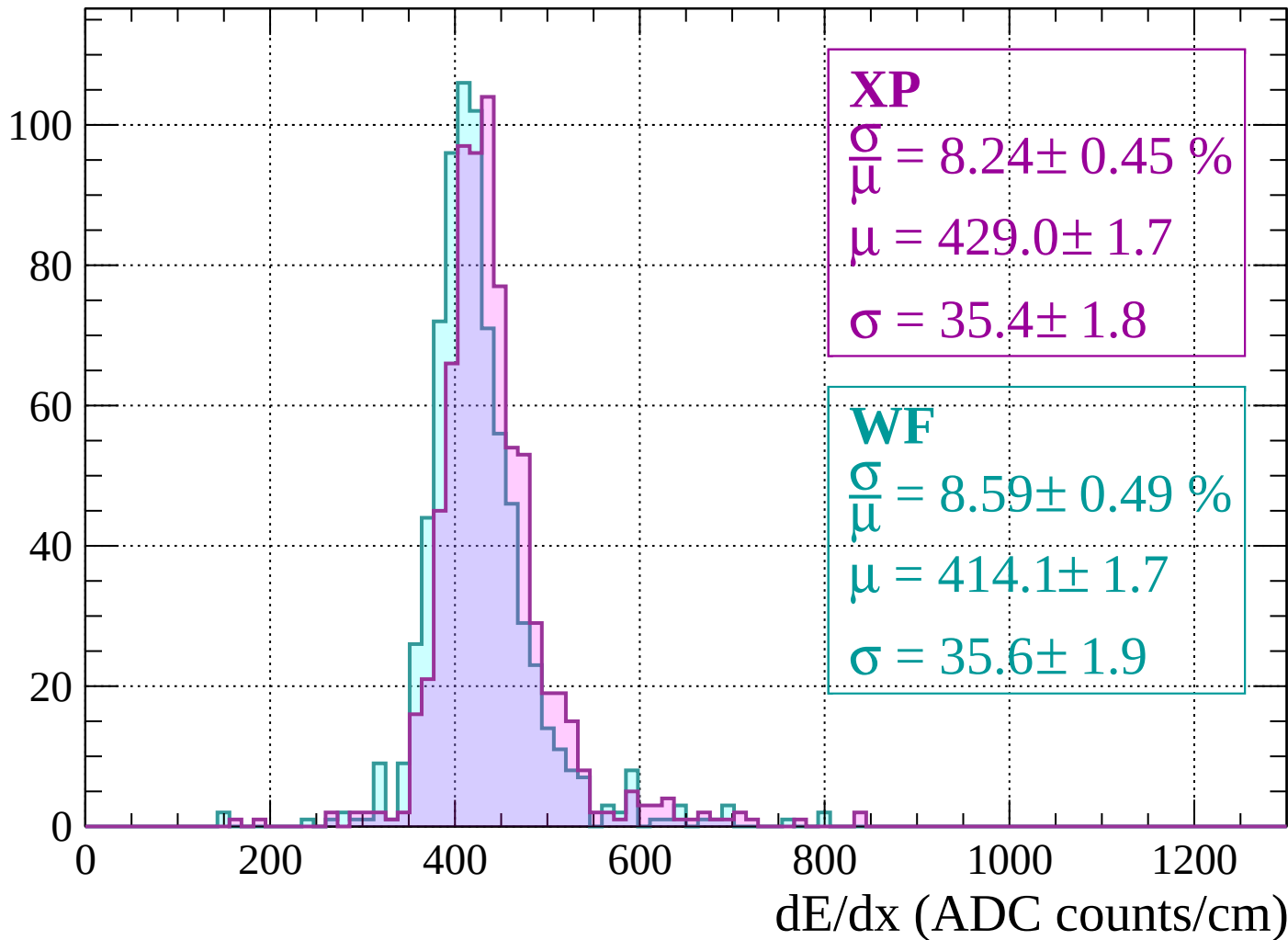


Energy loss | $-320 < p < -280$



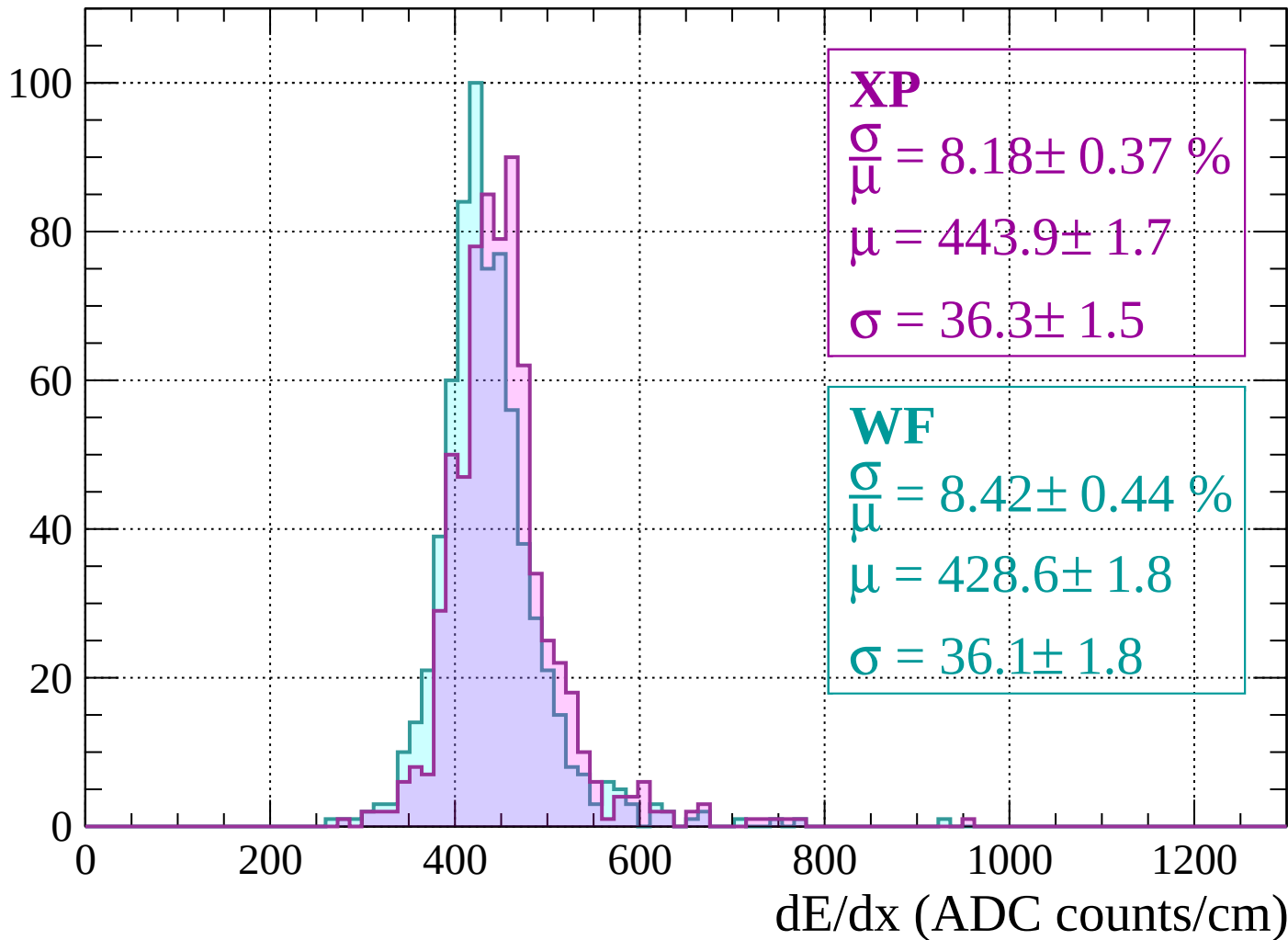
Energy loss | $-280 < p < -240$

Count



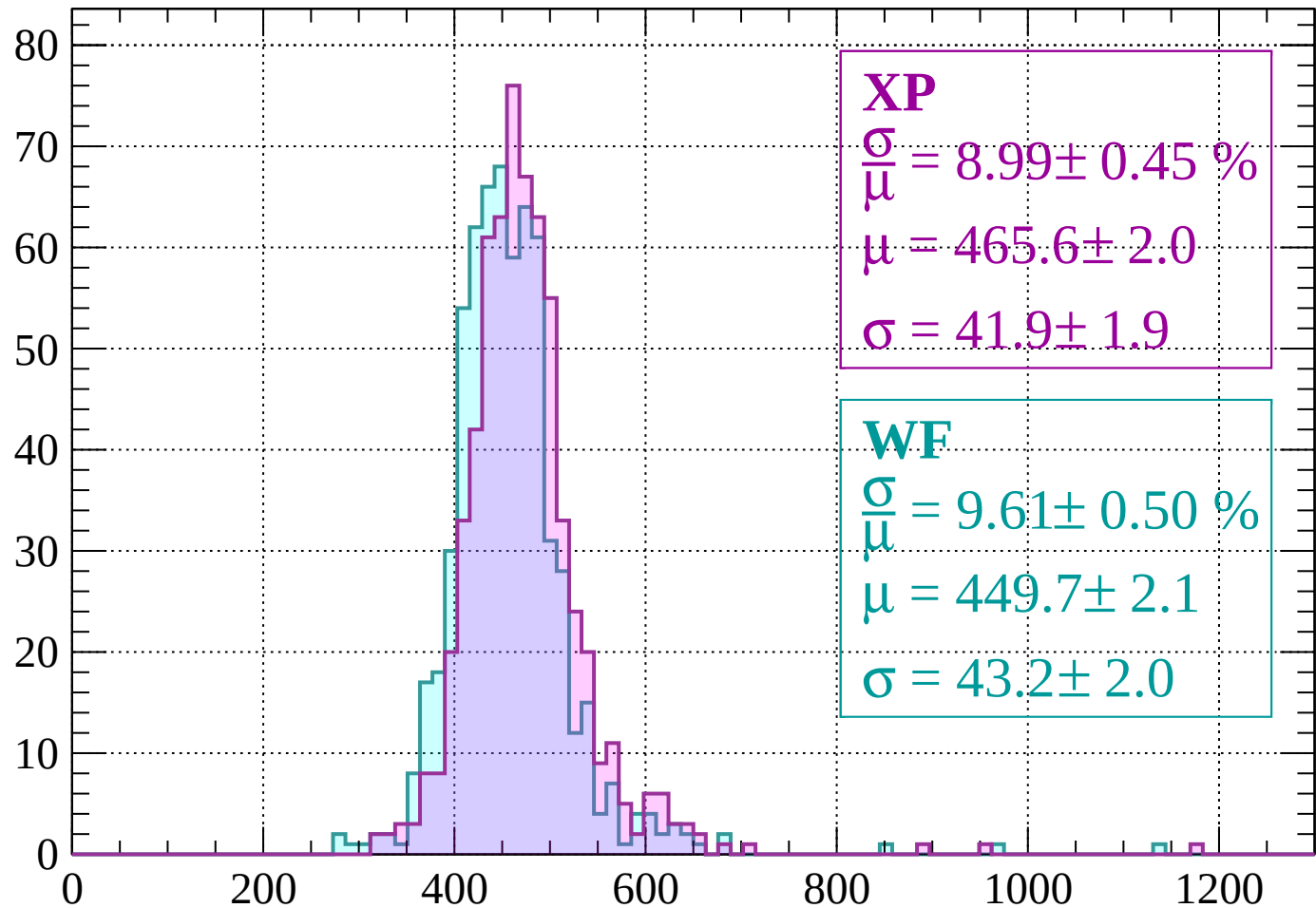
Energy loss | $-240 < p < -200$

Count



Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 8.99 \pm 0.45 \%$$

$$\mu = 465.6 \pm 2.0$$

$$\sigma = 41.9 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 9.61 \pm 0.50 \%$$

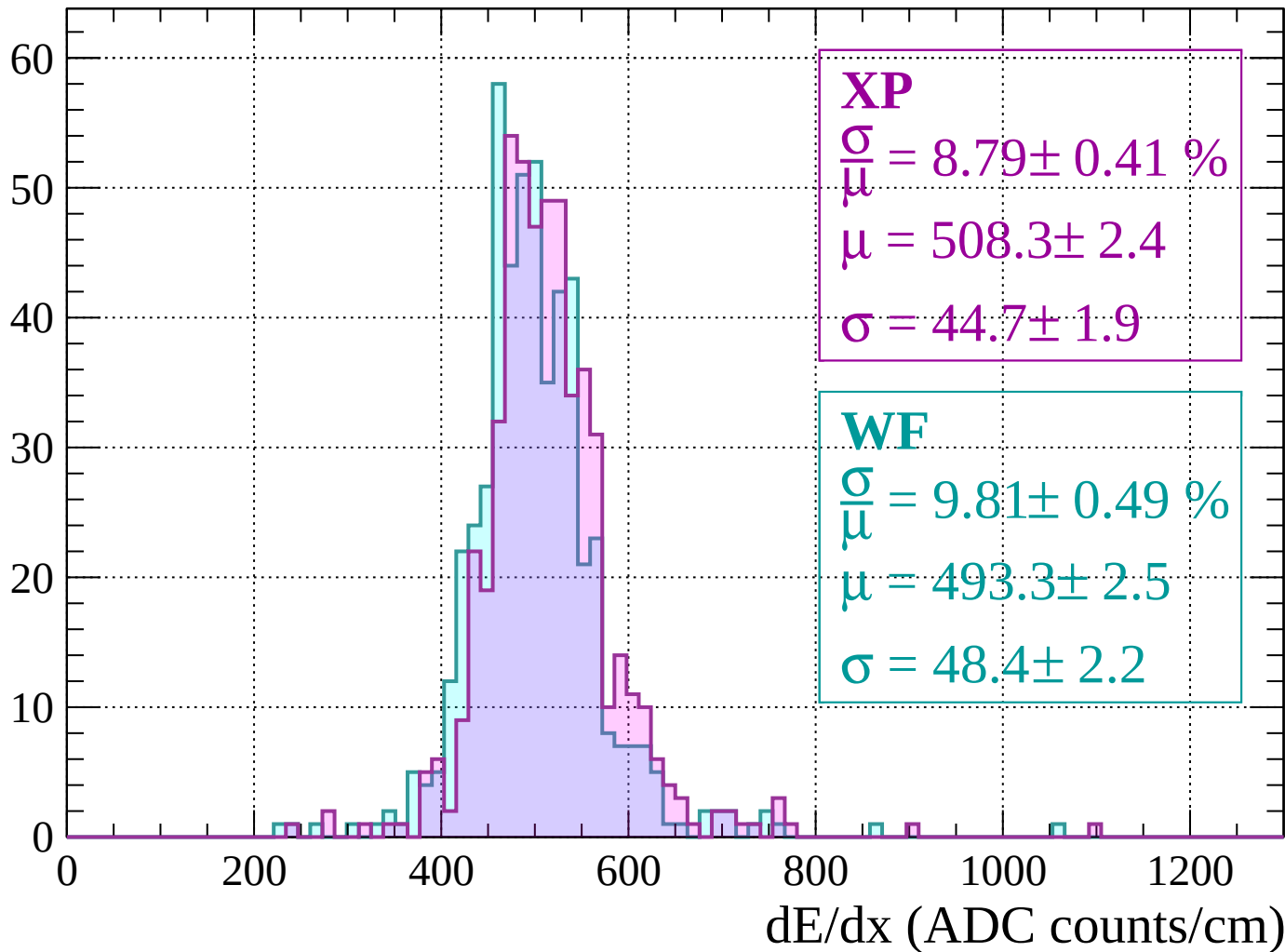
$$\mu = 449.7 \pm 2.1$$

$$\sigma = 43.2 \pm 2.0$$

dE/dx (ADC counts/cm)

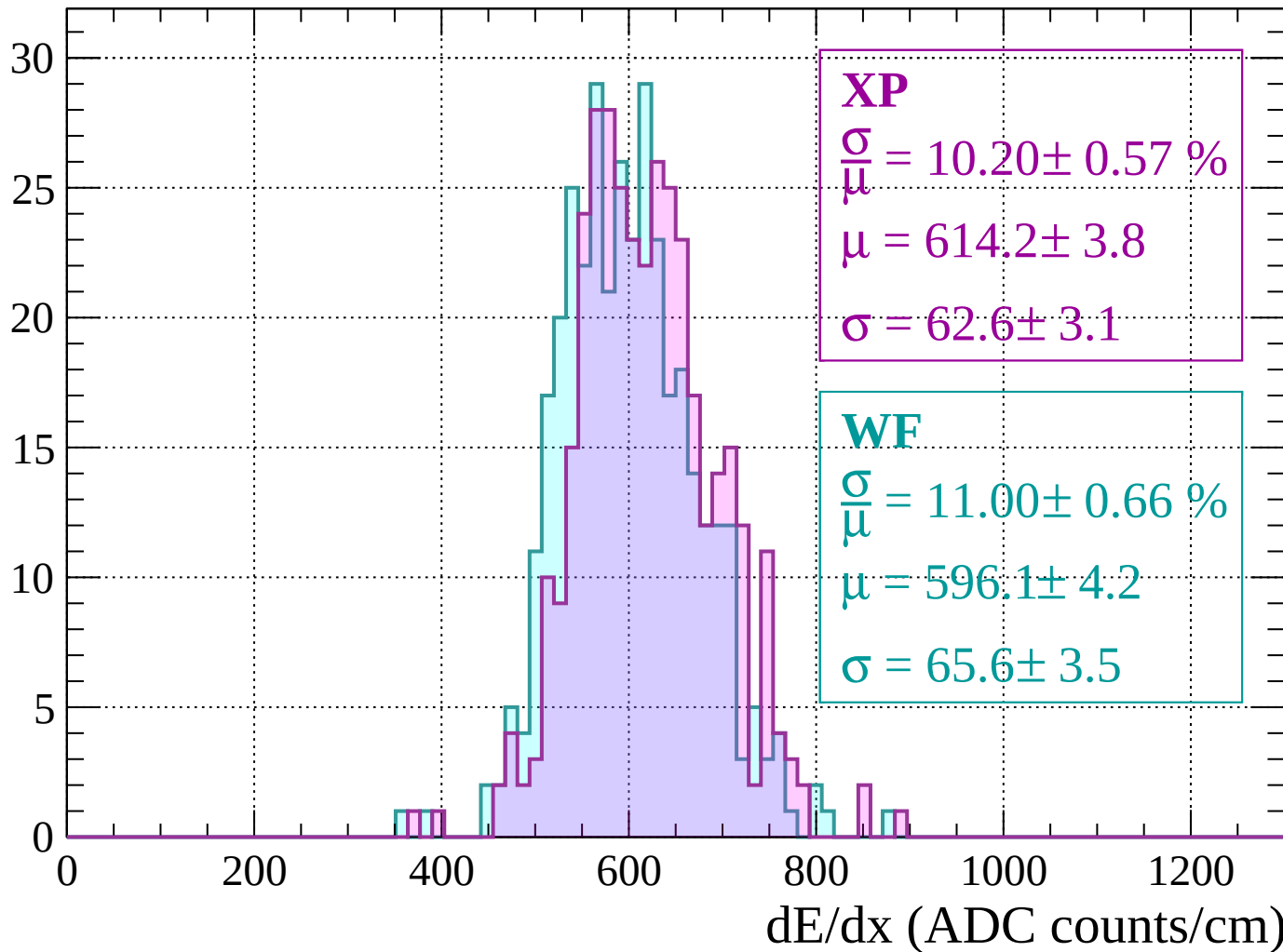
Energy loss | $-160 < p < -120$

Count



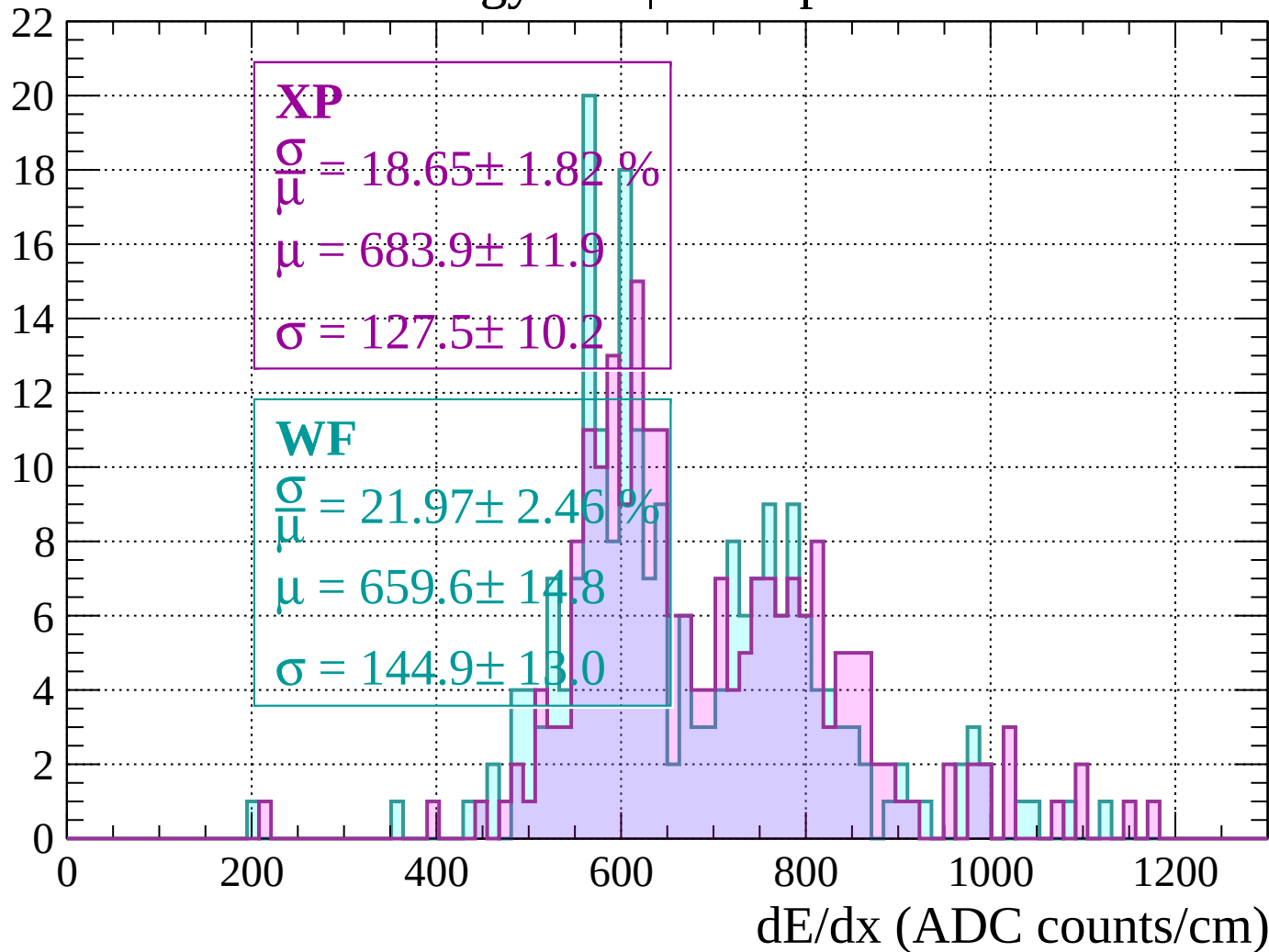
Energy loss | $-120 < p < -80$

Count



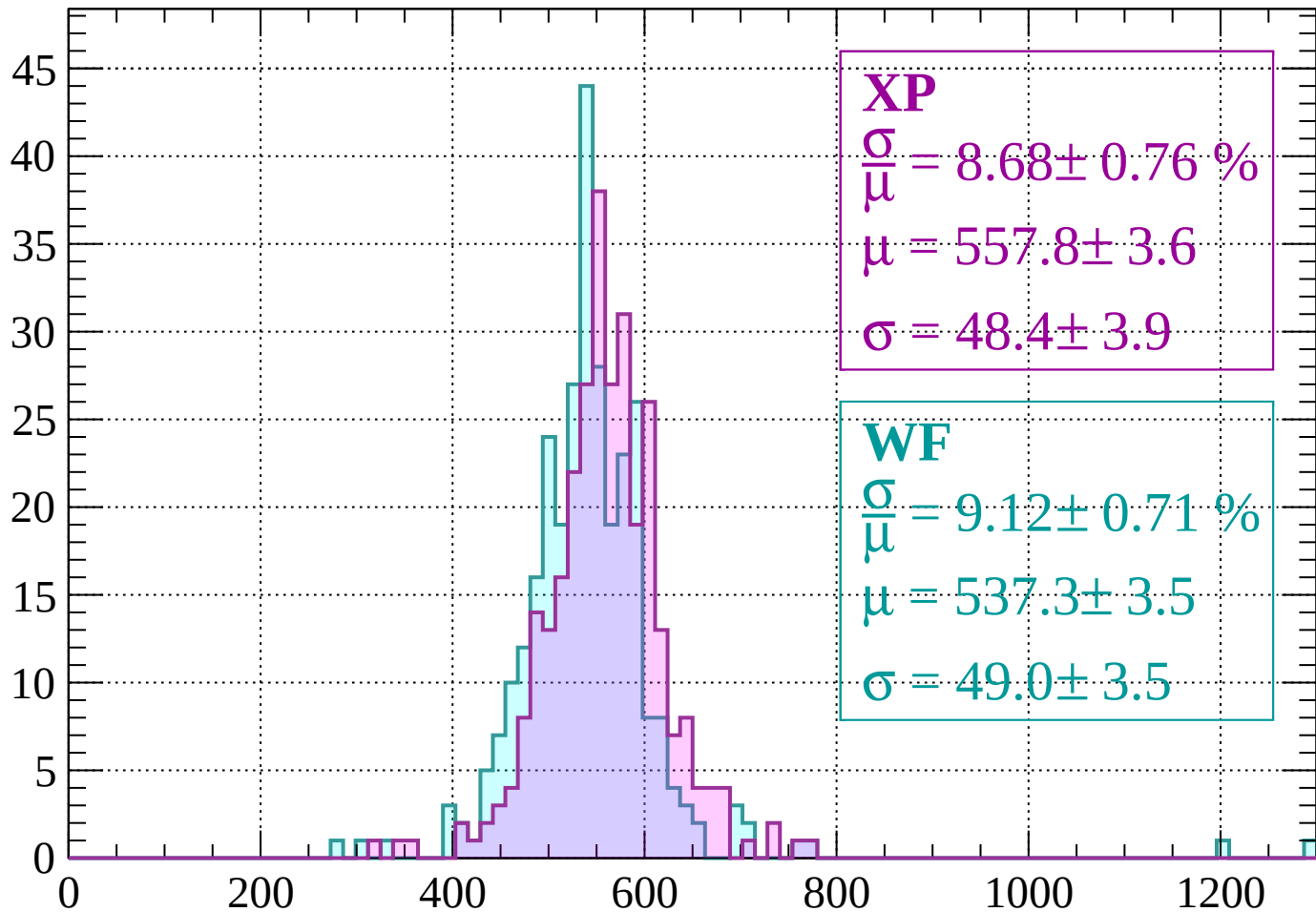
Energy loss | -80 < p < -40

Count



Energy loss | $-40 < p < 0$

Count



XP

$$\frac{\sigma}{\mu} = 8.68 \pm 0.76 \%$$

$$\mu = 557.8 \pm 3.6$$

$$\sigma = 48.4 \pm 3.9$$

WF

$$\frac{\sigma}{\mu} = 9.12 \pm 0.71 \%$$

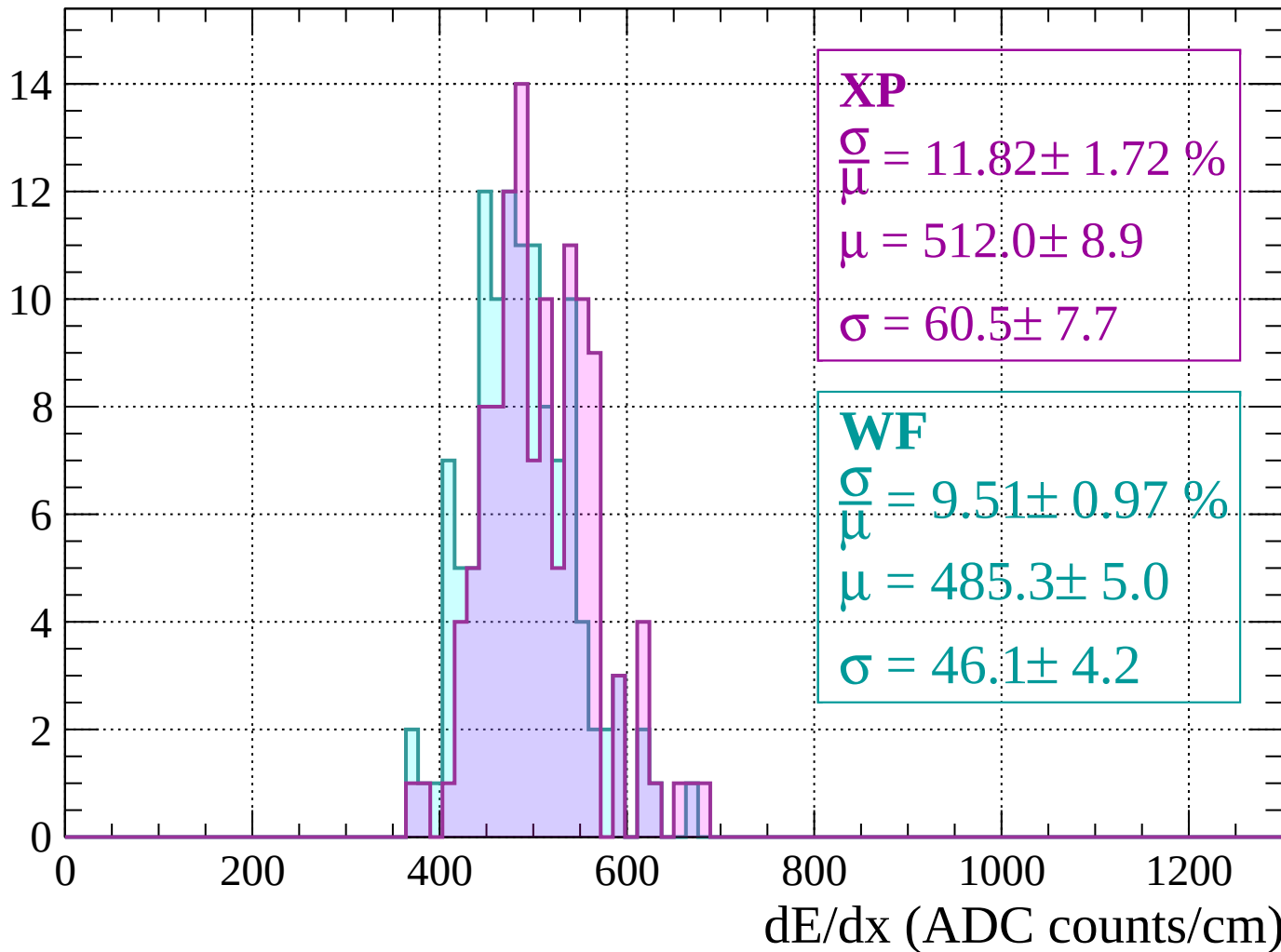
$$\mu = 537.3 \pm 3.5$$

$$\sigma = 49.0 \pm 3.5$$

dE/dx (ADC counts/cm)

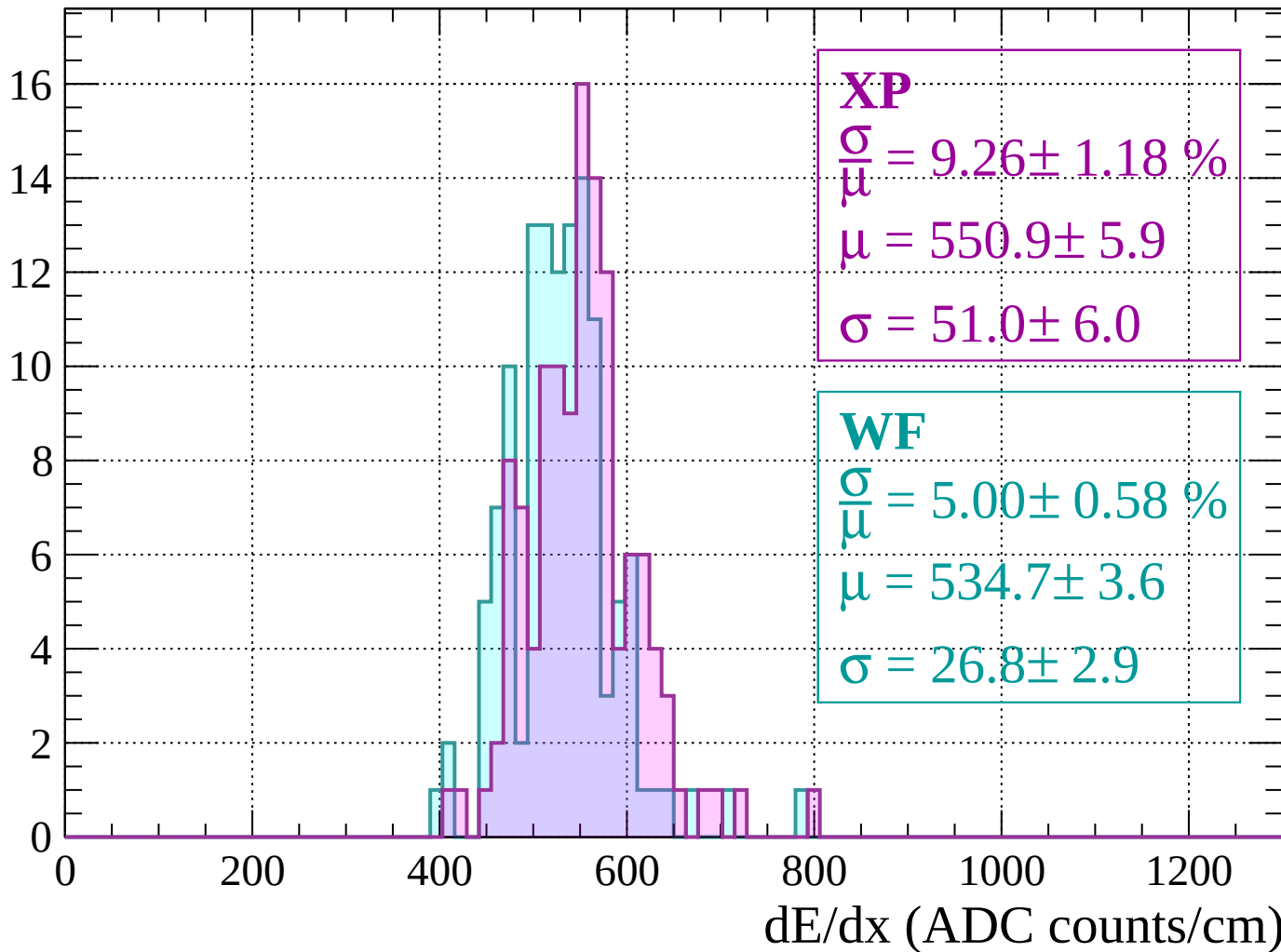
Energy loss | $0 < p < 40$

Count

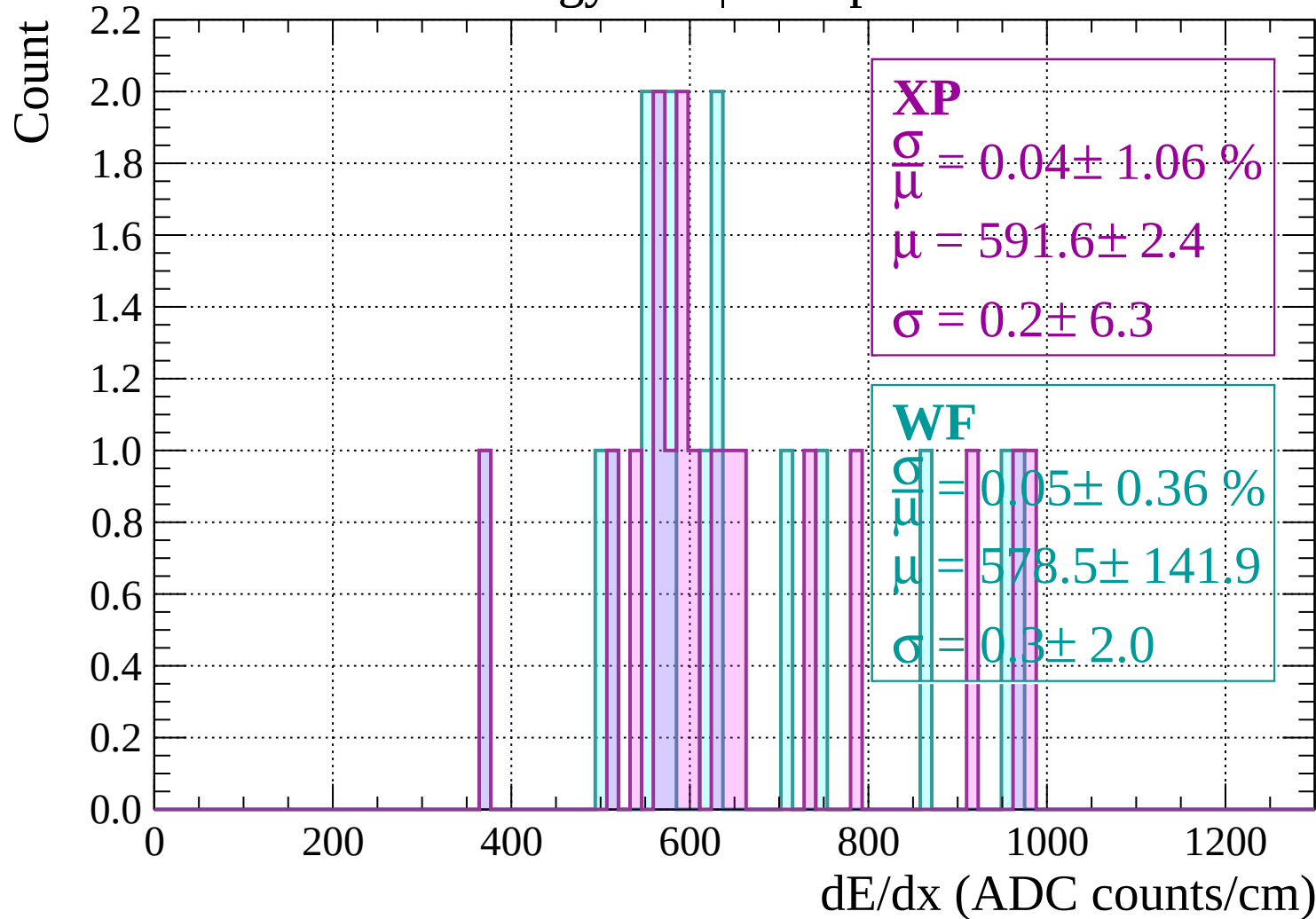


Energy loss | $40 < p < 80$

Count

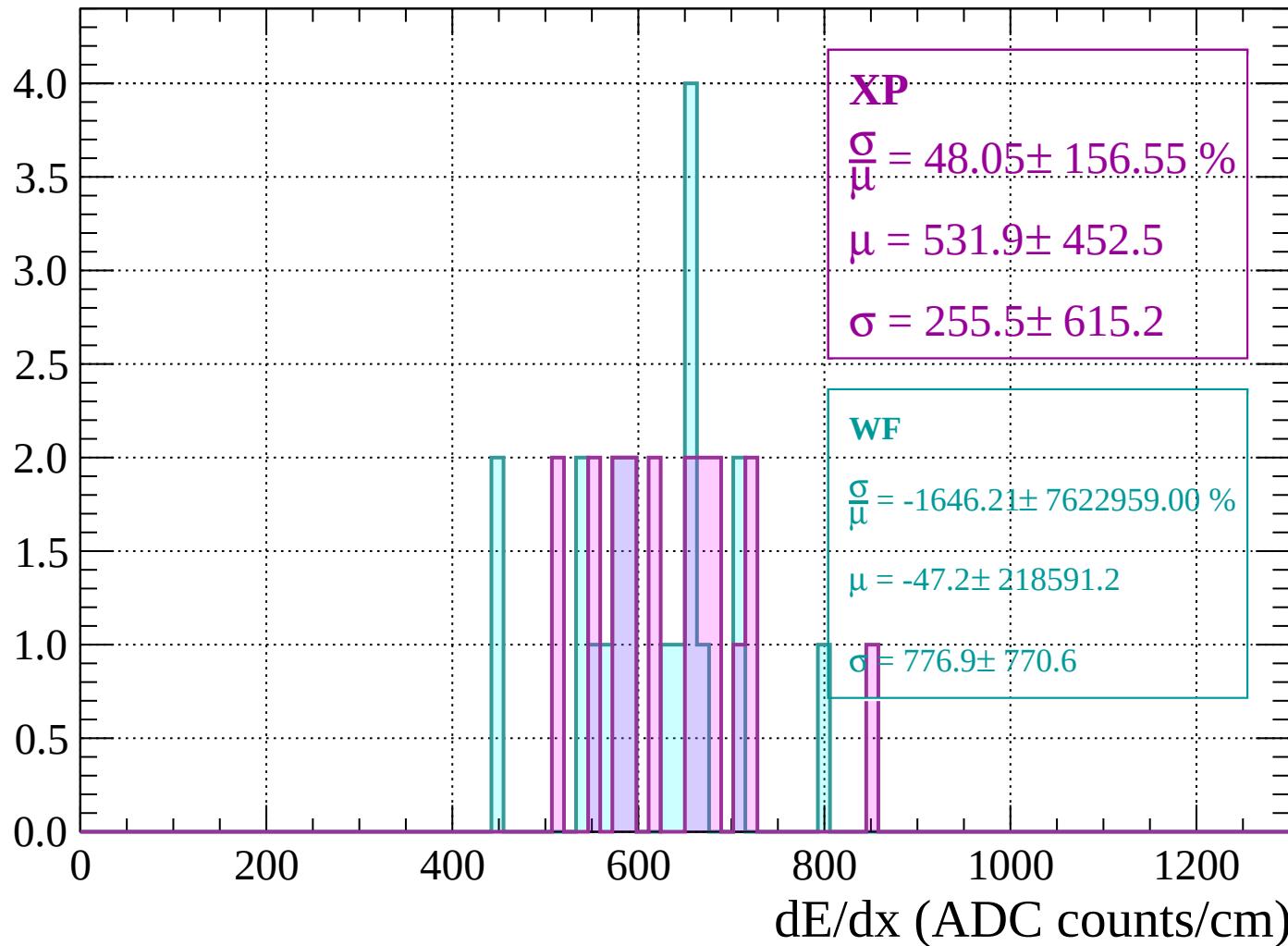


Energy loss | $80 < p < 120$



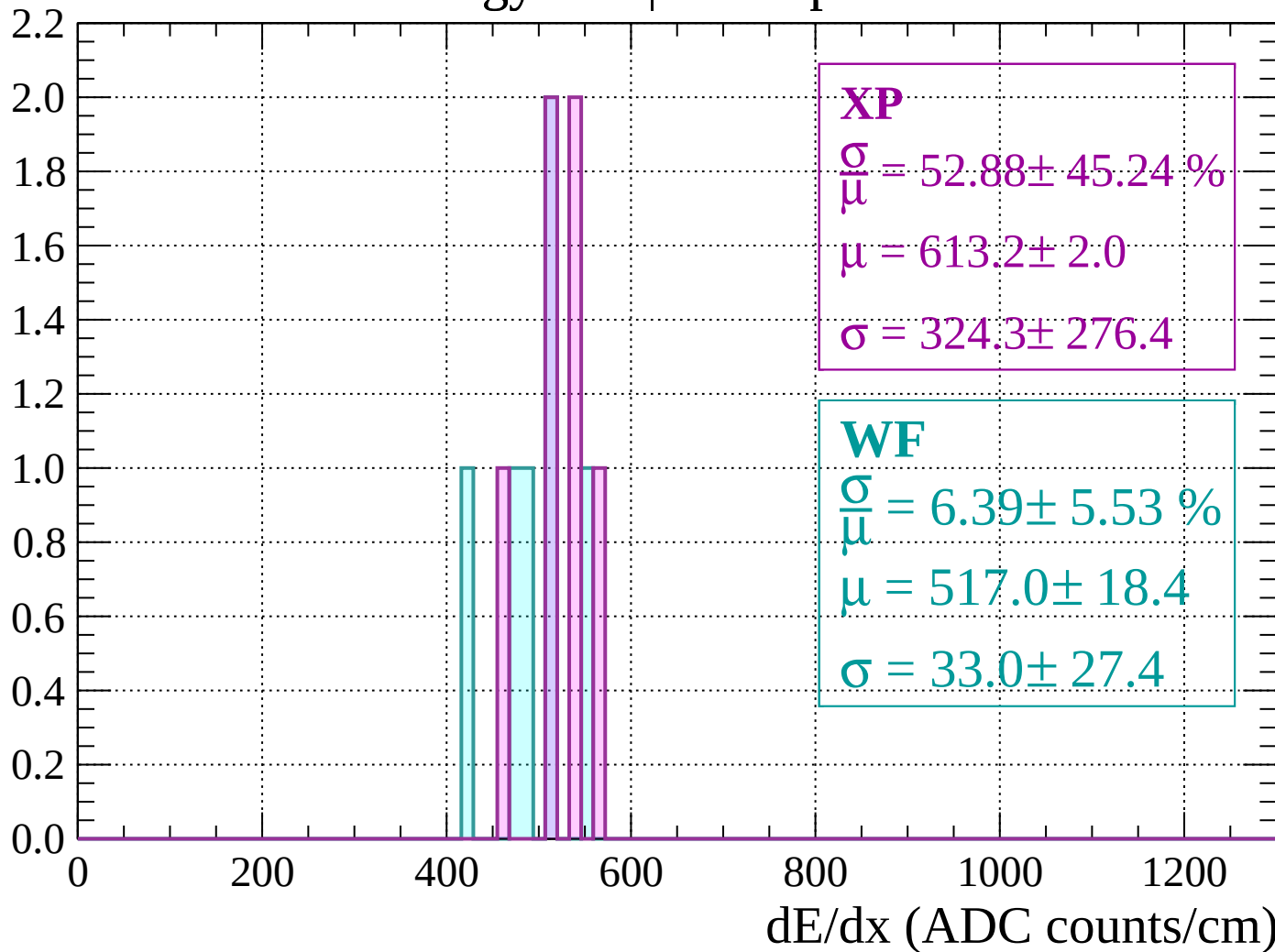
Energy loss | $120 < p < 160$

Count



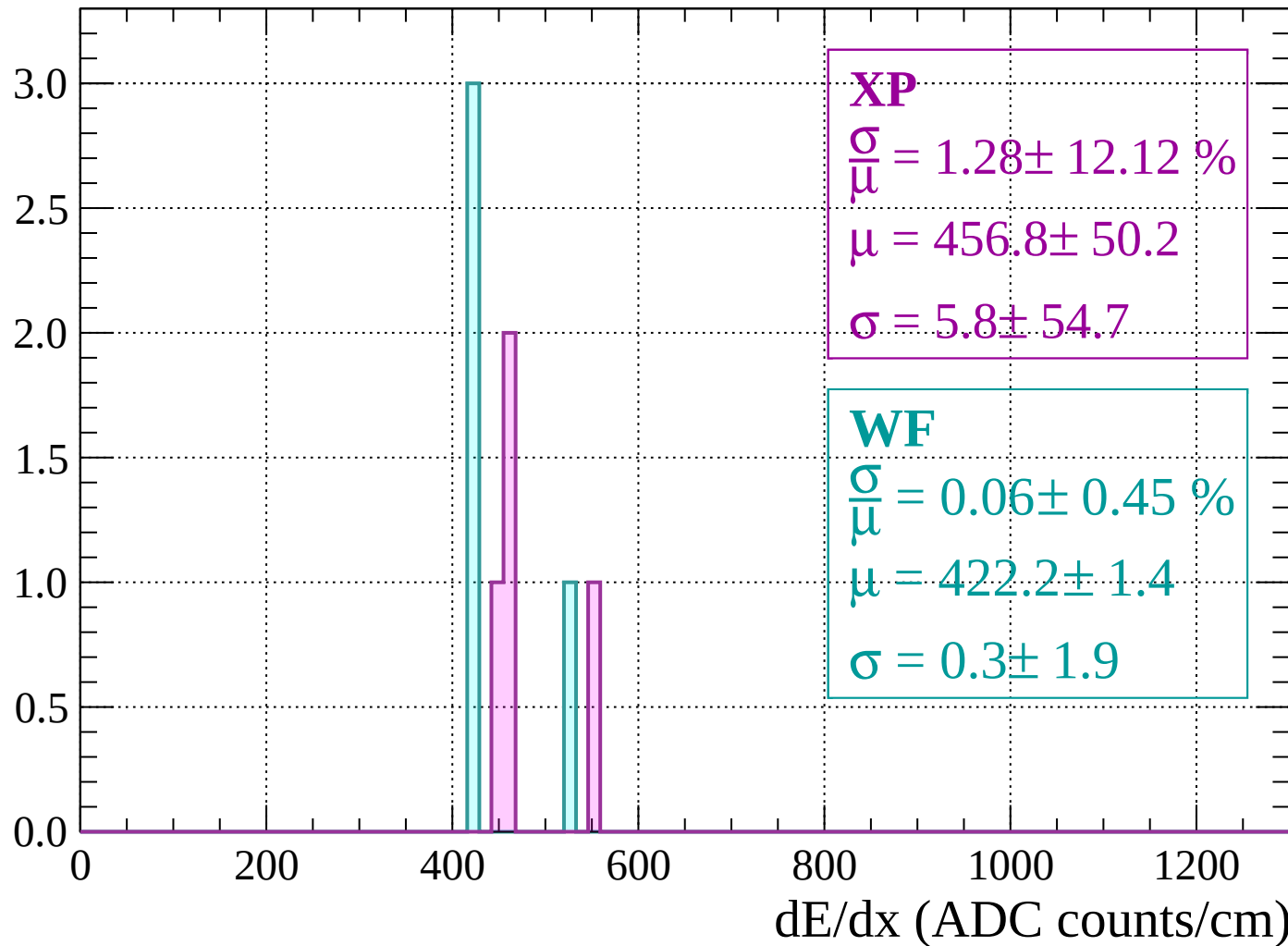
Energy loss | $160 < p < 200$

Count



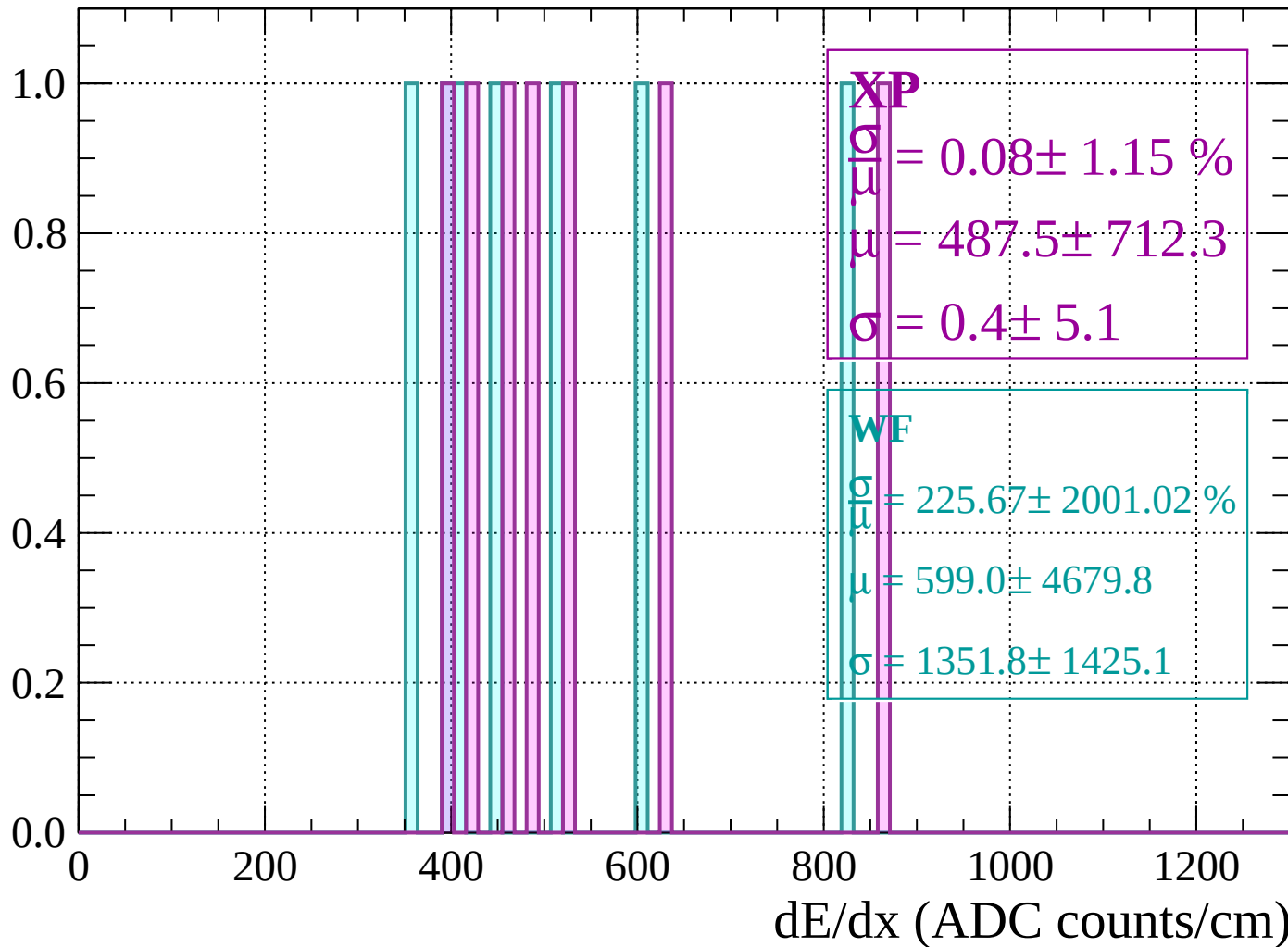
Energy loss | $200 < p < 240$

Count



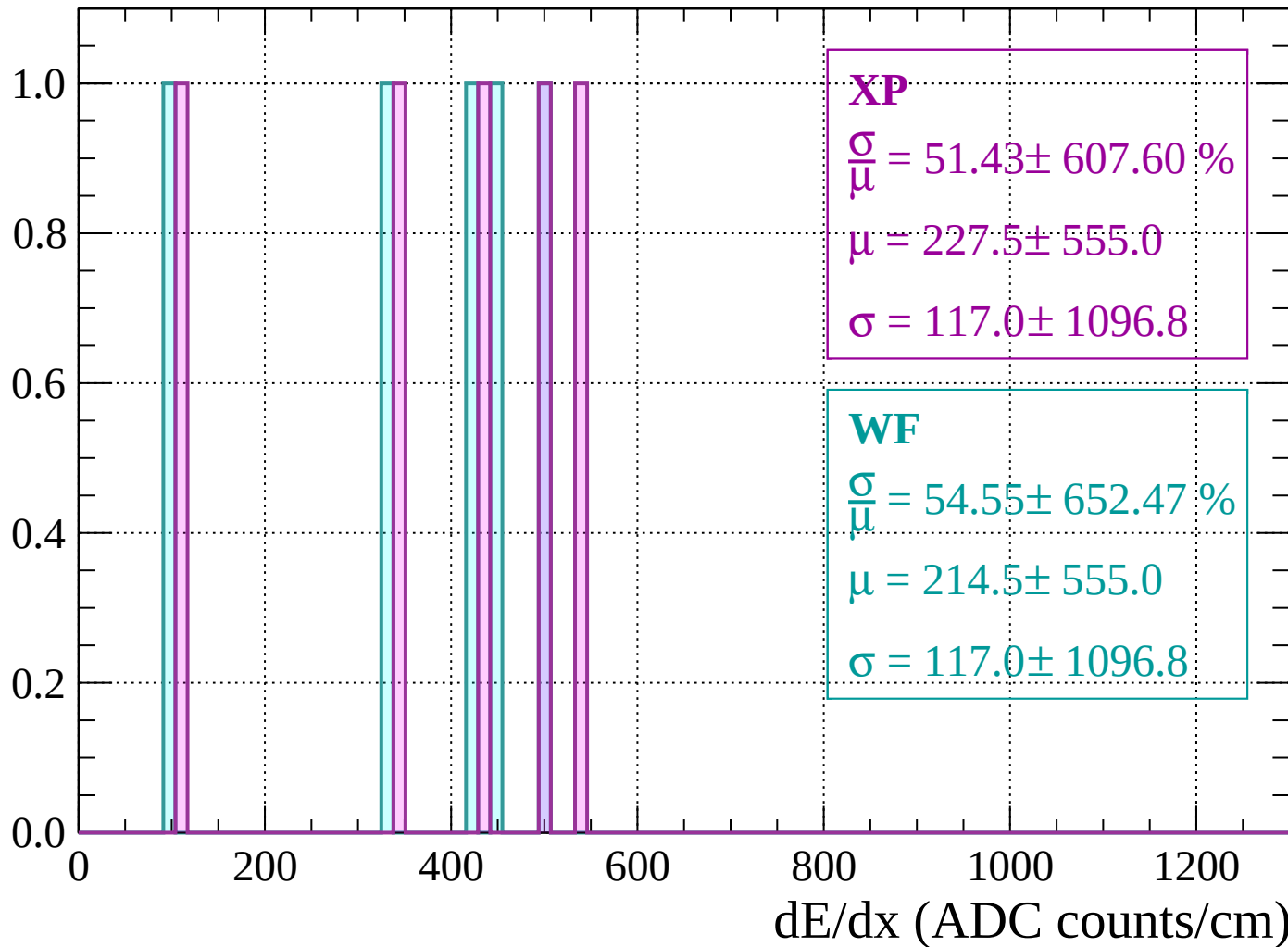
Energy loss | $240 < p < 280$

Count



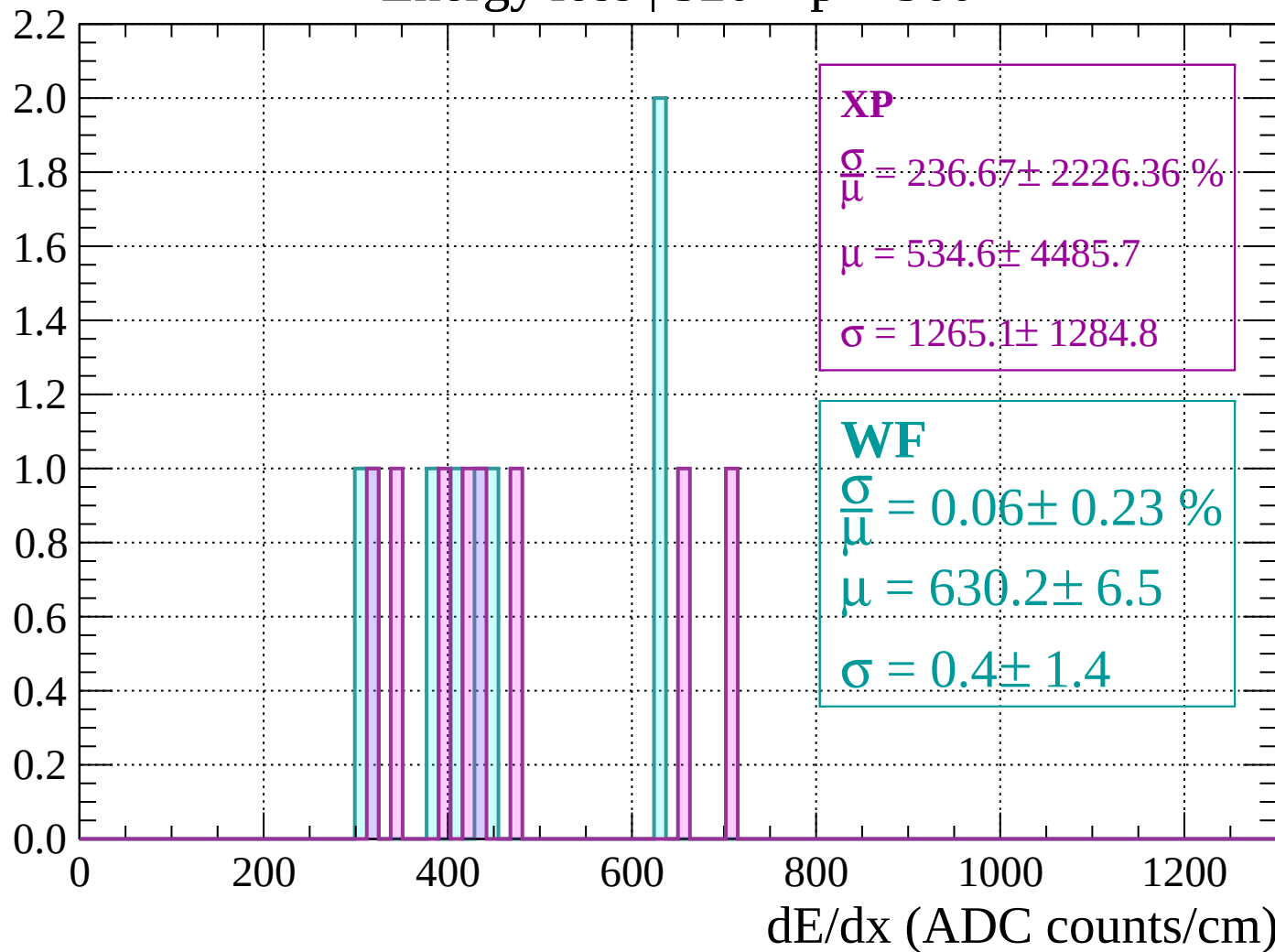
Energy loss | $280 < p < 320$

Count



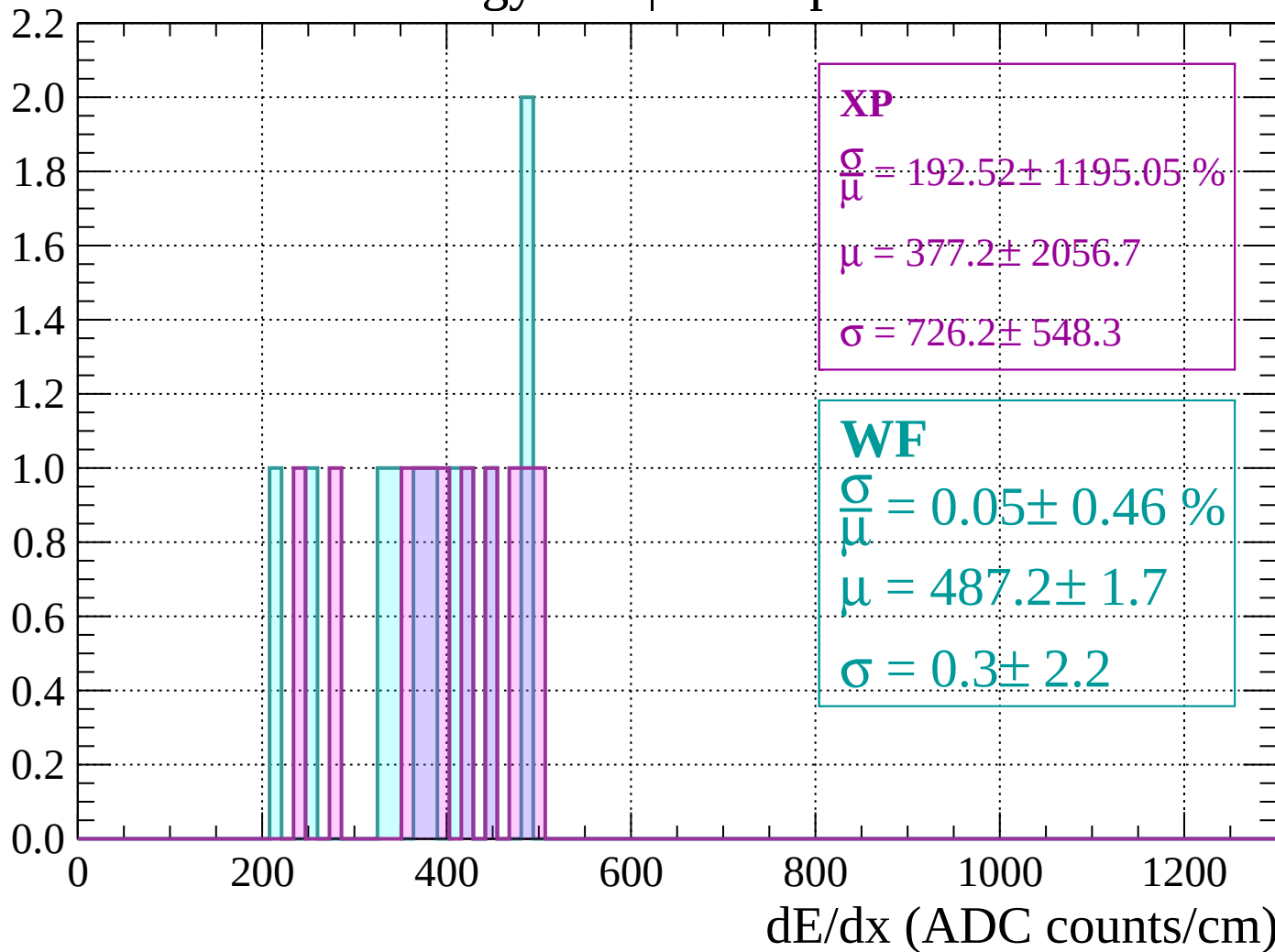
Energy loss | $320 < p < 360$

Count



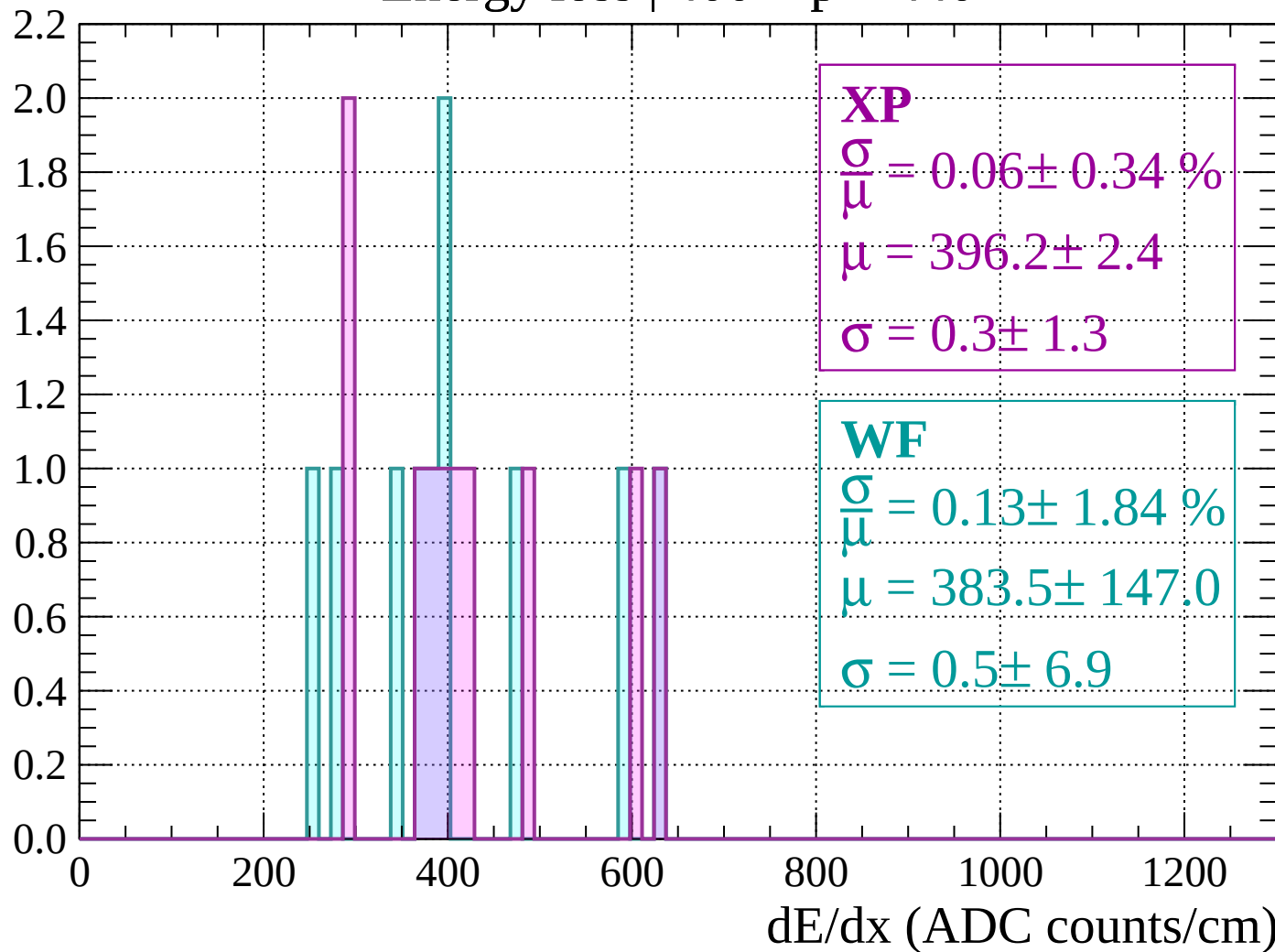
Energy loss | $360 < p < 400$

Count



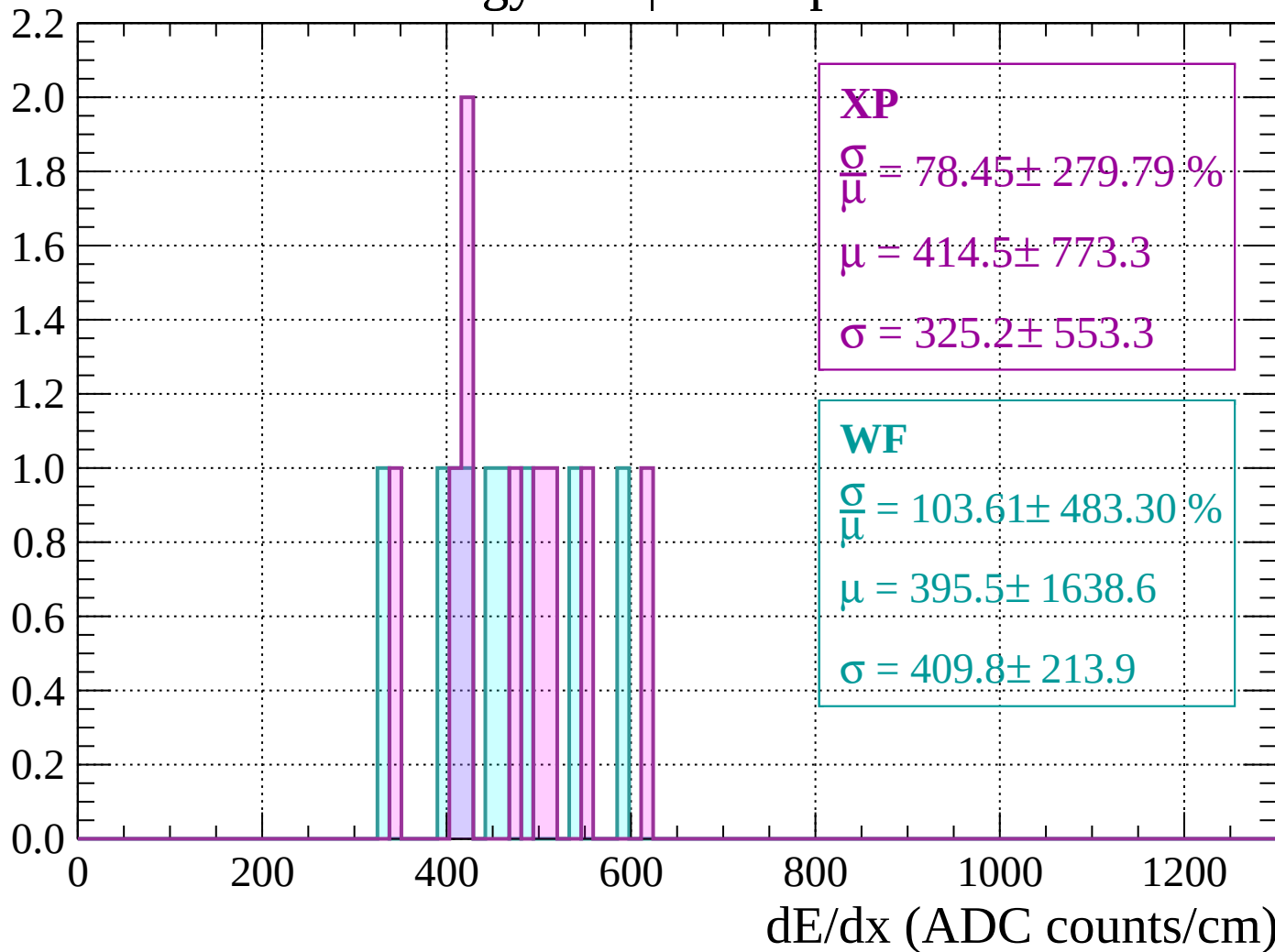
Energy loss | $400 < p < 440$

Count



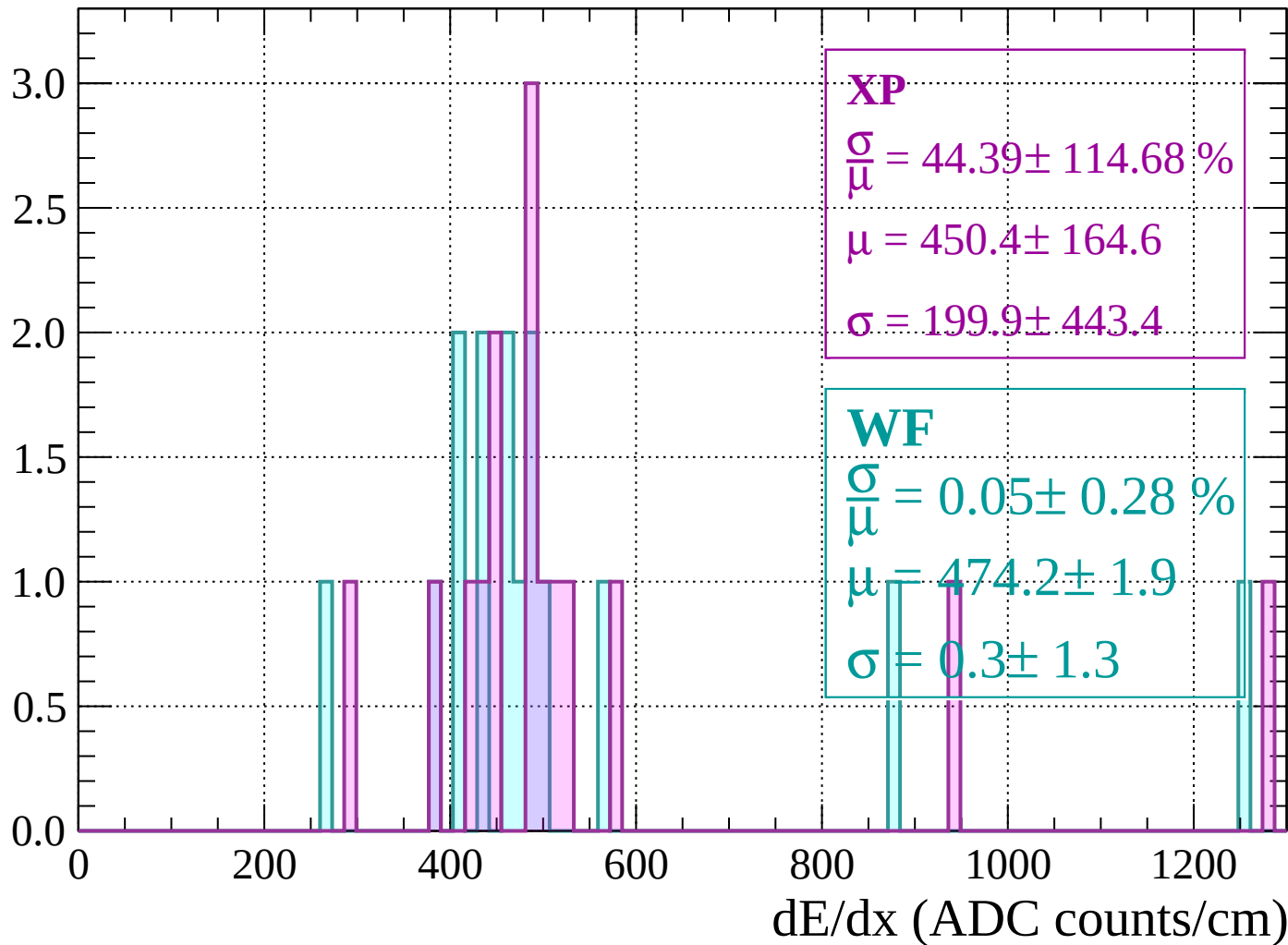
Energy loss | $440 < p < 480$

Count



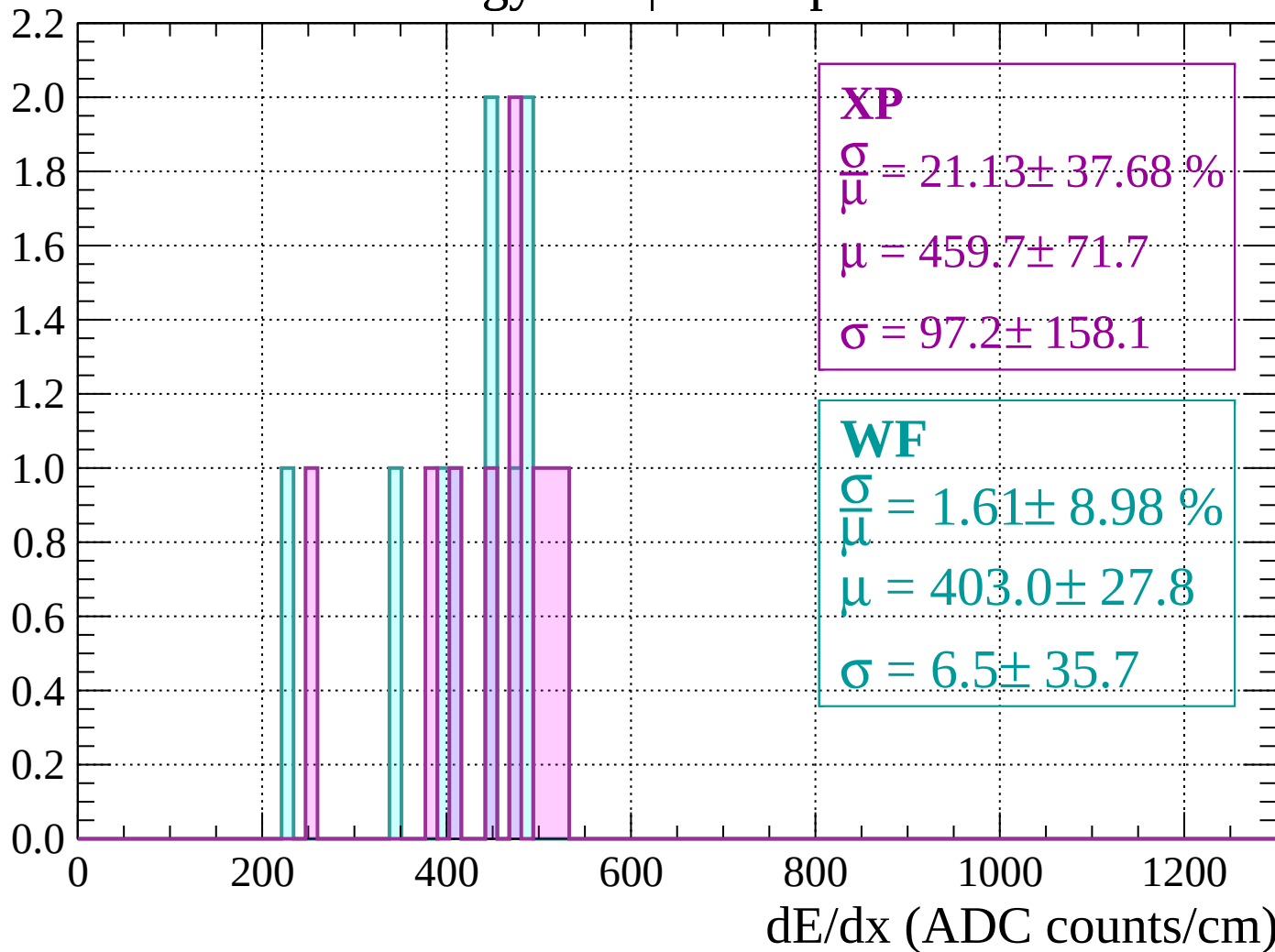
Energy loss | $480 < p < 520$

Count



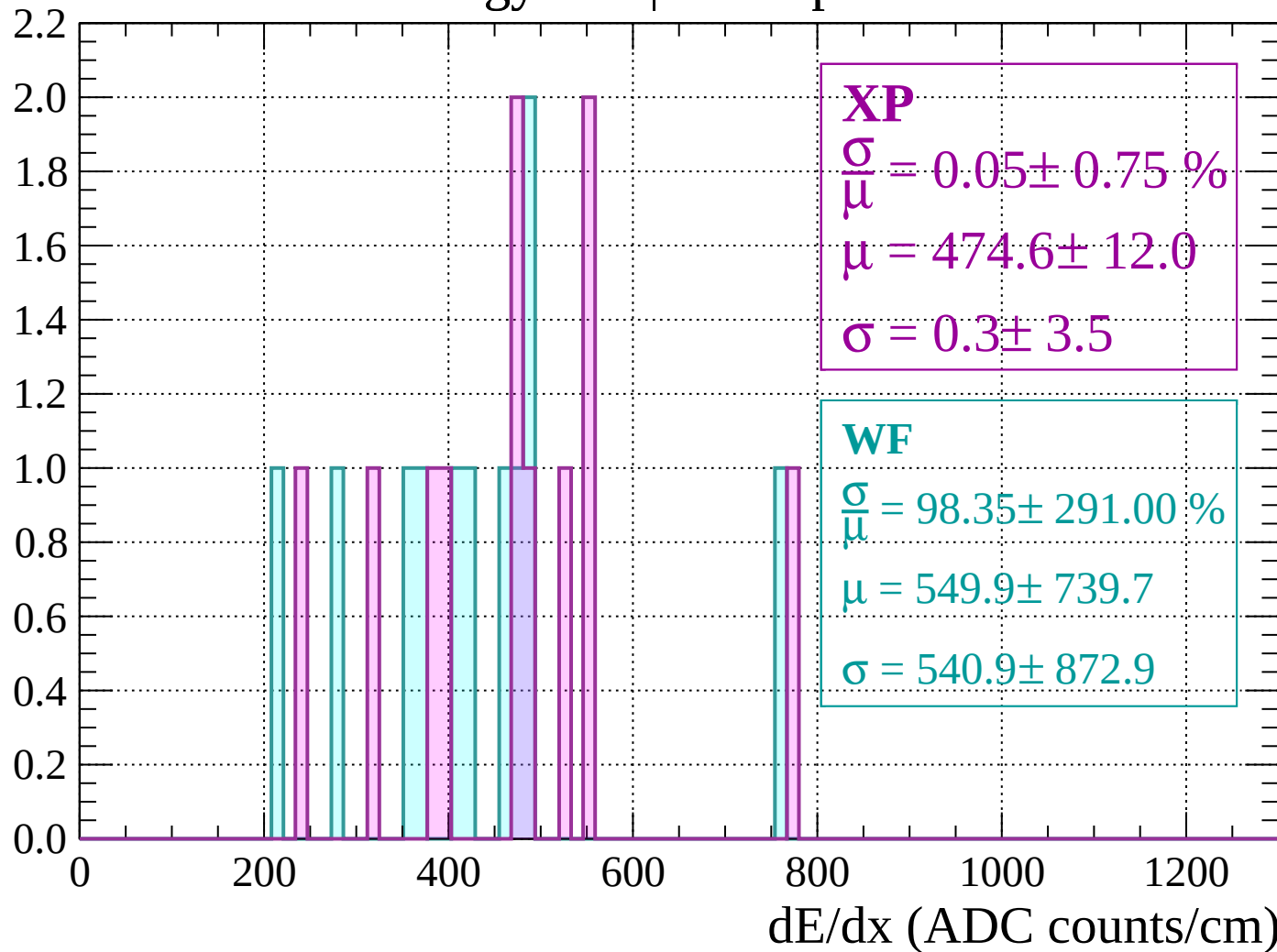
Energy loss | $520 < p < 560$

Count



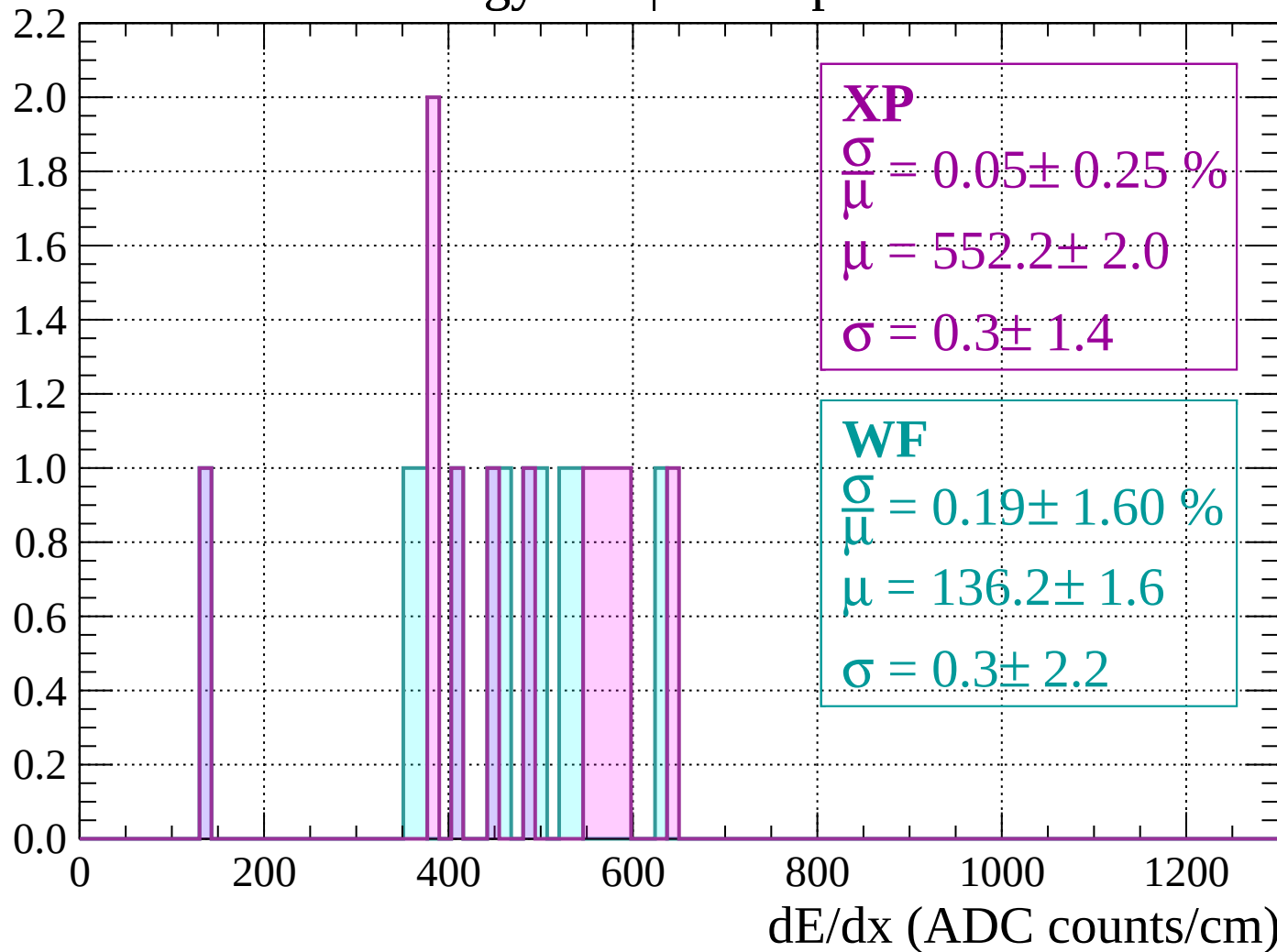
Energy loss | $560 < p < 600$

Count



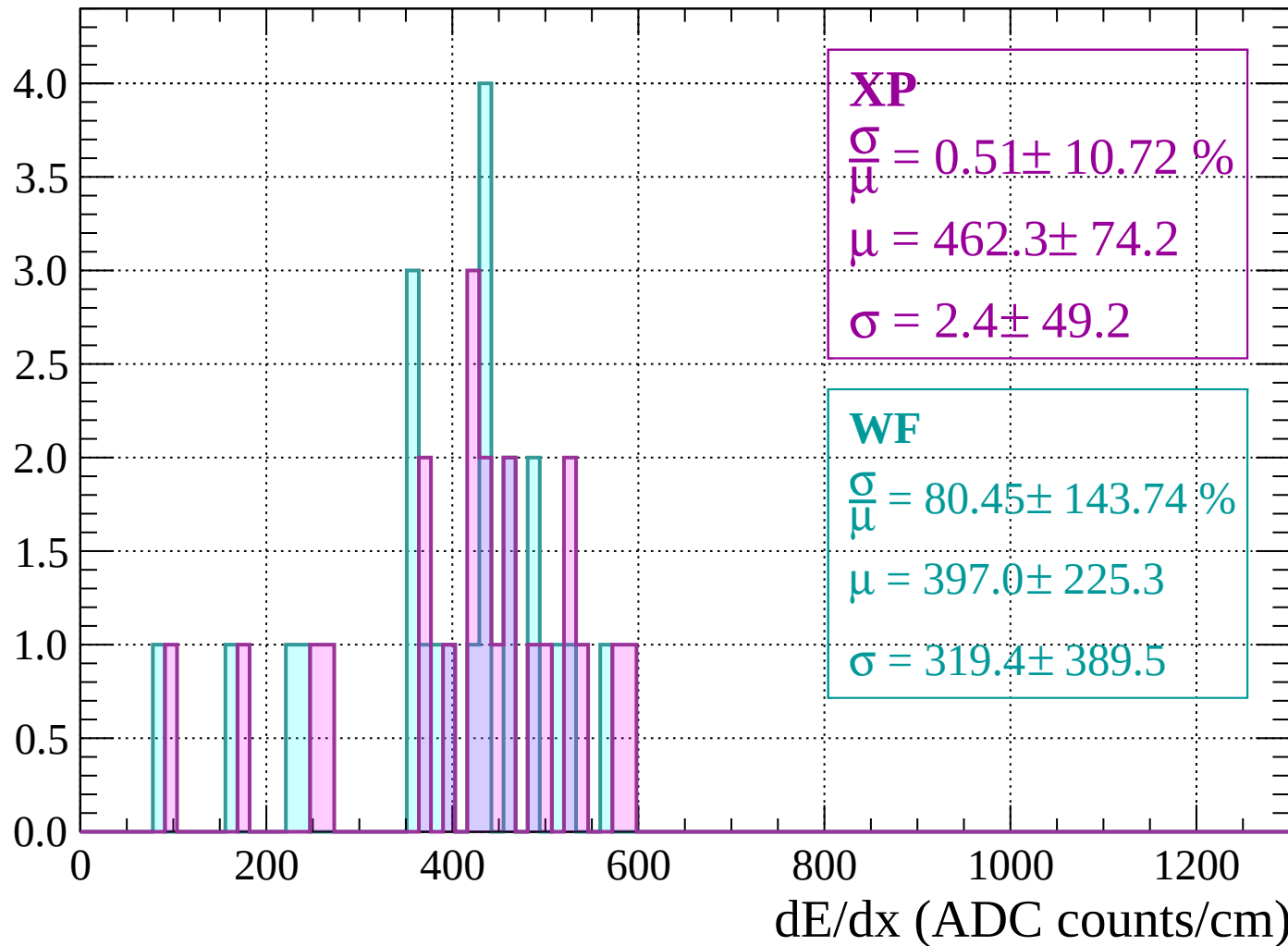
Energy loss | $600 < p < 640$

Count



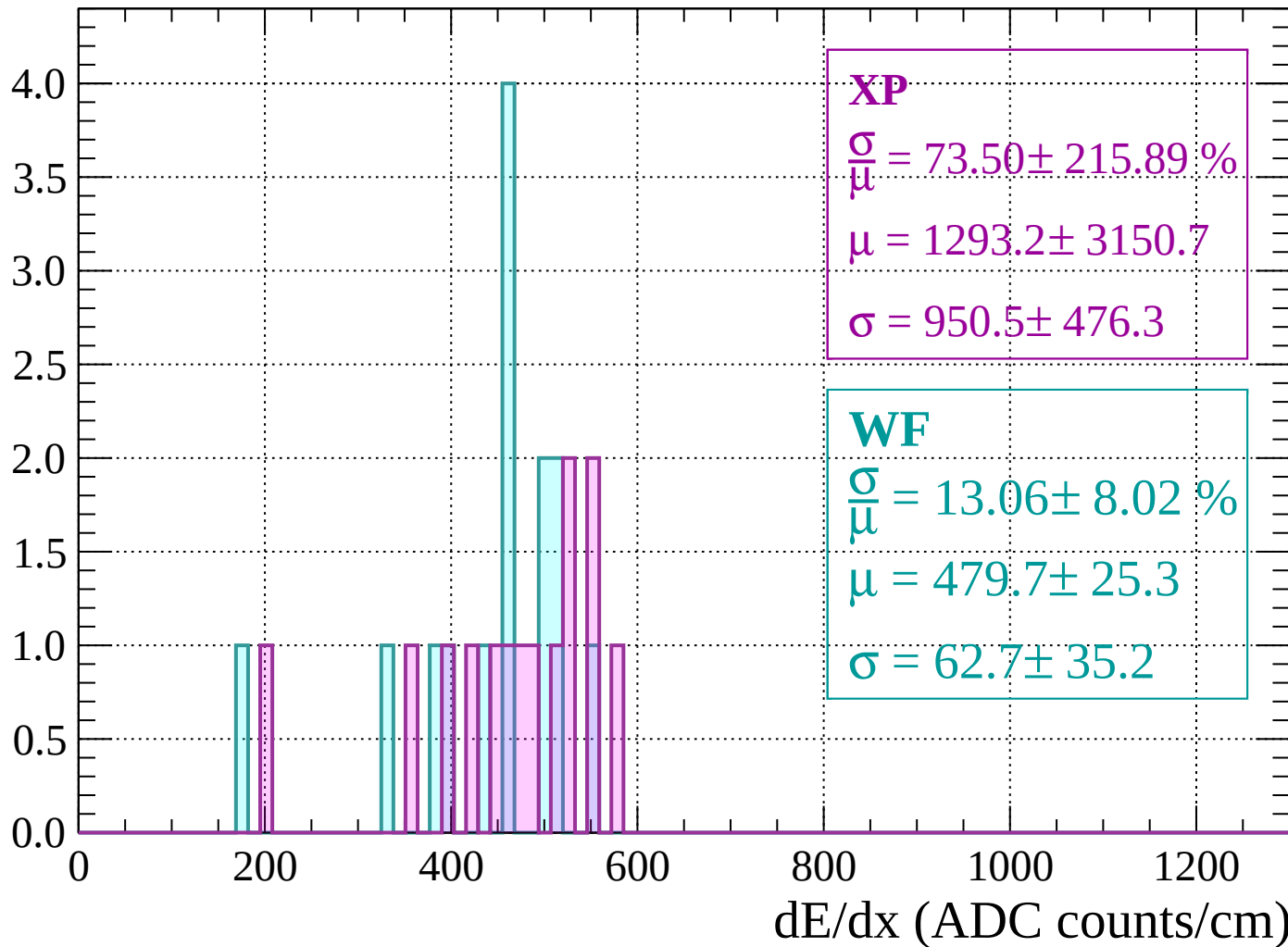
Energy loss | $640 < p < 680$

Count



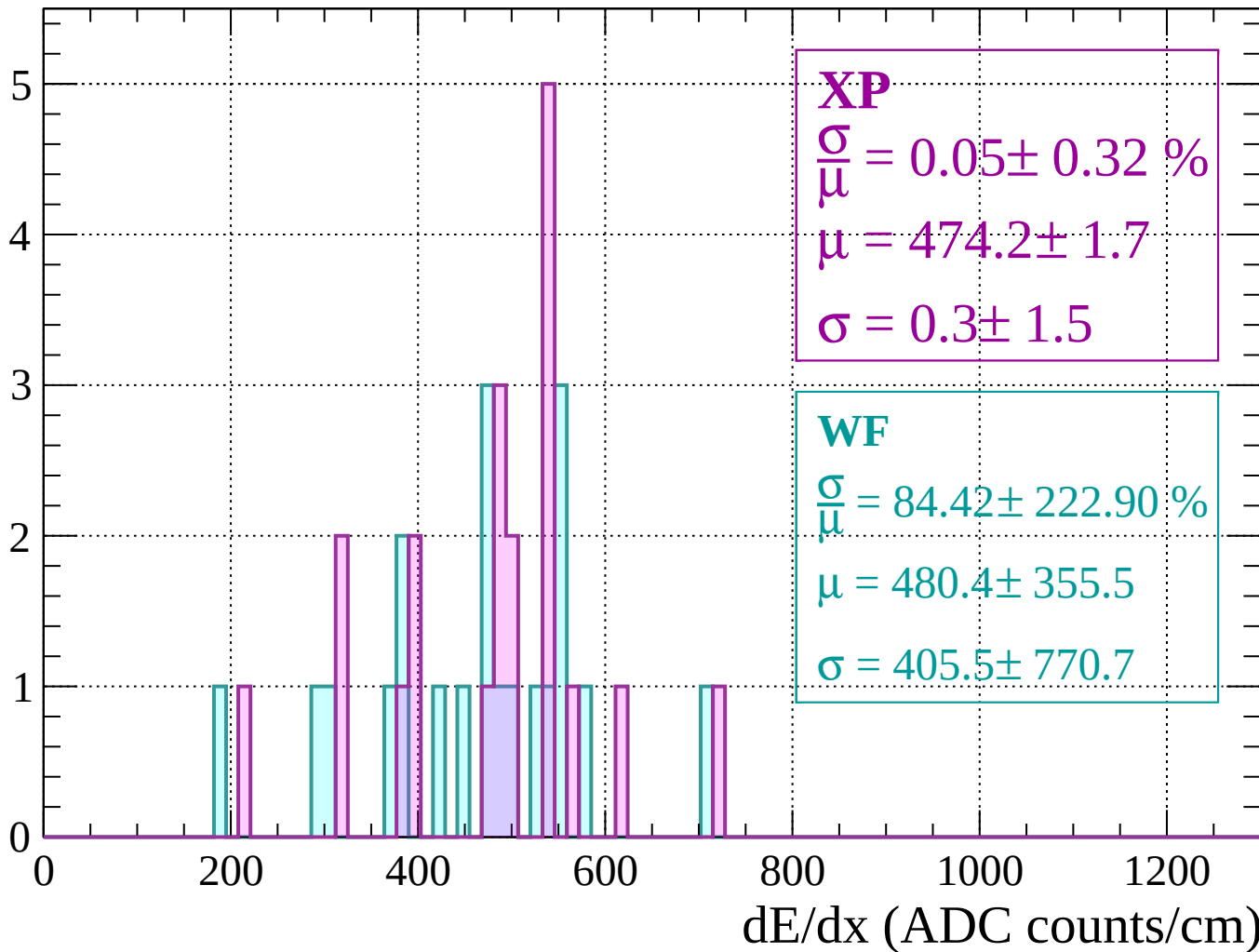
Energy loss | $680 < p < 720$

Count



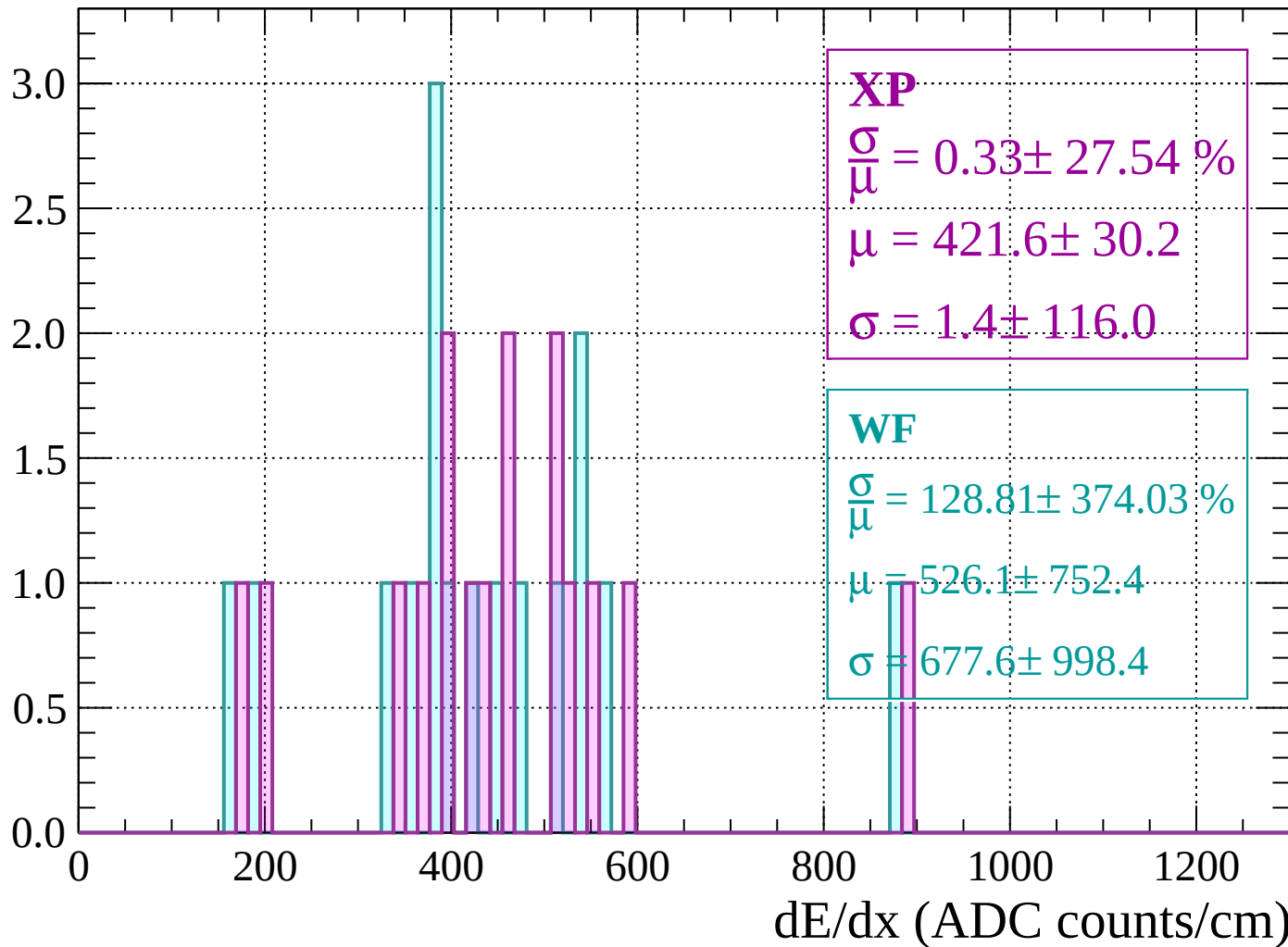
Energy loss | $720 < p < 760$

Count



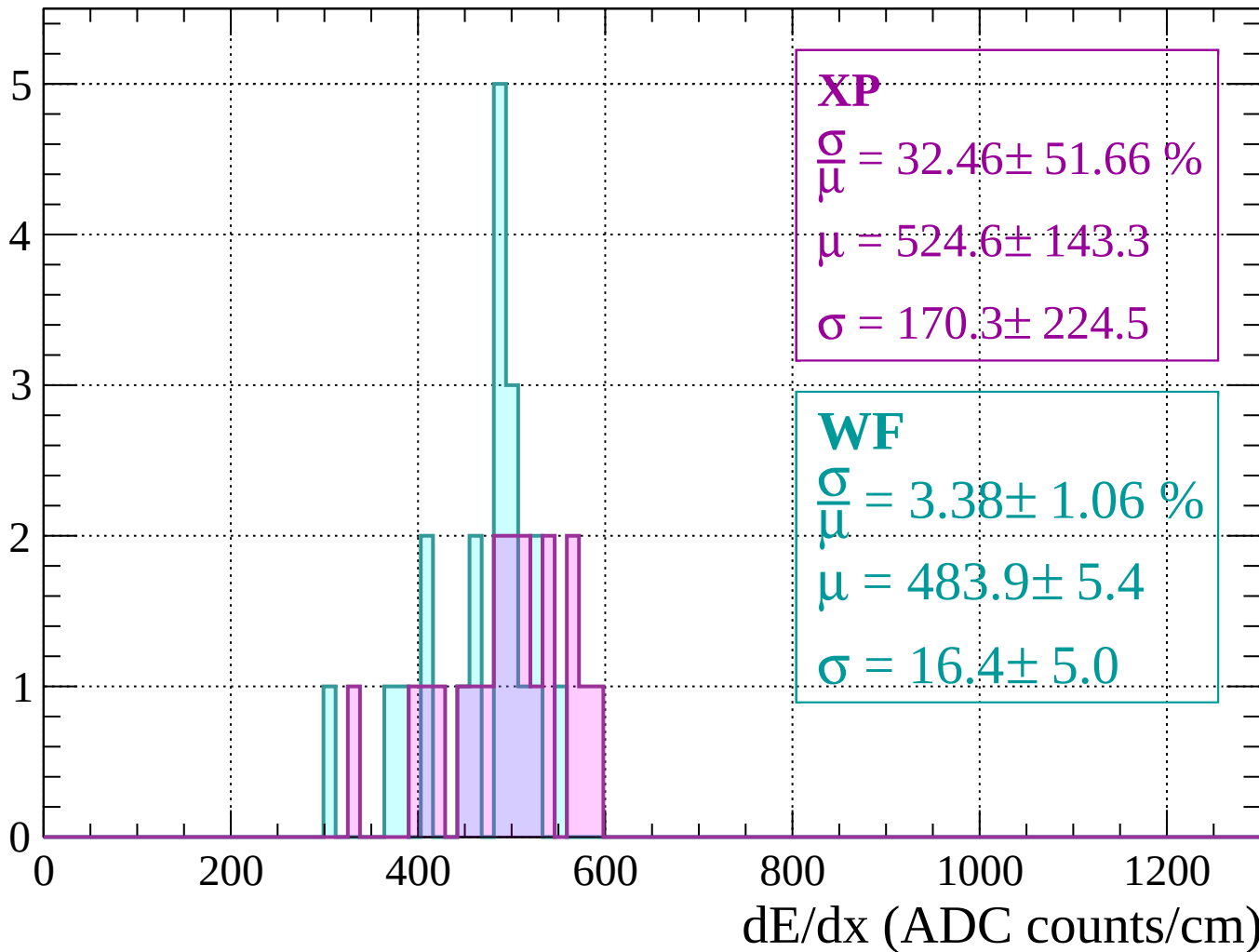
Energy loss | $760 < p < 800$

Count



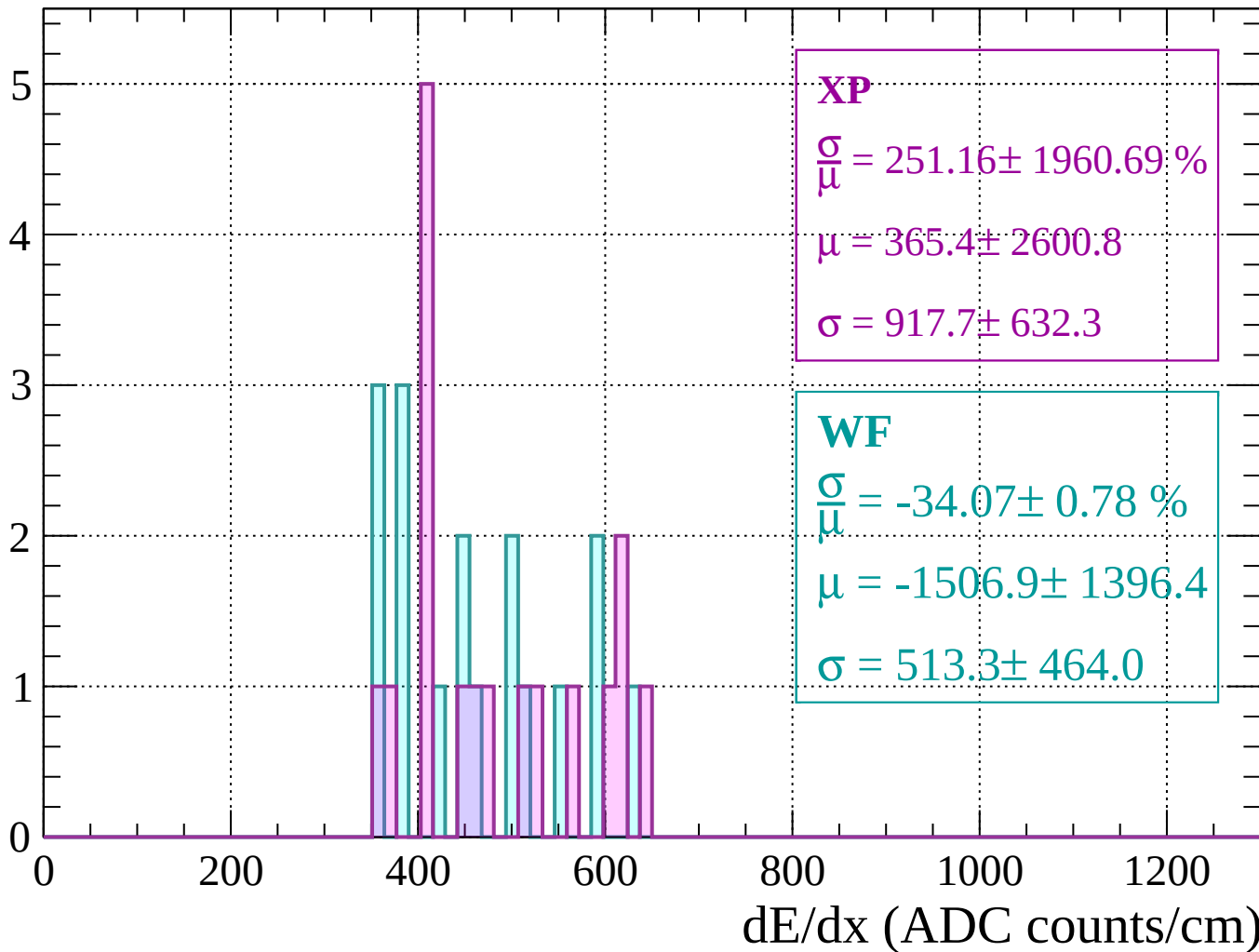
Energy loss | $800 < p < 840$

Count



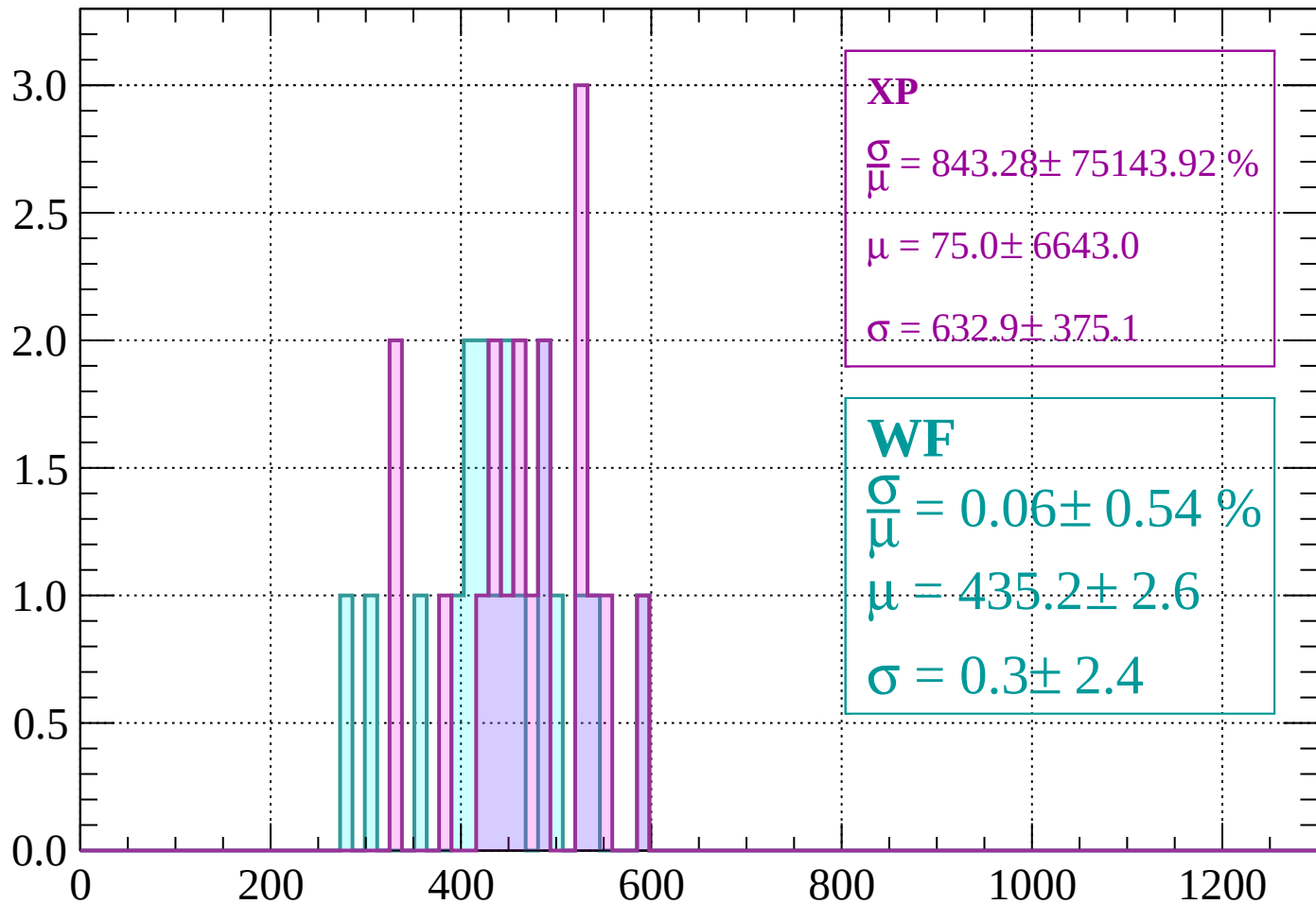
Energy loss | $840 < p < 880$

Count



Energy loss | $880 < p < 920$

Count



XP

$$\frac{\sigma}{\mu} = 843.28 \pm 75143.92 \%$$

$$\mu = 75.0 \pm 6643.0$$

$$\sigma = 632.9 \pm 375.1$$

WF

$$\frac{\sigma}{\mu} = 0.06 \pm 0.54 \%$$

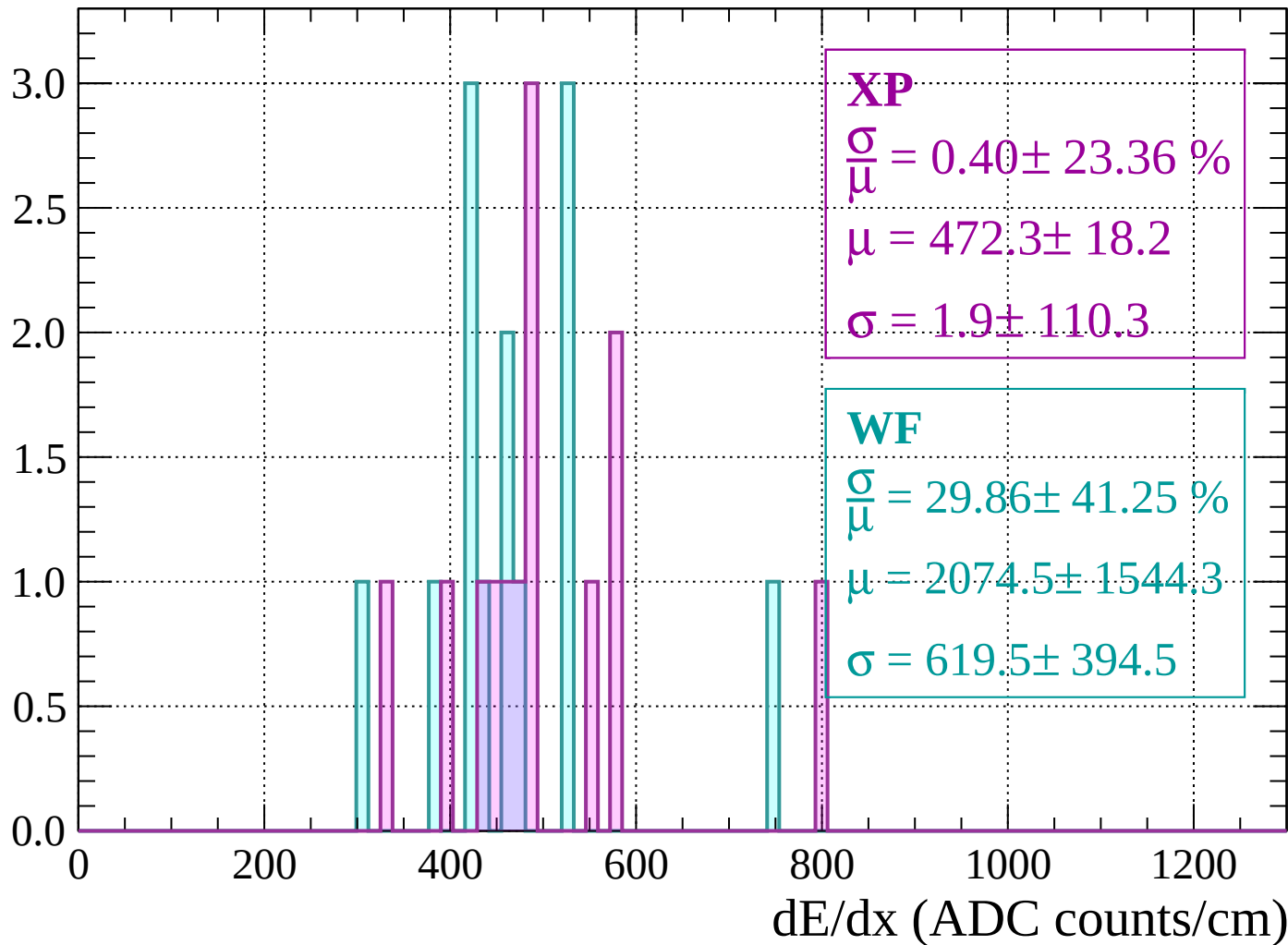
$$\mu = 435.2 \pm 2.6$$

$$\sigma = 0.3 \pm 2.4$$

dE/dx (ADC counts/cm)

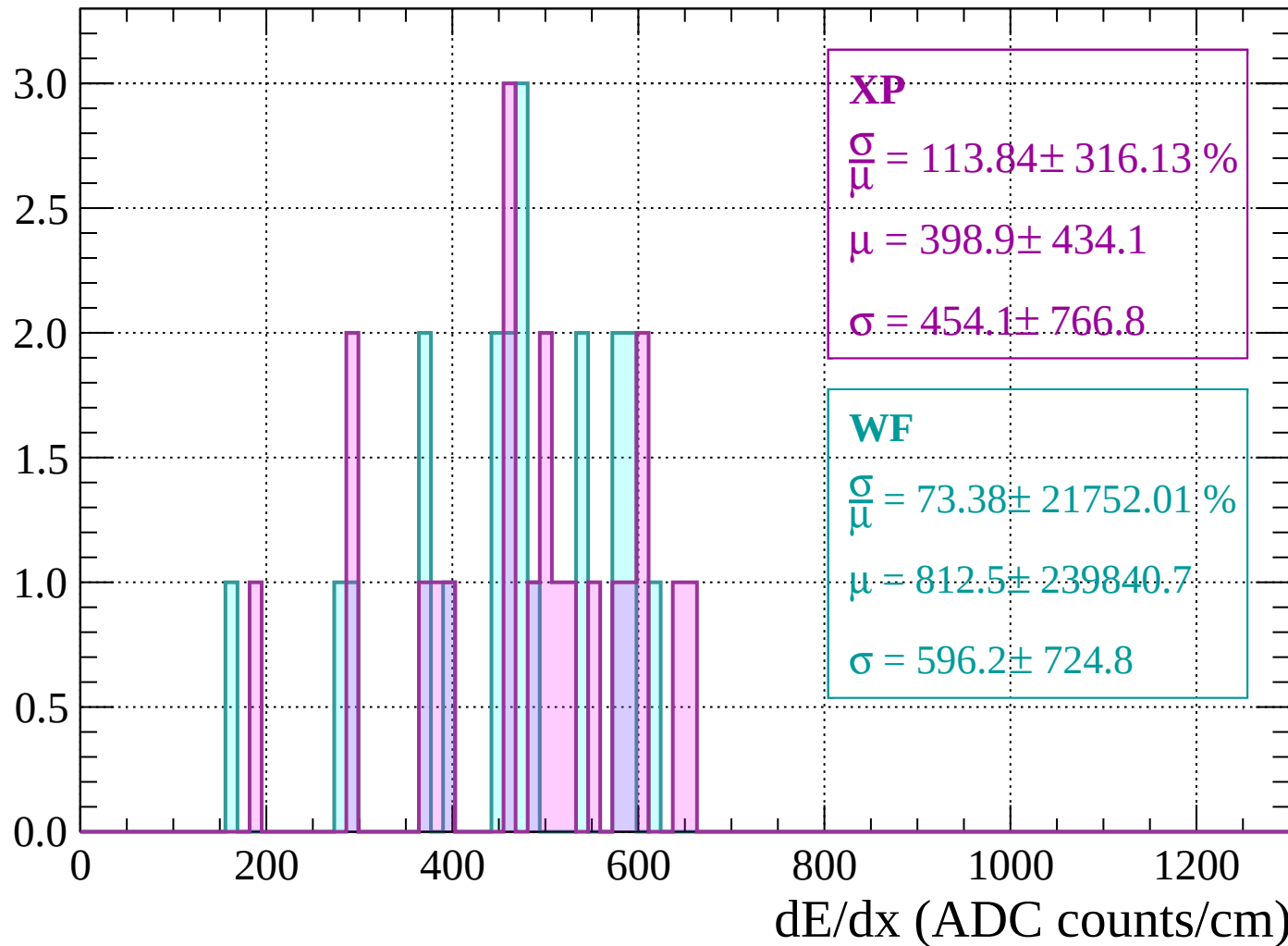
Energy loss | $920 < p < 960$

Count



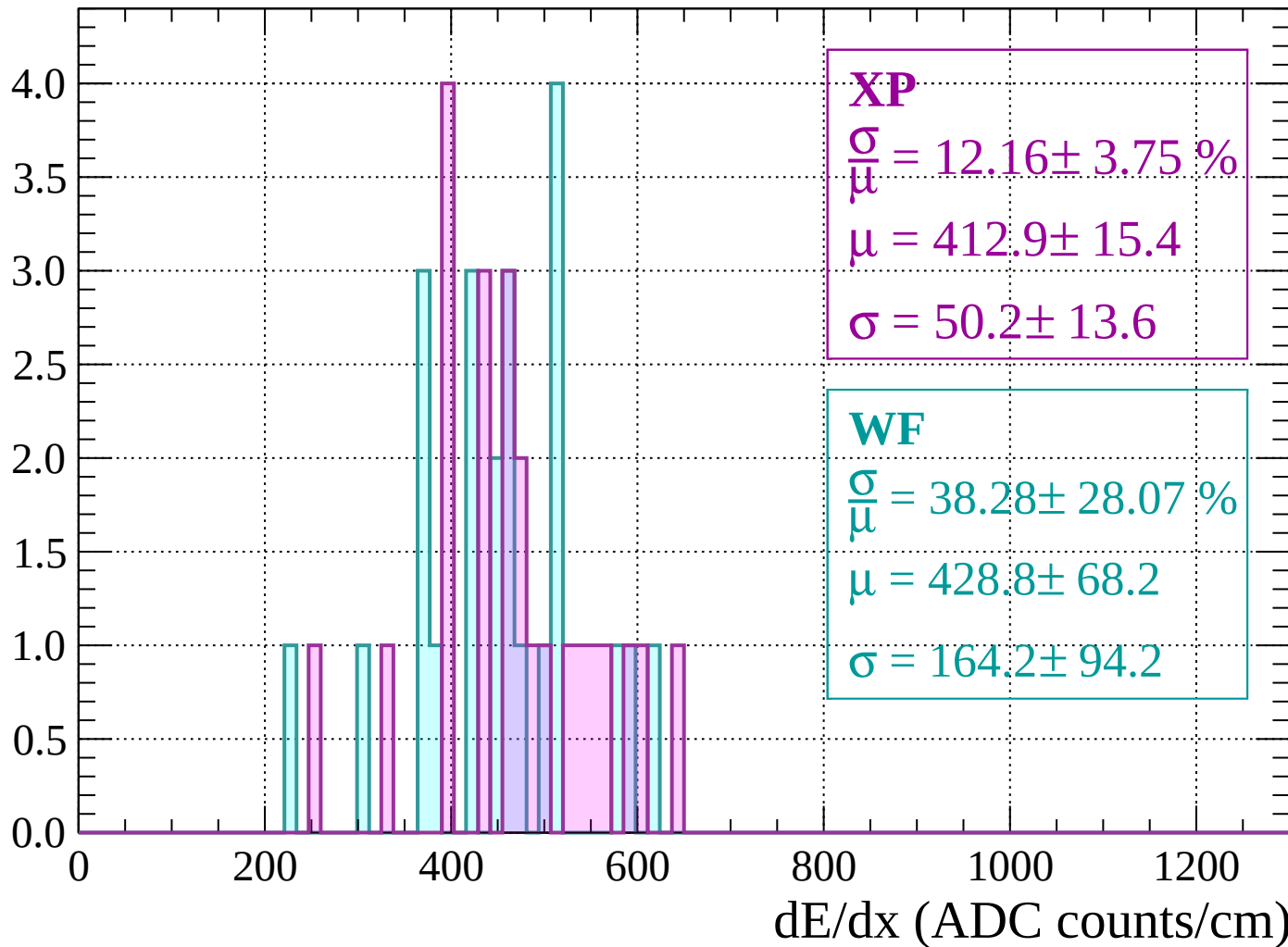
Energy loss | $960 < p < 1000$

Count



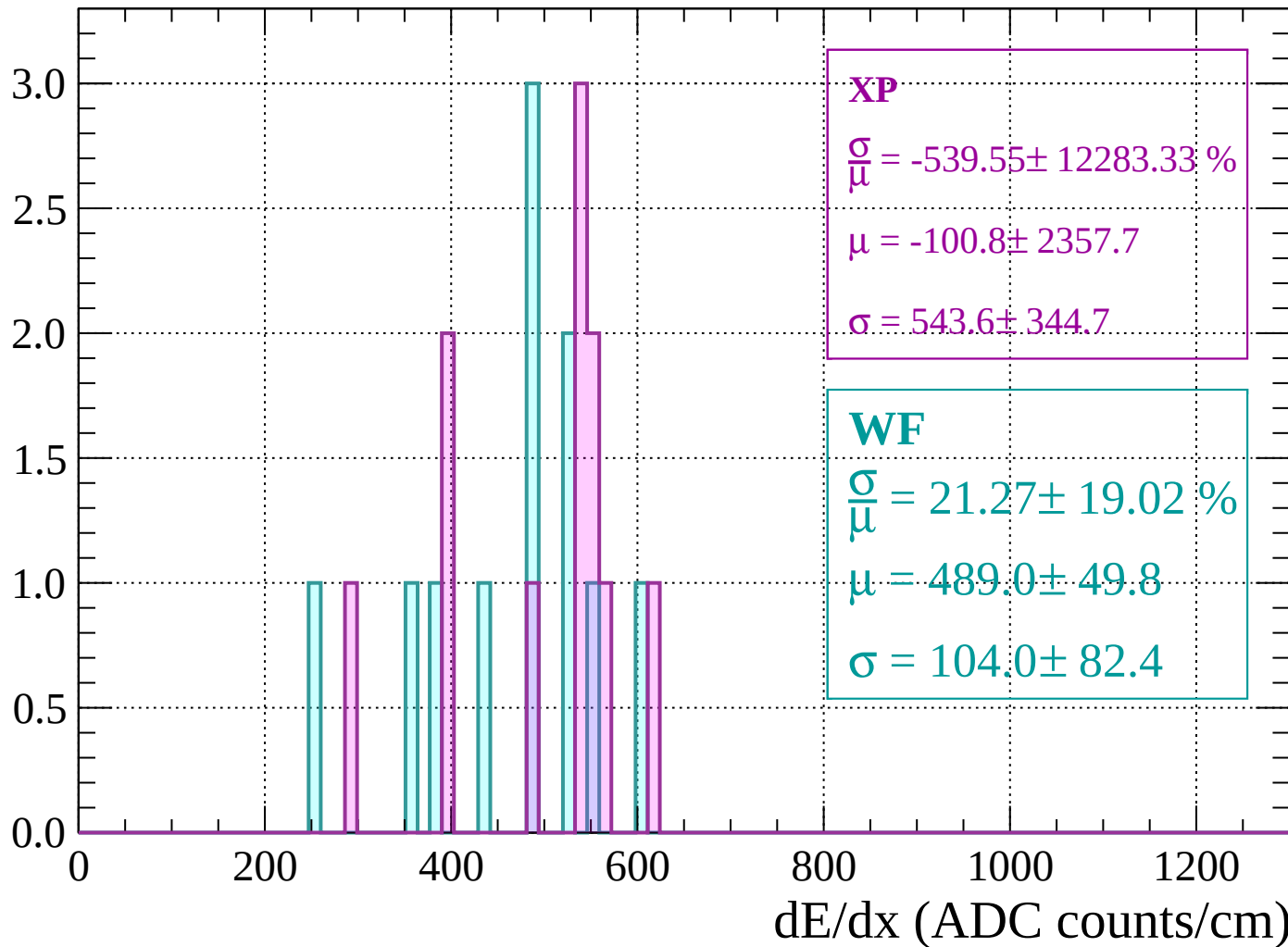
Energy loss | $1000 < p < 1040$

Count



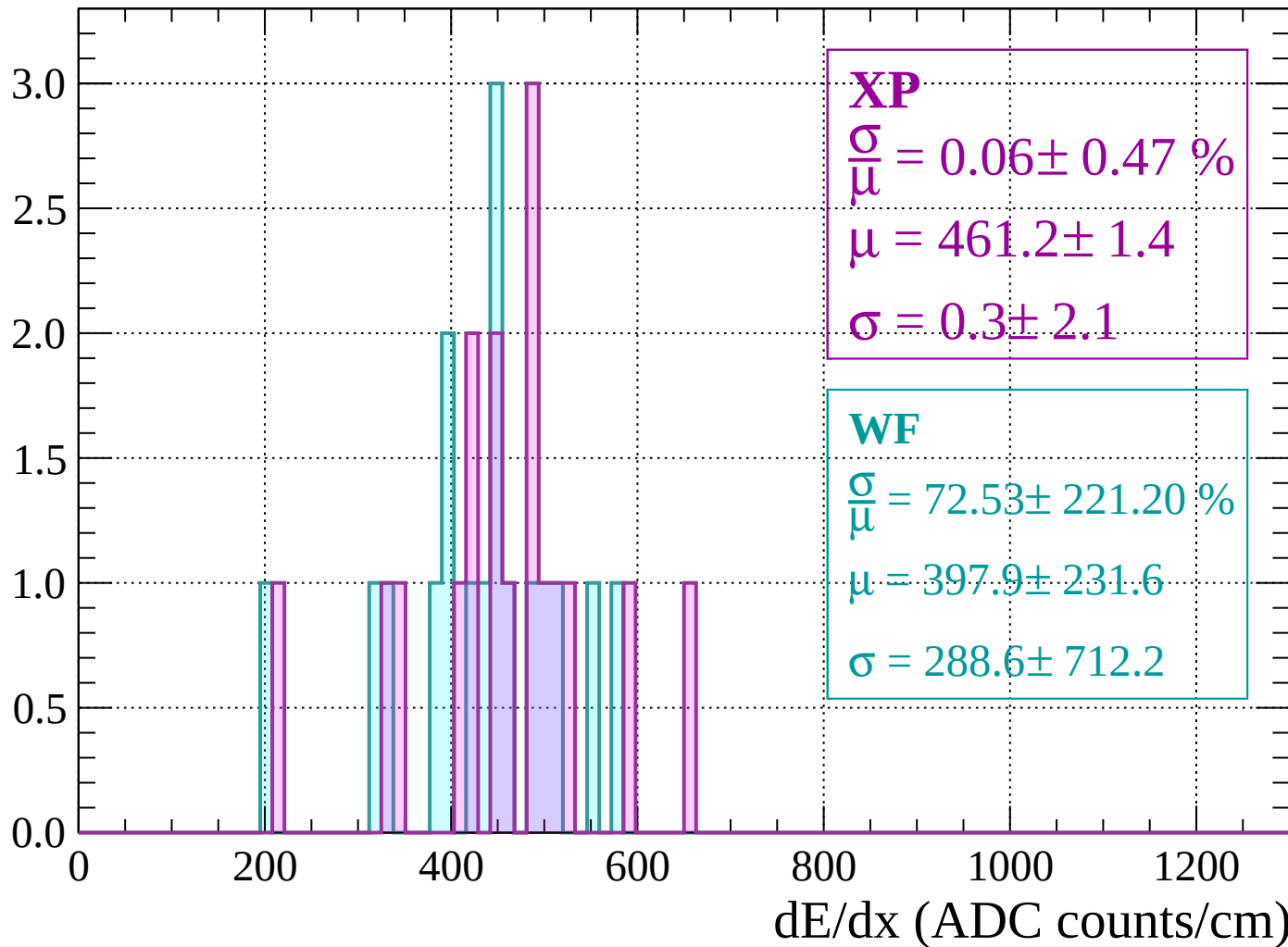
Energy loss | $1040 < p < 1080$

Count



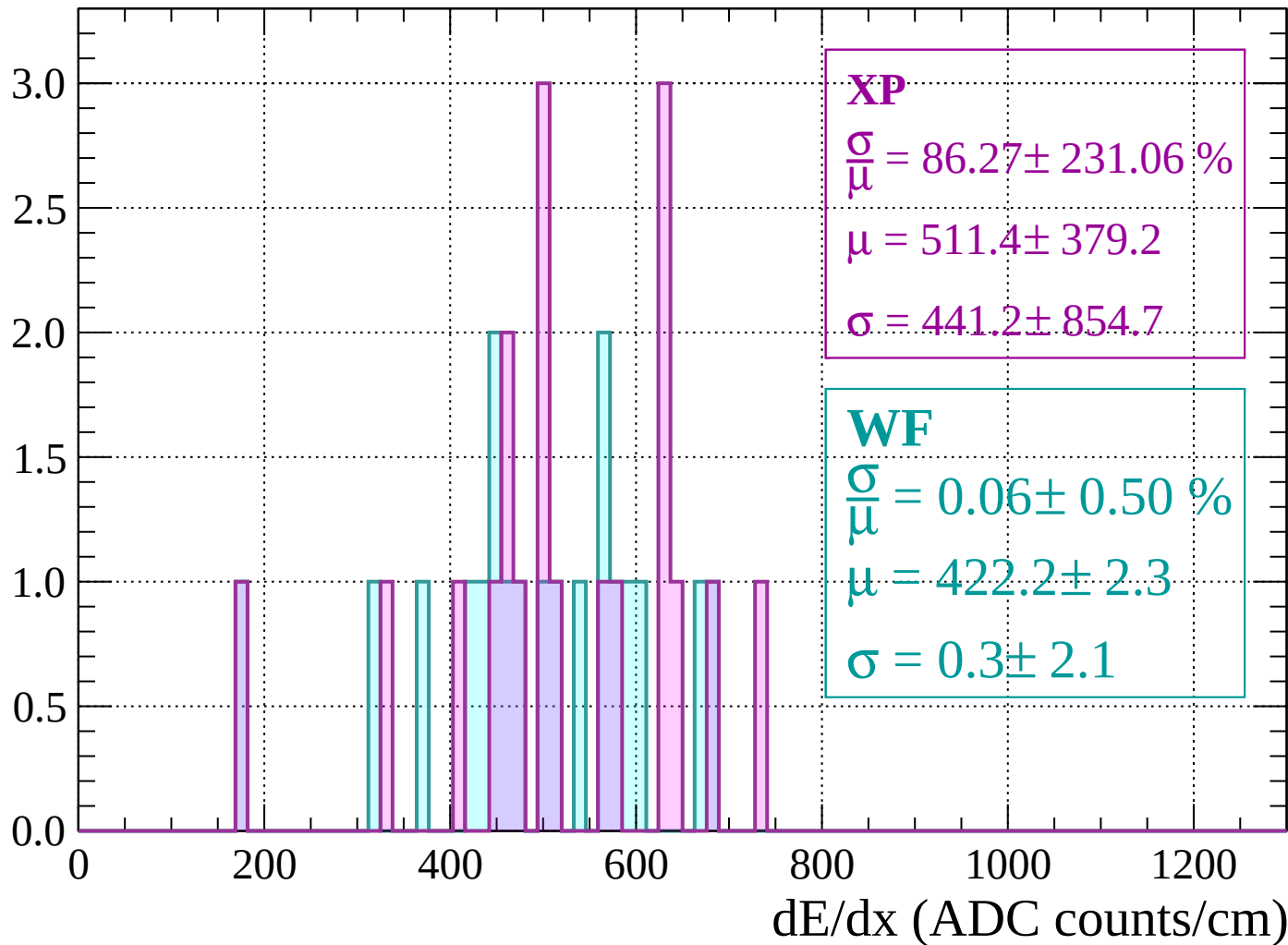
Energy loss | $1080 < p < 1120$

Count



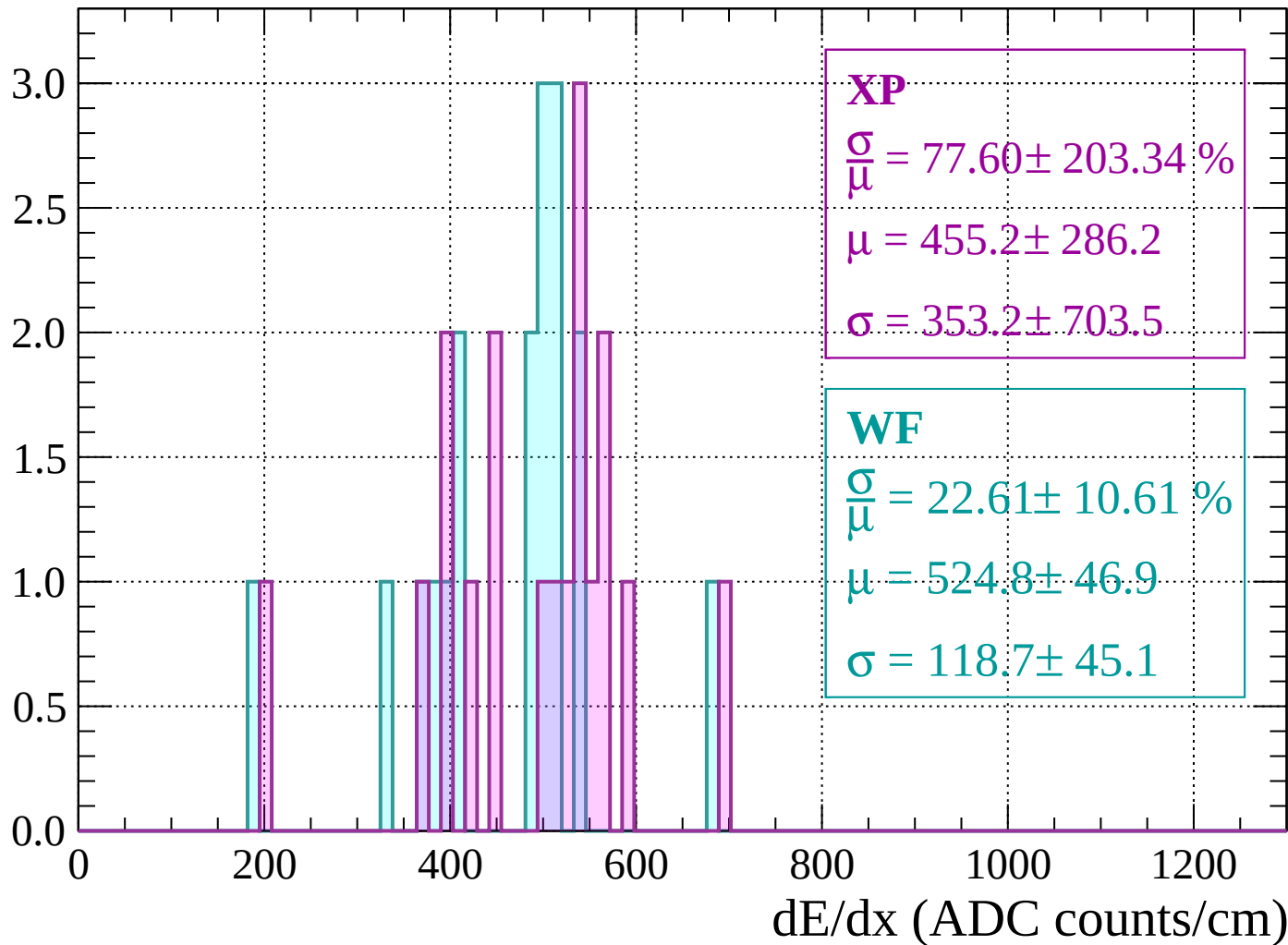
Energy loss | $1120 < p < 1160$

Count



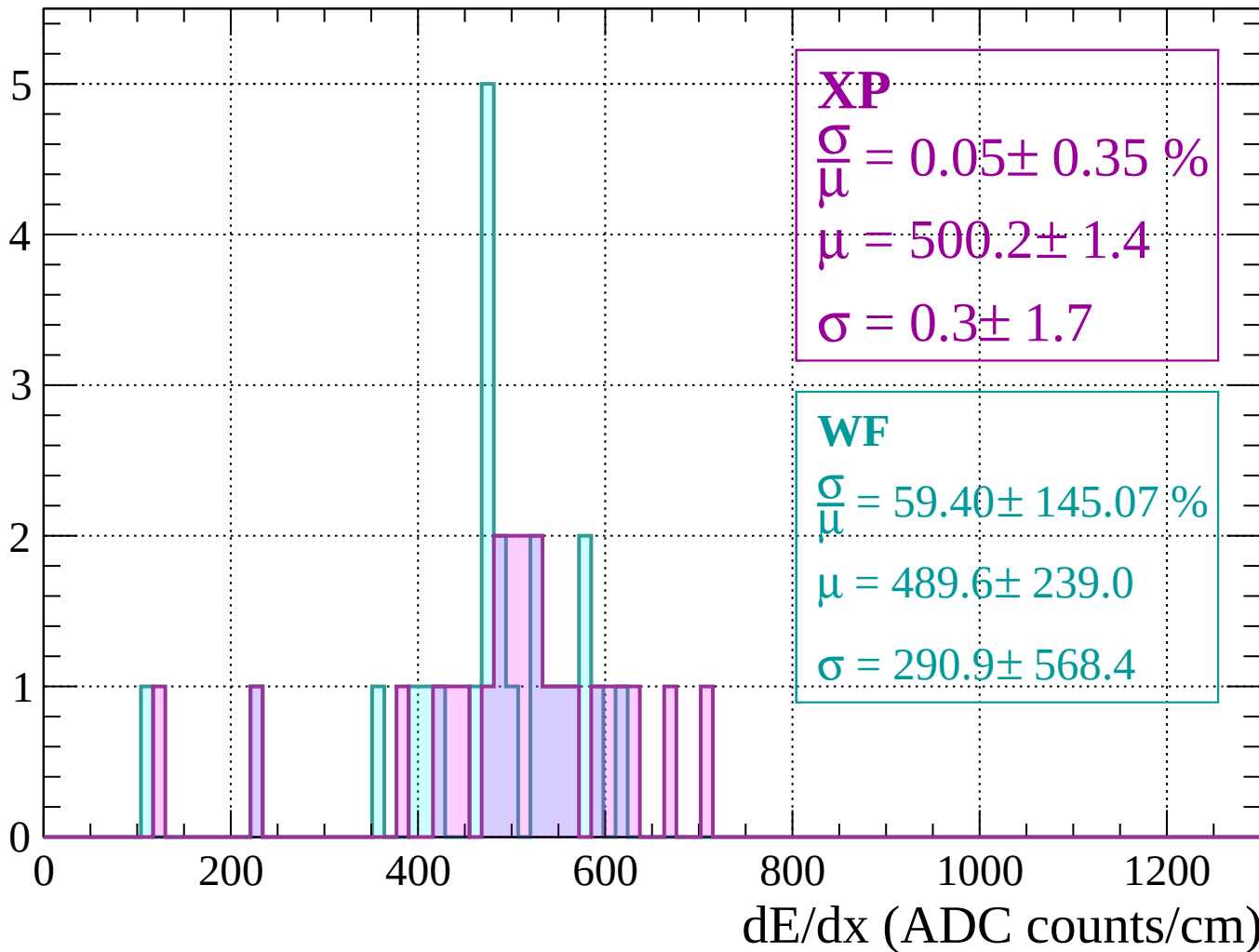
Energy loss | $1160 < p < 1200$

Count



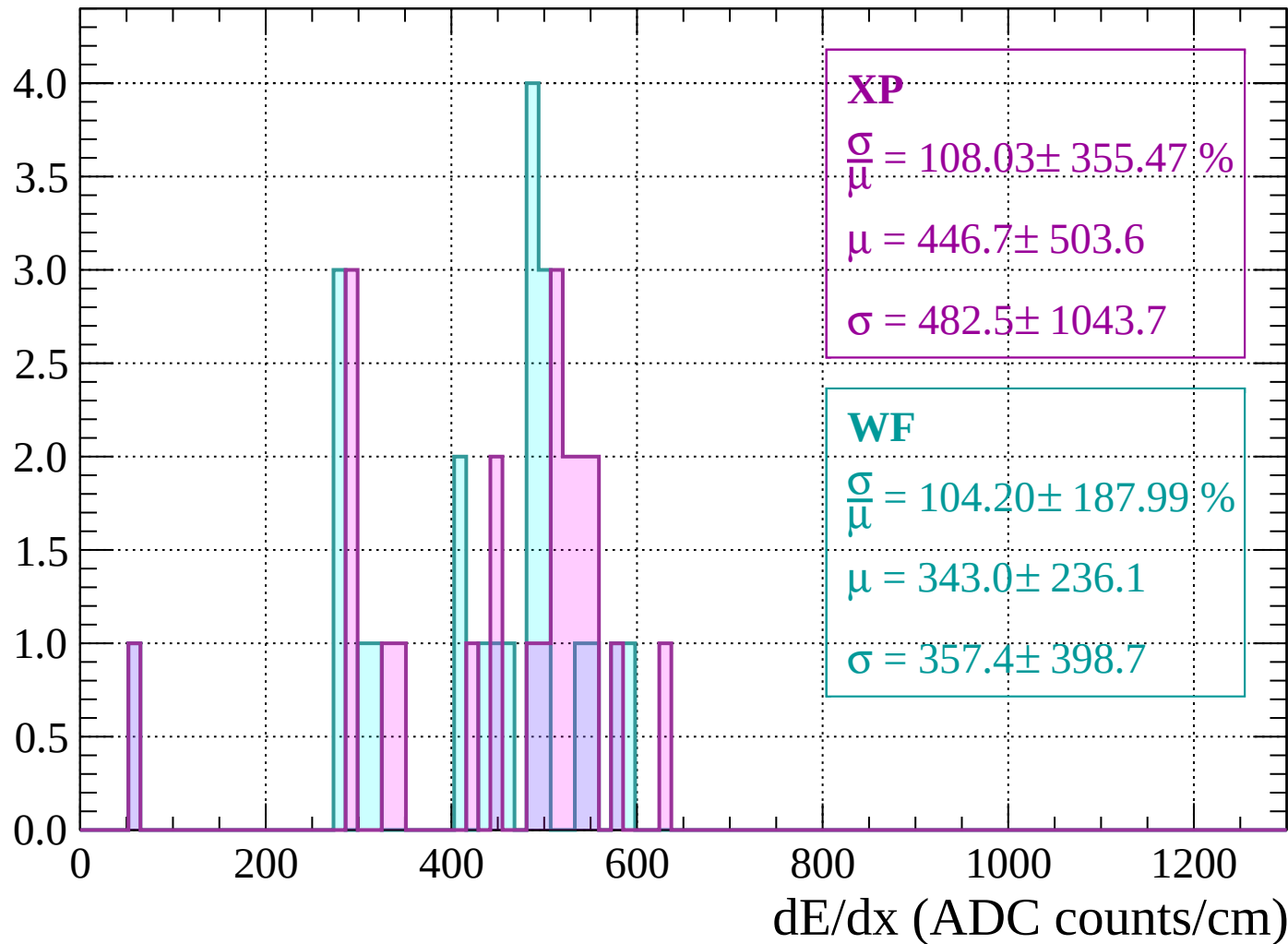
Energy loss | $1200 < p < 1240$

Count



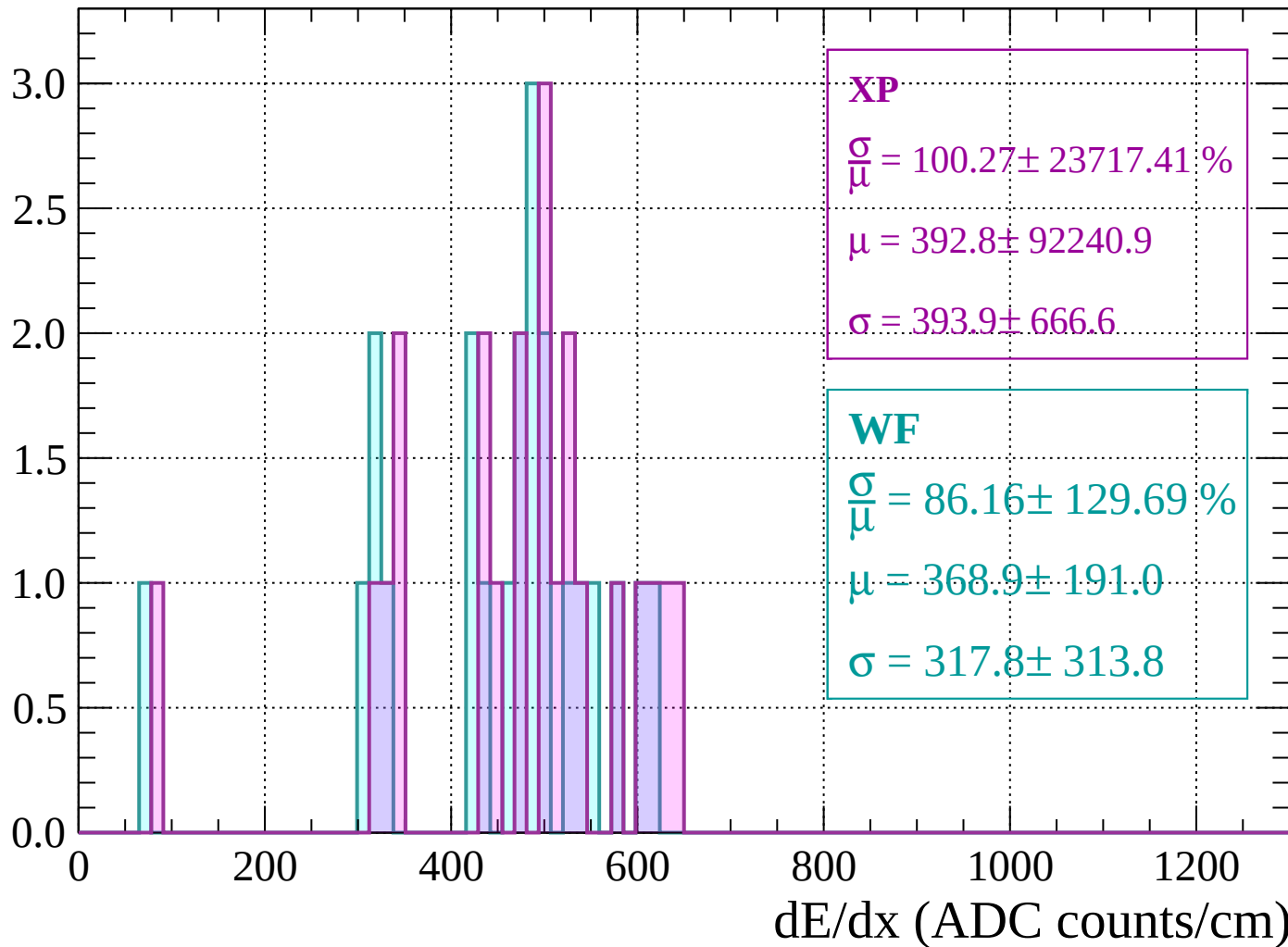
Energy loss | $1240 < p < 1280$

Count



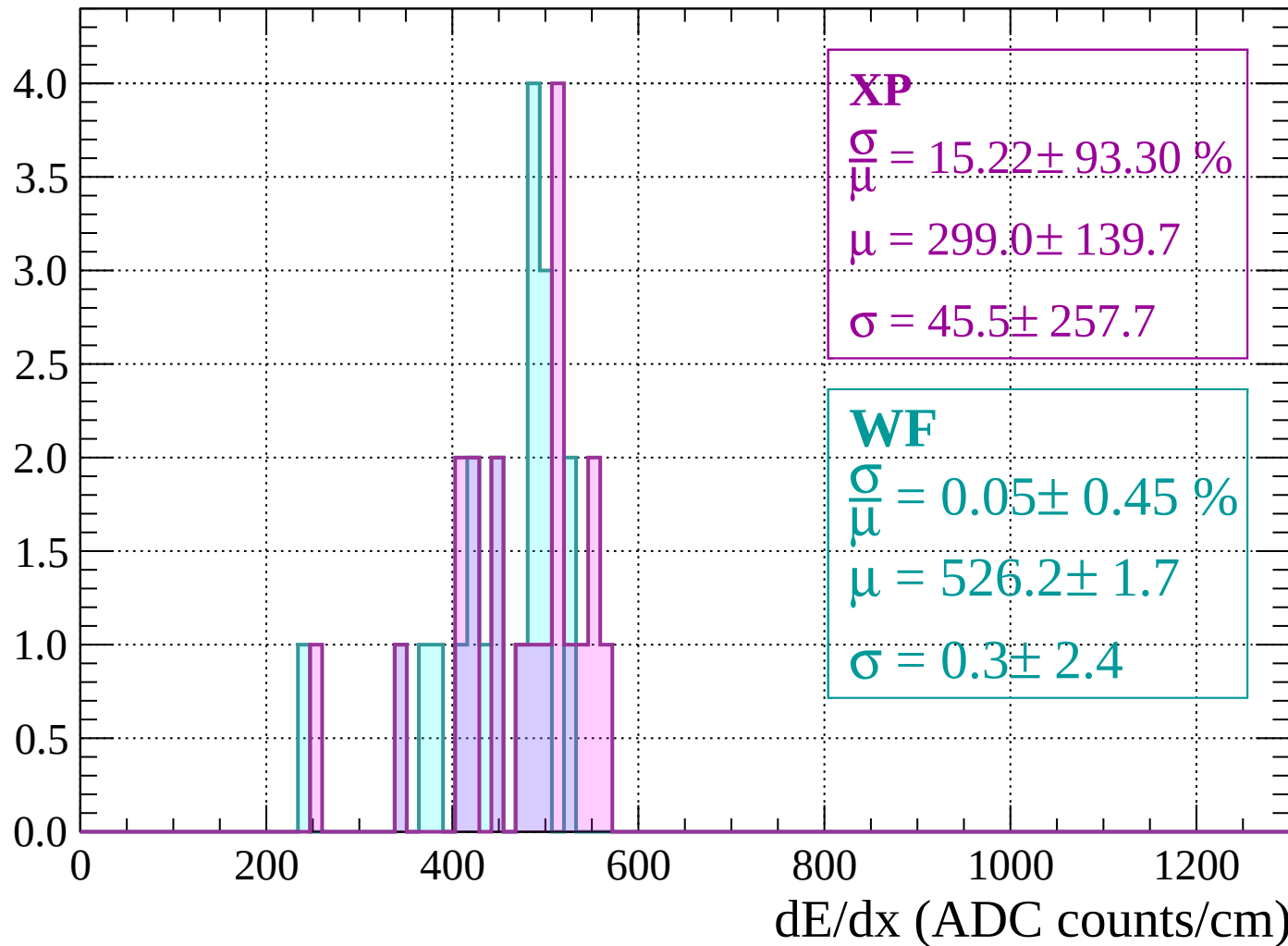
Energy loss | $1280 < p < 1320$

Count



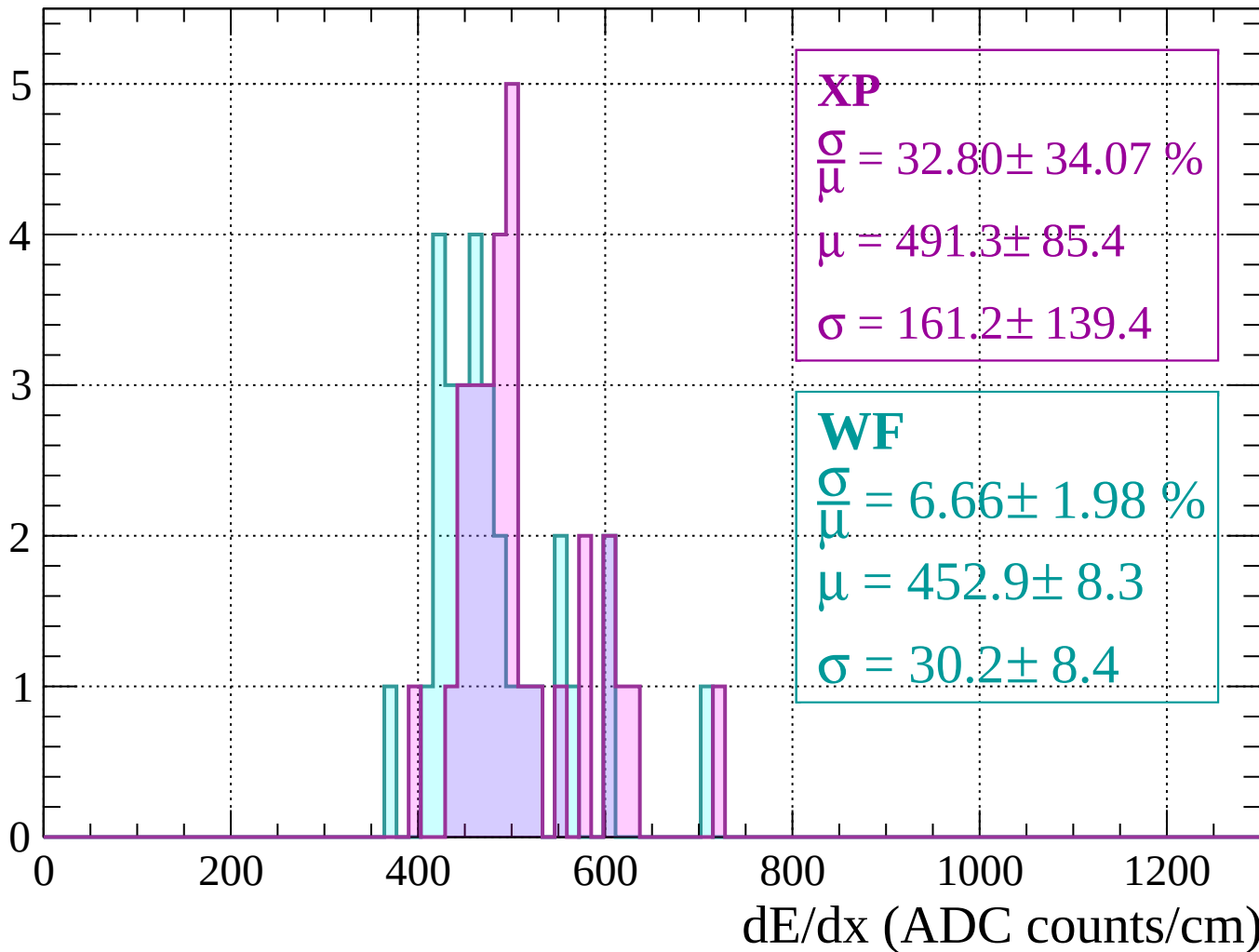
Energy loss | $1320 < p < 1360$

Count



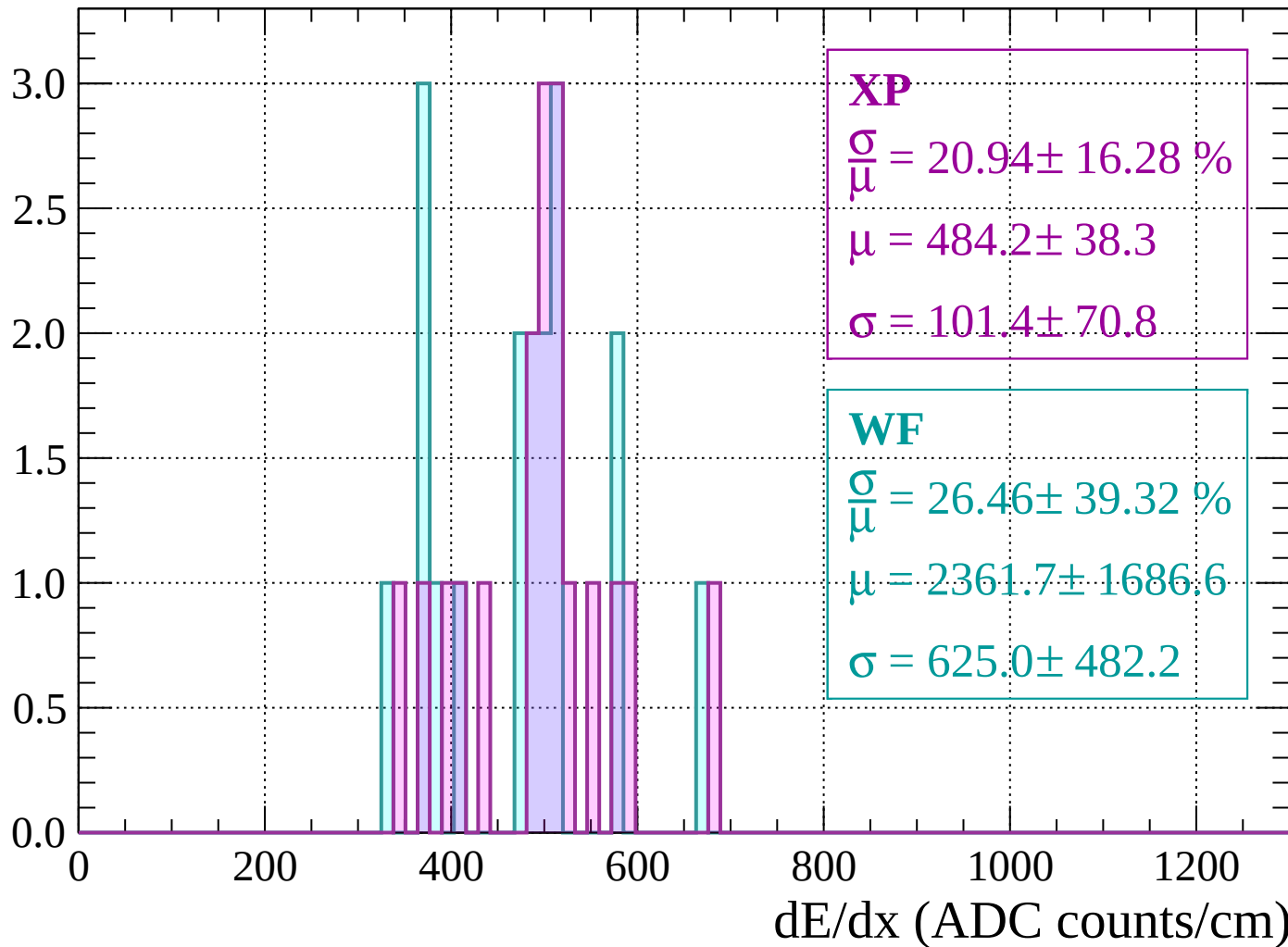
Energy loss | $1360 < p < 1400$

Count



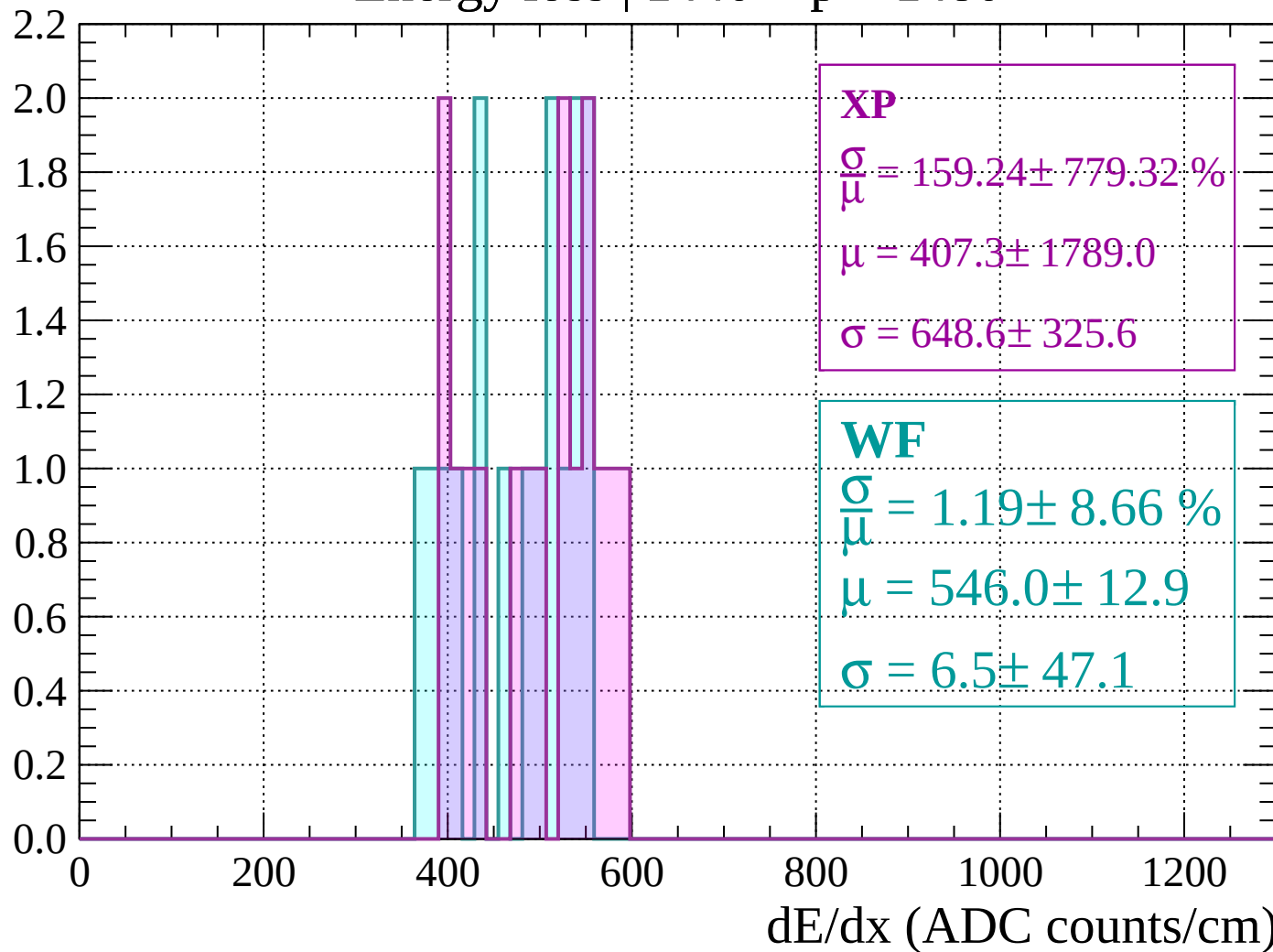
Energy loss | $1400 < p < 1440$

Count



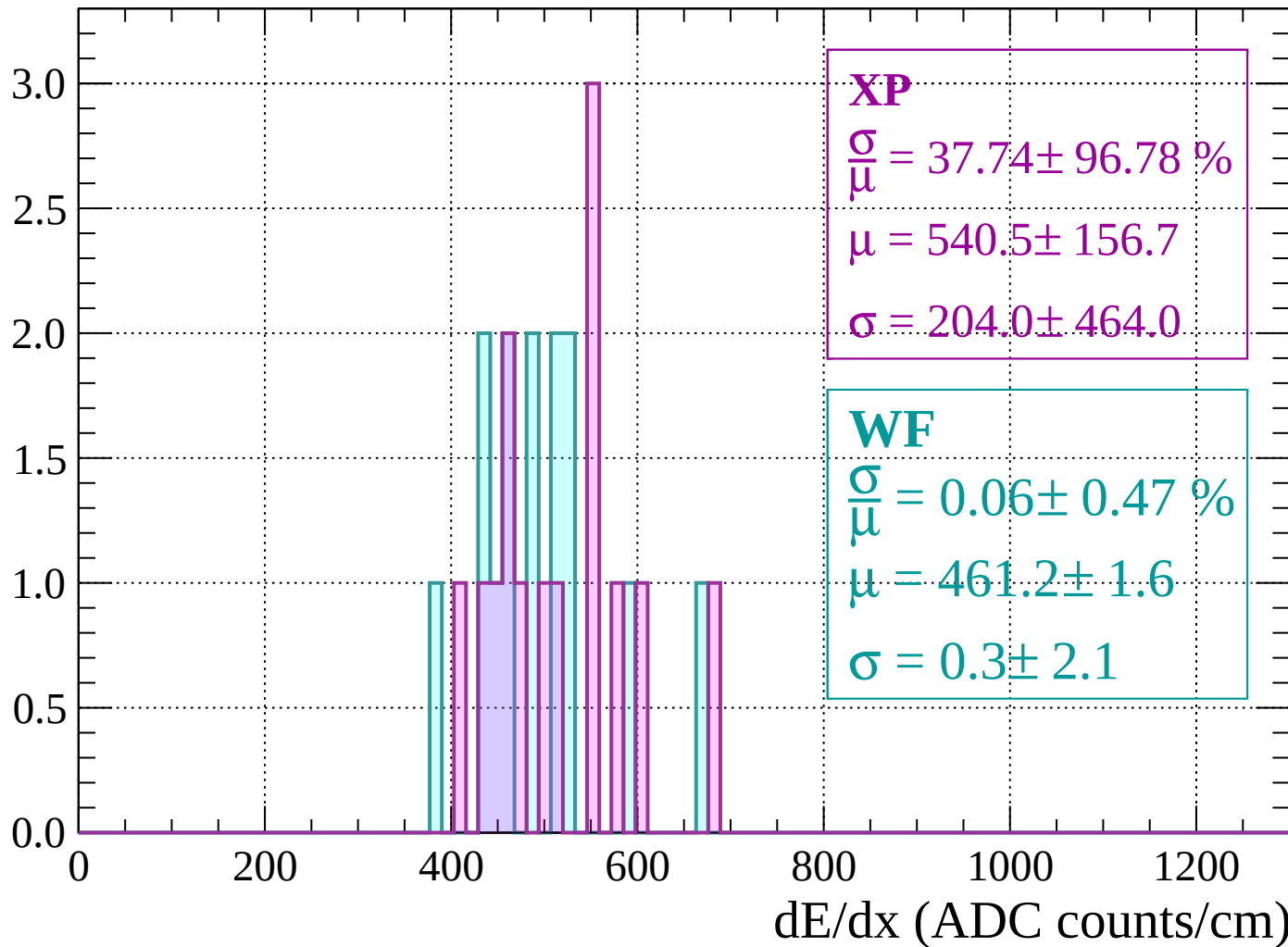
Energy loss | $1440 < p < 1480$

Count



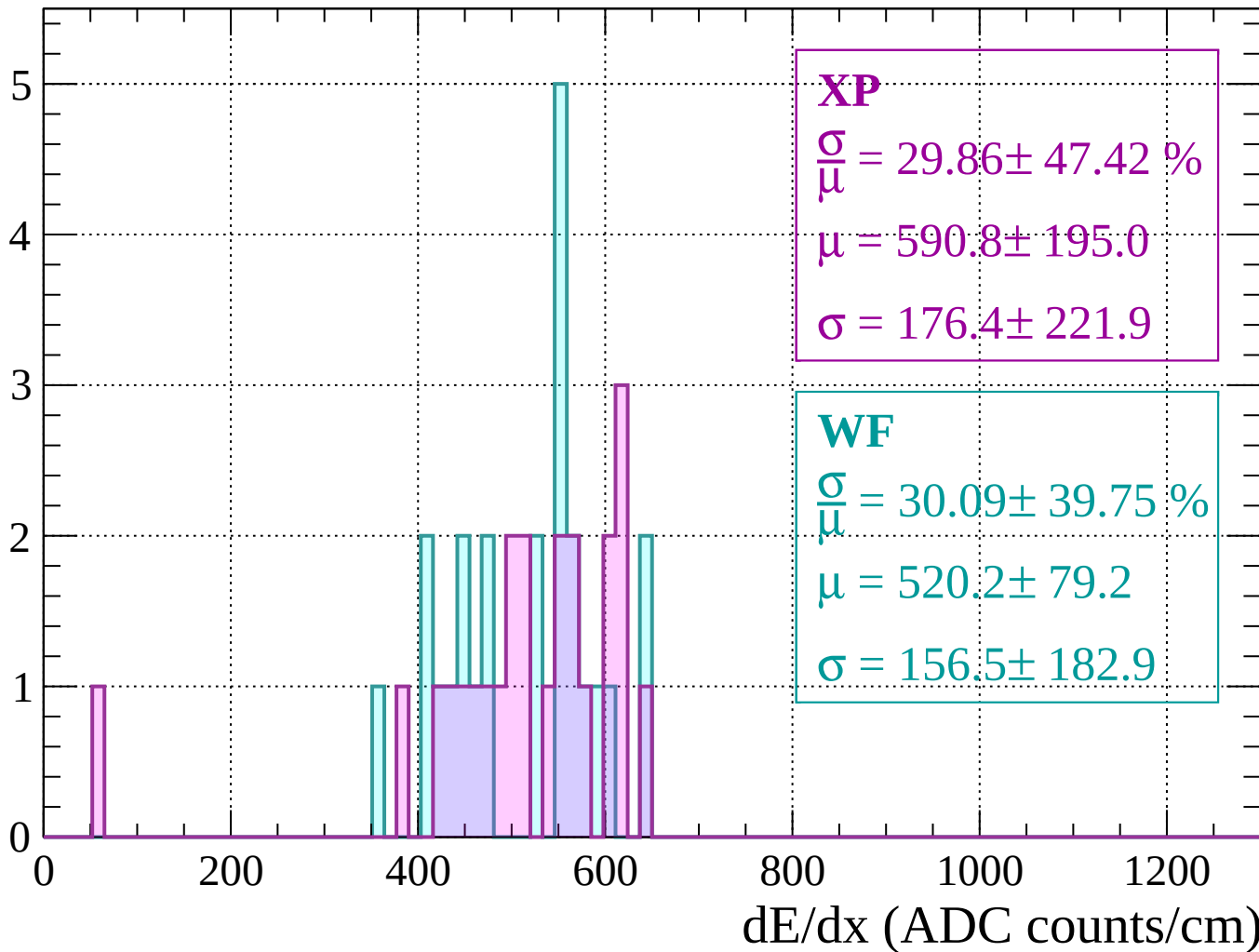
Energy loss | $1480 < p < 1520$

Count



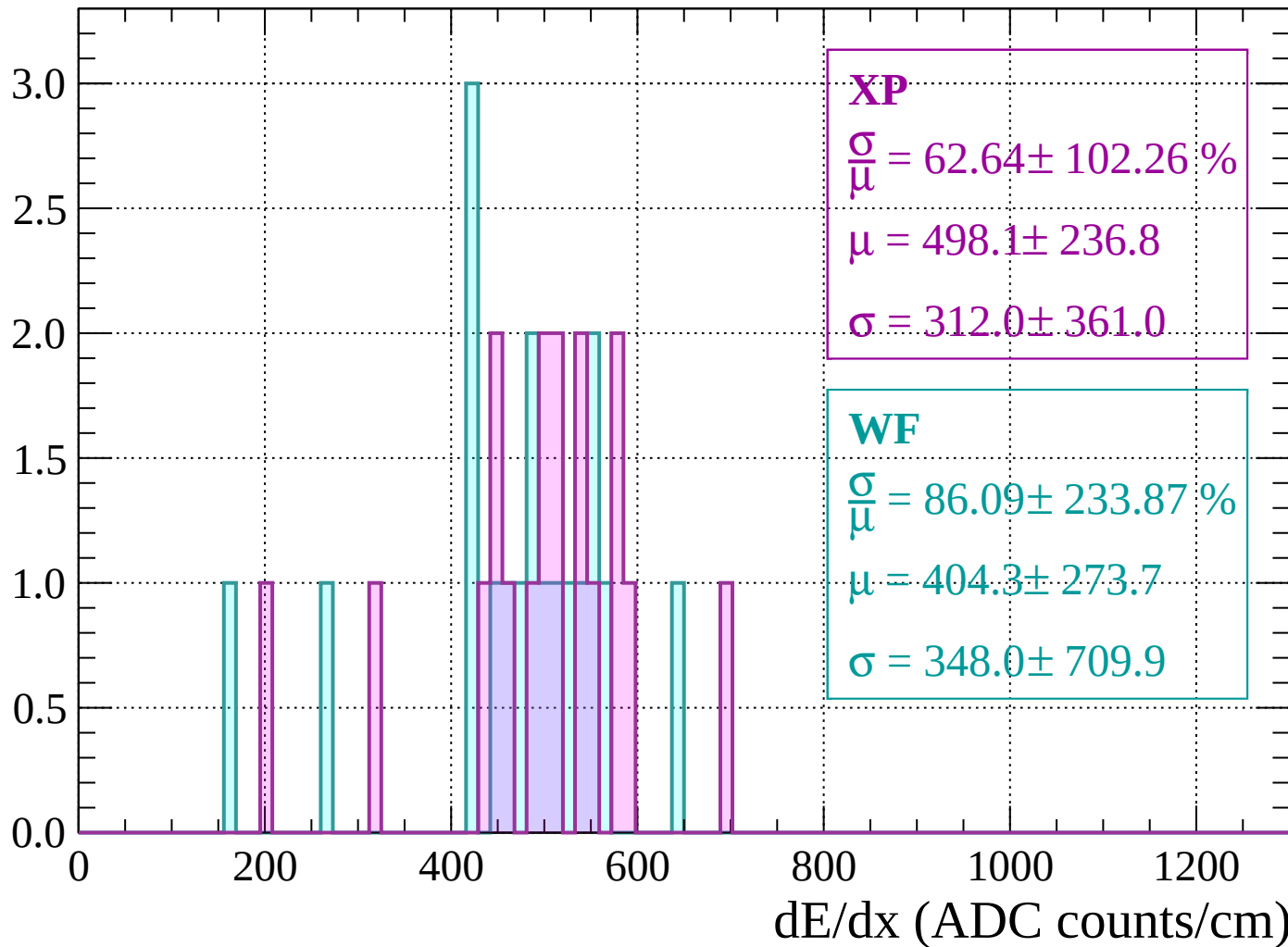
Energy loss | $1520 < p < 1560$

Count



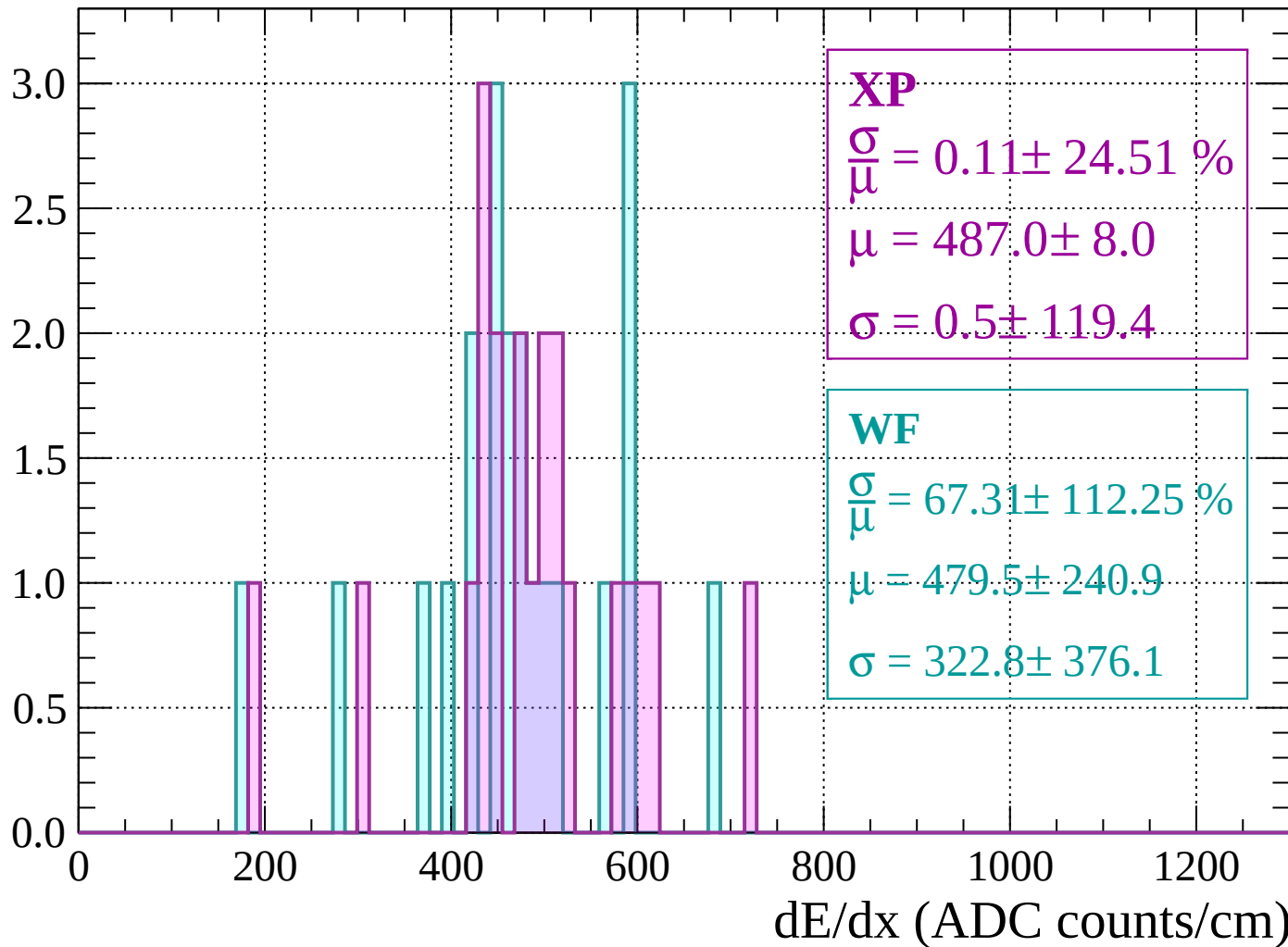
Energy loss | $1560 < p < 1600$

Count



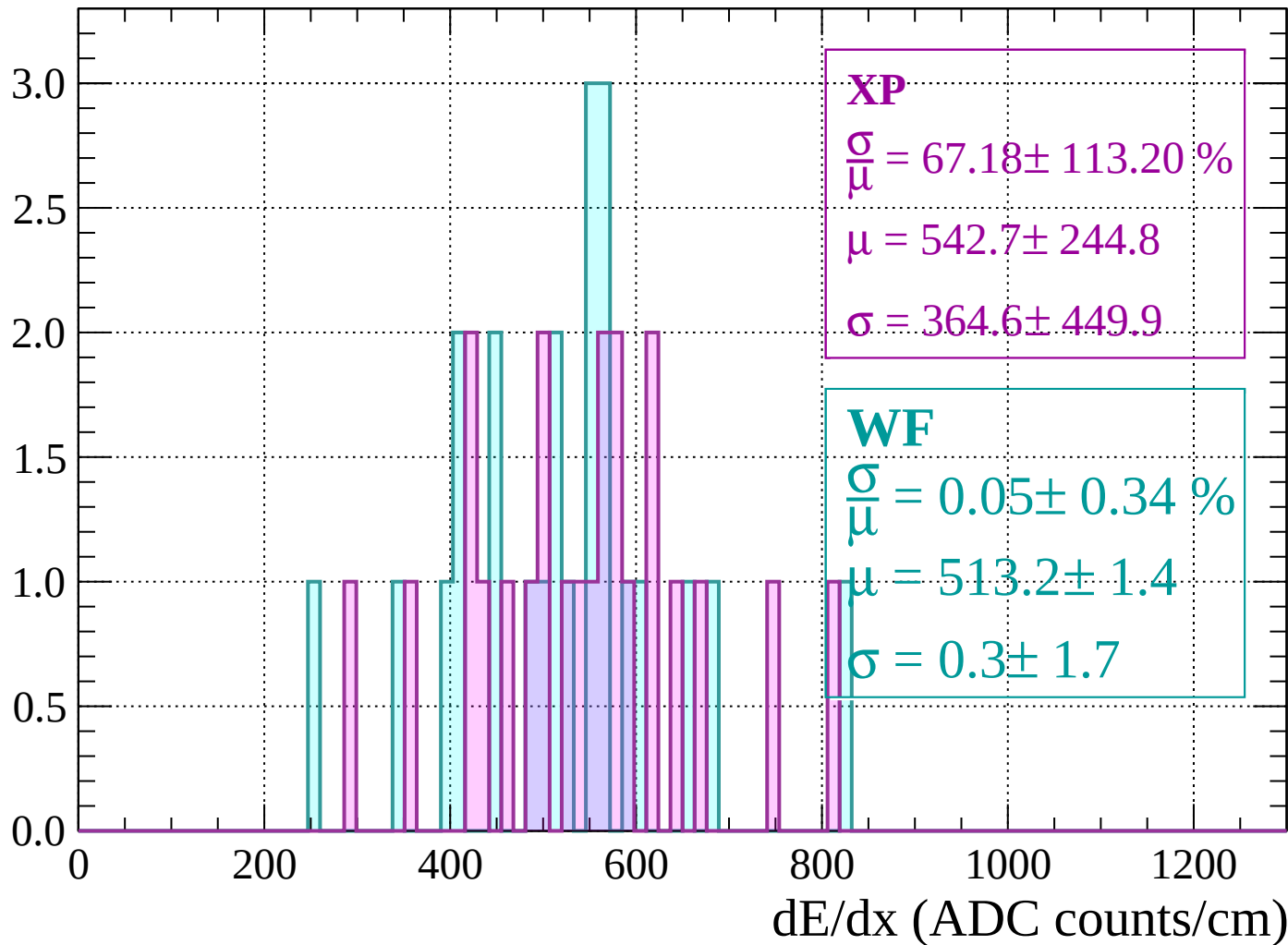
Energy loss | $1600 < p < 1640$

Count



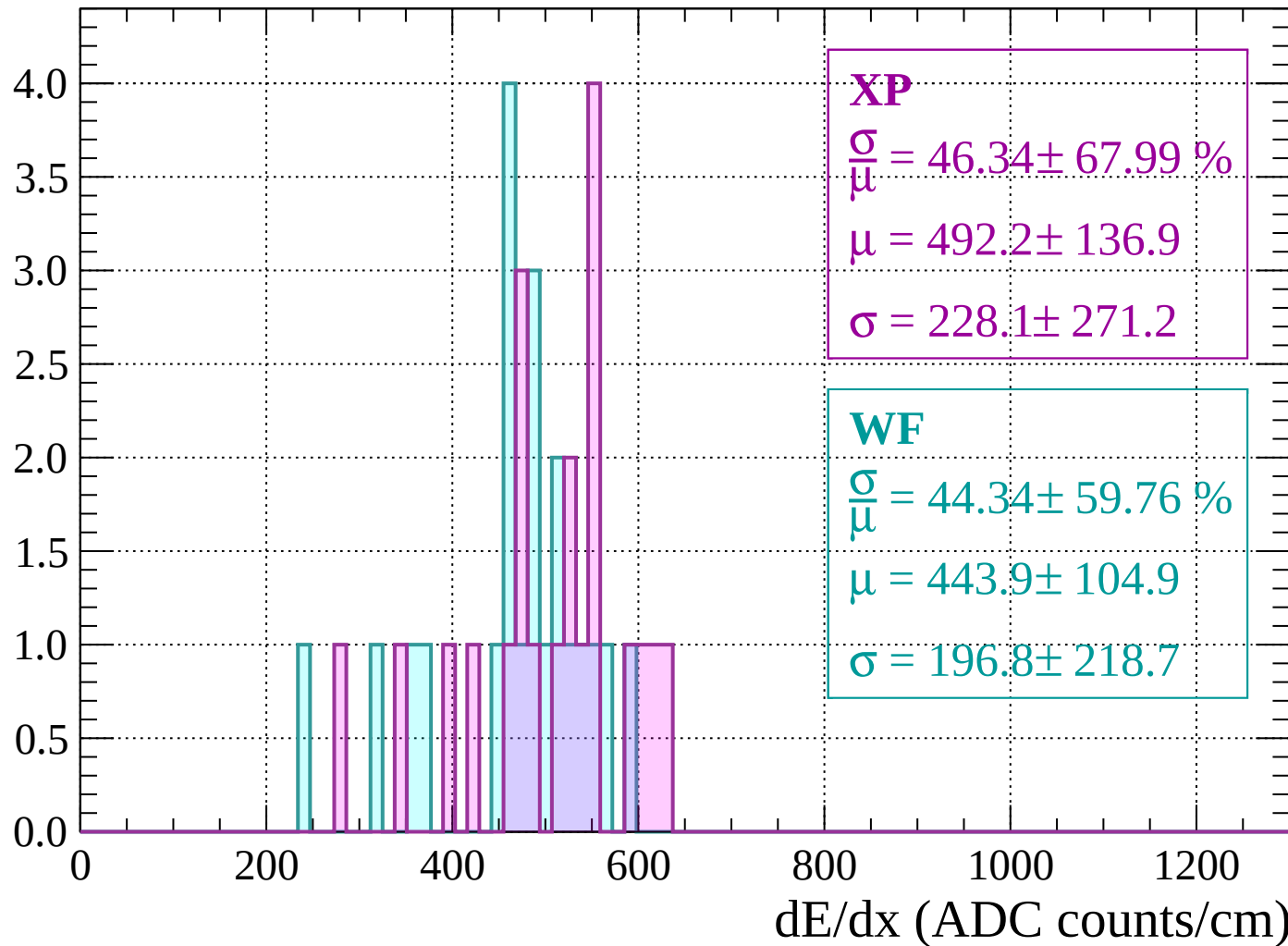
Energy loss | $1640 < p < 1680$

Count



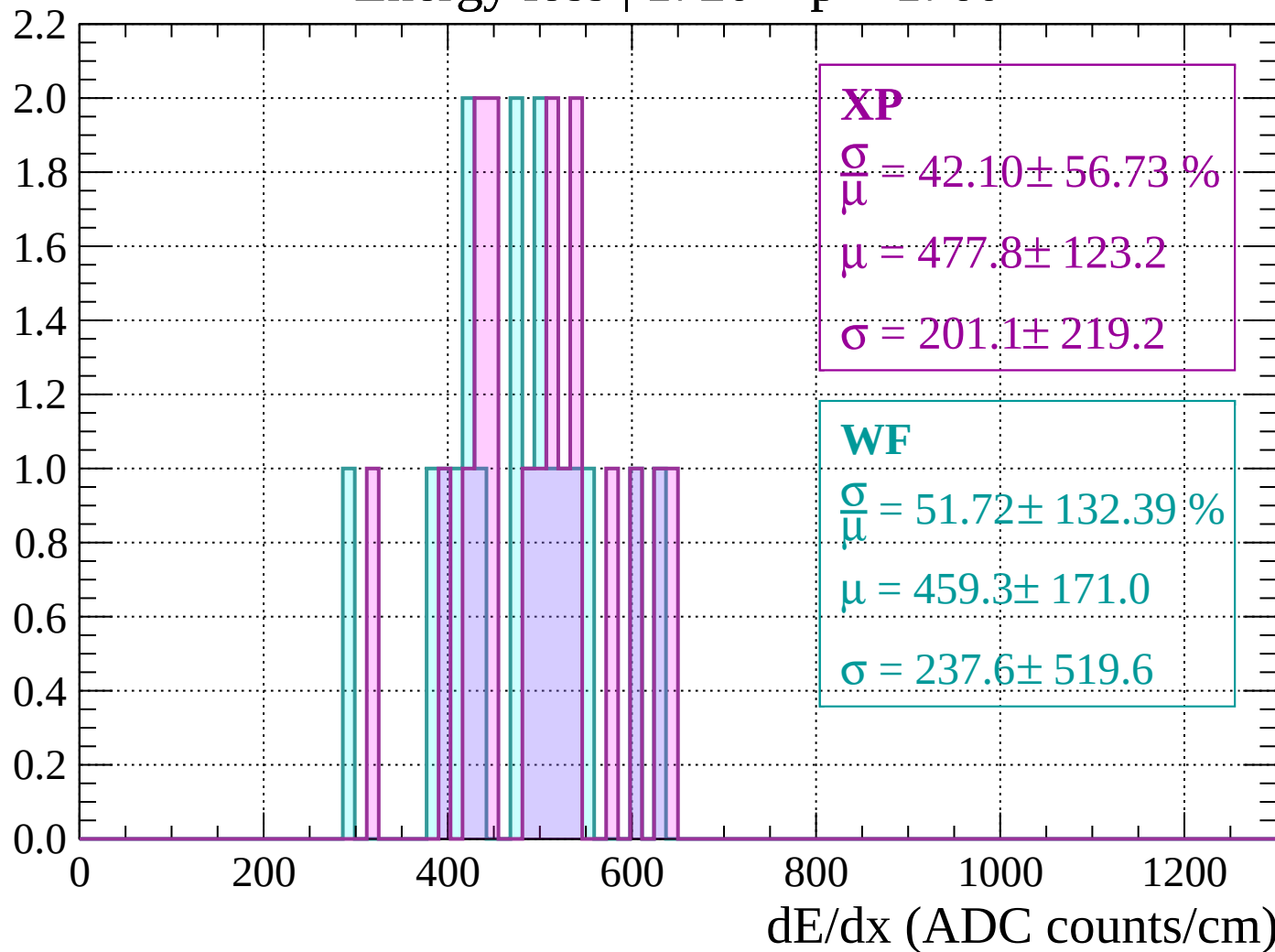
Energy loss | $1680 < p < 1720$

Count



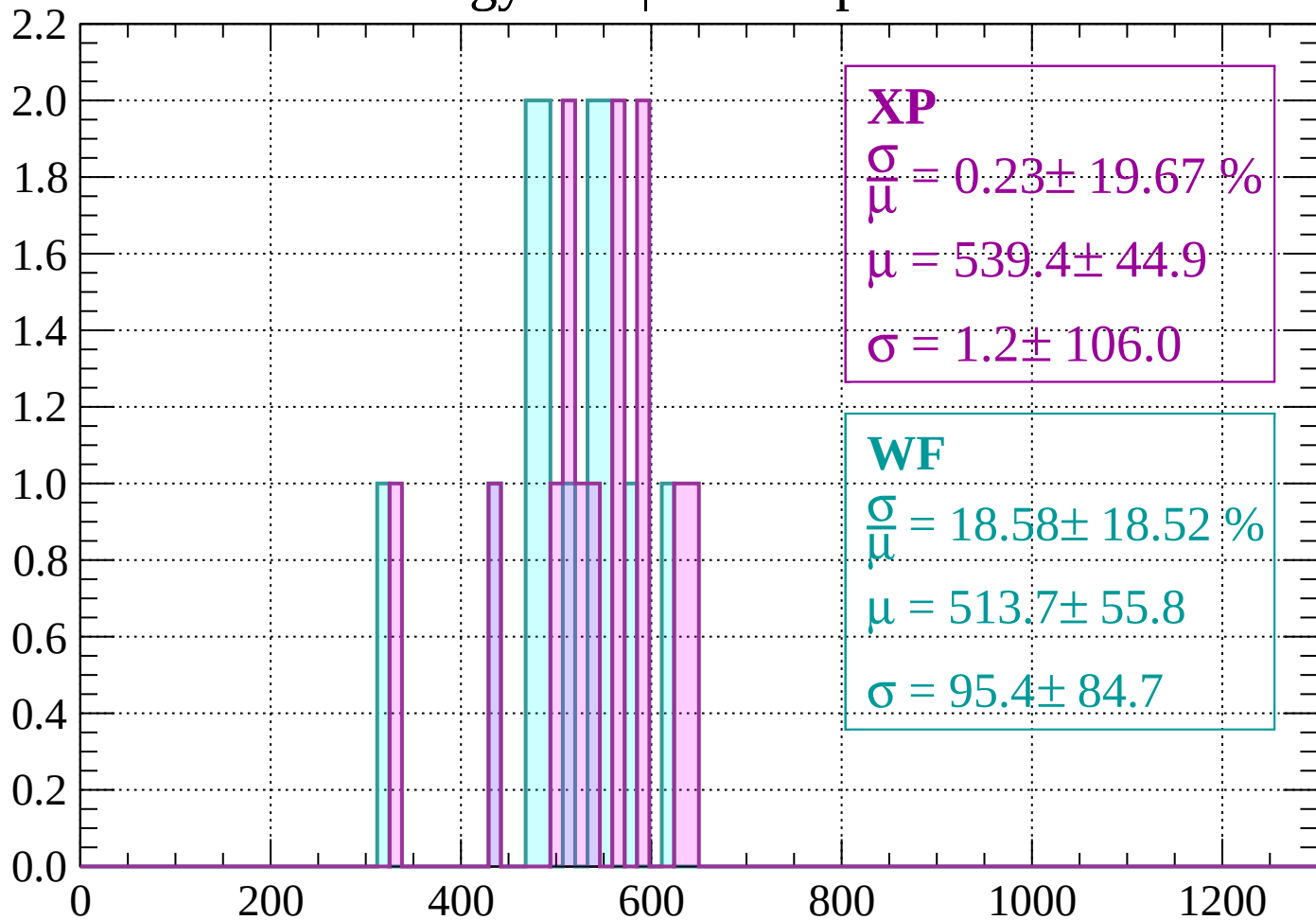
Energy loss | $1720 < p < 1760$

Count



Energy loss | $1760 < p < 1800$

Count



XP

$$\frac{\sigma}{\mu} = 0.23 \pm 19.67 \%$$

$$\mu = 539.4 \pm 44.9$$

$$\sigma = 1.2 \pm 106.0$$

WF

$$\frac{\sigma}{\mu} = 18.58 \pm 18.52 \%$$

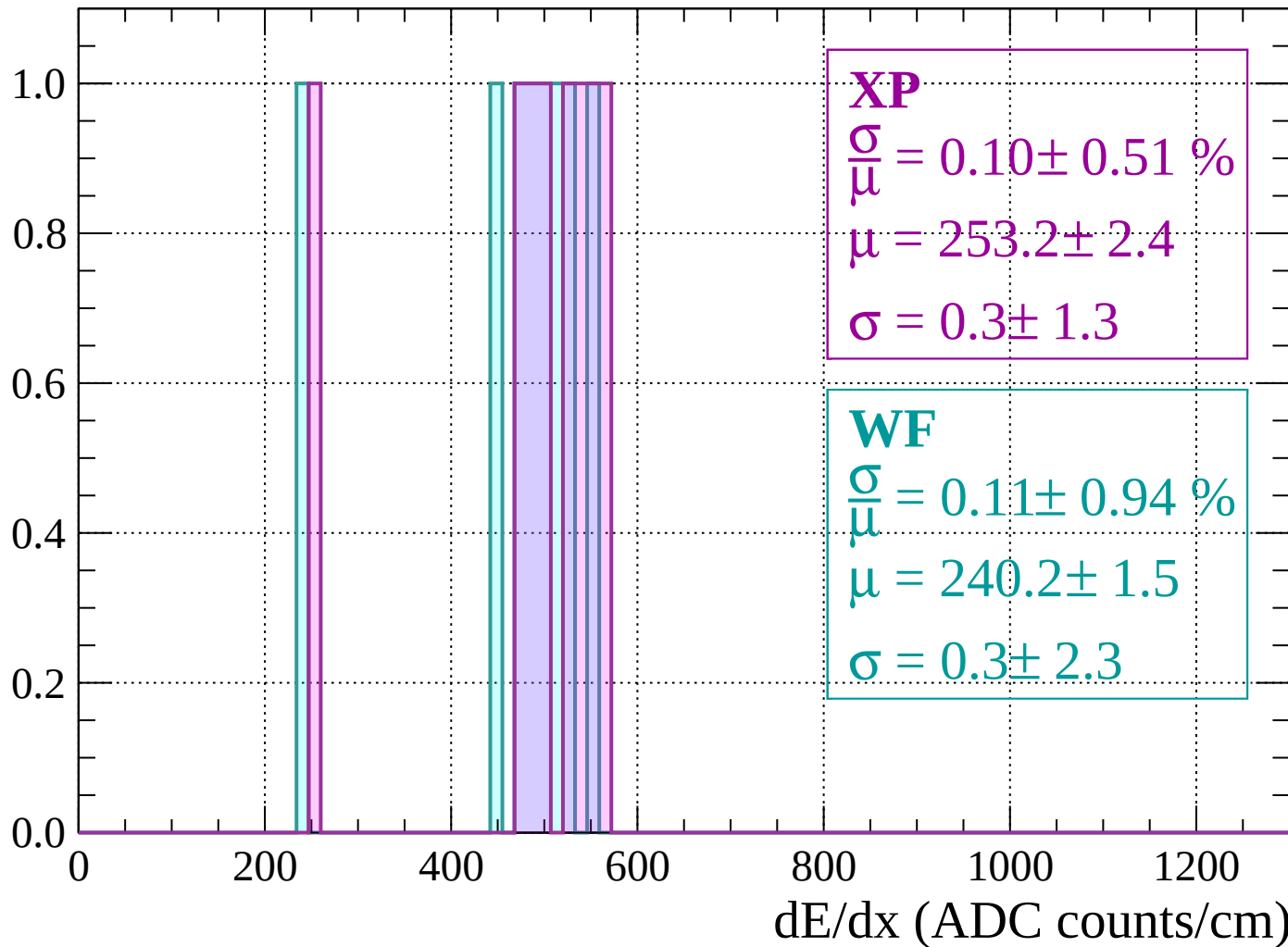
$$\mu = 513.7 \pm 55.8$$

$$\sigma = 95.4 \pm 84.7$$

dE/dx (ADC counts/cm)

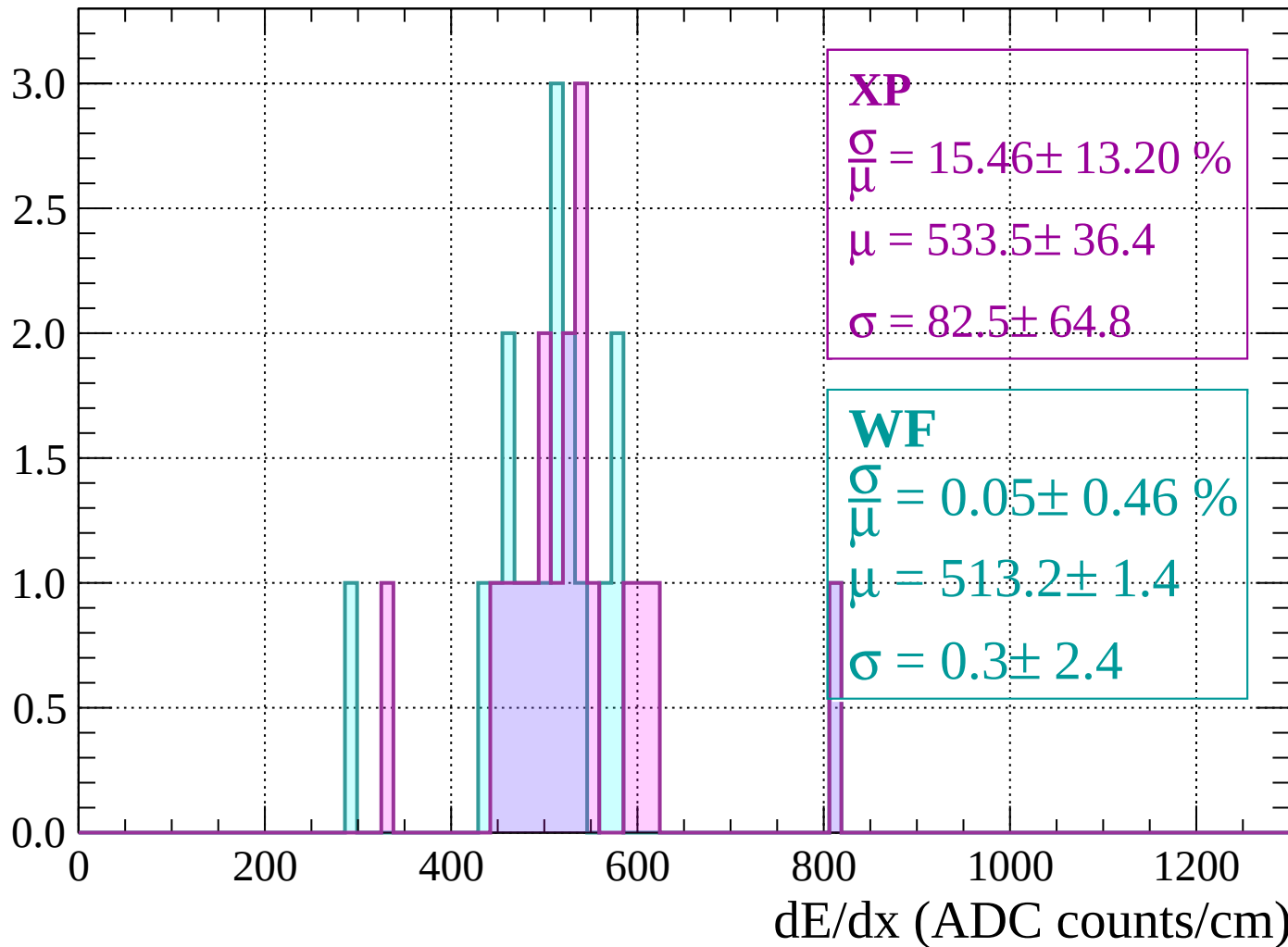
Energy loss | $1800 < p < 1840$

Count



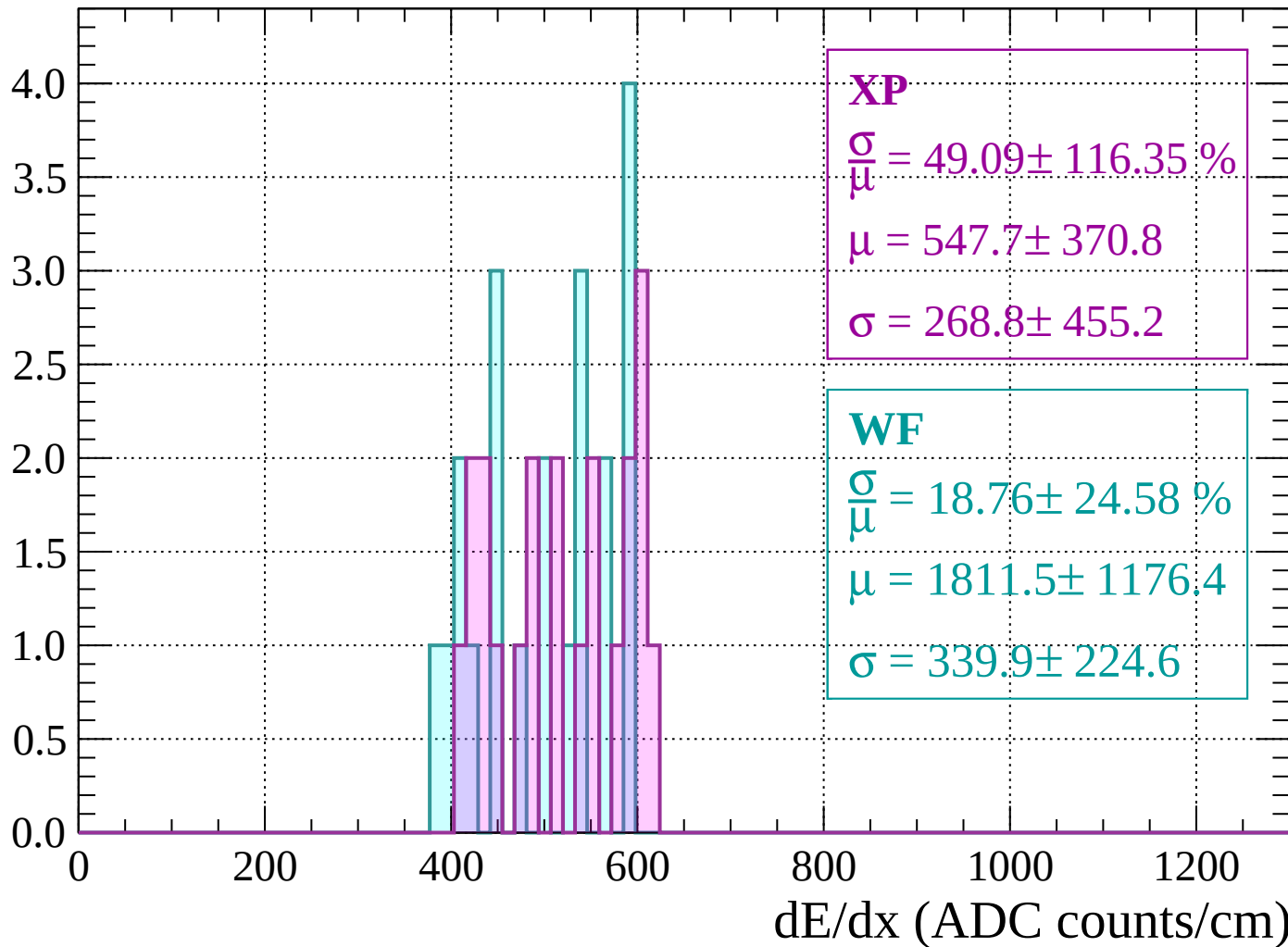
Energy loss | $1840 < p < 1880$

Count



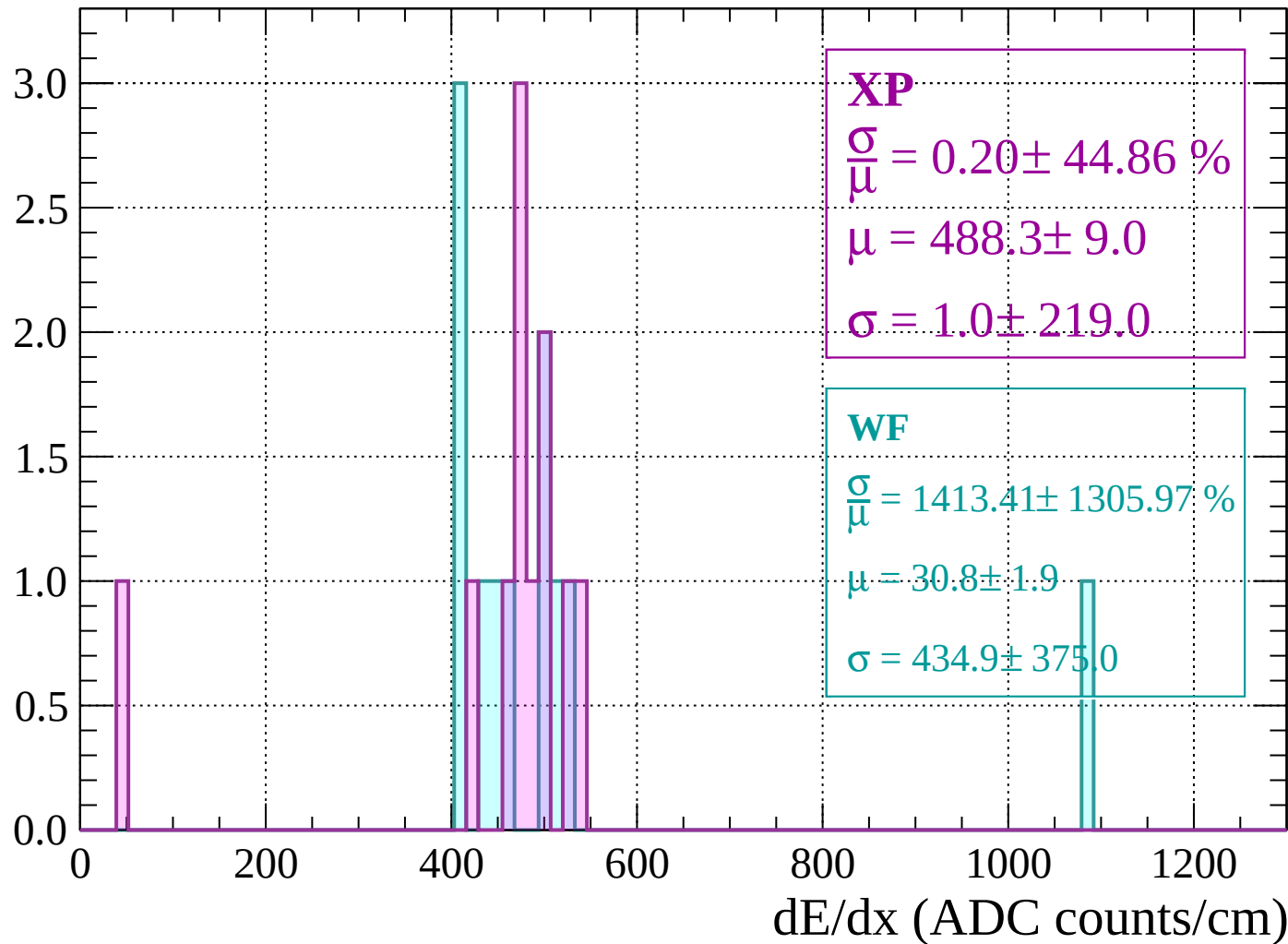
Energy loss | $1880 < p < 1920$

Count



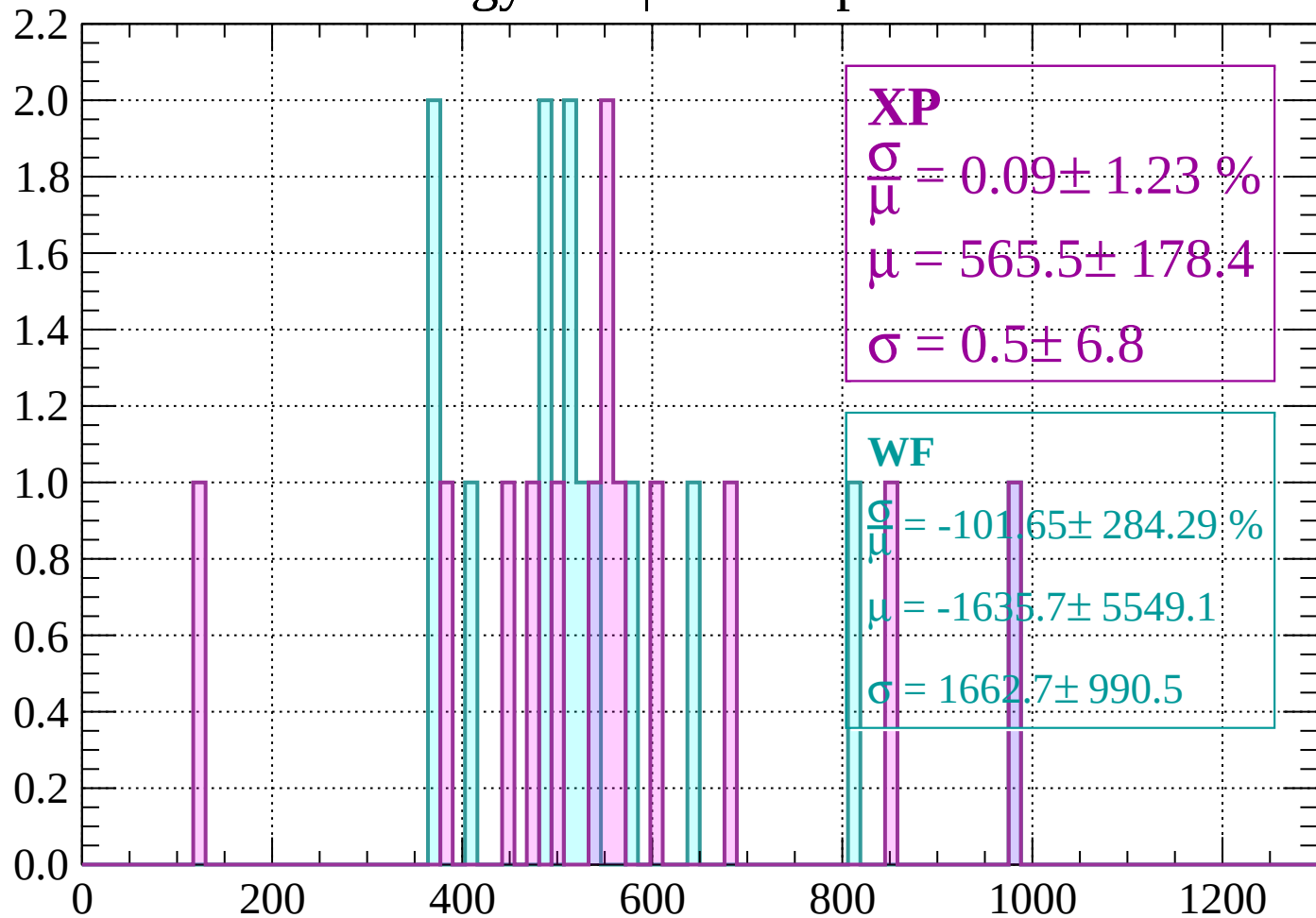
Energy loss | $1920 < p < 1960$

Count



Energy loss | 1960 < p < 2000

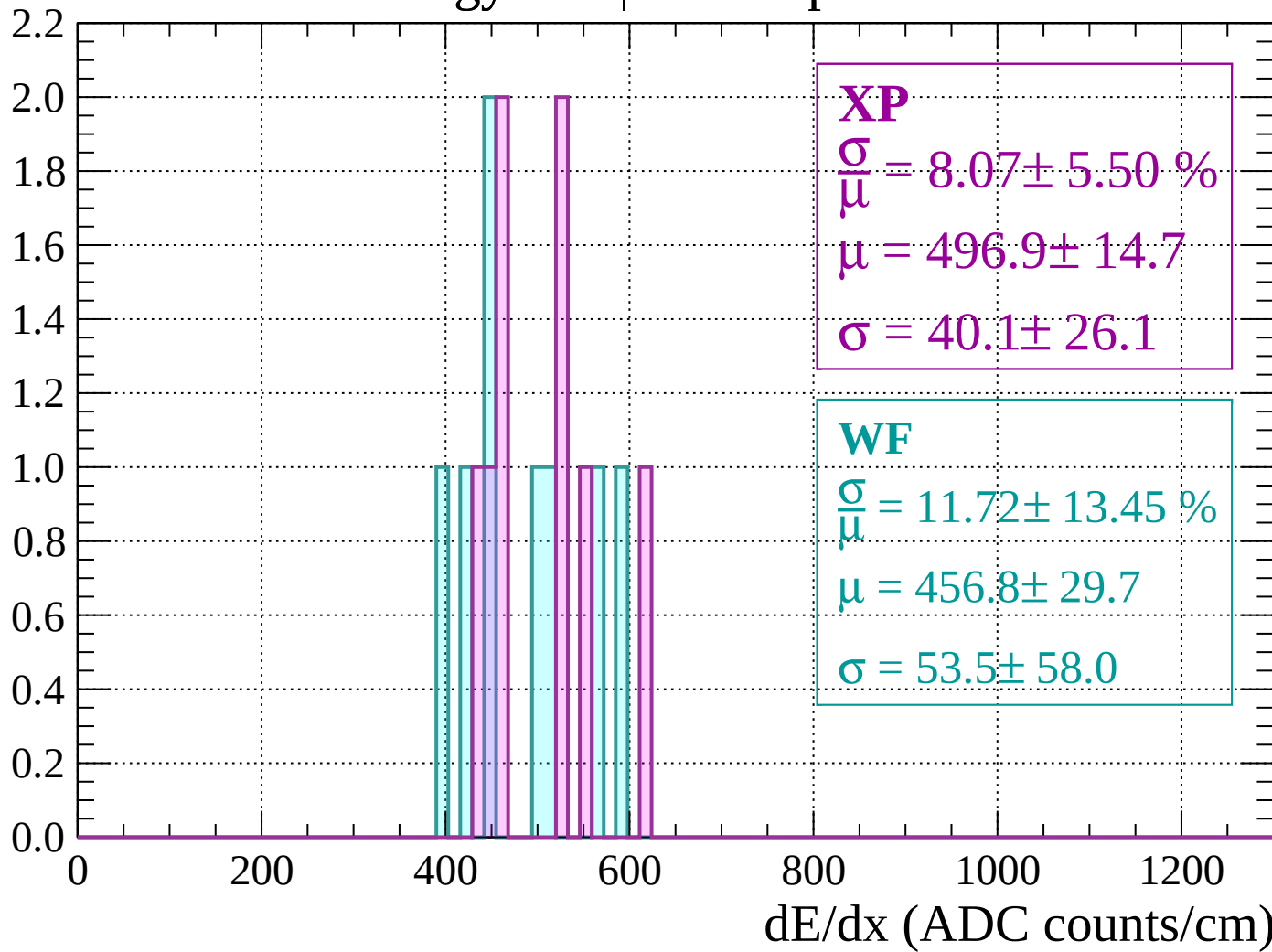
Count



dE/dx (ADC counts/cm)

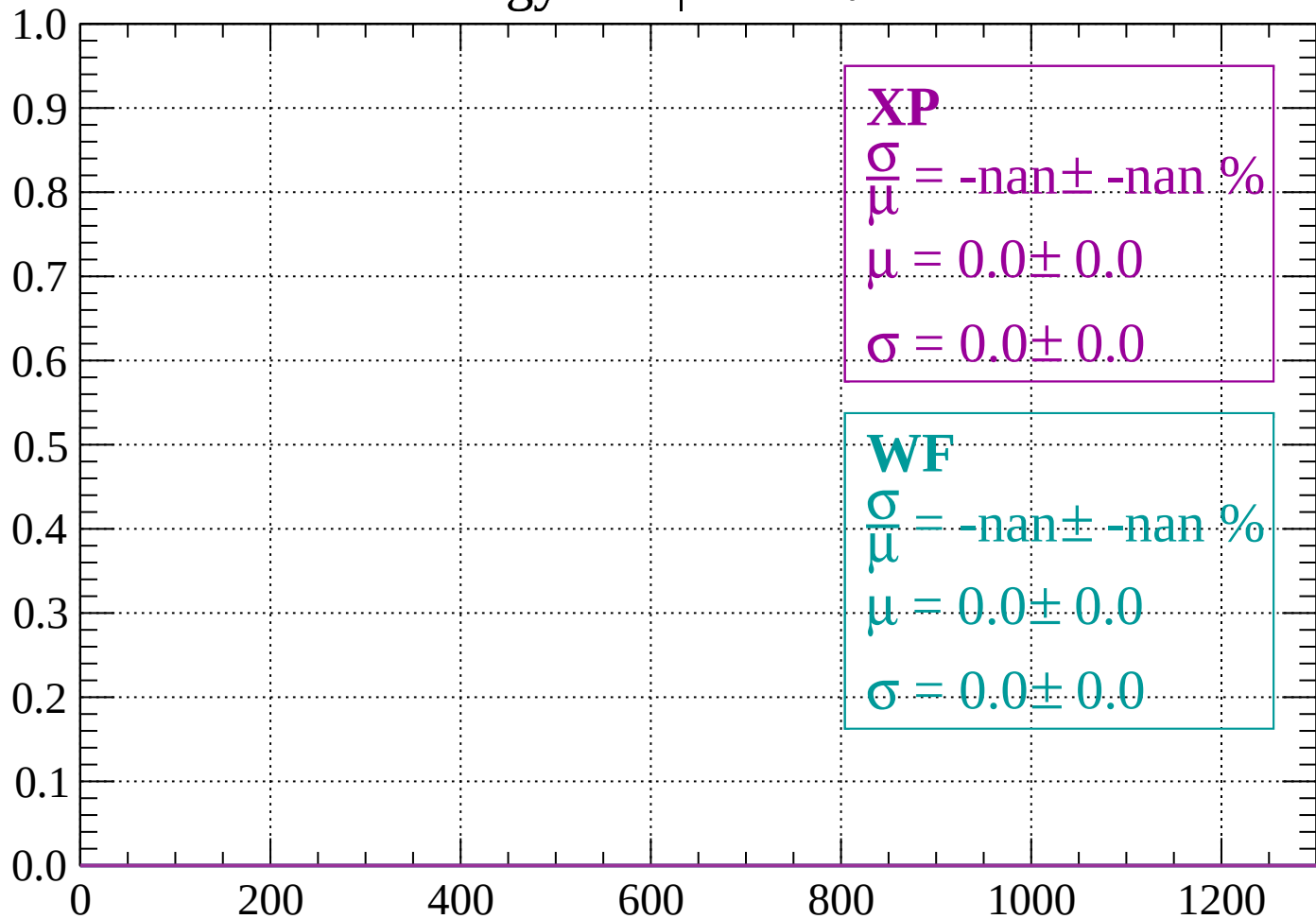
Energy loss | $2000 < p < 2040$

Count



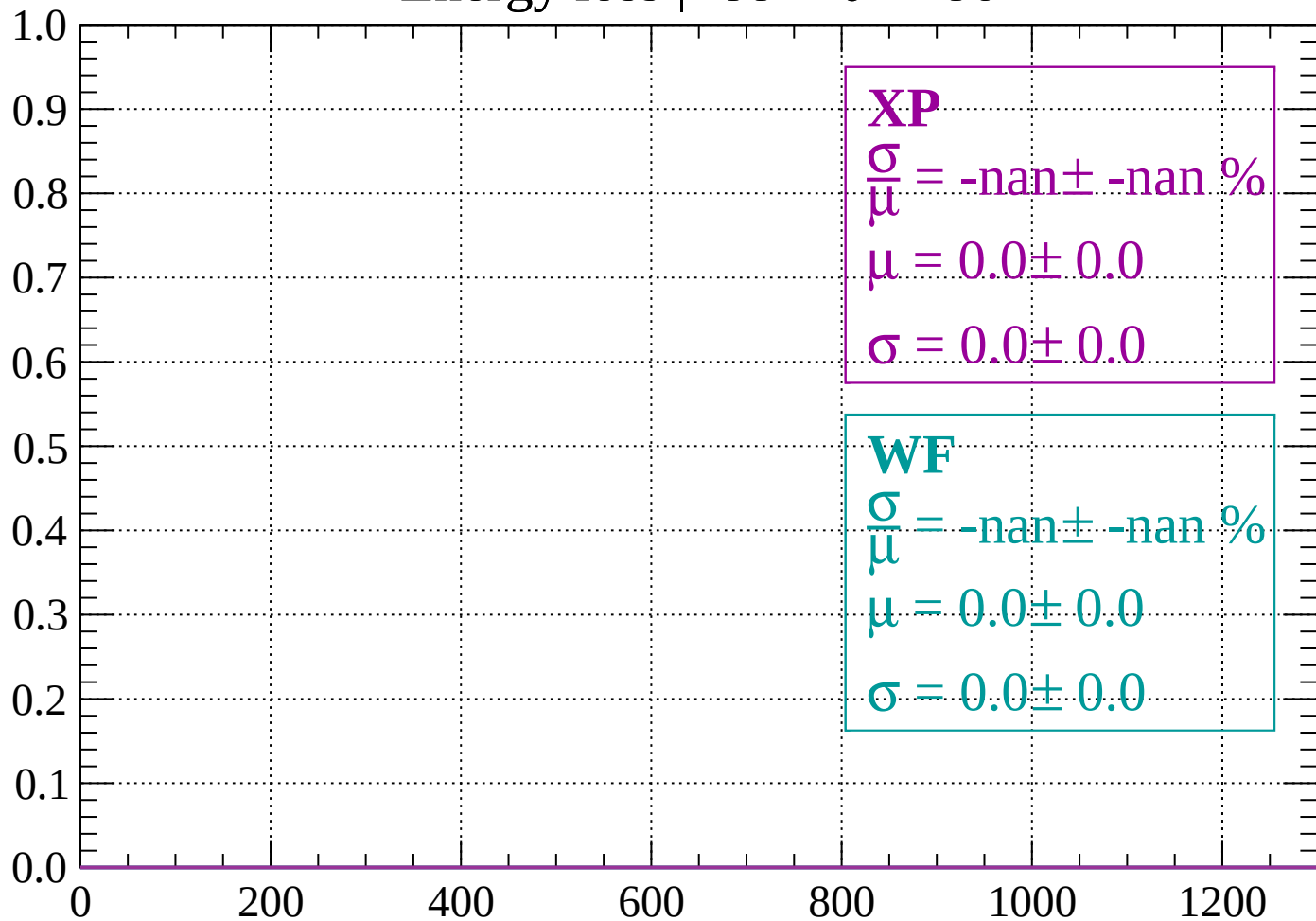
Energy loss | $-90 < \theta < -88$

Count



Energy loss | $-88 < \theta < -86$

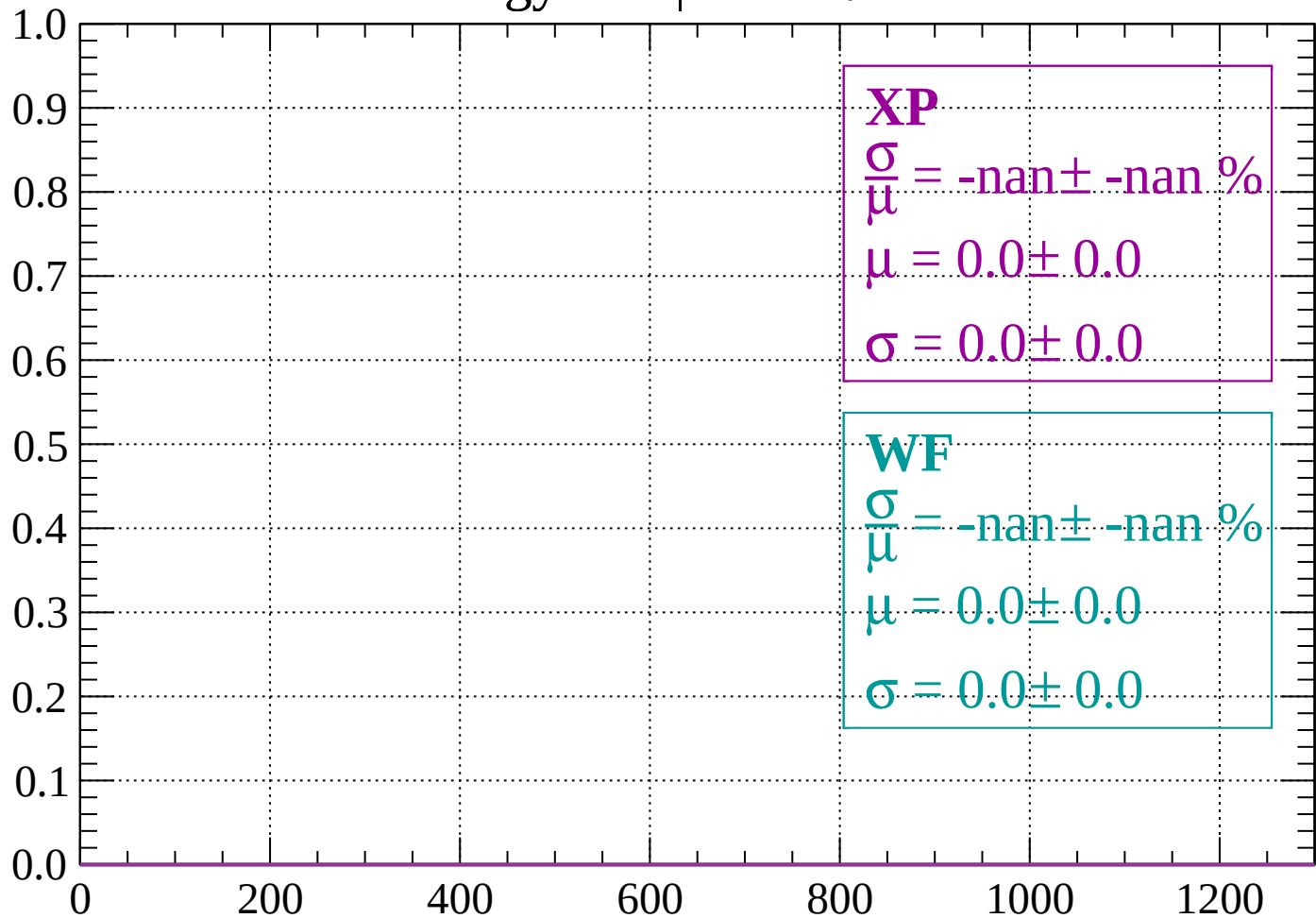
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

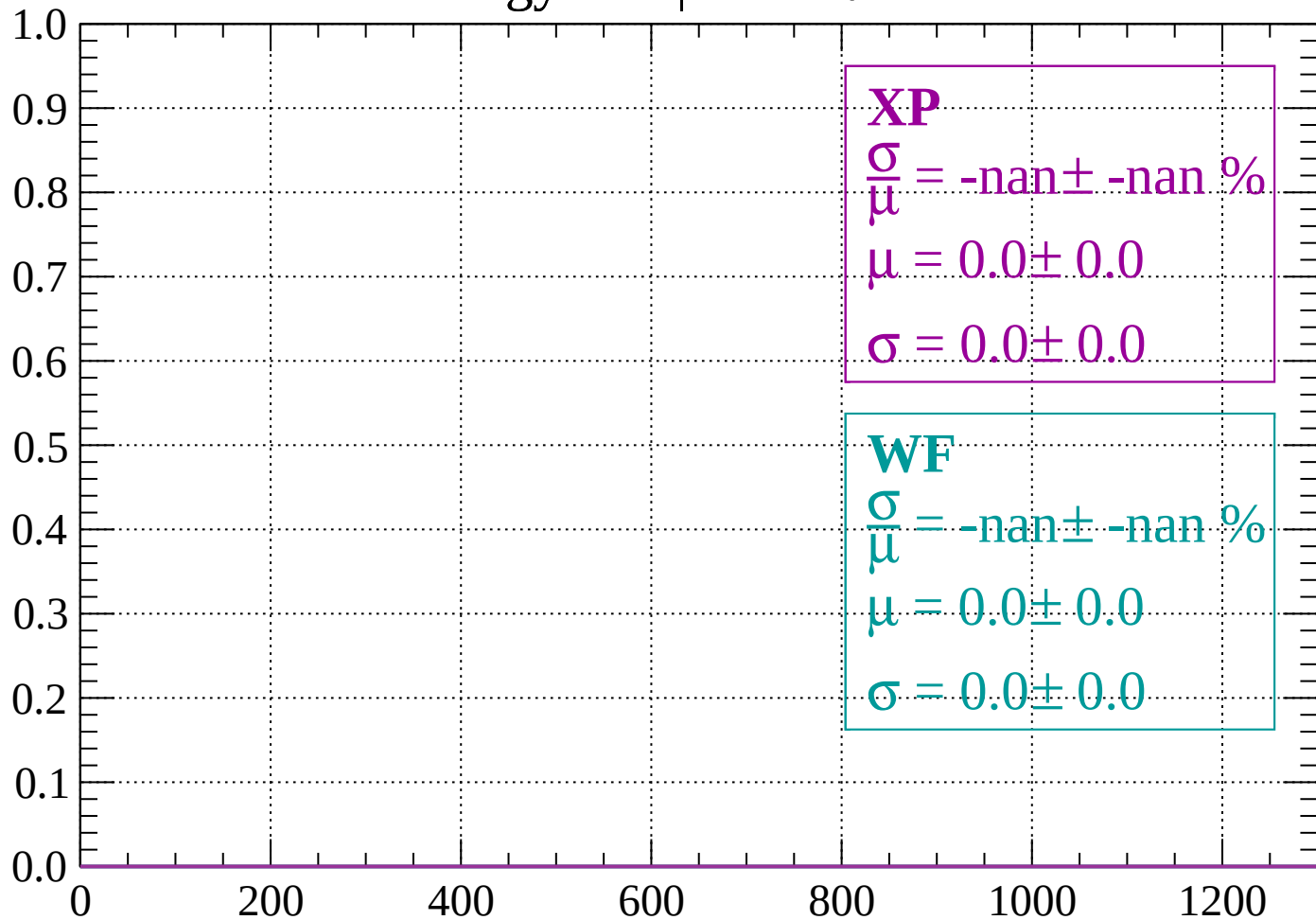
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

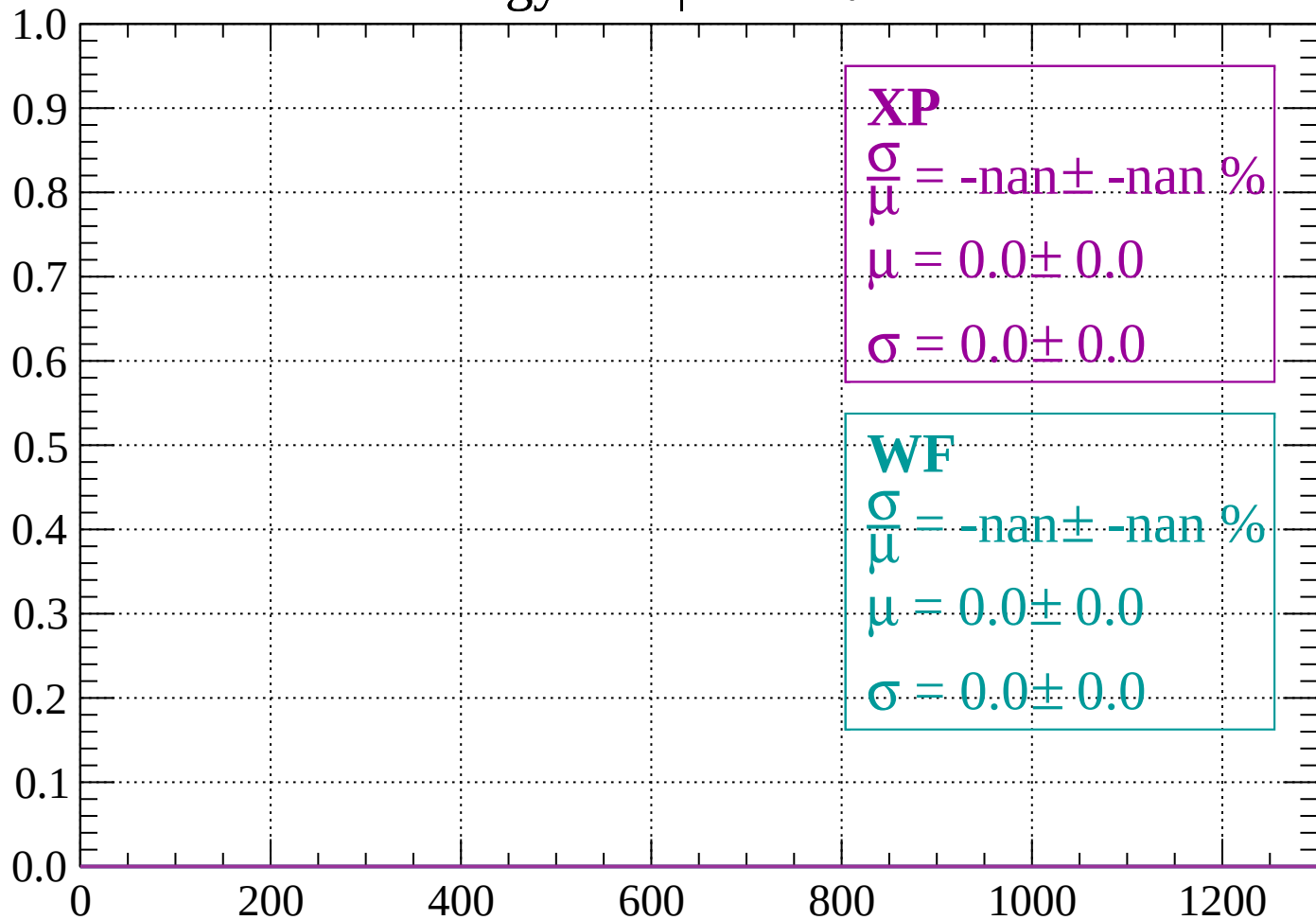
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

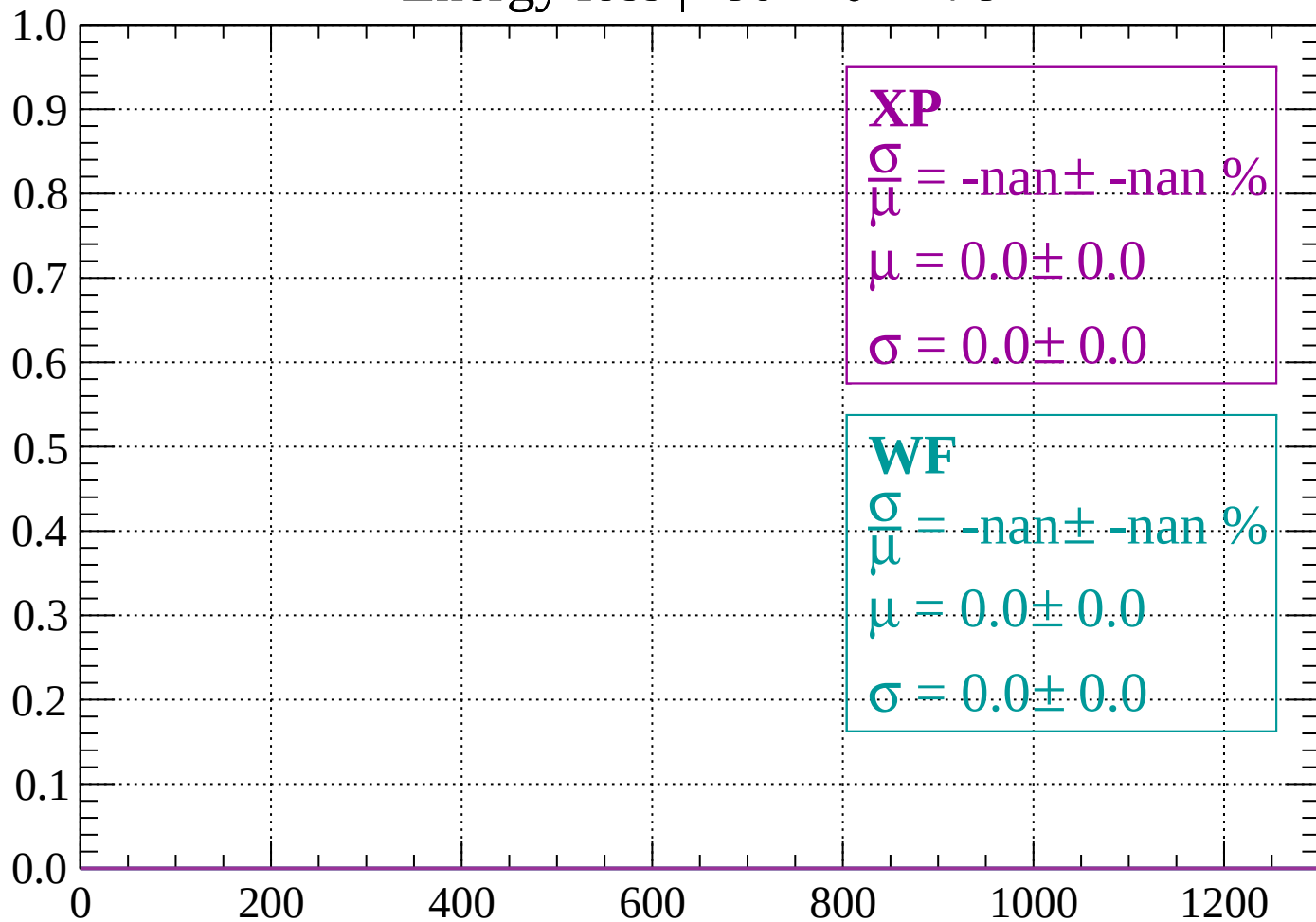
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

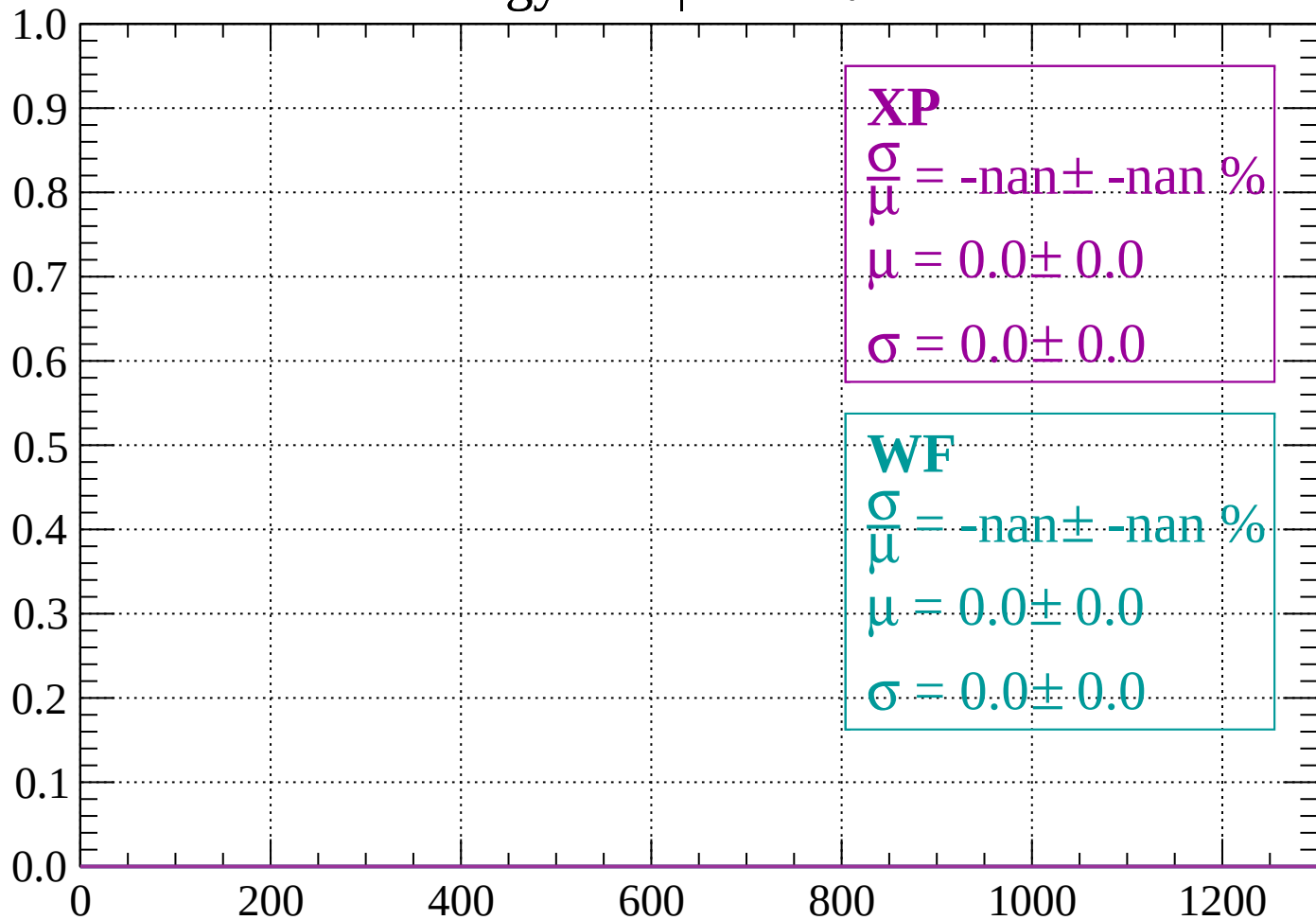
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

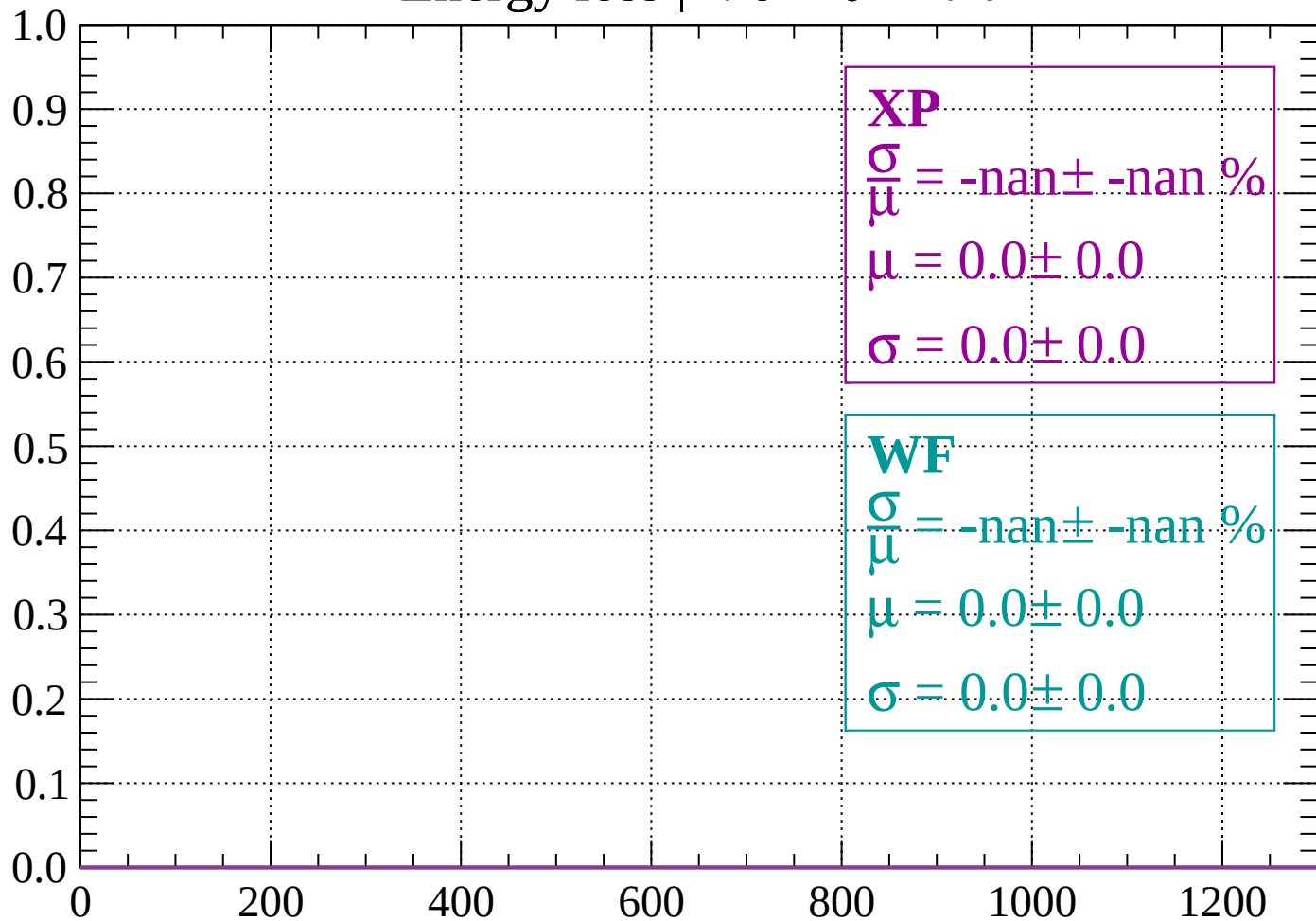
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

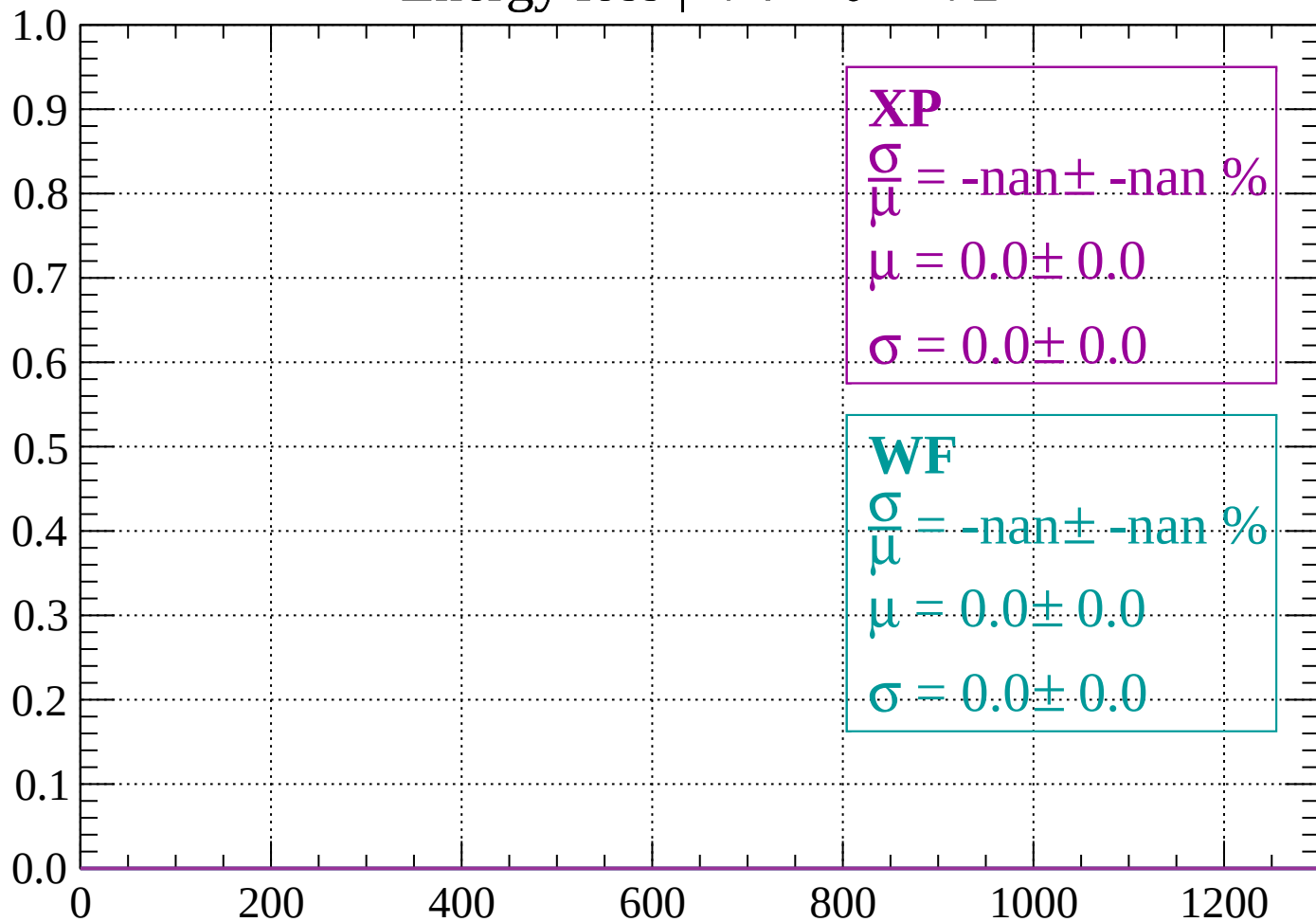
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

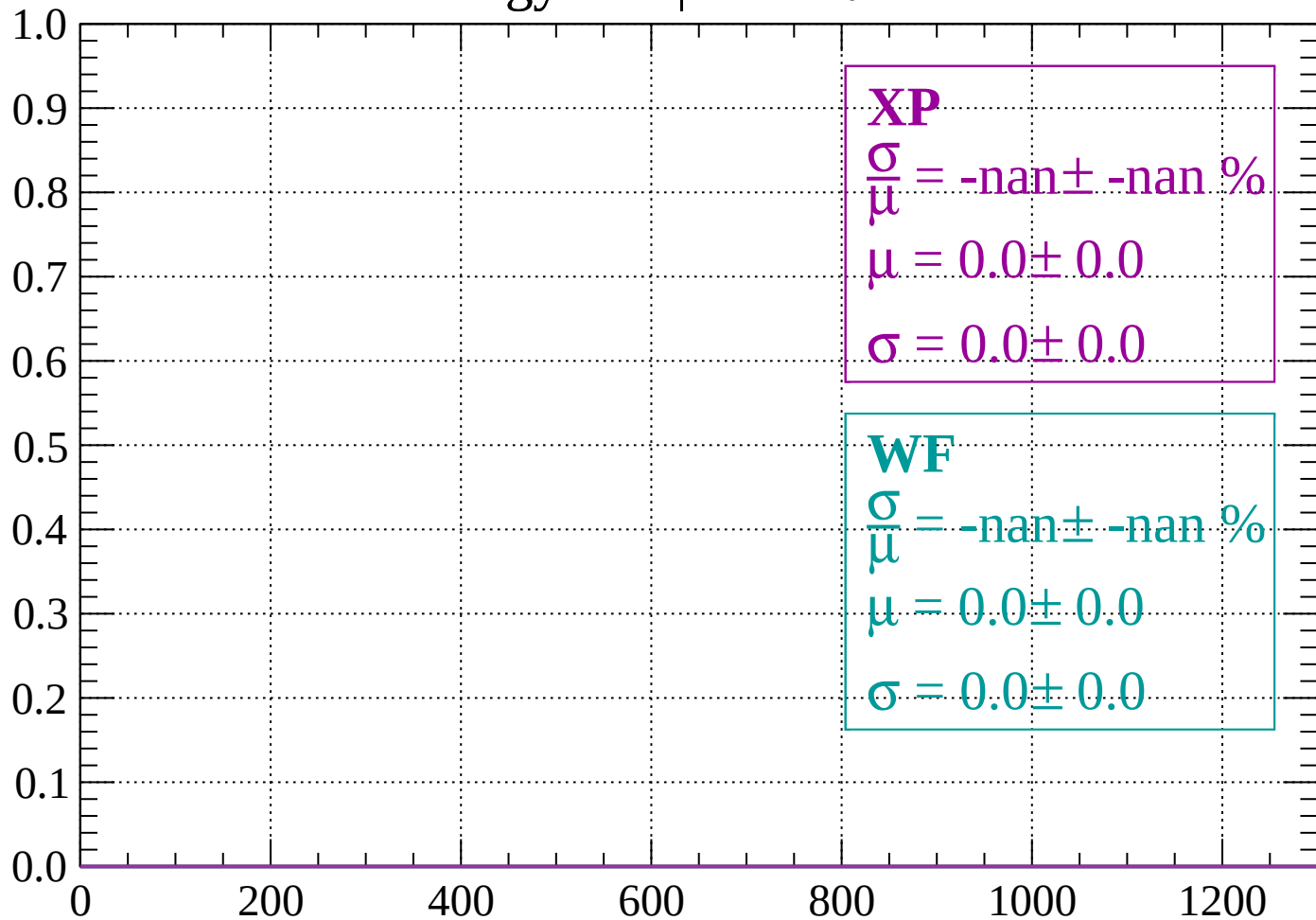
Count



dE/dx (ADC counts/cm)

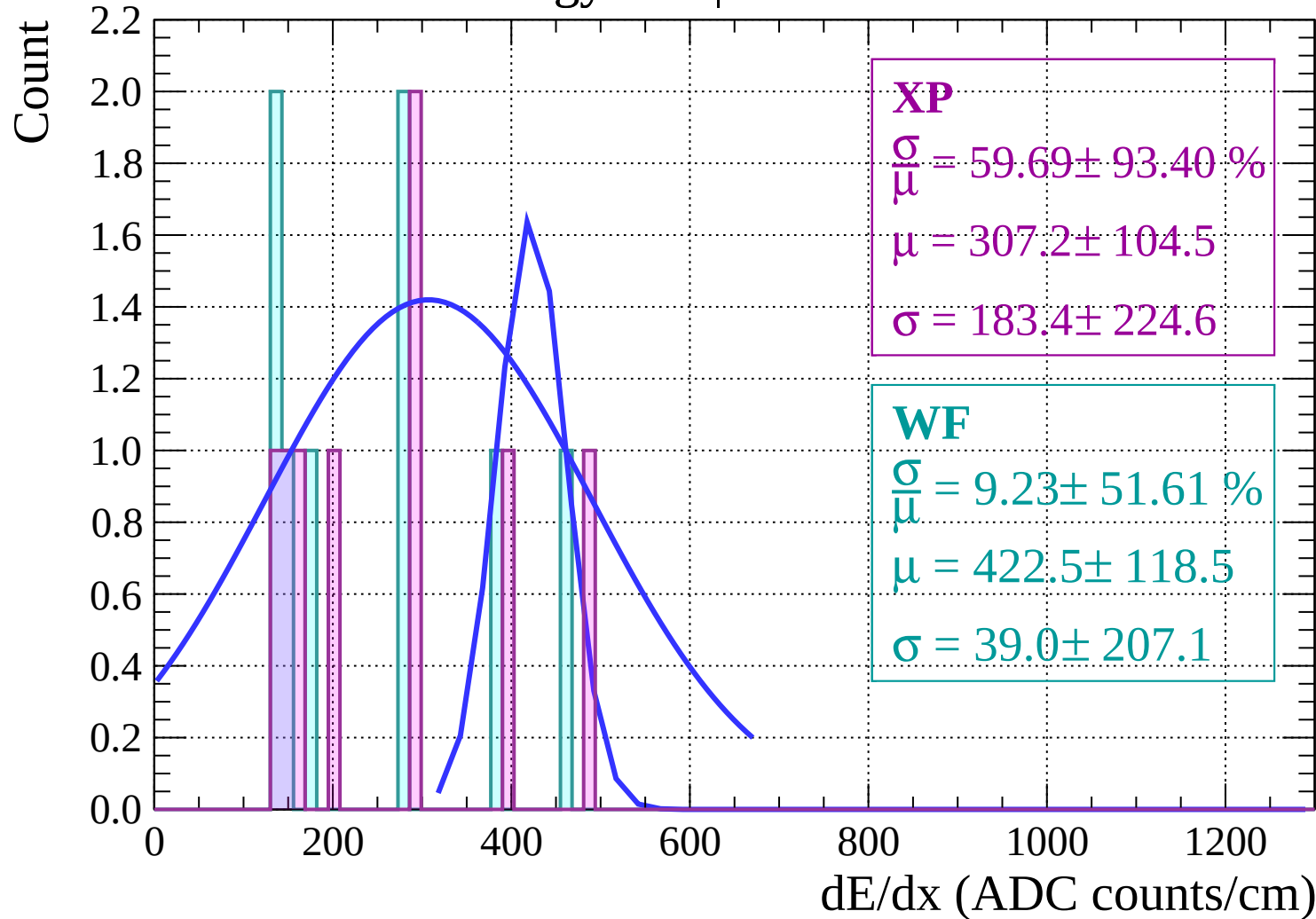
Energy loss | $-72 < \theta < -70$

Count



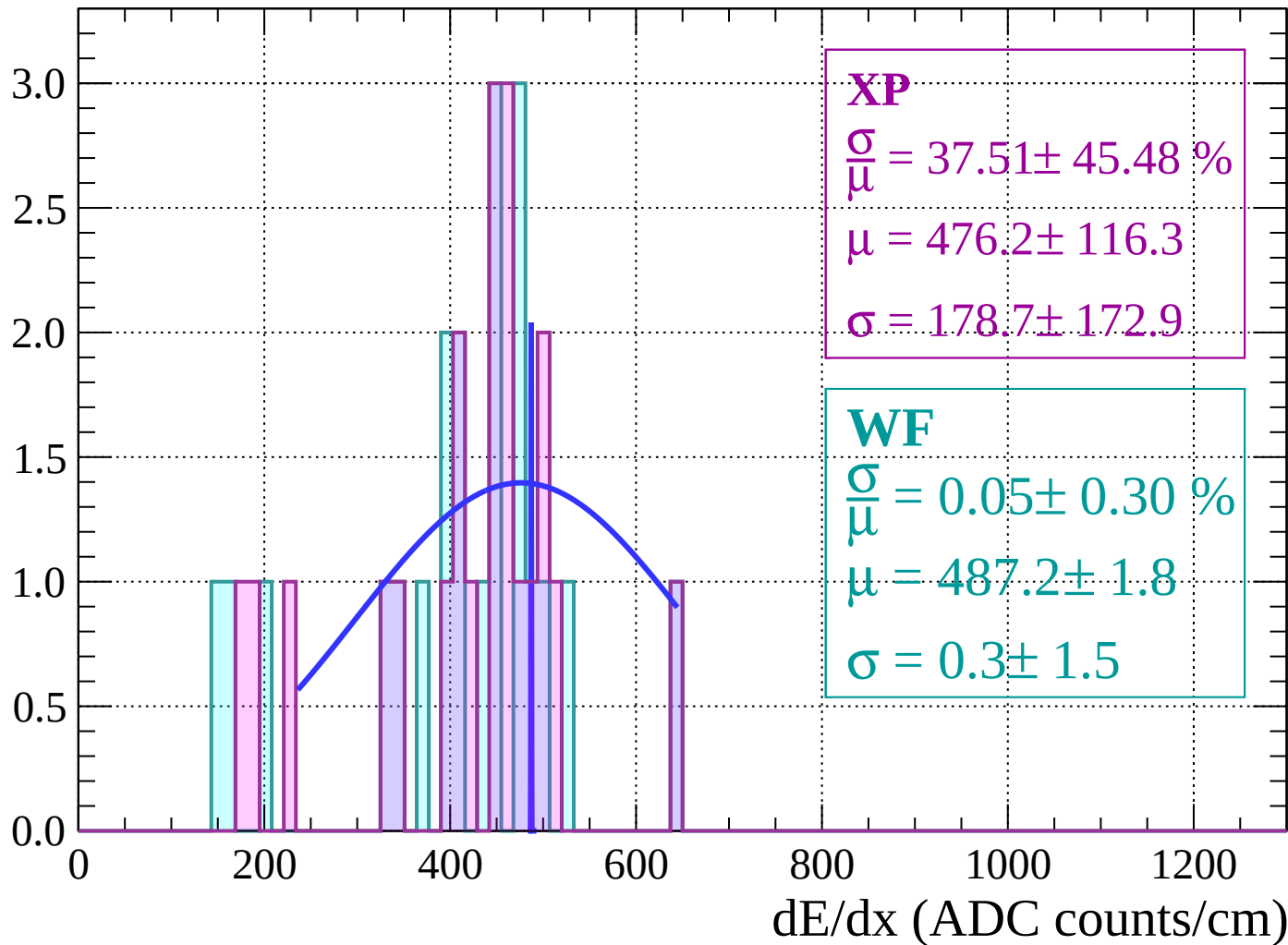
dE/dx (ADC counts/cm)

Energy loss | $-70 < \theta < -68$



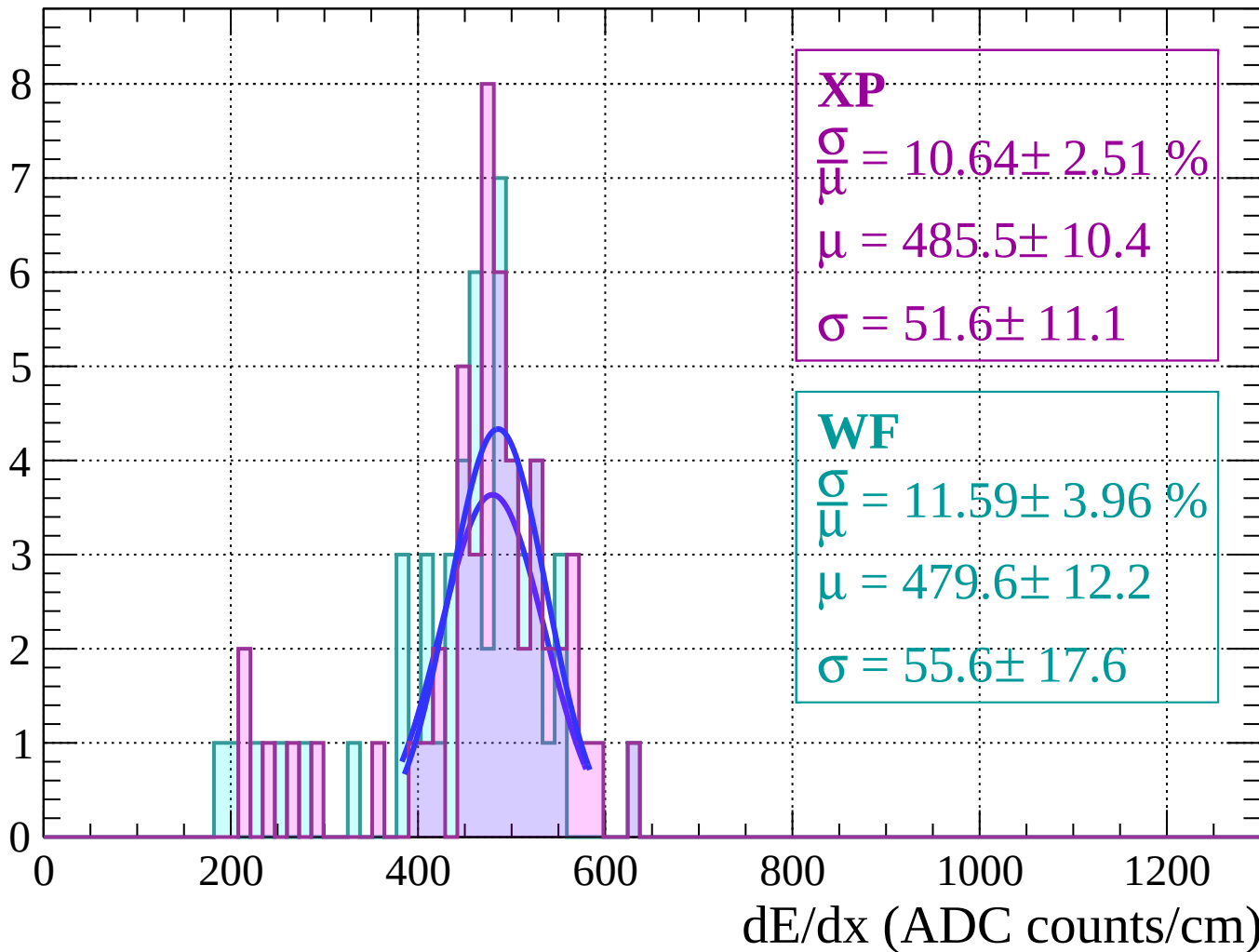
Energy loss | $-68 < \theta < -66$

Count



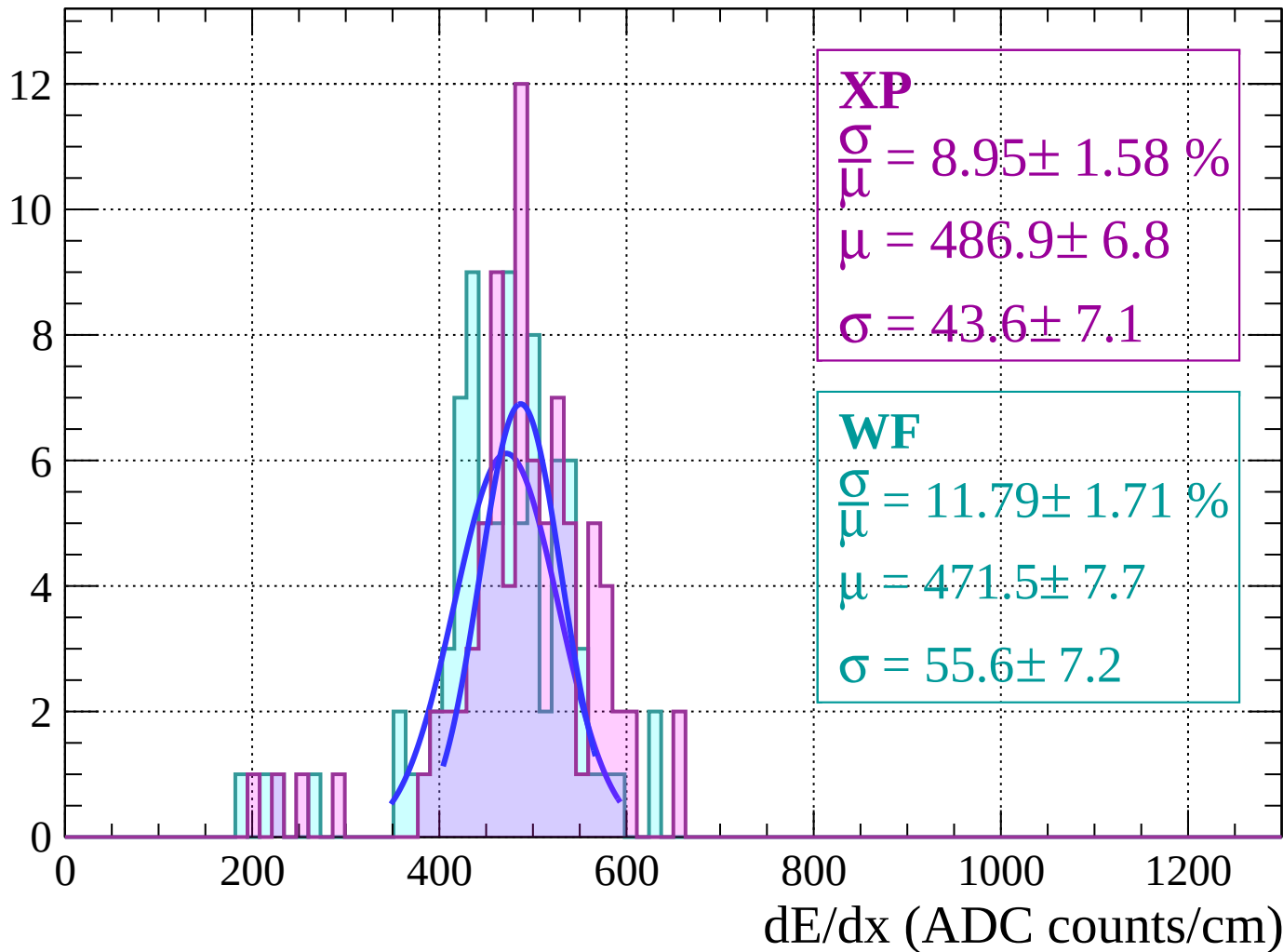
Energy loss | $-66 < \theta < -64$

Count



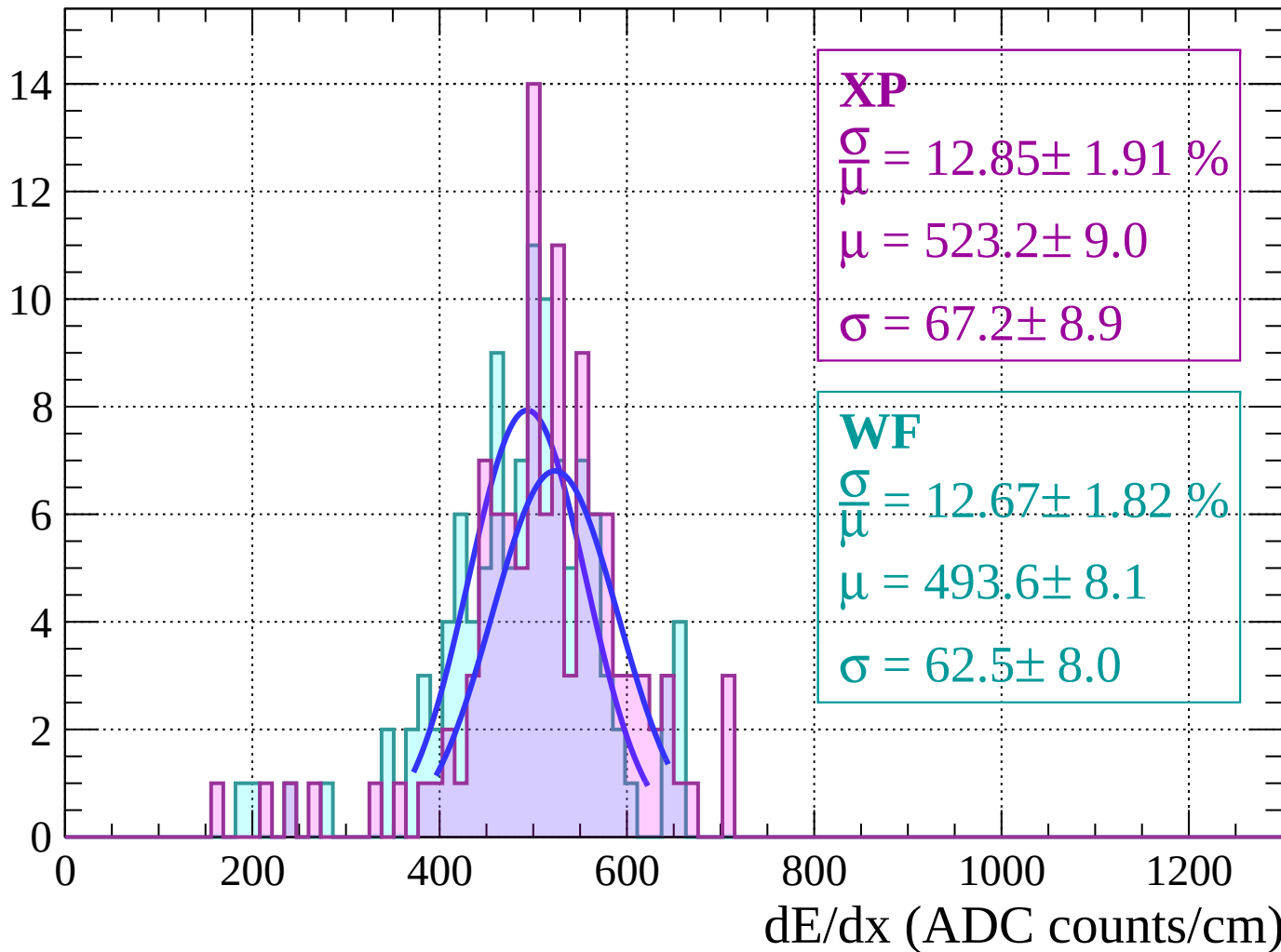
Energy loss | $-64 < \theta < -62$

Count



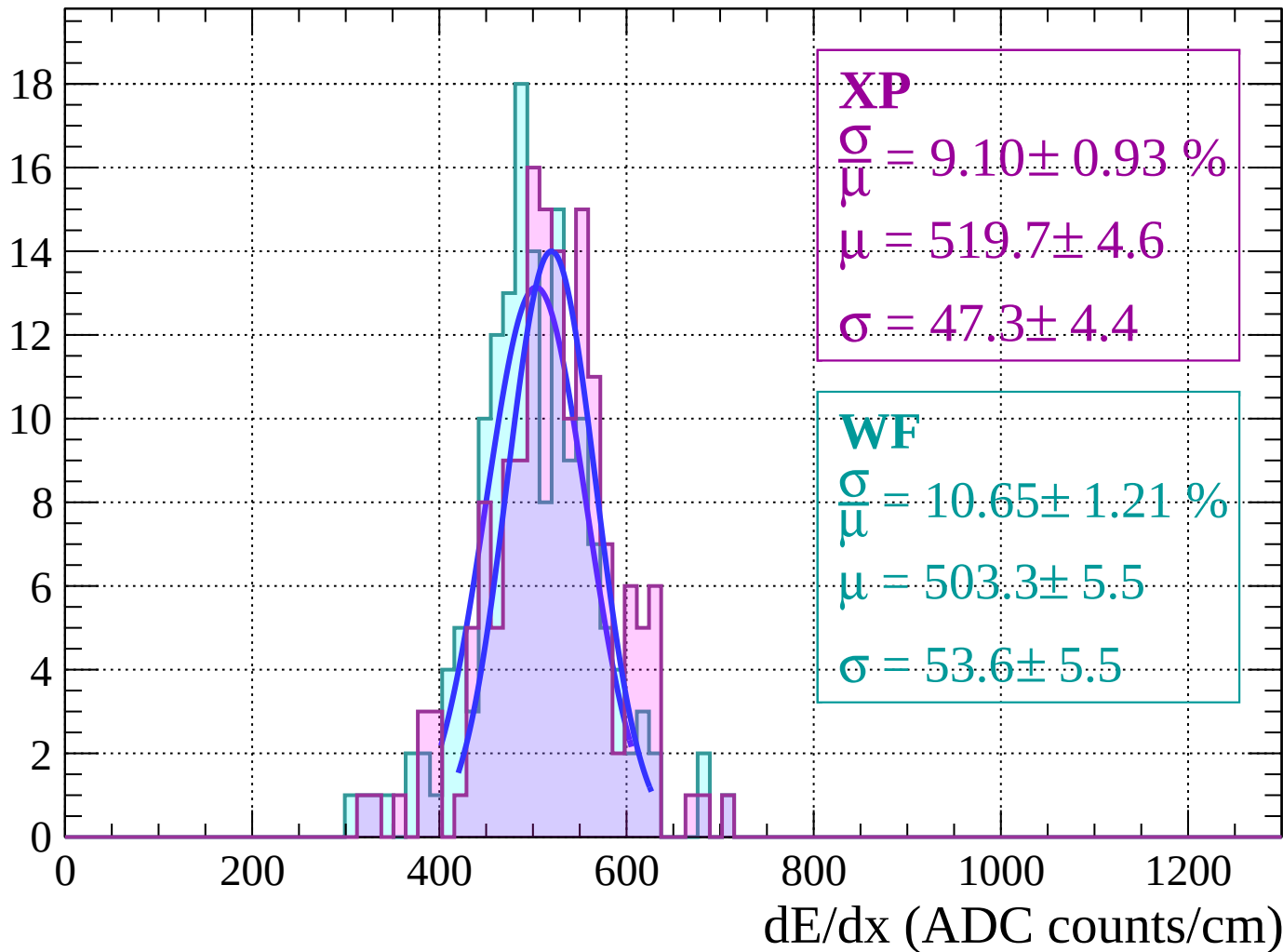
Energy loss | $-62 < \theta < -60$

Count



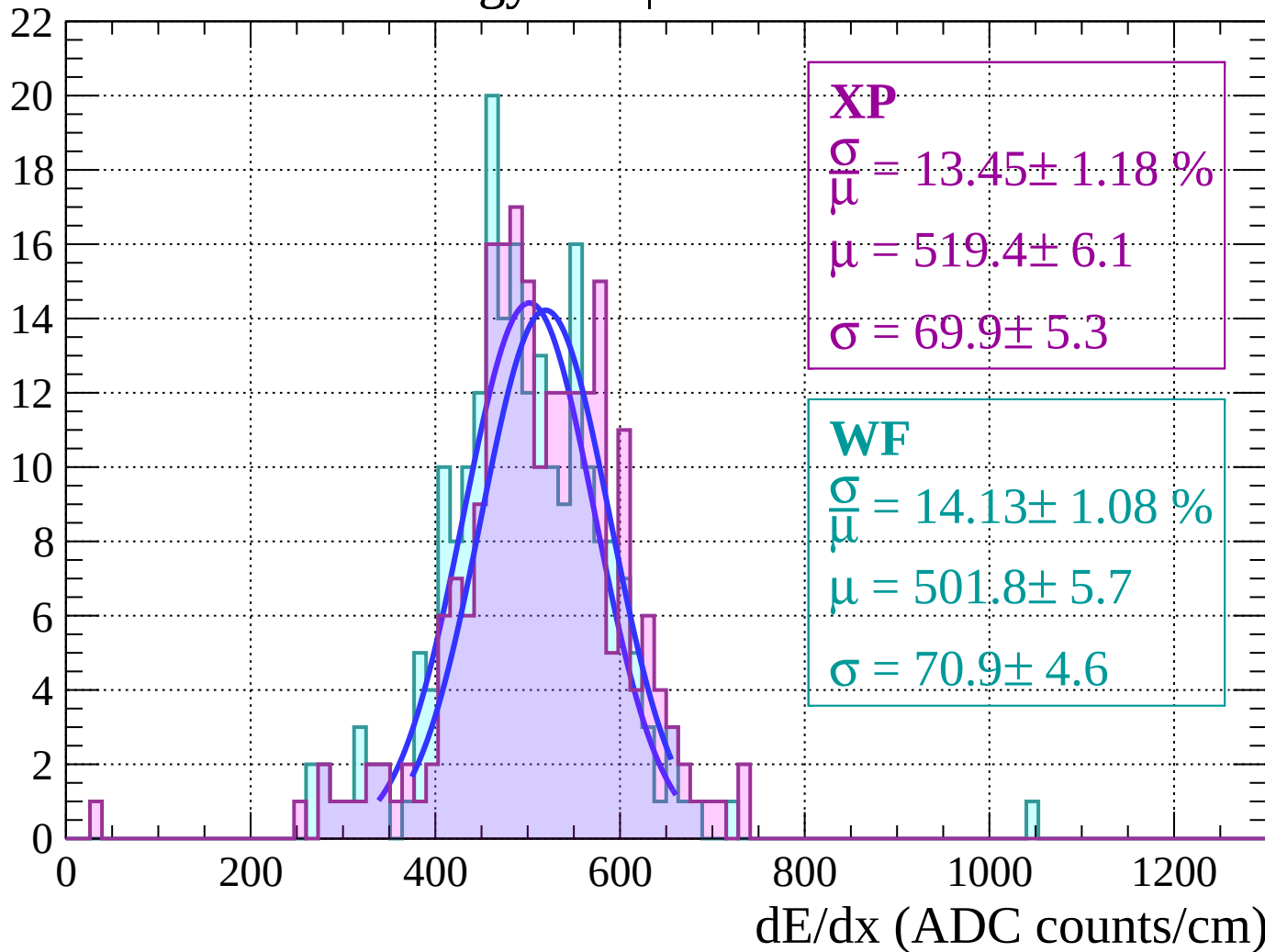
Energy loss | $-60 < \theta < -58$

Count



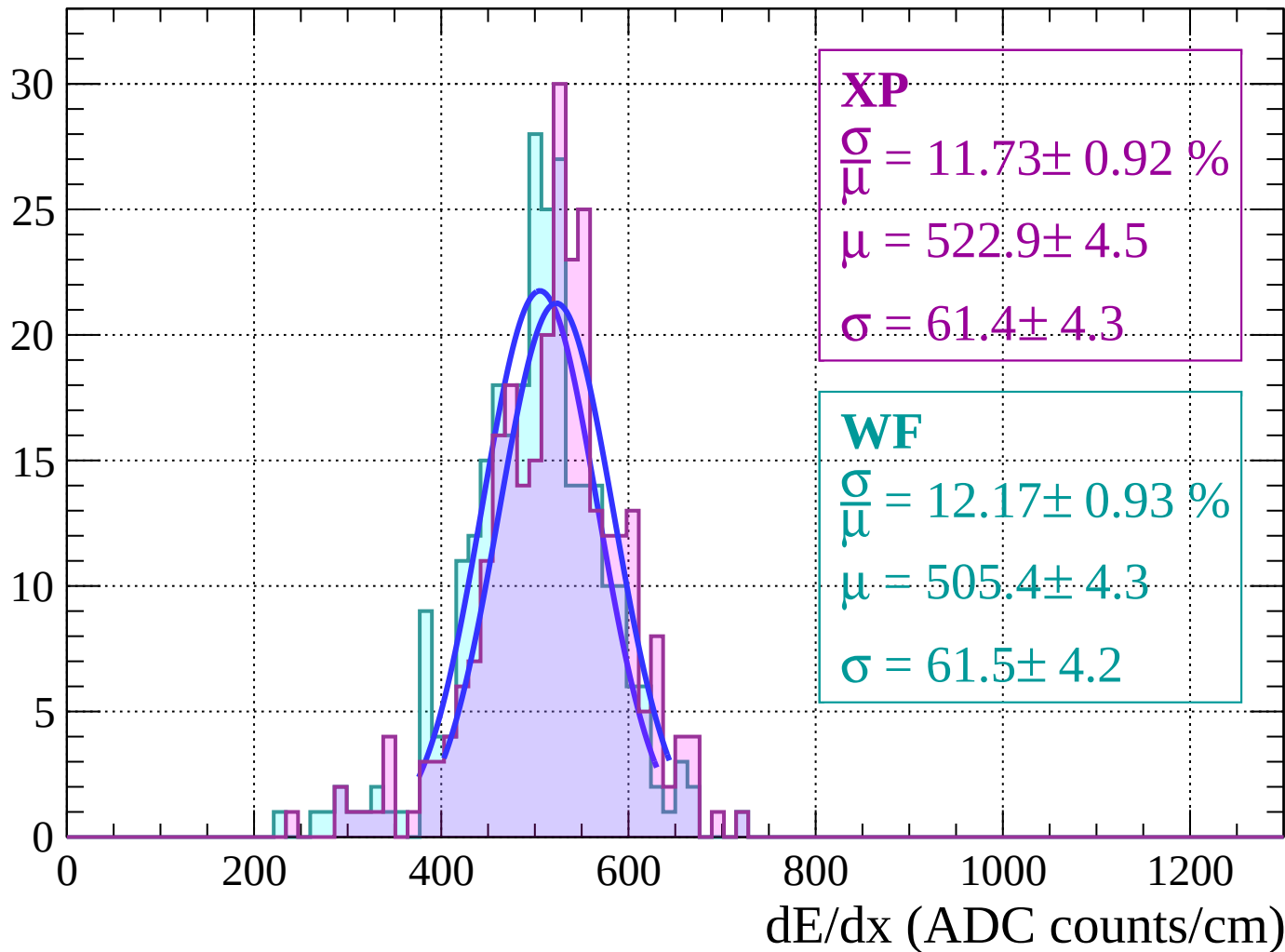
Energy loss | $-58 < \theta < -56$

Count



Energy loss $|-56 < \theta < -54$

Count



Energy loss $|-54 < \theta < -52$

Count

XP

$$\frac{\sigma}{\bar{\mu}} = 12.37 \pm 1.11 \%$$

$$\mu = 525.7 \pm 4.6$$

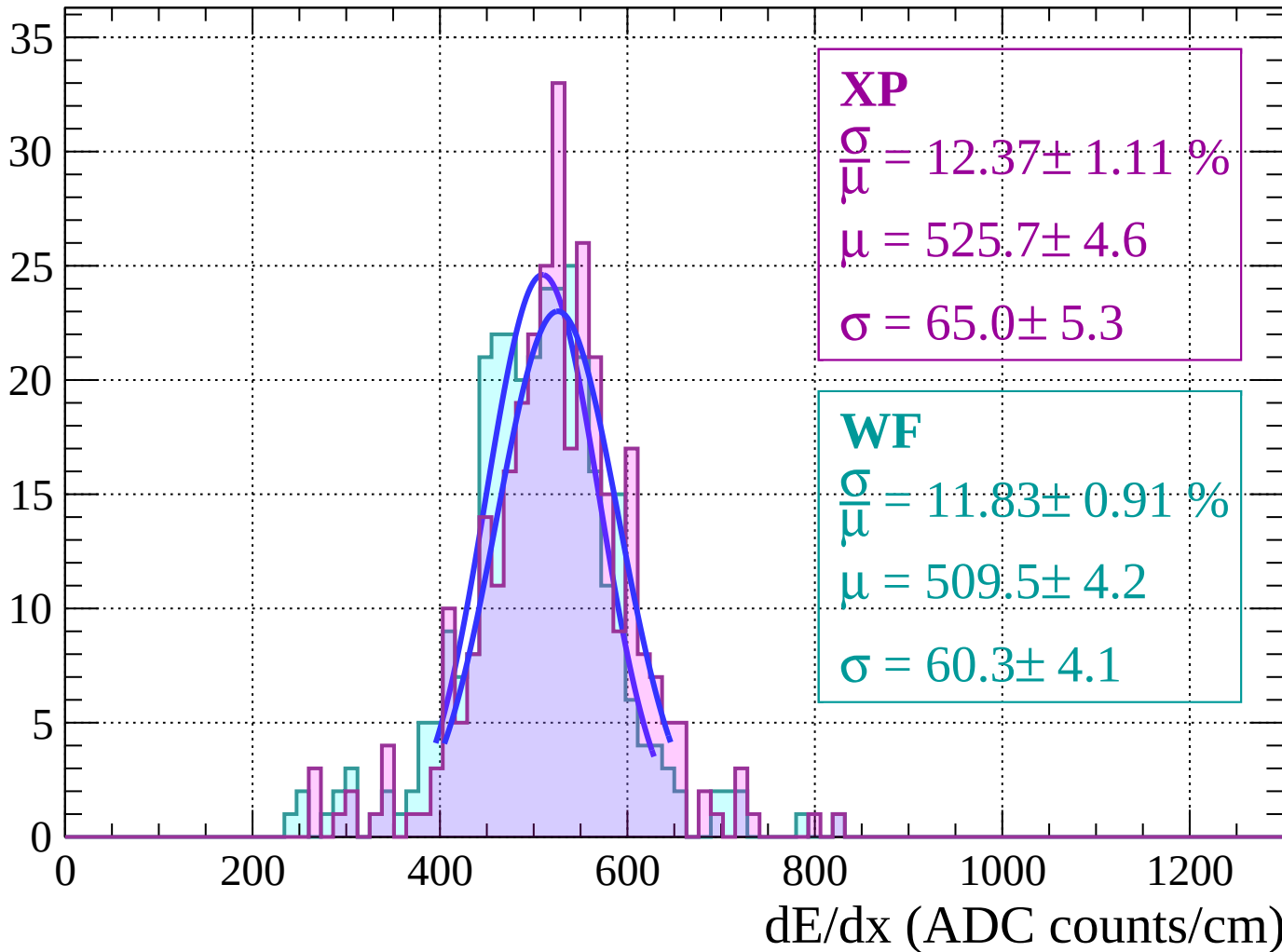
$$\sigma = 65.0 \pm 5.3$$

WF

$$\frac{\sigma}{\bar{\mu}} = 11.83 \pm 0.91 \%$$

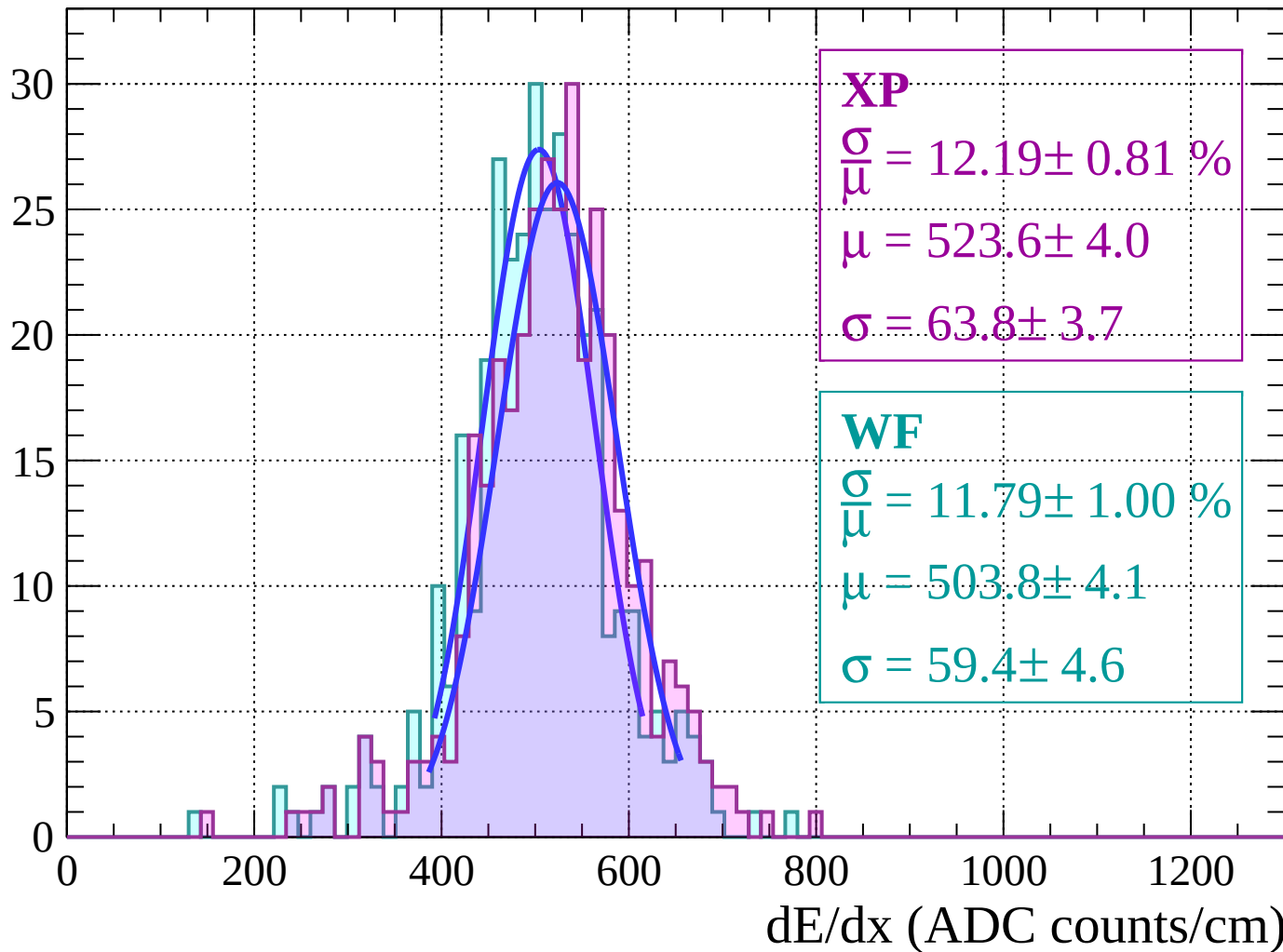
$$\mu = 509.5 \pm 4.2$$

$$\sigma = 60.3 \pm 4.1$$



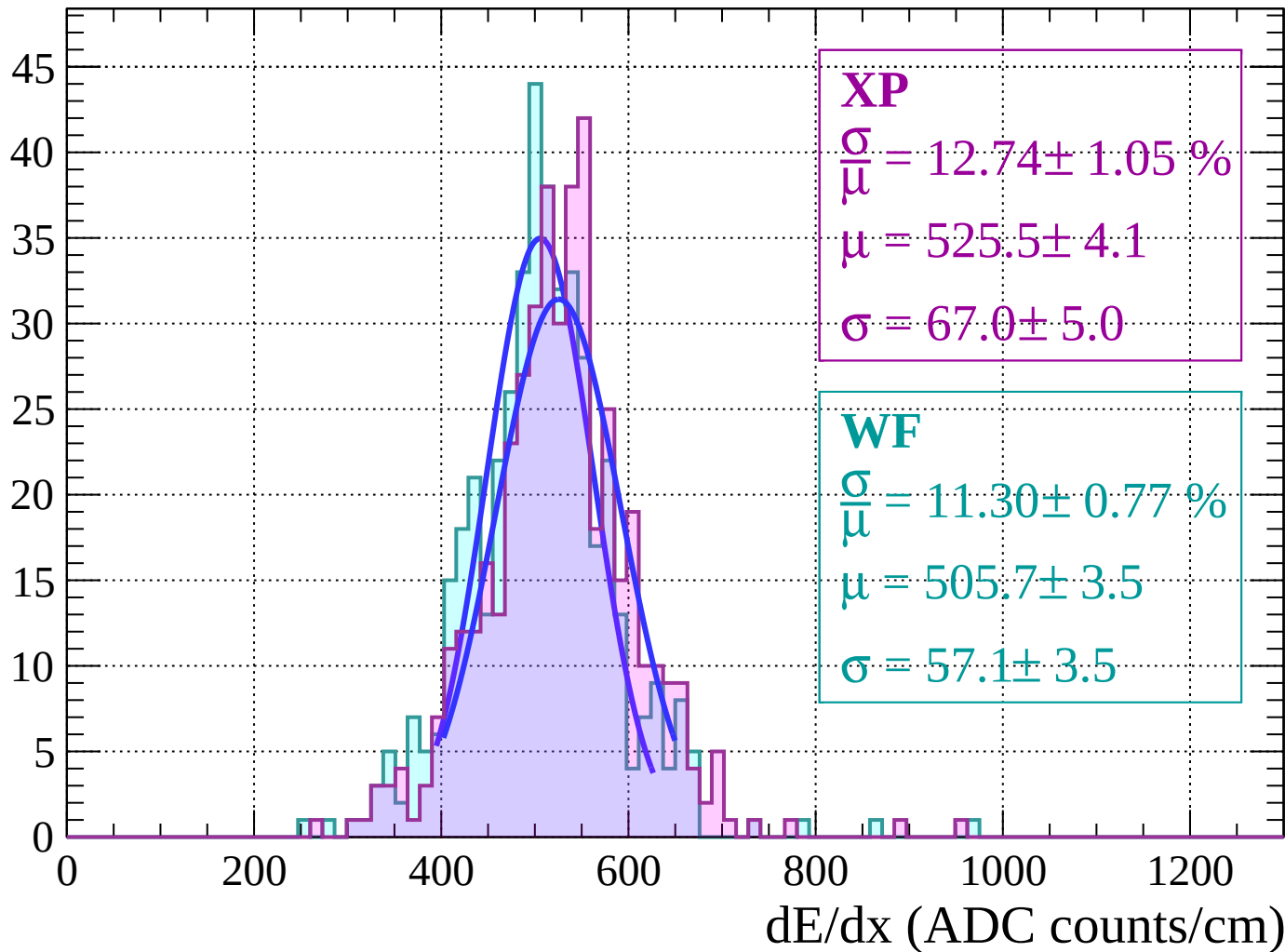
Energy loss | $-52 < \theta < -50$

Count



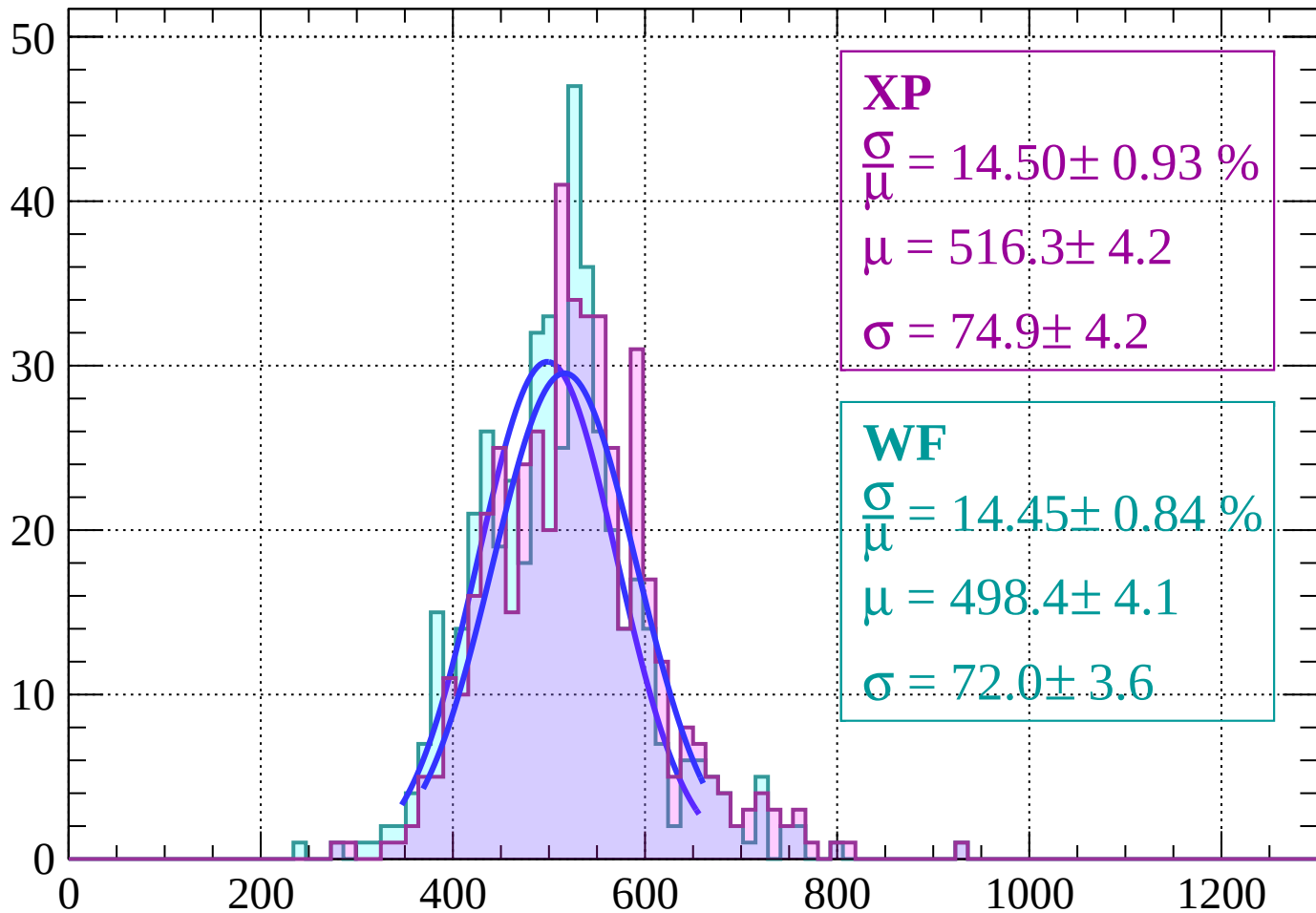
Energy loss | $-50 < \theta < -48$

Count



Energy loss | $-48 < \theta < -46$

Count



XP

$$\frac{\sigma}{\mu} = 14.50 \pm 0.93 \%$$

$$\mu = 516.3 \pm 4.2$$

$$\sigma = 74.9 \pm 4.2$$

WF

$$\frac{\sigma}{\mu} = 14.45 \pm 0.84 \%$$

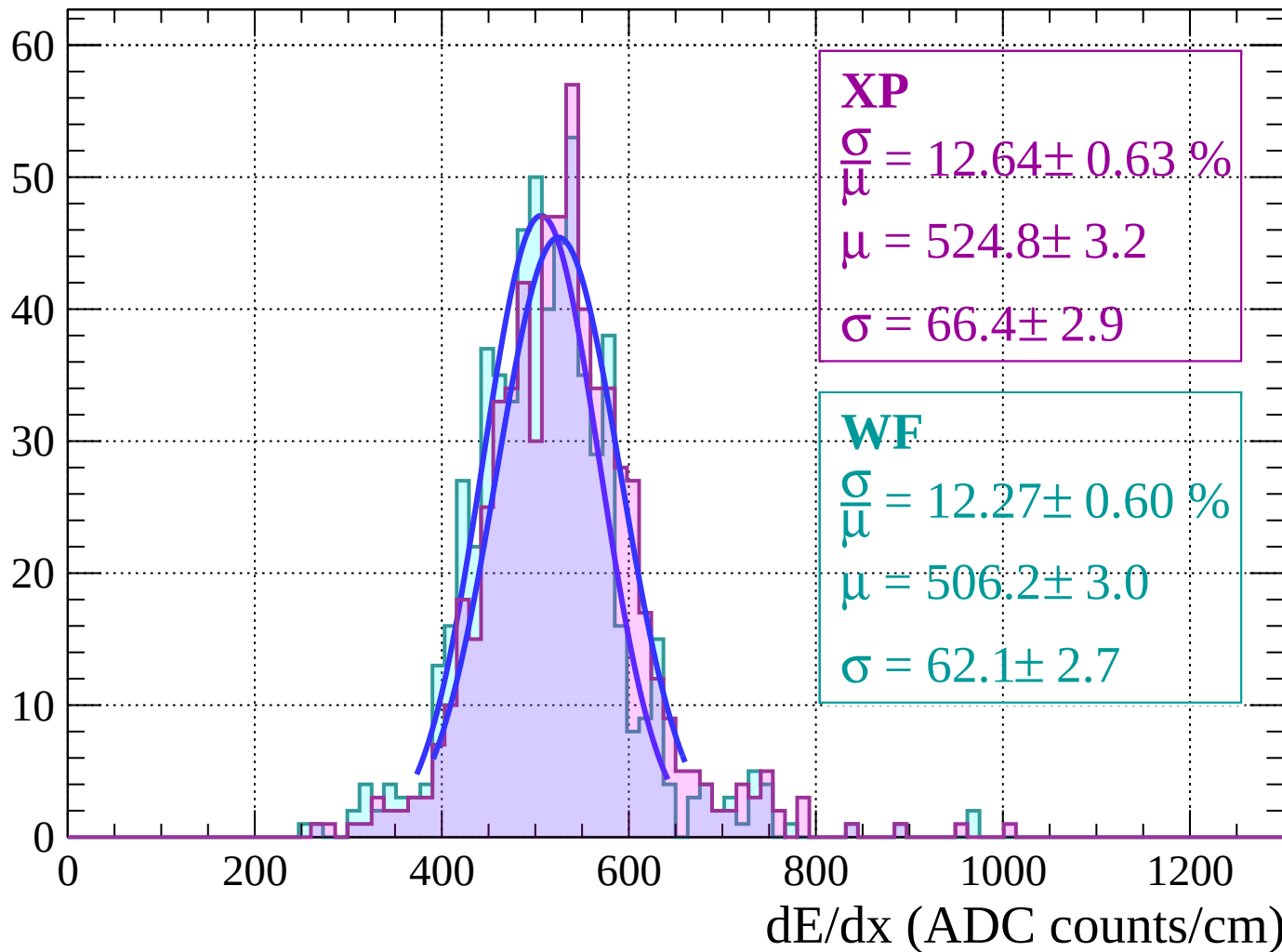
$$\mu = 498.4 \pm 4.1$$

$$\sigma = 72.0 \pm 3.6$$

dE/dx (ADC counts/cm)

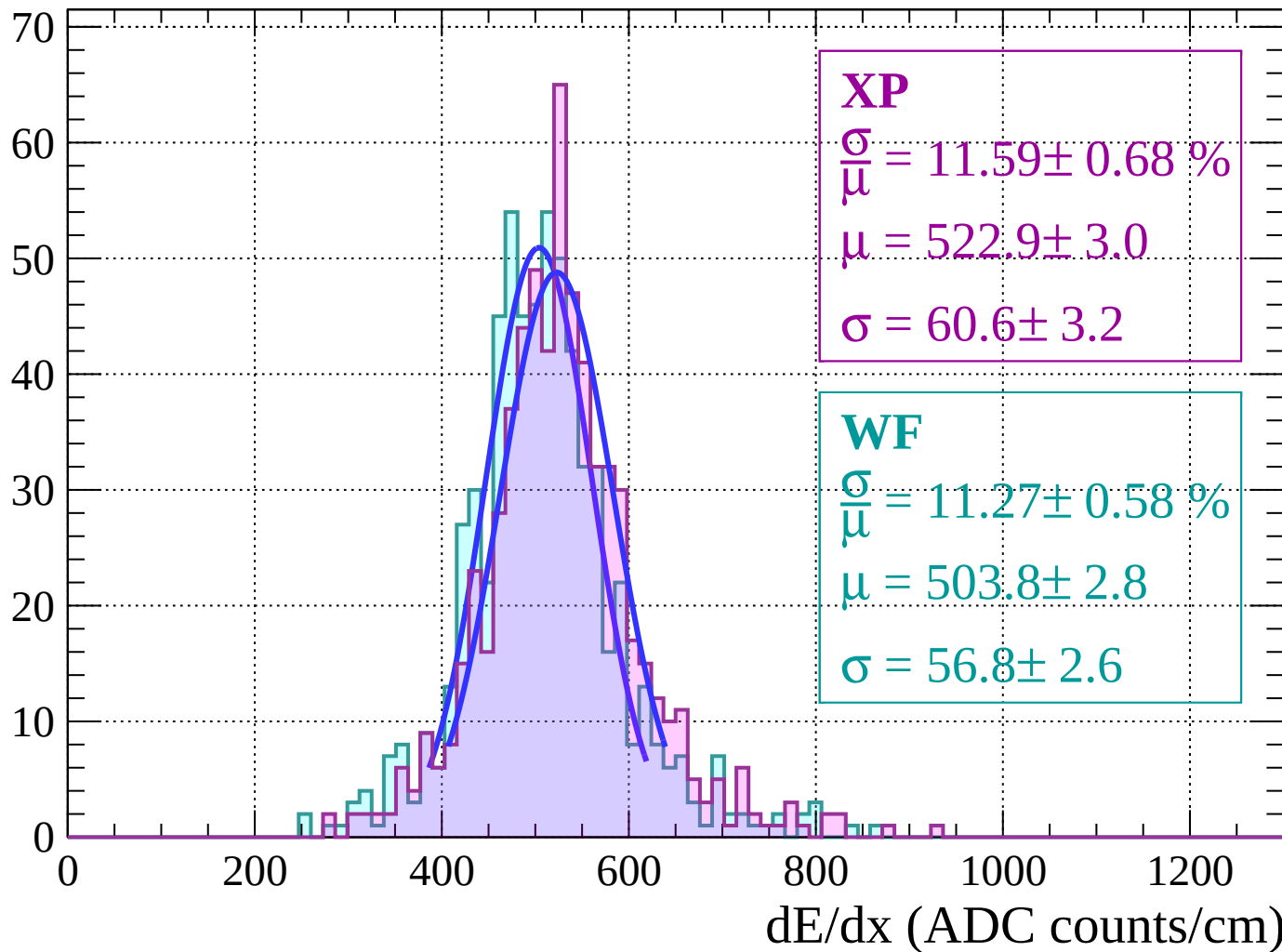
Energy loss $|-46 < \theta < -44$

Count



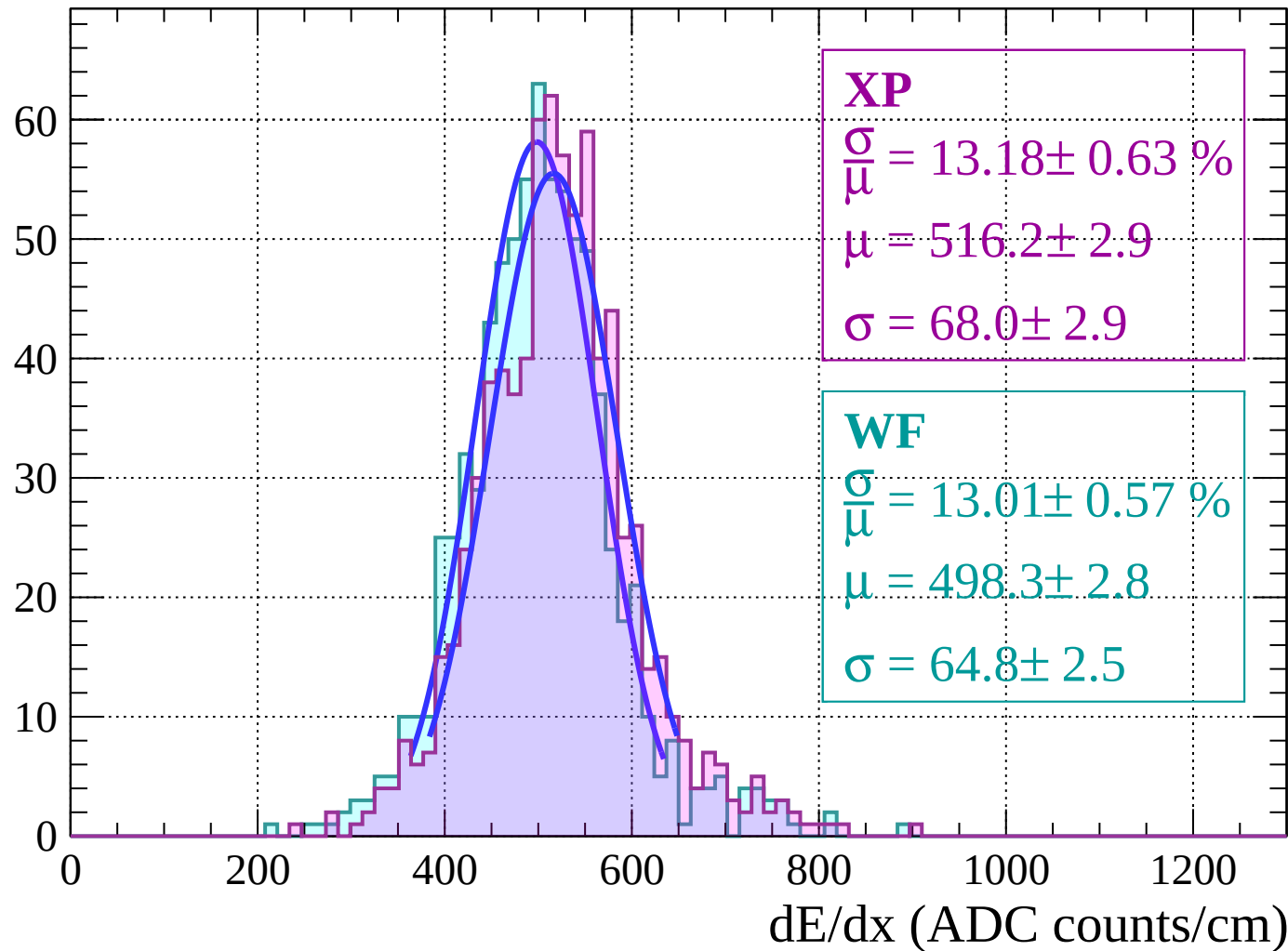
Energy loss | $-44 < \theta < -42$

Count



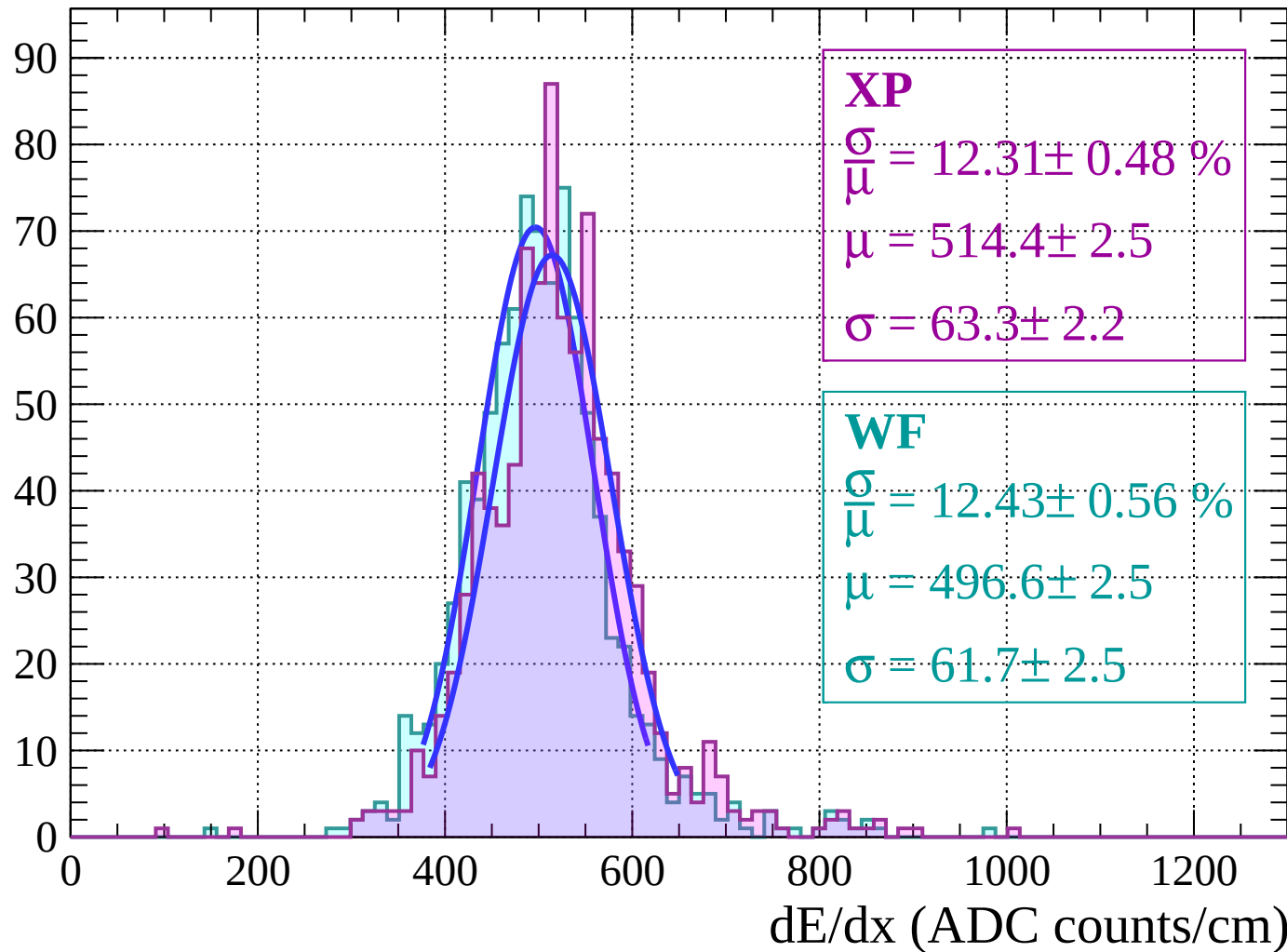
Energy loss | $-42 < \theta < -40$

Count



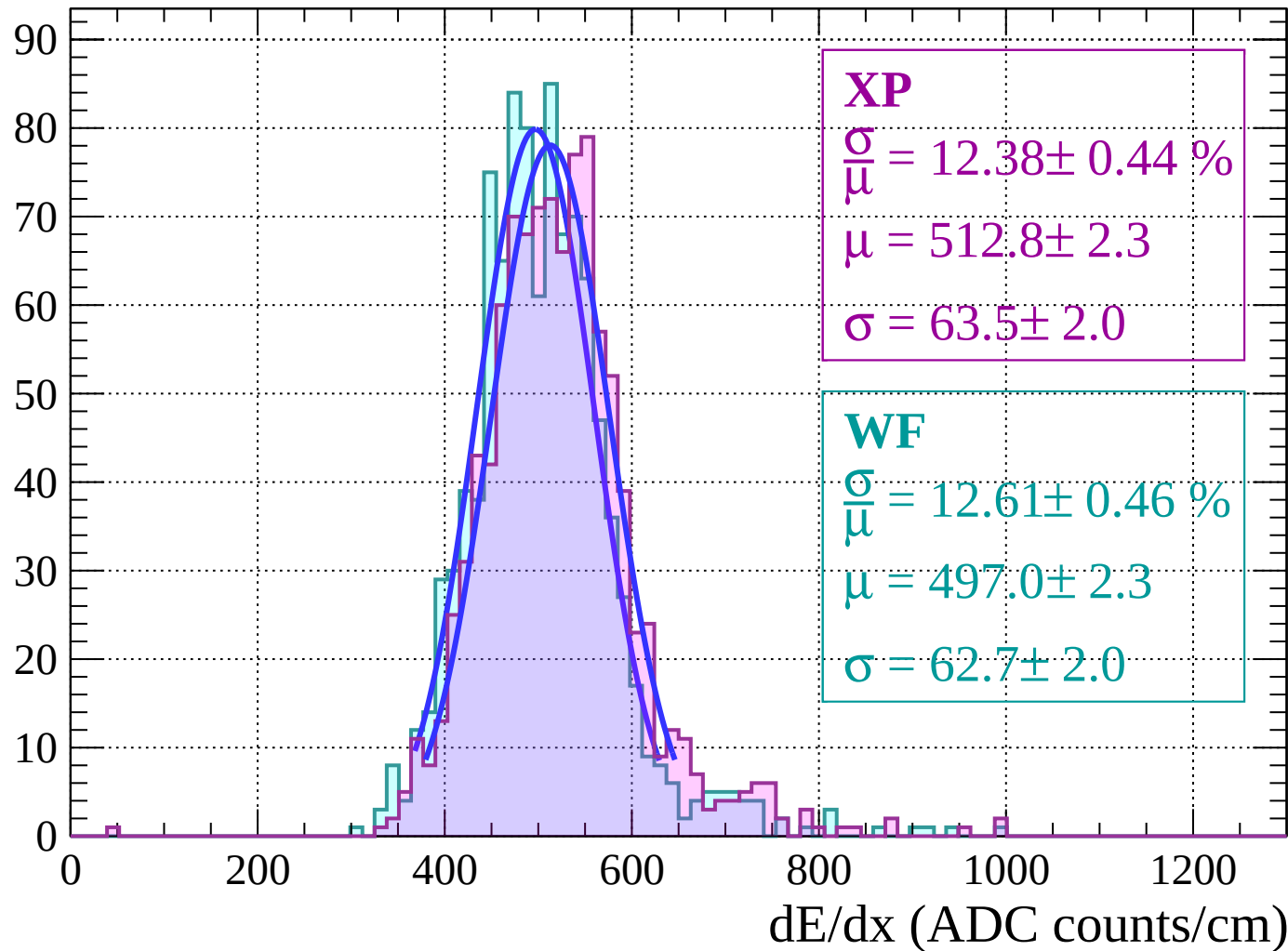
Energy loss | $-40 < \theta < -38$

Count



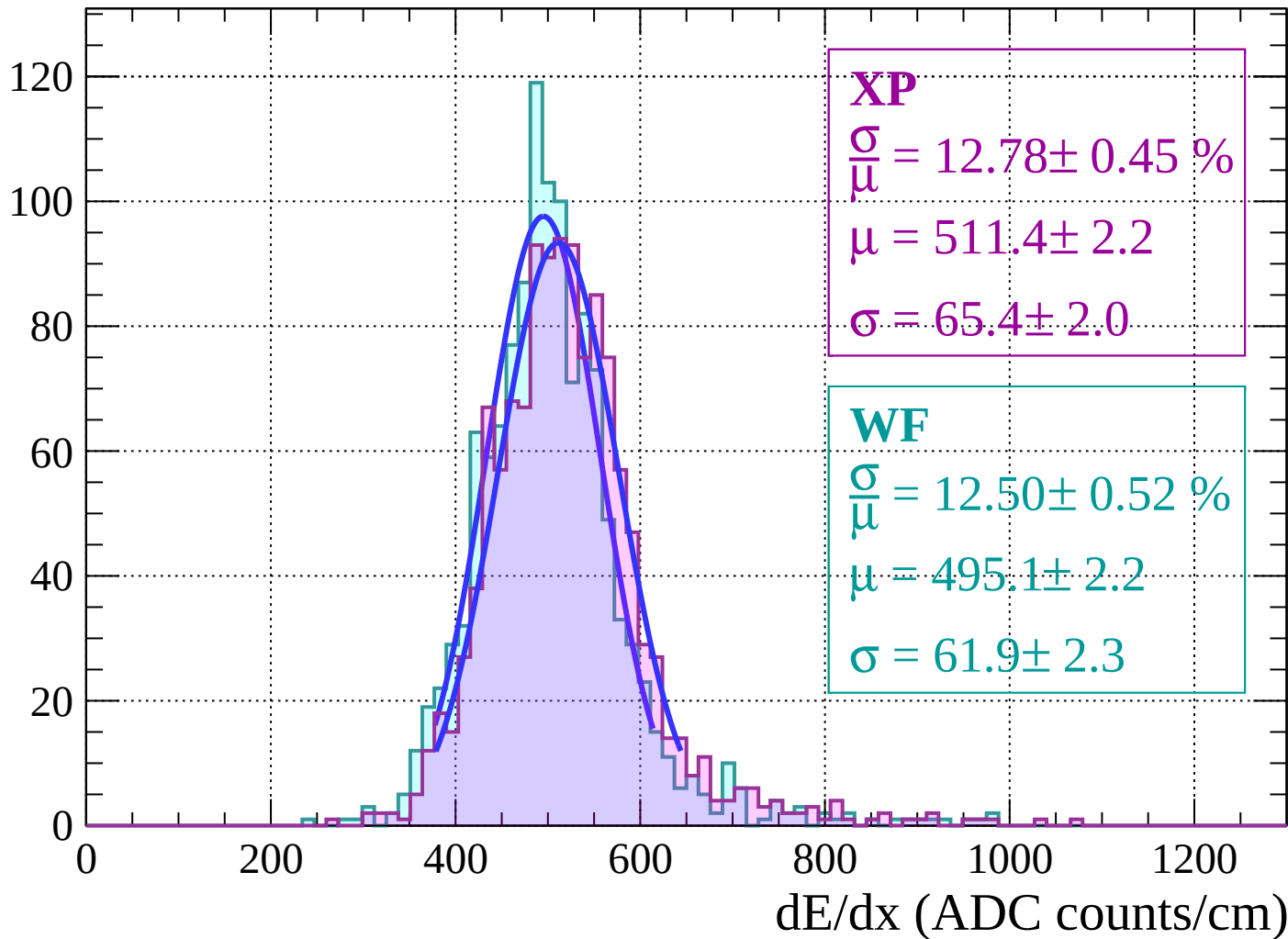
Energy loss | $-38 < \theta < -36$

Count



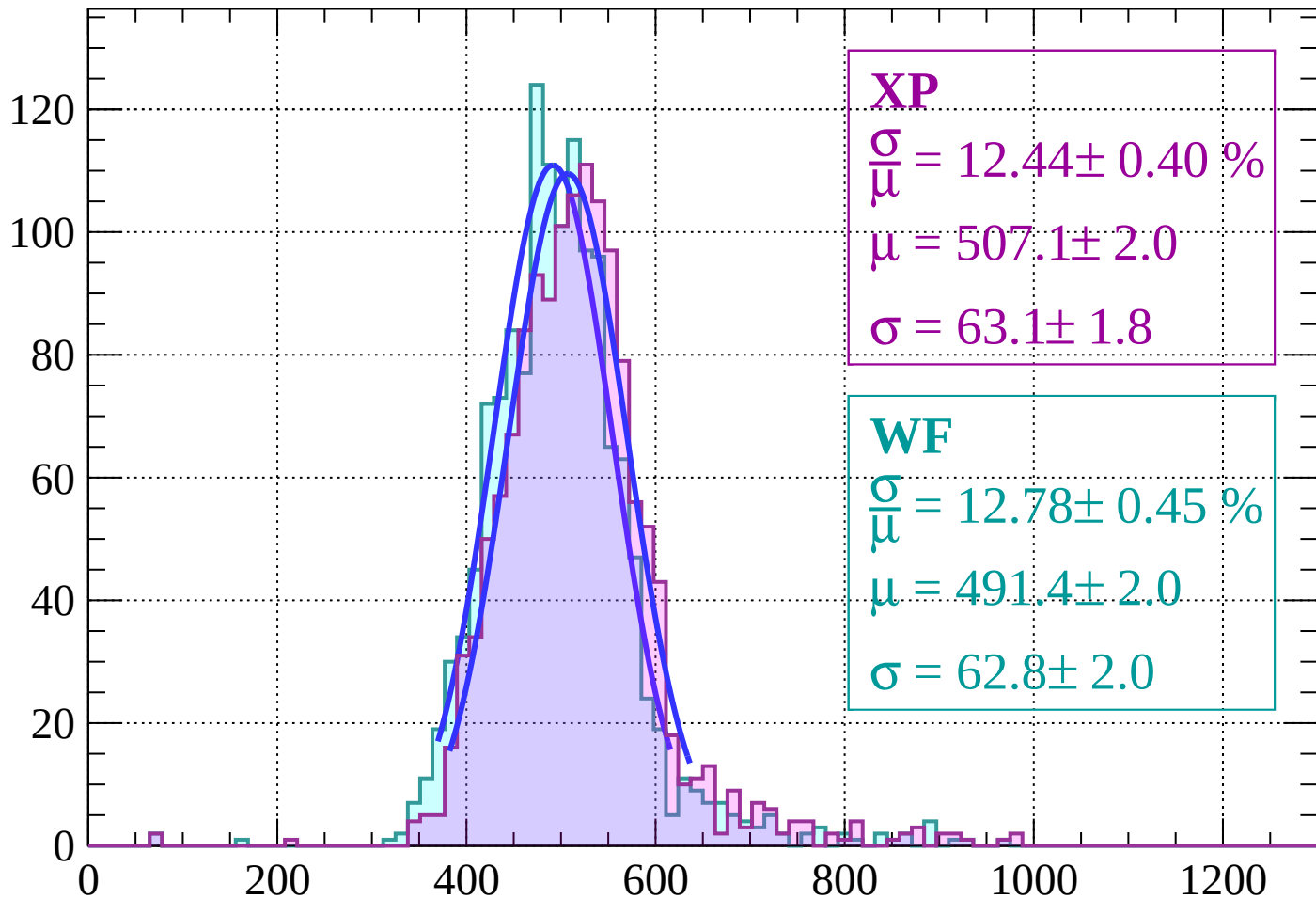
Energy loss $|-36 < \theta < -34$

Count



Energy loss $|-34 < \theta < -32$

Count



XP

$$\frac{\sigma}{\mu} = 12.44 \pm 0.40 \%$$

$$\mu = 507.1 \pm 2.0$$

$$\sigma = 63.1 \pm 1.8$$

WF

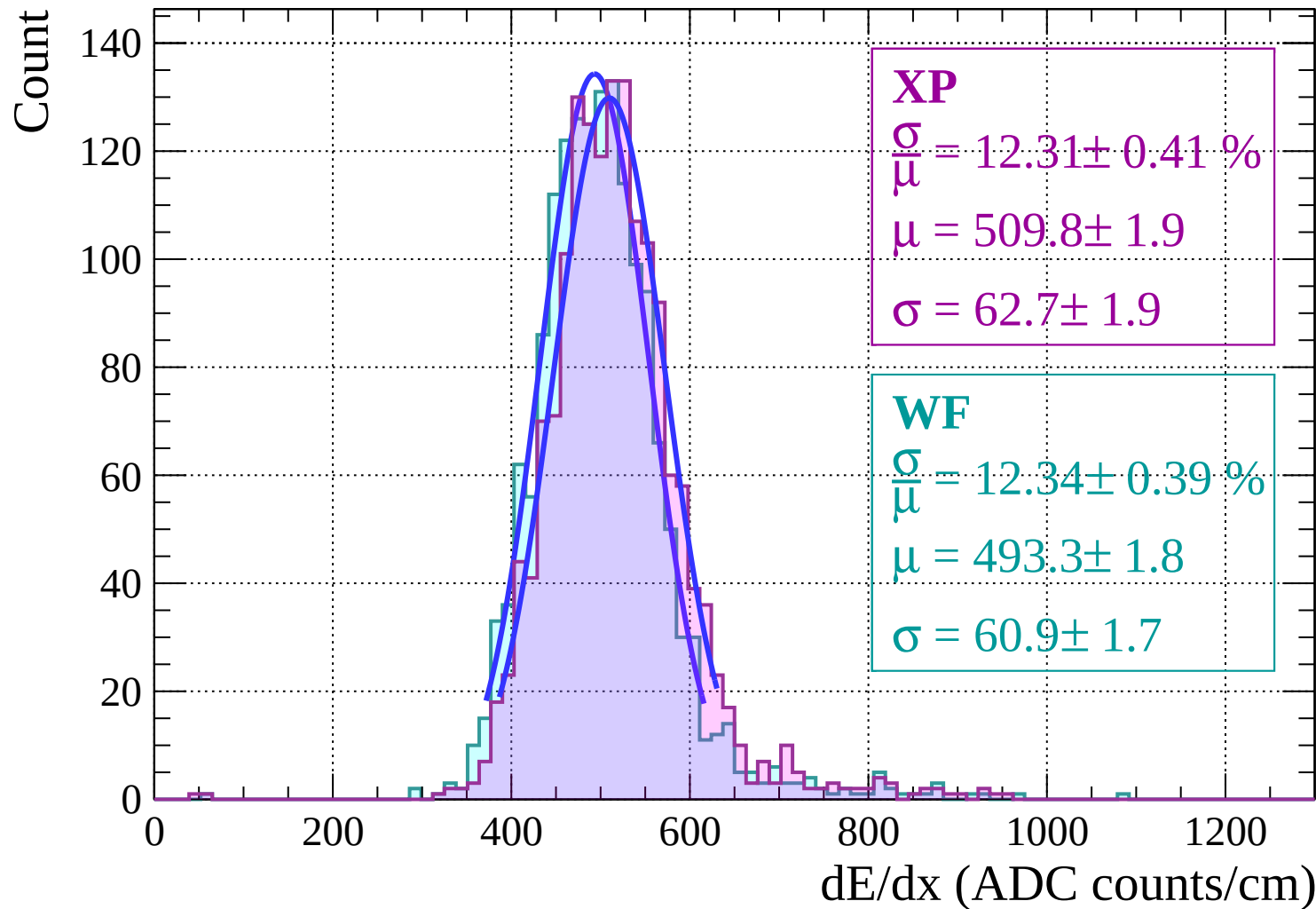
$$\frac{\sigma}{\mu} = 12.78 \pm 0.45 \%$$

$$\mu = 491.4 \pm 2.0$$

$$\sigma = 62.8 \pm 2.0$$

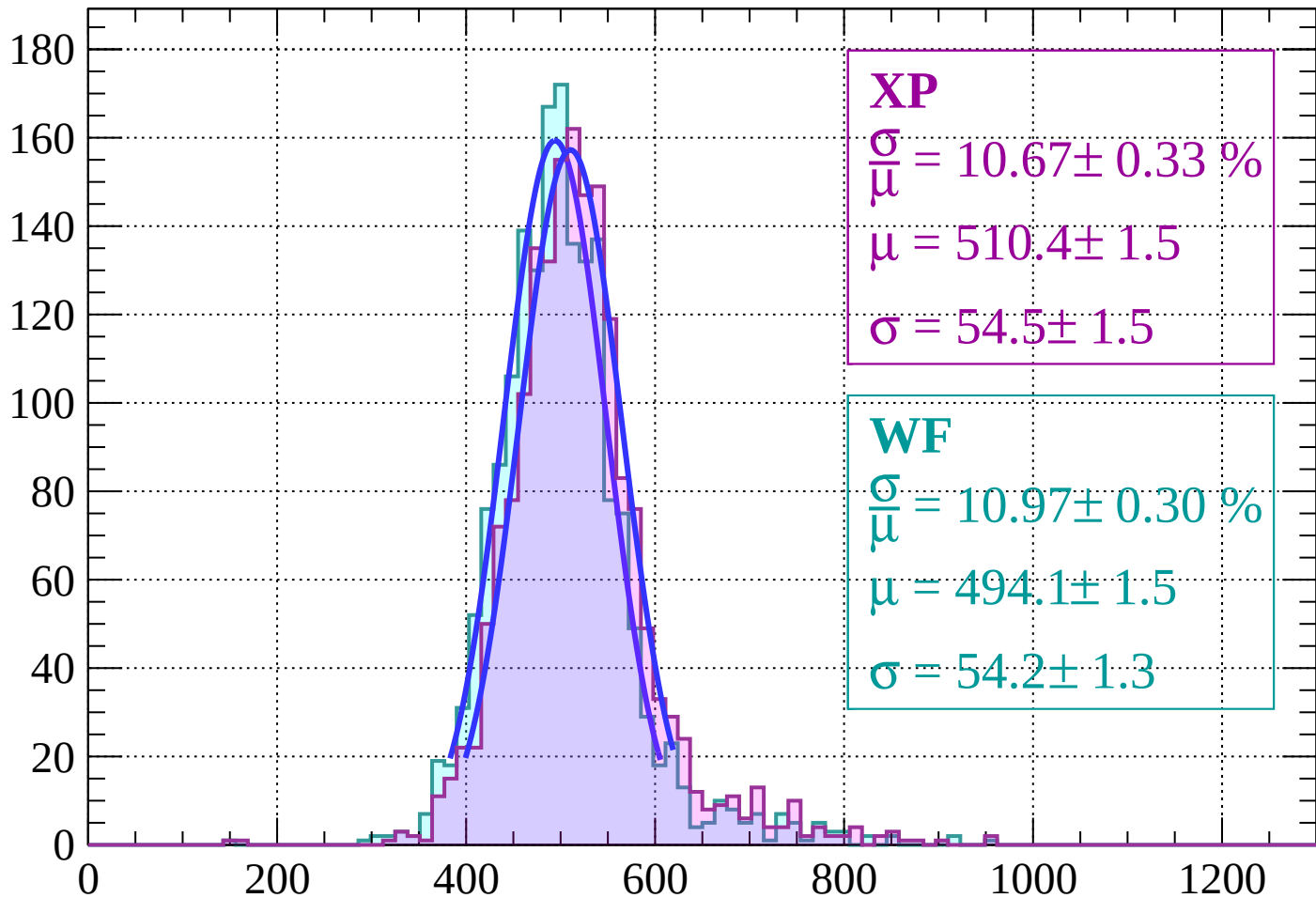
dE/dx (ADC counts/cm)

Energy loss | $-32 < \theta < -30$



Energy loss $|-30 < \theta < -28$

Count



XP

$$\frac{\sigma}{\mu} = 10.67 \pm 0.33 \%$$

$$\mu = 510.4 \pm 1.5$$

$$\sigma = 54.5 \pm 1.5$$

WF

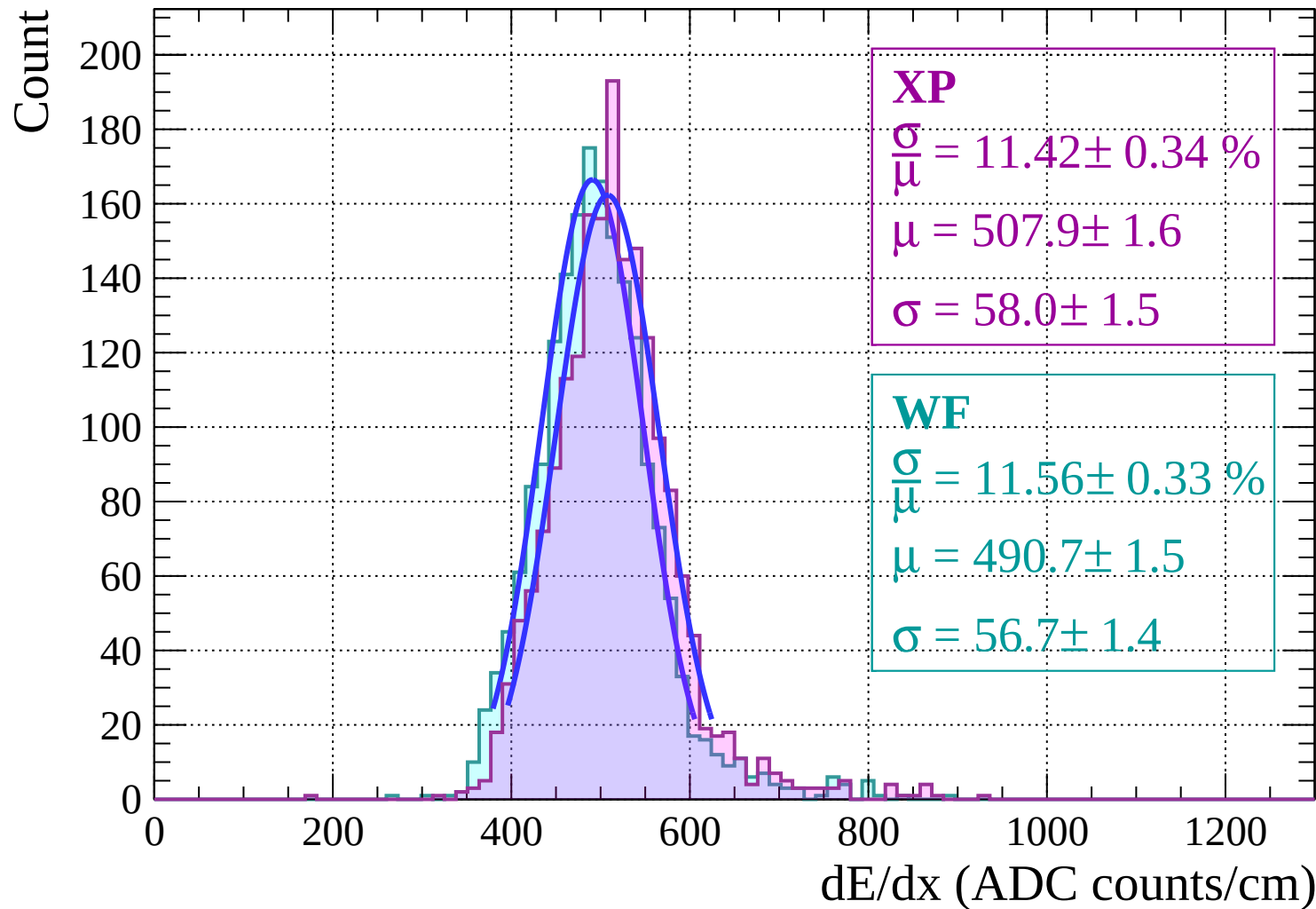
$$\frac{\sigma}{\mu} = 10.97 \pm 0.30 \%$$

$$\mu = 494.1 \pm 1.5$$

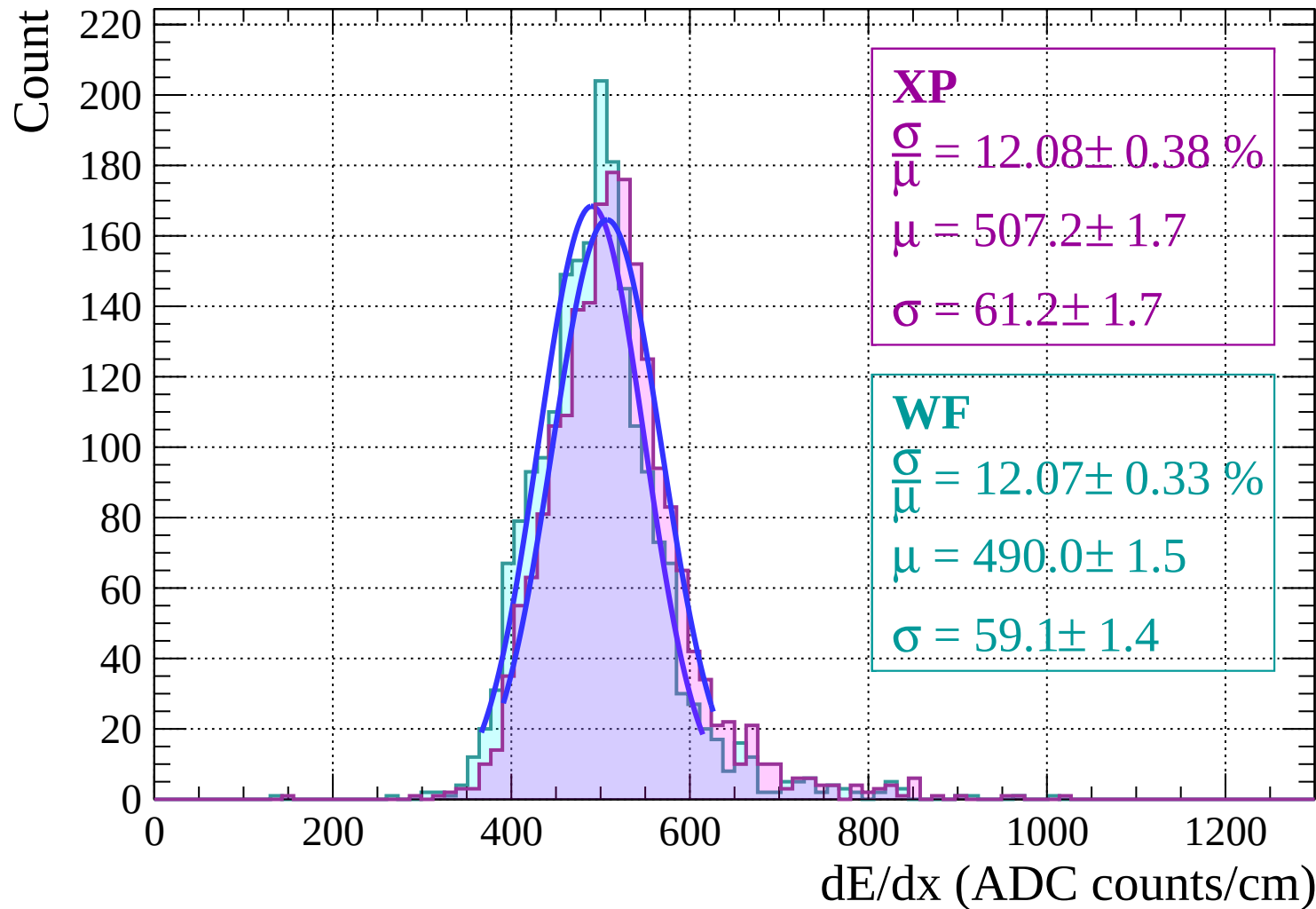
$$\sigma = 54.2 \pm 1.3$$

dE/dx (ADC counts/cm)

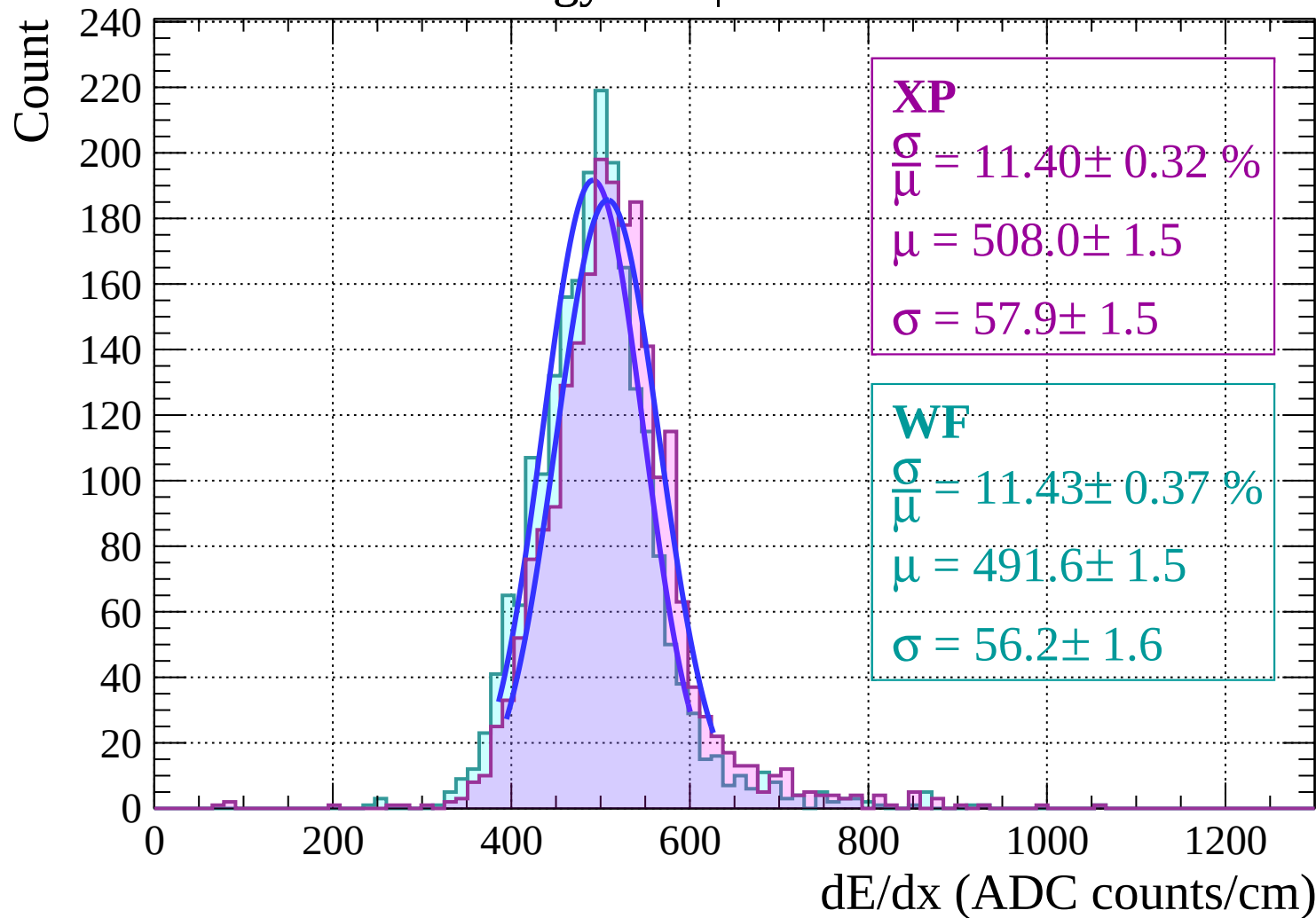
Energy loss | $-28 < \theta < -26$



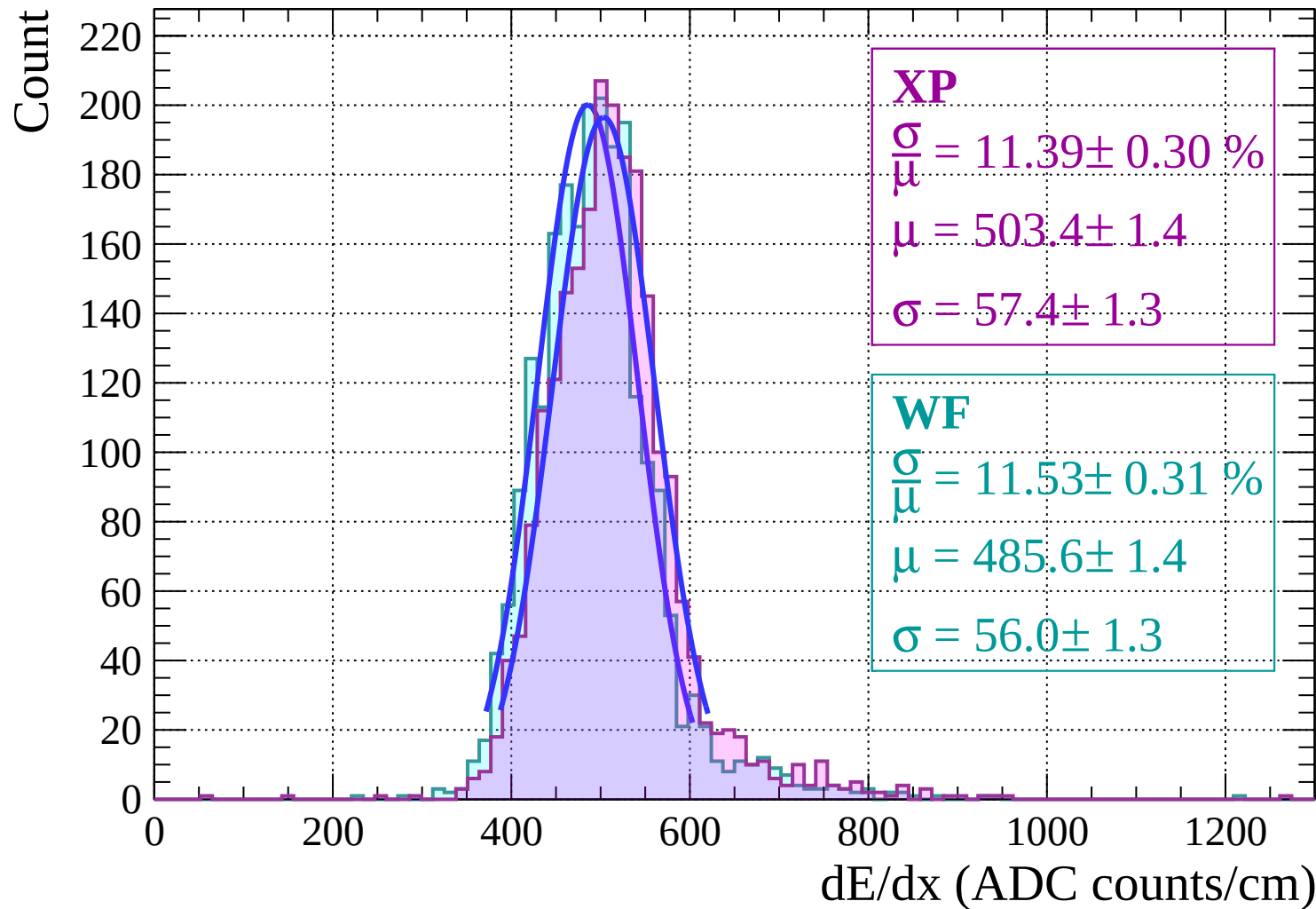
Energy loss | $-26 < \theta < -24$



Energy loss $|-24 < \theta < -22$

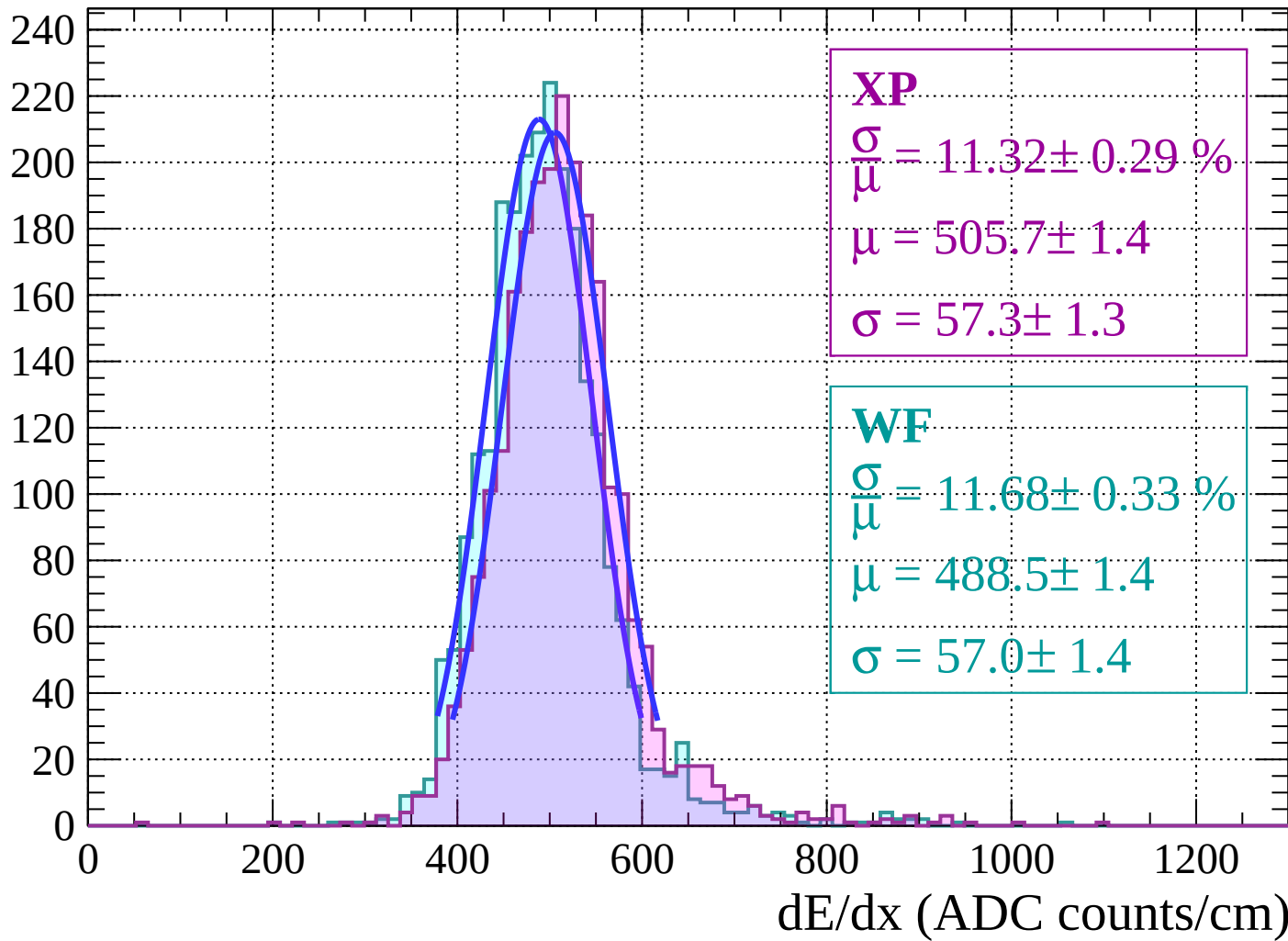


Energy loss | $-22 < \theta < -20$



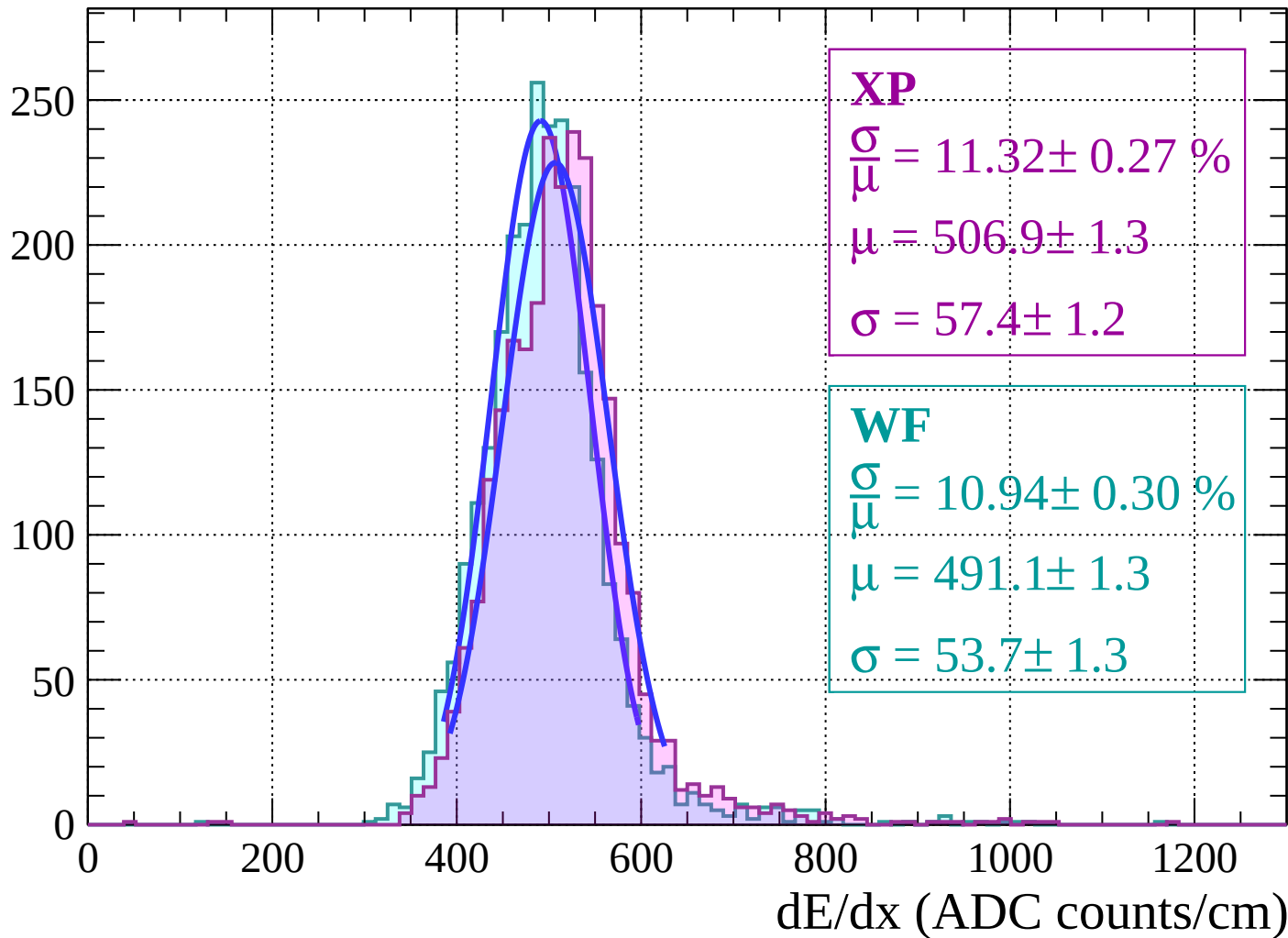
Energy loss | $-20 < \theta < -18$

Count

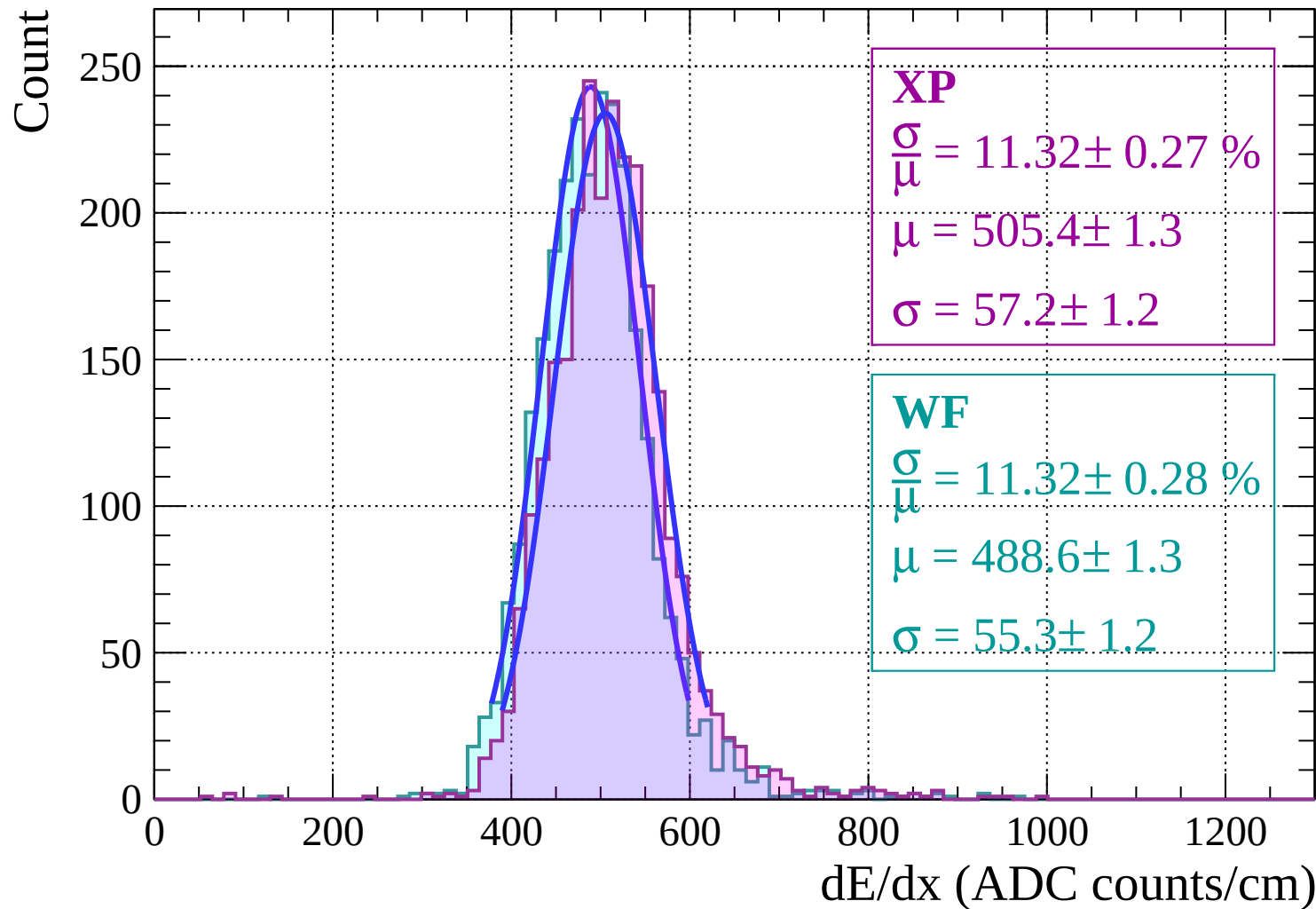


Energy loss $|-18 < \theta < -16$

Count

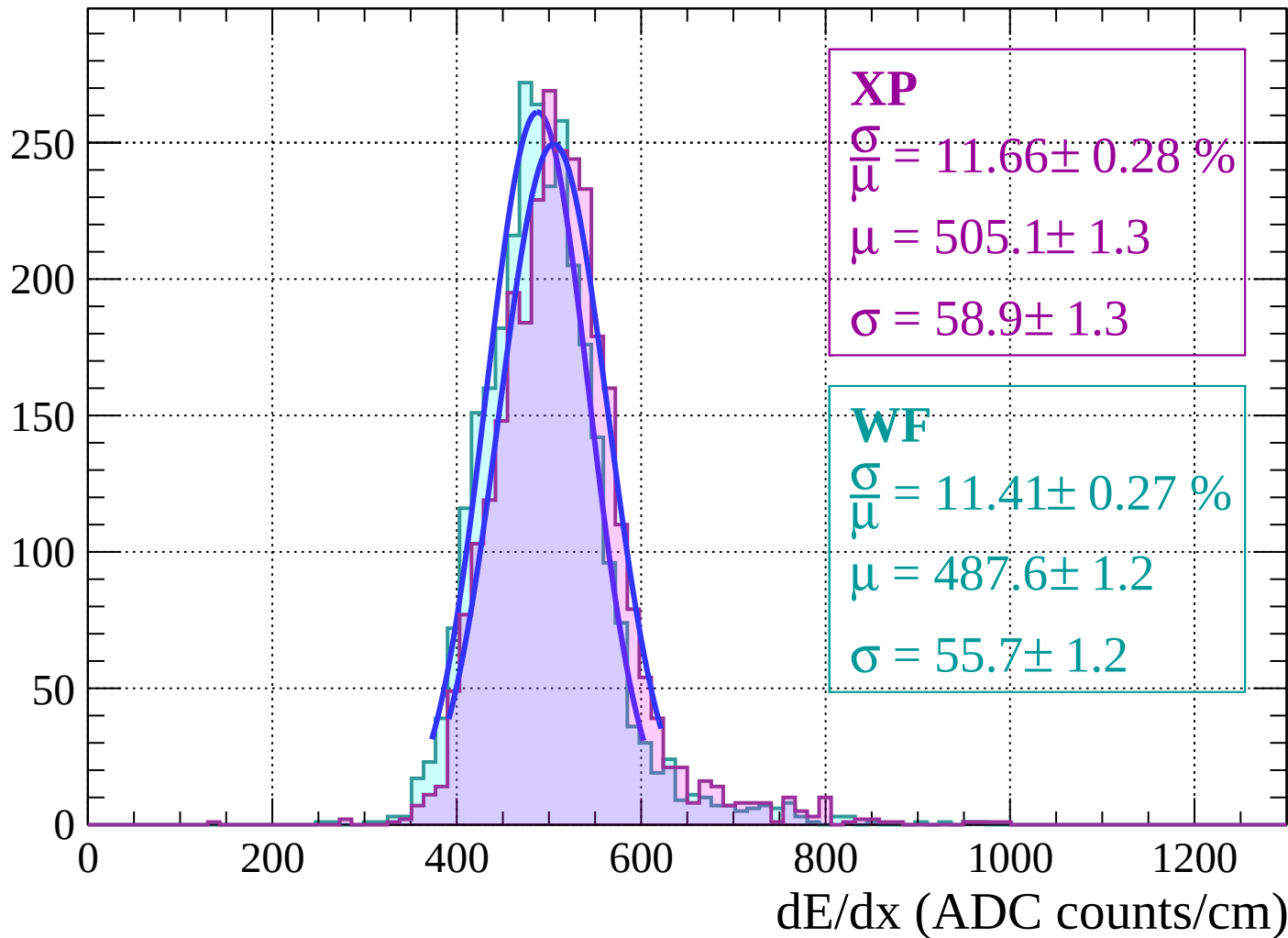


Energy loss $-16 < \theta < -14$

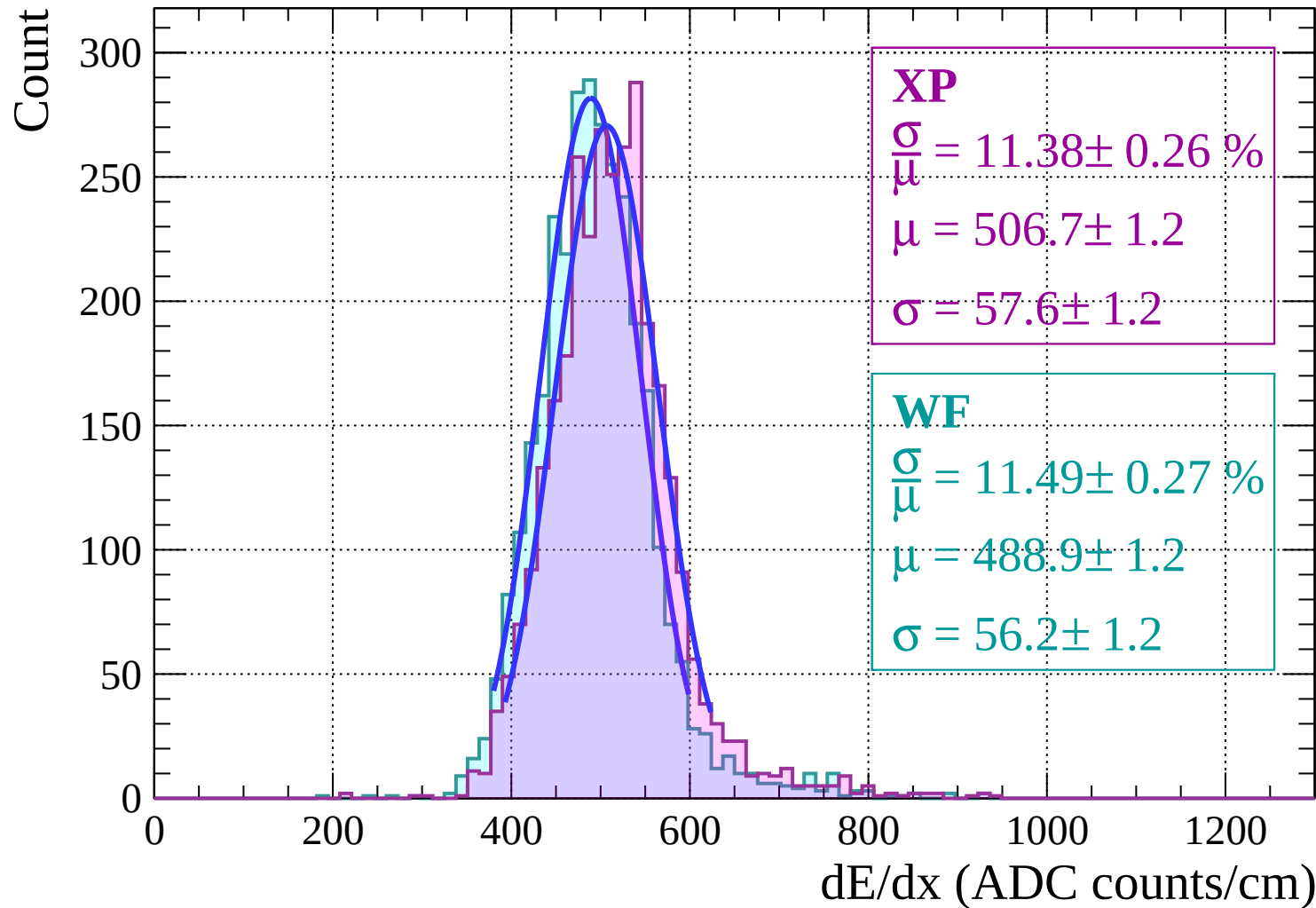


Energy loss $|-14 < \theta < -12$

Count



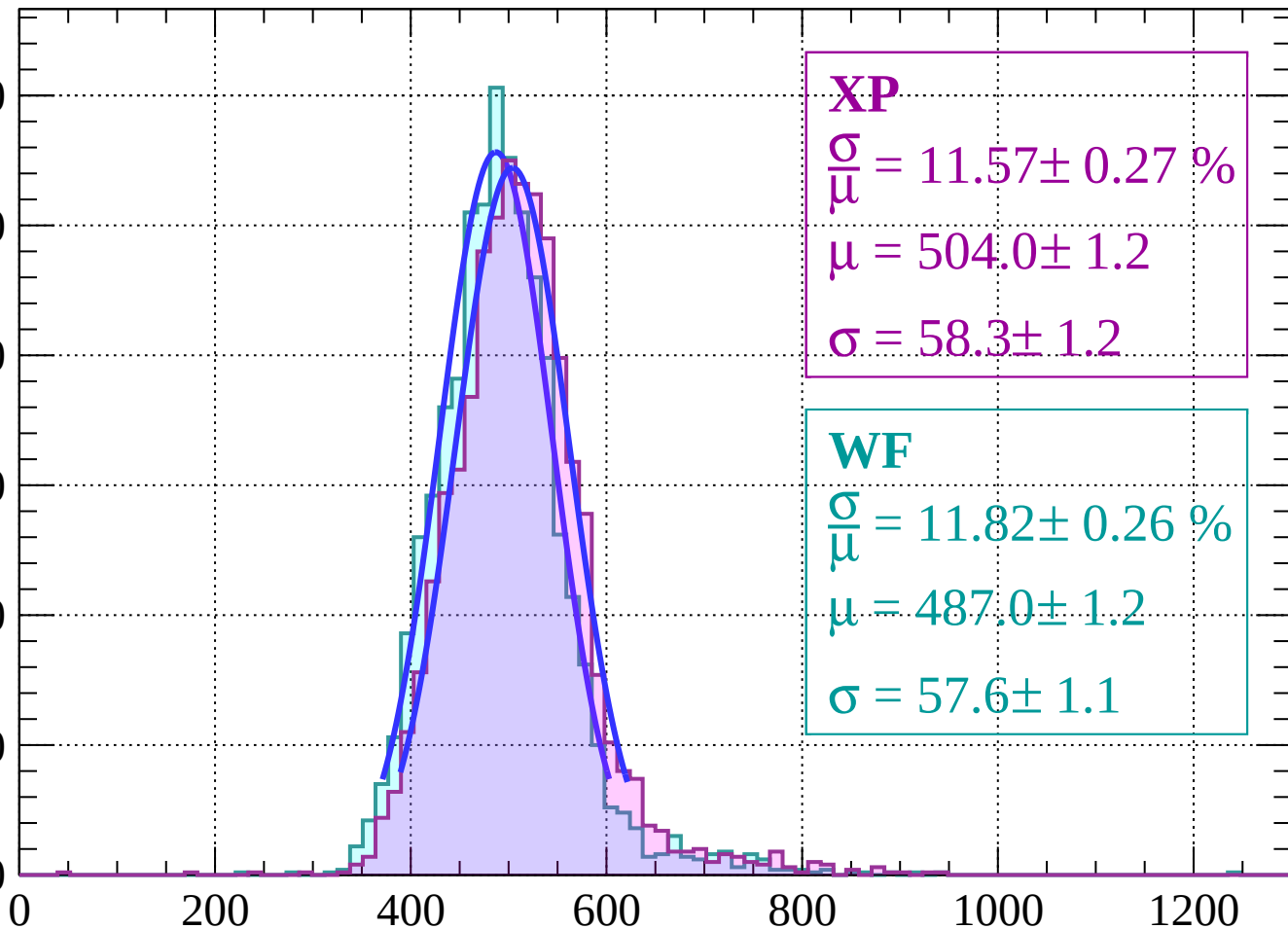
Energy loss | $-12 < \theta < -10$



Energy loss | $-10 < \theta < -8$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.57 \pm 0.27 \%$$

$$\mu = 504.0 \pm 1.2$$

$$\sigma = 58.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 11.82 \pm 0.26 \%$$

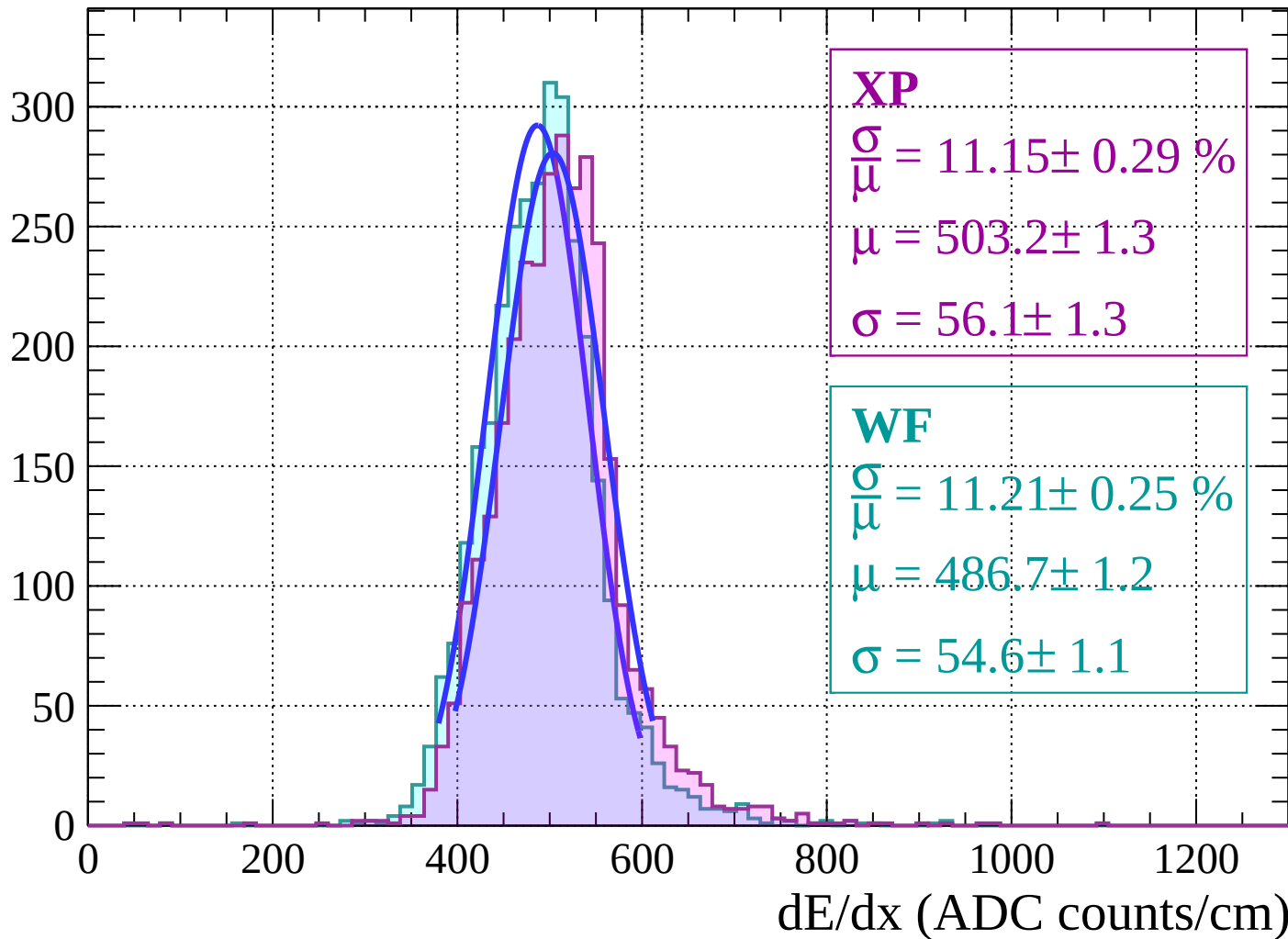
$$\mu = 487.0 \pm 1.2$$

$$\sigma = 57.6 \pm 1.1$$

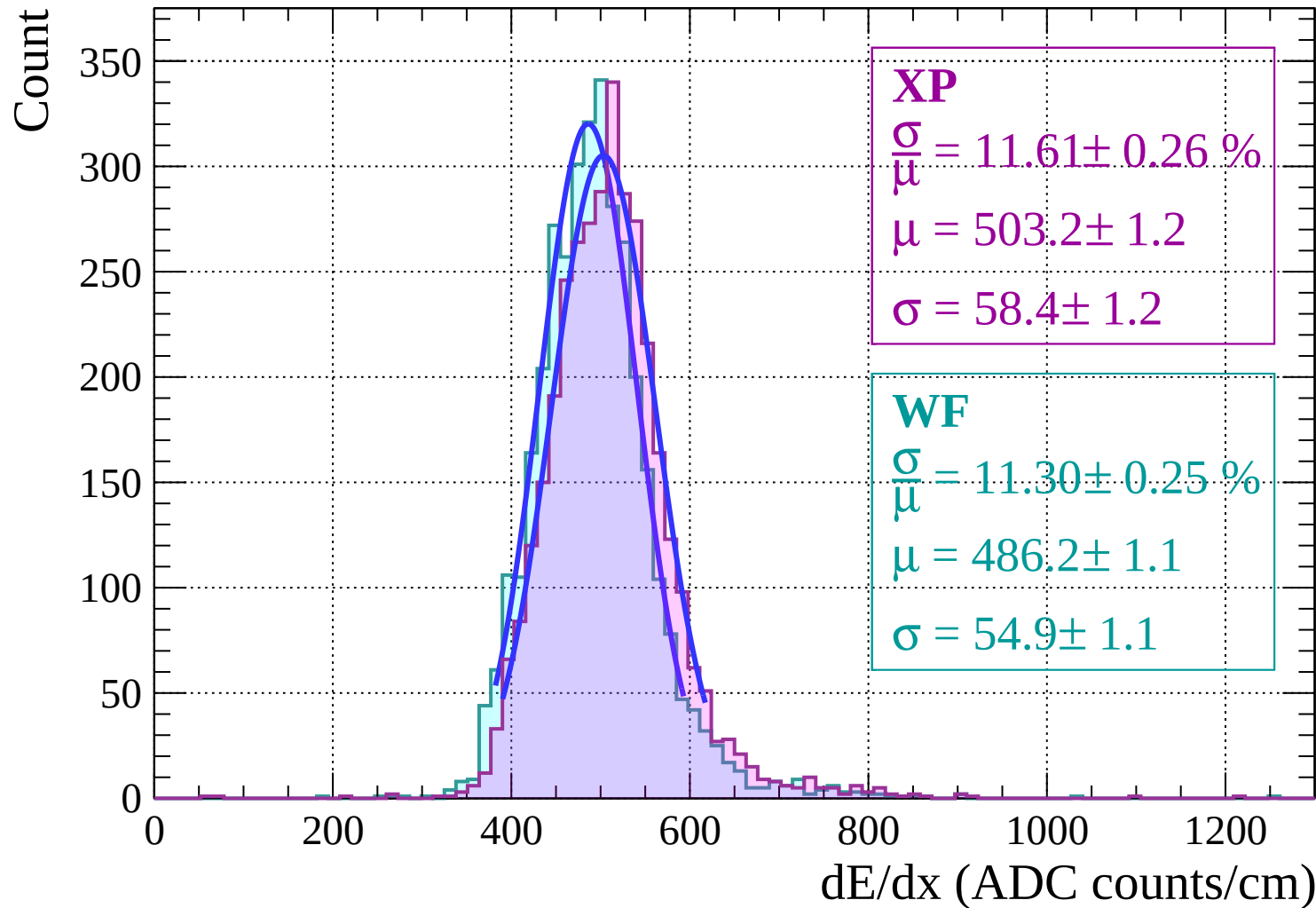
dE/dx (ADC counts/cm)

Energy loss $|-8 < \theta < -6$

Count

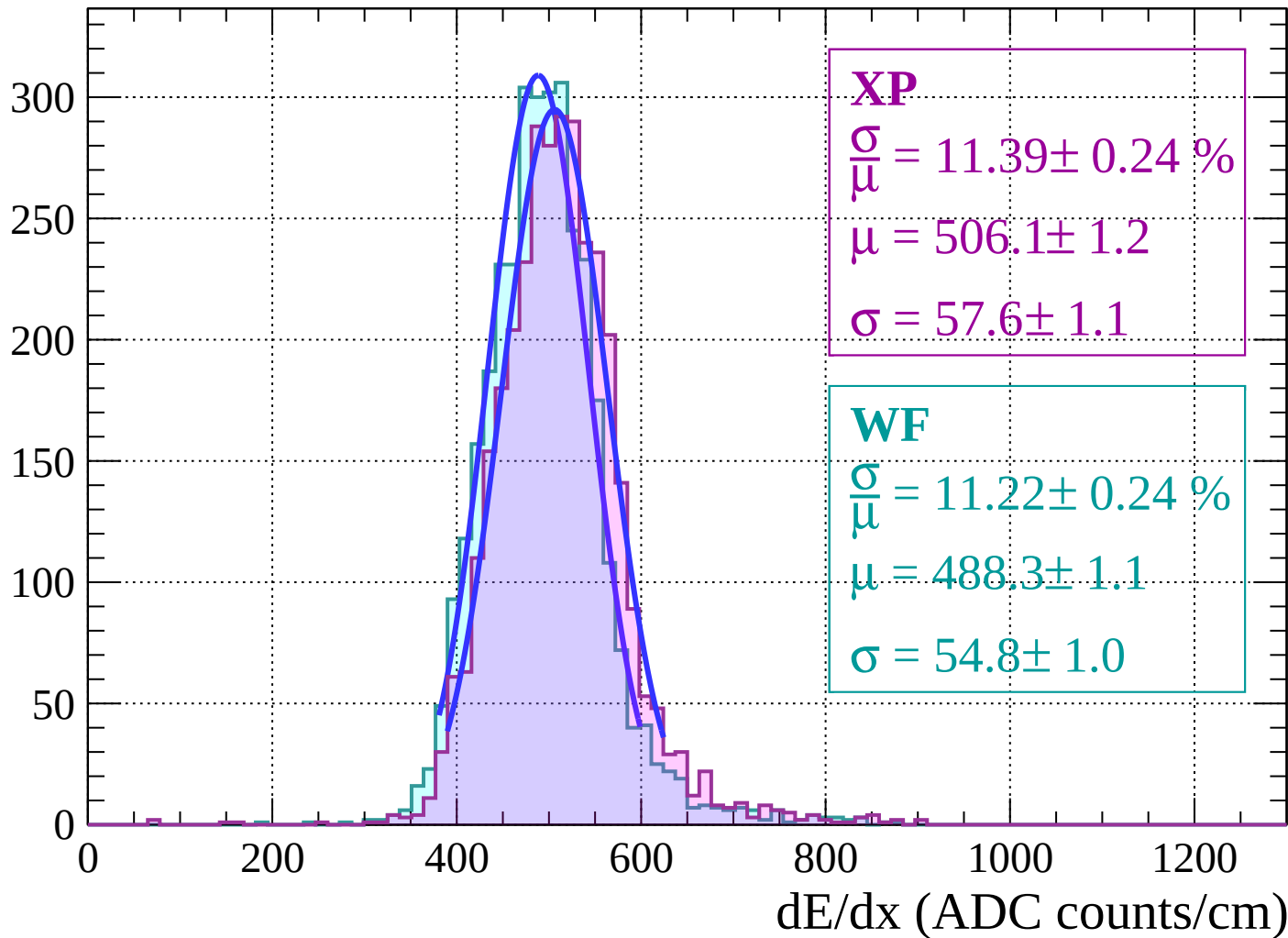


Energy loss $-6 < \theta < -4$



Energy loss $|-4 < \theta < -2$

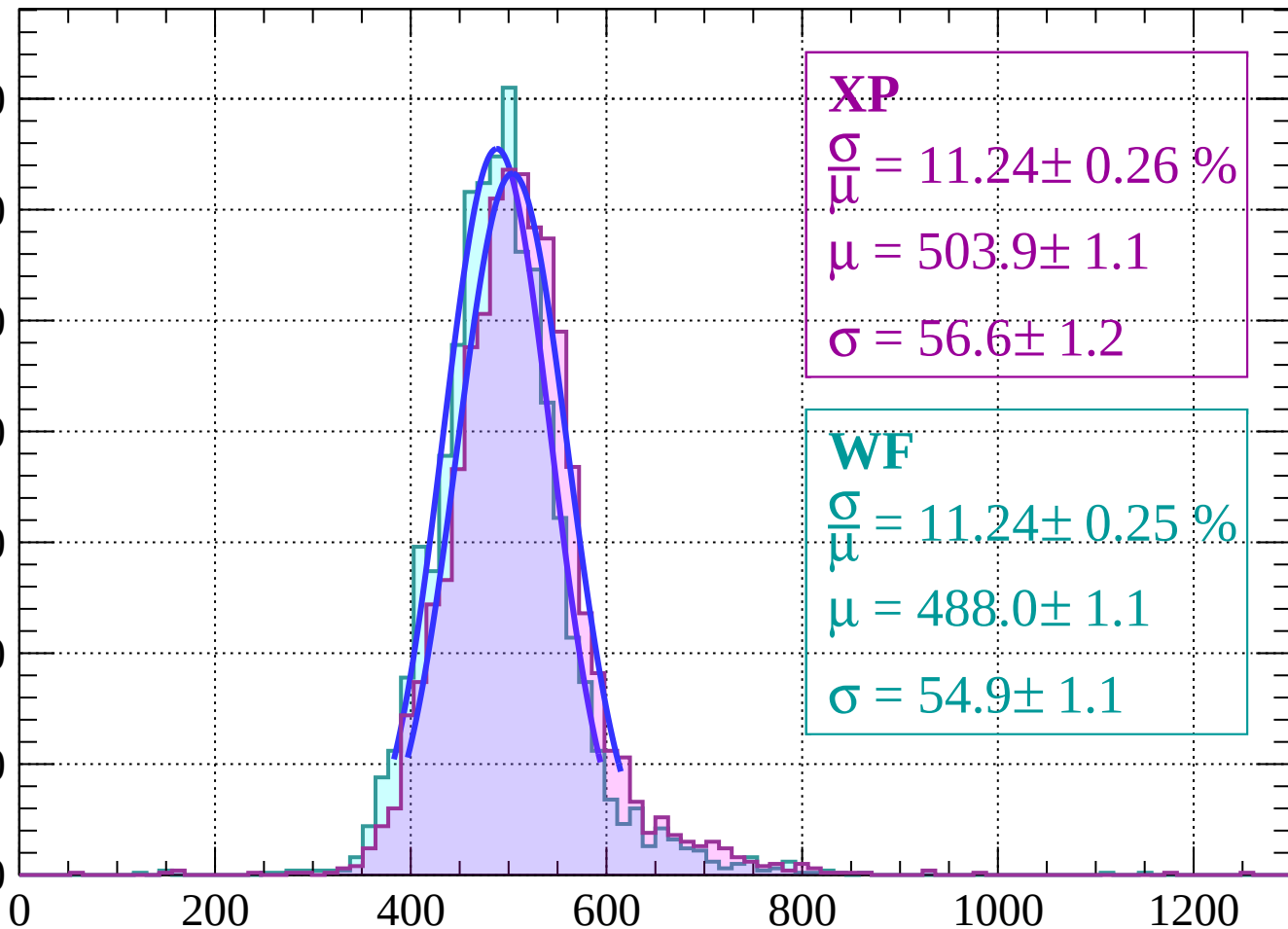
Count



Energy loss | $-2 < \theta < 0$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.24 \pm 0.26 \%$$

$$\mu = 503.9 \pm 1.1$$

$$\sigma = 56.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 11.24 \pm 0.25 \%$$

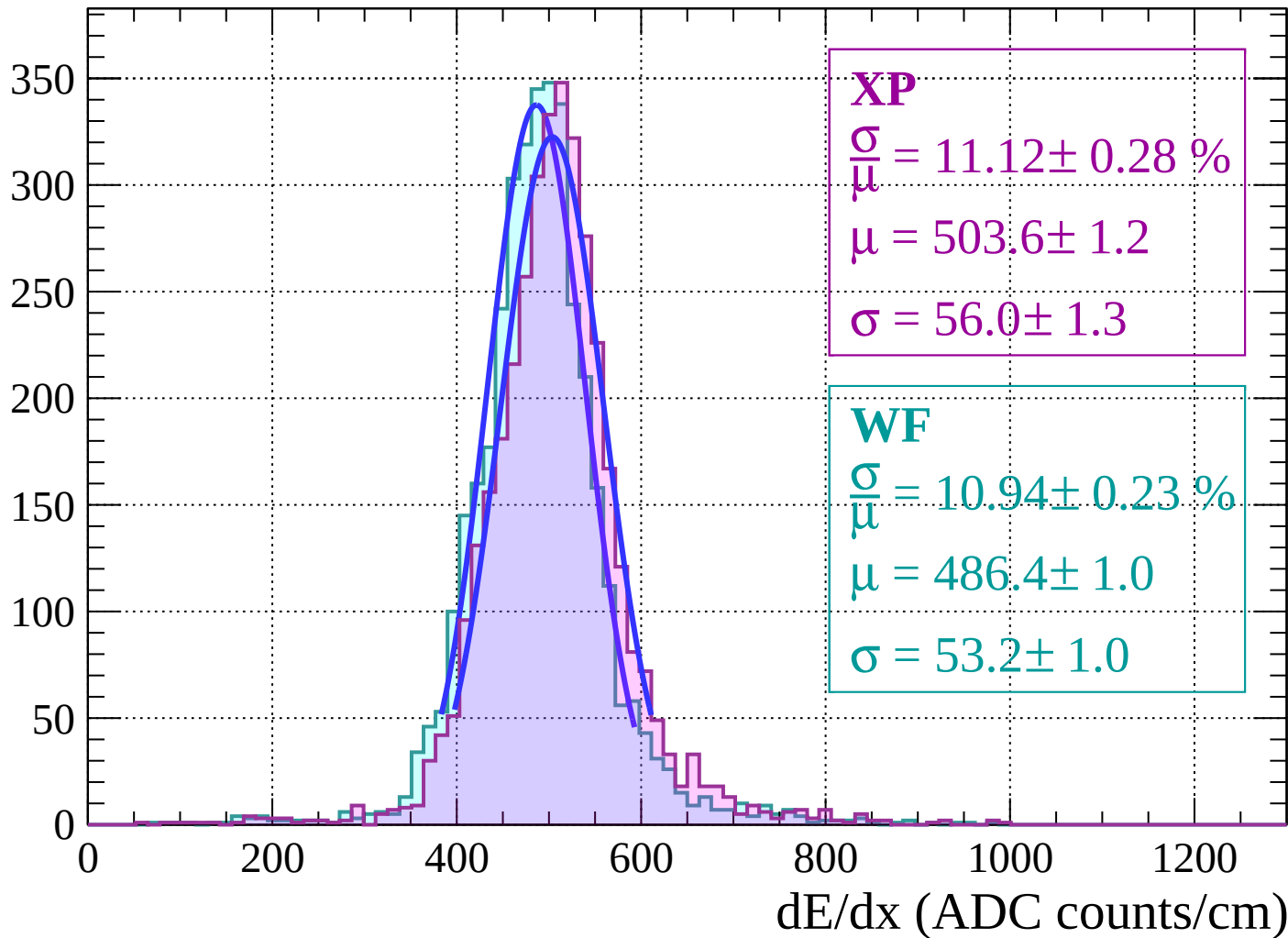
$$\mu = 488.0 \pm 1.1$$

$$\sigma = 54.9 \pm 1.1$$

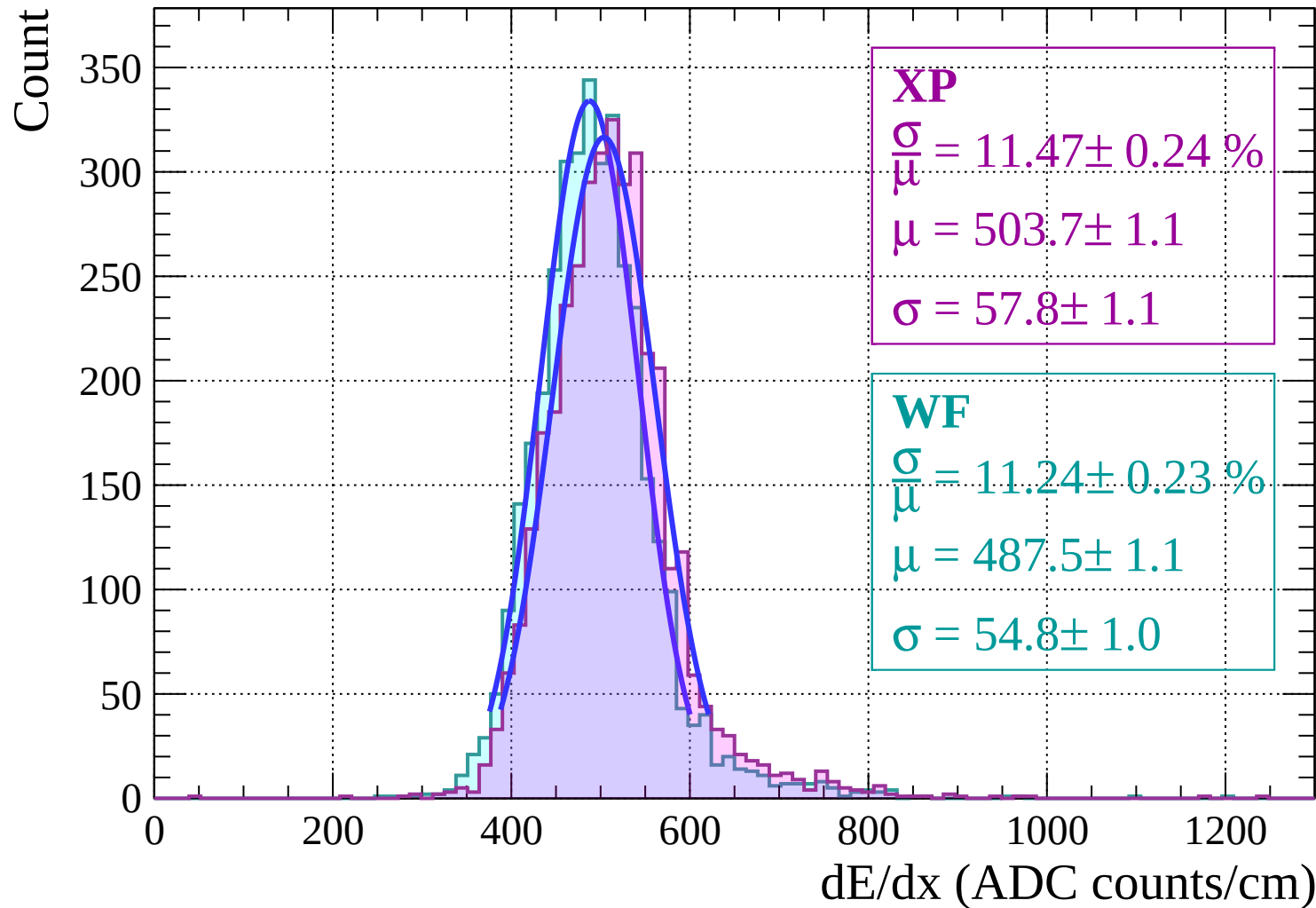
dE/dx (ADC counts/cm)

Energy loss | $0 < \theta < 2$

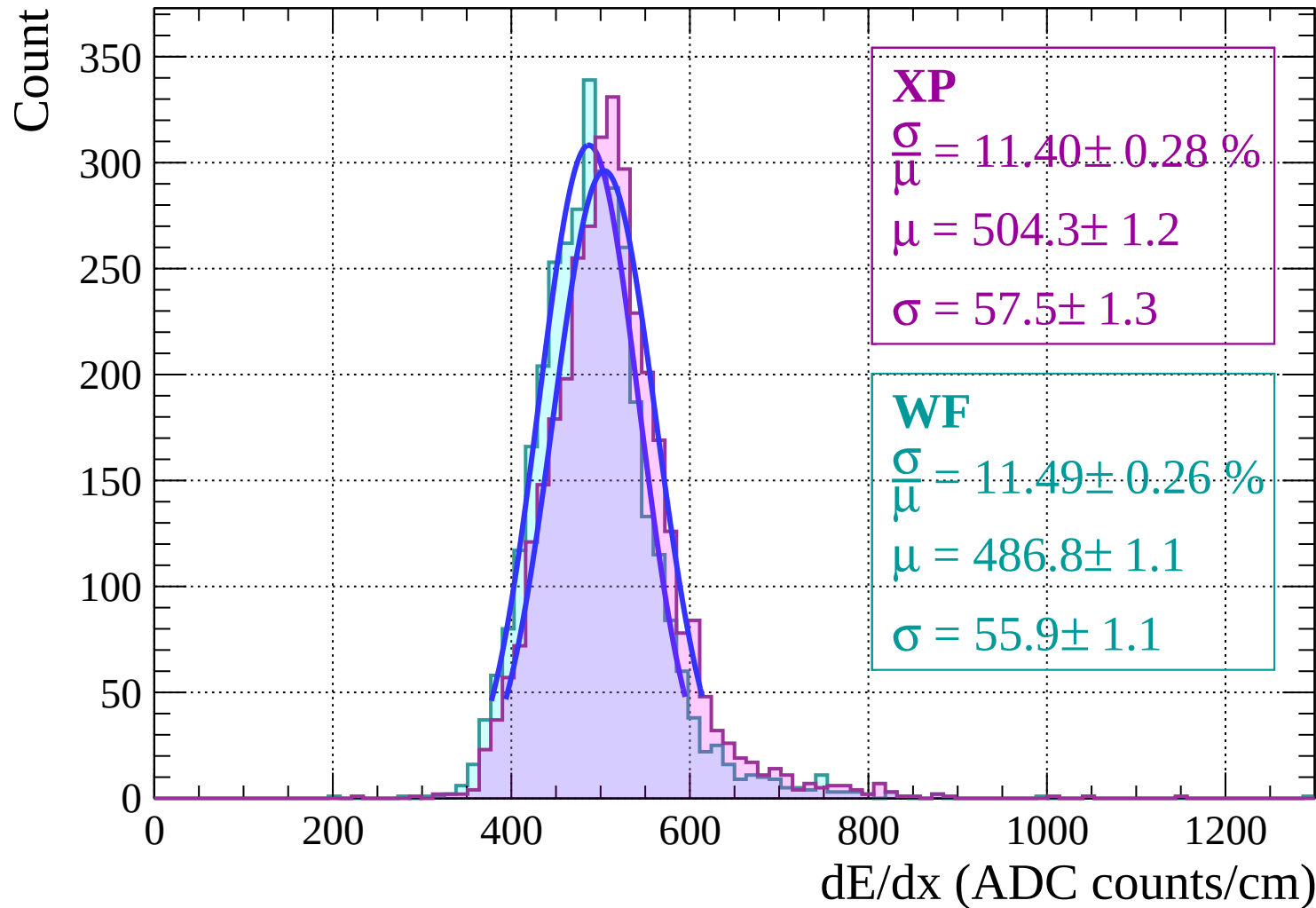
Count



Energy loss | $2 < \theta < 4$

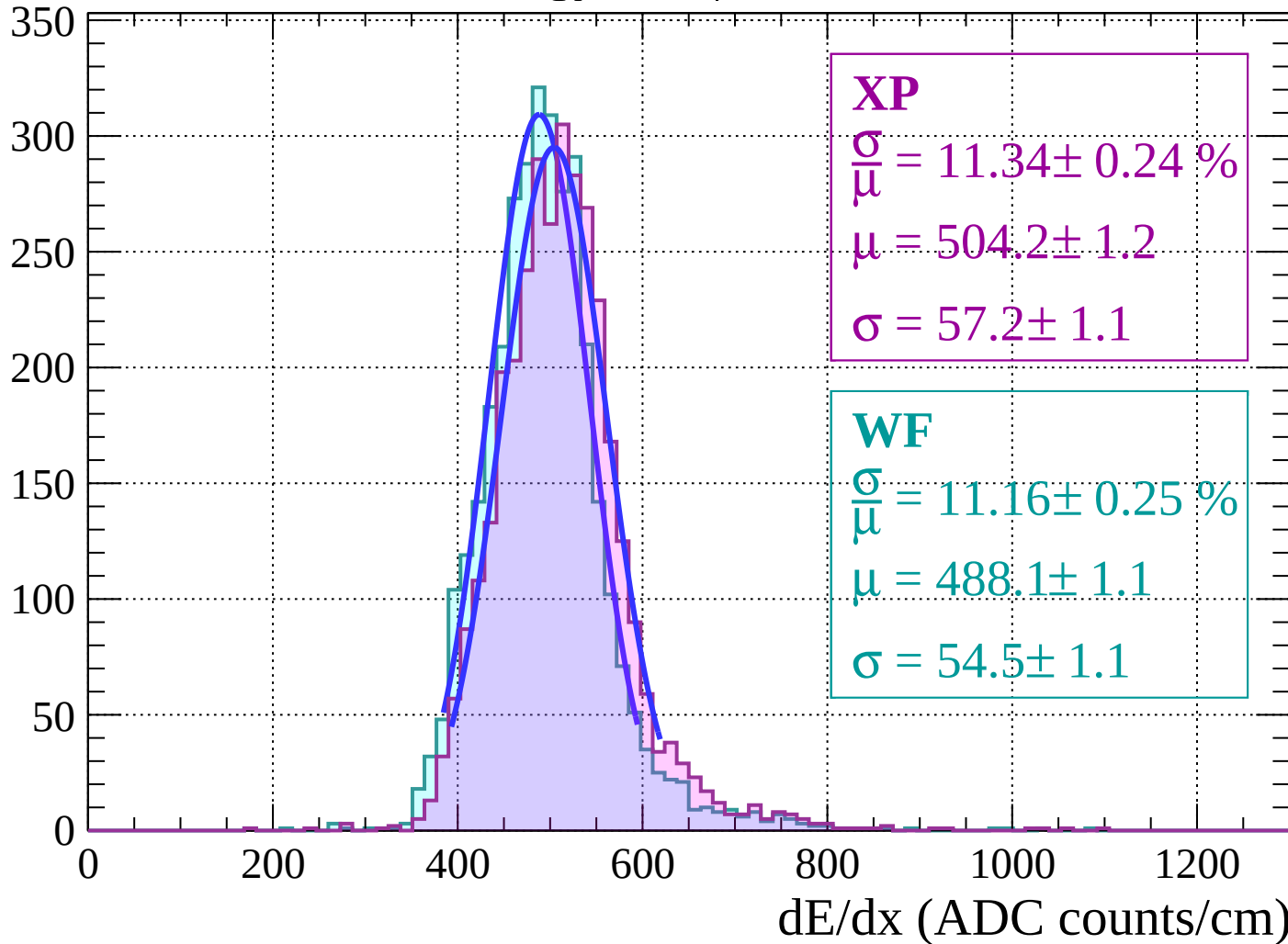


Energy loss | $4 < \theta < 6$



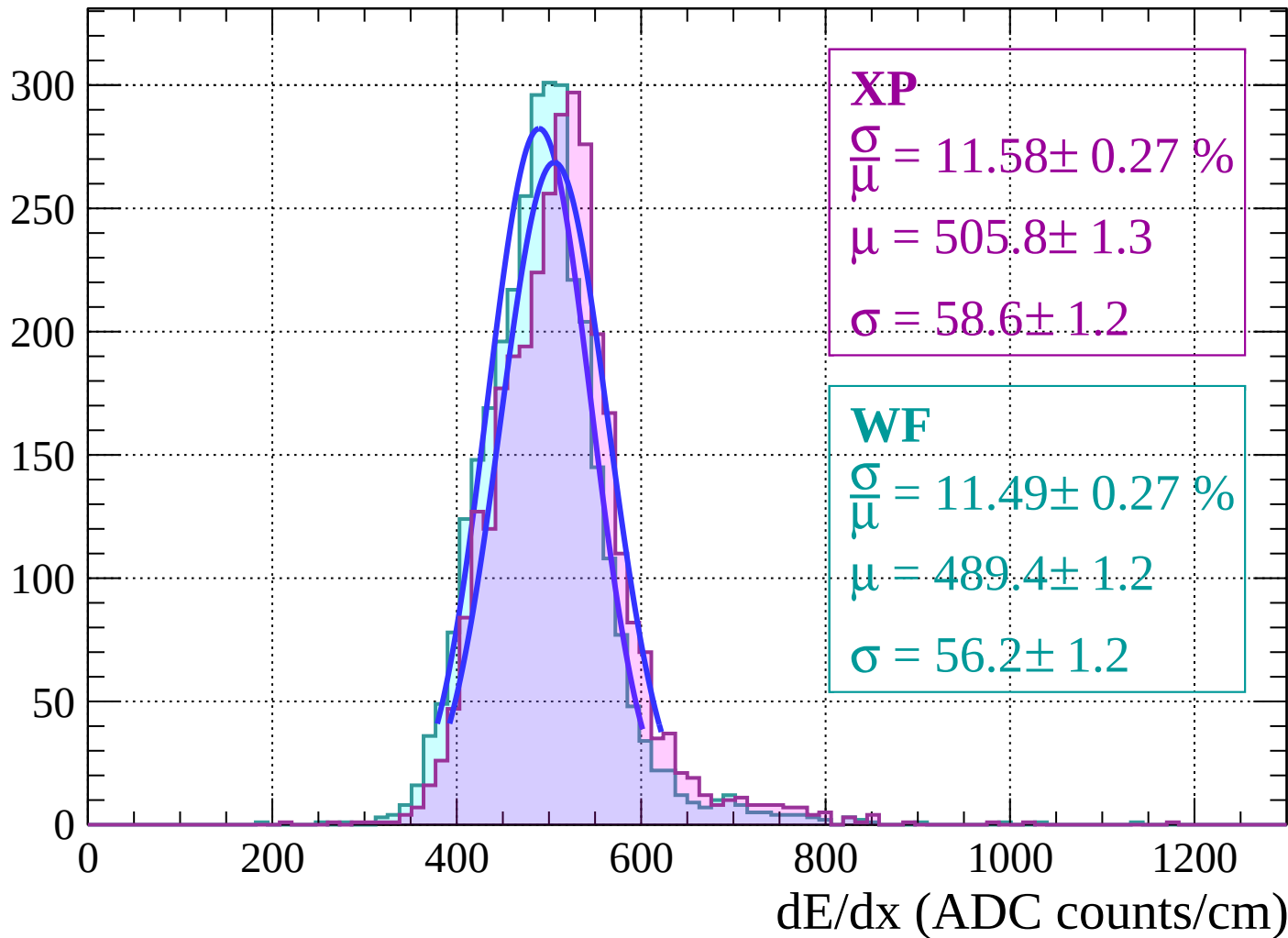
Energy loss | $6 < \theta < 8$

Count



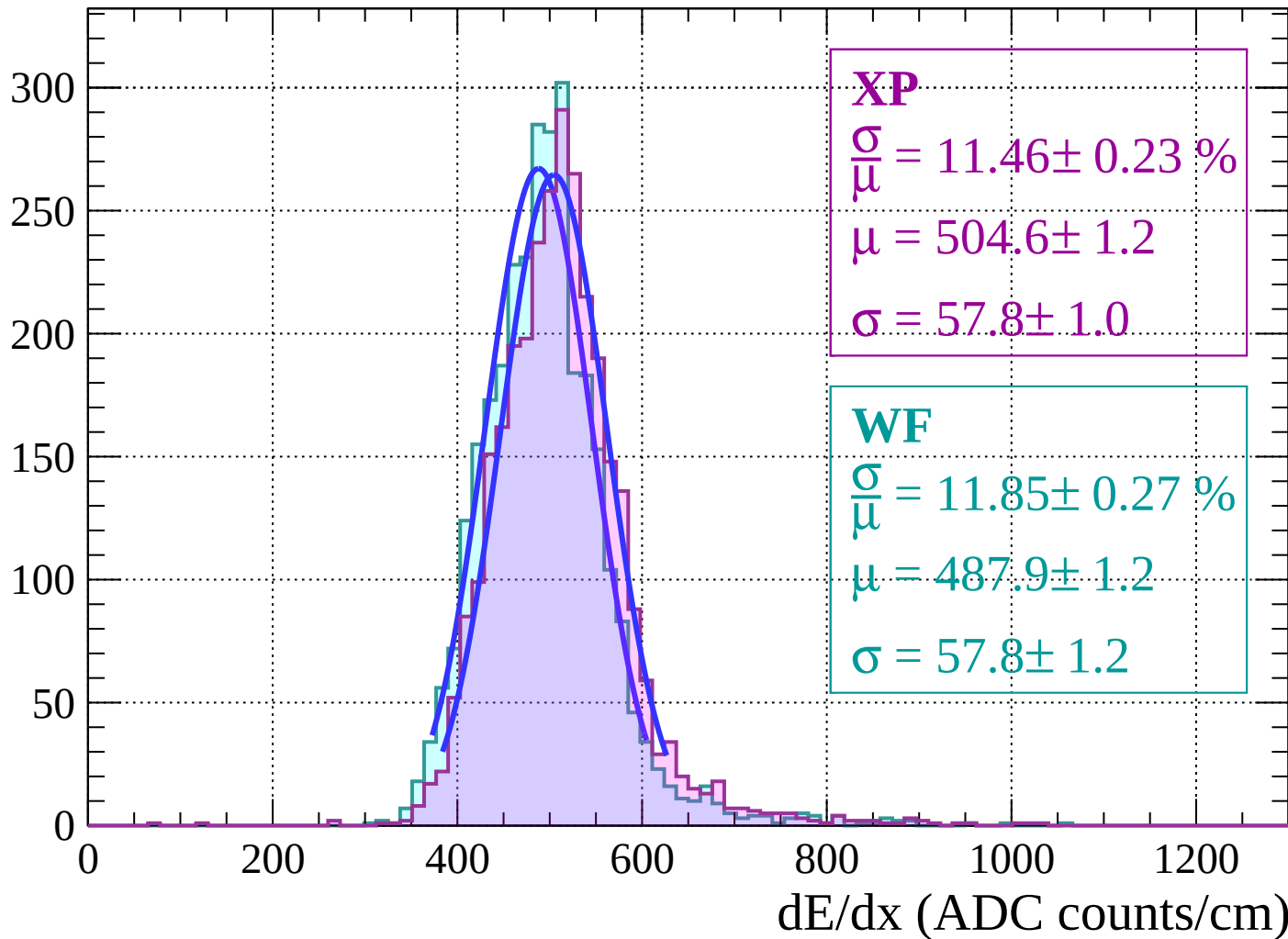
Energy loss | $8 < \theta < 10$

Count



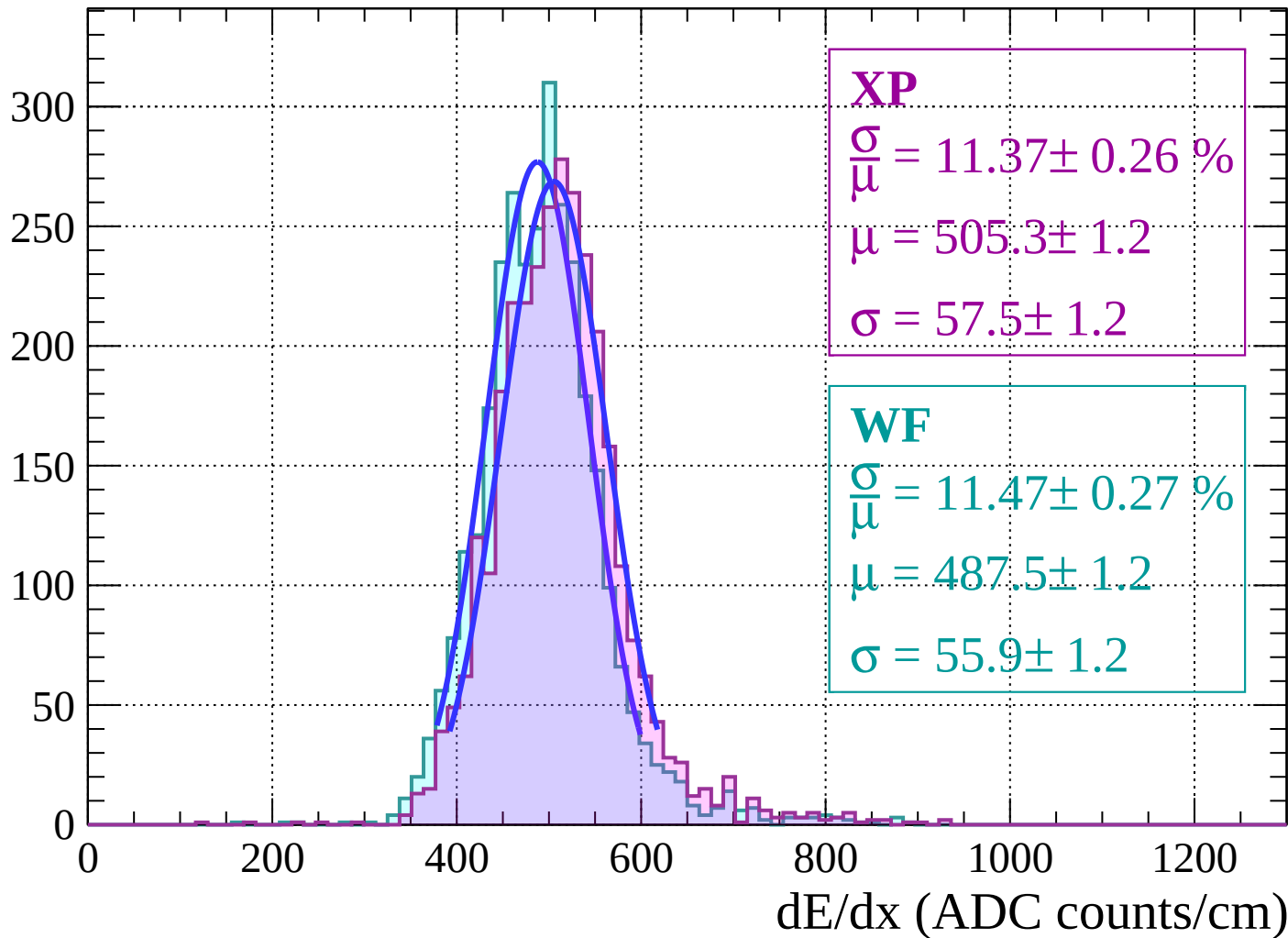
Energy loss | $10 < \theta < 12$

Count



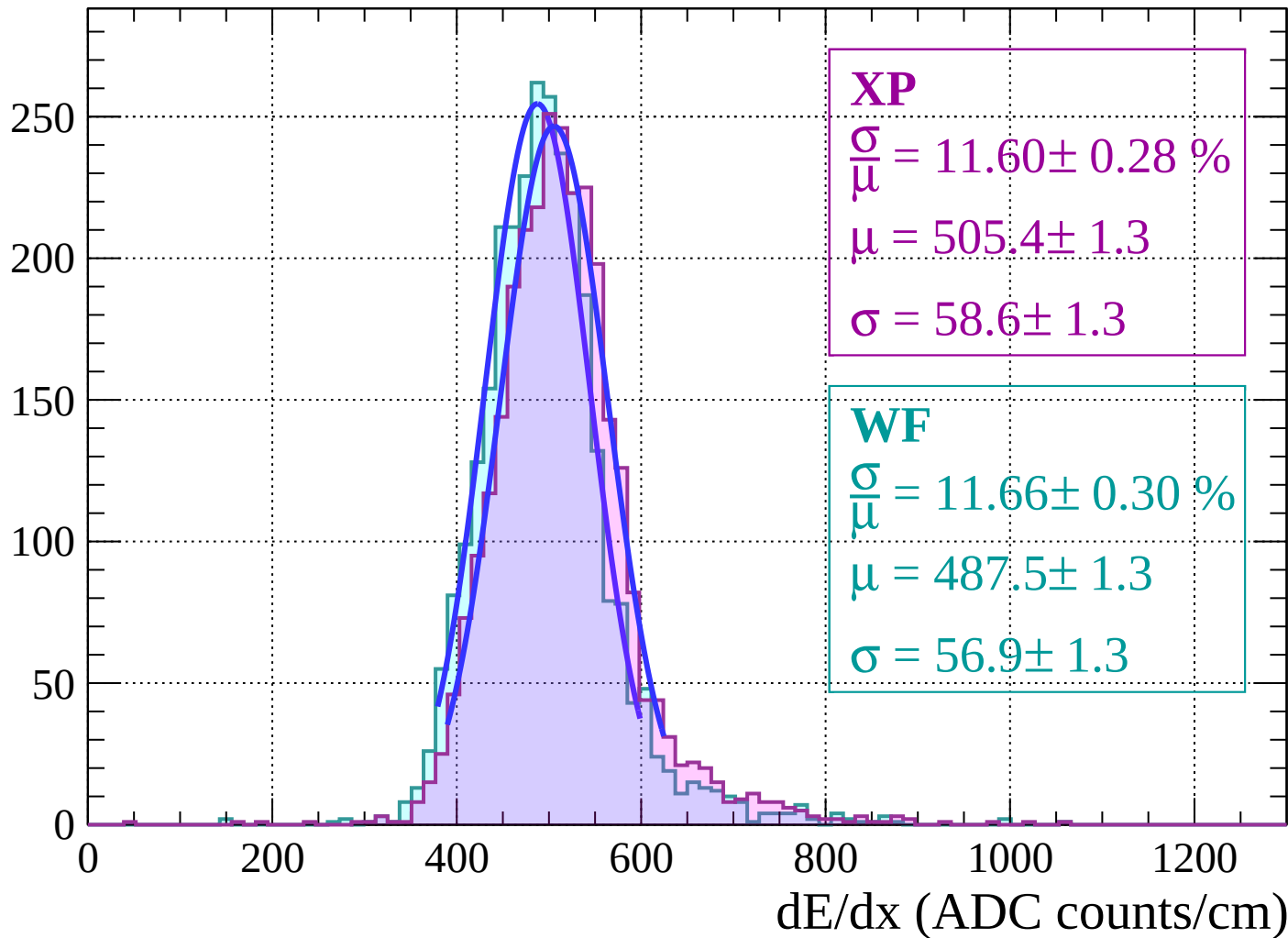
Energy loss | $12 < \theta < 14$

Count



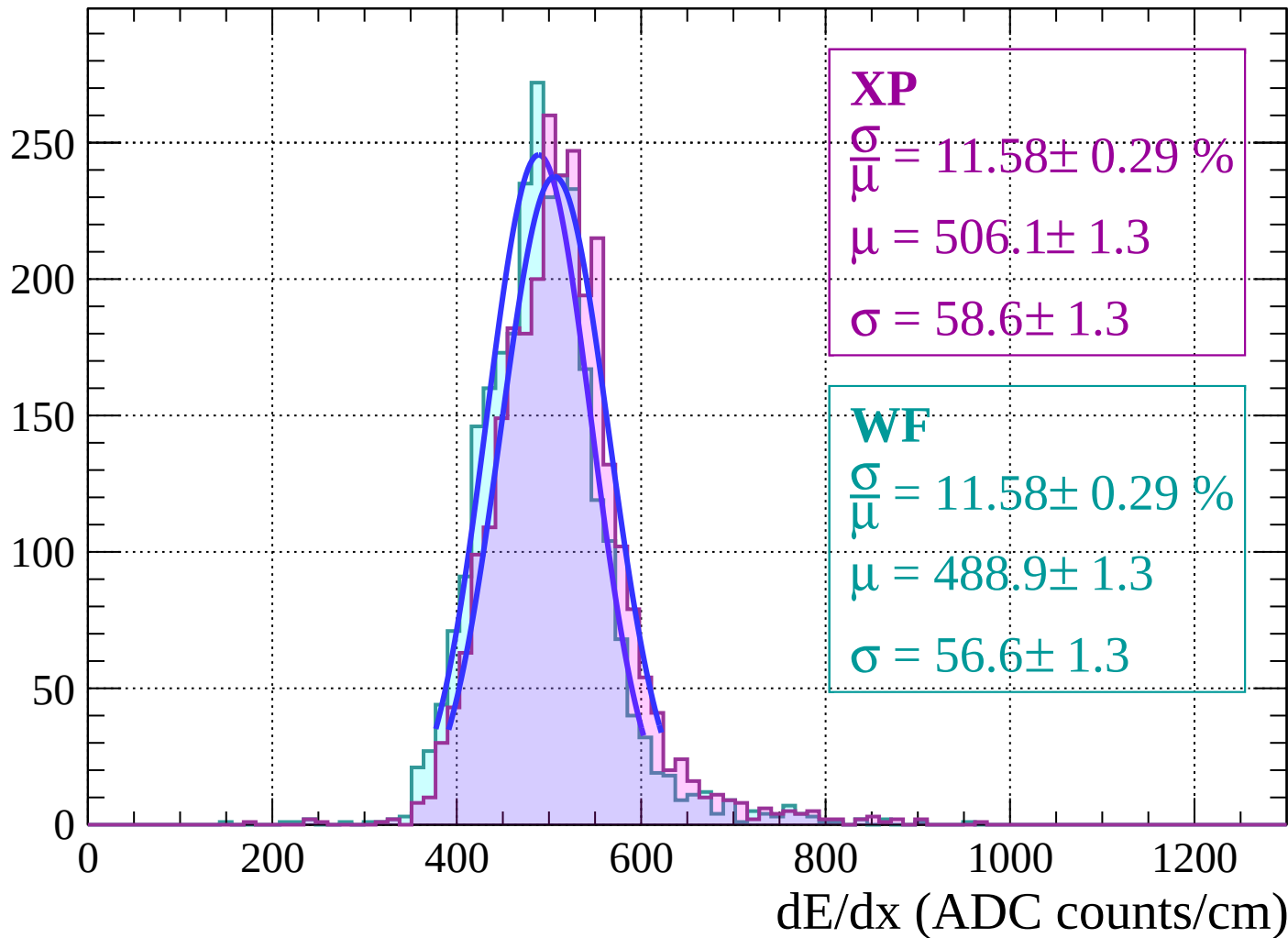
Energy loss | $14 < \theta < 16$

Count



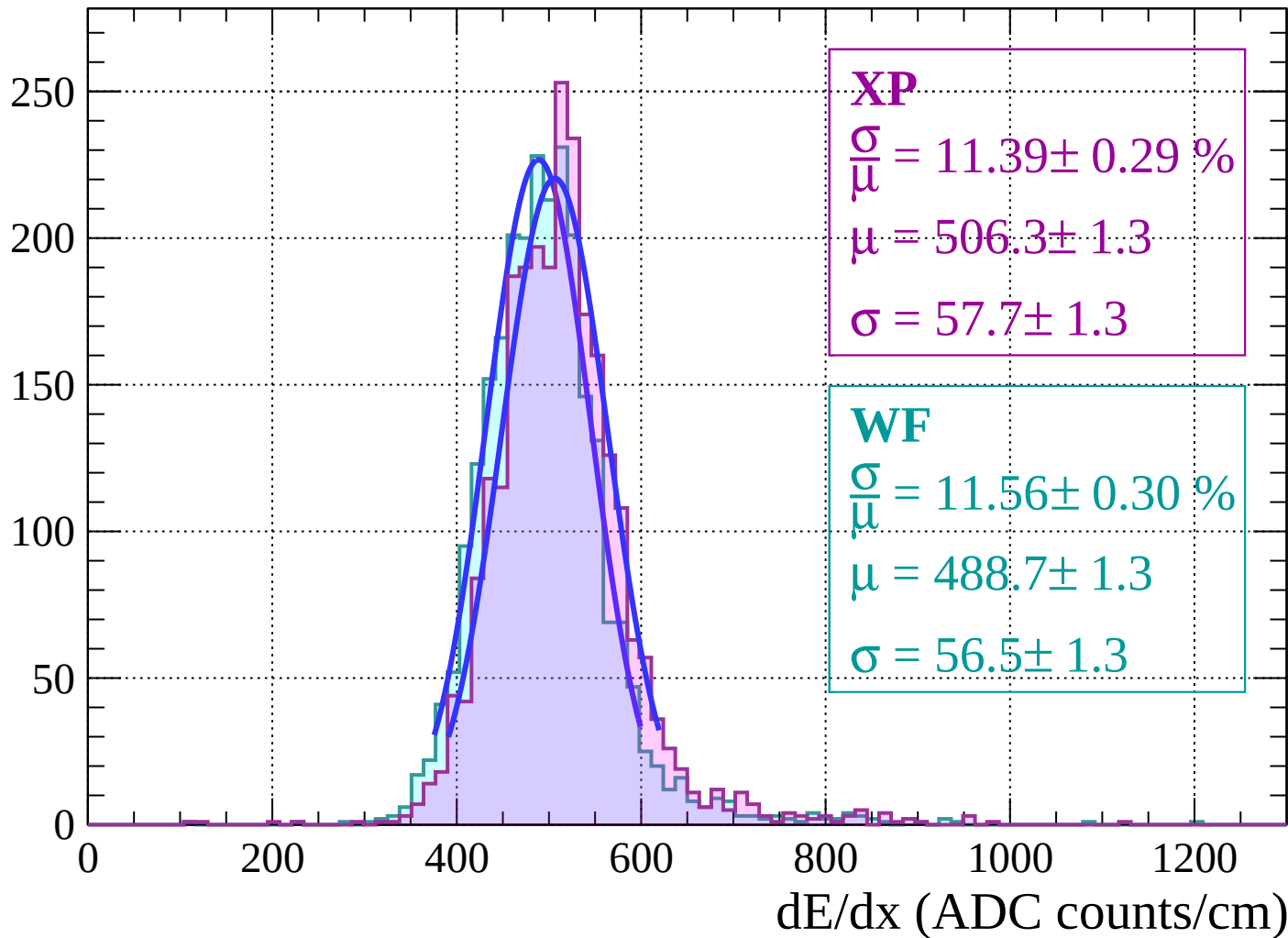
Energy loss | $16 < \theta < 18$

Count



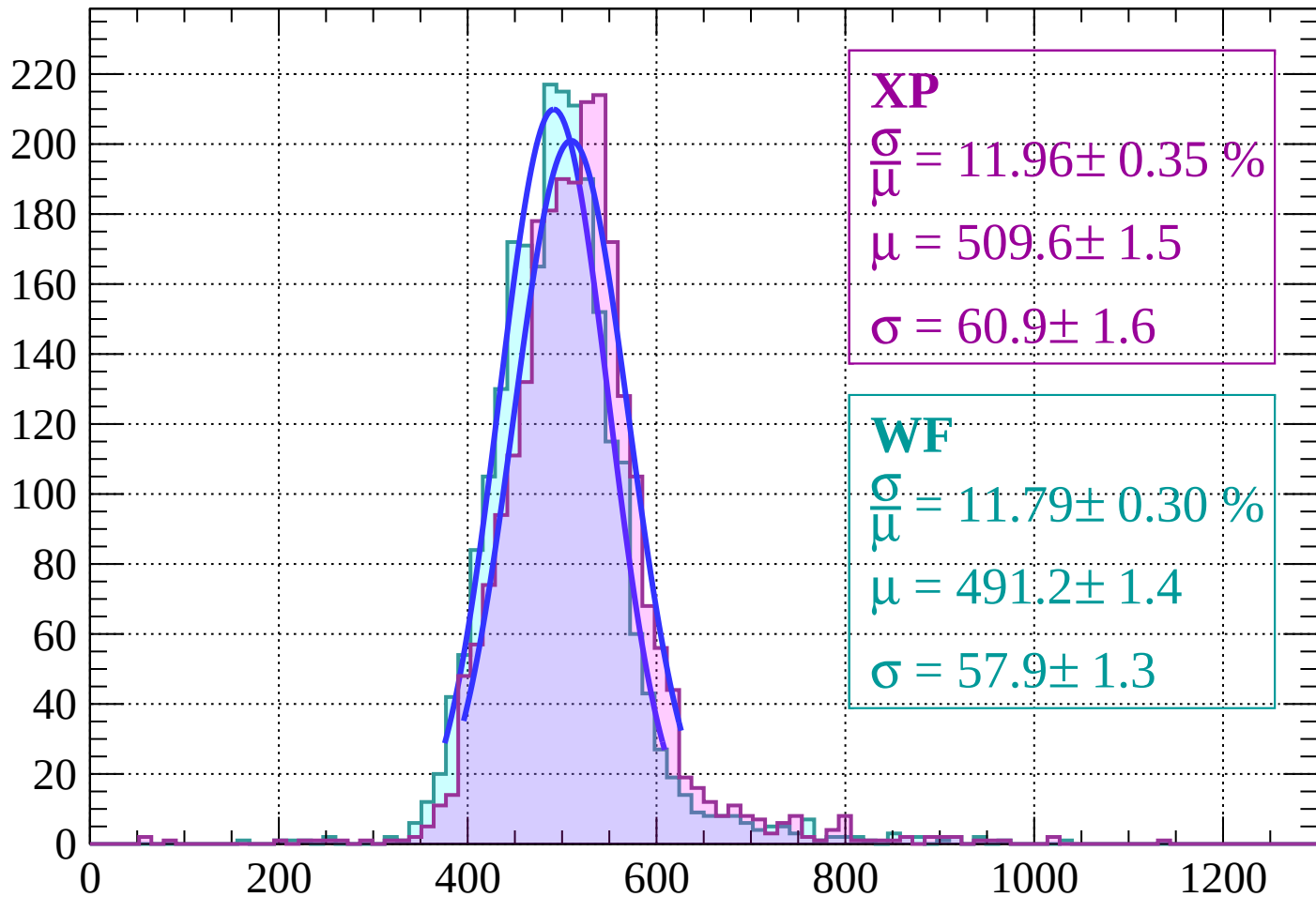
Energy loss | $18 < \theta < 20$

Count



Energy loss | $20 < \theta < 22$

Count



XP

$$\frac{\sigma}{\mu} = 11.96 \pm 0.35 \%$$

$$\mu = 509.6 \pm 1.5$$

$$\sigma = 60.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 11.79 \pm 0.30 \%$$

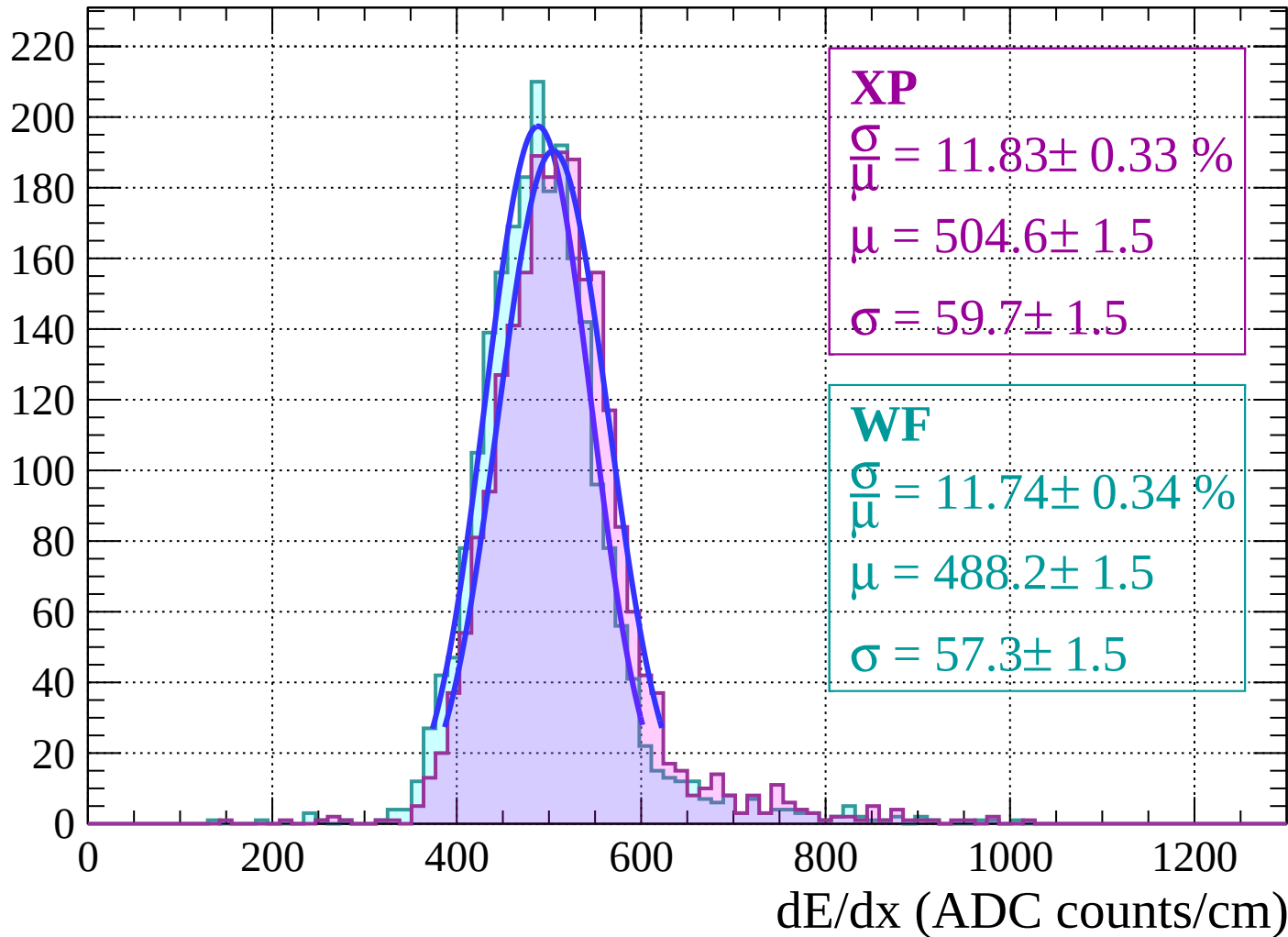
$$\mu = 491.2 \pm 1.4$$

$$\sigma = 57.9 \pm 1.3$$

dE/dx (ADC counts/cm)

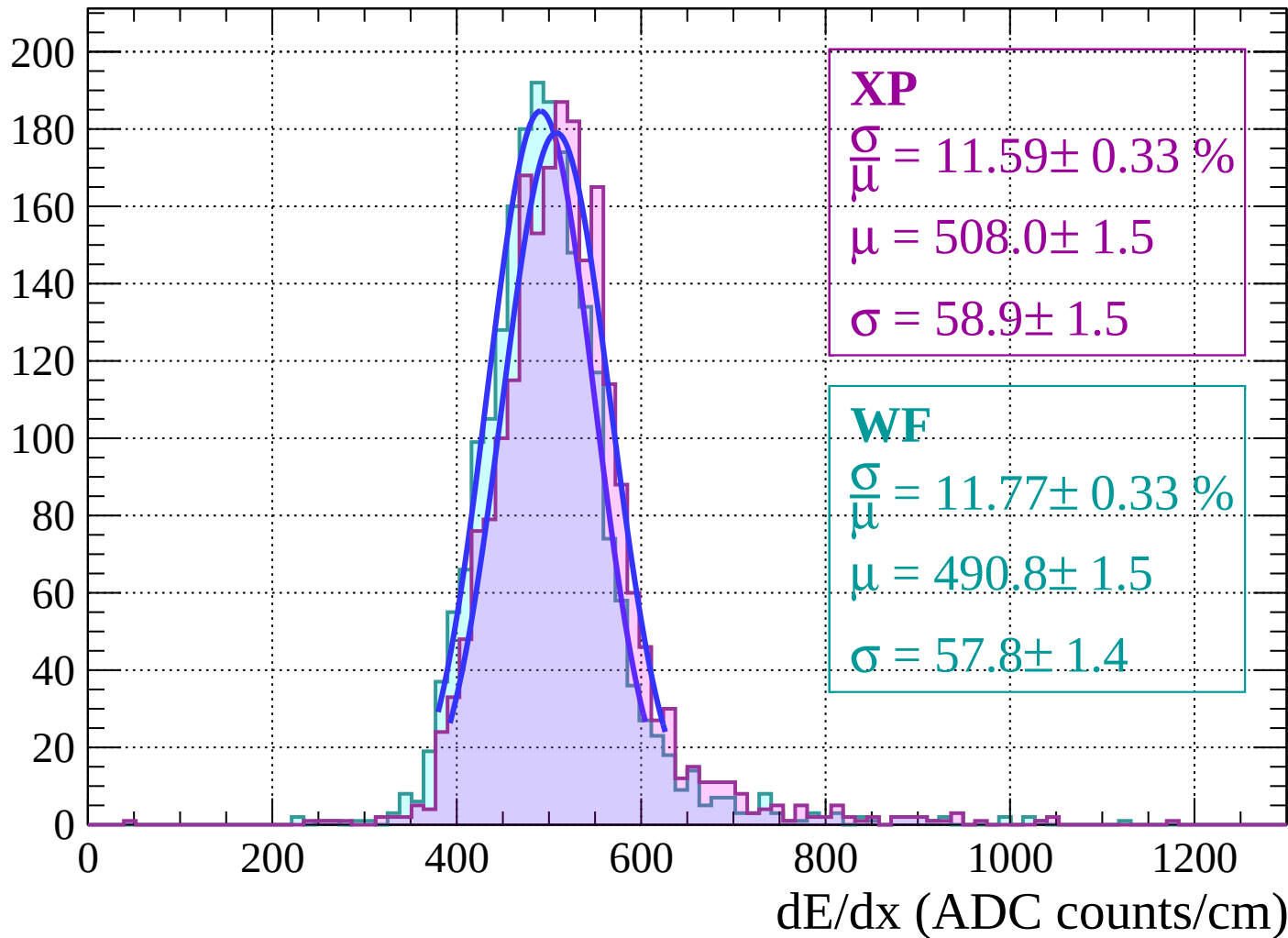
Energy loss | $22 < \theta < 24$

Count



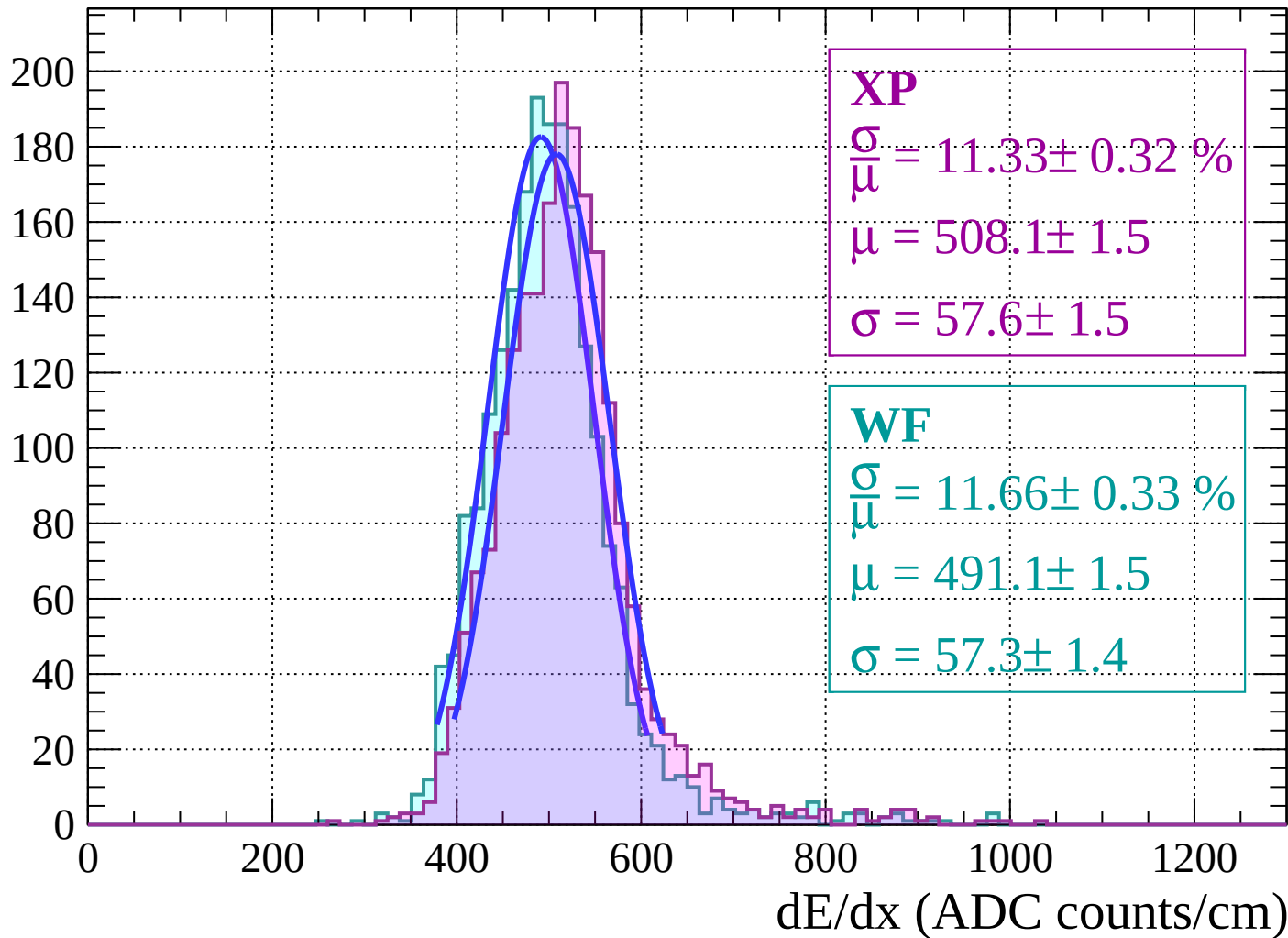
Energy loss | $24 < \theta < 26$

Count



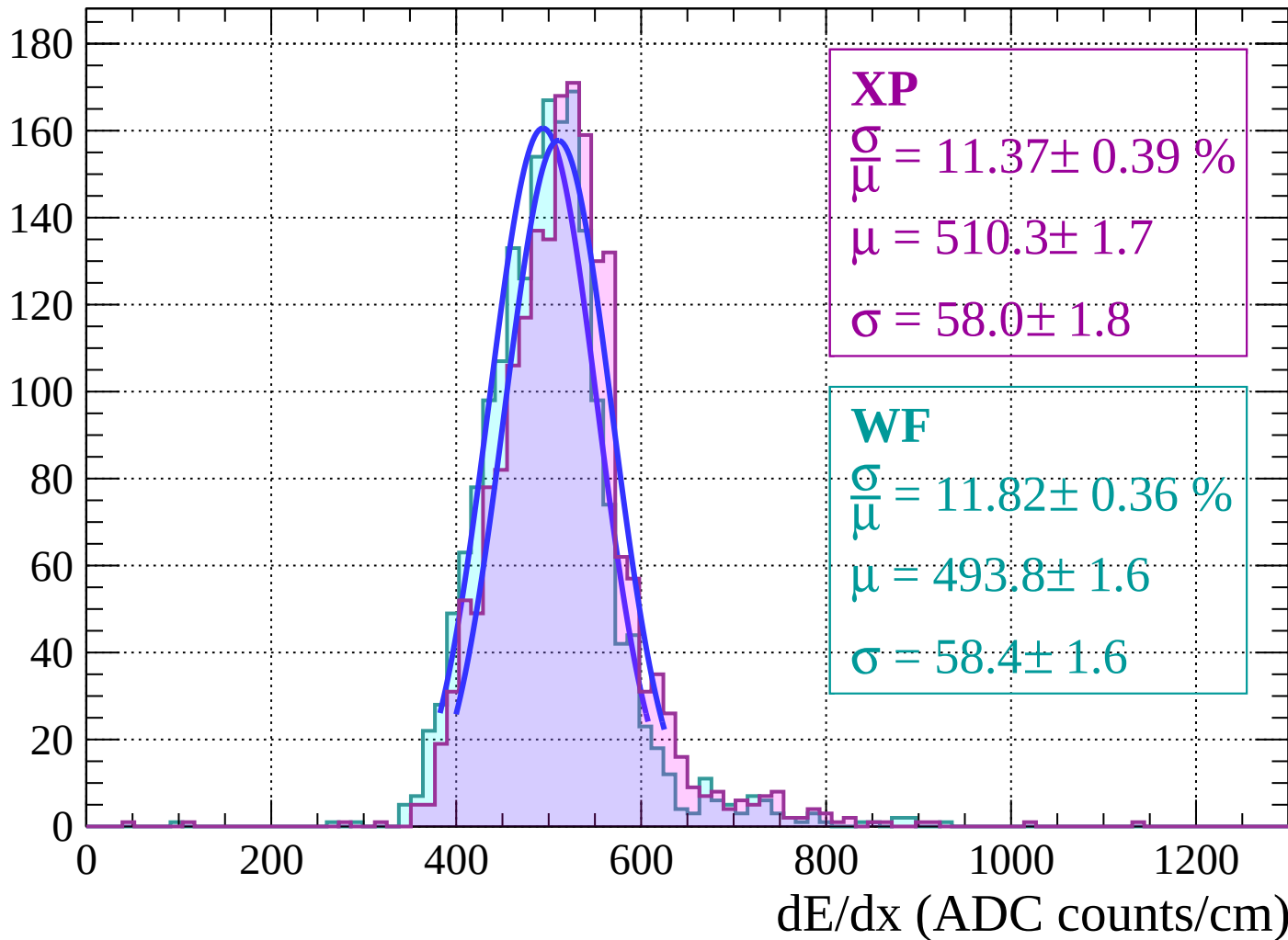
Energy loss | $26 < \theta < 28$

Count



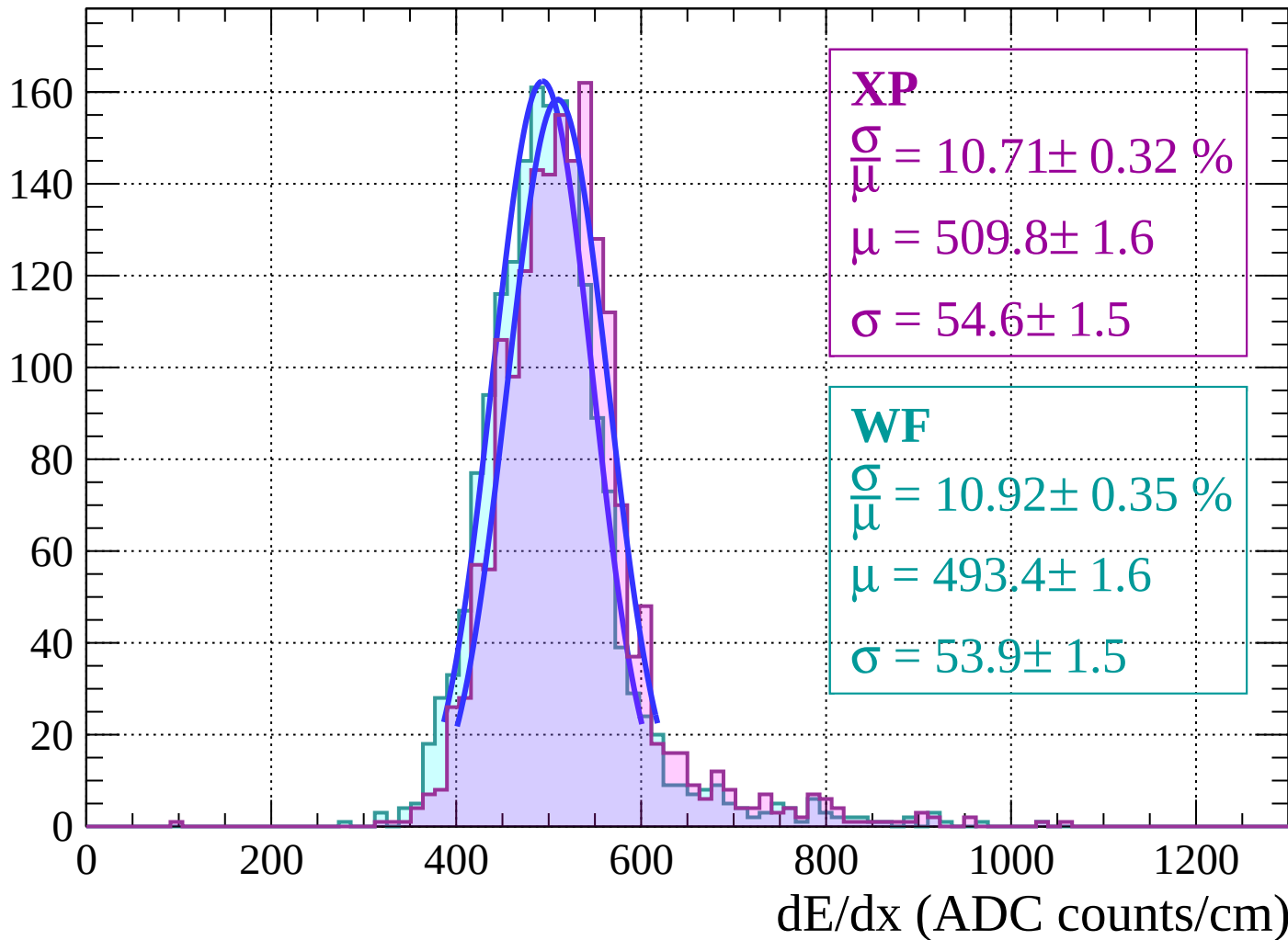
Energy loss | $28 < \theta < 30$

Count

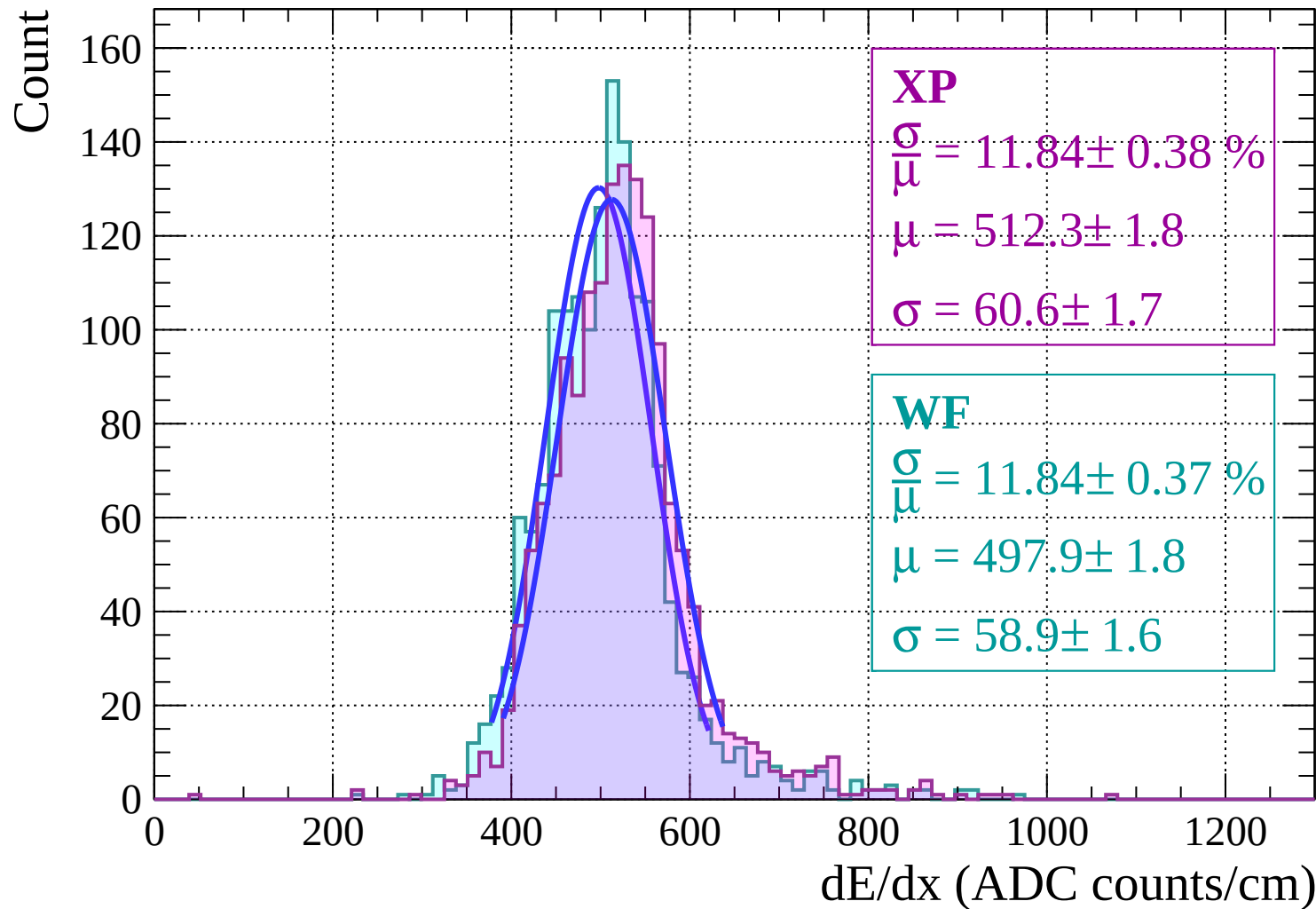


Energy loss | $30 < \theta < 32$

Count

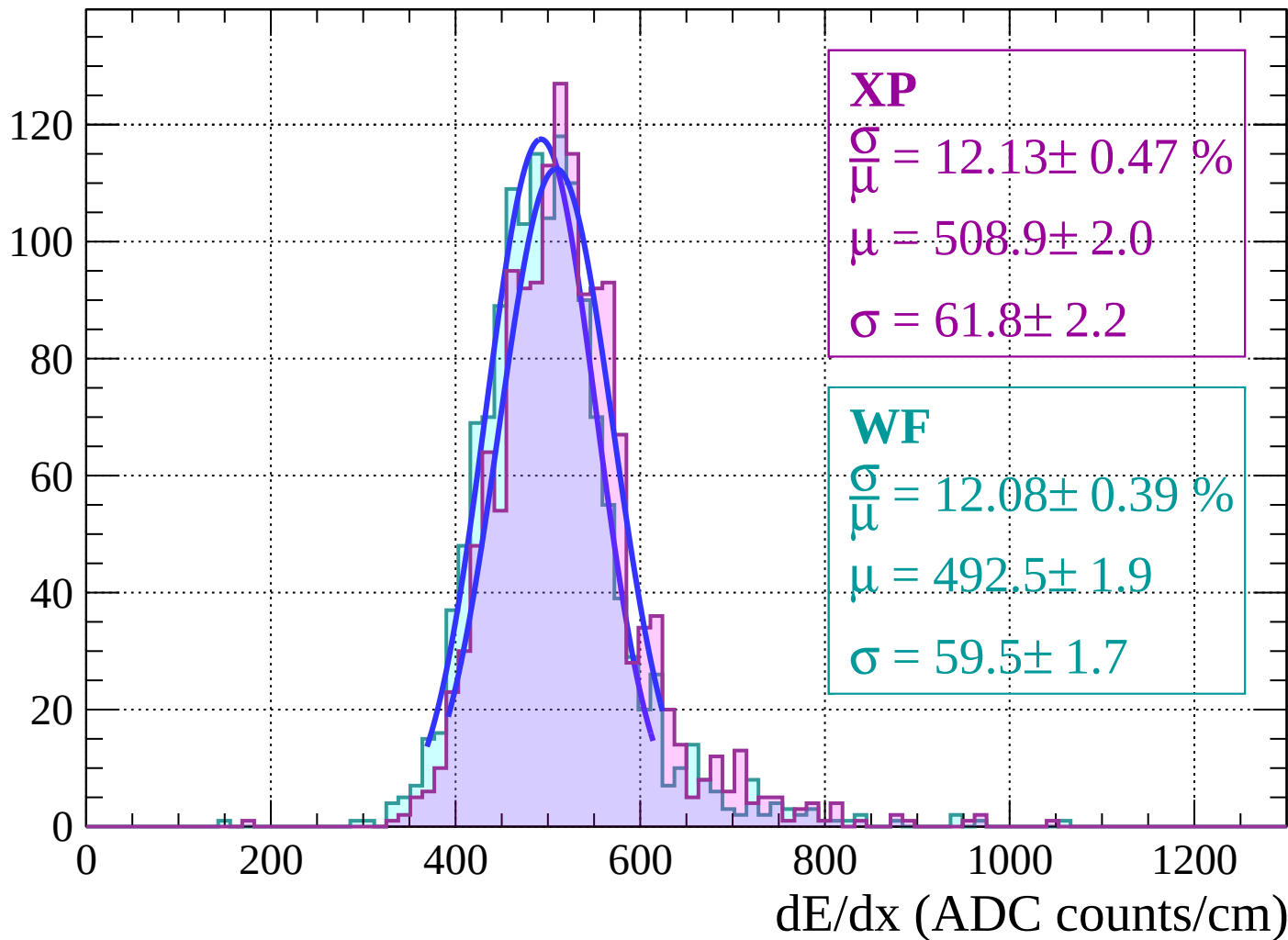


Energy loss | $32 < \theta < 34$

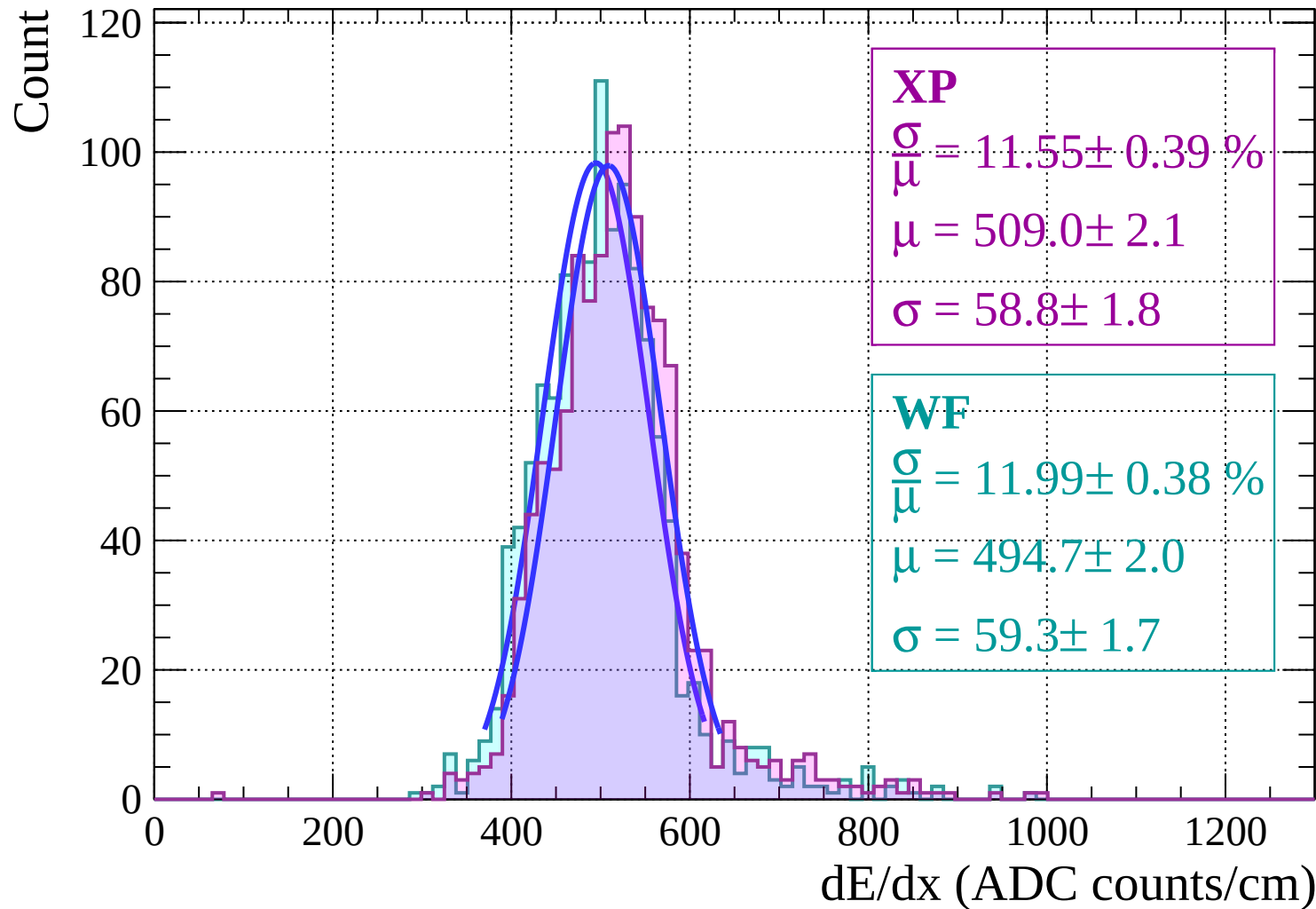


Energy loss | $34 < \theta < 36$

Count

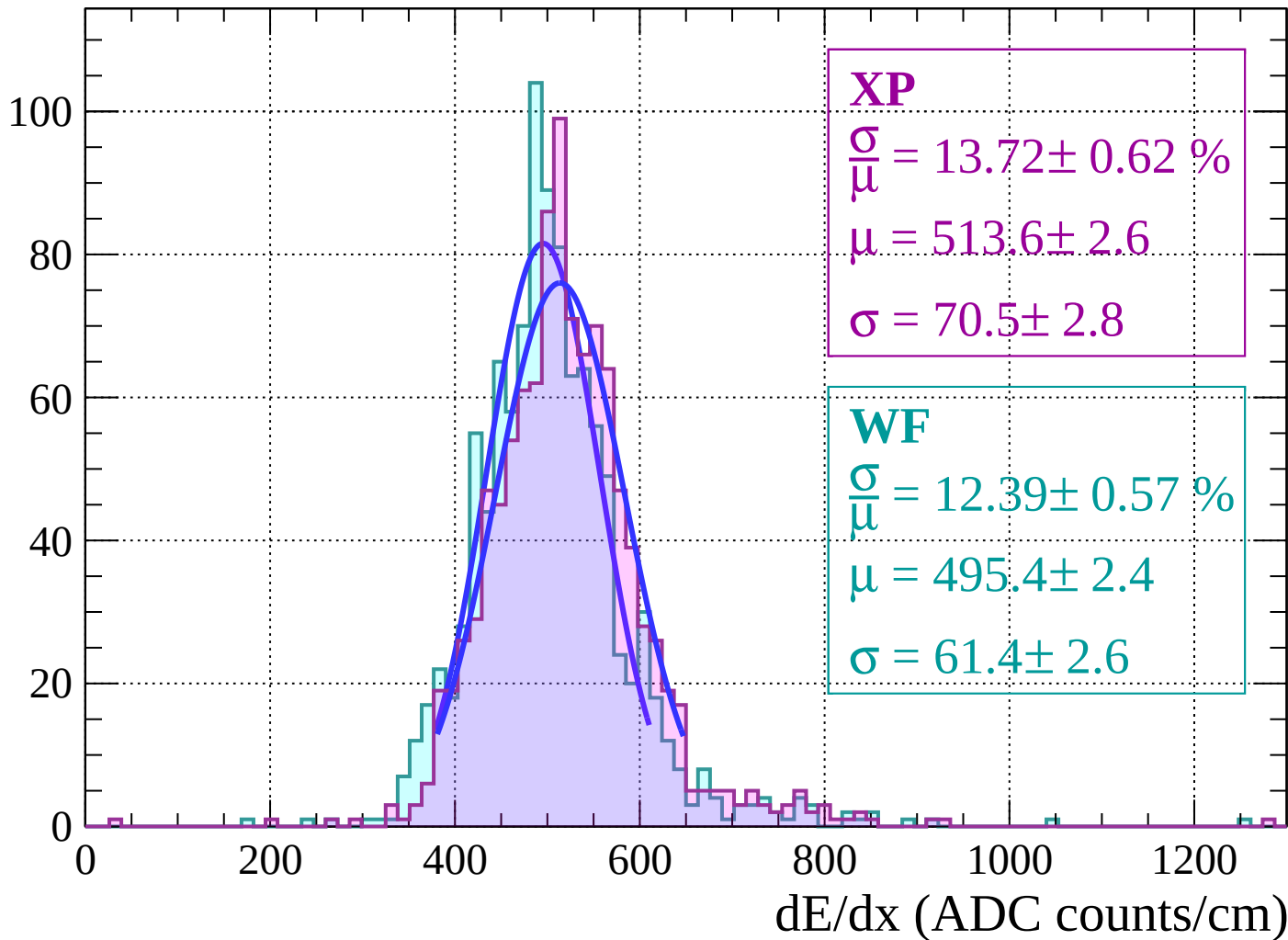


Energy loss | $36 < \theta < 38$



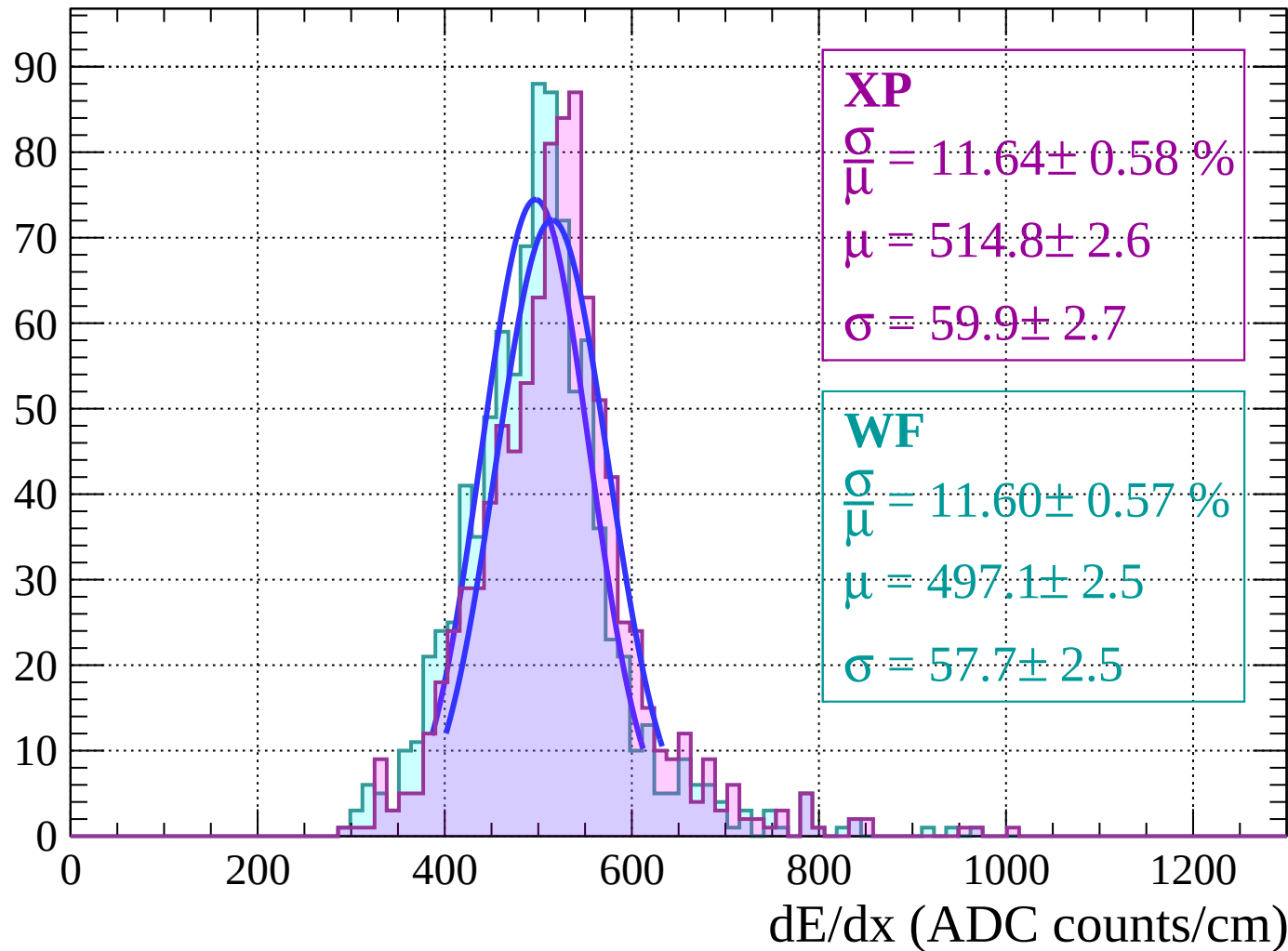
Energy loss | $38 < \theta < 40$

Count



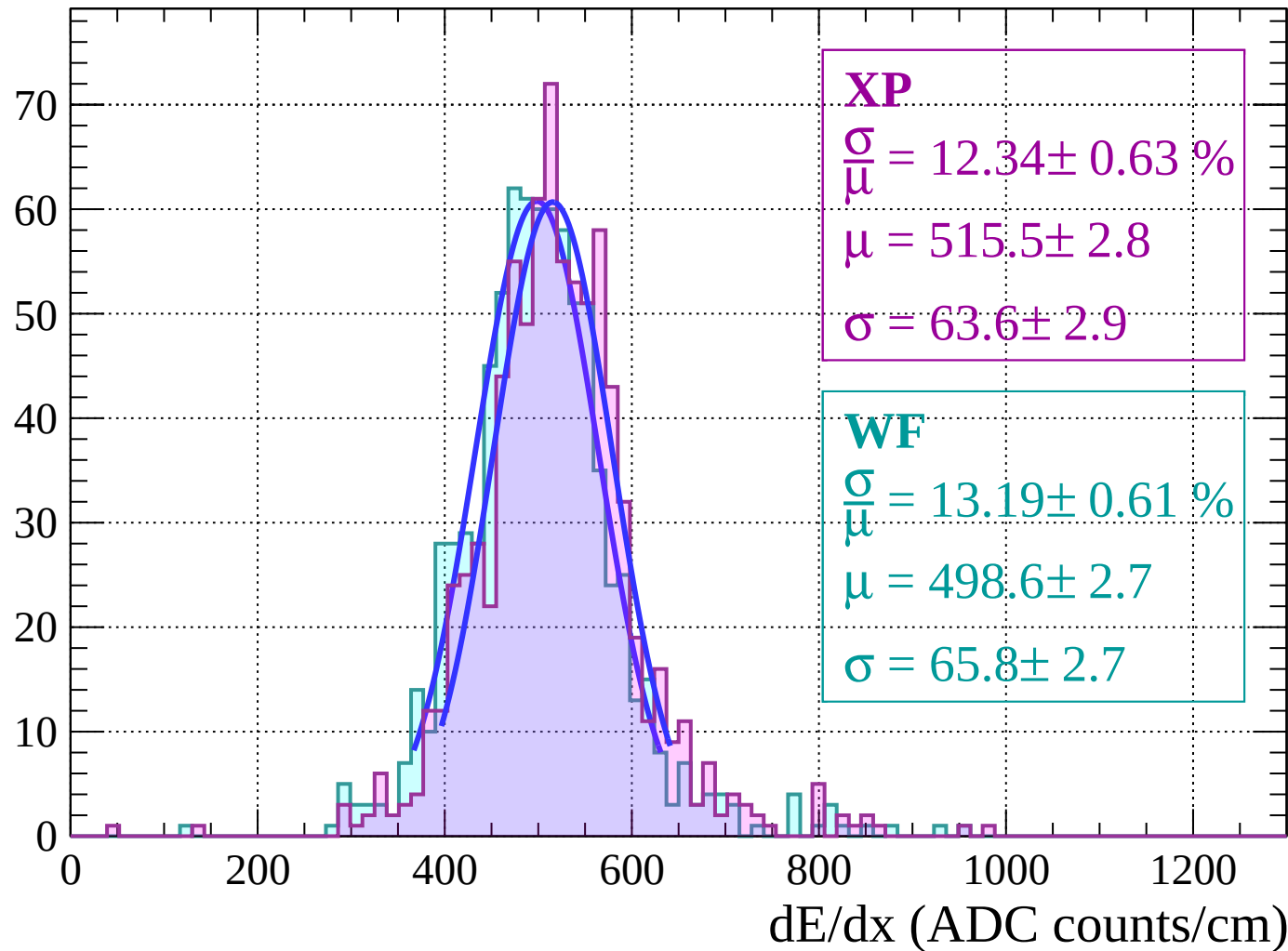
Energy loss | $40 < \theta < 42$

Count



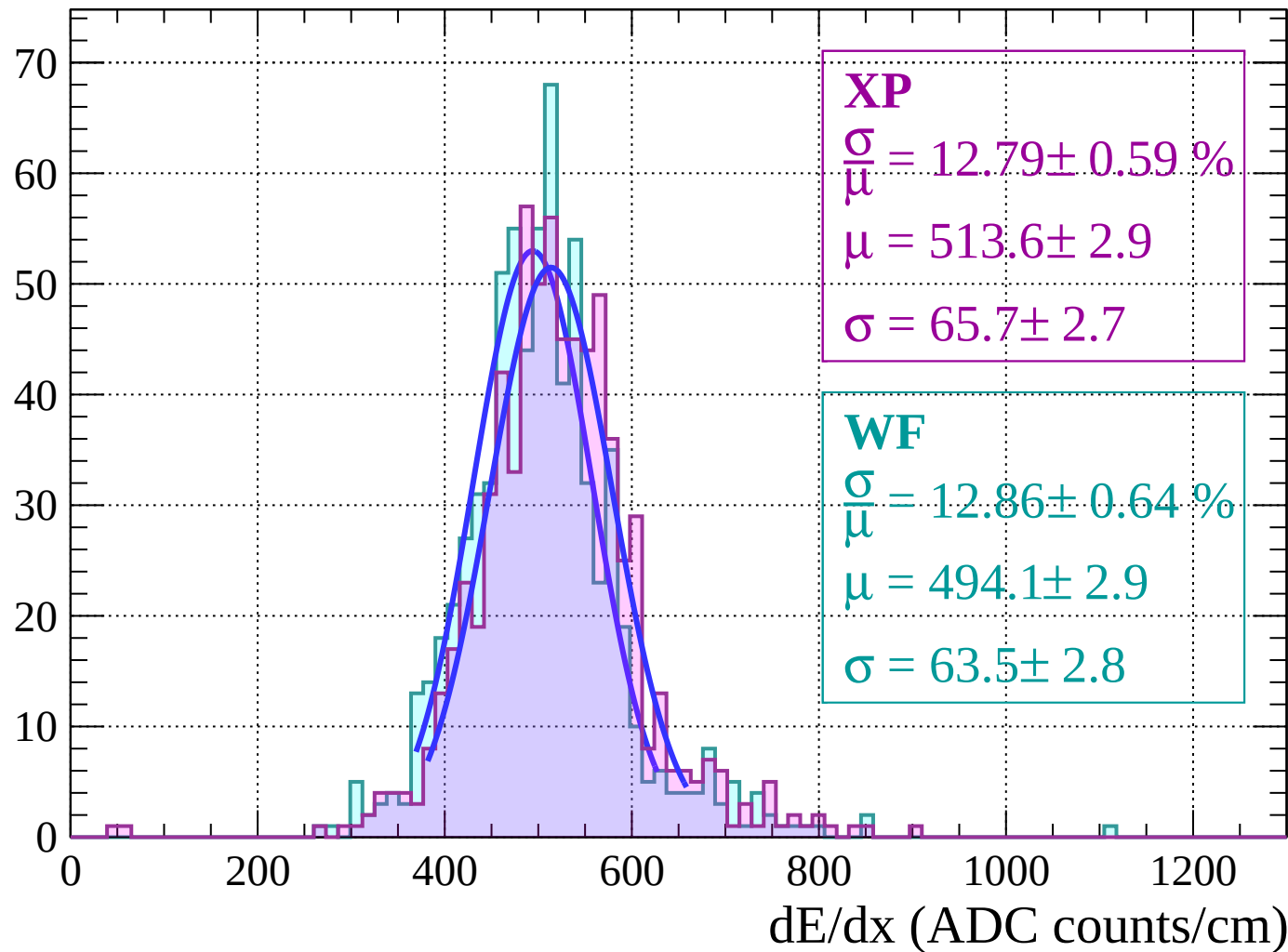
Energy loss | $42 < \theta < 44$

Count



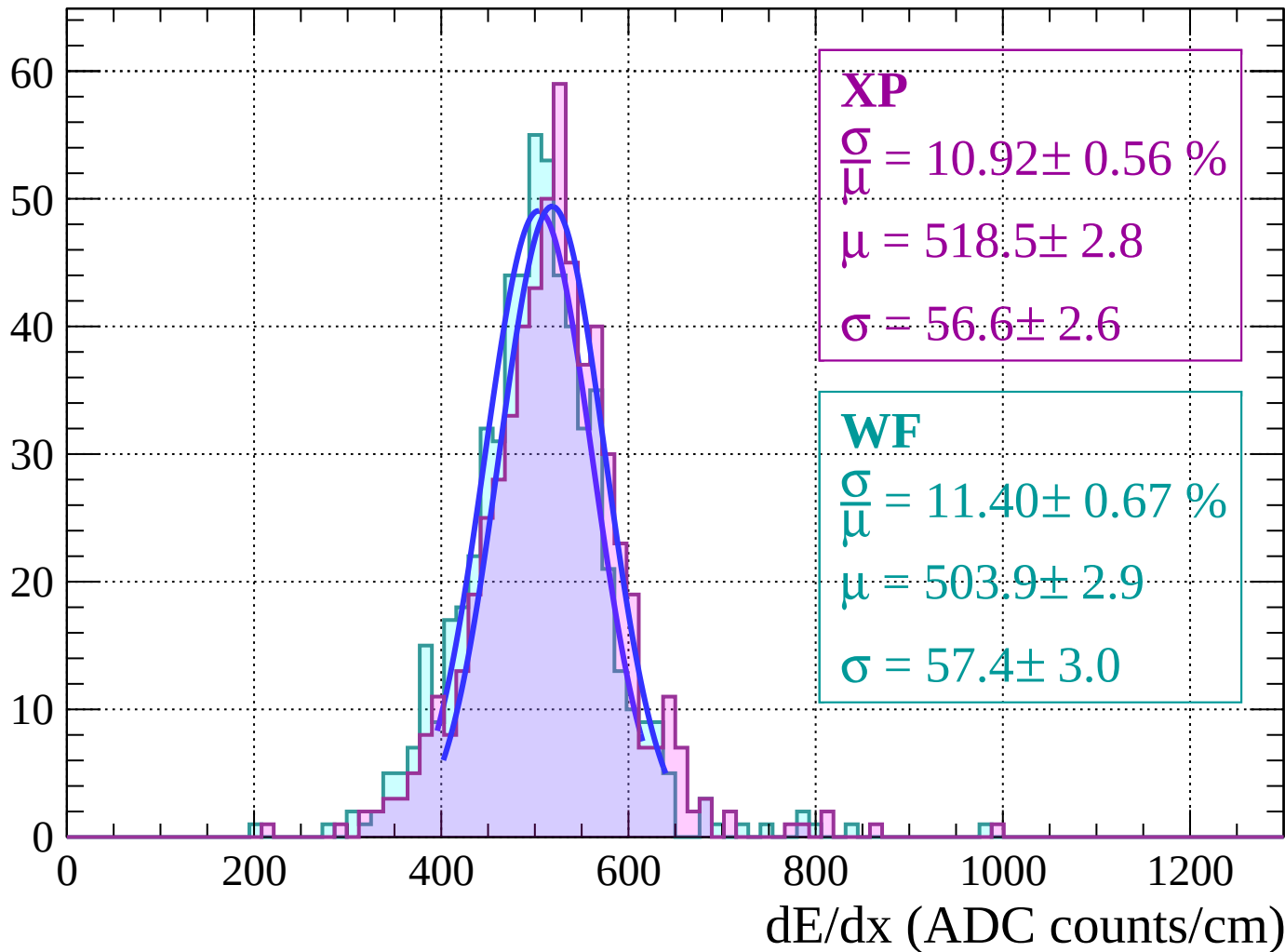
Energy loss | $44 < \theta < 46$

Count



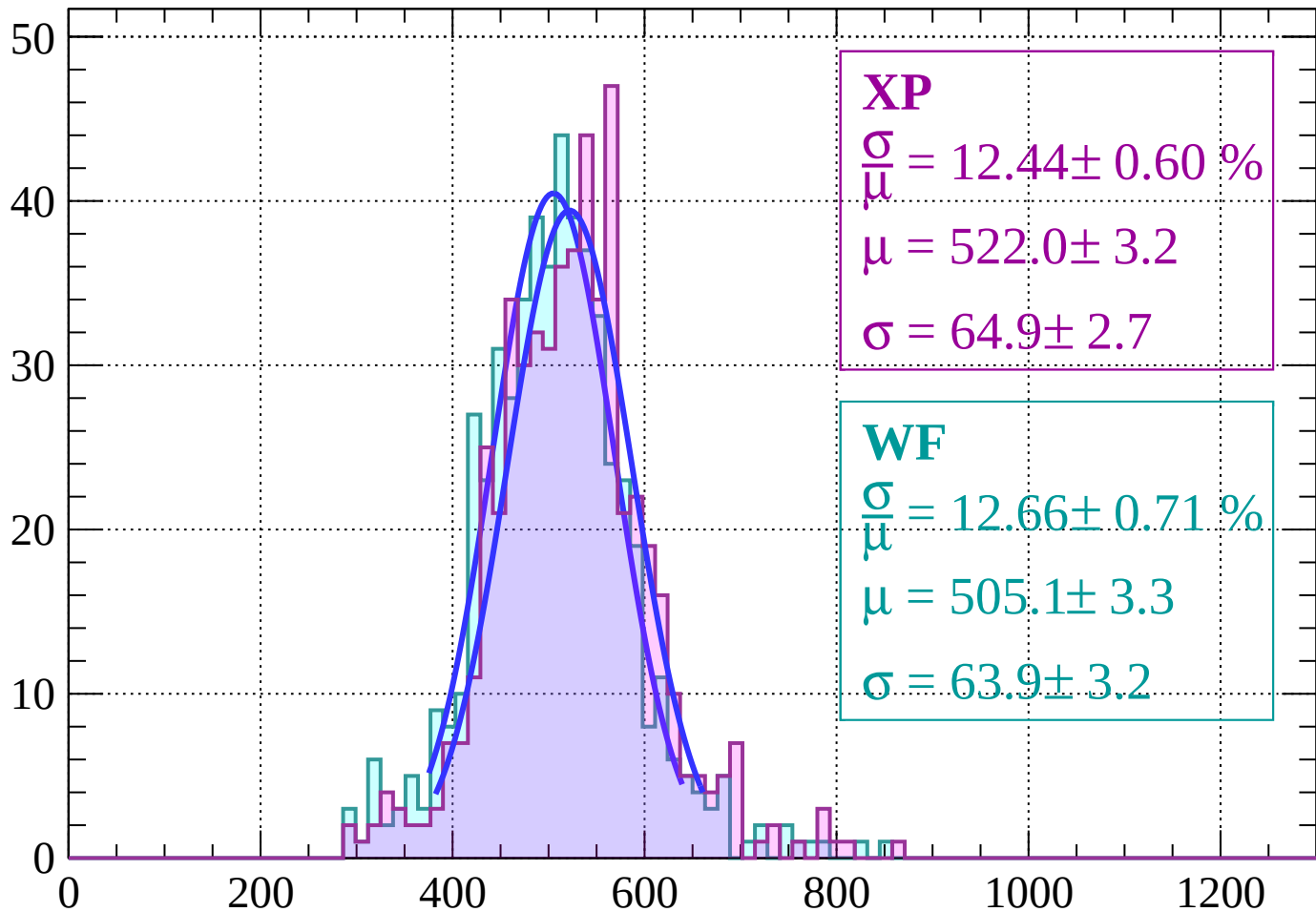
Energy loss | $46 < \theta < 48$

Count



Energy loss | $48 < \theta < 50$

Count



XP

$$\frac{\sigma}{\mu} = 12.44 \pm 0.60 \%$$

$$\mu = 522.0 \pm 3.2$$

$$\sigma = 64.9 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 12.66 \pm 0.71 \%$$

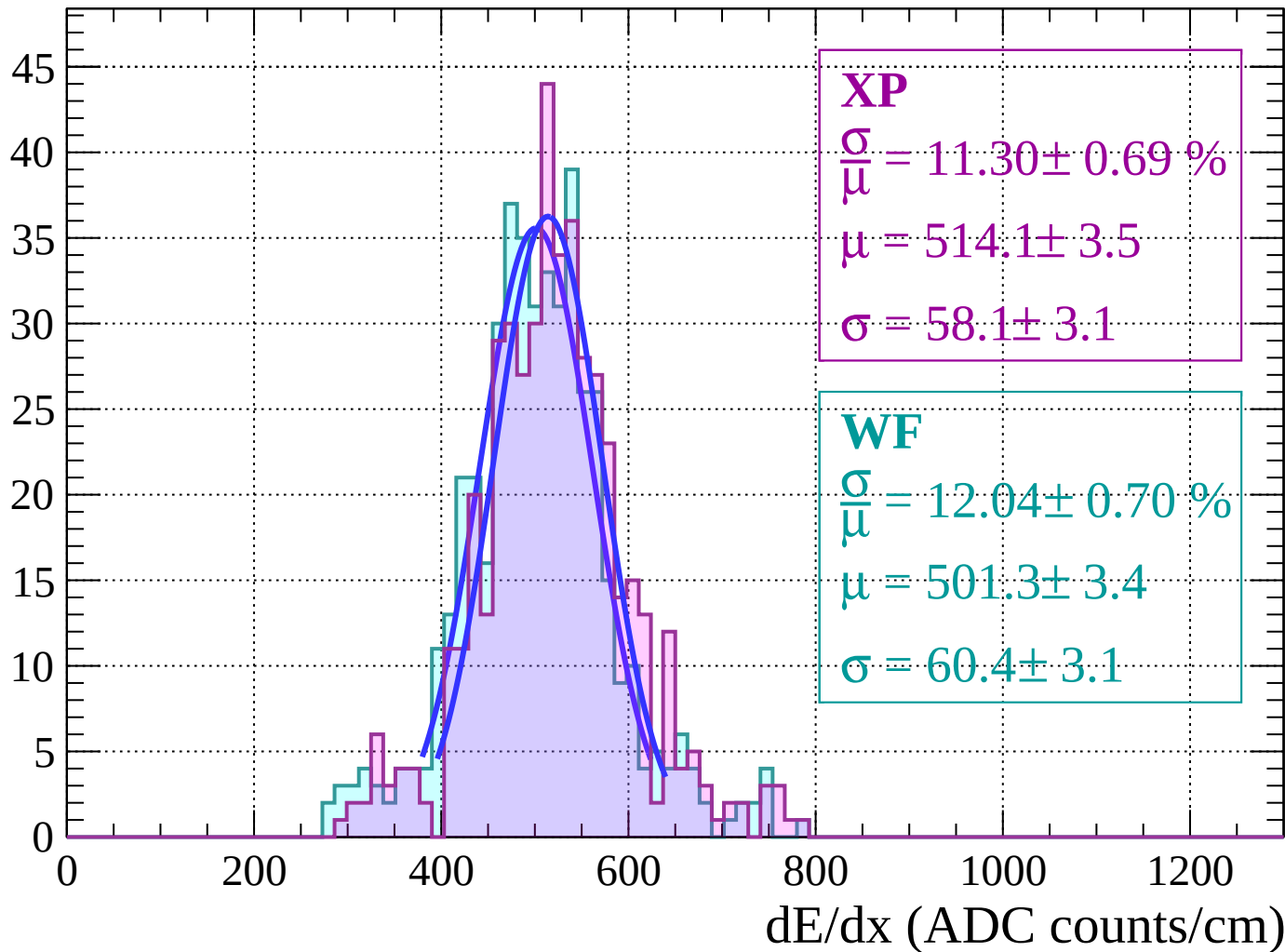
$$\mu = 505.1 \pm 3.3$$

$$\sigma = 63.9 \pm 3.2$$

dE/dx (ADC counts/cm)

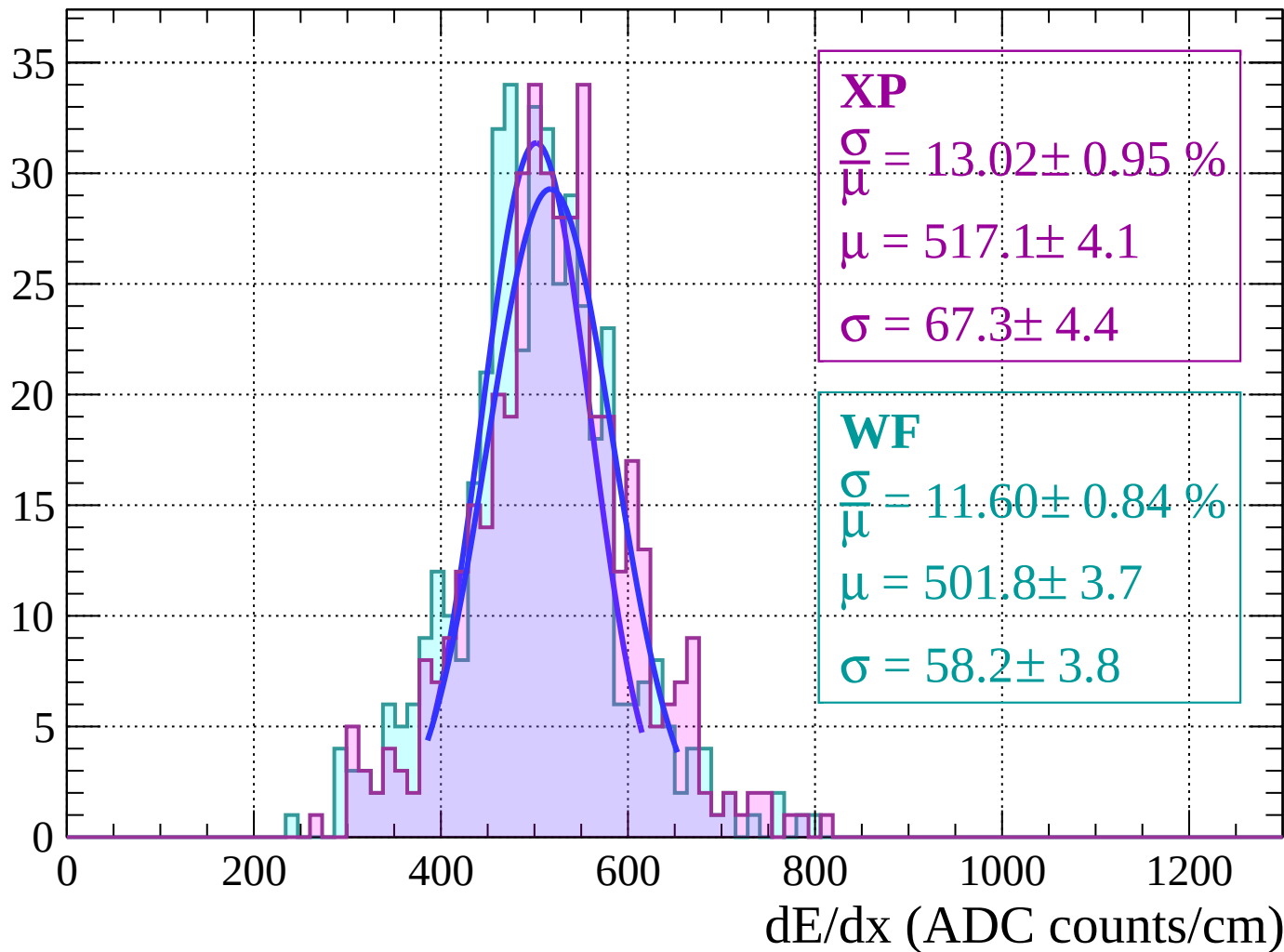
Energy loss | $50 < \theta < 52$

Count



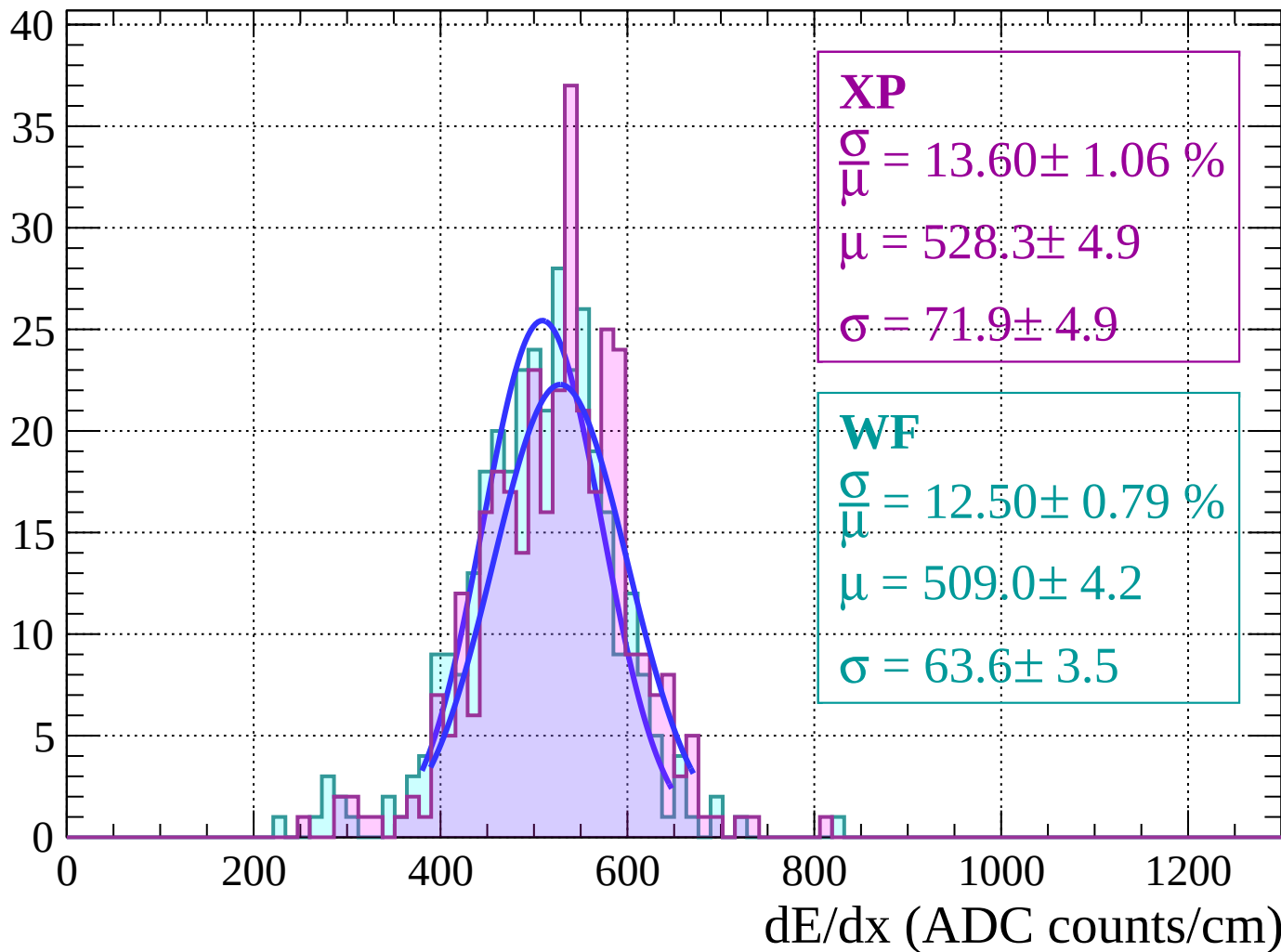
Energy loss | $52 < \theta < 54$

Count



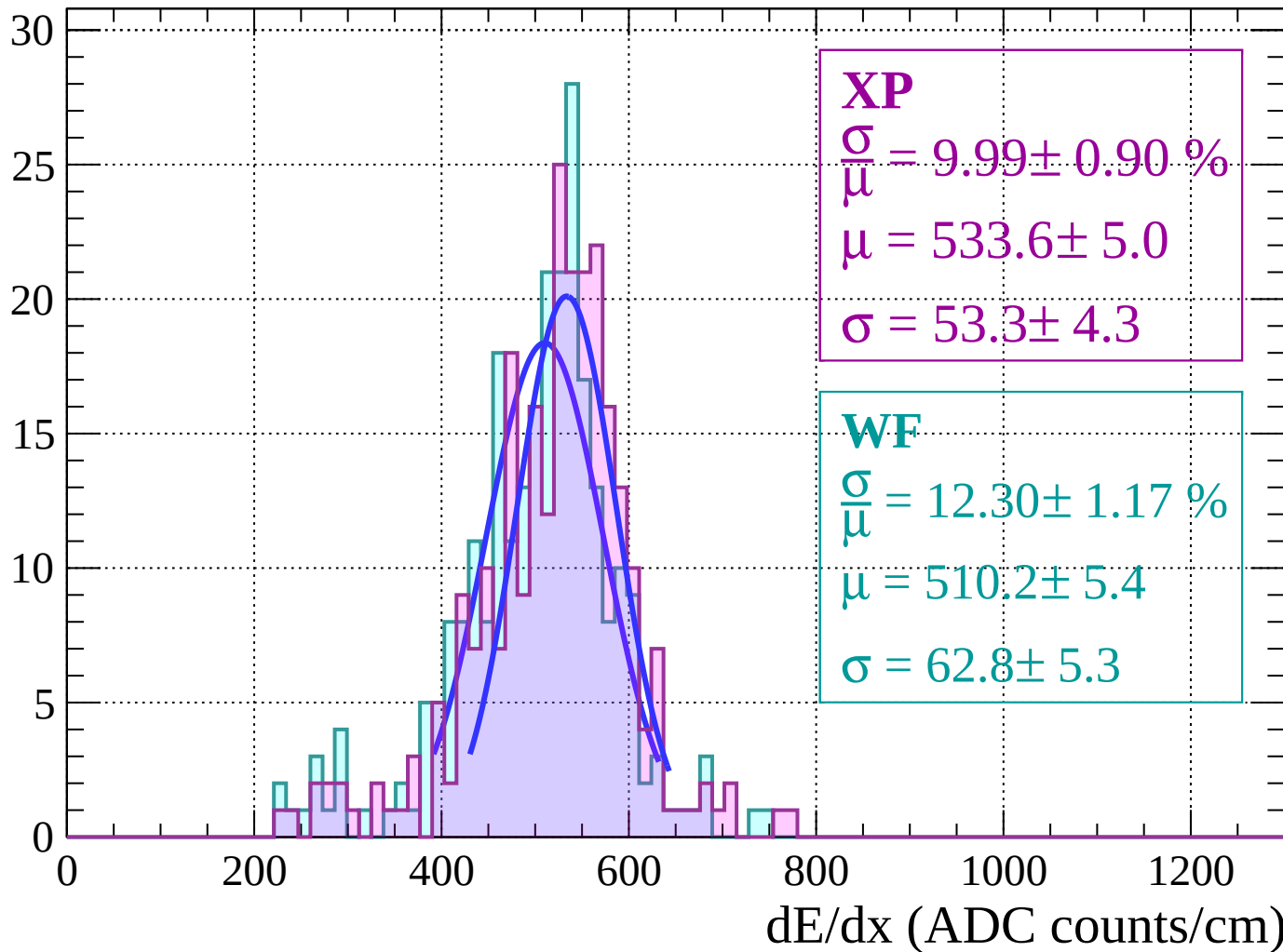
Energy loss | $54 < \theta < 56$

Count



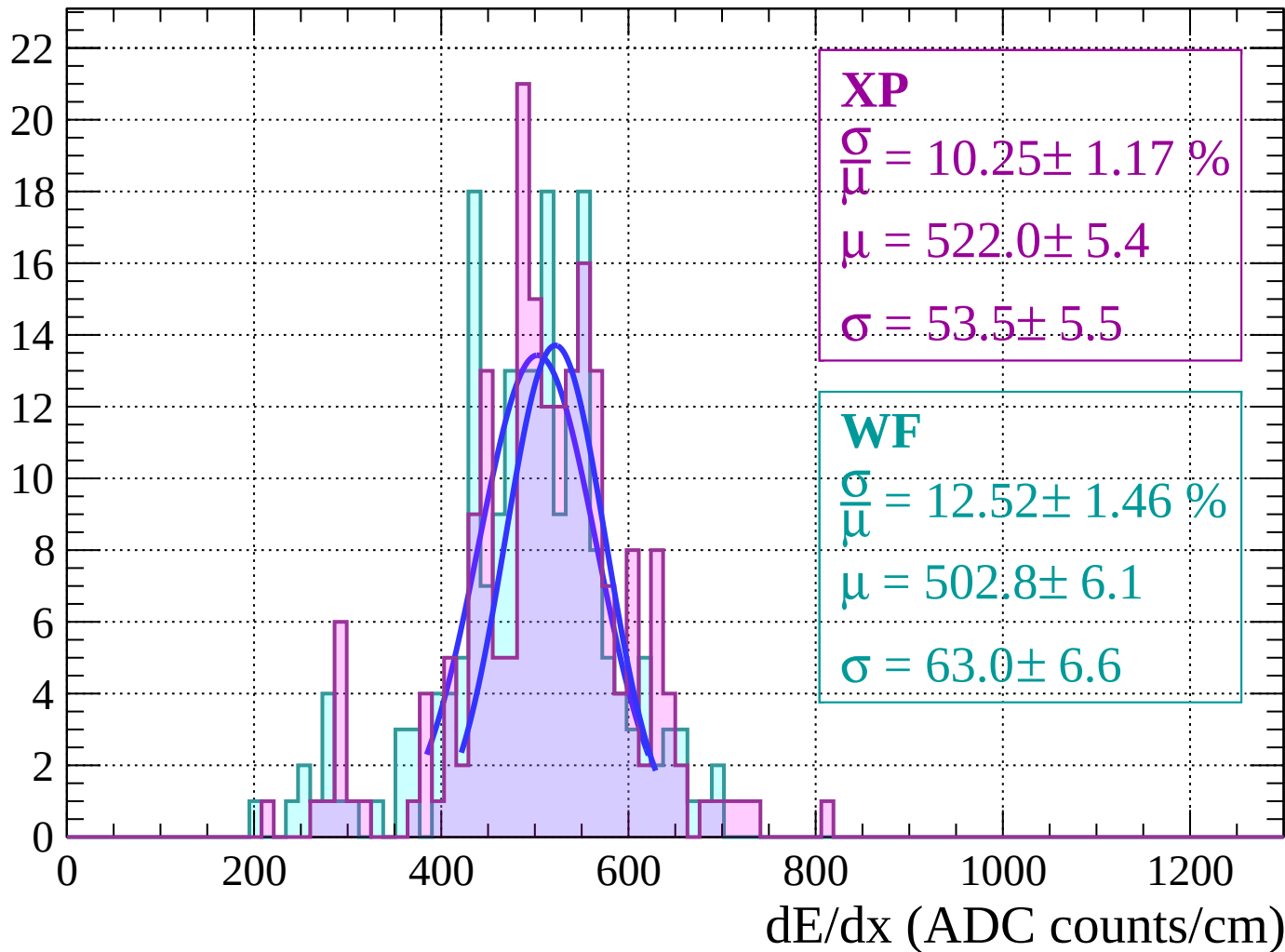
Energy loss | $56 < \theta < 58$

Count



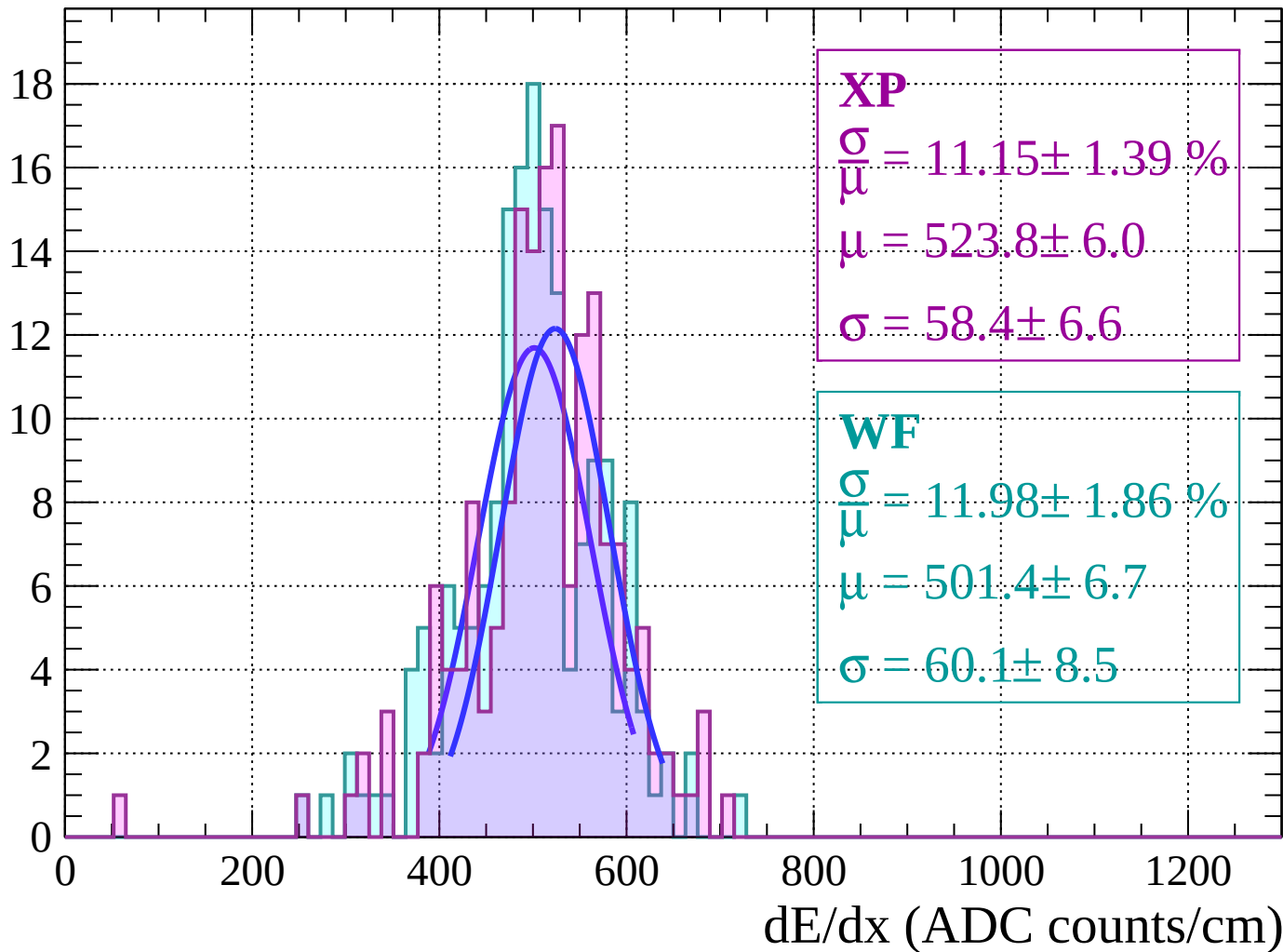
Energy loss | $58 < \theta < 60$

Count



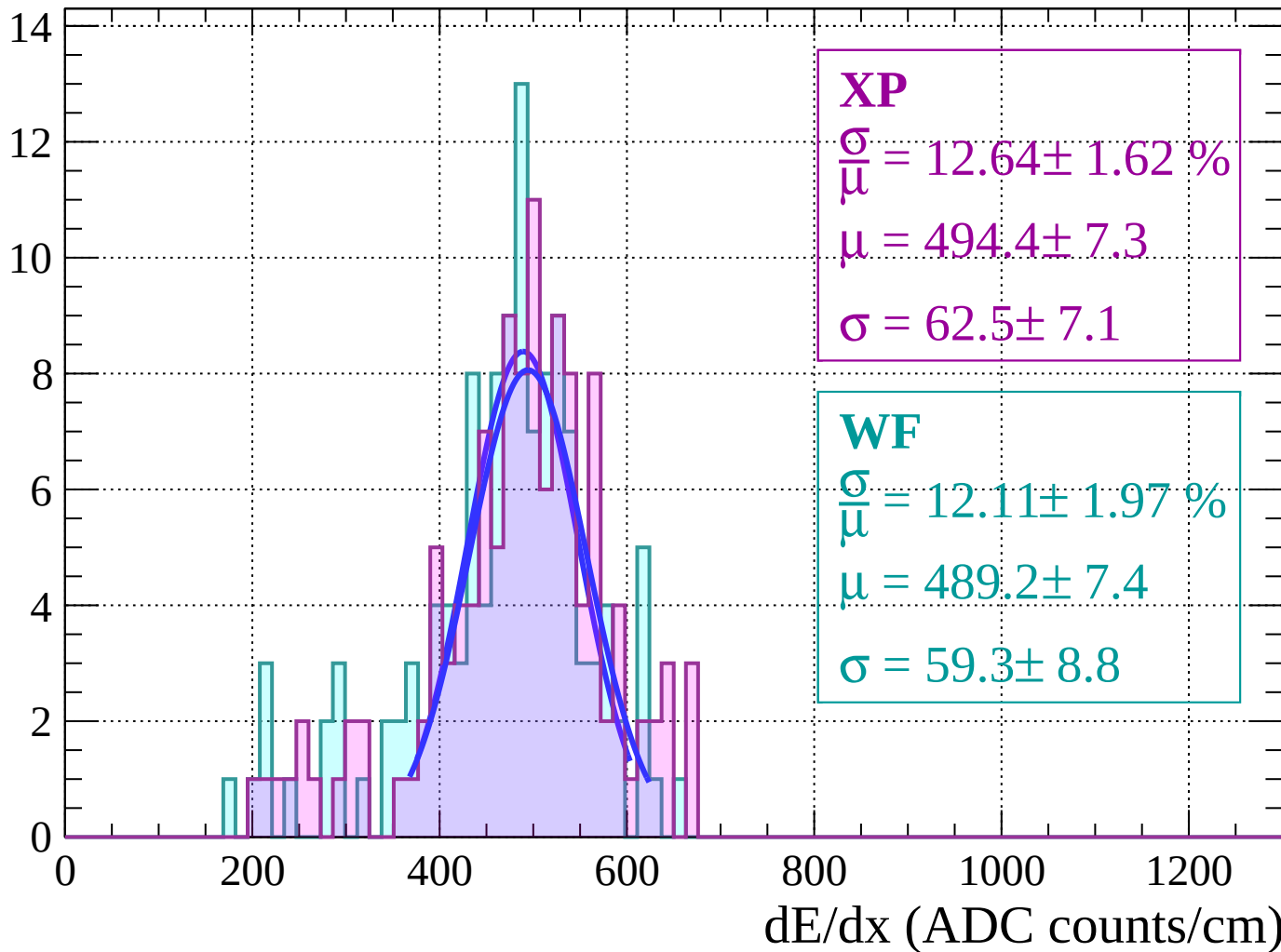
Energy loss | $60 < \theta < 62$

Count



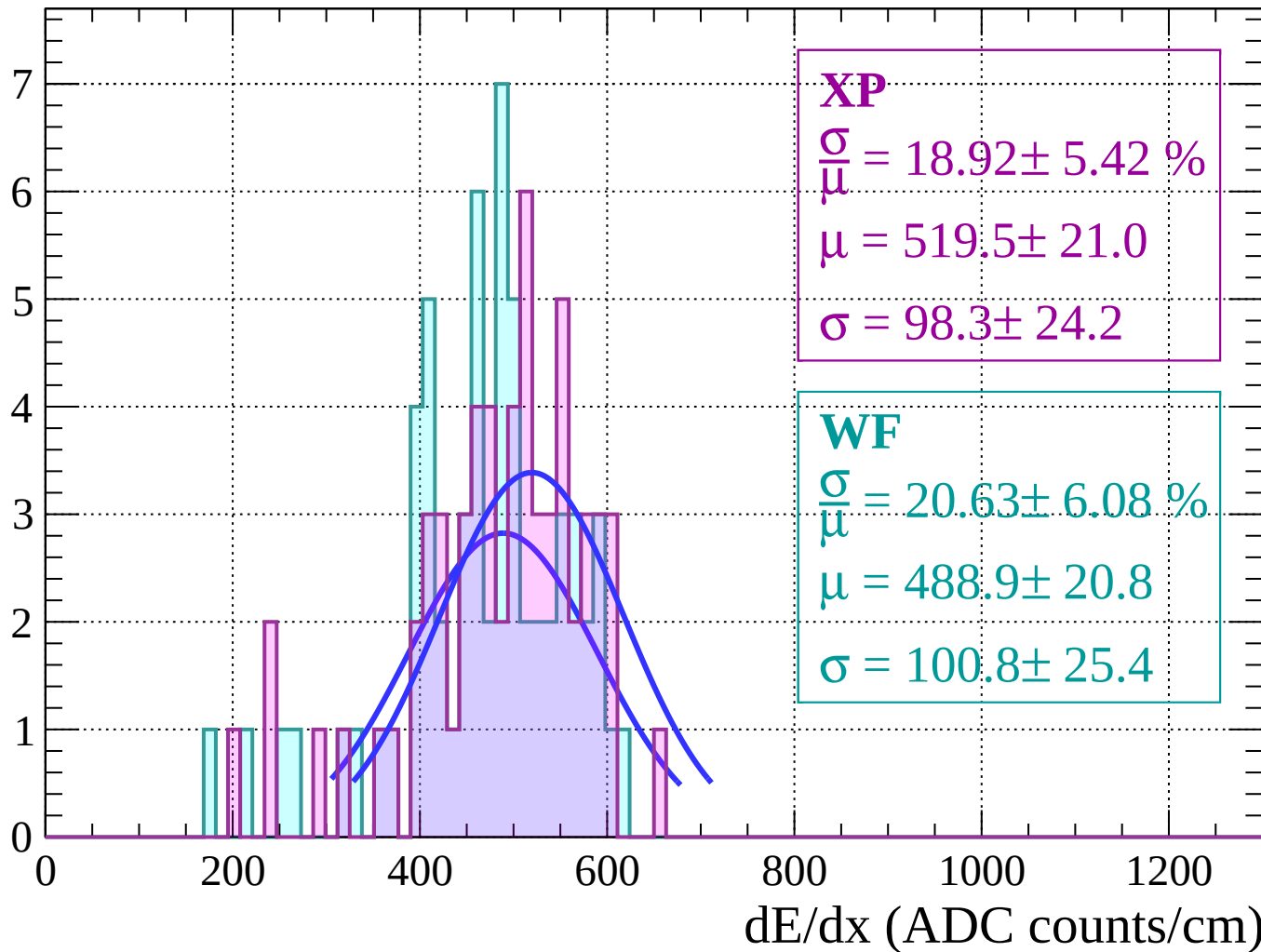
Energy loss | $62 < \theta < 64$

Count



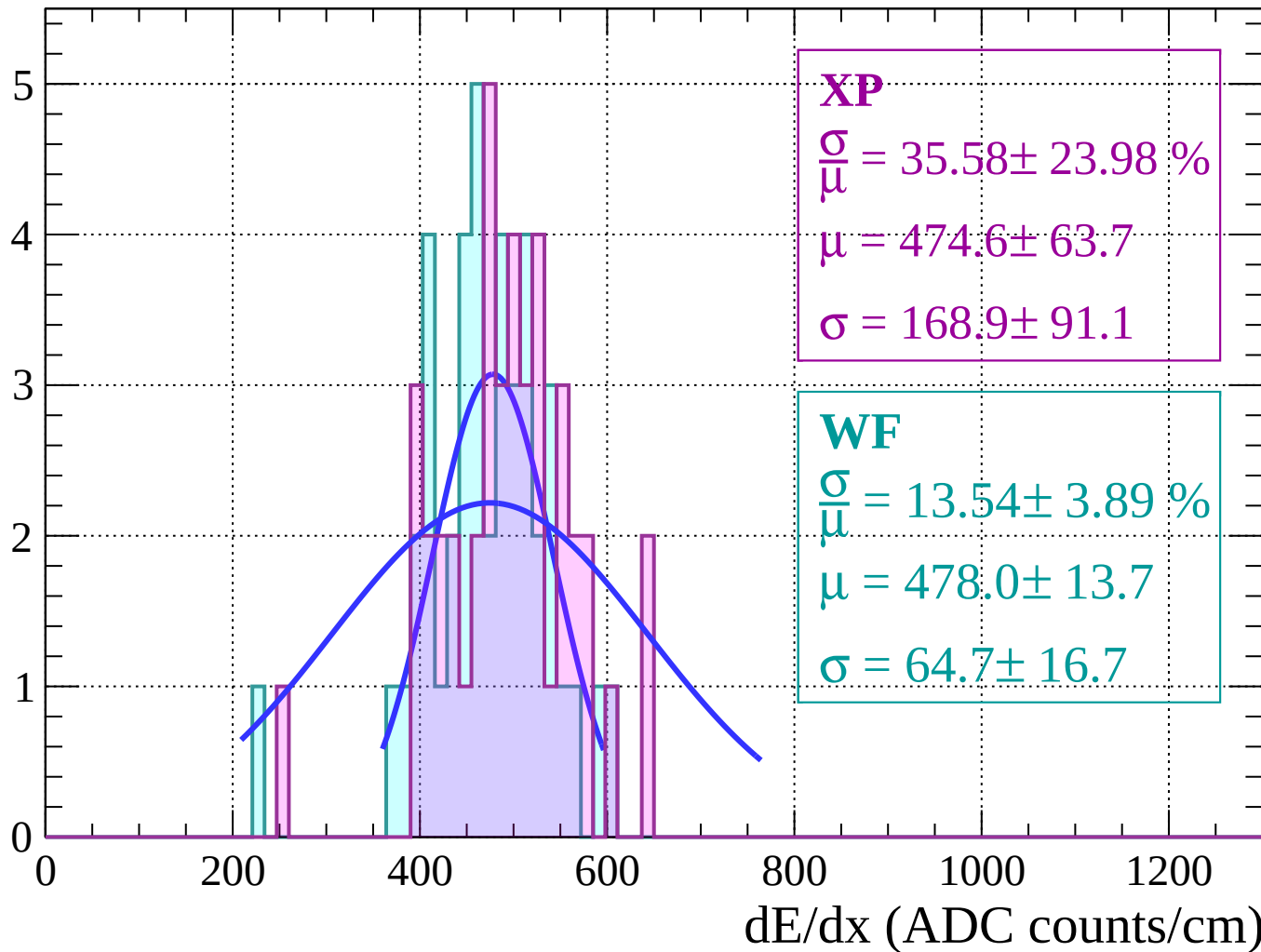
Energy loss | $64 < \theta < 66$

Count



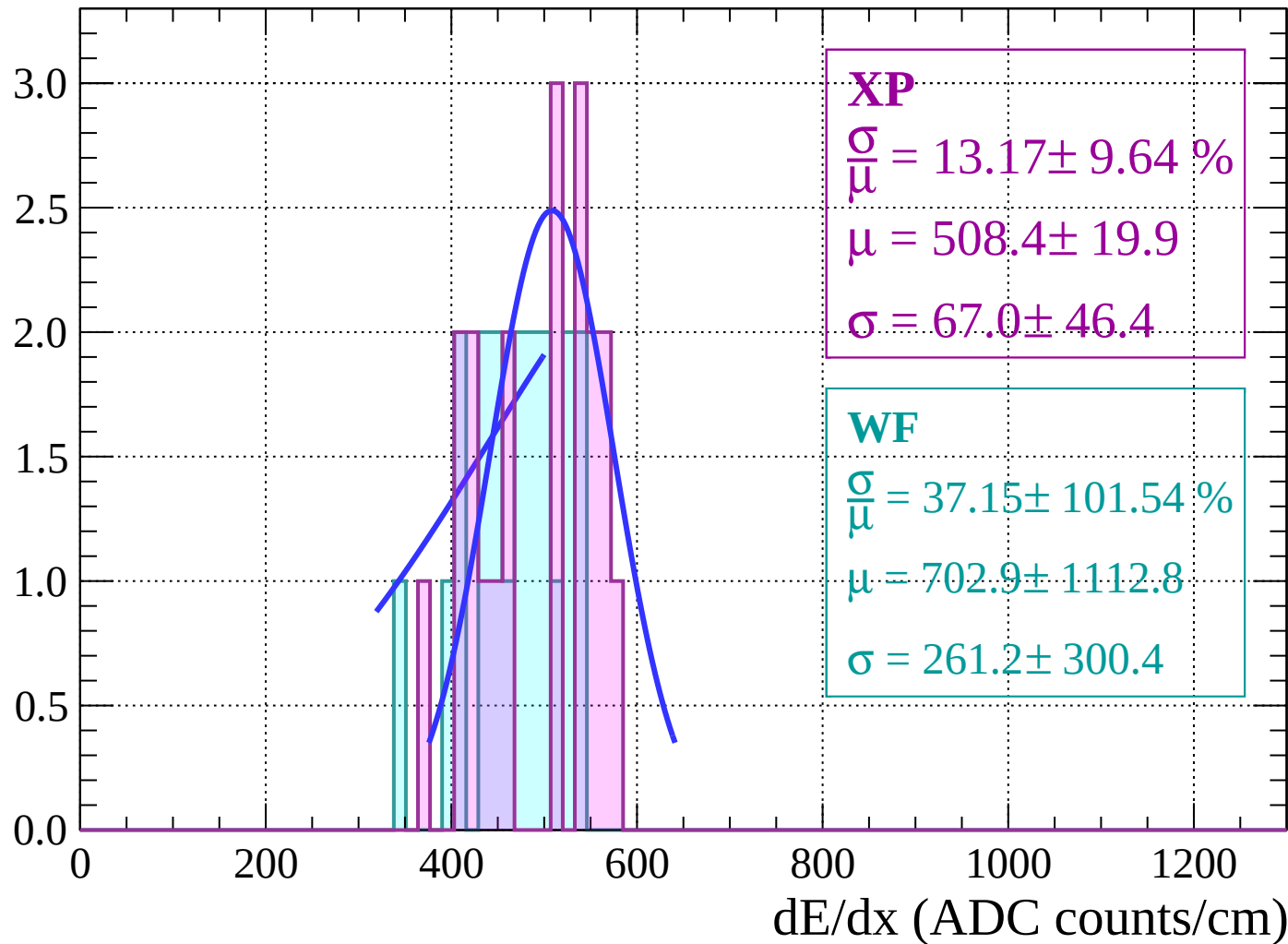
Energy loss | $66 < \theta < 68$

Count



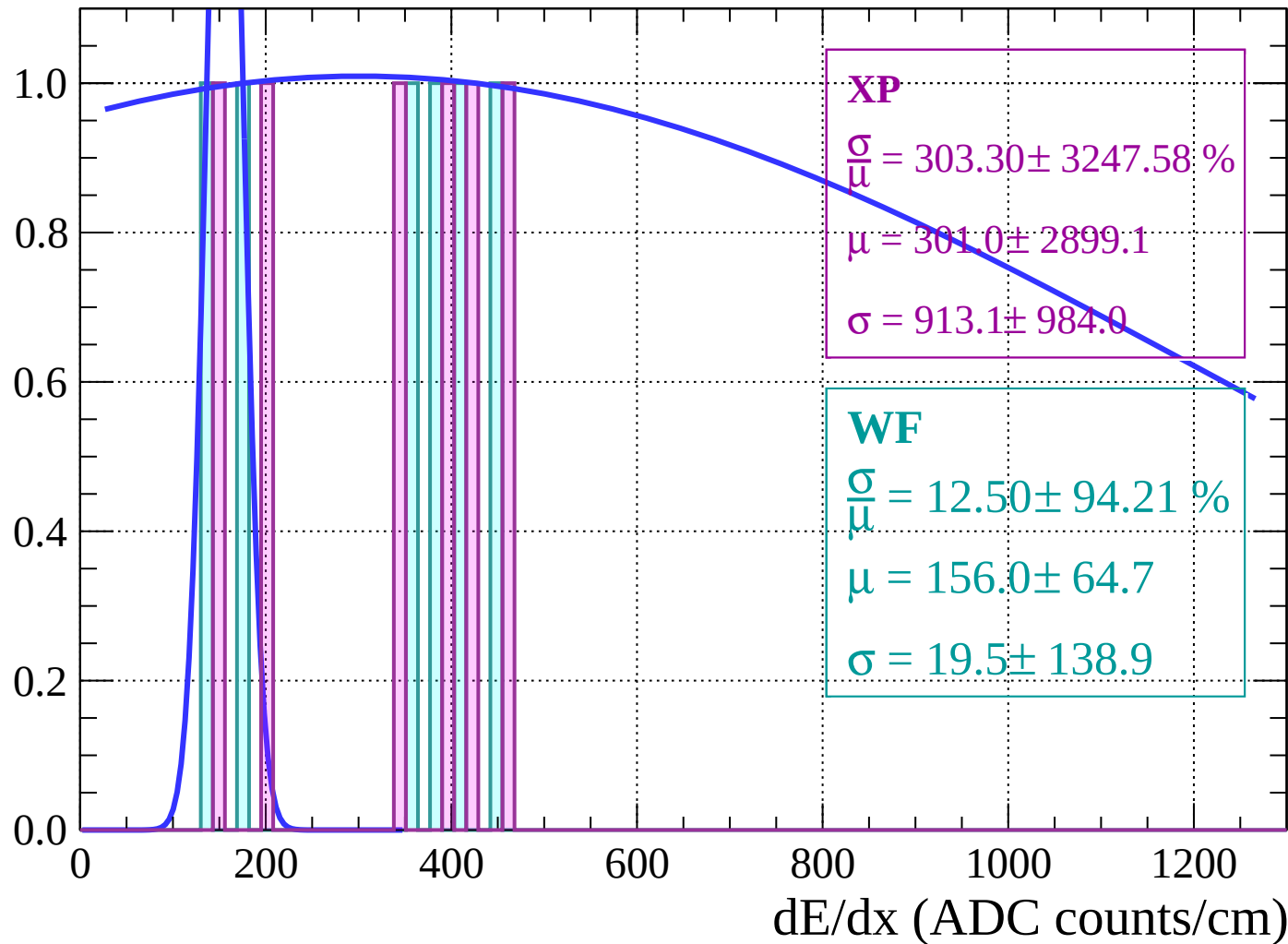
Energy loss | $68 < \theta < 70$

Count



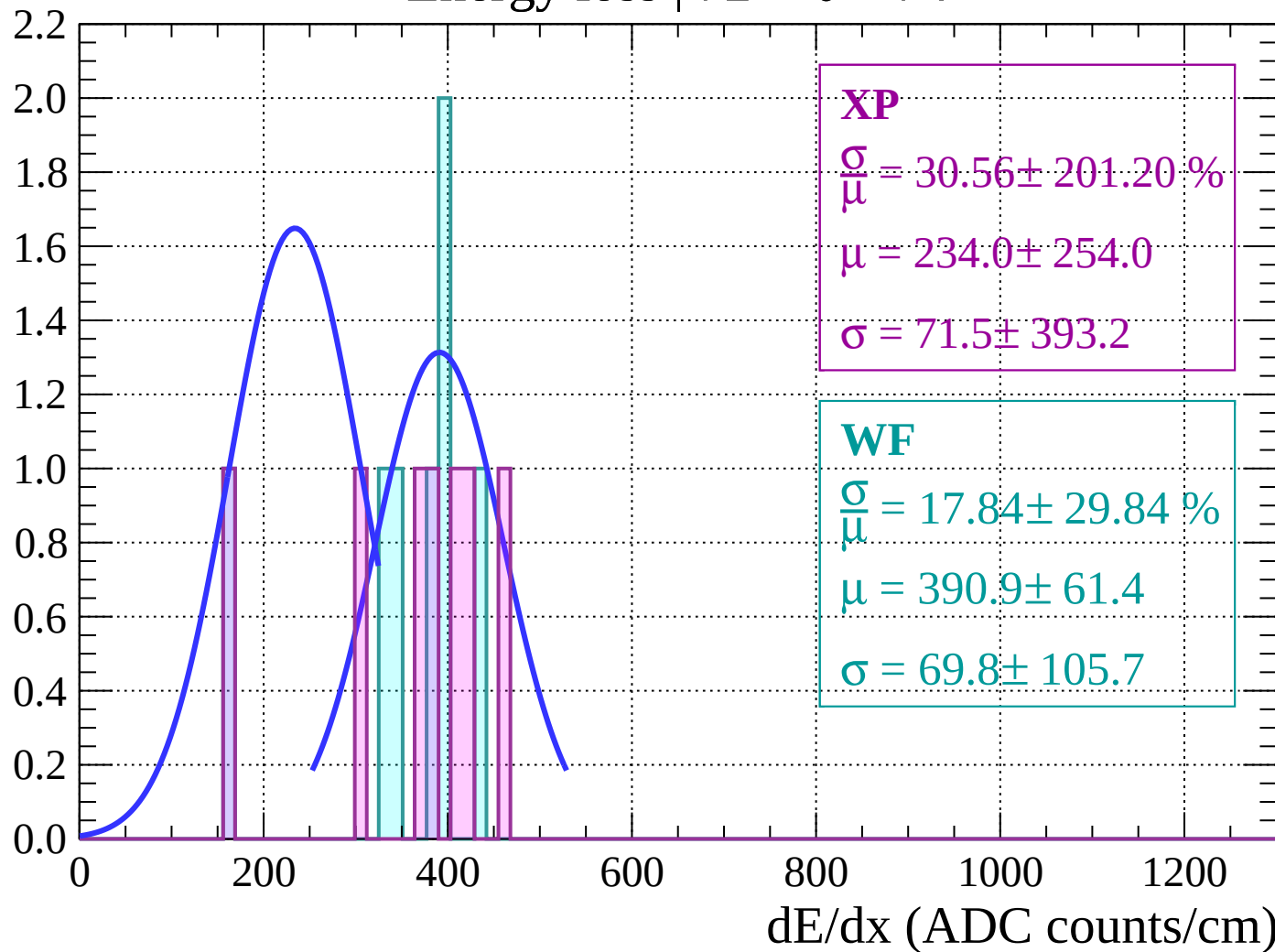
Energy loss | $70 < \theta < 72$

Count



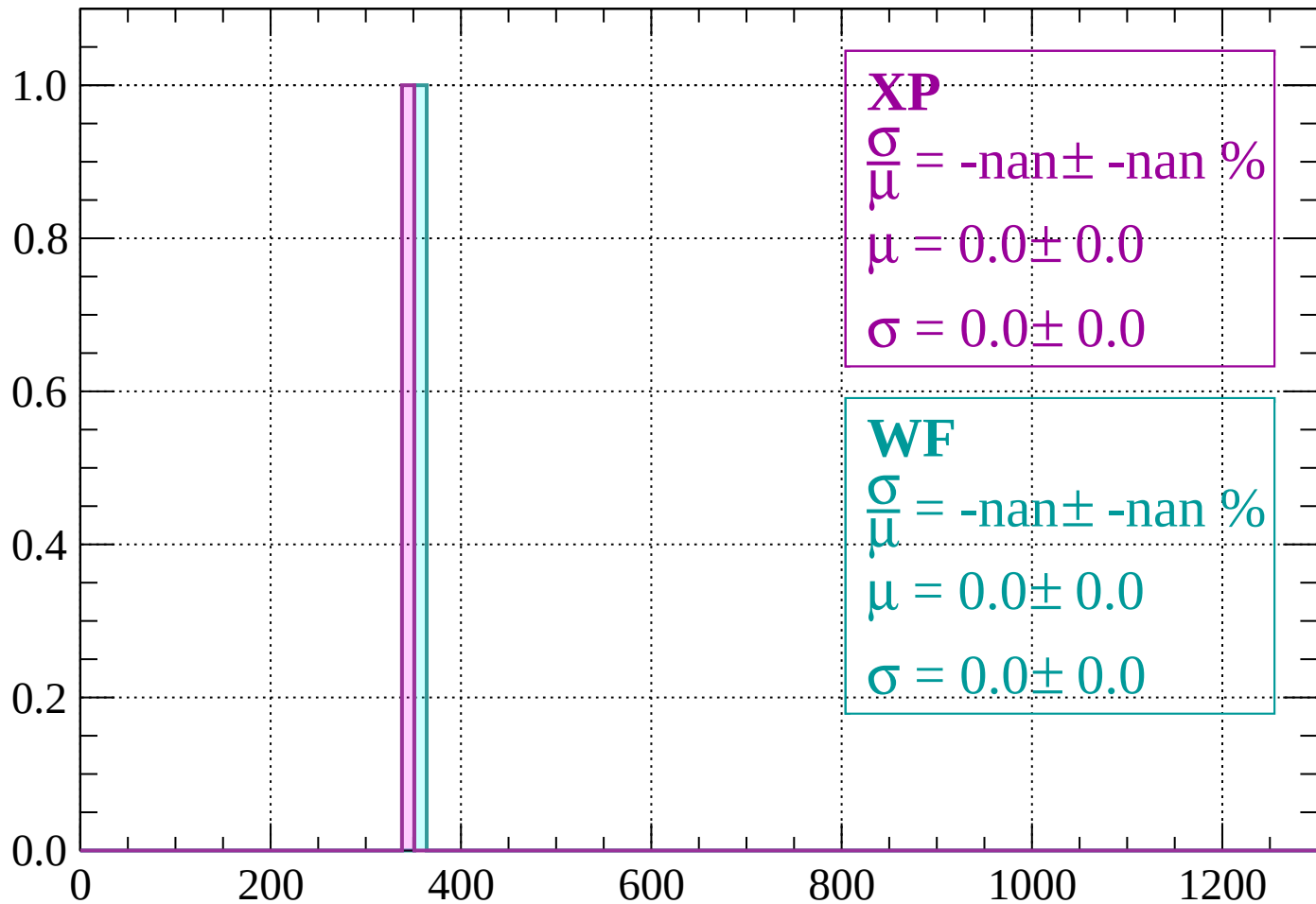
Energy loss | $72 < \theta < 74$

Count



Energy loss | $74 < \theta < 76$

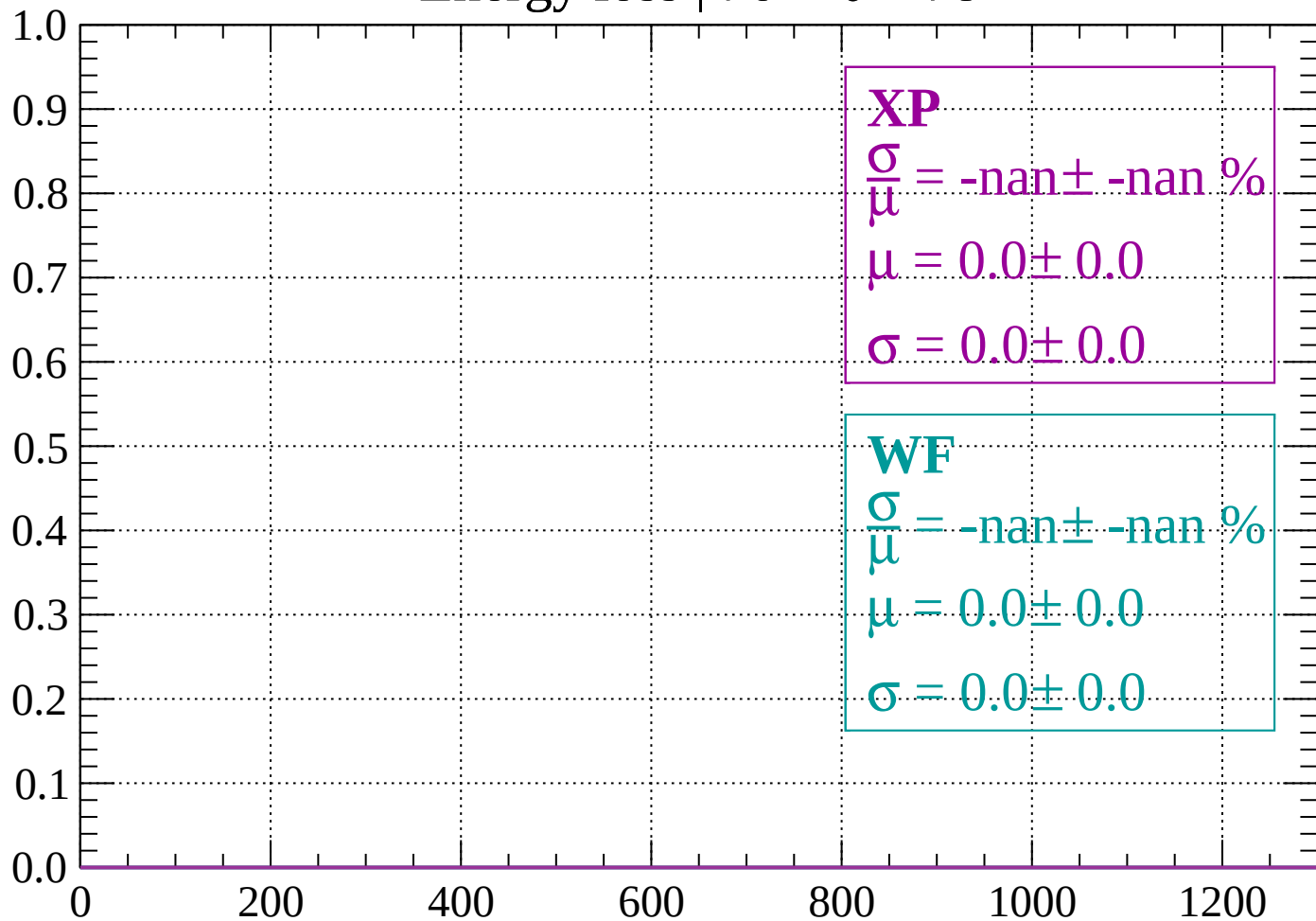
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

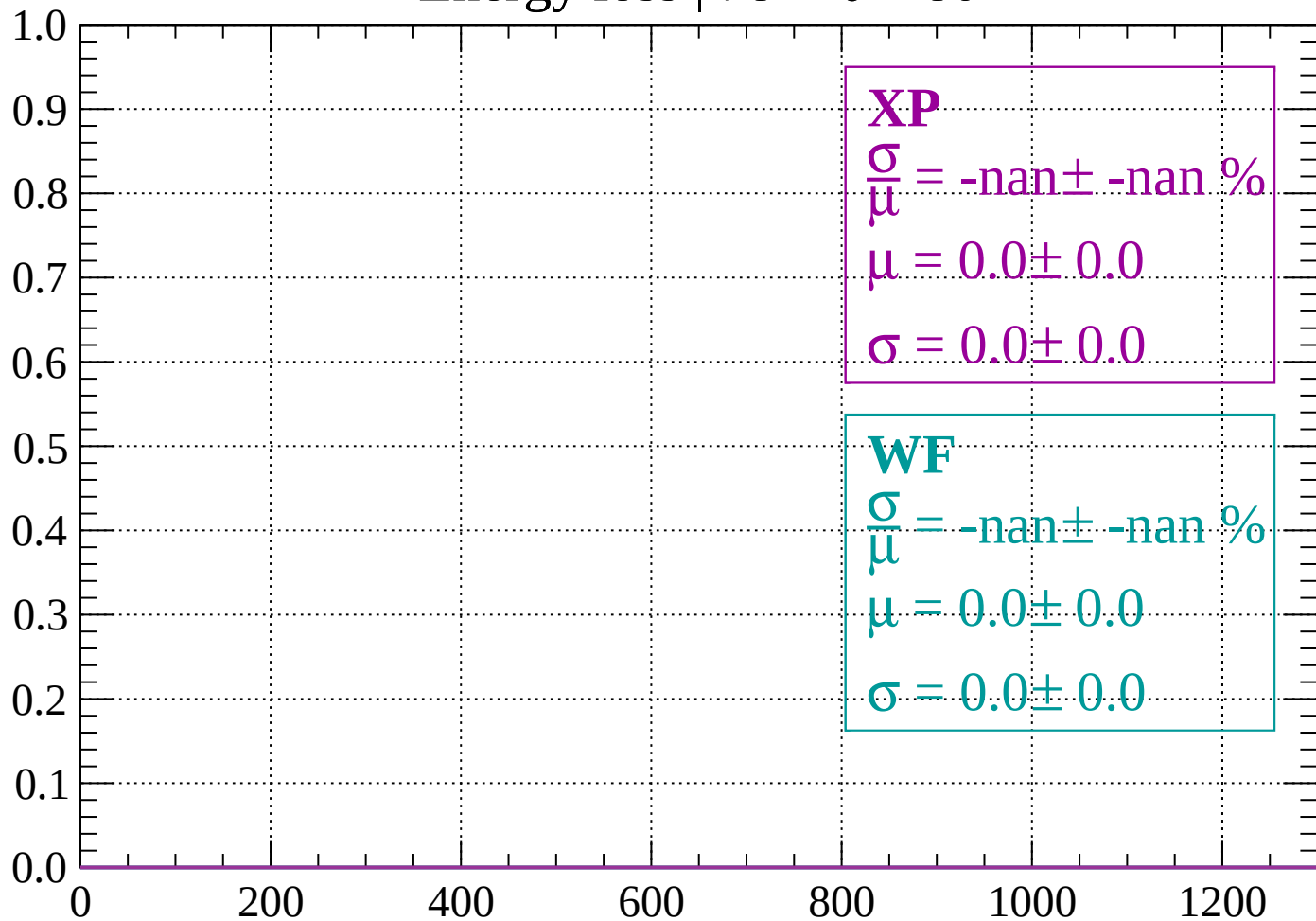
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

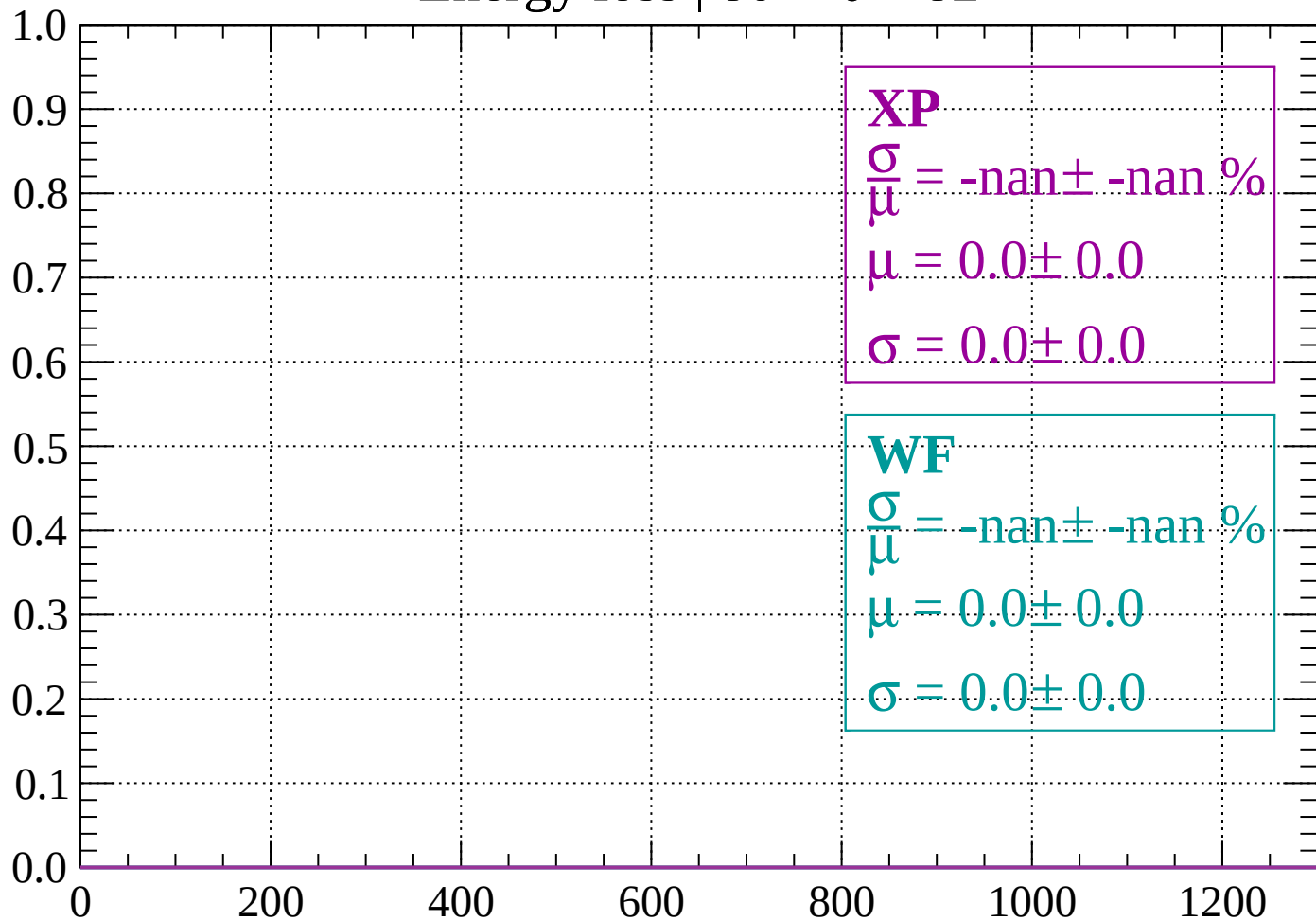
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

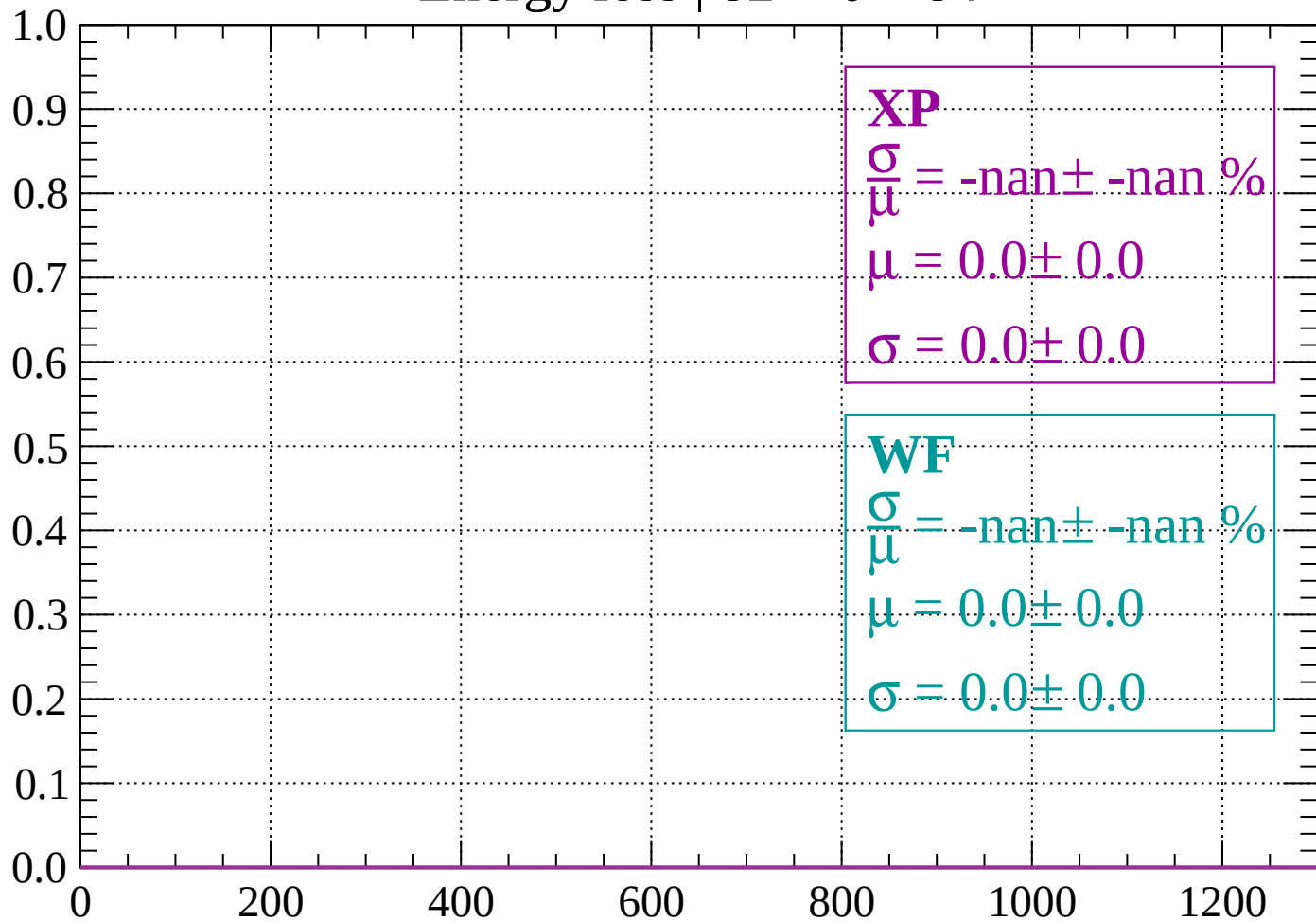
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

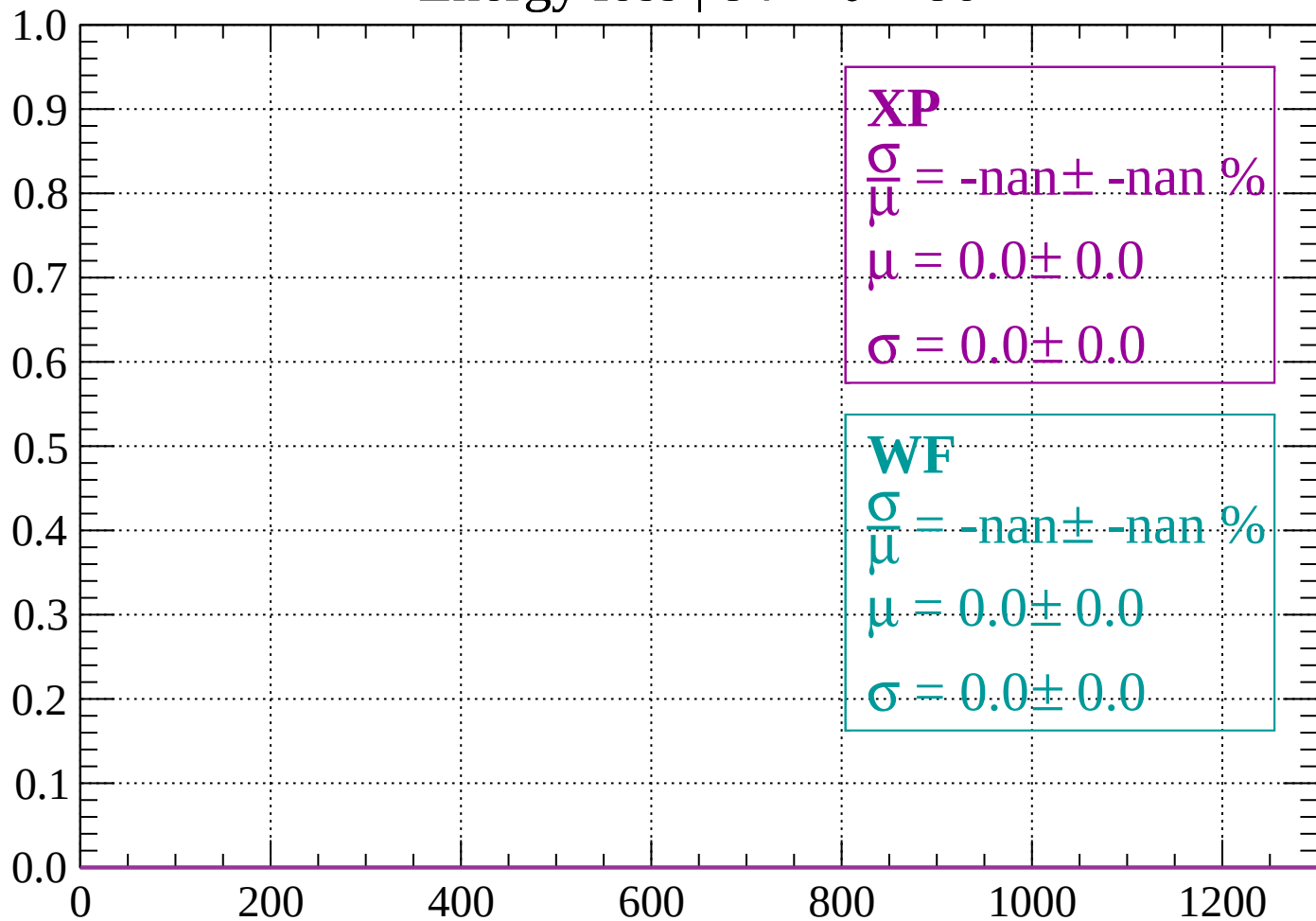
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

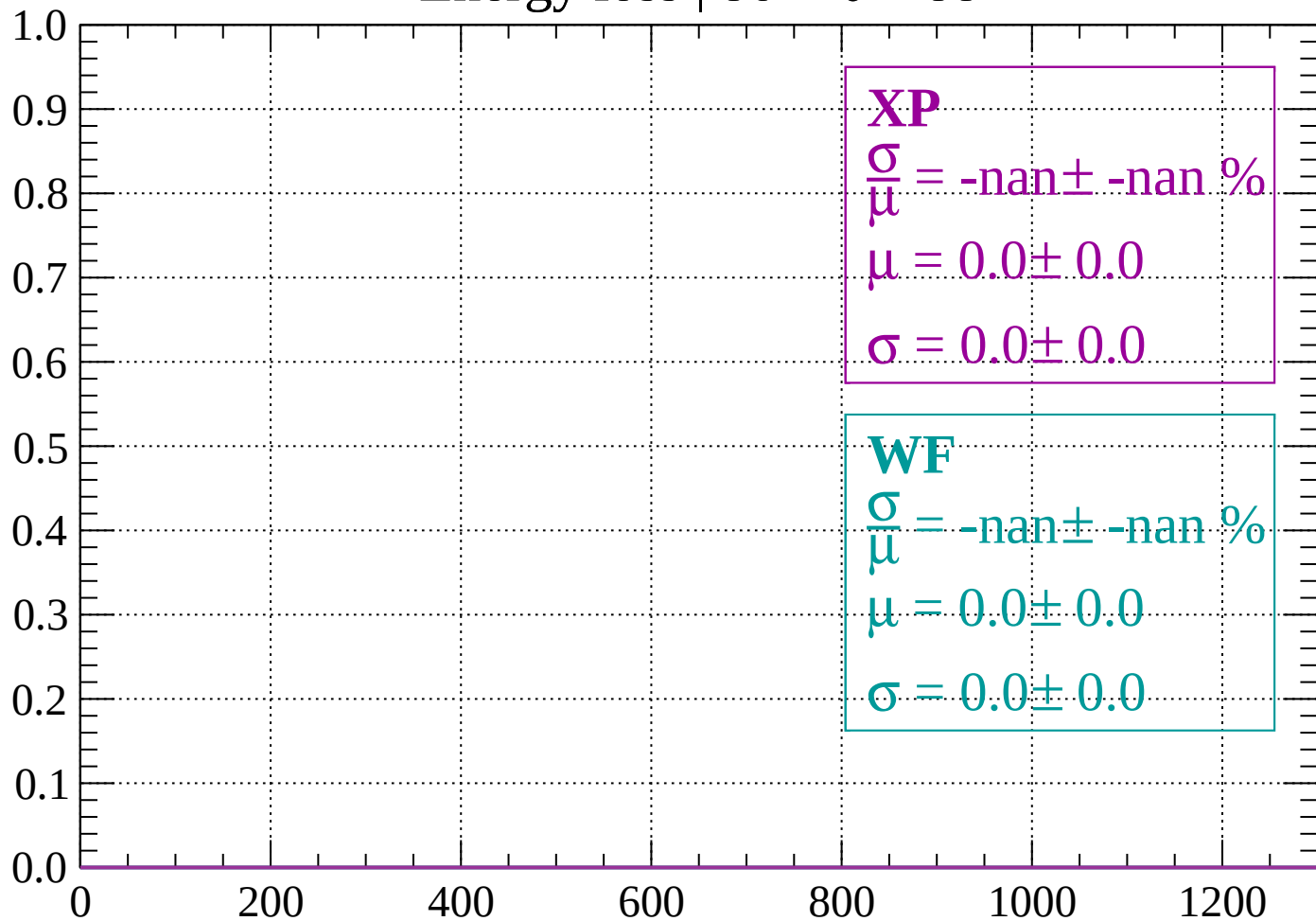
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

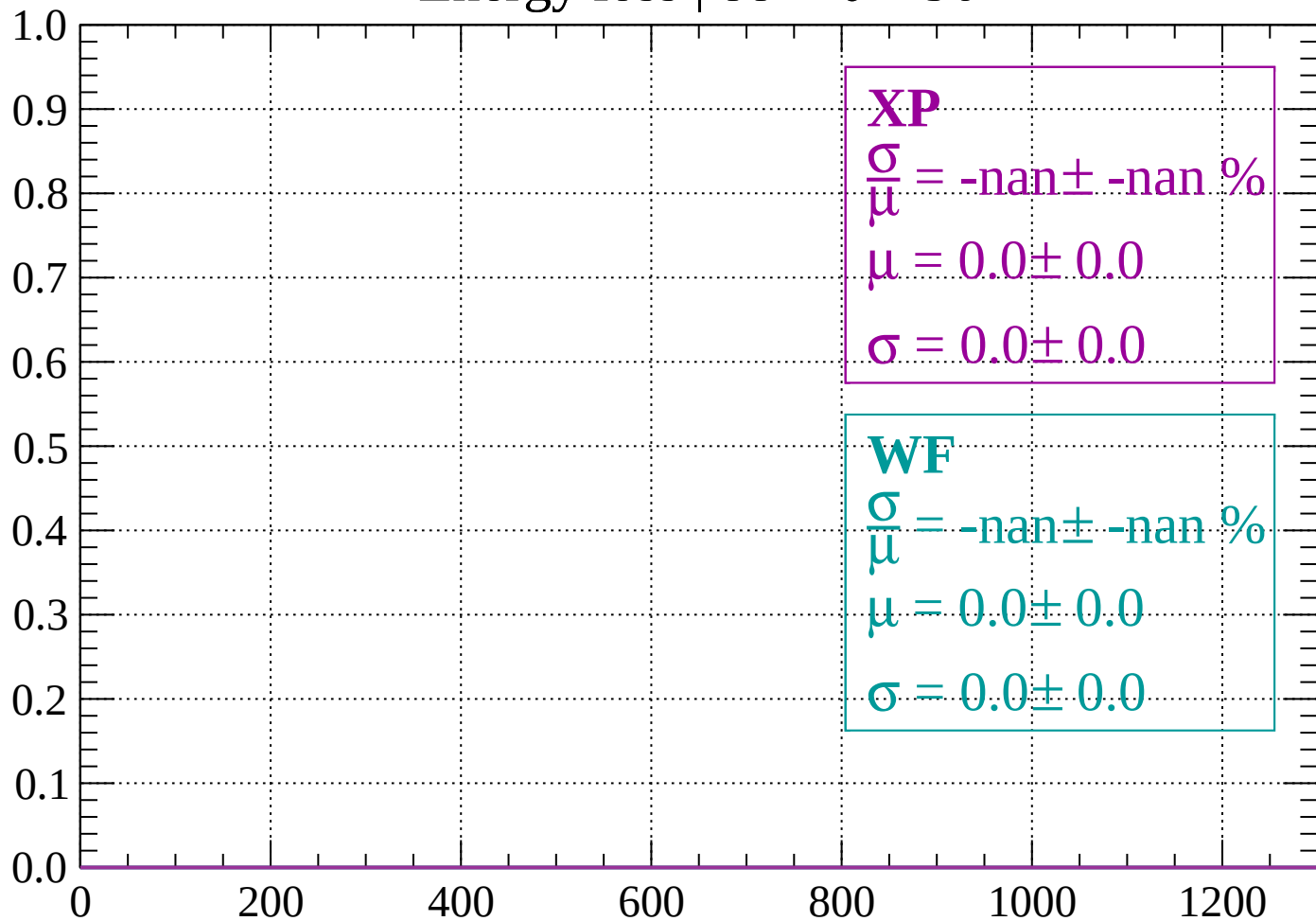
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

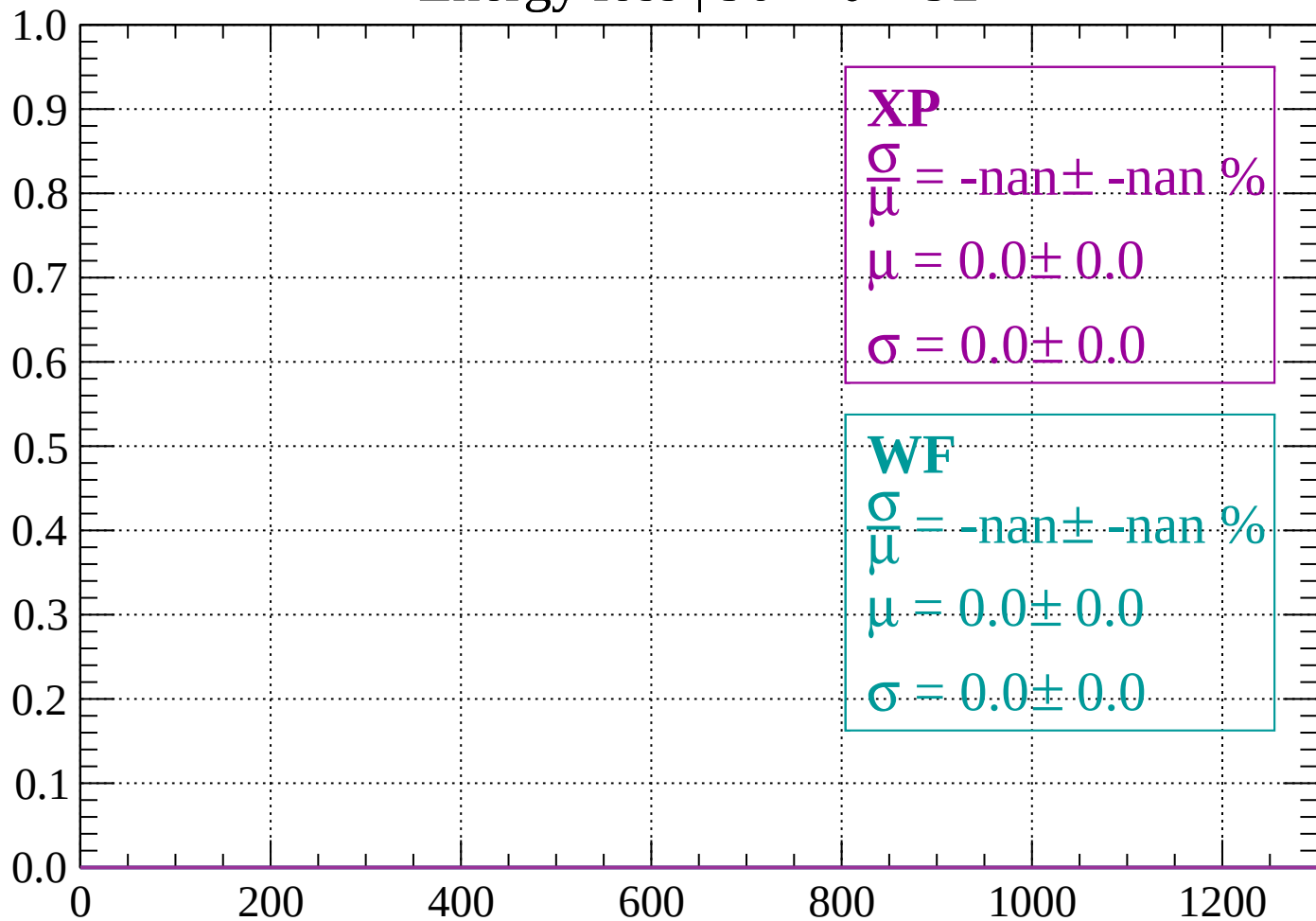
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

